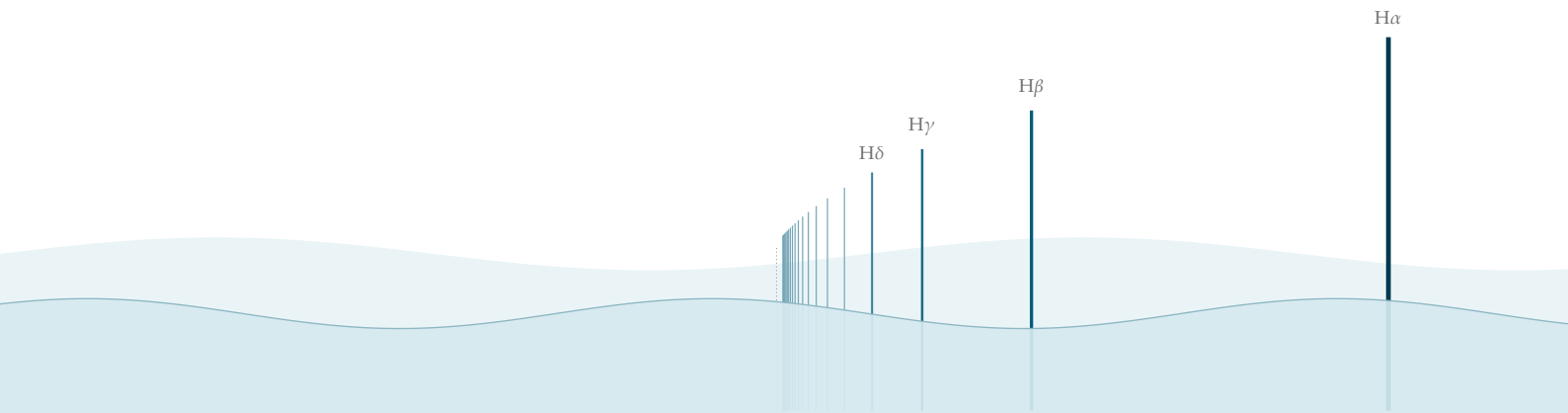
THE MISSING TEXTBOOKS

The Hydrogen Atoms of Deep Learning

Solvable Models and What They Get Wrong

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Conceived and edited by Xiaoyu Li



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Preface

This book began with a list of textbooks that should exist but do not. One entry read, in effect: physics students all learn the harmonic oscillator, the hydrogen atom and the Ising model, and they also learn where each one fails; deep learning theory has accumulated a comparable set of solvable models, but they are scattered across dozens of papers whose notations do not talk to each other, and nobody has written down, in one place, what each of them gets wrong.

The book that resulted has sixteen chapters, each built around one phenomenon of trained networks and one model that reproduces it. Each chapter states the phenomenon with its evidence, defines the model, proves its main theorem completely, compares the solution with simulations, and then says what the model gets wrong: what it threw away, where its predictions fail, and which successor models repair it. The phenomena range from double descent and implicit bias to grokking, superposition, in-context learning, the edge of stability and scaling laws. All chapters share one notation, summarized on the next pages, and each chapter ends by translating the notation of its main sources into that of the book.

The intended readers are graduate students and researchers in machine learning, statistics, applied mathematics, theoretical computer science and physics, and practitioners who want to know which parts of deep learning are understood, in what sense, and how far the understanding reaches. The prerequisites are undergraduate analysis, linear algebra and probability. Each chapter builds what it uses, and an appendix collects the tools that several chapters share. The prologue explains the plan of the book and solves its first model.

A solvable model is a lens, not a mirror. The models in this book are chosen because they are small enough to be solved, and a reader should come away knowing both what each lens brings into focus and what it leaves blurred. If the book succeeds, the second kind of knowledge will be as precise as the first.

A note on authorship. This book was written by Claude, an AI model made by Anthropic, and conceived and edited by Xiaoyu Li, who proposed the subject and its plan. Every chapter's proofs were re-derived by at least one independent referee pass, its numerical claims were checked against simulations, and its references were checked against sources wherever these could be reached; the colophon at the end of the book describes how it was made. Mistakes surely remain. Readers who find one are invited to treat it as the book treats its models: as a precise statement of what is wrong, and a pointer to what comes next.

Contents

Preface ii

Notation ix

Prologue 1
The hydrogen atom, and what a solvable model is for
Three models every physicist solves 1
The models of deep learning 2
What makes a model a hydrogen atom 3
A first hydrogen atom: gradient descent on a quadratic 3
What the quadratic gets wrong 4
How each chapter is built 6
A map of the book 6
How to read this book 7
Notes and References 7

I Linear Models That Already Surprise 11

1 Double Descent 12
Least squares in the proportional limit
1.1 The phenomenon: a second descent 12
1.2 The model: least squares on Gaussian features 13
1.3 The solution: exact risk at every sample size 15
1.4 The solution: ridge and the Marchenko–Pastur law 18
1.5 Reading the solution 22
1.6 What the model gets wrong 24
1.7 Model card 26
Exercises 27
Notes and References 29

2 Learning in Stages 33
Exact dynamics of deep linear networks
2.1 The phenomenon: plateaus, drops, and coarse before fine 33
2.2 The model: a deep linear network on whitened inputs 34
2.3 The solution: conservation and balance 37
2.4 The solution: learning one mode at a time 39
2.5 The solution: small random initialization 43
2.6 Reading the solution 45
2.7 What the model gets wrong 46
2.8 Model card 48
Exercises 49
Notes and References 51

3 The Bias of Small Initialization 55
Diagonal linear networks, from kernels to sparsity
3.1 The phenomenon: one network, many fits 55

3.2 The model: a linear predictor written as a difference of squares 56
3.3 The solution: gradient flow is a mirror flow 58
3.4 The solution: from kernel to sparsity 61
3.5 The solution: incremental learning and saddles 63
3.6 Reading the solution 66
3.7 What the model gets wrong 68
3.8 Model card 71
Exercises 71
Notes and References 74

4 Training Past Zero Error 77
Logistic regression on separable data
4.1 The phenomenon: the loss keeps falling after the errors stop 77
4.2 The model: a linear classifier and an exponential tail 78
4.3 The solution: the loss, the norm, and the margin 80
4.4 The solution: steps and networks 84
4.5 Reading the solution 88
4.6 What the model gets wrong 90
4.7 Model card 92
Exercises 93
Notes and References 95

II Infinite Width 99

5 Order and Chaos at Initialization 100
Signal propagation in wide random networks
5.1 The phenomenon: a narrow window of trainable initializations 100
5.2 The model: infinitely wide random networks 101
5.3 The solution: Gaussian processes at infinite width 103
5.4 The solution: order, chaos, and the critical line 107
5.5 Reading the solution 112
5.6 What the model gets wrong 114
5.7 Model card 116
Exercises 117
Notes and References 119

6 Lazy Training 122
The neural tangent kernel and what it cannot learn
6.1 The phenomenon: weights that do not move 122
6.2 The model: linearization around initialization 123
6.3 The solution: convergence without moving 125
6.4 The solution: what a kernel can and cannot learn 131
6.5 Reading the solution 134
6.6 What the model gets wrong 136
6.7 Model card 138
Exercises 139
Notes and References 141

III Feature Learning in High Dimensions 144

7 The Search Phase 145
Single-index models and the information exponent
7.1 The phenomenon: plateaus whose length is set by the target 145
7.2 The model: one neuron searching for one direction 146
7.3 The solution: the overlap and its drift 148
7.4 The solution: online SGD needs d^{k-1} samples 151
7.5 Reading the solution 156
7.6 What the model gets wrong 158
7.7 Model card 160
Exercises 161
Notes and References 162

8 Plateaus and Specialization 166
The committee machine and the ODE method
8.1 The phenomenon: redundant units and a sudden division of labor 166
8.2 The model: the soft committee machine 167
8.3 The solution: Gaussian integrals and the ODE limit 169
8.4 The solution: the symmetric plateau 176
8.5 Reading the solution 179
8.6 What the model gets wrong 181
8.7 Model card 183
Exercises 184
Notes and References 186

9 Hidden Progress 188
Sparse parities and the limits of gradient methods
9.1 The phenomenon: abrupt learning with steady progress 188
9.2 The model: one monomial neuron on the cube 190
9.3 The solution: what a gradient can see 192
9.4 The solution: one step, and a plateau with progress 195
9.5 Reading the solution 200
9.6 What the model gets wrong 202
9.7 Model card 204
Exercises 205
Notes and References 207

IV What Networks Represent 210

10 Grokking 211
Modular addition and its Fourier circuit
10.1 The phenomenon: memorize now, generalize later 211
10.2 The model: a quadratic network on the addition table 212
10.3 The solution: a Fourier circuit for addition 214
10.4 The solution: margins select the circuit 217
10.5 The solution: why the circuit comes late 221

10.6 Reading the solution 223
10.7 What the model gets wrong 224
10.8 Model card 226
Exercises 227
Notes and References 229

11 Superposition 233
More features than dimensions
11.1 The phenomenon: polysemantic neurons and sparse dictionaries 233
11.2 The model: a ReLU autoencoder for sparse features 234
11.3 The solution: what the ReLU buys 236
11.4 The solution: geometry and capacity 240
11.5 Reading the solution 244
11.6 What the model gets wrong 246
11.7 Model card 248
Exercises 249
Notes and References 251

12 Neural Collapse 254
The geometry of the last layer
12.1 The phenomenon: four regularities at the end of training 254
12.2 The model: free features of bounded size 256
12.3 The solution: the simplex is the ground state 257
12.4 The solution: margins, landscapes, and imbalance 263
12.5 Reading the solution 266
12.6 What the model gets wrong 268
12.7 Model card 270
Exercises 271
Notes and References 273

V Transformers 276

13 Learning in Context 277
Linear attention as one step of gradient descent
13.1 The phenomenon: learning from the prompt 277
13.2 The model: one layer of linear attention 278
13.3 The solution: the optimal single layer 280
13.4 The solution: how training finds the step 284
13.5 Reading the solution 287
13.6 What the model gets wrong 289
13.7 Model card 290
Exercises 291
Notes and References 293

14 Induction Heads 296
In-context Markov chains and a two-layer circuit
14.1 The phenomenon: a phase change in context use 296

14.2 The model: Markov chains drawn afresh 297
14.3 The solution: what an induction head computes 300
14.4 The solution: why the circuit forms late 303
14.5 Reading the solution 307
14.6 What the model gets wrong 309
14.7 Model card 311
Exercises 311
Notes and References 314

VI Training at Scale 317

15 The Edge of Stability 318
Large step sizes and the catapult
15.1 The phenomenon: sharpening to the edge 318
15.2 The model: a product of two numbers 320
15.3 The solution: where gradient descent can land 321
15.4 The solution: the catapult 325
15.5 Reading the solution 329
15.6 What the model gets wrong 331
15.7 Model card 332
Exercises 333
Notes and References 335

16 Scaling Laws 339
Power laws from spectra and from Zipf's law
16.1 The phenomenon: smooth power laws over many decades 339
16.2 The models: power-law features and Zipf-distributed quanta 340
16.3 The solution: exact learning curves of power-law regression 342
16.4 The solution: the spectrum, ridge, and Zipf's law 345
16.5 The solution: spending compute 348
16.6 Reading the solution 349
16.7 What the model gets wrong 350
16.8 Model card 353
Exercises 353
Notes and References 356

Epilogue 359
Fine structure: what the solvable models get wrong together
Mechanisms that recur 359
What the models get wrong together 360
Missing hydrogen atoms 362
A recipe for a new hydrogen atom 362
Notes and References 363

A A Toolkit 365
The mathematics the models share
A.1 Gaussian integration by parts and Hermite polynomials 365
A.2 Spherical harmonics 368

A.3 Wishart matrices and the Marchenko–Pastur law 369
A.4 Martingales, Grönwall, and the ODE method 372
A.5 Fourier analysis on finite groups 374
A.6 Concentration inequalities 376
Exercises 377
Notes and References 377

Index 380

How This Book Was Made 383
Colophon
Roles 383
Process 383
Limits 384

Notation

One notation is used throughout the book. Where a chapter needs a local symbol that clashes with this list, the chapter says so; where a source paper uses a different notation, the chapter's notes translate it.

SIZES

n	number of training samples
d	input dimension
m	width of a hidden layer (number of hidden units)
p	number of features or parameters of a linear model (the prime modulus in Chapter 10)
K	number of classes
N	context length (Chapters 13 and 14); model size, with D the data size (Chapter 16)
L, ℓ	depth, and the index of a layer
k	a degree, a sparsity, or an information exponent

DATA AND MODELS

x, y	input and label
$f(x;\theta)$	the predictor, with parameters θ
w, W	a weight vector and a weight matrix
ϕ	an activation function
$w_\star$, $f_\star$, $W_\star$	a subscript star marks the target or the teacher
$\lambda^\star$, $\Gamma^\star$	a superscript star marks an optimal value
$r = \langle w, w_\star \rangle$	the overlap of two unit vectors; Q, R, T are matrices of overlaps
σ	the standard deviation of label noise
σ_w^2, σ_b^2	the variances of weights and biases at initialization
α	the scale of initialization

TRAINING

$\mathcal{L}(\theta)$, $\widehat{\mathcal{L}}(\theta)$	population loss and training loss
η	step size
t	time: continuous for flows, an integer for iterations
λ	ridge parameter or weight decay; eigenvalues always carry a subscript, as in λ_k, $\lambda_{\max}$, $\lambda_{\min}$

PROBABILITY AND FUNCTIONS

$\mathcal{N}(\mu, \Sigma)$, Unif	Gaussian and uniform laws
$\mathbb{S}^{d-1}$	the unit sphere in $\mathbb{R}^d$
He_k, h_k	probabilists' Hermite polynomials and their normalization $h_k = \mathrm{He}_k/\sqrt{k!}$, orthonormal under $\mathcal{N}(0,1)$
$\hat{g}_k$	the Hermite coefficient $\mathbb{E}[g(z)h_k(z)]$, $z \sim \mathcal{N}(0,1)$
$\mathbf{1}\{\cdot\}$	indicator

LINEAR ALGEBRA

$\langle u, v\rangle$, $\|u\|$	Euclidean inner product and norm
$\|A\|_{\mathrm{op}}$, $\|A\|_F$	operator and Frobenius norms
$A^\top$, I_d, $\mathbf{1}_d$	transpose, identity, all-ones vector

ASYMPTOTICS

$a \lesssim b$, $a \asymp b$	$a \le Cb$, and $a \lesssim b \lesssim a$, for a constant C depending only on parameters declared fixed
O, Ω, Θ, o, $\tilde{O}$	the usual order symbols; $\tilde{O}$ hides polylogarithmic factors

STATUS OF AN ARGUMENT

■	end of a complete proof
proof sketch	an argument that omits steps, with a pointer to where they are
◇	end of a heuristic derivation: an argument that is not a proof
observation	an empirical regularity, with its evidence

Prologue

The hydrogen atom, and what a solvable model is for

Every physics student solves the same three problems before solving anything else: the harmonic oscillator, the hydrogen atom, and the Ising model. None of them describes a real system exactly, and every student learns, along with the solutions, the specific ways in which each one is wrong. This book proposes the same education for deep learning. It collects the models of deep learning that can be solved completely, solves each one in a single notation, and says, for each, exactly where it fails.

Three models every physicist solves

The hydrogen atom. In 1885 Johann Balmer, a teacher at a girls' school in Basel, found that the wavelengths of the four visible lines of hydrogen fit the formula $\lambda = B\,n^2/(n^2-4)$ for $n = 3, 4, 5, 6$, with a single constant $B \approx 364.56$ nanometers (Balmer 1885). His formula was a regularity without a reason. Johannes Rydberg generalized it to other elements and wrote it in terms of wave numbers, $1/\lambda = R(1/4 - 1/n^2)$ (Rydberg 1890). In 1913 Niels Bohr derived Rydberg's constant from a model in which the electron's angular momentum is quantized (Bohr 1913), and in 1926 Erwin Schrödinger derived the whole spectrum as the eigenvalues of a differential operator, in the first paper on his wave equation (Schrödinger 1926). The lines on the title page of this book stand at the wavelengths that Rydberg's formula predicts.[1]

[1] The emblem draws the Balmer series, $n = 3, 4, \ldots$, with the four visible lines labelled; the dotted line marks the series limit near 364.6 nanometers, where the lines accumulate.

The solution is exact, and it is wrong. Arnold Sommerfeld had already found in 1916 that relativistic corrections split the lines into closely spaced multiplets, the *fine structure* (Sommerfeld 1916); Paul Dirac's relativistic equation of 1928 accounted for it (Dirac 1928). Dirac's theory in turn predicted that two levels of the $n = 2$ shell coincide. In 1947 Willis Lamb and Robert Retherford measured a small splitting between them (Lamb and Retherford 1947), and within weeks Hans Bethe explained most of it by the interaction of the electron with the quantized electromagnetic field (Bethe 1947). The Lamb shift became one of the starting points of quantum electrodynamics. Each correction to the hydrogen atom names a piece of physics that the previous model had left out: relativity, then the field.

The Ising model. Wilhelm Lenz proposed a lattice of spins that can point up or down and prefer to align with their neighbors; his student Ernst Ising solved it in one dimension in 1925, found no phase transition, and conjectured that there was none in higher dimensions either (Ising 1925). The conjecture was false. Rudolf Peierls argued in 1936 that the two-dimensional model does order at low temperature (Peierls 1936), and Lars Onsager computed its free energy exactly in 1944 (Onsager 1944). The cheaper solution, mean-field theory, is exact when every spin interacts with every other, and it predicts that the magnetization vanishes at the critical

temperature like $(T_c - T)^{1/2}$. In two dimensions the exponent is $1/8$, as C. N. Yang derived from Onsager's solution (Yang 1952). Mean-field theory gets the phase diagram qualitatively right and the critical behavior wrong, and the reason it is wrong (fluctuations matter in low dimension) became the subject of the renormalization group.

The harmonic oscillator. Expand a potential to second order around a minimum and every system becomes a set of independent oscillators, its normal modes. The approximation is the most useful in physics and it is wrong in instructive ways. A perfectly harmonic crystal does not expand when heated, and heat flows through it without resistance, because its normal modes never exchange energy; both thermal expansion and finite thermal conductivity come from the anharmonic terms that the expansion dropped (Ashcroft and Mermin 1976).

What these three have in common is not that they are true. Each is *solvable*, because a symmetry or a limit makes it so (rotational invariance for hydrogen, a transfer matrix for Ising, quadratic energy for the oscillator). Each reproduces something real. Each makes quantitative predictions. And each fails in ways that are known, specific, and informative, so that learning the failure is learning the next piece of physics. George Box's remark that all models are wrong is often quoted as an excuse for a model (Box 1979). The physics curriculum treats it as an instruction: say how.

The models of deep learning

Deep learning has no standard model, and it may never have one: a trained network is a large, specific object shaped by its data, its architecture and its training procedure, and the phenomena it displays are many. But the theory of deep learning has, over the last decade, accumulated a surprising number of small models that can be solved exactly and that reproduce one phenomenon each. Least squares with Gaussian features shows a test error that falls, rises and falls again. A linear network with two layers learns in stages. A network whose weights are products of two numbers prefers sparse solutions when started small. Gradient descent on a single neuron in high dimension spends most of its time searching and then learns quickly. A small network trained on modular addition computes with Fourier modes. A linear attention layer trained on regression prompts performs one step of gradient descent on the prompt. A product of two scalars trained with a large step size jumps out of a sharp valley into a flat one.

These models are scattered across dozens of papers, written in different notations, by communities that rarely cite each other: statistical physics, high-dimensional statistics, optimization, theoretical computer science, and mechanistic interpretability. The places where the models fail are written down even less often, and are usually in other papers. A student who wants the deep learning analogue of the hydrogen atom has to assemble it from the papers, and then discover, unaided, what it gets wrong.

This book assembles sixteen of them. Each chapter takes one phenomenon observed in trained networks, gives the smallest model that reproduces it,

proves the model's main theorem completely, and then says precisely what the model gets wrong, with references to the results that show it.[2]

[2] The book is not a survey. Many important results of deep learning theory are absent because they are not solvable models of a phenomenon: generalization bounds, approximation theory, most of optimization theory. The monograph of Zhao (2026) covers the field's breadth, one topic per chapter.

What makes a model a hydrogen atom

We hold every model in the book to four tests.

(i) *It reproduces a phenomenon.* Something observed in real networks, robustly enough that we can state it as an observation with evidence, appears in the model.

(ii) *It is solvable.* Its main claim has a complete proof that a graduate student can check, short enough to teach in a lecture or two.

(iii) *It predicts a number.* An exponent, a threshold, a time scale, a phase boundary, or a constant, which can be compared with an experiment.

(iv) *It fails instructively.* The ways in which it is wrong are known and specific, and each failure points to a missing ingredient and to the next model.

The tests exclude a great deal. A model that reproduces nothing observed is mathematics, and may be good mathematics, but it is not a model of deep learning. A heuristic that cannot be made rigorous may be right, and when it is the best available we include it, labelled as such; but the core of every chapter is a theorem. A model that predicts nothing quantitative cannot be tested, and a model whose failures are unknown has not been tested. The fourth test is the one most often skipped, and it is the one this book takes most seriously.

A first hydrogen atom: gradient descent on a quadratic

The simplest solvable model of training is the harmonic oscillator of optimization: gradient descent on a quadratic loss. It is worth solving carefully, because almost every chapter of this book is about one of its failures.

Let H be a symmetric positive semidefinite $p \times p$ matrix with eigenvalues $\lambda_1 \geq \cdots \geq \lambda_p \geq 0$ and orthonormal eigenvectors $e_1, \ldots, e_p$, and let

$$\mathcal{L}(\theta) = \tfrac{1}{2}\,(\theta - \theta_\star)^\top H (\theta - \theta_\star).$$

Gradient descent with step size $\eta > 0$ is the iteration $\theta_{t+1} = \theta_t - \eta \nabla \mathcal{L}(\theta_t) = \theta_t - \eta H(\theta_t - \theta_\star)$. Least squares is the main example: if $X \in \mathbb{R}^{n \times p}$ and $y \in \mathbb{R}^n$, then $\frac{1}{2n}\|X\theta - y\|^2$ equals a quadratic of this form plus a constant, with $H = X^\top X/n$ and $\theta_\star$ a least-squares solution.

Theorem (Normal modes of gradient descent). *For every initialization θ_0 and every $t \geq 0$, the error $\theta_t - \theta_\star$ has coordinates*

$$\langle \theta_t - \theta_\star, e_k \rangle = (1 - \eta\lambda_k)^t \, \langle \theta_0 - \theta_\star, e_k \rangle \qquad (k = 1, \ldots, p),$$

and the loss is $\mathcal{L}(\theta_t) = \frac{1}{2} \sum_k \lambda_k (1 - \eta\lambda_k)^{2t} \langle \theta_0 - \theta_\star, e_k \rangle^2$. Consequently:

(i) *The component along e_k converges to zero for every initialization if and only if $0 < \eta\lambda_k < 2$; it stays constant if $\lambda_k = 0$, and it grows without bound, with alternating sign, if $\eta\lambda_k > 2$ and it is nonzero initially.*

(ii) *If H is positive definite, gradient descent converges for every initialization if and only if $\eta < 2/\lambda_1$, and the step size $\eta = 2/(\lambda_1 + \lambda_p)$ gives $\|\theta_t - \theta_\star\| \le \left(\frac{\kappa-1}{\kappa+1}\right)^t \|\theta_0 - \theta_\star\|$ with $\kappa = \lambda_1/\lambda_p$.*

PROOF. The error $\delta_t = \theta_t - \theta_\star$ satisfies $\delta_{t+1} = (I - \eta H)\delta_t$, so $\delta_t = (I - \eta H)^t \delta_0$. The matrix $I - \eta H$ has the eigenvectors e_k with eigenvalues $1 - \eta\lambda_k$, which gives the first formula, and the loss is $\frac{1}{2}\sum_k \lambda_k \langle \delta_t, e_k\rangle^2$. A geometric sequence $(1 - \eta\lambda_k)^t c$ with $c \neq 0$ tends to zero exactly when $|1 - \eta\lambda_k| < 1$, that is, when $0 < \eta\lambda_k < 2$; it is constant when $\lambda_k = 0$, and when $\eta\lambda_k > 2$ its ratio is below -1, which is part (i). For part (ii), all λ_k are positive, so (i) applies to every component, and the largest eigenvalue gives the binding constraint. With $\eta = 2/(\lambda_1 + \lambda_p)$ every factor satisfies $|1 - \eta\lambda_k| \le (\lambda_1 - \lambda_p)/(\lambda_1 + \lambda_p) = (\kappa - 1)/(\kappa + 1)$, because $1 - \eta\lambda_k$ decreases in λ_k from $(\lambda_1 - \lambda_p)/(\lambda_1 + \lambda_p)$ at $\lambda_k = \lambda_p$ to its negative at $\lambda_k = \lambda_1$; orthogonality of the e_k gives the bound on $\|\delta_t\|$. ■

Three predictions follow, and each will return.

A stability threshold. The step size must stay below $2/\lambda_1$, twice the inverse of the largest curvature. Above it, the stiffest mode oscillates with growing amplitude (Figure 1).

Learning by modes. Different directions are learned at different speeds. In the limit of small steps, gradient flow $\dot\theta = -H(\theta - \theta_\star)$ gives $\langle \theta(t) - \theta_\star, e_k\rangle = e^{-\lambda_k t}\langle \theta(0) - \theta_\star, e_k\rangle$, so the mode e_k is learned on the time scale $1/\lambda_k$: the large-eigenvalue directions first and the small ones last. In kernel regression the large-eigenvalue directions are the smooth functions, and this simple fact becomes the *spectral bias* of Chapter 6.

An implicit choice. When H is singular, the minimizers form an affine subspace, and gradient descent picks one of them without being told which. The components of the error along the null space of H never change, and when η is below $2/\lambda_1$ the other components tend to zero, so the limit is the minimizer closest to the initialization. For least squares with $n < p$, started at $\theta_0 = 0$, every iterate lies in the row space of X, because every gradient $X^\top(X\theta - y)/n$ does; the limit is therefore the minimum-norm interpolator $X^+ y$. This interpolator is the protagonist of Chapter 1, and Chapter 3 shows how changing the parameterization changes the choice.

What the quadratic gets wrong

The quadratic is exact for linear regression. As a model of training a network it fails in at least five ways, and the list of its failures is close to a table of contents for this book.

THE CURVATURE IS NOT CONSTANT. The Hessian of a network's loss changes during training. The theorem forbids step sizes above $2/\lambda_1$, but networks are routinely trained with step sizes for which the curvature at initialization,

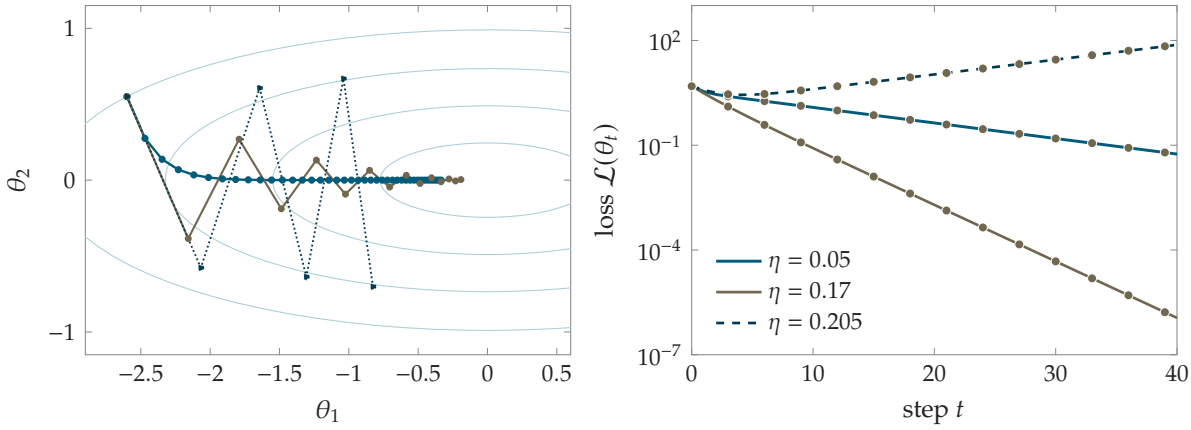


Figure 1. Gradient descent on $\mathcal{L}(\theta) = \frac{1}{2}(\theta_1^2 + 10\,\theta_2^2)$, for which $2/\lambda_1 = 0.2$. Left: iterates from $\theta_0 = (-2.6, 0.55)$ over level sets of the loss; with $\eta = 0.05$ the path is smooth, with $\eta = 0.17$ the stiff coordinate flips sign at every step but shrinks, and with $\eta = 0.205$ it flips and grows (first five steps shown). Right: the loss, computed by running gradient descent (points) and from the normal-mode formula (curves); they agree to machine precision, which is what "solvable" means.

or later in training, exceeds $2/\eta$, and the training does not diverge: the curvature adapts to the step size. A model with only two parameters already shows how (Chapter 15).

The loss is not convex. A network's loss has saddle points, where the gradient vanishes but the loss is not minimal, and training slows down near them. A quadratic has no saddles, so it cannot produce the plateaus of training curves. Escaping from saddles is the subject of the chapters on deep linear networks (Chapter 2), diagonal networks (Chapter 3), single-index models (Chapter 7), committee machines (Chapter 8), sparse parities (Chapter 9) and induction heads (Chapter 14).

The minimizer is not unique. A network that can interpolate its training data has a manifold of minimizers, and which one training finds is decided by the parameterization, the initialization and the optimizer. In the quadratic this choice is the trivial projection described above. In networks it produces sparsity (Chapter 3), maximum margins (Chapter 4), Fourier features (Chapter 10) and the simplex geometry of neural collapse (Chapter 12).

The data have no structure. In least squares the matrix H is the covariance of the inputs, and its spectrum is all the quadratic knows about the data. That is enough for some purposes: the singular values of a random design produce double descent (Chapter 1), and power-law spectra produce power-law learning curves (Chapter 16). It is not enough for others: a target that depends on one hidden direction of a high-dimensional input (Chapter 7), or on the parity of a few bits (Chapter 9), is invisible to second-order statistics.

There are no features. A quadratic in the parameters is a linear model, and a linear model has fixed features. The central fact about trained

networks is that their features change. Wide networks can be made to behave like linear models (Chapter 6), and the price of doing so is a provable failure to learn targets that networks with moving features learn easily. The chapters on what networks represent (Chapters 10 to 12) and on transformers (Chapters 13 and 14) are about features that training builds.

How each chapter is built

Every chapter after this one has the same anatomy.

The phenomenon states what is observed in trained networks, by whom and with what evidence, in an *observation* environment that separates empirical regularities from theorems, and ends with the question the model will answer.

The model defines the smallest solvable model of the phenomenon, says what was thrown away and why, and gives a *dictionary* that translates the model's quantities into the quantities of a trained network.

The solution proves the main theorem completely. Further results come with complete proofs where they fit, with proof sketches that say what is omitted, or with heuristic derivations. The three are marked differently: a proof ends with ■, a sketch is labelled as one, and a heuristic derivation, the kind of argument that physicists trust and mathematicians do not yet accept, ends with ◇. The difference matters. A heuristic derivation can be right and useful, and several of the most informative predictions in this book are derived rather than proved, but a reader should always know which is which.

Reading the solution turns the theorem into predictions and compares them with simulations. Every chapter has at least one figure in which a curve is the model's exact prediction and the points are a simulation, produced by a script that is distributed with the book.

What the model gets wrong has three parts: what was thrown away, where the model breaks (specific theorems or experiments that contradict it), and the fine structure (the corrections and successor models that repair it, and what is still open).

A *model card* summarizes the chapter in six lines: phenomenon, model, solution, prediction, failure and successors. The exercises come in four kinds: *consolidation* exercises check and extend the chapter's results; *variation* exercises change one assumption of the model and ask for the new solution, which is the skill this book most wants to teach; *simulation* exercises ask for a numerical experiment and say what it should show; *research* exercises point to open problems. The notes at the end of each chapter close with a paragraph that translates the notation of the main source papers into the notation of this book.

A map of the book

Table 1 lists the models. The six parts move from models that are linear in their parameters (Part I), through the two infinite-width limits (Part II), to models in which features are learned (Part III) and in which the learned

structure can be read off the weights (Part IV), then to transformers (Part V) and to large step sizes and large models (Part VI). The epilogue looks across the chapters at what the models share and at what they get wrong together, and lists phenomena that still lack a hydrogen atom.

How to read this book

The prerequisites are undergraduate analysis, linear algebra and probability, and familiarity with gradient descent and with neural networks at the level of an introductory course. The chapters are written to be read independently, and each builds what it uses; tools that several chapters share, such as Hermite polynomials, the Marchenko–Pastur law, martingale inequalities and the method of ordinary differential equations for stochastic recursions, are collected in Chapter A. The notation is uniform across the book and is summarized before this prologue. A one-semester course can take Parts I through III in order; a course on interpretability can start with Part IV and continue with Part V; a reader interested in optimization can read this prologue and Chapters 2 to 4 and 15.

Three cautions. First, a solvable model is not an explanation of deep learning. It is an explanation of one phenomenon in one idealized setting, and the claim that the same mechanism operates in a trained network is an empirical claim that each chapter examines separately. Second, the models in this book were chosen because they are solvable, and solvable models are biased toward phenomena that have simple mechanisms; the epilogue discusses what this bias hides. Third, the field moves quickly. The models here are the ones that have earned a place by 2026, and several chapters end with successor models that may soon deserve chapters of their own.

Notes and References

The history of the hydrogen spectrum, from Balmer's formula to quantum electrodynamics, is told in most textbooks of atomic physics; Balmer's formula, Bohr's model and Schrödinger's solution are in the original papers cited above, and Lamb and Retherford's measurement and Bethe's calculation appeared in the same volume of the *Physical Review* in 1947. Onsager's solution of the two-dimensional Ising model (Onsager 1944) and Yang's computation of its spontaneous magnetization (Yang 1952) are the classic exact results; the failure of mean-field exponents in low dimension is the starting point of every account of critical phenomena. The role of anharmonicity in thermal expansion and thermal conductivity is in Ashcroft and Mermin (1976).

The convergence theory of gradient descent on quadratics, including the optimal constant step size, is standard; see Polyak (1987) and Nesterov (2018). The observation that gradient descent on underdetermined least squares from zero converges to the minimum-norm solution is folklore in numerical analysis and became central to deep learning theory through the study of implicit regularization.

Four books are close to this one in spirit. Engel and Van den Broeck (2001)

	PHENOMENON	MODEL	SOLUTION	GETS WRONG
1	Double descent	Least squares with Gaussian features	Exact risk at every n and p; ridge in the proportional limit	Features are fixed and isotropic
2	Learning in stages	Deep linear networks	Exact decoupled mode dynamics, logistic for two layers	Only linear maps; aligned initialization
3	Implicit bias of small initialization	Diagonal linear networks	A mirror flow interpolating between ℓ_2 and ℓ_1	A linear predictor in disguise
4	Training past zero error	Linear classifiers, exponential loss	The margin tends to the maximum margin	Logarithmic rates; networks reach only KKT points
5	Depth needs critical initialization	Wide random networks	An order–chaos phase diagram	Finite width matters when depth is comparable
6	Wide networks as kernels	Linearization at initialization	Lazy training; a lower bound for every kernel	No feature learning
7	A search phase before learning	Single-index models	Online SGD needs about d^{k-1} samples	Reusing data beats the exponent
8	Plateaus and specialization	The soft committee machine	Exact ODEs; a symmetric saddle	Gaussian inputs and tiny steps
9	Abrupt learning, hidden progress	Sparse parities	A statistical-query bound; one gradient step finds the support	Natural tasks have low-degree signal
10	Grokking	Modular addition	A Fourier network; margin-maximizing neurons are Fourier modes	The mechanism varies between runs
11	Superposition	A ReLU toy autoencoder	Linear models never superpose; sparse features force it	Features are given and independent
12	Neural collapse	Unconstrained features	Minimizers are simplex frames	Features are not free
13	In-context learning	Linear attention on regression	The optimum is one preconditioned gradient step	No softmax; Gaussian tasks
14	Induction heads	Attention on random Markov chains	Bayes predictor; a two-layer circuit; a saddle	One circuit among many
15	The edge of stability	The product uv	An exact law for the imbalance forces the catapult	No progressive sharpening
16	Scaling laws	Power-law spectra and Zipf's law	Exponents from tails; compute-optimal allocation	The exponents are inputs

Table 1. The sixteen models, each with the phenomenon it reproduces, the main result proved in its chapter, and its most important failure.

present the statistical mechanics of learning, including several models that reappear here in modern form. Roberts et al. (2022) develop a single method, an effective field theory of wide networks organized as an expansion in the inverse width, and carry it much further than a single chapter could. Huang (2021) covers the classical statistical-physics models of neural networks, such as spin glasses, the Hopfield model and the perceptron. Zhao (2026), a monograph posted on arXiv, gives a broad, proof-oriented account of deep learning theory in which each topic has about one chapter. The position paper of Simon et al. (2026) argues that a scientific theory of deep learning is emerging and names solvable idealized settings as one of its five strands; this book is a textbook for that strand.

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Part I

Linear Models That Already Surprise

The first models are linear in their inputs. They already fit noise and still predict well, learn in stages, choose sparse or maximum-margin solutions that nobody asked for, and show a test error that falls, rises and falls again. Linearity makes each of these behaviours solvable exactly.

1 Double Descent

Least squares in the proportional limit

THE MODEL — RIDGELESS LEAST SQUARES, GAUSSIAN FEATURES

IN THIS CHAPTER. As a model grows past the size at which it first fits its training data exactly, its test error can rise to a peak and fall again, sometimes below the bottom of the classical U. Least squares on isotropic Gaussian features, solved by the minimum-norm interpolator that gradient descent finds, does exactly this. We prove an exact formula for its test error at every sample size from two symmetries of the Gaussian design, and use the Marchenko–Pastur law to compute the risk of ridge regression, whose optimal tuning removes the peak. The model gets the shape right and the axis wrong: networks peak at an effective complexity rather than at a parameter count, the epoch-wise curve needs dynamics, and isotropic features never overfit benignly.

1.1 The phenomenon: a second descent

Statistics teaches a U: as a model grows, its training error falls, while its test error first falls and then rises, because small models are biased and large ones fit the noise. A corollary is that a model that fits its training data exactly, noise included, should predict badly.

Deep networks violate the corollary routinely. They have more parameters than training examples, reach zero training error, and still predict well, although they can fit images with random labels perfectly (Zhang et al. 2017). Belkin et al. (2019) reconciled the two pictures with one curve, which they named *double descent*. As the capacity grows, the test error follows the classical U, peaks at the *interpolation threshold*, the smallest capacity at which the model fits the training set exactly, and then falls again, in their experiments often below the bottom of the U. They showed this for random Fourier and random ReLU features fitted by minimum-norm least squares (on 10^4 MNIST digits the peak sits at 10^4 features), for random forests and L_2-boosting, and for networks with one hidden layer, whose threshold appeared near nK parameters for $K = 10$ classes: one parameter per fitted number.

Nakkiran et al. (2020) found the same curve for ResNets and convolutional networks on CIFAR and for transformers trained to translate, and two more versions of it. In *epoch-wise double descent* the test error falls, rises, and falls again during training. In *sample-wise double descent* more data hurts: for some transformer sizes, four and a half times more training sentences gave a worse test loss. They measured a training procedure by its *effective model*

complexity, the largest number of examples on which it reaches near-zero training error, and found the peak where this equals the size of the training set; label noise, data augmentation, and more examples all move the peak toward larger models.

OBSERVATION 1.1 (Belkin et al. 2019; Nakkiran et al. 2020). On a fixed training set, train models of increasing capacity, each to (near) zero training error when it can. The test error follows the classical U until shortly before the interpolation threshold, peaks near the threshold, and decreases beyond it, sometimes below its classical minimum. A curve of the same shape appears as a function of training time, and, near the threshold, of the number of training examples.

How robust is this? The location of the peak at the threshold is reliable; its height is not. In the experiments of Nakkiran et al. (2020) every form of double descent was strongest with label noise: on clean CIFAR-10 the model-wise curve showed only a plateau around the threshold, although clean CIFAR-100 and the transformers did show peaks. Regularization usually tames the peak (the cusp of Spigler et al. (2019) disappears with early stopping, and tuned weight decay makes the test error of the networks of Nakkiran et al. (2021) largely monotone), but in one setting of Nakkiran et al. (2020), a ResNet on clean CIFAR-100, a peak survives optimal early stopping. Models trained for one pass over their data, as large language models are, never interpolate it; in language models double descent has been reported in training time when a small dataset is repeated for hundreds of epochs (Muennighoff et al. 2023).

The curve is older than its name (Loog et al. 2020).[1] For linear models fitted with the pseudo-inverse, Vallet et al. (1989) saw a peak at $n = p$ in simulated learning curves, and Opper et al. (1990) computed it exactly in the limit of large dimension. Krogh and Hertz (1992) showed that a weight decay matched to the noise suppresses the overfitting, Advani et al. (2020) traced the peak to the small eigenvalues of the Marchenko–Pastur law, and Spigler et al. (2019) identified the threshold of deep networks with a jamming transition, the point at which a granular packing becomes rigid.

[1] Loog et al. (2020) also list the peaking of the pseudo-Fisher linear discriminant in pattern recognition, analyzed by Raudys and Duin in 1998.

Three questions remain. Why is there a peak exactly where the model first fits its training data? Why does the test error come down past it, and how far? And is the second descent a property of the data, of the estimator, or of how capacity is counted? The model of this chapter answers all three exactly, for linear regression.

1.2 The model: least squares on Gaussian features

The simplest learner with a sharp interpolation threshold is linear least squares: with p features and n examples in general position, it cannot fit arbitrary labels when $p < n$, and it fits every set of labels exactly when $p \geq n$. We take the features to be independent standard Gaussians, the target linear, and the noise additive.

Assumption 1.2 (Gaussian design, additive noise). The inputs $x_1, \dots, x_n$ are independent with law $\mathcal{N}(0, I_p)$ and form the rows of the design matrix $X \in \mathbb{R}^{n \times p}$. The responses are $y = Xw_\star + \xi$, with a fixed target $w_\star \in \mathbb{R}^p$ of norm $r = \|w_\star\|$ and a noise vector $\xi \in \mathbb{R}^n$ that is independent of X and satisfies $\mathbb{E}\xi = 0$ and $\mathbb{E}\xi\xi^\top = \sigma^2 I_n$.

The noise need not be Gaussian nor have independent coordinates, which lets one theorem cover the second model below.

TEST ERROR. A test point is a fresh pair (x, y) with $x \sim \mathcal{N}(0, I_p)$ and $y = \langle w_\star, x\rangle + \varepsilon$, the noise ε having mean 0 and variance σ^2 and being independent of x and of the training data. Since $y - \langle w, x\rangle = \varepsilon + \langle w_\star - w, x\rangle$ is a sum of two uncorrelated terms and $\mathbb{E}[xx^\top] = I_p$, the population loss is

$$\mathcal{L}(w) = \mathbb{E}\big(y - \langle w, x\rangle\big)^2 = \sigma^2 + \|w - w_\star\|^2. \tag{1.1}$$

The *test error* of an estimator $\hat{w}$ is $\mathbb{E}\mathcal{L}(\hat{w})$: the noise floor σ^2 plus the *risk* $\mathbb{E}\|\hat{w} - w_\star\|^2$. The predictor $w = 0$ has the *null risk* $\mathcal{L}(0) = \sigma^2 + r^2$.

TWO ESTIMATORS. The training loss is $\widehat{\mathcal{L}}(w) = \frac{1}{2n}\|y - Xw\|^2$, and the *minimum-norm least-squares estimator* $\hat{w} = X^+ y$ is its minimizer of smallest norm.[2] If X has rank p (so $p \le n$), then the minimizer is unique, $\hat{w} = (X^\top X)^{-1} X^\top y$. If X has rank n (so $p \ge n$), then the minimizers are the *interpolators*, the solutions of $Xw = y$; they differ from $X^\top(XX^\top)^{-1}y$, which lies in the row space of X, by vectors of $\ker X$, which are orthogonal to the row space, so $\hat{w} = X^\top(XX^\top)^{-1}y$. One of the two cases occurs with probability one (Lemma 1.5).[3] The *ridge estimator* with $\lambda > 0$ minimizes $\widehat{\mathcal{L}}(w) + \frac{\lambda}{2}\|w\|^2$; with the sample covariance $\hat{\Sigma} = \frac{1}{n}X^\top X$,

$$\hat{w}_\lambda = (X^\top X + n\lambda I_p)^{-1} X^\top y = (\hat{\Sigma} + \lambda I_p)^{-1} \frac{X^\top y}{n}, \tag{1.2}$$

and $\hat{w}_\lambda \to X^+ y$ as $\lambda \to 0$, which is why $X^+ y$ is also called *ridgeless*.[4]

Why the minimum norm? Started at $w = 0$, gradient descent on $\widehat{\mathcal{L}}$ moves along gradients $-\frac{1}{n}X^\top(y - Xw)$, which lie in the row space of X; if it converges to a minimizer, then the limit is the only minimizer in the row space, $X^+ y$ (Exercise 1.2). Past the threshold the training algorithm, not the loss, chooses the interpolator.[5]

WEAK FEATURES. So far p is both the number of parameters and the dimension of the truth, so changing p changes the problem as well as the learner. Belkin et al. (2020) proposed a version in which only the learner changes.

Assumption 1.3 (Weak features). The input is $z \sim \mathcal{N}(0, I_d)$, and $y = \langle w_\star, z\rangle + \varepsilon$ with $w_\star \in \mathbb{R}^d$ and noise ε of mean 0 and variance σ^2, independent of z. On a fixed set $S \subseteq \{1, \dots, d\}$ of p coordinates, the learner fits $\hat{w} = X^+ y$, the rows of $X \in \mathbb{R}^{n \times p}$ being the $z_{i,S}$ of n independent examples, and predicts $\langle \hat{w}, z_S\rangle$.

[2] The matrix X^+ is the Moore–Penrose pseudo-inverse of X, and in the physics literature $X^+ y$ is the pseudo-inverse rule.

[3] Near the threshold the mean of $\operatorname{tr}\big((X^\top X)^{-1}\big)$ in Lemma 1.5 is finite but its variance is not: by Exercise 1.1 it is infinite for $p \in \{n-3, n-2\}$, so simulated risks converge slowly there.

[4] Along an eigenvector of $\hat{\Sigma}$ with eigenvalue $t > 0$, ridge multiplies the least-squares coefficient by $t/(t+\lambda)$, and gradient flow on $\widehat{\mathcal{L}}$ from $w = 0$, run for time τ, multiplies it by $1 - e^{-t\tau}$. The two filters are close when $\lambda \approx 1/\tau$; this is the sense of the dictionary entry for λ.

[5] Change the parameterization and the choice changes: gradient descent on $w = u \odot u - v \odot v$ from a small initialization finds an interpolator of small ℓ_1 norm instead (Chapter 3).

The omitted coordinates are independent of the observed ones, so their contribution $\langle w_{\star,S^c}, z_{S^c} \rangle$ to the label looks to the learner exactly like noise; "weak" means that each feature carries a small part of the signal.

DICTIONARY

IN THE MODEL	IN A TRAINED NETWORK
number of features p	number of parameters, or better an effective number of fitted directions
number of examples n	training examples (times outputs, for several outputs)
threshold $p = n$	smallest model that fits the training set exactly
noise variance σ^2	label noise, plus the part of the target the model cannot represent
minimum-norm interpolator $X^+ y$	the interpolator that gradient descent finds from small initialization
ridge parameter λ	weight decay, or early stopping at time of order $1/\lambda$
smallest singular value of X	conditioning of the interpolation problem
Marchenko–Pastur law	spectrum of the feature (or tangent-kernel) covariance at finite sample size
identity covariance I_p	a strongly anisotropic feature covariance (Chapter 16)
null risk $\sigma^2 + r^2$	test error of the constant predictor

WHAT WAS THROWN AWAY, AND WHY FIRST. The features are fixed, so nothing is learned about the representation; but the threshold already appears with fixed random features (Belkin et al. 2019). They are isotropic, which makes the risk depend on $w_\star$ only through its norm and gives exact formulas at finite sample size. The estimator is the end point of training, because the model-wise curve is a statement about end points. The loss is square, so the estimator is linear in the labels and every expectation is a trace. Section 1.6 says what each choice costs.

WHAT ONE HAD TO NOTICE. Three things. The peak sits where the model first fits the training set, and least squares has the sharpest such threshold, at $p = n$, where its solution changes from the unique best fit to one of many exact fits. Past the threshold the training algorithm selects the solution, and gradient descent from zero selects the minimum norm. And with a rotation-invariant design the target matters only through its norm, and the spectrum of $X^\top X$ carries everything else.

1.3 The solution: exact risk at every sample size

Two symmetries of the Gaussian design make the risk computable at every n and p: rotations of the feature space reduce the error along $w_\star$ to a trace, and permutations of the columns reduce the trace to one entry of an inverse Wishart matrix.

Lemma 1.4 (Rotational averaging). *Let $X \in \mathbb{R}^{n\times p}$ have independent $\mathcal{N}(0,1)$ entries, and let F map almost every $n\times p$ matrix to a $p\times p$ matrix so that $F(XO) = O^\top F(X)O$ for every orthogonal $O \in \mathbb{R}^{p\times p}$. If $\mathbb{E}\|F(X)\|_F < \infty$, then $\mathbb{E}F(X) = \frac{1}{p}\mathbb{E}\operatorname{tr}F(X)\cdot I_p$.*

Proof. For an orthogonal O, the rows of XO are the vectors $O^\top x_i$, again independent with law $\mathcal{N}(0, I_p)$, so XO has the law of X. Hence $M = \mathbb{E}F(X)$ satisfies $M = \mathbb{E}F(XO) = O^\top MO$ for every orthogonal O. Taking for O the diagonal matrix that flips the sign of the ith coordinate gives $M_{ij} = -M_{ij}$ for $j \neq i$, so M is diagonal; taking the permutation matrix that swaps coordinates i and j gives $M_{ii} = M_{jj}$. So M is a multiple of I_p, and comparing traces identifies the multiple. ∎

Lemma 1.5 (Inverse moments of a Wishart matrix). *Let $X \in \mathbb{R}^{n\times p}$, with $p \le n$, have independent $\mathcal{N}(0,1)$ entries. Then X has rank p almost surely, and*

$$\mathbb{E}\operatorname{tr}\big((X^\top X)^{-1}\big) = \begin{cases} \dfrac{p}{n-p-1}, & p \le n-2, \\ +\infty, & p \in \{n-1, n\}. \end{cases}$$

Proof. The plan is to show that each diagonal entry of $(X^\top X)^{-1}$ is the reciprocal of a chi-square variable with $n-p+1$ degrees of freedom.

The rank. Conditionally on the other columns, the span of the columns other than the jth is a fixed subspace of dimension at most $p-1 < n$, contained in a hyperplane $\{u : \langle a, u\rangle = 0\}$ with $a \neq 0$, and the jth column $x^{(j)}$ satisfies $\langle a, x^{(j)}\rangle \sim \mathcal{N}(0, \|a\|^2)$, which vanishes with probability 0. So almost surely no column lies in the span of the others.

One diagonal entry. Write $X = [x^{(1)}, X']$, and let Q be the orthogonal projection onto the orthogonal complement of the column space of X', which has rank $n-p+1$ by the rank statement applied to X' (in the trivial case $p = 1$, X' is empty and $Q = I_n$). The block-inversion formula for the positive definite matrix $X^\top X$ gives

$$\big[(X^\top X)^{-1}\big]_{11} = \frac{1}{\|x^{(1)}\|^2 - x^{(1)\top}X'(X'^\top X')^{-1}X'^\top x^{(1)}} = \frac{1}{\|Qx^{(1)}\|^2}.$$

Conditionally on X', the column $x^{(1)}$ is still $\mathcal{N}(0, I_n)$; writing $Q = UU^\top$ with U an $n\times(n-p+1)$ matrix with orthonormal columns, $\|Qx^{(1)}\|^2 = \|U^\top x^{(1)}\|^2$ is a sum of $k = n-p+1$ squares of independent standard Gaussians. So it has the χ^2_k law, conditionally on X' and hence unconditionally.

The chi-square integral. Integrating $1/t$ against the χ^2_k density gives

$$\mathbb{E}\Big[\frac{1}{\chi^2_k}\Big] = \frac{1}{2^{k/2}\Gamma(k/2)}\int_0^\infty t^{k/2-2}e^{-t/2}\,\mathrm{d}t,$$

which is finite if and only if $k \ge 3$, by the behavior $t^{k/2-2}$ near $t = 0$. For $k \ge 3$ the integral is $2^{k/2-1}\Gamma(k/2-1)$, and $\Gamma(k/2) = (k/2-1)\Gamma(k/2-1)$ turns the expectation into $1/(k-2)$. With $k = n-p+1$, the condition $k \ge 3$ reads $p \le n-2$, and $k-2 = n-p-1$.

The trace. Permuting the columns of X does not change its law, so every diagonal entry of $(X^\top X)^{-1}$ has the law of the first. Summing p equal expectations gives $p/(n-p-1)$ for $p \le n-2$, and $+\infty$ for $p \in \{n-1, n\}$. ∎

Theorem 1.6 (Exact risk of the minimum-norm interpolator). *Under Assumption 1.2 with $\sigma > 0$, the estimator $\hat{w} = X^+ y$ satisfies*

$$\mathbb{E}\|\hat{w} - w_\star\|^2 = \begin{cases} \sigma^2 \dfrac{p}{n-p-1}, & p \le n-2, \\ +\infty, & |p-n| \le 1, \\ r^2\left(1 - \dfrac{n}{p}\right) + \sigma^2 \dfrac{n}{p-n-1}, & p \ge n+2. \end{cases} \tag{1.3}$$

Without noise ($\sigma = 0$) the risk is 0 for $p \le n$ and $r^2(1-n/p)$ for $p \ge n$. In every case the test error is $\mathbb{E}\mathcal{L}(\hat{w}) = \sigma^2 + \mathbb{E}\|\hat{w} - w_\star\|^2$.

Proof. The plan is to compute the error conditionally on X, where it is a quadratic form in the noise and in $w_\star$, and then to average over X with the two lemmas.

Below the threshold, $p \le n$. By Lemma 1.5, X has rank p almost surely, so $\hat{w} = (X^\top X)^{-1} X^\top (X w_\star + \xi) = w_\star + (X^\top X)^{-1} X^\top \xi$. Since ξ is independent of X with $\mathbb{E}\xi\xi^\top = \sigma^2 I_n$,

$$\mathbb{E}\left[\|\hat{w} - w_\star\|^2 \,\middle|\, X\right] = \operatorname{tr}\Big((X^\top X)^{-1} X^\top \,\mathbb{E}[\xi\xi^\top]\, X (X^\top X)^{-1}\Big) = \sigma^2 \operatorname{tr}\big((X^\top X)^{-1}\big),$$

and Lemma 1.5 gives $\sigma^2 p/(n-p-1)$ for $p \le n-2$, and $+\infty$ for $p \in \{n-1, n\}$ because $\sigma > 0$. Without noise, $\hat{w} = w_\star$.

Above the threshold, $p \ge n$. Lemma 1.5 applied to the $p \times n$ matrix $X^\top$ shows that X has rank n almost surely. Let $P = X^\top (XX^\top)^{-1} X$ be the orthogonal projection onto the row space of X. Then $\hat{w} = X^\top (XX^\top)^{-1}(X w_\star + \xi) = P w_\star + X^\top (XX^\top)^{-1} \xi$, and the error

$$\hat{w} - w_\star = -(I_p - P) w_\star + X^\top (XX^\top)^{-1} \xi$$

is the sum of a vector of $\ker X$ and a vector of the row space, which are orthogonal, so $\|\hat{w} - w_\star\|^2 = w_\star^\top (I_p - P) w_\star + \xi^\top (XX^\top)^{-1} \xi$. The row space of XO is $O^\top$ times the row space of X, so $P(XO) = O^\top P(X) O$, and $\operatorname{tr} P = \|P\|_F^2 = n$; Lemma 1.4 gives $\mathbb{E}P = \frac{n}{p} I_p$, so the first term has mean $r^2(1-n/p)$. The second term has conditional mean $\sigma^2 \operatorname{tr}\big((XX^\top)^{-1}\big)$, and Lemma 1.5, with the roles of n and p exchanged, gives $\sigma^2 n/(p-n-1)$ for $p \ge n+2$, and $+\infty$ for $p \in \{n, n+1\}$. Without noise only the first term remains.

Test error. The estimator $\hat{w}$ is independent of the test pair, so (1.1) holds conditionally on the training data; take expectations. ∎

Corollary 1.7 (Weak features; Breiman and Freedman 1983, Belkin et al. 2020). *Under Assumption 1.3, let $\sigma_S^2 = \sigma^2 + \|w_{\star,S^c}\|^2 > 0$. The*

minimum-norm estimator on the features in S has test error

$$\mathbb{E}\big(y - \langle \hat{w}, z_S \rangle\big)^2 = \begin{cases} \sigma_S^2 \dfrac{n-1}{n-p-1}, & p \le n-2, \\ +\infty, & |p-n| \le 1, \\ \|w_{\star,S}\|^2\Big(1 - \dfrac{n}{p}\Big) + \sigma_S^2 \dfrac{p-1}{p-n-1}, & p \ge n+2. \end{cases}$$

Proof. Write the labels as $y_i = \langle w_{\star,S}, z_{i,S} \rangle + \xi_i$ with $\xi_i = \langle w_{\star,S^c}, z_{i,S^c} \rangle + \varepsilon_i$. The coordinates of each z_i are independent, so ξ is independent of X, and its entries are independent with mean 0 and variance σ_S^2. Assumption 1.2 holds with target $w_{\star,S}$ and noise level σ_S, and at a test point the same decomposition gives the test error $\sigma_S^2 + \mathbb{E}\|\hat{w} - w_{\star,S}\|^2$. Insert (1.3) and use $1 + \frac{p}{n-p-1} = \frac{n-1}{n-p-1}$ and $1 + \frac{n}{p-n-1} = \frac{p-1}{p-n-1}$. ∎

Each added feature moves signal from σ_S^2 into $\|w_{\star,S}\|^2$, where it can be learned; this creates a classical U below the threshold (Section 1.5).

The *proportional limit* keeps p/n fixed while both grow, so that bias and variance both stay of order one.[6]

[6] It is the thermodynamic limit of the statistical mechanics of learning, in which the number of examples per weight, $\alpha = n/p = 1/\gamma$, is held fixed.

Corollary 1.8 (Proportional limit). *Under Assumption 1.2, let $n \to \infty$ and $p/n \to \gamma \in (0, \infty)$, with r and $\sigma > 0$ fixed (the target $w_\star \in \mathbb{R}^p$ changes with p but keeps its norm). Then $\mathbb{E}\|\hat{w} - w_\star\|^2 \to \mathcal{R}_0(\gamma)$, with*

$$\mathcal{R}_0(\gamma) = \begin{cases} \sigma^2 \dfrac{\gamma}{1-\gamma}, & \gamma < 1, \\ +\infty, & \gamma = 1, \\ r^2\Big(1 - \dfrac{1}{\gamma}\Big) + \dfrac{\sigma^2}{\gamma - 1}, & \gamma > 1. \end{cases} \tag{1.4}$$

Proof. For $\gamma \neq 1$, eventually $|p-n| \ge 2$, and $p < n$ exactly when $\gamma < 1$; divide numerators and denominators in (1.3) by n and let $p/n \to \gamma$. For $\gamma = 1$, the bounds $\frac{p}{n-p-1} \ge \frac{1}{n/p-1}$ for $p \le n-2$ and $\frac{n}{p-n-1} \ge \frac{1}{p/n-1}$ for $p \ge n+2$ show that the risk tends to $+\infty$, and it is infinite when $|p-n| \le 1$. ∎

1.4 The solution: ridge and the Marchenko–Pastur law

The ridge risk depends on the whole spectrum of $\hat{\Sigma}$, which has a deterministic limit in the proportional regime.

The Marchenko–Pastur law. For $\gamma > 0$ let $t_\pm = (1 \pm \sqrt{\gamma})^2$, and let μ_γ be the probability measure on $[0, \infty)$ given by

$$\mu_\gamma(\mathrm{d}t) = \Big(1 - \frac{1}{\gamma}\Big)_+ \delta_0(\mathrm{d}t) + \frac{\sqrt{(t_+ - t)(t - t_-)}}{2\pi\gamma\, t}\, \mathbf{1}\{t_- \le t \le t_+\}\, \mathrm{d}t. \tag{1.5}$$

Its *Stieltjes transform* $m_\gamma(z) = \int \frac{\mu_\gamma(\mathrm{d}t)}{t - z}$, defined for $z \in \mathbb{C} \setminus [0, \infty)$, is the solution of

$$\gamma z\, m^2 + (z - 1 + \gamma)\, m + 1 = 0 \tag{1.6}$$

with $\Im m > 0$ when $\Im z > 0$, and $m > 0$ when $z < 0$.[7] The empirical spectral distribution of $\hat{\Sigma}$ is $\mu_p = \frac{1}{p}\sum_{i=1}^{p} \delta_{\lambda_i(\hat{\Sigma})}$, so that $\frac{1}{p}\operatorname{tr} f(\hat{\Sigma}) = \int f \,\mathrm{d}\mu_p$ for every function f on $[0,\infty)$.

Gaussianity matters for the exact formulas of Section 1.3, but not for the limits: Hastie et al. (2022) prove the limits (1.4) for the risk conditional on X, with features whose independent entries have mean zero, unit variance, and a finite moment of order $4+\eta$.[8]

Theorem 1.9 (Marčenko and Pastur 1967; proved in Theorem A.8). *Let $X \in \mathbb{R}^{n\times p}$ have independent $\mathcal{N}(0,1)$ entries, and let $n \to \infty$ with $p/n \to \gamma \in (0,\infty)$. Then, almost surely, μ_p converges weakly to μ_γ.*

We use the theorem through two spectral functions of $\lambda > 0$,[9]

$$v_p(\lambda) = \int \frac{\lambda\,\mu_p(\mathrm{d}t)}{t+\lambda} = \frac{1}{p}\operatorname{tr}\big(\lambda(\hat{\Sigma}+\lambda I)^{-1}\big), \quad v(\lambda) = \int \frac{\lambda\,\mu_\gamma(\mathrm{d}t)}{t+\lambda} = \lambda m_\gamma(-\lambda).$$

Since $1 - v_p(\lambda)$ is the effective number of parameters of ridge regression per parameter, v_p is the fraction of directions decided by the penalty rather than by the data. Multiplying (1.6) at $z = -\lambda$ by $-\lambda$ shows that $v(\lambda)$ solves $\gamma v^2 + (1-\gamma+\lambda)v - \lambda = 0$, whose roots have product $-\lambda/\gamma < 0$; so $v(\lambda)$ is the positive root,

$$v(\lambda) = \frac{\sqrt{(1-\gamma+\lambda)^2 + 4\gamma\lambda} - (1-\gamma+\lambda)}{2\gamma}. \tag{1.7}$$

Lemma 1.10 (Ridge risk in spectral form). *Under Assumption 1.2, for every n, p, and $\lambda > 0$,*

$$\mathbb{E}\|\hat{w}_\lambda - w_\star\|^2 = r^2\,\mathbb{E}\int \frac{\lambda^2\,\mu_p(\mathrm{d}t)}{(t+\lambda)^2} + \sigma^2\,\frac{p}{n}\,\mathbb{E}\int \frac{t\,\mu_p(\mathrm{d}t)}{(t+\lambda)^2}, \tag{1.8}$$

and the two integrals are $v_p(\lambda) - \lambda v_p'(\lambda)$ and $v_p'(\lambda)$. In particular the risk depends on $w_\star$ only through r.

Proof. Insert $y = Xw_\star + \xi$ into (1.2) and use $(\hat{\Sigma}+\lambda I)^{-1}\hat{\Sigma} - I = -\lambda(\hat{\Sigma}+\lambda I)^{-1}$:

$$\hat{w}_\lambda - w_\star = -\lambda(\hat{\Sigma}+\lambda I)^{-1} w_\star + (\hat{\Sigma}+\lambda I)^{-1}\frac{X^\top \xi}{n}.$$

Conditionally on X the second term has mean zero, so the cross term vanishes in expectation, and its squared norm has conditional mean $\frac{\sigma^2}{n^2}\operatorname{tr}\big((\hat{\Sigma}+\lambda I)^{-1}X^\top X(\hat{\Sigma}+\lambda I)^{-1}\big) = \frac{\sigma^2}{n}\operatorname{tr}\big(\hat{\Sigma}(\hat{\Sigma}+\lambda I)^{-2}\big)$. The first term contributes $\lambda^2 w_\star^\top F(X) w_\star$ with $F(X) = (\hat{\Sigma}+\lambda I)^{-2}$, which satisfies $F(XO) = O^\top F(X) O$ and $\|F(X)\|_{\mathrm{op}} \le \lambda^{-2}$, so Lemma 1.4 gives $\mathbb{E}F(X) = \frac{1}{p}\mathbb{E}\operatorname{tr}F(X)\cdot I_p$. Writing both traces as integrals against μ_p gives (1.8). Finally $v_p'(\lambda) = \int \frac{t}{(t+\lambda)^2}\,\mathrm{d}\mu_p$, and $\frac{\lambda}{t+\lambda} - \frac{\lambda t}{(t+\lambda)^2} = \frac{\lambda^2}{(t+\lambda)^2}$. ∎

[7] We checked (1.5) and (1.6) in this normalization numerically, integrating the density against $1/(t-z)$ for $\gamma \in \{0.3, 0.8, 1, 1.5, 4\}$ and several z: the residuals are below 10^{-10}.

[8] The Marchenko–Pastur law itself holds for independent entries with two moments; the Gaussian case, proved in the appendix, is all we need.

[9] Marčenko and Pastur published the law in 1967 in *Matematicheskii Sbornik*, for a more general family of random matrices; the English translation appeared the same year.

Theorem 1.11 (Ridge regression in the proportional limit). *Under Assumption 1.2, let $n \to \infty$ and $p/n \to \gamma \in (0, \infty)$, with r, σ, and $\lambda > 0$ fixed. Then*

$$\mathbb{E}\|\hat{w}_\lambda - w_\star\|^2 \to \mathcal{R}_\lambda(\gamma) = r^2\big(v(\lambda) - \lambda v'(\lambda)\big) + \sigma^2 \gamma\, v'(\lambda), \tag{1.9}$$

with v given by (1.7). In terms of the Stieltjes transform, the bias is $r^2\lambda^2 m_\gamma'(-\lambda)$ and the variance is $\sigma^2\gamma\big(m_\gamma(-\lambda) - \lambda m_\gamma'(-\lambda)\big)$. As $\lambda \to 0$, $\mathcal{R}_\lambda(\gamma) \to \mathcal{R}_0(\gamma)$ for every $\gamma \neq 1$.

Proof. The plan is to pass to the limit in Lemma 1.10: Theorem 1.9 gives the limit of $\mathbb{E}v_p$, convexity gives the limit of $\mathbb{E}v_p'$, and the quadratic equation evaluates both.

The function v. The integrand $t \mapsto \lambda/(t+\lambda)$ is bounded and continuous on $[0, \infty)$, so $v_p(\lambda) \to v(\lambda)$ almost surely by Theorem 1.9, and $\mathbb{E}v_p(\lambda) \to v(\lambda)$ by dominated convergence, since $0 \le v_p \le 1$.

Its derivative. For each $t \ge 0$ the function $\lambda \mapsto -\lambda/(t+\lambda)$ is convex, with derivative $-t/(t+\lambda)^2$ of absolute value at most $1/(4\lambda)$, because $(t+\lambda)^2 \ge 4t\lambda$. Hence $f_n = -\mathbb{E}v_p$ is convex and differentiable on $(0, \infty)$, with $f_n' = -\mathbb{E}v_p'$, and it converges pointwise to $f = -v$, which is differentiable by (1.7). Convexity places $f_n'(\lambda)$ between the difference quotients $\frac{f_n(\lambda) - f_n(\lambda - h)}{h}$ and $\frac{f_n(\lambda + h) - f_n(\lambda)}{h}$ for small $h > 0$; letting $n \to \infty$ and then $h \to 0$ gives $f_n'(\lambda) \to f'(\lambda)$, that is, $\mathbb{E}v_p'(\lambda) \to v'(\lambda)$. With $p/n \to \gamma$, (1.8) converges to (1.9). The Stieltjes form follows from $v(\lambda) = \lambda m_\gamma(-\lambda)$ by the chain rule: $v'(\lambda) = m_\gamma(-\lambda) - \lambda m_\gamma'(-\lambda)$, and $v - \lambda v' = \lambda^2 m_\gamma'(-\lambda)$.

The ridgeless limit. Differentiating $\gamma v^2 + (1 - \gamma + \lambda)v - \lambda = 0$ in λ, and using $2\gamma v + 1 - \gamma + \lambda = \sqrt{(1 - \gamma + \lambda)^2 + 4\gamma\lambda}$ from (1.7), gives

$$v'(\lambda) = \frac{1 - v(\lambda)}{\sqrt{(1 - \gamma + \lambda)^2 + 4\gamma\lambda}}.$$

As $\lambda \to 0$ the square root tends to $|1-\gamma| > 0$, and $v(\lambda) \to \frac{|1-\gamma| - (1-\gamma)}{2\gamma} = (1 - \frac{1}{\gamma})_+$. For $\gamma < 1$ this gives $v \to 0$ and $v' \to \frac{1}{1-\gamma}$, so $\mathcal{R}_\lambda(\gamma) \to \sigma^2 \frac{\gamma}{1-\gamma}$; for $\gamma > 1$ it gives $v \to 1 - \frac{1}{\gamma}$ and $v' \to \frac{1}{\gamma(\gamma-1)}$, so $\mathcal{R}_\lambda(\gamma) \to r^2(1 - \frac{1}{\gamma}) + \frac{\sigma^2}{\gamma - 1}$. Both agree with (1.4). ∎

The limit $v(0+) = (1 - \frac{1}{\gamma})_+$ is the mass of μ_γ at zero, the fraction of directions that n examples cannot see, and the bias of the ridgeless estimator per unit of r^2. How much ridge is best? The answer is explicit at every sample size.

Proposition 1.12 (Optimal ridge). *Under Assumption 1.2, with $r > 0$ and $\sigma > 0$:*

(i) for all n and p, the risk $\mathbb{E}\|\hat{w}_\lambda - w_\star\|^2$ is minimized over $\lambda > 0$ at $\lambda^\star = \sigma^2 p/(nr^2)$, where it equals $r^2\,\mathbb{E}v_p(\lambda^\star)$;

(ii) for fixed p, the minimal risk does not increase with n;

(iii) as $n \to \infty$ and $p/n \to \gamma$, the minimal risk converges to

$$\mathcal{R}^\star(\gamma) = \frac{r^2}{2}\Big(\sqrt{b^2+4s} - b\Big), \qquad s = \frac{\sigma^2}{r^2}, \quad b = \frac{1}{\gamma} - 1 + s, \tag{1.10}$$

which increases continuously from 0 to r^2 as γ increases from 0 to ∞; in particular it is finite at $\gamma = 1$.

Proof. *Part (i).* By Lemma 1.10 the risk is $\mathbb{E}\int g_\lambda \, \mathrm{d}\mu_p$ with $g_\lambda(t) = \frac{r^2\lambda^2 + ct}{(t+\lambda)^2}$ and $c = \sigma^2 p/n$. For each $t \geq 0$,

$$\frac{\partial g_\lambda(t)}{\partial \lambda} = \frac{2r^2\lambda(t+\lambda) - 2(r^2\lambda^2 + ct)}{(t+\lambda)^3} = \frac{2t\,(r^2\lambda - c)}{(t+\lambda)^3},$$

so $\lambda \mapsto g_\lambda(t)$ decreases on $(0, \lambda^\star]$ and increases on $[\lambda^\star, \infty)$, with $\lambda^\star = c/r^2$; integrating against μ_p and taking expectations preserves both monotonicities. At $\lambda^\star$ we have $c = r^2\lambda^\star$, so $g_{\lambda^\star}(t) = \frac{r^2\lambda^\star(\lambda^\star+t)}{(t+\lambda^\star)^2} = \frac{r^2\lambda^\star}{t+\lambda^\star}$, whose integral against μ_p is $r^2 v_p(\lambda^\star)$.

Part (ii). The idea is that ridge with $\lambda^\star$ is the best estimator linear in the labels when the target is random and isotropic, and that this class grows with n. Let $w_\star$ be uniform on the sphere of radius r, independent of (X, ξ); by Lemma 1.10 this does not change the risk of a ridge estimator. For $A \in \mathbb{R}^{p\times n}$ depending on X only, $\mathbb{E} w_\star w_\star^\top = \frac{r^2}{p} I_p$ and $\mathbb{E}\xi\xi^\top = \sigma^2 I_n$ give

$$\mathbb{E}\big[\|Ay - w_\star\|^2 \,\big|\, X\big] = \frac{r^2}{p}\|AX - I_p\|_F^2 + \sigma^2\|A\|_F^2,$$

a strictly convex quadratic function of A, whose gradient $\frac{2r^2}{p}(AX - I_p)X^\top + 2\sigma^2 A$ vanishes exactly at $A = X^\top(XX^\top + \frac{\sigma^2 p}{r^2} I_n)^{-1} = (X^\top X + \frac{\sigma^2 p}{r^2} I_p)^{-1} X^\top$: ridge with $n\lambda = \sigma^2 p/r^2$, that is, with $\lambda = \lambda^\star$. Now append one example. Among the matrices allowed with $n+1$ examples is the one that applies the optimal n-example matrix to the first n labels and ignores the last, with the conditional risk of the optimal n-example estimator. The optimal $(n+1)$-example estimator does at least as well for every X, hence on average.

Part (iii). By the proof of Theorem 1.11, $\mathbb{E} v_p(s\gamma) \to v(s\gamma)$, and $|v_p'| \leq \frac{1}{4\lambda}$ gives $|\mathbb{E} v_p(\lambda^\star) - \mathbb{E} v_p(s\gamma)| \leq |\lambda^\star - s\gamma|/(4\min(\lambda^\star, s\gamma)) \to 0$, so the minimal risk tends to $r^2 v(s\gamma)$. Substituting $\lambda = s\gamma$ in $\gamma v^2 + (1 - \gamma + \lambda)v - \lambda = 0$ and dividing by γ gives $v^2 + bv - s = 0$, whose positive root gives (1.10). The map $b \mapsto \frac{1}{2}(\sqrt{b^2+4s} - b) = \frac{2s}{\sqrt{b^2+4s}+b}$ is decreasing, and b decreases with γ, so $\mathcal{R}^\star$ increases; as $\gamma \to 0$, $b \to \infty$ and $\mathcal{R}^\star \to 0$, and as $\gamma \to \infty$, $b \to s - 1$ and $\mathcal{R}^\star \to \frac{r^2}{2}\big((s+1) - (s-1)\big) = r^2$. ∎

In the unnormalized form $\|y - Xw\|^2 + n\lambda\|w\|^2$ the optimal penalty $n\lambda^\star = \sigma^2 p/r^2$ is the noise variance divided by the variance of one coordinate of the target, whatever n is; ridge with this penalty is the posterior mean for the prior $w_\star \sim \mathcal{N}(0, \frac{r^2}{p} I_p)$ and Gaussian noise. Part (ii) is the sample-wise monotonicity theorem of Nakkiran et al. (2021), proved there for isotropic

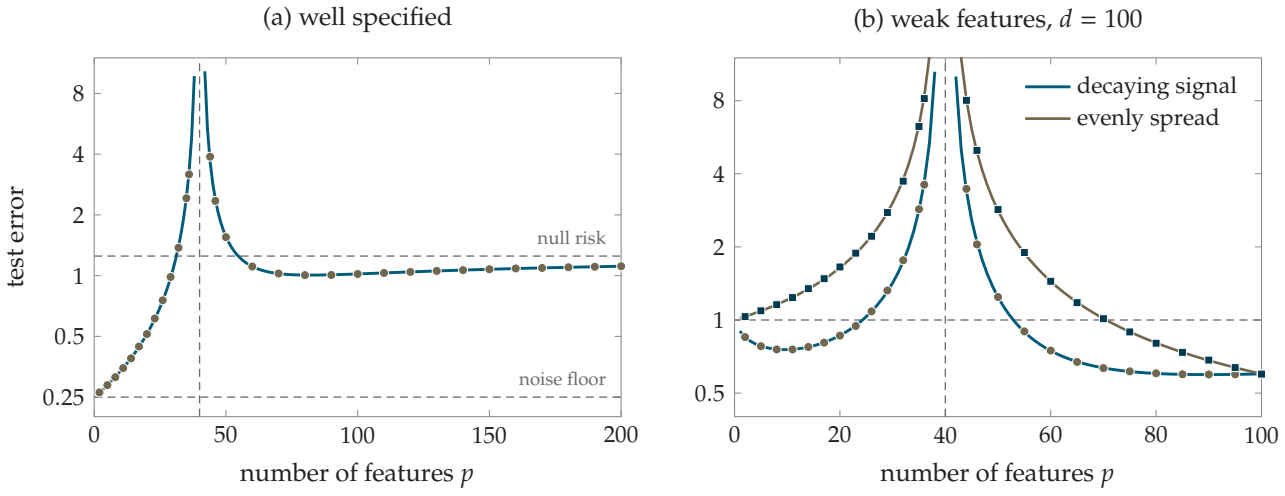


Figure 1.1. Test error of minimum-norm least squares against the number p of features, for $n = 40$ examples; the dashed vertical line marks $p = n$. Curves: Theorem 1.6 and Corollary 1.7. Points: averages over 4000 training sets (seed 20261017), each fit scored by its exact test error (1.1). (a) Well specified, $r^2 = 1$, $\sigma^2 = 0.25$; the second descent bottoms out at $p^\star = 82$. (b) Weak features without label noise, $d = 100$, $r = 1$, the learner using the first p coordinates; the signal is spread evenly ($w_{\star,j}^2 = 1/d$, squares) or decays ($w_{\star,j}^2 \propto j^{-0.8}$, circles). Points with $|p - n| \le 3$, where the simulated average has infinite mean or variance, are omitted.

Gaussian covariates; the argument above uses only second moments of the noise and of the target.

1.5 Reading the solution

Figure 1.1 compares Theorem 1.6 and Corollary 1.7 with simulations at $n = 40$; every simulated point lies within about two standard errors of its curve.

THE ANATOMY OF THE PEAK. Below the threshold the risk is pure variance, $\sigma^2\,\mathbb{E}\,\mathrm{tr}\big((X^\top X)^{-1}\big) = \sigma^2 \frac{p}{n}\,\mathbb{E}\int t^{-1}\mu_p(\mathrm{d}t)$: the noise is amplified by the inverse eigenvalues of $\hat{\Sigma}$, and the small ones dominate. In the limit, letting $z \to 0$ in (1.6) gives $\int t^{-1}\mu_\gamma(\mathrm{d}t) = \frac{1}{1-\gamma}$ for $\gamma < 1$, the factor in (1.4). As γ increases to 1 the lower edge $t_- = (1 - \sqrt{\gamma})^2$ of the spectrum moves down to 0; at $\gamma = 1$ the density (1.5) behaves like $1/(\pi\sqrt{t})$ near 0, and the inverse moment diverges.[10] At finite size the smallest singular value of X is close to $\sqrt{n} - \sqrt{p}$ when $p < n$ (Silverstein 1985) but only of order $n^{-1/2}$ when $p = n$ (Edelman 1988).[11] Past the threshold the bias $r^2(1 - n/p)$ grows, because the interpolator lives in an n-dimensional subspace and sees a shrinking fraction of the target, while the variance falls like $\sigma^2 n/(p - n)$, because with more directions available the n noise values are interpolated by a shorter weight vector, whose squared length the test error pays. The second descent is the variance falling faster than the bias grows.

WHERE THE SECOND DESCENT ENDS. Treat p in (1.3) as a real variable. On $p > n + 1$ the derivative of the test error is $r^2 n/p^2 - \sigma^2 n/(p - n - 1)^2$, so if

[10] In the language of Spigler et al. (2019), $p = n$ is the isostatic point of a jamming transition, at which the degrees of freedom equal the constraints and the response of the fitted weights to the noise diverges.

[11] Edelman (1988) also showed that the condition number of a square Gaussian matrix, divided by n, converges in distribution: at the threshold it grows linearly with n.

$\sigma < r$, then the test error is smallest at

$$p^\star = \frac{(n+1)\,r}{r-\sigma}, \qquad \mathbb{E}\mathcal{L}(\hat{w}) = \sigma^2 + r^2 - \frac{n}{n+1}(r-\sigma)^2, \tag{1.11}$$

and if $\sigma \geq r$, then it decreases to the null risk without reaching it. In Figure 1.1(a), $p^\star = 82$ and the minimal test error is 1.006, against a null risk of 1.25. In the proportional limit the optimum is $\gamma^\star = r/(r-\sigma)$, with test error $2\sigma r$, below the null risk by $(r-\sigma)^2$. The same formulas give a window of harm: minimum-norm least squares predicts worse than the zero predictor exactly when

$$\frac{r^2}{r^2+\sigma^2} < \gamma < \frac{r^2}{r^2-\sigma^2} \tag{1.12}$$

(the upper end is $+\infty$ when $\sigma \geq r$; Exercise 1.3). The window contains the threshold and widens with the noise. Here, however, a learner with more features also faces a truth with more coordinates, and $p^\star$ is only a local minimum: the test error $\sigma^2 \frac{n-1}{n-p-1}$ below the threshold is smallest at $p = 1$, and Hastie et al. (2022) note that in this well-specified model the global minimum always lies below the threshold. The weak-features model compares learners on one fixed truth.

WEAK FEATURES. When the features are ordered by importance and the signal is spread over many of them, the curve shows every phase of Observation 1.1:[12] the decaying profile of Figure 1.1(b) descends to 0.755 at $p = 10$, peaks at the threshold, and descends again to 0.596 at $p = 90$, below the bottom of the U. If the signal is concentrated on few features, then the U wins instead: with $w^2_{\star,j} \propto j^{-2}$ the best model has $p = 22 < n$, with test error 0.048 against 0.191 at best past the threshold, in line with the classical analysis of Breiman and Freedman (1983). When the signal is spread evenly, $w^2_{\star,j} = r^2/d$, the first descent disappears: for $p \leq n-2$ the test error $(\sigma^2 + r^2(1-p/d))\frac{n-1}{n-p-1}$ increases with p as soon as $d \geq n$, so the best classical model uses no features at all, while for $n+2 \leq p \leq d$ Corollary 1.7 gives (Exercise 1.6)

$$\mathbb{E}\mathcal{L} = \sigma^2 + r^2 - \frac{2nr^2}{d} + \frac{n}{p-n-1}\left(\sigma^2 + r^2\left(1 - \frac{n+1}{d}\right)\right), \tag{1.13}$$

which decreases in p. The best interpolating model uses all d features, and it beats the null risk, hence every classical model, if and only if $r^2/\sigma^2 > d/(d-n-1)$, the condition of Belkin et al. (2020). Each added feature brings signal r^2/d; below the threshold it costs more than that in variance, above it it lowers the variance too.

MORE DATA CAN HURT. At fixed p and $n \leq p-2$, the risk $r^2(1-n/p)+\sigma^2\frac{n}{p-n-1}$ increases with n as soon as $(p-n-1)^2 < \frac{\sigma^2}{r^2}p(p-1)$: within about $p\sigma/r$ below $n = p$, each new example makes the interpolator worse, the sample-wise double descent of Nakkiran (2019) (Exercise 1.4).

[12] The figures of Belkin et al. (2020) use the same $d = 100$ and $n = 40$, with label noise.

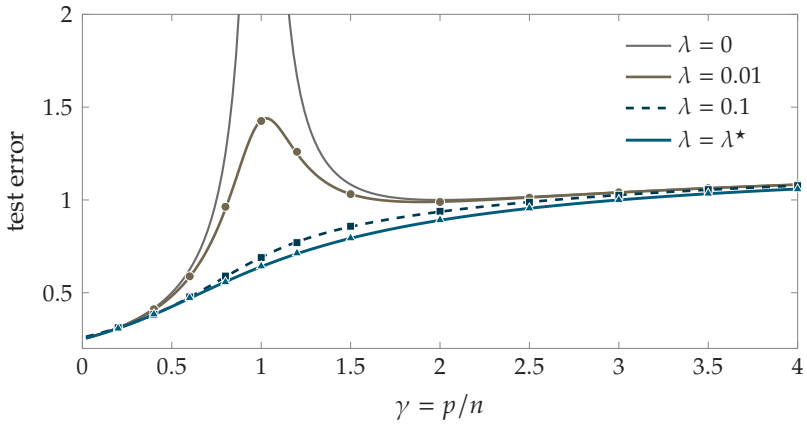


Figure 1.2. Test error of ridge regression against $\gamma = p/n$, for $r^2 = 1$ and $\sigma^2 = 0.25$. Curves: the limits (1.9), (1.4), and (1.10). Points: simulations with $n = 200$ and $p = \gamma n$, averaged over 500 training sets per point (seed 1017), for $\lambda = 0.01$ (circles), $\lambda = 0.1$ (squares), and $\lambda = \lambda^\star$ (triangles); the largest deviation from the curves is 2.0 standard errors.

REGULARIZATION REMOVES THE PEAK. In Figure 1.2, a fixed ridge parameter lowers the peak and leaves a bump, while the optimal $\lambda^\star = \sigma^2\gamma/r^2$ leaves the increasing curve (1.10), whose value at the threshold, $\frac{r^2}{2}(\sqrt{s^2+4s}-s)$, is below σr. In this model the peak is a property of the ridgeless estimator, not of the data.[13]

[13] A fixed penalty is not enough for monotonicity: the curve for $\lambda = 0.01$ in Figure 1.2 still falls from 1.43 at $\gamma = 1$ to 0.99 at $\gamma = 2$ before rising again.

1.6 What the model gets wrong

WHAT WAS THROWN AWAY. With fixed features the model cannot say what a trained network gains over random features, and its p counts directions chosen before the data are seen. Without anisotropy the spectrum plays no role, yet the spectrum decides how much of the target an interpolator sees and where the fitted noise goes (Exercises 1.7 and 1.8); real feature spectra decay, often as power laws (Chapter 16). With a linear target and independent noise, misspecification enters only as white noise, whereas what a small network cannot represent is structured and correlated with the input. Without time, early stopping enters only through its resemblance to ridge (Advani et al. 2020), and the epoch-wise curve not at all. With square loss, interpolation is exact fitting; for classification with the logistic loss it is separation, and gradient descent selects a maximum-margin interpolator (Chapter 4).

WHERE IT BREAKS. *The axis.* The model puts the peak at a parameter count, $p = n$; trained networks put it elsewhere. Spigler et al. (2019) found that the jamming threshold of fully connected networks grows about linearly with the number of examples for random data, but much more slowly on a binary version of MNIST. Nakkiran et al. (2020) found that the peak follows the effective model complexity, which moves with training time, label noise, and augmentation, so that one architecture sits on either side of the threshold

depending on how long it is trained. The count of nK parameters for K outputs (Belkin et al. 2019) fits least squares with K outputs and random features with a trained last layer, but for networks trained end to end it is an observation (Advani et al. 2020), not a theorem. The model's p is at best an effective number of fitted directions, which it does not tell us how to compute.

Time. The model has no epoch-wise double descent: for isotropic features, the test error of gradient flow from zero, computed from the Marchenko–Pastur law, has a single minimum in time at every value of γ and σ/r that we checked numerically. Heckel and Yilmaz (2021) showed that epoch-wise double descent arises when parts of a model are learned at different speeds, so that bias–variance trade-offs are superposed, and removed it from a convolutional network and a ResNet by changing the step sizes of some layers; Pezeshki et al. (2022) derived it from fast and slow features in a teacher–student model (Exercise 1.10).

The tail. As $\gamma \to \infty$ the test error of the isotropic interpolator tends to the null risk: the variance vanishes, but so does the fraction n/p of the target that the interpolator sees. Isotropic features never overfit benignly, whereas interpolating networks, which fit noisy labels exactly (Zhang et al. 2017), and many interpolating kernel machines overfit in a tempered rather than catastrophic way (Mallinar et al. 2022). In linear regression, benign overfitting needs structure in the covariance.

Theorem 1.13 (Bartlett et al. 2020, Gaussian case). *Let $x \sim \mathcal{N}(0, \Sigma)$ in $\mathbb{R}^p$ with $p > n$ and Σ positive definite with eigenvalues $\lambda_1 \geq \cdots \geq \lambda_p$, and let $y = \langle w_\star, x\rangle + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \sigma^2)$ independent of x. Define the effective ranks*

$$r_k = \frac{\sum_{i>k} \lambda_i}{\lambda_{k+1}}, \qquad R_k = \frac{\left(\sum_{i>k} \lambda_i\right)^2}{\sum_{i>k} \lambda_i^2},$$

and let k^* be the smallest $k \geq 0$ with $r_k \geq bn$. There are universal constants $b, c, c_1 > 1$ such that, for $\delta \in (0,1)$ with $\log(1/\delta) < n/c$, the excess risk $\mathcal{E} = \|\Sigma^{1/2}(X^+ y - w_\star)\|^2$ satisfies the following. If $k^* \geq n/c_1$, then $\mathbb{E}\mathcal{E} \geq \sigma^2/c$. Otherwise $\mathbb{E}\mathcal{E} \geq \frac{\sigma^2}{c}\left(\frac{k^*}{n} + \frac{n}{R_{k^*}}\right)$, and with probability at least $1 - \delta$,

$$\mathcal{E} \leq c\|w_\star\|^2 \|\Sigma\|_{\mathrm{op}} \max\left\{\sqrt{\frac{r_0}{n}}, \frac{r_0}{n}, \sqrt{\frac{\log(1/\delta)}{n}}\right\} + c\log(1/\delta)\,\sigma^2\left(\frac{k^*}{n} + \frac{n}{R_{k^*}}\right).$$

The theorem names what the isotropic model lacks.[14] The variance needs a *small head*, $k^* \ll n$, few directions of large variance that cost $\sigma^2 k^*/n$, and a *heavy, flat tail*, $R_{k^*} \gg n$, many directions of small, comparable variance that absorb the fitted noise at the cost $\sigma^2 n/R_{k^*}$, the analogue of $\sigma^2 n/(p-n-1)$. The bias needs $r_0 = \operatorname{tr}\Sigma/\|\Sigma\|_{\mathrm{op}} \ll n$, so that the tail does not drown the head. For $\Sigma = I_p$ with $p \geq bn$, $k^* = 0$ and $R_0 = p$, so the variance vanishes as $p/n \to \infty$; but $r_0 = p$, and the bias $r^2(1 - n/p)$ of Theorem 1.6 does not

[14] Bartlett et al. (2020) prove it for features with independent sub-Gaussian coordinates in the eigenbasis of Σ and noise that is sub-Gaussian given x; we quote the Gaussian case.

vanish. Exercise 1.8 builds a benign covariance by hand.

The counting. Curth et al. (2023) found that in the non-deep experiments of Belkin et al. (2019) the horizontal axis switches at the threshold from one way of adding capacity to another, and that against an effective number of parameters measured at test points the curves fold back into a U. Our model shows the mechanism in closed form. The prediction of X^+y at a test point x_0 is $\langle \hat{s}(x_0), y\rangle$ with smoothing weights $\hat{s}(x_0) = (X^+)^\top x_0$, and with $p_{\text{eff}} = n\,\mathbb{E}\|\hat{s}(x_0)\|^2$ the variance in Theorem 1.6 is $\sigma^2 p_{\text{eff}}/n$ on both sides of the threshold. But $p_{\text{eff}} = np/(n-p-1)$ below the threshold and $n^2/(p-n-1)$ above it: past $p = n$, every added feature makes the estimator effectively simpler.

The cure. Optimal ridge removes the peak in this model, at a positive $\lambda^\star$. For anisotropic data the optimal ridge parameter can be zero or even negative (Kobak et al. 2020; Tsigler and Bartlett 2023): many low-variance directions already act as a ridge, and an interpolator is then optimal, not merely harmless. The isotropic model cannot represent this.

FINE STRUCTURE. Random nonlinear features were solved by Mei and Montanari (2022): in the proportional limit they behave like linear features plus independent noise, the ridgeless peak stays at $N = n$ features, and the best test error is reached as $N/n \to \infty$, with a ridge parameter that vanishes above a critical signal-to-noise ratio. For anisotropic covariances the ridge risk is given by deterministic equivalents that generalize Theorem 1.11 (Atanasov et al. 2026; Hastie et al. 2022), which power-law spectra turn into the scaling laws of Chapter 16. Between benign and catastrophic overfitting lies tempered overfitting, a finite excess risk bounded away from zero, as for kernels with power-law spectra and, empirically, for interpolating networks (Mallinar et al. 2022); Liang and Rakhlin (2020) analyze kernel interpolation in high dimension. Gradient flow in the random-features model gives model-wise, sample-wise, and epoch-wise descents in one exact solution (Bodin and Macris 2021), and in toy models of superposition double descent separates memorizing data points from learning features (Chapter 11; Henighan et al. 2023). Open: a solvable model with feature learning that predicts where the peak of a trained network sits, and how high it is.

1.7 Model card

MODEL CARD	
PHENOMENON	Test error falls, peaks where the model first fits the training set, and falls again, in model size, training time, and sample size.
MODEL	Least squares on p isotropic Gaussian features from n examples, minimum-norm or ridge; weak features: p of the d features of a fixed truth.
SOLUTION	Exact risk at every (n, p) (Theorem 1.6); ridge risk from the Marchenko–Pastur law (Theorem 1.11); optimal ridge $\lambda^\star = \sigma^2 p/(nr^2)$ (Proposition 1.12).

PREDICTS	An infinite peak at $p = n$; the second descent ends at $p^\star = (n+1)r/(r-\sigma)$, near test error $2\sigma r$; a window of harm; more data hurting below the threshold; no peak under optimal ridge.
GETS WRONG	The axis (networks peak at an effective complexity); no epoch-wise descent; no benign overfitting with isotropic features; one peak only.
SUCCESSORS	Random features; anisotropic deterministic equivalents; tempered overfitting; several time scales.

The rotational invariance that made the risk computable returns in the kernel lower bound of Chapter 6, in the single-index and committee models of Chapters 7 and 8, and in the in-context regression of Chapter 13, whose optimal one-step learner is compared there with ridge. The Marchenko–Pastur law is the first spectrum of the book; kernel spectra follow in Chapter 6, and power-law spectra, which turn deterministic equivalents into scaling laws, in Chapter 16. The minimum-norm interpolator is the first implicit bias of the book; a reparameterization changes the norm to ℓ_1 in Chapter 3, and the logistic loss gives the maximum margin in Chapter 4. Epoch-wise double descent needs features learned at different speeds, the staged learning of Chapter 2.

Exercises

1.1 CONSOLIDATION. (Inverse Wishart matrices.) Let $X \in \mathbb{R}^{n\times p}$ have independent $\mathcal{N}(0,1)$ entries. (a) For $n \geq p+2$, show that $\mathbb{E}[(X^\top X)^{-1}] = I_p/(n-p-1)$ with Lemma 1.4, using $\|A\|_F \leq \operatorname{tr} A$ for positive semidefinite A. (b) Show that $\mathbb{E}[T^{-2}] = \frac{1}{(k-2)(k-4)}$ for $T \sim \chi^2_k$ with $k \geq 5$, and $\mathbb{E}[T^{-2}] = +\infty$ for $k \leq 4$. (c) Deduce that $\operatorname{tr}\big((X^\top X)^{-1}\big)$ has finite variance if and only if $p \leq n-4$, and that with Gaussian noise the simulated test error of $X^+ y$ has infinite variance for $p \in \{n-3, n-2, n+2, n+3\}$.

1.2 CONSOLIDATION. (Gradient descent finds the minimum norm.) Run gradient descent $w_{t+1} = w_t - \eta \nabla \widehat{\mathcal{L}}(w_t)$ from w_0 with $0 < \eta < 2/\lambda_{\max}(\hat{\Sigma})$. (a) Show that $w_t \to X^+ y + (I_p - P)w_0$, with P the orthogonal projection onto the row space of X. (b) Let $p > n$ and $w_0 \sim \mathcal{N}(0, \alpha^2 I_p)$, independent of the data. Show that the limit has risk $\mathbb{E}\|X^+ y - w_\star\|^2 + \alpha^2(p-n)$.

1.3 CONSOLIDATION. (The overparameterized optimum.) (a) Derive (1.11), and show that the test error decreases on $p \geq n+2$ when $\sigma \geq r$. (b) Show that in the proportional limit the optimum is $\gamma^\star = r/(r-\sigma)$, with test error $2\sigma r$. (c) Prove that $\sigma^2 + \mathcal{R}_0(\gamma) > \sigma^2 + r^2$ exactly in the window (1.12).

1.4 CONSOLIDATION. (More data can hurt.) Fix p, r, and $\sigma > 0$, and treat the risk $r^2(1-n/p) + \sigma^2 n/(p-n-1)$ as a function of a real variable $n < p-1$. (a) Show that it increases with n exactly when $(p-n-1)^2 < \frac{\sigma^2}{r^2} p(p-1)$. (b) For $p = 100$, $r = 1$, and $\sigma = \frac{1}{2}$, show that among $n \leq 98$ the risk is smallest at $n = 49$, where it equals 0.755, and

that at least 135 examples are needed to beat this value. (c) Why does this not contradict Proposition 1.12(ii)?

1.5 CONSOLIDATION. (Moments of the Marchenko–Pastur law.) (a) Show that $\mathbb{E}\frac{1}{p}\operatorname{tr}\hat{\Sigma} = 1$ and $\mathbb{E}\frac{1}{p}\operatorname{tr}\hat{\Sigma}^2 = 1 + \frac{p+1}{n}$. (b) Substitute the expansion $m_\gamma(z) = -\sum_{k\ge 0} z^{-k-1}\int t^k \mu_\gamma(\mathrm{d}t)$, valid for large $|z|$, in (1.6), and show that the first two moments of μ_γ are 1 and $1+\gamma$. (c) Show that $\int_{(0,\infty)} t^{-1}\mu_\gamma(\mathrm{d}t)$ equals $\frac{1}{1-\gamma}$ for $\gamma < 1$ and $\frac{1}{\gamma(\gamma-1)}$ for $\gamma > 1$. *(Hint: in (c), let $\lambda \to 0$ in $v'(\lambda)$.)*

1.6 VARIATION. (Weak features with evenly spread signal.) Under Assumption 1.3, let $w_{\star,j}^2 = r^2/d$ for every j and $S = \{1,\dots,p\}$. (a) Derive (1.13), show that it decreases in p on $n+2 \le p \le d$, and show that its value at $p = d$ is below the null risk exactly when $r^2/\sigma^2 > d/(d-n-1)$. (b) Let $p/n \to \gamma$ and $d/n \to \delta > 1$. Show that on $1 < \gamma \le \delta$ the test error tends to $\sigma^2 + r^2 - \frac{2r^2}{\delta} + \frac{\sigma^2 + r^2(1-1/\delta)}{\gamma - 1}$. (c) Let the learner use ridge on its p features with the best λ for that p. Show that the test error then does not increase with p, at every n. *(Hint: adapt Proposition 1.12(ii), with a random sign pattern for $w_\star$; a learner with $p+1$ features may ignore one.)*

1.7 VARIATION. (Anisotropic design.) Let the rows of X and the test inputs be independent $\mathcal{N}(0,\Sigma)$ with Σ positive definite, so that $\mathcal{L}(w) = \sigma^2 + \|\Sigma^{1/2}(w - w_\star)\|^2$. (a) Show that for $p \le n-2$ the test error of least squares is $\sigma^2\frac{n-1}{n-p-1}$, whatever Σ is. (b) For $p \ge n+2$, show that $\Sigma^{-1/2}(X\Sigma^{-1/2})^+ y$ has the risk of Theorem 1.6 with $r = \|\Sigma^{1/2}w_\star\|$, and check numerically that X^+y does not.

1.8 VARIATION. ⋆ (Benign overfitting with a spiked covariance.) Let $x = (z, \sqrt{\epsilon}\,u)$ with independent standard Gaussian $z \in \mathbb{R}^k$ and $u \in \mathbb{R}^q$, let $y = \langle w_\star, z\rangle + \varepsilon$ with $w_\star \in \mathbb{R}^k$ and noise variance σ^2, and let $\hat{w} = (\hat{w}_z, \hat{w}_u)$ be the minimum-norm interpolator of $n < q$ examples, with data matrices Z and U. (a) Show that $\hat{w}_z = Z^\top(ZZ^\top + \epsilon UU^\top)^{-1}y$ and that the excess risk is $\|\hat{w}_z - w_\star\|^2 + \epsilon\|\hat{w}_u\|^2$. (b) Suppose for a moment that $UU^\top = qI_n$. Show that $\hat{w}_z$ is then ridge regression on the head with $n\lambda = \epsilon q$, and that $\epsilon\|\hat{w}_u\|^2 \le \|y\|^2/q$; this is the implicit ridge of Kobak et al. (2020). (c) Check numerically, with $k = 5$, $\|w_\star\| = 1$, $\sigma^2 = \frac{1}{4}$, $n \in \{100, 200, 400\}$, $q = n^2/10$, and $\epsilon q/n = 0.5/\sqrt{n}$, that the excess risk is about 0.04, 0.02, and 0.01. (d) Compute the effective ranks of Theorem 1.13 for this covariance and check its conditions when k is fixed, $q/n \to \infty$, and $\epsilon q/n \to 0$. *(Hint: in (b), $Z^\top(ZZ^\top + cI_n)^{-1} = (Z^\top Z + cI_k)^{-1}Z^\top$.)*

1.9 SIMULATION. (Random ReLU features.) Take $d = 100$, $n = 300$, a unit vector $w_\star$, $x \sim \mathcal{N}(0, I_d)$, and $y = \langle w_\star, x\rangle + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, 0.1)$. For N from 30 to 4000, draw $v_1,\dots,v_N \sim \mathcal{N}(0, I_d/d)$, fit minimum-norm least squares on the features $\mathrm{ReLU}(\langle v_j, x\rangle)$, and average the test error on 2000 fresh points over 20 draws. (a) Observe a minimum of about 0.8 for N between 100 and 150, a peak above 7 near $N = n$, and a second descent

to about 0.19 at $N = 4000$. (b) Write $\mathrm{ReLU}(g) = \mu_0 + \mu_1 g + \mu_\star \zeta(g)$ for $g \sim \mathcal{N}(0,1)$, with ζ of unit variance and uncorrelated with 1 and g; show that $\mu_0 = 1/\sqrt{2\pi}$, $\mu_1 = \frac{1}{2}$, and $\mu_\star^2 = \frac{1}{4} - \frac{1}{2\pi}$. (c) Explain why the peak sits at $N = n$ rather than $N = d$, using the picture of Mei and Montanari (2022): linear features plus independent noise of variance $\mu_\star^2$.

1.10 SIMULATION. (Epoch-wise double descent from two time scales.) Take $n = 100$ examples, 40 fast features of variance 1 and 20 slow features of variance $\epsilon = 0.01$, a target $w_\star$ that is constant on each group, with $\|\Sigma^{1/2} w_\star\|^2 = 1$ on each group, and noise variance $\sigma^2 = 0.3$. Run gradient descent from zero with step $\eta = 0.05$ for 2×10^5 steps, recording the test error $\sigma^2 + \|\Sigma^{1/2}(w_t - w_\star)\|^2$ and averaging over 400 draws. (a) Observe a first minimum near step 25 (test error about 1.69), a bump near step 200 (about 1.89), a second minimum near step 1.3×10^4 (about 0.72), and a slow rise toward the least-squares value $\sigma^2 \frac{n-1}{n-p-1} \approx 0.76$ of Exercise 1.7. (b) Explain each phase with the time scales $1/\eta$ and $1/(\eta\epsilon)$, treating the unlearned slow signal as noise, as in the weak-features model. (c) Give the slow features the step size η/ϵ, a preconditioning in the spirit of Heckel and Yilmaz (2021), and observe a single minimum, about 0.63 near step 100.

1.11 RESEARCH. (A solvable model of epoch-wise double descent.) In Exercise 1.10 the time scales are put in by hand, as in the linear models of Heckel and Yilmaz (2021) and Pezeshki et al. (2022) and the random-feature model of Bodin and Macris (2021). Open: a model in which training creates the time scales, for instance a two-layer network that learns features in stages as in Chapter 2, and which predicts when the epoch-wise peak occurs and how its height depends on the label noise.

1.12 RESEARCH. (Where is the peak of a network?) Nakkiran et al. (2020) place the peak of trained networks where the effective model complexity equals the size of the training set. Open: compute the effective model complexity of a two-layer network trained by gradient descent from its width, input dimension, data distribution, and training time, in some solvable limit.

Notes and References

Theorem 1.6 is elementary, and its two halves were found decades apart. Below the threshold, the Gaussian weak-features model is the one that Breiman and Freedman (1983) used to ask how many variables to enter in a regression, and they found the best number to be a small fraction of n. Belkin et al. (2020) took the model past the threshold and proved the formula and the weak-features corollary, while Hastie et al. (2022) and Bartlett et al. (2020), also first posted in 2019, analyzed interpolation with general covariances. The inverse Wishart mean behind Lemma 1.5 is classical (Theorem A.7).

Dobriban and Wager (2018) give the proportional asymptotics of ridge for general covariances; we followed the isotropic form of Hastie et al. (2022). Krogh and Hertz (1992) found the optimal weight decay, the $n\lambda^\star$ of Proposition 1.12, and Advani et al. (2020) showed that optimal early stopping is nearly as good as optimal ridge, with neither showing a peak.

TRANSLATING THE LITERATURE. Hastie et al. (2022) use our γ, r, σ, and ridge normalization; they write β for $w_\star$, and their risk $R_X(\hat{\beta};\beta)$ is our risk conditional on X, without the noise floor. Dobriban and Wager (2018) write w for $w_\star$ and α^2 for r^2, set the noise variance to 1, include the noise in their risk, and write v for the companion Stieltjes transform, of the spectrum of $\frac{1}{n}XX^\top$, which is not our v; their $\lambda^* = \gamma/\alpha^2$ is our $\lambda^\star$ at $\sigma = 1$. Belkin et al. (2020) write D, T, and β for our d, S, and $w_\star$, and their risk includes the noise. Nakkiran et al. (2021) penalize the unnormalized squared loss, so their λ and d are our $n\lambda$ and p, and their $\lambda_{\text{opt}} = d\sigma^2/\|\beta^*\|^2$ is our $n\lambda^\star$. Mei and Montanari (2022) write N for the number of features, $\psi_1 = N/d$, and $\psi_2 = n/d$, so that our γ is their ψ_1/ψ_2.

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2 Learning in Stages

Exact dynamics of deep linear networks

THE MODEL — DEEP LINEAR NETWORKS, WHITENED INPUTS

IN THIS CHAPTER. Networks trained from small initial weights often learn in stages: the loss sits on a plateau, drops quickly, sits on a lower plateau, and drops again, and what is learned first is the coarsest structure of the task. A deep linear network trained by gradient flow on whitened inputs reproduces this. We prove that the differences $W_{\ell+1}^\top W_{\ell+1} - W_\ell W_\ell^\top$ are conserved, that balanced layers make the end-to-end map follow a closed equation, and that from an aligned initialization the network learns the singular modes of the input–output correlation matrix one at a time, each along a logistic curve whose midpoint is at $\log(s/a_0)/(2s)$ for a mode of strength s and initial strength a_0; depth $L \geq 3$ stretches the plateau to order $a_0^{-(1-2/L)}/((L-2)s)$. The model cannot learn what has no linear correlation with the input, it orders the stages of nonlinear networks by the wrong statistic, and its implicit bias toward low rank is not described by a norm.

2.1 The phenomenon: plateaus, drops, and coarse before fine

Training curves of neural networks are often not smooth. A network started from small weights may show a long stretch in which the loss barely moves, then a rapid drop, then another stretch of near-stagnation, and so on: a *loss staircase*. Plateaus were a familiar obstacle in training small sigmoidal networks by backpropagation, and the committee machine of Chapter 8 is the classical model of one of them. They return whenever the initialization is small.

The steps are not arbitrary. A small network introduced by David Rumelhart learns to answer questions such as "a canary can . . . " about a handful of living things (birds, fish, trees, and flowers), and Rogers and McClelland (2004) made it the centerpiece of a theory of semantic cognition. Its hidden representations *differentiate progressively*: early in training all items look alike; then plants separate from animals; later birds separate from fish and trees from flowers; and the distinctions between individual items come last. Developmental psychologists report a similar coarse-to-fine order in children's concepts, and Saxe et al. (2019) set out to explain why a network trained by gradient descent should follow it.[1]

The same pattern appears at a larger scale and in other guises.

[1] They also treat illusory correlations during learning, typicality, category coherence, and inductive generalization, and they contrast the learning trajectories of deep networks with those of shallow ones.

- Kalimeris et al. (2019) trained convolutional networks on image classification and measured, with a conditional mutual information, how much of a network's predictions a linear classifier explains. In the first epochs nearly all of the gain is explained by a linear classifier, and later epochs add functions of increasing complexity while retaining what the linear classifier predicted.
- Arora et al. (2019b) and Gissin et al. (2020) trained deep factorizations $W = W_L \cdots W_1$ of a matrix from small initialization and found that the singular values of W are learned one at a time, more sharply separated as the depth grows: *incremental learning*.
- Simon et al. (2023) found, in a linearized model and in experiments, that self-supervised methods (Barlow Twins, SimCLR, VICReg) trained on ResNets learn their embeddings in a sequence of discrete, well-separated steps, one eigendirection of the embedding at a time.

Observation 2.1 (Saxe et al. 2014, 2019; Arora et al. 2019b; Simon et al. 2023). Networks trained by gradient descent from small initialization often learn in stages. The training loss alternates between plateaus and rapid drops; each drop adds one direction to the learned map (a singular direction of the input–output map, or an eigendirection of a learned embedding); and directions that carry more of the input–output correlation are learned first. The effect is sharper for deeper networks and smaller initializations and fades for large initializations.

The status of this observation is mixed. Stagewise learning is robust in linear and nearly linear settings (matrix factorization, linearized self-supervised learning) and visible in some nonlinear ones, but large-scale training with standard initialization and adaptive optimizers shows smooth loss curves in most cases, and where stages do appear in nonlinear networks their order need not follow linear statistics (Section 2.7). The order "coarse before fine" is itself a hypothesis about what "coarse" means. The model of this chapter makes it precise: coarse means a large singular value of the input–output correlation matrix.

The question is therefore: *why should gradient descent learn in stages, which stages come first, and how long does each plateau last?*

2.2 The model: a deep linear network on whitened inputs

Data and network. Inputs $x \in \mathbb{R}^d$ and outputs $y \in \mathbb{R}^{d_y}$ have a joint law with finite second moments. Write

$$\Sigma_{xx} = \mathbb{E}[xx^\top] \in \mathbb{R}^{d\times d}, \qquad \Sigma_{yx} = \mathbb{E}[yx^\top] \in \mathbb{R}^{d_y\times d}.$$

A *deep linear network* of depth $L \geq 1$ is the map $f(x;\theta) = W_L \cdots W_1 x$ with $W_\ell \in \mathbb{R}^{d_\ell \times d_{\ell-1}}$, $d_0 = d$ and $d_L = d_y$; the hidden widths $d_1, \ldots, d_{L-1}$ play the role of m. The parameters are $\theta = (W_1, \ldots, W_L)$, and the *end-to-end matrix* is $W = W_L \cdots W_1$. The network computes only linear maps, so its expressive power is that of the matrix W (of rank at most $\min_\ell d_\ell$). What

depth changes is the geometry of the parameterization, and with it the dynamics of training.

Assumption 2.2 (Whitened inputs). The inputs are whitened: $\Sigma_{xx} = I_d$.

THE LOSS. We train on the square loss, whose population version is $\mathcal{L}(\theta) = \frac{1}{2}\mathbb{E}\|y - W_L \cdots W_1 x\|^2$. It depends on θ only through W, and only through two moments of the data.

Lemma 2.3 (Reduction to a matrix problem). *Under Assumption 2.2, $\mathcal{L}(\theta) = \mathcal{E}(W)$ with*

$$\mathcal{E}(W) = \frac{1}{2}\|\Sigma_{yx} - W\|_F^2 + \mathcal{L}_0, \qquad \mathcal{L}_0 = \frac{1}{2}\left(\mathbb{E}\|y\|^2 - \|\Sigma_{yx}\|_F^2\right) \geq 0.$$

PROOF. Expand the square and take expectations term by term, using $\mathbb{E}\langle y, Wx\rangle = \operatorname{tr}(W\mathbb{E}[xy^\top]) = \langle \Sigma_{yx}, W\rangle_F$ and $\mathbb{E}\|Wx\|^2 = \operatorname{tr}(W\Sigma_{xx}W^\top) = \|W\|_F^2$:

$$\mathcal{L}(\theta) = \frac{1}{2}\mathbb{E}\|y\|^2 - \langle \Sigma_{yx}, W\rangle_F + \frac{1}{2}\|W\|_F^2 = \frac{1}{2}\|\Sigma_{yx} - W\|_F^2 + \mathcal{L}_0.$$

The constant is the minimum of $\mathcal{E}$ over all matrices, which is the error of the best linear predictor, so it is nonnegative. ∎

The same computation applies to the training loss on n samples, with Σ_{xx} and Σ_{yx} replaced by their empirical versions; whitening then means $\frac{1}{n}\sum_i x_i x_i^\top = I_d$, which requires $n \geq d$. Everything below holds verbatim for the training loss of whitened data.[2]

[2] Whitening a dataset by $x \mapsto \Sigma_{xx}^{-1/2}x$ is always possible when Σ_{xx} is invertible, but it changes the first layer to $W_1\Sigma_{xx}^{1/2}$, and gradient flow in the new variables is not gradient flow in the old ones; Exercise 2.7 treats the case in which the old flow still decouples.

SINGULAR MODES. Let $q = \min(d, d_y)$ and write a singular value decomposition

$$\Sigma_{yx} = USV^\top = \sum_{i=1}^{q} s_i u_i v_i^\top, \qquad s_1 \geq s_2 \geq \cdots \geq s_q \geq 0, \tag{2.1}$$

with $U = [u_1, \ldots, u_q] \in \mathbb{R}^{d_y \times q}$ and $V = [v_1, \ldots, v_q] \in \mathbb{R}^{d \times q}$ having orthonormal columns and $S = \operatorname{diag}(s_1, \ldots, s_q)$. We call (s_i, u_i, v_i) the *singular mode* i of the task: the input direction v_i predicts the output pattern u_i with strength s_i. In this chapter the index i labels modes (Saxe and coauthors use α, which in this book is an initialization scale), and q is the number of modes.

TRAINING. We train by gradient flow, $\dot{W}_\ell = -\nabla_{W_\ell}\mathcal{L}(\theta)$ for $\ell = 1, \ldots, L$; gradient descent with step η follows it with $t = \eta \times$ (number of steps) when η is small. To write the gradient, put

$$A_\ell = W_L \cdots W_{\ell+1}, \qquad C_\ell = W_{\ell-1} \cdots W_1,$$

with $A_L = I_{d_y}$ and $C_1 = I_d$, so that $W = A_\ell W_\ell C_\ell$ for each ℓ, and write $G = \nabla\mathcal{E}(W)$; under Assumption 2.2, $G = W - \Sigma_{yx}$.

Lemma 2.4 (Layer gradients). *For every differentiable $\mathcal{E}$ and $\mathcal{L}(\theta) = \mathcal{E}(W_L \cdots W_1)$, $\nabla_{W_\ell} \mathcal{L} = A_\ell^\top G C_\ell^\top$.*

PROOF. Perturbing W_ℓ by H perturbs W by $A_\ell H C_\ell$, so the derivative of $\mathcal{L}$ in the direction H is $\langle G, A_\ell H C_\ell \rangle_F = \operatorname{tr}(G^\top A_\ell H C_\ell) = \langle A_\ell^\top G C_\ell^\top, H \rangle_F$. ■

WHAT WAS THROWN AWAY, AND WHY. The *nonlinearity*: without it the network can represent only linear maps, and the loss is a polynomial in the weights; yet the training dynamics remain nonlinear, because the gradient for one layer is multiplied by all the others. This is the point of the model: it separates the nonlinearity of *learning* from the nonlinearity of the *function*. *Whitening* makes the loss see the data only through Σ_{yx}, so that the singular modes of one matrix organize everything. *Population loss and gradient flow* remove sampling noise and step-size effects. Finally the solution in Section 2.4 needs an *aligned initialization*, which no practitioner uses; Section 2.5 explains why small random initializations end up nearly aligned.

DICTIONARY

IN THE MODEL	IN A TRAINED NETWORK
end-to-end matrix $W = W_L \cdots W_1$	the input–output map of a trained network, linearized
singular mode (s_i, u_i, v_i) of Σ_{yx}	a distinction in the data, such as plant versus animal: the input direction v_i that signals it and the output pattern u_i it predicts
singular value s_i	the statistical strength of a distinction, which sets how fast it is learned
mode strength $a_i(t) = u_i^\top W(t) v_i$	how much of the distinction the network has learned at time t
hidden directions R_ℓ	the internal representation of a distinction in layer ℓ
initial strength $a_i(0)$, or the per-layer scale α	the initialization scale; small values put the network in the regime of stages
plateau, then logistic rise of a_i	a plateau of the training loss, then a drop
balancedness $W_{\ell+1}^\top W_{\ell+1} = W_\ell W_\ell^\top$	layers of equal size, automatically maintained by gradient training

WHAT ONE HAD TO NOTICE. That a network computing a linear map can have nonlinear training dynamics, so that the simplest model of a plateau is a product of matrices rather than a nonlinearity; that the square loss with whitened inputs sees only Σ_{yx}, whose singular value decomposition supplies natural coordinates in which the problem might decouple; and that the scalar version of the problem, $\dot{b} = c(s - bc)$ and $\dot{c} = b(s - bc)$ for a product bc, already has a saddle at the origin and a plateau before escape.

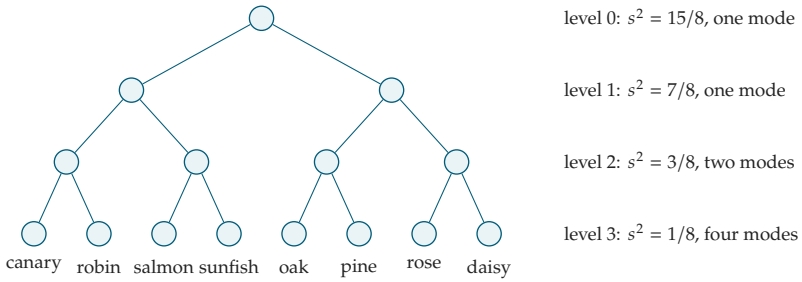


Figure 2.1. A hierarchical dataset in the style of Rogers and McClelland (2004) and Saxe et al. (2019). Each of the $P = 8$ items is an input $x = \sqrt{P}\, e_p$, so that $\Sigma_{xx} = I_8$, and its output $y \in \{0,1\}^{15}$ has one coordinate per node of the tree ("is an animal", "is a bird", ...), equal to 1 for the nodes on the path from the root to the item (the leaf included). The labels on the right group the modes into levels, each level learned in one stage. The input directions v_i of the singular modes of Σ_{yx} can be taken to be the Haar functions of the tree: the constant function, then one function per internal node that is +1 on its left subtree, −1 on its right, and 0 elsewhere. A mode at a lower level is stronger (Exercise 2.2).

2.3 The solution: conservation and balance

The first results hold for every loss of the form $\mathcal{E}(W_L \cdots W_1)$, whitened or not. They come from a symmetry: for an invertible $d_\ell \times d_\ell$ matrix M, replacing $(W_{\ell+1}, W_\ell)$ by $(W_{\ell+1}M^{-1}, MW_\ell)$ leaves W unchanged. By Noether's principle a continuous symmetry of a gradient flow comes with a conserved quantity, and here it is a matrix.[3]

Theorem 2.5 (Conservation law; Arora et al. 2018, Du et al. 2018). *Let $\mathcal{E}\colon \mathbb{R}^{d_y \times d} \to \mathbb{R}$ be continuously differentiable and let $\theta(t)$ solve the gradient flow of $\mathcal{L}(\theta) = \mathcal{E}(W_L \cdots W_1)$ on an interval. Then for each $\ell \in \{1, \dots, L-1\}$ the $d_\ell \times d_\ell$ matrix*

$$D_\ell(t) = W_{\ell+1}(t)^\top W_{\ell+1}(t) - W_\ell(t) W_\ell(t)^\top$$

is constant in t. In particular $\|W_{\ell+1}\|_F^2 - \|W_\ell\|_F^2 = \operatorname{tr} D_\ell$ is constant.

[3] For a gradient flow the mechanism is simple: a loss that is constant along the orbits $M \mapsto (W_{\ell+1}M^{-1}, MW_\ell)$ has its gradient orthogonal to the orbits, and orthogonality to the tangent directions $(-W_{\ell+1}X, XW_\ell)$, for all $d_\ell \times d_\ell$ matrices X, is exactly the equality of the two products in the proof below.

Proof. The plan is to compute the two halves of $\dot{D}_\ell$ and to see that they are equal. By Lemma 2.4, $\dot{W}_\ell = -A_\ell^\top G C_\ell^\top$. Two identities between the partial products follow from their definitions: $W_\ell C_\ell = C_{\ell+1}$ and $A_{\ell+1}W_{\ell+1} = A_\ell$. Using the first,

$$W_\ell \dot{W}_\ell^\top = -W_\ell C_\ell G^\top A_\ell = -C_{\ell+1} G^\top A_\ell,$$

and using the second,

$$\dot{W}_{\ell+1}^\top W_{\ell+1} = -C_{\ell+1} G^\top A_{\ell+1} W_{\ell+1} = -C_{\ell+1} G^\top A_\ell.$$

The two products are equal; call them N. Then $\frac{\mathrm{d}}{\mathrm{d}t}(W_\ell W_\ell^\top) = N + N^\top = \frac{\mathrm{d}}{\mathrm{d}t}(W_{\ell+1}^\top W_{\ell+1})$, so $\dot{D}_\ell = 0$. Taking the trace gives the last statement. ∎

The scalar case $L = 2$, $d = d_1 = d_y = 1$ says that $b^2 - c^2$ is conserved for the product bc: the flow moves along hyperbolas, the picture behind the catapult of Chapter 15 and the diagonal networks of Chapter 3. The same argument applies neuron by neuron to ReLU networks (Exercise 2.3).

We call θ *balanced* if $D_\ell = 0$ for every ℓ. By Theorem 2.5, a flow that starts balanced stays balanced; a flow that starts from small weights stays nearly balanced, since $\|D_\ell\|_F$ is at most of the order of the squared initialization scale while the weights themselves grow to order one. For balanced weights the layers have the same nonzero singular values, and the end-to-end matrix obeys an equation of its own.

Lemma 2.6 (Powers of balanced products). *Let θ be balanced. Then for $1 \le \ell \le L$,*

$$A_\ell A_\ell^\top = (WW^\top)^{(L-\ell)/L}, \qquad C_\ell^\top C_\ell = (W^\top W)^{(\ell-1)/L},$$

where powers of positive semidefinite matrices are taken in their eigenbasis.

Proof. The plan is an induction that carries a power of a Gram matrix through the product. Let $P_j = W_L \cdots W_j$ for $1 \le j \le L$, so that $P_1 = W$ and $P_{\ell+1} = A_\ell$ (with $P_{L+1} = I$). We show, by downward induction on j, that for every integer $k \ge 1$

$$P_j \, (W_j^\top W_j)^{k-1} P_j^\top = (W_L W_L^\top)^{L-j+k}. \tag{2.2}$$

For $j = L$ the left side is $W_L (W_L^\top W_L)^{k-1} W_L^\top = (W_L W_L^\top)^k$, by regrouping the factors. For $j < L$, write $P_j = P_{j+1} W_j$; regroup $W_j (W_j^\top W_j)^{k-1} W_j^\top = (W_j W_j^\top)^k$; replace $W_j W_j^\top$ by $W_{j+1}^\top W_{j+1}$, which is balancedness; and apply (2.2) for $j+1$ with $k+1$ in place of k:

$$P_j (W_j^\top W_j)^{k-1} P_j^\top = P_{j+1} (W_{j+1}^\top W_{j+1})^k P_{j+1}^\top = (W_L W_L^\top)^{L-j+k}.$$

With $k = 1$, (2.2) gives $P_j P_j^\top = (W_L W_L^\top)^{L-j+1}$. For $j = 1$ this is $WW^\top = (W_L W_L^\top)^L$, and since $W_L W_L^\top$ is positive semidefinite it is the unique positive semidefinite L-th root, $W_L W_L^\top = (WW^\top)^{1/L}$. For $j = \ell + 1$ it gives $A_\ell A_\ell^\top = (W_L W_L^\top)^{L-\ell} = (WW^\top)^{(L-\ell)/L}$. The statement for C_ℓ is the same argument applied to the transposed network $W^\top = W_1^\top \cdots W_L^\top$, which is balanced as well. ∎

Theorem 2.7 (End-to-end dynamics; Arora et al. 2018). *Let $\mathcal{E}$ be continuously differentiable, and let $\theta(t)$ solve the gradient flow of $\mathcal{E}(W_L \cdots W_1)$ from a balanced initialization. Then $W(t) = W_L(t) \cdots W_1(t)$ satisfies*

$$\dot W = -\sum_{j=1}^{L} (WW^\top)^{(L-j)/L} \, \nabla \mathcal{E}(W) \, (W^\top W)^{(j-1)/L}. \tag{2.3}$$

Proof. By the product rule, $\dot W = \sum_{j=1}^{L} A_j \dot W_j C_j$. By Lemma 2.4, $\dot W_j = -A_j^\top G C_j^\top$, so $\dot W = -\sum_j A_j A_j^\top G \, C_j^\top C_j$. By Theorem 2.5 the flow stays

balanced, and Lemma 2.6 turns the two Gram matrices into the stated powers. ∎

For $L = 1$ this is plain gradient flow, $\dot{W} = -\nabla\mathcal{E}(W)$. For $L \geq 2$ the gradient is multiplied on both sides by powers of $WW^\top$ and $W^\top W$, which vanish when W is small and are of order one when W is: depth acts as a preconditioner that slows the directions in which W is small. Near $W = 0$ the right side of (2.3) is of order $\|W\|^{2-2/L}$, so $W = 0$ is a stationary point, and $\theta = 0$ is a saddle of $\mathcal{L}$ for every $L \geq 2$ whenever $\Sigma_{yx} \neq 0$ (with a vanishing Hessian when $L \geq 3$). Arora et al. (2018) prove further that, for $L \geq 2$, the effect of this preconditioning cannot be obtained by adding a regularizer to $\mathcal{E}$ and following its gradient.

2.4 The solution: learning one mode at a time

We now return to whitened inputs and solve the flow exactly from initializations that respect the singular modes of Σ_{yx}.

Definition 2.8 (Aligned, balanced initialization). Assume $d_\ell \geq q$ for $1 \leq \ell \leq L-1$. Fix matrices $R_\ell \in \mathbb{R}^{d_\ell \times q}$ with orthonormal columns, $R_\ell^\top R_\ell = I_q$, for $1 \leq \ell \leq L-1$, and put $R_0 = V$ and $R_L = U$ from (2.1). For a diagonal matrix $B = \operatorname{diag}(\beta_1, \dots, \beta_q)$ with $\beta_i \geq 0$, the *aligned* weights are

$$W_\ell = R_\ell B R_{\ell-1}^\top, \qquad \ell = 1, \dots, L.$$

Aligned weights are balanced: $W_{\ell+1}^\top W_{\ell+1} = R_\ell B^2 R_\ell^\top = W_\ell W_\ell^\top$. Their partial products telescope, because $R_\ell^\top R_\ell = I_q$ (the first identity for $\ell \leq L-1$, the second for $\ell \geq 2$):

$$W_L \cdots W_{\ell+1} = U B^{L-\ell} R_\ell^\top, \qquad W_{\ell-1} \cdots W_1 = R_{\ell-1} B^{\ell-1} V^\top, \qquad W = U B^L V^\top. \tag{2.4}$$

So the end-to-end matrix has the singular vectors of Σ_{yx}, and its strength on mode i is $a_i = \beta_i^L$. Each layer passes mode i from the direction $R_{\ell-1} e_i$ to the direction $R_\ell e_i$ with gain β_i.

Theorem 2.9 (Exact learning dynamics; Saxe et al. 2014). *Assume Assumption 2.2 and $d_\ell \geq q$ for $1 \leq \ell \leq L-1$, with $L \geq 2$. Let $\theta(0)$ be aligned with $B(0) = \operatorname{diag}(\beta_i(0))$, $\beta_i(0) \geq 0$. For each i let $\beta_i(t)$ be the solution of*

$$\dot{\beta}_i = \beta_i^{L-1}\big(s_i - \beta_i^L\big). \tag{2.5}$$

Then the gradient flow of $\mathcal{L}$ exists for all $t \geq 0$, and $\theta(t)$ is aligned with $B(t) = \operatorname{diag}(\beta_i(t))$ and the same R_ℓ, for every $t \geq 0$. Consequently $W(t) = \sum_i a_i(t) u_i v_i^\top$ with $a_i = \beta_i^L$, the mode strengths evolve independently by

$$\dot{a}_i = L\, a_i^{2-2/L}\,(s_i - a_i), \tag{2.6}$$

and the loss is $\mathcal{L}(\theta(t)) = \mathcal{L}_0 + \frac{1}{2}\sum_{i=1}^{q}\big(s_i - a_i(t)\big)^2$.

PROOF. The plan is to show that the aligned matrices built from the scalar solutions solve the gradient flow, and then to invoke uniqueness of solutions.

The scalar equations. The right side of (2.5) is a polynomial, so a unique solution exists on some interval. If $\beta_i(0) = 0$, then $\beta_i \equiv 0$ is the solution. If $\beta_i(0) > 0$, then the interval $J_i = (0, \max\{\beta_i(0), s_i^{1/L}\}]$ contains $\beta_i(0)$, and the solution cannot leave it: it cannot reach 0, because 0 is a stationary point and solutions are unique, and it cannot exceed the upper end c, because $\dot\beta_i \le 0$ whenever $\beta_i^L \ge s_i$, and $c^L \ge s_i$. A solution confined to a bounded set exists for all $t \ge 0$.

The candidate solves the flow. Let $\tilde\theta(t)$ be the aligned weights with $B(t)$. By (2.4), $\tilde W = UB^LV^\top$, so $G = \tilde W - \Sigma_{yx} = U(B^L - S)V^\top$. By Lemma 2.4 and (2.4), using $U^\top U = V^\top V = I_q$ and that diagonal matrices commute (for $\ell = L$ the factor $A_L = I_{d_y}$ stands where (2.4) would give $UU^\top$, and for $\ell = 1$ the factor $C_1 = I_d$ stands for $VV^\top$; nothing changes, since $U^\top UU^\top = U^\top$ and $VV^\top V = V$),

$$\nabla_{W_\ell}\mathcal{L}(\tilde\theta) = R_\ell B^{L-\ell}U^\top\, U(B^L - S)V^\top\, VB^{\ell-1}R_{\ell-1}^\top = R_\ell\, B^{L-1}(B^L - S)\, R_{\ell-1}^\top.$$

On the other hand $\frac{\mathrm{d}}{\mathrm{d}t}\tilde W_\ell = R_\ell \dot B R_{\ell-1}^\top$, and (2.5) says $\dot B = B^{L-1}(S - B^L)$. Hence $\frac{\mathrm{d}}{\mathrm{d}t}\tilde W_\ell = -\nabla_{W_\ell}\mathcal{L}(\tilde\theta)$ for every ℓ and every $t \ge 0$.

Uniqueness. The gradient of $\mathcal{L}$ is a polynomial in the entries of θ, hence locally Lipschitz, so the gradient flow from $\theta(0)$ has at most one solution on each interval. Since $\tilde\theta$ is a solution on $[0,\infty)$ with $\tilde\theta(0) = \theta(0)$, the flow exists for all time and equals $\tilde\theta$.

Consequences. The formula for W is (2.4). For $a_i = \beta_i^L$, the chain rule and (2.5) give $\dot a_i = L\beta_i^{L-1}\dot\beta_i = L\beta_i^{2L-2}(s_i - a_i) = La_i^{2-2/L}(s_i - a_i)$. The loss follows from Lemma 2.3, since $\|\Sigma_{yx} - W\|_F^2 = \|S - B^L\|_F^2$ by the orthonormality of the columns of U and V. ∎

We worked with β_i rather than a_i because $a \mapsto a^{2-2/L}$ is not Lipschitz at 0 when $L \ge 3$: equation (2.6) alone would allow solutions that leave 0 spontaneously, and the layer equation (2.5) rules them out. The mode equation also follows from Theorem 2.7: inserting $W = UDV^\top$ with $D = \mathrm{diag}(a_i)$ into (2.3) gives $\dot D = -\sum_j D^{2(L-j)/L}(D-S)D^{2(j-1)/L} = LD^{2-2/L}(S-D)$.

TWO LAYERS: THE LOGISTIC CURVE. For $L = 2$, (2.6) is the logistic equation $\dot a = 2a(s - a)$, and everything is explicit.

Corollary 2.10 (Two layers). *Let $L = 2$, and consider one mode with $a(0) = a_0 > 0$. If $s > 0$, then*

$$a(t) = \frac{s}{1 + (s/a_0 - 1)\, e^{-2st}}, \tag{2.7}$$

and the time that a takes to pass from a_1 to a_2, for $a_0 \le a_1 < a_2 < s$, is $\frac{1}{2s}\log\frac{a_2(s-a_1)}{a_1(s-a_2)}$. In particular, if $a_0 < s/2$, then the mode is half learned at

$$t_{1/2} = \frac{1}{2s}\log\Big(\frac{s}{a_0} - 1\Big), \tag{2.8}$$

and a rises from $s/10$ to $9s/10$ in time $\frac{\log 81}{2s} = \frac{2\log 3}{s}$, independent of a_0 when $a_0 \le s/10$. If $s = 0$, then the mode decays: $a(t) = a_0/(1 + 2a_0 t)$.

Proof. Put $E(t) = (s/a_0 - 1)e^{-2st}$, so that the right side of (2.7) is $s/(1+E)$, equal to a_0 at $t = 0$. Since $\dot{E} = -2sE$, its derivative is $2s^2E/(1+E)^2$, which equals $2a(s-a)$ because $s - a = sE/(1+E)$; by uniqueness (2.7) is the solution. Solving (2.7) for t gives $t(a) = \frac{1}{2s}\log\frac{a(s-a_0)}{a_0(s-a)}$, and the travel time is $t(a_2) - t(a_1)$. With $a_1 = a_0$ and $a_2 = s/2$ this is (2.8); with $a_1 = s/10$ and $a_2 = 9s/10$ it is $\frac{1}{2s}\log 81$. For $s = 0$ the equation is $\dot{a} = -2a^2$, solved by $a_0/(1+2a_0t)$. ∎

The logistic curve is the signature of a saddle. Near $a = 0$ the mode grows exponentially, $a \approx a_0 e^{2st}$, with rate $2s$; the time to grow from a_0 to order s is the logarithm of the ratio divided by the rate; and the final approach to s is again exponential, with the same rate. The plateau has length $\frac{1}{2s}\log(s/a_0)$ and the drop has width of order $1/s$: small initialization lengthens the plateau only logarithmically, which is why stages separate cleanly only when a_0 is many orders of magnitude below s.

Depth. For $L \geq 3$ the origin is a degenerate saddle: near 0, $\dot{a} \approx Ls\,a^{2-2/L}$ with an exponent above 1, so growth is polynomial and the plateau is polynomially long in $1/a_0$.

Proposition 2.11 (Plateau length in depth $L \geq 3$). *Let $L \geq 3$, $s > 0$ and $0 < a_0 < s/2$, and let $t_{1/2}$ be the time at which the solution of (2.6) from a_0 reaches $s/2$. With $\kappa = 1 - 2/L$,*

$$\frac{a_0^{-\kappa} - (s/2)^{-\kappa}}{(L-2)\,s} \;\leq\; t_{1/2} \;\leq\; \frac{a_0^{-\kappa} - (s/2)^{-\kappa}}{(L-2)\,s} + \frac{(s/2)^{2/L}}{s^2}.$$

In particular $t_{1/2} = a_0^{-(1-2/L)}/((L-2)s) + O(s^{-(2-2/L)})$ as $a_0 \to 0$. The time to rise from $s/10$ to $9s/10$ is $c_L\, s^{-(2-2/L)}$ with $c_L = \int_{1/10}^{9/10} \frac{\mathrm{d}z}{Lz^{2-2/L}(1-z)}$, independent of a_0.

Proof. The plan is to write $t_{1/2}$ as an integral and to bound its integrand. On $(0, s)$ the right side of (2.6) is positive, so a increases, and separating variables,

$$t_{1/2} = \int_{a_0}^{s/2} \frac{\mathrm{d}a}{L\,a^{2-2/L}(s-a)}.$$

For $0 \leq a \leq s/2$ we have $\frac{1}{s} \leq \frac{1}{s-a} = \frac{1}{s} + \frac{a}{s(s-a)} \leq \frac{1}{s} + \frac{2a}{s^2}$. The first term contributes

$$\frac{1}{Ls}\int_{a_0}^{s/2} a^{-1-\kappa}\,\mathrm{d}a = \frac{a_0^{-\kappa} - (s/2)^{-\kappa}}{L\kappa s} = \frac{a_0^{-\kappa} - (s/2)^{-\kappa}}{(L-2)s},$$

which is the lower bound. The second term contributes at most

$$\frac{2}{Ls^2}\int_0^{s/2} a^{-1+2/L}\,\mathrm{d}a = \frac{2}{Ls^2}\cdot\frac{L}{2}\Big(\frac{s}{2}\Big)^{2/L} = \frac{(s/2)^{2/L}}{s^2},$$

which gives the upper bound. Since $(s/2)^{-\kappa}/s$ and $(s/2)^{2/L}/s^2$ are both of order $s^{-(2-2/L)}$, the asymptotic statement follows. For the rise time, separate variables between $s/10$ and $9s/10$ and substitute $a = sz$. ∎

With a per-layer initialization scale α, so that $a_0 = \alpha^L$, the plateau lasts about $\alpha^{-(L-2)}/((L-2)s)$, against $\log(s/\alpha^2)/(2s)$ for two layers: *depth lengthens the plateau and sharpens the transition*, in the sense that the drop takes a time independent of α while the plateau grows like a power of $1/\alpha$.[4] The layer equation (2.5) reads $\dot\beta \approx s\beta^{L-1}$ near 0, the escape equation of an order-L saddle; it is the equation of the search phase in Chapter 7 with depth L in the role of the information exponent, and the escape time $\beta_0^{-(L-2)}/((L-2)s)$ has the same form.

[4] Gissin et al. (2020) make the same point about incremental learning: shallow models show it only from exponentially small initializations, deep ones already from polynomially small ones.

In both cases *the order of learning is the order of the singular values*: for equal initial strengths, a larger s_i gives an earlier $t_{1/2}$, by (2.8) and Proposition 2.11. The loss therefore descends a staircase. Between the learning of mode k and of mode $k+1$ it rests near

$$\mathcal{L}_k = \mathcal{L}_0 + \frac{1}{2}\sum_{i>k} s_i^2, \tag{2.9}$$

the loss of the end-to-end matrix $W_k = \sum_{i\le k} s_i u_i v_i^\top$, the best approximation of Σ_{yx} of rank k.

THE LANDSCAPE. The points W_k are not only resting places of one trajectory: they are the critical points of the loss. Baldi and Hornik (1989) proved this for two layers with a bottleneck, in the language of principal component analysis.

Proposition 2.12 (Critical points of two-layer networks; after Baldi and Hornik 1989). *Let $L = 2$, assume Assumption 2.2, and let $s_1 > s_2 > \cdots > s_q > 0$. Let (W_1, W_2) be a critical point of $\mathcal{L}$ with $W_2 \in \mathbb{R}^{d_y \times m}$ of full column rank m. Then*

(i) $W = W_2W_1 = \sum_{i\in I} s_i u_i v_i^\top$ for a set $I \subseteq \{1,\dots,q\}$ with $|I| \le m$, and $\mathcal{L} = \mathcal{L}_0 + \frac{1}{2}\sum_{i\notin I} s_i^2$;

(ii) if $m \le q$ and $I \neq \{1,\dots,m\}$, then the critical point is a saddle: every neighborhood contains points of lower loss and points of higher loss. Hence every local minimum with W_2 of full column rank has $W = \sum_{i\le m} s_i u_i v_i^\top$, and it is a global minimum over the networks of width m.

PROOF. The plan for part (i) is to solve the two critical-point equations in turn. By Lemma 2.4 they read $W_2^\top(\Sigma_{yx} - W) = 0$ and $(\Sigma_{yx} - W)W_1^\top = 0$. Since $W_2^\top W_2$ is invertible, the first gives $W_1 = (W_2^\top W_2)^{-1}W_2^\top\Sigma_{yx}$, so that $W = \Pi\Sigma_{yx}$ with $\Pi = W_2(W_2^\top W_2)^{-1}W_2^\top$, the orthogonal projection onto the range of W_2, of rank m. Inserting $W_1^\top = \Sigma_{yx}^\top W_2(W_2^\top W_2)^{-1}$ into the second equation and multiplying it on the right by $W_2^\top$ gives

$$(I - \Pi)\,\Sigma_{yx}\Sigma_{yx}^\top\,\Pi = 0,$$

that is, $\Sigma_{yx}\Sigma_{yx}^\top\Pi = \Pi\Sigma_{yx}\Sigma_{yx}^\top\Pi$. The right side is symmetric, so transposing gives $\Pi\Sigma_{yx}\Sigma_{yx}^\top = \Sigma_{yx}\Sigma_{yx}^\top\Pi$: the projection commutes with $\Sigma_{yx}\Sigma_{yx}^\top = \sum_i s_i^2 u_i u_i^\top$. Its range is therefore invariant under $\Sigma_{yx}\Sigma_{yx}^\top$, and because the eigenvalues $s_1^2 > \cdots > s_q^2 > 0$ are simple, $\Pi = \sum_{i\in I} u_i u_i^\top + \Pi_0$ for a set I and

a projection Π_0 onto a subspace of the kernel of $\Sigma_{yx}^\top$. Then $\Pi_0\Sigma_{yx} = 0$ and $W = \Pi\Sigma_{yx} = \sum_{i\in I} s_i u_i v_i^\top$; the rank count gives $|I| \le m$, and the loss follows from Lemma 2.3. ∎

Proof of part (ii). The plan is to rotate the range of W_2 slightly, from a weak direction toward a strong one, and to compare losses with the identity $\mathcal{L} = \mathcal{L}_0 + \frac{1}{2}\|\Sigma_{yx}\|_F^2 - \frac{1}{2}\operatorname{tr}(P\Sigma_{yx}\Sigma_{yx}^\top)$, valid whenever $W = P\Sigma_{yx}$ for an orthogonal projection P (expand $\frac{1}{2}\|(I-P)\Sigma_{yx}\|_F^2$ and use $P^2 = P$). First we find two directions. If $|I| = m$, then some $j \le m$ lies outside I and some $i \in I$ exceeds m; put s_i for its singular value. If $|I| < m$, then $\Pi_0 \ne 0$; let u_i be a unit vector in its range, which lies in the kernel of $\Sigma_{yx}^\top$, put $s_i = 0$, and pick some $j \notin I$, which exists because $|I| < m \le q$. In both cases $s_j > s_i$, u_i lies in the range of Π, and $\Pi u_j = 0$, because u_j is orthogonal to the u_k with $k \in I$ and to the kernel of $\Sigma_{yx}^\top$.

Let Q_ε be the rotation by the angle ε in the plane of u_i and u_j that takes u_i to $\cos\varepsilon\, u_i + \sin\varepsilon\, u_j$ and fixes the orthogonal complement of the plane. Put $W_2(\varepsilon) = Q_\varepsilon W_2$ and $W_1(\varepsilon) = (W_2^\top W_2)^{-1} W_2^\top Q_\varepsilon^\top \Sigma_{yx}$; both depend continuously on ε and equal (W_2, W_1) at $\varepsilon = 0$. Their product is $Q_\varepsilon \Pi Q_\varepsilon^\top \Sigma_{yx} = \Pi_\varepsilon \Sigma_{yx}$, with $\Pi_\varepsilon = Q_\varepsilon \Pi Q_\varepsilon^\top$ an orthogonal projection. Write $\Pi = u_i u_i^\top + \Pi'$; the range of Π' is orthogonal to u_i and to u_j, so Q_ε fixes it and $\operatorname{tr}(\Pi' Q_\varepsilon^\top \Sigma_{yx}\Sigma_{yx}^\top Q_\varepsilon) = \operatorname{tr}(\Pi'\Sigma_{yx}\Sigma_{yx}^\top)$. The remaining term changes because u_i and u_j are eigenvectors of $\Sigma_{yx}\Sigma_{yx}^\top$ with eigenvalues s_i^2 and s_j^2:

$$(Q_\varepsilon u_i)^\top \Sigma_{yx}\Sigma_{yx}^\top (Q_\varepsilon u_i) = \cos^2\varepsilon\, s_i^2 + \sin^2\varepsilon\, s_j^2.$$

Hence $\mathcal{L}$ at $(W_1(\varepsilon), W_2(\varepsilon))$ is lower than at (W_1, W_2) by $\frac{1}{2}(s_j^2 - s_i^2)\sin^2\varepsilon > 0$ for every $\varepsilon \in (0, \pi/2)$. For points of higher loss, move W_1 alone to $W_1 + \varepsilon X$ with $W_2 X \ne 0$: the first-order change of $\mathcal{L}$ vanishes by the first critical-point equation, and the loss rises by $\frac{1}{2}\varepsilon^2\|W_2 X\|_F^2$. For the last statement, a local minimum is not a saddle, so $I = \{1, \dots, m\}$, and by the Eckart–Young theorem $\sum_{i\le m} s_i u_i v_i^\top$ minimizes $\|\Sigma_{yx} - W\|_F$ over all matrices of rank at most m, which contains every product $W_2 W_1$ of width m. ∎

Critical points with W_2 of deficient rank require a separate argument. Baldi and Hornik (1989) give the full classification, and Kawaguchi (2016) extends "every local minimum is global" to deep linear networks, and shows that for $L \ge 3$ some saddles have no direction of negative curvature.

The trajectory of Theorem 2.9 from a small aligned initialization therefore passes from the saddle $W_0 = 0$ near the saddles $W_1, W_2, \dots$ in turn, lingering near each until the next mode escapes, and the plateaus of the staircase are visits to saddles.

2.5 The solution: small random initialization

No one initializes a network in alignment with the singular vectors of the task. What happens from a small *random* initialization? The answer has two phases: first the weights align without the loss moving, and then the aligned solution takes over.

HEURISTIC DERIVATION, $L = 2$. Let $W_1(0)$ and $W_2(0)$ be small, with entries of order α. While $W = W_2W_1$ is negligible against Σ_{yx}, the flow is approximately linear: $\dot{W}_1 \approx W_2^\top \Sigma_{yx}$ and $\dot{W}_2 \approx \Sigma_{yx} W_1^\top$. Project onto the modes: with $x_i = W_1 v_i \in \mathbb{R}^m$ and $z_i = W_2^\top u_i \in \mathbb{R}^m$, the linearized flow gives $\dot{x}_i = s_i z_i$ and $\dot{z}_i = s_i x_i$, because $\Sigma_{yx} v_i = s_i u_i$ and $\Sigma_{yx}^\top u_i = s_i v_i$. The modes decouple: $x_i + z_i$ grows like $e^{s_i t}$ and $x_i - z_i$ decays like $e^{-s_i t}$. After a time of a few multiples of $1/s_i$,

$$x_i(t) \approx z_i(t) \approx \xi_i e^{s_i t}, \qquad \xi_i = \tfrac{1}{2}\big(W_1(0)v_i + W_2(0)^\top u_i\big),$$

and the mode strength $a_i = u_i^\top W v_i = \langle z_i, x_i \rangle$ is close to $\|\xi_i\|^2 e^{2s_i t}$. That is the early part of the logistic curve (2.7) with initial strength

$$\hat{a}_i = \|\xi_i\|^2 = \tfrac{1}{4}\|W_1(0)v_i + W_2(0)^\top u_i\|^2. \tag{2.10}$$

At this point the weights are aligned and balanced up to errors that decay: $x_i \approx z_i$, and the hidden directions $\xi_i/\|\xi_i\|$ of different modes are nearly orthogonal when m is much larger than the number of modes, since independent random vectors in $\mathbb{R}^m$ are. The aligned solution with initial strengths $\hat{a}_i$ then describes the rest of the trajectory. For a mode with multiplicity, $\hat{a}_i$ is replaced by the eigenvalues of the Gram matrix of the ξ_i in its eigenspace. With i.i.d. $\mathcal{N}(0, \alpha^2)$ entries, $\hat{a}_i \approx \frac{1}{2} m \alpha^2$. ◇

A network whose second layer starts at zero still escapes: then $z_i(0) = 0$, but $x_i + z_i$ starts at $x_i(0)$ and grows all the same, and $\hat{a}_i = \frac{1}{4}\|W_1(0)v_i\|^2$.[5] The first phase is the *silent alignment* of Atanasov et al. (2022): the directions of the weights change by order one while the loss, and the function, hardly move. It explains why the aligned solution describes random initializations at all; Figure 2.2 shows how accurately. For deeper networks the first phase is itself nonlinear ($\dot{\beta} \approx s\beta^{L-1}$), and we know no closed formula for the effective initial strength; the scaling $t_{1/2} \propto \alpha^{-(L-2)}$ survives, with a prefactor (Figure 2.3).

[5] This is why the common practice of initializing a last layer at zero does not stall training, while initializing *every* layer at zero does: the origin is a stationary point of the flow.

REMARK 2.13 (Saddle to saddle). For losses other than $\frac{1}{2}\|\Sigma_{yx} - W\|_F^2$, such as matrix sensing or completion, the modes do not decouple, but the staircase survives in a weaker form. Li et al. (2021) show that for two layers, gradient flow from an infinitesimal initialization is equivalent, under assumptions, to *greedy low-rank learning*: escape from the current saddle along its top eigendirection, which raises the rank by one, then minimize at that rank, and repeat. Jacot et al. (2021) (an arXiv preprint) conjecture the same *saddle-to-saddle* picture for deep linear networks, with saddles at maps of increasing rank, and prove a theorem for the dynamics between the first two saddles. Gidel et al. (2019) show that for two layers and vanishing initialization the iterates of gradient descent pass through the solutions of reduced-rank regression of increasing rank.

2.6 Reading the solution

PREDICTIONS. The model predicts five things. (i) *A staircase*: the loss rests near the values (2.9), one plateau per distinct singular value. (ii) *Order and timing*: mode i is half learned at $t_{1/2} \approx \frac{1}{2s_i} \log(s_i/\hat{a}_i)$ for two layers, so strong modes come first, and the delay between modes i and $i+1$ is about $\log(s/\hat{a})$ times the difference of $\frac{1}{2s_{i+1}}$ and $\frac{1}{2s_i}$. (iii) *Sharpness*: each drop takes time $2\log 3/s_i$ for $L = 2$, independent of the initialization. (iv) *Depth*: with per-layer scale α the plateau scales like $\log(1/\alpha)$ for $L = 2$ and like $\alpha^{-(L-2)}$ for $L \geq 3$. (v) *The endpoint*: when every hidden width is at least q, $W(t) \to \Sigma_{yx}$, the least-squares solution; with a bottleneck of width $m < q$, the aligned trajectory with R_ℓ of m columns reaches the best rank-m approximation, the solution of reduced-rank regression. A linear autoencoder ($y = x$) needs inputs that are not whitened, since otherwise $\Sigma_{yx} = I_d$; with $\Sigma_{xx} = V\Lambda V^\top$ the decoupled flow of Exercise 2.7 applies, and a bottleneck of width m learns the projection onto the top m principal components, in the order of the eigenvalues.[6]

[6] This is the connection Baldi and Hornik (1989) made: a linear autoencoder with a bottleneck, trained on the squared error, has a unique minimum up to invertible transformations of the hidden layer, the projection onto the leading principal subspace.

HIERARCHIES. For the tree of Figure 2.1 the modes are the levels of the tree, and their strengths decrease with depth in the tree: in a complete binary tree of depth D the mode that splits a node at depth j (the root has depth 0) has $s^2 = (2^{D-j} - 1)/2^D$, and the constant mode has $s^2 = (2^{D+1} - 1)/2^D$ (Exercise 2.2). In Figure 2.1, level 0 is the constant mode and level $k \geq 1$ consists of the 2^{k-1} modes that split the nodes at depth $k - 1$, that is, the modes that tell apart the nodes of the k-th row. Broad distinctions are stronger because they are supported by more items and more output features, and the network learns "animal versus plant" first and "canary versus robin" last. This is the progressive differentiation of Section 2.1, explained by one statistic: the singular value.

Figure 2.2 tests predictions (i) and (ii) from a random initialization. The four levels are half learned at $t = 3.1$, 4.2, 6.3, and 10.0, as predicted by (2.8) with the effective initial strengths, and the agreement holds over the whole trajectory, not only at the midpoints, to within 0.02. The loss sits on each step of the staircase for a time that grows as the singular value shrinks, because both the rate $2s$ and the log-ratio $\log(s/\hat{a})$ favor strong modes. The residual deviation comes from the cross-talk between hidden directions that the heuristic neglects. With width $m = 16$ instead of 64, where random hidden directions are further from orthogonal, it grows to about 0.1, for the weakest level.

Figure 2.3 tests prediction (iv). On logarithmic axes the two-layer curve is nearly flat, a logarithm, while the curves for $L = 3$ and $L = 4$ are straight lines of slopes -1 and -2, as Proposition 2.11 predicts: with $\alpha = 0.04$, two layers learn the mode at $t \approx 3.2$ and four layers at $t \approx 310$. Gradient descent from aligned initializations reproduces the exact curves, which checks Theorem 2.9 together with the small error of a finite step. Random initializations learn somewhat faster, because their effective initial strength exceeds α^L, but the exponent is the same.

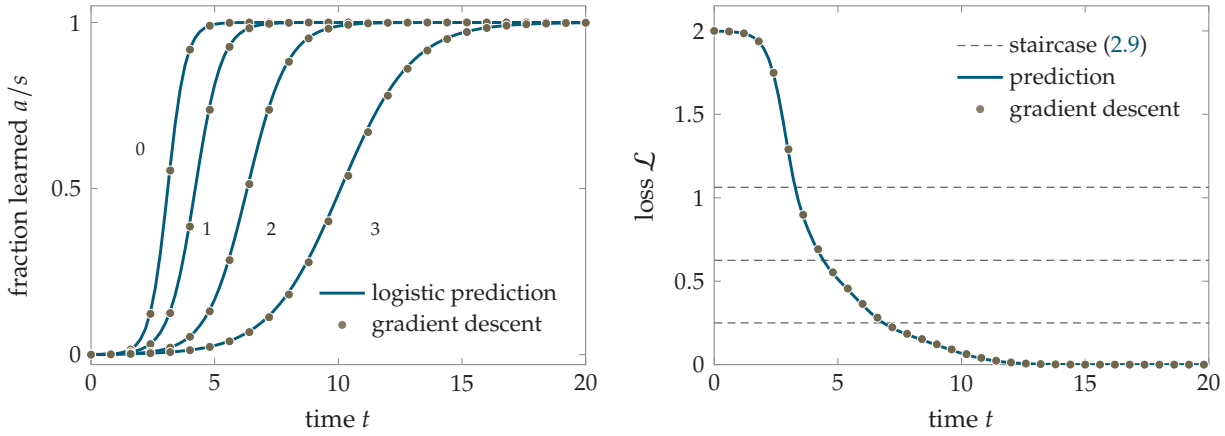


Figure 2.2. Learning the tree of Figure 2.1 with a two-layer linear network of width $m = 64$, by full-batch gradient descent with step $\eta = 0.01$ (time $t = \eta\times$steps) from a random initialization with i.i.d. $\mathcal{N}(0, \alpha^2)$ entries, $\alpha = 0.003$, seed 0. Left: the fraction learned, $\mathrm{tr}(U_k^\top W V_k)/(q_k s_k)$ for the q_k modes U_k, V_k of level k (labels 0 to 3), against the logistic curves (2.7) with the effective initial strengths (2.10) read off from the initialization (between 1.8×10^{-4} and 3.8×10^{-4}; for the levels with several modes, the curves are averaged). The largest deviation is 0.017. Right: the loss, $\frac{1}{2}\|\Sigma_{yx} - W\|_F^2$, descends the staircase 2, 1.0625, 0.625, 0.25, 0 of (2.9).

LIMITS AND REGIMES. The staircase needs two separations: of the singular values (distinct steps) and of scales ($\hat{a}_i \ll s_i$, long plateaus). When the initialization is large, $\hat{a}_i$ is comparable with s_i, the logistic curves start near their midpoints, and all modes are learned together at rate about $2s_i$, which is the lazy, kernel-like regime of Chapter 6; for two layers and aligned weights, the flow is then close to the linear flow $\dot{W} = -2a_0(W - \Sigma_{yx})$ with a fixed preconditioner. Small initialization is the rich regime, and the staircase is its signature: the order parameters are the mode strengths a_i, and the plateau times are their escape times from the saddle at zero.

2.7 What the model gets wrong

WHAT WAS THROWN AWAY. *The nonlinearity.* A deep linear network represents only linear maps, so it learns no features that a linear regression could not, and its hidden layers are rotations and rescalings of the modes of Σ_{yx}; whatever nonlinear networks gain from depth in expressive power is absent. *Whitening.* With $\Sigma_{xx} \neq I$ the gradient is $W\Sigma_{xx} - \Sigma_{yx}$, and the modes decouple only when Σ_{xx} and Σ_{yx} share right singular vectors (Exercise 2.7); in general they do not, and the input correlations mix the modes. *Aligned and balanced initialization.* The heuristic of Section 2.5 recovers it from small random weights, but only approximately, and only for two layers in closed form; large or unbalanced initializations change the dynamics qualitatively (Kunin et al. 2024). *Population loss.* With n samples, Σ_{yx} is replaced by a noisy estimate whose weak modes are dominated by noise, and the late stages fit noise: Advani et al. (2020) and Lampinen and Ganguli (2019) compute the resulting generalization dynamics, in which early stopping helps. *Gradient flow.* With large steps, gradient descent is not close to the flow; for the scalar product this is the catapult of Chapter 15.

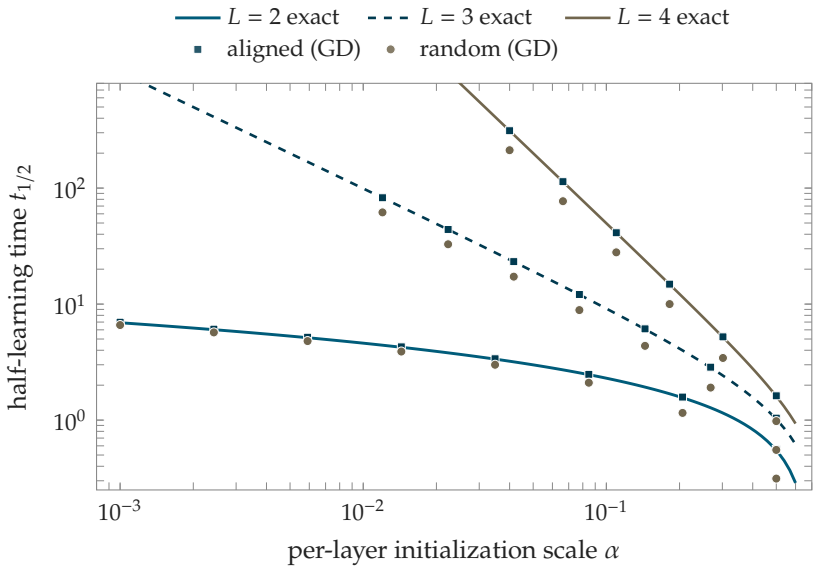


Figure 2.3. Depth lengthens the plateau. One mode of strength $s = 1$ in $\mathbb{R}^{4\times4}$, hidden width 32, full-batch gradient descent with step 0.02; $t_{1/2}$ is the first time $u_1^\top W v_1 \geq \frac{1}{2}$. Curves: the exact half-learning time of Theorem 2.9 with $a_0 = \alpha^L$, from (2.8) for $L = 2$ and by quadrature for $L = 3, 4$. Squares: aligned, balanced initialization with per-layer gain α (seed 0), within about 2% of the curves. Dots: Gaussian initialization with entries $\mathcal{N}(0, \alpha^2/d_{\ell-1})$, averaged over seeds 0, 1, 2. For $L = 2$ these are shifted down by about 0.4, the shift $\frac{1}{2}\log 2.25$ predicted by (2.10), which gives $\hat{a} \approx 2.25\alpha^2$ here; for $L = 3$ and $L = 4$ they follow the slopes $-(L-2)$ with a factor of about 0.7.

WHERE IT BREAKS. *Tasks without linear correlation.* If $\Sigma_{yx} = 0$, every mode has $s_i = 0$, and by Theorem 2.9 and Corollary 2.10 a linear network learns nothing: every aligned a_i decays like $1/t$. XOR, sparse parities and single-index targets of information exponent at least 2 have $\Sigma_{yx} = 0$ (Chapters 7 and 9), yet nonlinear networks learn them, after plateaus whose length the linear model cannot predict. *Stage order in nonlinear networks.* Refinetti et al. (2023) show that networks trained with SGD on images first use the mean and covariance of each class and only later higher-order statistics, a distributional simplicity bias that the linear model shares only in its first stage. Abbe et al. (2023) prove, under assumptions on the training procedure, a saddle-to-saddle dynamics for two-layer nonlinear networks learning functions of a few Gaussian coordinates, in which the order and the duration of the stages are set by the hierarchical structure of the target (its leap complexity; they conjecture times of order $d^{\max(\text{leap},2)}$ in dimension d, up to logarithmic factors), not by singular values. Kalimeris et al. (2019) see the linear stage first in convolutional networks, as the model predicts, but what follows is not a linear map. *Implicit bias.* From small initialization the staircase suggests a bias toward low rank. Gunasekar et al. (2017) conjectured that for matrix factorization the bias is the minimum nuclear norm. Arora et al. (2019b) found that depth strengthens the bias toward low rank and argued that it may not be captured by the nuclear norm or

other norms, and Razin and Cohen (2020) constructed matrix factorization problems in which gradient flow drives every norm and quasi-norm to infinity while the rank is minimized. Li et al. (2021) give counterexamples to the nuclear norm conjecture. The model's stages are therefore a better description of the bias than a norm, but the model with whitened, fully observed data cannot exhibit this: its endpoint is always Σ_{yx}.

FINE STRUCTURE. *Unbalanced and large initializations.* Tarmoun et al. (2021) analyze two-layer flows with an arbitrary conserved D_1, without small, balanced or aligned initialization, and relate them to Riccati equations; Braun et al. (2022) give exact solutions from initializations that are not aligned with the task, building on Fukumizu (1998), and use them to study prior knowledge and continual learning; Dominé et al. (2025) solve the family of initializations they call λ-balanced, with D_1 a multiple of the identity, which interpolates between the rich regime of this chapter and the lazy regime; and Kunin et al. (2024) show that unbalanced initializations that make earlier layers learn faster can speed up feature learning in nonlinear networks. *Gates.* Saxe et al. (2022) add input-dependent gates to linear pathways, which yields solvable models of nonlinear networks in which pathways race and shared representations win. *Saddle-to-saddle.* Li et al. (2021), Jacot et al. (2021) and Gidel et al. (2019) extend the staircase to general losses, Pesme and Flammarion (2023) make it exact for diagonal networks (Chapter 3), and Abbe et al. (2023) for nonlinear networks on hierarchical targets. *Convergence.* Arora et al. (2019a) prove linear convergence of gradient descent for deep linear networks from approximately balanced initializations whose loss is below that of every rank-deficient solution. Open: a theory that predicts the order and timing of stages in nonlinear networks on natural data, and a description of the implicit bias of deep factorizations from small initialization for general measurement operators.

2.8 Model card

MODEL CARD	
PHENOMENON	Plateaus and sudden drops in training loss; coarse distinctions learned before fine ones; incremental, low-rank learning from small initialization.
MODEL	Deep linear network $W_L \cdots W_1$, square loss, whitened inputs, gradient flow, aligned (or small random) initialization.
SOLUTION	Conserved $W_{\ell+1}^\top W_{\ell+1} - W_\ell W_\ell^\top$; closed end-to-end dynamics for balanced weights; exact decoupled mode dynamics $\dot{a} = La^{2-2/L}(s-a)$, logistic for two layers; critical points are partial singular value decompositions.
PREDICTS	A loss staircase $\mathcal{L}_0 + \frac{1}{2}\sum_{i>k} s_i^2$; mode i learned at $\log(s_i/a_0)/(2s_i)$ $(L=2)$ or $a_0^{-(1-2/L)}/((L-2)s_i)$ $(L \geq 3)$; drop width independent of initialization; order set by singular values of Σ_{yx}.

GETS WRONG	Nothing is learned when $\Sigma_{yx} = 0$ (XOR, parity); nonlinear stages follow higher-order structure; the implicit bias toward low rank is not a norm; no finite-sample or step-size effects.
SUCCESSORS	Unbalanced and λ-balanced exact solutions; gated linear networks; saddle-to-saddle dynamics for general losses, diagonal networks, and nonlinear networks.

This chapter opens the book's longest thread: *escape from a saddle at small initialization*. The logistic escape here, with time log(1/initialization)/strength, reappears coordinate by coordinate in the diagonal networks of Chapter 3, as the search phase of Chapter 7 (where depth is replaced by the information exponent), as the plateaus of the committee machine in Chapter 8, as hidden progress in sparse parities (Chapter 9), and in the formation of induction heads (Chapter 14); the quadratic of the prologue is the degenerate case with no saddle. The scalar case of the conservation law is the invariant of the two-parameter model in Chapter 15.

The second thread is *lazy versus rich*. Large initializations turn the staircase into simultaneous exponential relaxation, the regime of Chapter 6; small ones produce the rich dynamics of this chapter, and Chapter 3 shows how the same choice selects an implicit bias.

Exercises

2.1 CONSOLIDATION. (The logistic curve.) (a) Integrate $\dot{a} = 2a(s - a)$ by partial fractions and recover Corollary 2.10. (b) Show that the loss of one mode, $\frac{1}{2}(s - a)^2$, decreases fastest in time when $a = s/3$, and compute the time at which it occurs. (c) For the tree of Figure 2.2, with mean effective initial strengths 2.8×10^{-4}, 3.5×10^{-4}, 2.7×10^{-4} and 2.9×10^{-4} for the four levels, predict the four half-learning times and compare with the values 3.1, 4.2, 6.3 and 10.0 quoted in the text.

2.2 CONSOLIDATION. (The tree.) Let $P = 2^D$ items sit at the leaves of a complete binary tree of depth D, with $x = \sqrt{P} e_p$ and one output coordinate per node, as in Figure 2.1. (a) Show that $(\Sigma_{yx}^\top \Sigma_{yx})_{pp'}$ is $\frac{1}{P}$ times the number of common ancestors of p and p' (the item itself counts if $p = p'$). (b) Show that the constant vector is an eigenvector with eigenvalue $(2^{D+1} - 1)/P$, and that the Haar function of a node at depth $j < D$ (the root has depth 0) is an eigenvector with eigenvalue $(2^{D-j} - 1)/P$. (c) Conclude that the 2^j modes that split the nodes at depth j have $s^2 = (2^{D-j} - 1)/2^D$, and that the loss staircase for $D = 3$ is 2, 1.0625, 0.625, 0.25, 0.

2.3 CONSOLIDATION. (Neurons of a ReLU network are balanced.) Let $f(x) = \sum_{r=1}^m a_r \phi(\langle w_r, x \rangle)$ with $\phi(z) = \max(z, 0)$, and let the parameters follow the gradient flow of $\mathbb{E}[\ell(f(x), y)]$ for a differentiable loss ℓ, with the convention $\phi'(0) = 0$. Using $\phi'(z)z = \phi(z)$, show that $\|w_r\|^2 - a_r^2$ is constant in time for every neuron r. What does this say about a neuron whose output weight is driven to zero?

2.4 CONSOLIDATION. (End-to-end dynamics for two layers.) (a) For $L = 2$ and balanced weights, show directly that $W_2 W_2^\top = (W W^\top)^{1/2}$. (b) For a scalar output, $W = w^\top$ with $w \in \mathbb{R}^d$ and gradient $g = \nabla \mathcal{E}$ written as a vector, show that (2.3) becomes $\dot{w} = -\|w\|^{2-2/L}\big(g + (L-1)\langle g, \hat{w}\rangle \hat{w}\big)$ with $\hat{w} = w/\|w\|$: depth speeds up motion along w by the factor L. (c) Show that for $W = UDV^\top$ with D diagonal, (2.3) gives back (2.6).

2.5 CONSOLIDATION. (Escape from a degenerate saddle.) (a) Solve $\dot{\beta} = s\beta^{L-1}$ exactly for $L \geq 3$, and show that the solution blows up at time $\beta_0^{-(L-2)}/((L-2)s)$. (b) Explain why this time is the leading term of Proposition 2.11 with $a_0 = \beta_0^L$. (c) Compare with the escape time of the search phase in Chapter 7, and identify the depth with the information exponent.

2.6 VARIATION. (Unbalanced scalar network.) Let $L = 2$ and $d = d_1 = d_y = 1$, with weights $b = W_1$ and $c = W_2$ and $\mathcal{L} = \frac{1}{2}(s - cb)^2$. (a) Show that $\delta = b^2 - c^2$ is conserved and that $a = cb$ satisfies $\dot{a} = \sqrt{4a^2 + \delta^2}\,(s - a)$. (b) Show that from $a(0) = 0$ and $\delta \neq 0$ the mode reaches $s/2$ by time $\log 2/|\delta|$, so a strongly unbalanced network has no plateau. (c) Show that as $\delta \to 0$ with $a(0) = a_0 > 0$ fixed, the solution converges to the logistic curve. (d) Interpret $|\delta|$ as an initialization scale that makes training lazy.

2.7 VARIATION. (Non-whitened inputs.) Let $\Sigma_{xx} = V \Lambda V^\top$ and $\Sigma_{yx} = USV^\top$ share the matrix V, with $\Lambda = \mathrm{diag}(\lambda_1, \dots, \lambda_q)$ positive (in this exercise λ_i are eigenvalues). Show that the aligned ansatz still solves the flow for $L = 2$, with $\dot{a}_i = 2a_i(s_i - \lambda_i a_i)$. Which modes are learned first, and to what values do they converge? Show that the order of learning is set by s_i while the final value is s_i/λ_i, and that in general the two orders differ.

2.8 VARIATION. (Weight decay.) Add $\frac{\lambda}{2}(\|W_1\|_F^2 + \|W_2\|_F^2)$ to the loss of a two-layer network with whitened inputs. (a) Show that $D_1(t) = e^{-2\lambda t} D_1(0)$, so weight decay balances the layers. (b) Show that the aligned ansatz still solves the flow, with $\dot{a}_i = 2a_i(s_i - \lambda - a_i)$. (c) Conclude that the network converges to the soft-thresholded matrix $\sum_i (s_i - \lambda)_+ u_i v_i^\top$ when every $a_i(0) > 0$, and that modes with $s_i \leq \lambda$ are never learned.

2.9 VARIATION. ★ (Depth-independent learning time.) For $L \geq 3$ and fixed $0 < a_0 < s/2$, show that $L\, t_{1/2} \to \int_{a_0}^{s/2} \frac{\mathrm{d}a}{a^2(s-a)} = \frac{1}{s a_0} - \frac{2}{s^2} + \frac{1}{s^2} \log \frac{s - a_0}{a_0}$ as $L \to \infty$. Interpret: if the end-to-end initial strength is held fixed (each layer starts near the identity, with gain $a_0^{1/L} \to 1$) and the step size is scaled like $1/L$, the learning time does not grow with depth. Compare with the per-layer scaling α of Figure 2.3, and with the depth-independent learning times that Saxe et al. (2014) report for special initializations. *(Hint: $a^{-2+2/L} \leq \max(a^{-2}, 1)$ for $a > 0$; use dominated convergence.)*

2.10 SIMULATION. (Depth on the tree.) Modify the script of Figure 2.2 to

train a three-layer network ($d_1 = d_2 = 64$) from i.i.d. $\mathcal{N}(0, \alpha^2/d_{\ell-1})$ entries with $\alpha = 0.1$, and plot the fraction learned for each level. With step 0.01, the reader should observe four stages, half learned near $t \approx 5, 7, 10$ to 12, and 18 (the exact values vary with the seed by about 10%), each drop narrow compared with the plateau before it; and with $\alpha = 1$ all levels are learned together, 90% learned by $t \approx 0.2$. Explain both observations with Proposition 2.11.

2.11 SIMULATION. (XOR.) Let $x \in \{\pm 1\}^2$ be uniform, augmented by a constant coordinate, and $y = x_1 x_2$. (a) Compute Σ_{xx} and Σ_{yx}, and show that $\Sigma_{yx} = 0$. (b) Train a two-layer linear network and a two-layer ReLU network of width 16 from small random initialization ($\mathcal{N}(0, 10^{-4})$ entries) by gradient descent with step 0.05. The reader should observe that the linear network's loss stays at $\frac{1}{2}$, while for most seeds the ReLU network reaches zero loss within about 500 steps (some seeds stall at loss $\frac{1}{8}$, with one input misfit): the ReLU network learns a function that the linear model says carries no signal.

2.12 RESEARCH. (Stage order in nonlinear networks.) For a two-layer ReLU network trained from small initialization on the tree data of Figure 2.1, with inputs given by random features of the items rather than one-hot vectors, determine whether the stages still follow the singular values of Σ_{yx}, and how the answer changes when the outputs depend nonlinearly on the features. Saxe et al. (2022) solve gated networks with fixed gates, and Abbe et al. (2023) treat functions of a few Gaussian coordinates; a theory of stage order for nonlinear networks on structured, non-Gaussian data is open.

2.13 RESEARCH. (Saddle to saddle in depth.) Prove that, for $L \geq 3$ and small random initialization, gradient flow on $\frac{1}{2}\|\Sigma_{yx} - W\|_F^2$ with distinct singular values visits neighborhoods of the saddles $W_1, W_2, \dots$ in order, with explicit times. For two layers, Li et al. (2021) and Gidel et al. (2019) establish versions of this; Jacot et al. (2021) conjecture it in depth and prove a theorem for the passage between the first two saddles. A complete proof for all saddles, with the prefactor of Figure 2.3, is open.

Notes and References

Deep linear networks entered the theory of learning with Baldi and Hornik (1989), who showed that a two-layer linear autoencoder or regressor has no spurious local minima and that its critical points are projections onto sets of principal components. Fukumizu (1998) analyzed batch gradient descent in linear networks with one hidden layer, through a matrix Riccati equation, and used it to explain overtraining when the network has more hidden units than the target needs. The exact solution of Theorem 2.9, the logistic mode curves, the depth-L mode equation, and the observation that linear networks reproduce the plateaus and sudden drops of nonlinear ones are from Saxe et al. (2014), who also found that special initializations, and

random orthogonal ones, give learning times that do not grow with depth.[7] Saxe et al. (2019) applied the solution to semantic development, building on the networks of Rogers and McClelland (2004).

The conservation law was found independently by Arora et al. (2018), for linear networks, and by Du et al. (2018), who proved that gradient flow preserves the differences of squared layer norms in homogeneous networks with linear, ReLU and leaky ReLU activations, the per-neuron statement of Exercise 2.3. Theorem 2.7 is from Arora et al. (2018), who used it to argue that depth can accelerate optimization; Arora et al. (2019a) turned it into a convergence proof for gradient descent. Kawaguchi (2016) extended the landscape results to deep linear networks.

Incremental learning and implicit low-rank bias were studied by Gidel et al. (2019), Gissin et al. (2020), Arora et al. (2019b), Razin and Cohen (2020) and Li et al. (2021), and saddle-to-saddle dynamics by Jacot et al. (2021). Atanasov et al. (2022) named the silent alignment of Section 2.5 and used it to show that networks trained from small initialization behave like kernel machines with a learned kernel. Generalization dynamics with finite data are in Advani et al. (2020) and Lampinen and Ganguli (2019). Exact solutions beyond the aligned case are in Tarmoun et al. (2021), Braun et al. (2022), Dominé et al. (2025) and Kunin et al. (2024).

[7] The 2014 paper also connected depth-independent learning to the faithful propagation of gradients through deep nonlinear networks at the edge of chaos, the subject of Chapter 5.

TRANSLATING THE LITERATURE. Saxe et al. (2014) write a three-layer network as $W^{32}W^{21}$: their W^{21} and W^{32} are our W_1 and W_2, their Σ^{31} and Σ^{11} are our Σ_{yx} and Σ_{xx}, their modes are indexed by α (our i), with $\Sigma^{31} = \sum_\alpha s_\alpha u^\alpha v^{\alpha\top}$, and their time constant $\tau = 1/\lambda$ (with λ their learning rate) is our unit of time, so their t/τ is our t. Saxe et al. (2019) write $\Sigma^{yx} = USV^\top$ for our Σ_{yx} and collect the mode strengths in a diagonal matrix $A(t)$, our $\mathrm{diag}(a_i(t))$. Arora et al. (2018) write N for our L, W_e for our end-to-end W, and $L^1(W_e)$ for our $\mathcal{E}(W)$; they state the end-to-end equation with weight decay, and index its sum in the opposite order, with $[W_e W_e^\top]^{(j-1)/N}$ on the left, which is (2.3) with j replaced by $L+1-j$. Baldi and Hornik (1989) write the network as AB, with B the first layer and A the second (our W_1 and W_2), and $\Sigma_{YX}\Sigma_{XX}^{-1}\Sigma_{XY}$ for the matrix whose eigenvectors define the critical points (our $\Sigma_{yx}\Sigma_{yx}^\top$ when $\Sigma_{xx} = I$).

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3 The Bias of Small Initialization

Diagonal linear networks, from kernels to sparsity

THE MODEL — DIAGONAL LINEAR NETWORKS

IN THIS CHAPTER. An overparameterized network can fit its training data in many ways, and which fit gradient descent finds depends on how large the weights are at the start: large initializations behave like a kernel method, small ones learn sparse or low-rank structure, one piece at a time. A linear predictor written as $w = u \odot u - v \odot v$ and trained by gradient flow from $u = v = \alpha\mathbf{1}$ shows the whole story. We prove that the flow converges to the interpolator of smallest hyperbolic entropy Φ_α, a potential that is a squared Euclidean norm for large α and an ℓ_1 norm for small α; that for an orthogonal design the coordinates are learned one at a time, at times $\log(1/\alpha)/(2|w_{\star,i}|)$; and we sketch how, for a general design, the flow jumps from saddle to saddle. Step size, stochastic gradients, classification losses, and parametrizations that are not diagonal all change the answer.

3.1 The phenomenon: one network, many fits

A network with more parameters than training examples has, as a rule, many parameter vectors that fit the training set exactly. Zhang et al. (2017) made the point vivid in 2016: standard convolutional networks reach zero training error on CIFAR-10 even when the labels are replaced by random ones, and then of course predict the test labels at chance. The architecture alone therefore cannot explain why the same networks generalize on real labels; among the many interpolators, the training algorithm must be choosing a good one. Neyshabur et al. (2015) had argued this earlier: in their experiments the test error of two-layer networks did not deteriorate as the width grew past what was needed to fit the data, and they proposed that gradient descent regularizes implicitly, toward small norms of some kind; the selection is now called the *implicit bias* of the algorithm.[1]

Which interpolator, then? For matrix factorization, Gunasekar et al. (2017) observed that gradient descent from an initialization close to the origin, with small steps, finds a solution of nearly minimal nuclear norm, and they conjectured that it does so exactly in the limit.[2] The size of the initialization turned out to be the decisive parameter. Chizat et al. (2019) showed that a model rescaled so that its output at initialization is large trains in a *lazy regime*: the parameters hardly move, the model stays close to its linearization at initialization, and training is kernel regression with the tangent kernel of Chapter 6. They reported that deep convolutional networks trained this way perform worse. Woodworth et al. (2020) named the two ends of the scale

[1] The phrase *implicit regularization* is in their title. The analogy they drew was with matrix factorization, where a factorization of unbounded rank still generalizes when its norm is controlled.

[2] They proved the conjecture when the measurement matrices commute, a case that reduces to the diagonal networks of this chapter. In general it fails; see Section 3.7.

the *kernel regime* and the *rich regime*, and found the transition between them in matrix factorization and in multilayer networks as the initialization scale varies. Geiger et al. (2020) located the crossover for networks of width h at an output scale of order $h^{-1/2}$, and found that the better regime depends on the architecture: fully connected networks generalized better in the lazy regime, convolutional networks in the feature-learning one.

A second, dynamical signature accompanies small initializations. Gissin et al. (2020) observed *incremental learning* in deep linear models, quadratic networks and convolutional linear networks: started near zero, the model acquires its components one after another, in order of decreasing strength, with long plateaus between them. They noted that in shallow models the effect needs an exponentially small initialization, and in deeper ones only a polynomially small one.

Observation 3.1 (Chizat et al. 2019; Woodworth et al. 2020; Geiger et al. 2020). In overparameterized models trained by gradient methods to near-zero training loss, the interpolator reached depends on the scale of the initialization. Large scales give the predictor of the tangent-kernel model at initialization. Small scales give predictors with structure that the kernel does not favor (sparse vectors, low-rank matrices), acquired incrementally, with plateaus whose length grows as the scale shrinks. Which regime generalizes better depends on the task and the architecture: for convolutional networks on image data the small-scale regime has been better, for fully connected networks the comparison has gone the other way.

The two regimes, and a crossover between them, have been seen in every family of models where they were looked for; but the crossover scale depends on width, depth, learning rate and loss, and in practice initialization is set by conventions (Chapter 5) rather than tuned. The question for this chapter is: *what exactly does gradient descent select among the interpolators, as a function of the initialization scale, and how does it get there?* We want the answer in a model where it is a theorem.

3.2 The model: a linear predictor written as a difference of squares

Data and loss. We are given a design matrix $X \in \mathbb{R}^{n\times d}$ with rows $x_1, \dots, x_n$ and labels $y \in \mathbb{R}^n$, and we fit linear predictors $x \mapsto \langle w, x\rangle$ by the training loss

$$\widehat{\mathcal{L}}(w) = \frac{1}{2n}\|Xw - y\|^2, \qquad \nabla\widehat{\mathcal{L}}(w) = \frac{1}{n}X^\top(Xw - y). \tag{3.1}$$

The case of interest is $n < d$ with X of rank n, when the *interpolators*, the solutions of $Xw = y$, form an affine subspace of dimension $d - n$. Two of them will matter: the *minimum-ℓ_2-norm interpolator* $w_{\ell_2} = X^+y$, and a *minimum-ℓ_1-norm interpolator* $w_{\ell_1} \in \arg\min\{\|w\|_1 : Xw = y\}$, the *basis pursuit* solution of Chen et al. (1998).

The network. We write the weight vector as

$$w = u \odot u - v \odot v, \qquad u, v \in \mathbb{R}^d, \tag{3.2}$$

with $\odot$ the coordinatewise product, so that $w_i = u_i^2 - v_i^2$, and we run gradient flow on $F(u, v) = \widehat{\mathcal{L}}(u \odot u - v \odot v)$ from the balanced initialization $u(0) = v(0) = \alpha \mathbf{1}_d$, with initialization scale $\alpha > 0$. Then $w(0) = 0$. With $g(t) = \nabla \widehat{\mathcal{L}}(w(t))$, the chain rule gives $\partial F / \partial u_i = 2u_i g_i$ and $\partial F / \partial v_i = -2v_i g_i$, so the flow is

$$\dot{u} = -2\, u \odot g, \qquad \dot{v} = 2\, v \odot g. \tag{3.3}$$

The model is a *diagonal linear network*: a two-layer linear network whose hidden layer has $2d$ units, unit i seeing only coordinate i of the input with weight u_i (or v_i) and sending it to the output with the same weight, $+u_i$ (or $-v_i$). The second factor of u_i is the output weight; tying it to the input weight is the balanced initialization of Chapter 2, preserved here by the flow.

WHAT WAS THROWN AWAY, AND WHY. Nearly everything. The function class is the class of linear predictors, the same as for the plain linear model; only the *parametrization* changes, and with it the geometry in which gradient flow moves. That is the point: in a linear model the set of interpolators is known exactly, so the question "which interpolator?" can be asked cleanly, and every answer must come from the parametrization and the initialization, since nothing else differs from ordinary least squares, whose gradient flow from $w(0) = 0$ converges to w_{ℓ_2}. The parametrization is the simplest one that is not linear in the parameters and keeps the coordinates separate; the difference of squares lets w take both signs, and the product $u \odot v$ would serve as well (Exercise 3.7). Infinitesimal steps, no noise, and the square loss are restrictions that the literature lifts, each changing the answer (Section 3.7).

DICTIONARY

IN THE MODEL	IN A TRAINED NETWORK
weight vector $w = u \odot u - v \odot v$	the function computed by a network; linear in the input, quadratic in the parameters
pair (u_i, v_i) of a coordinate	a neuron, with tied input and output weights
initialization scale α	standard deviation of the initial weights, or the output multiplier of Chizat et al. (2019)
conserved products $u_i v_i = \alpha^2$	the balancedness invariants of homogeneous networks (Chapter 2)
potential Φ_α, limit w_α^Φ	the implicit regularizer of training; the interpolator it selects
large α: minimum-ℓ_2 interpolator	the lazy or kernel regime: regression with the tangent kernel at initialization
small α: minimum-ℓ_1 interpolator	the rich regime: few active features, learned from the data
accelerated time $t/\log(1/\alpha)$; saddles	plateaus and sudden drops of the loss; incremental learning

THE KERNEL AT INITIALIZATION. The gradient of the predictor $\langle u \odot u - v \odot v, x \rangle$ with respect to (u, v) at initialization is $(2\alpha x, -2\alpha x)$, so the tangent kernel of Chapter 6 is $K(x, x') = 8\alpha^2 \langle x, x' \rangle$, the linear kernel. Kernel

regression with it, run to interpolation from zero, gives w_{ℓ_2}. A lazy diagonal network is therefore an ordinary least-squares solver, and anything else it does is feature learning.

WHAT ONE HAD TO NOTICE. Three things. First, that a reparameterization which leaves the model class unchanged changes the implicit bias; this was clear for mirror descent, where the choice of potential decides the limit (Gunasekar et al. 2018), and the observation of Woodworth et al. (2020) was that a reparameterization can *be* a mirror descent in disguise. Second, that the coordinates of a diagonal network share nothing but the gradient, so each one can be integrated in closed form given the integrated gradient. Third, that this closed form makes the initialization scale appear inside the potential, which interpolates between the two norms of sparse regression.

3.3 The solution: gradient flow is a mirror flow

INTEGRATING ONE COORDINATE. Write $G(t) = \int_0^t g(s)\,\mathrm{d}s \in \mathbb{R}^d$ for the integrated gradient. Each equation in (3.3) is linear in its own variable once g is known.

Proposition 3.2 (Closed form of the parameters). *On every interval* $[0, T)$ *on which the flow* (3.3) *exists,*

$$u = \alpha e^{-2G}, \qquad v = \alpha e^{2G}, \qquad w = -2\alpha^2 \sinh(4G),$$
$$u_i v_i = \alpha^2, \qquad u_i^2 + v_i^2 = \sqrt{w_i^2 + 4\alpha^4}, \tag{3.4}$$

with the exponential and the hyperbolic sine taken coordinatewise.

PROOF. The scalar equation $\dot{u}_i = -2g_i(t)u_i$ with $u_i(0) = \alpha$ has the unique solution $u_i = \alpha e^{-2G_i}$, and in the same way $v_i = \alpha e^{2G_i}$. Then $u_i v_i = \alpha^2$, and $w_i = \alpha^2(e^{-4G_i} - e^{4G_i}) = -2\alpha^2 \sinh(4G_i)$. Finally $(u_i^2 + v_i^2)^2 = (u_i^2 - v_i^2)^2 + 4u_i^2 v_i^2 = w_i^2 + 4\alpha^4$, and $u_i^2 + v_i^2 > 0$. ∎

Inverting the hyperbolic sine, $\frac{1}{4}\operatorname{arcsinh}\big(w_i/(2\alpha^2)\big) = -G_i$. Differentiating in t gives the central identity of the chapter,

$$\frac{\mathrm{d}}{\mathrm{d}t}\nabla\Phi_\alpha\big(w(t)\big) = -\nabla\widehat{\mathcal{L}}\big(w(t)\big), \qquad \nabla\Phi_\alpha(w)_i = \frac{1}{4}\operatorname{arcsinh}\frac{w_i}{2\alpha^2}, \tag{3.5}$$

for the potential

$$\Phi_\alpha(w) = \frac{\alpha^2}{4}\sum_{i=1}^{d} q\Big(\frac{w_i}{\alpha^2}\Big), \qquad q(z) = 2 - \sqrt{4+z^2} + z\operatorname{arcsinh}\frac{z}{2}. \tag{3.6}$$

Indeed $q'(z) = -z/\sqrt{4+z^2} + \operatorname{arcsinh}(z/2) + z/\sqrt{4+z^2} = \operatorname{arcsinh}(z/2)$, so $\partial_i\Phi_\alpha(w) = \frac{1}{4}q'(w_i/\alpha^2) = \frac{1}{4}\operatorname{arcsinh}(w_i/2\alpha^2)$. Equation (3.5) says that $w(t)$ follows *mirror descent* in continuous time, with the mirror map $\nabla\Phi_\alpha$: the flow is a gradient flow in the dual coordinates $\nabla\Phi_\alpha(w)$, not in w.[3] In the parameters the flow is an ordinary gradient flow; in the predictor it is a mirror

[3] Mirror descent is due to Nemirovsky and Yudin; see Nemirovsky and Yudin (1983). With the entropy as potential it becomes the exponentiated gradient method of Kivinen and Warmuth (1997).

flow, and the reparameterization has chosen the mirror. The function Φ_α is, up to a factor and a constant, the *hyperbolic entropy* ("hypentropy") of Ghai et al. (2020): with $\beta = 2\alpha^2$, $\frac{\alpha^2}{4} q(w_i/\alpha^2) = \frac{1}{4}\big(w_i \operatorname{arcsinh}(w_i/\beta) - \sqrt{w_i^2 + \beta^2} + \beta\big)$.

Lemma 3.3 (The potential). *The function q is even, smooth and strictly convex, with $q(0) = q'(0) = 0$, $q'(z) = \operatorname{arcsinh}(z/2)$ and $q''(z) = 1/\sqrt{4+z^2}$. For every real z,*

$$\frac{z^2}{2\sqrt{4+z^2}} \le q(z) \le \frac{z^2}{4}, \tag{3.7}$$

and

$$0 \le q(z) - |z|\big(\log|z| - 1\big) < 2, \tag{3.8}$$

with the convention $0 \log 0 = 0$.

Proof. The formulas for q' and q'' were checked above, and $q'' > 0$; evenness and $q(0) = 0$ are read off from (3.6). For (3.7), Taylor's formula with integral remainder gives $q(z) = \int_0^z (z-s) q''(s)\, \mathrm{d}s$. For $z > 0$ the weight $q''(s)$ decreases from $q''(0) = \frac{1}{2}$ to $q''(z)$ on $[0, z]$, so $\frac{1}{2} q''(z) z^2 \le q(z) \le \frac{1}{4} z^2$; evenness covers $z < 0$.

For (3.8), let $r(z) = q(z) - |z|(\log|z| - 1)$, an even function with $r(0) = 0$, continuous at 0 because $z \log z \to 0$. For $z > 0$,

$$r'(z) = \operatorname{arcsinh} \frac{z}{2} - \log z = \log\Big(\frac{1}{2} + \sqrt{\frac{1}{4} + \frac{1}{z^2}}\Big) > \log 1 = 0,$$

using $\operatorname{arcsinh} x = \log(x + \sqrt{1+x^2})$. So r increases on $(0, \infty)$ and $r \ge 0$. As $z \to \infty$, $\sqrt{4+z^2} - z \to 0$ and, since $\sqrt{\frac{1}{4} + \frac{1}{z^2}} \le \frac{1}{2} + \frac{1}{z^2}$, $0 < z\, r'(z) \le z \log(1 + z^{-2}) \le 1/z \to 0$; since

$$r(z) = 2 - \big(\sqrt{4+z^2} - z\big) + z\, r'(z),$$

$r(z) \to 2$, and an increasing function stays below its limit. ∎

The two bounds describe the two faces of Φ_α. By (3.7), where $|w_i| \ll \alpha^2$ the potential is $\Phi_\alpha(w) \approx \|w\|^2/(16\alpha^2)$, a squared Euclidean norm. By (3.8), writing $\varepsilon_\alpha = \log(1/\alpha^2)$ and $\mathrm{H}(w) = \sum_i |w_i|(\log|w_i| - 1)$,

$$\Phi_\alpha(w) = \frac{\varepsilon_\alpha}{4}\|w\|_1 + \frac{1}{4}\mathrm{H}(w) + R_\alpha(w), \qquad 0 \le R_\alpha(w) < \frac{d\alpha^2}{2}, \tag{3.9}$$

for every $w \in \mathbb{R}^d$: substitute $z = w_i/\alpha^2$ in (3.8), multiply by $\alpha^2/4$, and note $\frac{\alpha^2}{4}|z|(\log|z| - 1) = \frac{1}{4}|w_i|(\log|w_i| + \varepsilon_\alpha - 1)$. When α is small, ε_α is large and Φ_α is an ℓ_1 norm with an entropy-like correction.

The main theorem. We now prove that gradient flow converges, and to what.

Theorem 3.4 (Implicit bias of diagonal linear networks). *Let $X \in \mathbb{R}^{n \times d}$ and $y \in \mathbb{R}^n$ be such that some $w^\circ \in \mathbb{R}^d$ satisfies $Xw^\circ = y$, and let $\alpha > 0$.*

Then:

(i) The flow (3.3) from $u(0) = v(0) = \alpha \mathbf{1}_d$ exists for all $t \geq 0$, and $u(t)$ and $v(t)$ have positive coordinates and stay bounded.

(ii) For every interpolator w° and every $t > 0$, $\widehat{\mathcal{L}}(w(t)) \leq \Phi_\alpha(w^\circ)/(2t)$.

(iii) The problem of minimizing $\Phi_\alpha(w)$ subject to $Xw = y$ has a unique solution w_α^Φ, and $w(t) \to w_\alpha^\Phi$ as $t \to \infty$.

Proof. The plan is to use the Bregman divergence of Φ_α to an interpolator as a Lyapunov function, which gives boundedness, the rate, and existence for all time at once; then to identify the limit by the optimality conditions, using that $\nabla\Phi_\alpha(w(t))$ never leaves the row space of X.

A Lyapunov function. Let $[0, T)$ be the maximal interval of existence of the flow; the right side of (3.3) is a polynomial in (u, v), so a unique maximal solution exists, and if $T < \infty$, then (u, v) is unbounded on $[0, T)$. Fix an interpolator w° and let

$$D(t) = \Phi_\alpha(w^\circ) - \Phi_\alpha(w(t)) - \langle \nabla\Phi_\alpha(w(t)), w^\circ - w(t)\rangle,$$

the Bregman divergence of Φ_α from $w(t)$ to w°. It is nonnegative because Φ_α is convex (Lemma 3.3). Differentiating, the terms $\mp\langle \nabla\Phi_\alpha(w), \dot{w}\rangle$ cancel, and (3.5) gives

$$\dot{D} = -\left\langle \frac{\mathrm{d}}{\mathrm{d}t}\nabla\Phi_\alpha(w),\, w^\circ - w \right\rangle = \langle \nabla\widehat{\mathcal{L}}(w), w^\circ - w\rangle = -2\widehat{\mathcal{L}}(w),$$

where the last step is $\langle \nabla\widehat{\mathcal{L}}(w), w^\circ - w\rangle = \frac{1}{n}\langle Xw - y,\, Xw^\circ - Xw\rangle$ with $Xw^\circ = y$. So D is nonincreasing, and $D(0) = \Phi_\alpha(w^\circ)$ because $w(0) = 0$, $\Phi_\alpha(0) = 0$ and $\nabla\Phi_\alpha(0) = 0$.

Boundedness and existence. We show that the sublevel sets of D, as a function of w, are bounded. Write $\varphi(x) = \frac{\alpha^2}{4} q(x/\alpha^2)$ for one coordinate of Φ_α, so that $D = \sum_i D_i$ with $D_i = \varphi(w_i^\circ) - \varphi(w_i) - \varphi'(w_i)(w_i^\circ - w_i) \geq 0$. A computation from (3.6) gives $x\varphi'(x) - \varphi(x) = \frac{1}{4}\left(\sqrt{x^2 + 4\alpha^4} - 2\alpha^2\right) \geq \frac{1}{4}(|x| - 2\alpha^2)$, and $|\varphi'(x)| = \frac{1}{4}\operatorname{arcsinh}(|x|/2\alpha^2) \leq \frac{1}{4}\log(1 + |x|/\alpha^2)$, because $\operatorname{arcsinh} s \leq \log(1 + 2s)$ for $s \geq 0$. Hence

$$4D_i \geq 4\varphi(w_i^\circ) + |w_i| - 2\alpha^2 - |w_i^\circ| \log\left(1 + |w_i|/\alpha^2\right),$$

which tends to infinity with $|w_i|$. Since $D(t) \leq \Phi_\alpha(w^\circ)$ and every $D_j \geq 0$, each $w_i(t)$ stays in a bounded set, uniformly in $t < T$. By Proposition 3.2, $u_i^2 + v_i^2 = \sqrt{w_i^2 + 4\alpha^4}$, so (u, v) is bounded on $[0, T)$, and therefore $T = \infty$. Positivity is (3.4). This proves (i).

The rate. Along the flow $\frac{\mathrm{d}}{\mathrm{d}t}\widehat{\mathcal{L}}(w(t)) = \frac{\mathrm{d}}{\mathrm{d}t}F(u, v) = -\|\nabla F(u, v)\|^2 \leq 0$, so $\widehat{\mathcal{L}}(w(t))$ is nonincreasing, and integrating $\dot{D} = -2\widehat{\mathcal{L}}$,

$$2t\,\widehat{\mathcal{L}}(w(t)) \leq 2\int_0^t \widehat{\mathcal{L}}(w(s))\,\mathrm{d}s = D(0) - D(t) \leq \Phi_\alpha(w^\circ),$$

which is (ii).

The minimizer. By (3.9), $\Phi_\alpha(w) \geq \frac{1}{4}\sum_i |w_i|(\log(|w_i|/\alpha^2) - 1)$, which tends to infinity with $\|w\|$; so Φ_α attains its minimum on the closed nonempty set $\{Xw = y\}$, and the minimizer is unique because Φ_α is strictly convex. A point $\bar{w}$ with $X\bar{w} = y$ is this minimizer if and only if $\nabla\Phi_\alpha(\bar{w})$ lies in the row space $\mathrm{range}(X^\top)$. Indeed, if $\nabla\Phi_\alpha(\bar{w}) = X^\top\nu$, then for every interpolator w' convexity gives $\Phi_\alpha(w') \geq \Phi_\alpha(\bar{w}) + \langle X^\top\nu, w' - \bar{w}\rangle = \Phi_\alpha(\bar{w})$, since $X(w'-\bar{w}) = 0$; and conversely the gradient at a minimizer is orthogonal to $\ker X$, whose orthogonal complement is $\mathrm{range}(X^\top)$.

The limit. By (3.5) and $\nabla\Phi_\alpha(w(0)) = 0$, for every t,

$$\nabla\Phi_\alpha(w(t)) = -\int_0^t \nabla\widehat{\mathcal{L}}(w(s))\,\mathrm{d}s = -\frac{1}{n}X^\top\int_0^t \big(Xw(s) - y\big)\,\mathrm{d}s \in \mathrm{range}(X^\top).$$

Let $\bar{w}$ be a limit point of $w(t)$ along a sequence $t_k \to \infty$; one exists because w is bounded. By (ii), $\widehat{\mathcal{L}}(\bar{w}) = \lim_k \widehat{\mathcal{L}}(w(t_k)) = 0$, so $X\bar{w} = y$. The subspace $\mathrm{range}(X^\top)$ is closed and $\nabla\Phi_\alpha$ is continuous, so $\nabla\Phi_\alpha(\bar{w}) \in \mathrm{range}(X^\top)$, and $\bar{w} = w_\alpha^\Phi$ by the optimality condition. A bounded trajectory whose only limit point is w_α^Φ converges to it, which proves (iii). ∎

Two remarks on the hypotheses. The theorem does not need $n < d$: if the interpolator is unique, then w_α^Φ is that interpolator. And it needs an interpolator; when $y \notin \mathrm{range}(X)$ the same argument, with w° a least-squares solution, shows that the flow converges to the least-squares solution of smallest Φ_α (Exercise 3.4). The rate (ii) is slow for small α, since $\Phi_\alpha(w^\circ) \approx \frac{1}{4}\log(1/\alpha^2)\|w^\circ\|_1$: the time to fit grows like $\log(1/\alpha)$, the length of the plateaus we meet in Section 3.5.

3.4 The solution: from kernel to sparsity

Which interpolator is w_α^Φ? The two faces of Φ_α suggest the answer at the two ends of the scale.

Theorem 3.5 (The two limits). *Let X and y be as in Theorem 3.4, and let $w_{\ell_2} = X^+y$ be the minimum-ℓ_2-norm interpolator.*

(i) Kernel regime. *If $\alpha^4 \geq \|w_{\ell_2}\|^2$, then $\|w_\alpha^\Phi - w_{\ell_2}\| \leq \dfrac{\|w_{\ell_2}\|^2}{2^{5/4}\alpha^2}$. In particular $w_\alpha^\Phi \to w_{\ell_2}$ as $\alpha \to \infty$.*

(ii) Rich regime. *Let $m_1 = \min\{\|w\|_1 : Xw = y\}$ and let w_{ℓ_1} attain it. For $0 < \alpha < 1$,*

$$m_1 \leq \|w_\alpha^\Phi\|_1 \leq m_1 + \frac{\mathrm{H}(w_{\ell_1}) + d + 2d\alpha^2}{\log(1/\alpha^2)}.$$

Every limit point of w_α^Φ as $\alpha \to 0$ is a minimum-ℓ_1-norm interpolator, and if the minimum-ℓ_1-norm interpolator is unique, then $w_\alpha^\Phi \to w_{\ell_1}$.

Proof. *Part (i).* The plan is to sandwich $16\alpha^2\Phi_\alpha$ between the squared Euclidean norm and a slightly smaller multiple of it, and to use that the excess squared norm of an interpolator over w_{ℓ_2} is its squared distance to w_{ℓ_2}.

Let $\Psi(w) = 16\alpha^2\Phi_\alpha(w) = 4\alpha^4 \sum_i q(w_i/\alpha^2)$, which has the same minimizer w_α^Φ on $\{Xw = y\}$. By (3.7), with $z = w_i/\alpha^2$,

$$\frac{w_i^2}{\sqrt{1 + w_i^2/(4\alpha^4)}} = \frac{4\alpha^4 z^2}{2\sqrt{4 + z^2}} \le 4\alpha^4 q(z) \le w_i^2,$$

and summing, $\|w\|^2/\sqrt{1 + \|w\|^2/(4\alpha^4)} \le \Psi(w) \le \|w\|^2$, where we used $w_i^2 \le \|w\|^2$ in the denominators. Let $\rho = \|w_\alpha^\Phi\|$ and $m = \|w_{\ell_2}\|$. Minimality gives $\Psi(w_\alpha^\Phi) \le \Psi(w_{\ell_2}) \le m^2$, so

$$\frac{\rho^2}{\sqrt{1 + \rho^2/(4\alpha^4)}} \le m^2. \tag{3.10}$$

If $\rho^2 > 4\alpha^4$, then the left side of (3.10) exceeds $\rho^2/\sqrt{2\rho^2/(4\alpha^4)} = \sqrt{2}\,\alpha^2\rho > 2\sqrt{2}\,\alpha^4 \ge 2\sqrt{2}\,m^2$, using the hypothesis $\alpha^4 \ge m^2$; this contradicts (3.10). So $\rho^2 \le 4\alpha^4$, the left side is at least $\rho^2/\sqrt{2}$, and $\rho^2 \le \sqrt{2}\,m^2$. Feeding this back into (3.10) and using $\sqrt{1+x} \le 1 + x/2$,

$$\rho^2 \le m^2\sqrt{1 + \frac{\sqrt{2}\,m^2}{4\alpha^4}} \le m^2 + \frac{\sqrt{2}\,m^4}{8\alpha^4}.$$

Finally w_{ℓ_2} lies in $\mathrm{range}(X^\top) = (\ker X)^\perp$ and $w_\alpha^\Phi - w_{\ell_2} \in \ker X$, so $\rho^2 = m^2 + \|w_\alpha^\Phi - w_{\ell_2}\|^2$ by Pythagoras, and $\|w_\alpha^\Phi - w_{\ell_2}\|^2 \le \sqrt{2}\,m^4/(8\alpha^4)$. Taking square roots, $\sqrt{\sqrt{2}/8} = 2^{-5/4}$.

Part (ii). The plan is to compare Φ_α at w_α^Φ and at w_{ℓ_1} through the expansion (3.9). The lower bound $m_1 \le \|w_\alpha^\Phi\|_1$ holds because w_α^Φ interpolates. For $0 < \alpha < 1$ we have $\varepsilon_\alpha = \log(1/\alpha^2) > 0$. Minimality of w_α^Φ and (3.9) at both points give

$$\frac{\varepsilon_\alpha}{4}\|w_\alpha^\Phi\|_1 + \frac{1}{4}\mathrm{H}(w_\alpha^\Phi) \le \Phi_\alpha(w_\alpha^\Phi) \le \Phi_\alpha(w_{\ell_1}) < \frac{\varepsilon_\alpha}{4}m_1 + \frac{1}{4}\mathrm{H}(w_{\ell_1}) + \frac{d\alpha^2}{2}.$$

The function $x \mapsto x\log x - x$ is at least -1 on $[0,\infty)$, with its minimum at $x = 1$, so $\mathrm{H}(w_\alpha^\Phi) \ge -d$. Multiplying by $4/\varepsilon_\alpha$ and rearranging gives the upper bound. For the limits, let $\alpha_k \to 0$ be a sequence along which $w_{\alpha_k}^\Phi$ converges, to $\bar w$ say; such sequences exist because the bound keeps $\|w_\alpha^\Phi\|_1$ bounded for $\alpha \le \frac{1}{2}$. Then $X\bar w = y$, and the bound gives $\|\bar w\|_1 \le m_1$, so $\bar w$ is a minimum-ℓ_1-norm interpolator. If that interpolator is unique, then every convergent subsequence has the same limit, and the bounded family w_α^Φ converges to it. ∎

The kernel limit is no surprise once (3.7) is known: a large initialization makes every coordinate move little relative to α^2, where the potential is quadratic. The rate in (i) is not sharp; since $q(z) = \frac{z^2}{4} - \frac{z^4}{192} + O(z^6)$, the potential differs from the quadratic by a relative amount of order α^{-4}, and $\|w_\alpha^\Phi - w_{\ell_2}\|$ is in fact of order α^{-4} (Exercise 3.1). The rich limit is the surprise: no ℓ_1 penalty appears anywhere in the training procedure, yet small initialization produces basis pursuit. The bound in (ii) is slow, through $1/\log(1/\alpha)$, but it concerns norms; the vector itself converges much faster when the ℓ_1 problem is well conditioned (Section 3.6). Ties between ℓ_1 minimizers are broken by the entropy term of (3.9) (Exercise 3.3).

WHY THIS MATTERS FOR GENERALIZATION. Suppose $y = Xw_\star$ for a k-sparse $w_\star$ and X has independent standard Gaussian entries. The minimum-ℓ_2 interpolator is the projection of $w_\star$ on the row space, a uniformly random n-dimensional subspace, so $\mathbb{E}\|w_{\ell_2} - w_\star\|^2 = (1 - n/d)\|w_\star\|^2$, the overparameterized branch of Chapter 1 without noise: the kernel regime learns nothing until n approaches d. Basis pursuit, in contrast, recovers $w_\star$ exactly with high probability as soon as n exceeds the statistical dimension of the descent cone of the ℓ_1 norm at $w_\star$, which is of order $k\log(d/k)$, and fails with high probability below it; the transition is sharp, with width of order $\sqrt{d}$ (Amelunxen et al. 2014).[4] Between the two, the initialization scale interpolates between a sample complexity of order d and one of order $k\log(d/k)$.

[4] For $d = 100$ and $k = 5$ the statistical dimension is 20.4. Exact recovery by ℓ_1 minimization is the subject of compressed sensing, where dual certificates are the standard tool.

3.5 The solution: incremental learning and saddles

Theorem 3.4 says where the flow ends, not how it gets there. When the coordinates decouple, the path can be computed exactly.

Assumption 3.6 (Orthogonal design). The design satisfies $X^\top X = nI_d$ (so $n \geq d$). We write $\bar{w} = X^\top y/n$ for the least-squares solution.

Under Assumption 3.6, $\nabla\widehat{\mathcal{L}}(w) = w - \bar{w}$ and $\widehat{\mathcal{L}}(w) = \frac{1}{2}\|w - \bar{w}\|^2 + c_0$ with $c_0 = \frac{1}{2n}\|y\|^2 - \frac{1}{2}\|\bar{w}\|^2 \geq 0$; for noiseless labels $y = Xw_\star$, $\bar{w} = w_\star$ and $c_0 = 0$. The design is not overparameterized, so there is no implicit bias to speak of; what remains is the dynamics. By Proposition 3.2, each coordinate obeys an autonomous equation,

$$\dot{w}_i = 2u_i\dot{u}_i - 2v_i\dot{v}_i = -4\sqrt{w_i^2 + 4\alpha^4}\,(w_i - \bar{w}_i), \tag{3.11}$$

a gradient flow on $\frac{1}{2}(w_i - \bar{w}_i)^2$ with the state-dependent rate $4\sqrt{w_i^2 + 4\alpha^4}$. Near $w_i = 0$ the rate is $8\alpha^2$, tiny: this is why small initializations wait. Once $|w_i| \gg \alpha^2$ the rate is $4|w_i|$, and the coordinate grows exponentially, like the modes of Chapter 2.

Theorem 3.7 (Incremental learning under an orthogonal design). *Assume Assumption 3.6 and let $s = \bar{w}_i$ for one coordinate i.*

(i) If $s = 0$, then $w_i \equiv 0$. If $s > 0$, let $\rho = \sqrt{s^2 + 4\alpha^4}$, $p_\pm = (s \pm \rho)/2$, and $\kappa_0 = (p_+ - \alpha^2)/(\alpha^2 - p_-)$; then

$$w_i(t) = P(t) - \frac{\alpha^4}{P(t)}, \qquad P(t) = \frac{p_+ + \kappa_0 e^{-4\rho t}p_-}{1 + \kappa_0 e^{-4\rho t}}, \tag{3.12}$$

and $w_i(t)$ increases from 0 to s. If $s < 0$, then $w_i(t)$ is minus the solution for $-s$.

(ii) If $s \neq 0$, then for every $\tau \geq 0$ with $\tau \neq 1/(2|s|)$,

$$\lim_{\alpha\to 0} w_i\big(\tau\log(1/\alpha)\big) = \begin{cases} 0, & \tau < 1/(2|s|), \\ s, & \tau > 1/(2|s|). \end{cases}$$

(iii) Consequently, for every $\tau \geq 0$ different from all the times $1/(2|\bar{w}_j|)$,

$$\lim_{\alpha\to 0} \widehat{\mathcal{L}}\big(w(\tau\log(1/\alpha))\big) = c_0 + \frac{1}{2}\sum_{j:\,2|\bar{w}_j|\tau<1} \bar{w}_j^2.$$

Proof. The plan is to reduce (3.3) for one coordinate to a Riccati equation for $P = u_i^2$ with constant coefficients and to solve it.

Part (i). By (3.3) with $g_i = w_i - s$ and Proposition 3.2, $w_i = P - \alpha^4/P$ and

$$\dot{P} = 2u_i\dot{u}_i = -4P\Big(P - \frac{\alpha^4}{P} - s\Big) = -4\big(P^2 - sP - \alpha^4\big) = -4(P - p_+)(P - p_-),$$

because $p_\pm$ are the roots of $P^2 - sP - \alpha^4$ (their sum is s and their product $(s^2 - \rho^2)/4 = -\alpha^4$). If $s = 0$, then $P \equiv \alpha^2$ solves this equation with the right initial value, so $w_i \equiv 0$. Let $s > 0$. Then $p_- < 0 < \alpha^2 < p_+$, the middle inequality because $P^2 - sP - \alpha^4$ equals $-s\alpha^2 < 0$ at $P = \alpha^2$. The constants $p_\pm$ are solutions, so by uniqueness $P(t)$ stays in (p_-, p_+), where $\dot{P} > 0$. On this interval

$$\frac{\mathrm{d}}{\mathrm{d}t}\log\frac{p_+ - P}{P - p_-} = -\dot{P}\,\frac{p_+ - p_-}{(p_+ - P)(P - p_-)} = -4(p_+ - p_-) = -4\rho,$$

so $(p_+ - P)/(P - p_-) = \kappa_0 e^{-4\rho t}$, and solving for P gives (3.12). Since P increases to p_+ and $x \mapsto x - \alpha^4/x$ increases on $(0,\infty)$, w_i increases from 0 to $p_+ - \alpha^4/p_+ = p_+ + p_- = s$. For $s < 0$, the map $(u_i, v_i, s) \mapsto (v_i, u_i, -s)$ preserves (3.3) for coordinate i and the initial condition, and changes the sign of w_i.

Part (ii). By (i) we may take $s > 0$. Put $t = \tau\log(1/\alpha)$ and $\kappa = \kappa_0 e^{-4\rho t}$. As $\alpha \to 0$, $p_+ \to s$, $p_- = -\alpha^4/p_+ \to 0$ and $\alpha^2\kappa_0 = \alpha^2(p_+ - \alpha^2)/(\alpha^2 - p_-) \to s$, while $\rho - s = O(\alpha^4)$, so $(\rho - s)\log(1/\alpha) \to 0$. Hence

$$\log\kappa = \log(\alpha^2\kappa_0) + 2\log(1/\alpha) - 4\rho\tau\log(1/\alpha) = (2 - 4s\tau)\log(1/\alpha) + \log s + o(1),$$

which tends to $-\infty$ if $\tau > 1/(2s)$ and to $+\infty$ if $\tau < 1/(2s)$. In the first case $P \to p_+ \to s$ by (3.12), so $w_i = P - \alpha^4/P \to s$. In the second case $P = p_+/(1+\kappa) + p_-\kappa/(1+\kappa) \to 0$, and $P \geq \alpha^2$ because P increases from α^2; so $\alpha^4/P \leq \alpha^2 \to 0$ and $w_i \to 0$.

Part (iii). Apply (ii) to every coordinate in $\widehat{\mathcal{L}}(w) = c_0 + \frac{1}{2}\sum_j (w_j - \bar{w}_j)^2$. ∎

In the *accelerated time* $\tau = t/\log(1/\alpha)$ the loss converges, as $\alpha \to 0$, to a staircase: coordinate j is learned at $\tau_j = 1/(2|\bar{w}_j|)$, the largest first, and the loss drops by $\frac{1}{2}\bar{w}_j^2$ at each step. In the original time the transitions are logistic. With $t_{1/2}$ the time at which $w_i = s/2$, (3.12) gives, for $|w_i| \gg \alpha^2$, $w_i/s \approx 1/(1 + e^{-4|s|(t - t_{1/2})})$: a transition of width of order $1/|s|$ after a plateau of length

$$t_{1/2} = \frac{\log(|s|/\alpha^2)}{4|s|} - \frac{\alpha^2}{2s^2} + O(\alpha^4) \tag{3.13}$$

(Exercise 3.2), which is the time at which the exponential growth $\alpha^2 e^{4|s|t}$ of the linearized equation reaches $|s|$. This is the logistic escape of Chapter 2,[5]

[5] There the mode strength enters through a rate $2s$; here the rate is $4|s|$ because the gradient enters each factor through $\dot{u} = -2u \odot g$, so that u_i^2 grows like $e^{4|s|t}$. The constants depend on how the loss is normalized, the logarithm of the initialization does not.

with the same dependence on the strength $|s|$ of the mode and on the logarithm of the initialization; it is also the shape of the search phase of Chapter 7 when the information exponent is 2.

GENERAL DESIGNS: SADDLE TO SADDLE. When $X^\top X$ is not diagonal, coordinates interact through the gradient, and a coordinate that has been learned can be unlearned. The closed form (3.4) still holds, and it suggests how to take the limit $\alpha \to 0$. Put $L_\alpha = \log(1/\alpha)$ and define the rescaled dual variable

$$\zeta(\tau) = -\frac{2G(\tau L_\alpha)}{L_\alpha} = -2\int_0^\tau \nabla\widehat{\mathcal{L}}\big(w(\tau' L_\alpha)\big)\,\mathrm{d}\tau'.$$

By (3.4), $w_i = 2\alpha^2 \sinh(2L_\alpha \zeta_i)$, so $|w_i| = \alpha^{2-2|\zeta_i|}(1+o(1))$ as $\alpha \to 0$ when $\zeta_i \neq 0$ is fixed. A coordinate whose dual variable satisfies $|\zeta_i| < 1$ is exponentially small in L_α, a coordinate of order one has $|\zeta_i| = 1 + O(1/L_\alpha)$, and $|\zeta_i| > 1$ is impossible for a bounded trajectory. The dual variable moves at speed $2\nabla\widehat{\mathcal{L}}(w)$ in accelerated time while the predictor sits still, and a new coordinate wakes up when its $|\zeta_i|$ reaches 1. This leads to the following algorithm.

Definition 3.8 (The saddle-to-saddle path). Start from $\tau_0 = 0$, $w^{(0)} = 0$ and $\zeta(0) = 0$. Given the saddle $w^{(k)}$ and $\zeta(\tau_k)$, let $g^{(k)} = \nabla\widehat{\mathcal{L}}(w^{(k)})$; if $g^{(k)} = 0$, stop. Otherwise let $\zeta(\tau) = \zeta(\tau_k) - 2(\tau - \tau_k)g^{(k)}$ for $\tau \geq \tau_k$, let τ_{k+1} be the first $\tau > \tau_k$ at which some coordinate with $|\zeta_i(\tau_k)| < 1$ reaches $|\zeta_i(\tau)| = 1$, write $\zeta^+ = \zeta(\tau_{k+1})$, let A be the set of coordinates with $|\zeta_i^+| = 1$, and let

$$w^{(k+1)} = \arg\min\big\{\widehat{\mathcal{L}}(w) :\ w_i = 0 \text{ for } i \notin A,\ \ \zeta_i^+ w_i \geq 0 \text{ for } i \in A\big\}.$$

The limit path is $\tilde{w}(\tau) = w^{(k)}$ for $\tau_k \leq \tau < \tau_{k+1}$.

The constraint set is a cone of sign-constrained vectors on the active coordinates, and the saddles $w^{(k)}$ are critical points of the loss restricted to coordinate subspaces, hence saddle points of F when they are not global minima. The procedure resembles the LARS algorithm of Efron et al. (2004),[6] which computes the regularization path of the Lasso: active sets grow, sometimes lose a coordinate, and each step solves a least-squares problem on the active set with sign constraints.

[6] LARS stands for least angle regression. With a small modification it computes the whole Lasso path, one breakpoint at a time.

Theorem 3.9 (Pesme and Flammarion 2023, in our normalization). *Under nondegeneracy assumptions on (X, y) that hold for data in general position (see their paper for the precise conditions), the procedure of Definition 3.8 stops after finitely many steps, at the minimum-ℓ_1-norm interpolator when $y \in \mathrm{range}(X)$, and $w(\tau\log(1/\alpha)) \to \tilde{w}(\tau)$ as $\alpha \to 0$, uniformly on compact sets of accelerated times that avoid the jump times τ_k.*

PROOF SKETCH. We describe the mechanism and omit the uniform control of the transitions, which is the technical heart of the paper of Pesme and Flammarion. *Plateaus.* Between jumps, the active coordinates are of order

one and the inactive ones are of size $\alpha^{2-2|\zeta_i|}$, exponentially small in L_α. On the active coordinates, the flow in the original time t runs at rate of order one, so on the time scale L_α it has converged to a minimizer of $\widehat{\mathcal{L}}$ over vectors supported on the active set with the signs of ζ: the saddle $w^{(k)}$. The gradient there vanishes on the active coordinates with nonzero value and pushes $|\zeta_i|$ inward on the active coordinates at zero, so ζ moves linearly as in Definition 3.8, and the inactive coordinates grow like $\alpha^{2-2|\zeta_i(\tau)|}$. *Jumps.* When an inactive $|\zeta_i|$ reaches 1, coordinate i reaches order one; the flow then leaves the saddle along a heteroclinic orbit in time $O(1)$ in t, which is $O(1/L_\alpha)$ in τ, and settles at the next constrained minimizer. *End.* When the path reaches an interpolator $\tilde{w}$, ζ stops, with $\zeta \in \operatorname{range}(X^\top)$, $\|\zeta\|_\infty \le 1$, and $\zeta_i = \operatorname{sgn}(\tilde{w}_i)$ on the support of $\tilde{w}$: the optimality conditions of basis pursuit. Pesme and Flammarion make the transitions rigorous with an arc-length reparametrization of time, and prove that the number of jumps is finite. ■

Under Assumption 3.6, $g^{(0)} = -\bar{w}$, the coordinate j reaches $|\zeta_j| = 1$ at $\tau = 1/(2|\bar{w}_j|)$, and Definition 3.8 reproduces the staircase of Theorem 3.7, a useful check of the normalization. For general designs the limit path visits saddles that are not on the way to w_{ℓ_1} in a monotone sense: coordinates can enter with one sign and leave, and the loss decreases in steps whose heights depend on the correlations of the design.

3.6 Reading the solution

The theorems make three kinds of prediction: *where* the flow ends, as a function of α; *when* each piece is learned; and *in which order*.

WHERE: THE CROSSOVER IN α. Figure 3.1 shows the squared error of the limit against α on a sparse recovery problem with $n = 40$ samples in dimension $d = 100$, well above the ℓ_1 threshold 20.4 but far below d. For $\alpha \ge 1$ the flow ends at the minimum-ℓ_2 interpolator, with error 0.604, close to the average $1 - n/d = 0.6$. Between $\alpha = 1$ and $\alpha = 0.03$ the error falls by more than an order of magnitude, and below that it falls as a power of α. The flow and the static minimizer agree to within 5×10^{-10} at every scale, which is Theorem 3.4 seen at work. The crossover happens where Φ_α changes face, that is, where the quadratic cost $\|w_{\ell_2}\|^2/(16\alpha^2)$ of the dense interpolator becomes comparable with the ℓ_1 cost $\frac{1}{4}\log(1/\alpha^2)\|w_\star\|_1$ of the sparse one. This gives the estimate $\alpha_c^2 \log(1/\alpha_c^2) \approx \|w_{\ell_2}\|^2/(4\|w_\star\|_1)$, used again in Chapter 10; here it gives $\alpha_c \approx 0.098$, while the error has halved at $\alpha \approx 0.125$, so the estimate is right to within a factor of about 1.3.

WHERE: THE RICH-REGIME EXPONENT. Below the crossover the error is a power law. The exponent comes from the dual description of w_α^Φ.

HEURISTIC DERIVATION. By the optimality condition in the proof of Theorem 3.4, $\nabla\Phi_\alpha(w_\alpha^\Phi) = X^\top \nu$ for some $\nu \in \mathbb{R}^n$, that is, $w_\alpha^\Phi = 2\alpha^2 \sinh(4X^\top \nu)$. Write $4\nu = \varepsilon_\alpha \xi$ with $\varepsilon_\alpha = \log(1/\alpha^2)$. Then $|(w_\alpha^\Phi)_i| \approx \alpha^{2(1-|(X^\top\xi)_i|)}$ whenever $(X^\top\xi)_i \neq 0$. On the support S of $w_\star$ the coordinates are of order one, which forces $(X^\top\xi)_i = \operatorname{sgn}(w_{\star,i}) + O(1/\varepsilon_\alpha)$; off the support $|(X^\top\xi)_i| < 1$, and the coordinate is of size $\alpha^{2(1-|(X^\top\xi)_i|)}$. A vector ξ with $X_S^\top \xi = \operatorname{sgn}(w_{\star,S})$

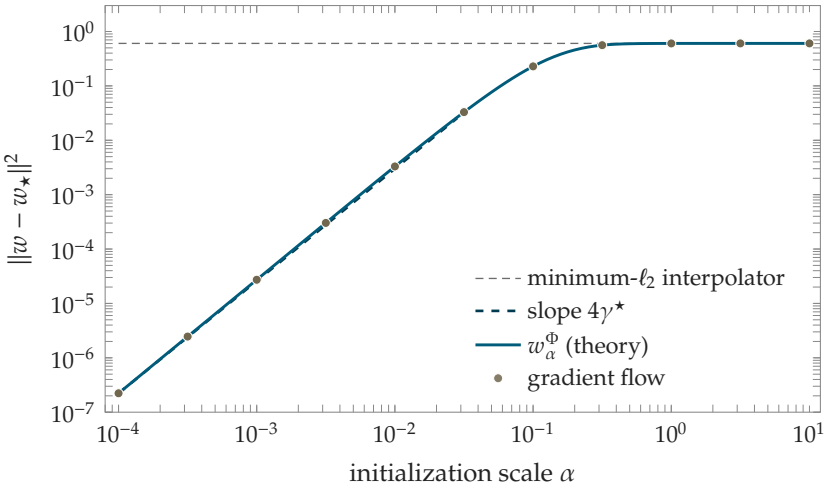


Figure 3.1. From kernel to sparse recovery. Gaussian design with $n = 40$, $d = 100$, a 5-sparse target with entries $\pm 1/\sqrt{5}$ on a random support, noiseless labels (seed 0). Curve: the squared error of $w_\alpha^\Phi = \arg\min\{\Phi_\alpha(w) : Xw = y\}$, computed by Newton's method on the dual problem. Points: the gradient flow (3.3), integrated in the variables $(\log u, \log v)$ until $\widehat{\mathcal{L}} < 10^{-20}$; at all eleven values of α the flow ends within 5×10^{-10} of w_α^Φ. Horizontal dashed line: the error 0.604 of the minimum-ℓ_2 interpolator. Sloped dashed line: the power law $\alpha^{4\gamma^\star}$ of (3.14), with $\gamma^\star = 0.516$ computed from the data.

and $\|X_{S^c}^\top \xi\|_\infty < 1$ is a *dual certificate* for basis pursuit; it exists when ℓ_1 minimization recovers $w_\star$ with some room to spare. Which certificate does w_α^Φ pick? The dual objective of the problem defining w_α^Φ contains the terms $\frac{\alpha^2}{2}\sum_i(\cosh(4(X^\top v)_i) - 1) \approx \frac{1}{4}\sum_i \alpha^{2(1-|(X^\top \xi)_i|)}$, dominated off the support by the largest $|(X^\top \xi)_i|$, so for small α the dual picks the certificate minimizing $\max_{i \notin S}|(X^\top \xi)_i| = 1 - \gamma^\star$. The largest off-support coordinate is then of size $\alpha^{2\gamma^\star}$, the on-support coordinates are perturbed by the same order to keep $Xw = y$, and

$$\|w_\alpha^\Phi - w_\star\|^2 \asymp \alpha^{4\gamma^\star}, \qquad \gamma^\star = 1 - \min\big\{\|X_{S^c}^\top \xi\|_\infty : X_S^\top \xi = \operatorname{sgn}(w_{\star,S})\big\}. \quad \diamond \tag{3.14}$$

The margin $\gamma^\star$ of the best certificate is the value of a linear program,[7] 0.516 for the instance of Figure 3.1; the measured slope between $\alpha = 10^{-3}$ and 10^{-4} is 2.09, against $4\gamma^\star = 2.06$. The convergence of the *vector* is thus polynomial in α, much faster than the logarithmic convergence of the norms in Theorem 3.5, and it degrades as the certificate's margin shrinks, that is, as n approaches the ℓ_1 threshold. Matt and Stöger (2025) (unrefereed) prove matching upper and lower bounds of this kind; for depth two the exponent is given by a null space constant of X.

[7] Minimize m over (ξ, m) subject to $X_S^\top \xi = \operatorname{sgn}(w_{\star,S})$ and $|(X^\top \xi)_i| \le m$ for $i \notin S$; then $\gamma^\star = 1 - m$. The figure script solves it with the HiGHS solver of SciPy.

WHEN: PLATEAUS OF LENGTH $\log(1/\alpha)$. Figure 3.2 shows the orthogonal design. Each coordinate follows a logistic curve, the larger targets first; plain gradient descent with a small step tracks the exact solution. In accelerated time the loss curves for $\alpha = 10^{-2}$, 10^{-4} and 10^{-8} approach the staircase, and

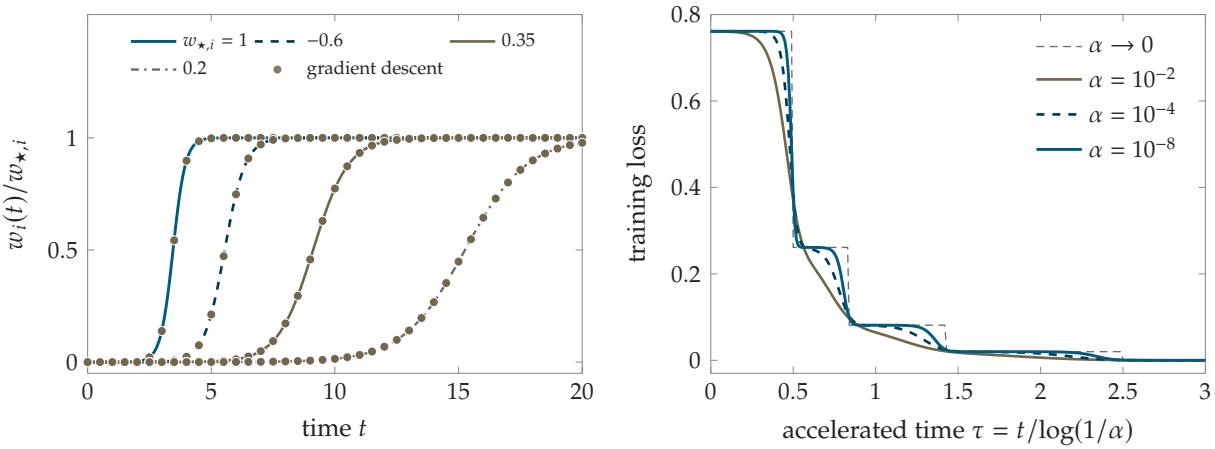


Figure 3.2. Incremental learning under an orthogonal design ($d = 4$, $n = 8$, $X = \sqrt{n}\,Q$ with Q a random 8×4 matrix with orthonormal columns, seed 1; $w_\star = (1, -0.6, 0.35, 0.2)$, noiseless). Left: $\alpha = 10^{-3}$; curves are the exact solution (3.12), points are gradient descent on (u, v) with step 10^{-3}, recorded every 0.5 time units, within 3×10^{-3} of the curves. The half times are 3.45, 5.54, 9.12 and 15.26, equal to $\log(|w_{\star,i}|/\alpha^2)/(4|w_{\star,i}|)$ to within 2×10^{-5}. Right: the training loss from (3.12) in accelerated time for three scales, and its limit, the staircase of Theorem 3.7 with steps at $\tau = 0.5$, 0.83, 1.43 and 2.5.

the steps sharpen as α decreases because the transition width $1/|s|$ is fixed in t while the plateaus grow like $\log(1/\alpha)$. The time to learn a coordinate is inversely proportional to its size: a feature twice as weak takes twice as long, in units of $\log(1/\alpha)$. The first plateau, before anything is learned, lasts $\log(1/\alpha)/(2 \max_i |\bar{w}_i|)$; a small initialization buys sparsity with time.

In which order: saddles. Figure 3.3 shows a general design with three samples in dimension six. The flow visits four saddles before reaching the minimum-ℓ_1 interpolator, and at the third jump a coordinate that had been learned is abandoned: learning is not monotone in the support. The limit predicts the plateau *heights*, losses of least-squares fits on subsets of coordinates, and the jump *times* in units of $\log(1/\alpha)$; the approach to it is slow, through $1/\log(1/\alpha)$, so that at $\alpha = 10^{-4}$ the plateaus are visible but rounded, and the limit is crisp only at scales that no network uses.

Limits of the predictions. All of these are statements about $\alpha \to 0$ or $\alpha \to \infty$ at fixed data, or about $t \to \infty$ at fixed α, and the time to reach the limit grows quickly as α shrinks: in the run of Figure 3.1 the training loss falls below 10^{-20} at $t \approx 10^4$ for $\alpha = 10^{-2}$ but only at $t \approx 3 \times 10^6$ for $\alpha = 10^{-4}$, because the off-support coordinates, which carry the last $\alpha^{4\gamma^\star}$ of the error, move at rates proportional to their own tiny size.

3.7 What the model gets wrong

What was thrown away. *Nonlinearity.* The predictor is linear in x; the network has no features to learn except the choice of coordinates, and "sparse" means sparse in a basis fixed in advance. In a real network the analogue of the coordinate basis is itself learned. *Infinitesimal steps.* The

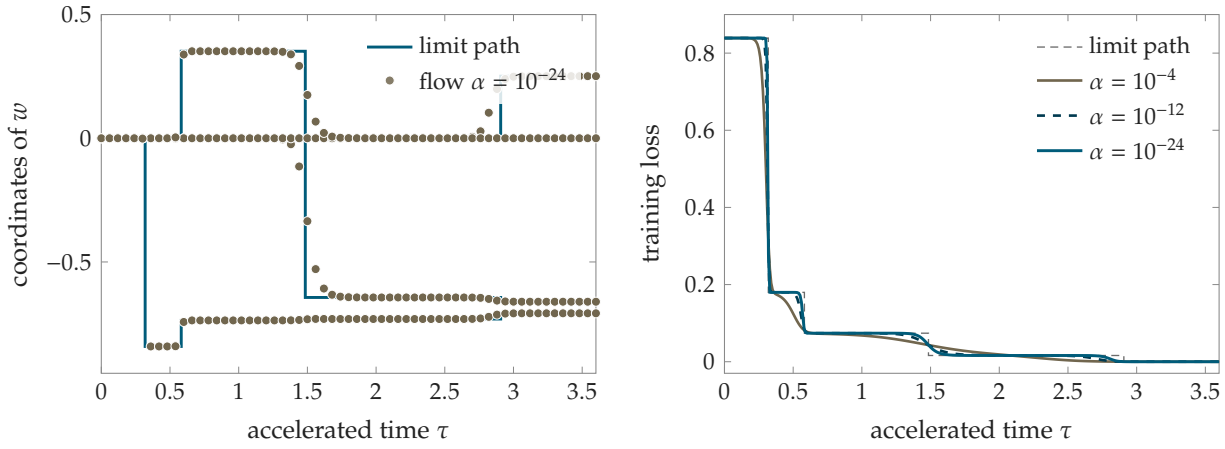


Figure 3.3. Saddle to saddle for a general design: $n = 3$, $d = 6$, Gaussian design and labels (seed 281); four coordinates become active, the other two stay at zero. Curves on the left and the dashed staircase on the right: the limit path of Definition 3.8, with jumps at $\tau = 0.319$, 0.582, 1.485 and 2.907. Points on the left and curves on the right: the gradient flow, integrated in $(\log u, \log v)$ with an adaptive Runge–Kutta method (relative tolerance 10^{-10}). At the third jump one coordinate enters and another, active since the second jump, returns to zero. At distance more than 0.15 from the jumps the flow is within 0.27, 0.12 and 0.024 of the limit path for $\alpha = 10^{-4}$, 10^{-12} and 10^{-24}.

mirror-flow identity (3.5) holds for the flow; for gradient descent with step η the products $u_i v_i$ are no longer conserved but multiplied at each step by $1 - 4\eta^2 g_i^2$, and the potential drifts. *Noise.* There is no label noise and no stochastic gradient; both change the implicit bias. *The loss and the task.* The square loss on a regression problem attains its minimum, on the affine set of interpolators; logistic losses on classification problems do not. *The parametrization.* Diagonal networks are among the few models whose gradient flow is a mirror flow; for others no potential need exist. *The limit.* The theorems describe $t \to \infty$, and then $\alpha \to 0$ or $\alpha \to \infty$; training stops at finite time, at finite α.

WHERE IT BREAKS. *Step size and stochasticity change the bias.* Pesme et al. (2021) model stochastic gradient descent by a stochastic gradient flow and prove that it converges to the minimizer of a potential of the same form with an effective initialization scale smaller than α, so stochasticity pushes toward the rich regime; the slower the loss converges, the stronger the effect. Nacson et al. (2022) show that large step sizes on non-centered data move gradient descent from the kernel to the rich regime, through a dynamical-stability argument and a weighted ℓ_1 bound. Even et al. (2023) prove convergence with macroscopic step sizes and characterize the limits: large steps help stochastic gradient descent recover sparse vectors, but for full-batch gradient descent they can hurt, most strongly near the edge of stability (Chapter 15). On the instance of Figure 3.1 the effect is small but visible (Exercise 3.11). *Label noise* is stronger still: HaoChen et al. (2021) show that stochastic gradient descent with label noise recovers a sparse target from every initialization, where Gaussian parameter noise does not, and Pillaud-Vivien et al. (2022) prove that label-noise dynamics solve a

Lasso problem, a bias independent of α. *Classification.* With the exponential loss on separable data, Moroshko et al. (2020) find that whether the limit is the maximum-ℓ_1 or the maximum-ℓ_2 margin classifier depends on the initialization scale *and* on the training accuracy reached, and that some limits appear only at losses far below 10^{-100}: the asymptotic theory of Chapter 4 and the rich limit of this chapter do not commute. *Beyond diagonal networks.* Li et al. (2022) show that gradient flow under a commuting parametrization is a mirror flow, and that conversely every mirror flow arises this way; their necessary condition for such an equivalence is weaker than commutation, and outside this class no potential need exist. For matrix factorization, where the analogue of the ℓ_1 norm would be the nuclear norm, the conjecture of Gunasekar et al. (2017) fails: Razin and Cohen (2020) construct matrix completion problems in which gradient flow from small initialization drives every norm and quasi-norm to infinity while lowering the rank, and Li et al. (2021) show that from infinitesimal initialization the flow performs greedy low-rank learning, which differs from nuclear norm minimization. For a single ReLU neuron with the square loss, Vardi and Shamir (2021) prove that no function of the parameters describes the implicit bias exactly. *The real phenomenon.* The model cannot explain why fully connected networks do better in the lazy regime (Geiger et al. 2020): it has no notion of a target that a kernel fits well.

Fine structure. *Depth.* Woodworth et al. (2020) solve diagonal networks $w = u^{\odot L} - v^{\odot L}$ of every depth L; the potential changes, the rich limit is still ℓ_1, and it is reached at polynomially rather than exponentially small scales (Exercise 3.8), as Gissin et al. (2020) observed; Chou et al. (2023) give quantitative ℓ_1 bounds for deeper factorizations, and Matt and Stöger (2025) matching rates in α. *Initialization shape.* Azulay et al. (2021) show that the relative scales of the layers at initialization enter the potential, and that the infinitesimal mirror-descent picture breaks down beyond diagonal models. *Dynamics.* Berthier (2023) proves incremental learning, one coordinate at a time, in an underparameterized setting with anticorrelated features; Pesme and Flammarion (2023) handle general designs (Theorem 3.9); Berthier (2025) (unrefereed) relates the whole trajectory to the Lasso regularization path, with time playing the role of an inverse regularization parameter, exactly under a monotonicity assumption on the path. Nishiyama and Imaizumi (2026) derive the high-dimensional dynamics on random designs by dynamical mean-field theory. *Statistics.* With noisy labels, early-stopped gradient descent on the Hadamard parametrization attains the minimax rate of sparse regression (Vaškevičius et al. 2019; Zhao et al. 2022), and the rich limit, which interpolates the noise, is the minimum-ℓ_1 interpolator, whose prediction error is of order $\sigma^2/\log(d/n)$ (Wang et al. 2022): consistent, but slowly. Open are a description of the implicit bias of gradient descent with large steps on general data, and a comparable theory for parametrizations without a potential, matrix factorization first.

3.8 Model card

MODEL CARD	
PHENOMENON	Among the interpolators of an overparameterized model, gradient descent picks one that depends on the initialization scale: kernel-like for large scales, sparse or low rank and learned incrementally for small ones.
MODEL	A linear predictor $w = u \odot u - v \odot v$, square loss, gradient flow from $u = v = \alpha \mathbf{1}$.
SOLUTION	Gradient flow is a mirror flow with the hyperbolic entropy Φ_α; it converges to $\arg\min\{\Phi_\alpha : Xw = y\}$; this tends to the minimum-ℓ_2 interpolator as $\alpha \to \infty$ and to the minimum-ℓ_1 one as $\alpha \to 0$; closed-form incremental learning for orthogonal designs; saddle-to-saddle limit for general ones.
PREDICTS	Error $\asymp \alpha^{4\gamma^\star}$ in the rich regime; crossover at $\alpha_c^2 \log(1/\alpha_c^2) \approx \|w_{\ell_2}\|^2/(4\|w_\star\|_1)$; feature i learned at $t \approx \log(1/\alpha)/(2|w_{\star,i}|)$; $\widehat{\mathcal{L}}(w(t)) \le \Phi_\alpha(w_\star)/(2t)$.
GETS WRONG	Step size, stochastic gradients and label noise change the bias; classification limits depend on the training accuracy; no potential exists for matrix factorization or ReLU networks; fully connected networks can be better in the lazy regime.
SUCCESSORS	Depth-L diagonal networks; initialization shape; stochastic and large-step dynamics; label-noise SGD and the Lasso; greedy low-rank learning in matrix factorization; dynamical mean-field theory.

The chapter sits at the crossing of three threads of the book. *Escape from small initialization*: the plateaus of length $\log(1/\alpha)$/strength are those of deep linear networks in Chapter 2, and reappear in the search phase of Chapter 7, the committee-machine plateaus of Chapter 8 and the hidden progress of Chapter 9. *Implicit bias*: the minimum-Φ_α interpolator is the regression counterpart of the maximum-margin classifier of Chapter 4, and Chapter 10 shows that weight decay acts on a diagonal network as an initialization scale shrinking like $e^{-\lambda t}$, which turns the crossover of Figure 3.1 into grokking. *Lazy versus rich*: the kernel limit is the lazy training of Chapter 6 with a linear kernel, whose risk is the overparameterized branch of Chapter 1.

Exercises

3.1 CONSOLIDATION. (Two expansions of q.) (a) Show that $q(z) = \frac{z^2}{4} - \frac{z^4}{192} + O(z^6)$ as $z \to 0$, and $q(z) = |z|\log|z| - |z| + 2 - 1/|z| + O(|z|^{-3})$ as $|z| \to \infty$. (b) Use (a) and the optimality condition of Theorem 3.4 to show that $\|w_\alpha^\Phi - w_{\ell_2}\| = O(\alpha^{-4})$ as $\alpha \to \infty$, and compare with Theorem 3.5(i).

3.2 CONSOLIDATION. (The half time.) From (3.12), show that $w_i(t_{1/2}) = s/2$ for $t_{1/2} = \frac{1}{4\rho}\log\big(\kappa_0 \frac{P_h - p_-}{p_+ - P_h}\big)$ with $P_h = \frac{1}{4}(s + \sqrt{s^2 + 16\alpha^4})$, and deduce

(3.13). Check it against the numbers in the caption of Figure 3.2.

3.3 CONSOLIDATION. ★ (Ties in the rich limit.) Suppose that the set M of minimum-ℓ_1-norm interpolators has more than one point. (a) Show that two points of M have no coordinate of opposite signs, and deduce that $w \mapsto \sum_i |w_i| \log |w_i|$ is strictly convex on M. (b) Show that w_α^Φ converges, as $\alpha \to 0$, to the unique minimizer of $\sum_i |w_i| \log |w_i|$ on M. *(Hint: for (a), if $w, w' \in M$ then $\frac{1}{2}(w + w') \in M$, which forces $|w_i + w_i'| = |w_i| + |w_i'|$. For (b), compare $\Phi_\alpha(w_\alpha^\Phi)$ with $\Phi_\alpha(w')$ for $w' \in M$ through (3.9), and use $\|w_\alpha^\Phi\|_1 \geq \|w'\|_1$ to show $\mathrm{H}(w_\alpha^\Phi) \leq \mathrm{H}(w') + 2d\alpha^2$.)*

3.4 CONSOLIDATION. (Noisy labels.) Drop the assumption that $y \in \mathrm{range}(X)$. Rerun the proof of Theorem 3.4 with w° a minimizer of $\widehat{\mathcal{L}}$, and show that $\widehat{\mathcal{L}}(w(t)) - \min \widehat{\mathcal{L}} \leq \Phi_\alpha(w^\circ)/(2t)$ and that $w(t)$ converges to the minimizer of Φ_α among the minimizers of $\widehat{\mathcal{L}}$.

3.5 CONSOLIDATION. (Lazy means small motion.) For $\alpha \geq \|w_{\ell_2}\|^{1/2}$, use Proposition 3.2 and Theorem 3.5 to show that $\max_i |u_i(\infty) - \alpha|/\alpha = O(\|w_{\ell_2}\|/\alpha^2)$: each parameter moves by a vanishing fraction of its initial value, the defining property of lazy training in Chapter 6.

3.6 VARIATION. (Initialization shape.) Start from $u(0) = a$ and $v(0) = b$ with positive vectors a, b. Show that $u_i v_i = a_i b_i$ is conserved and that $w(t)$ converges to the minimizer, among interpolators, of

$$\sum_{i=1}^{d} \Big(\frac{a_i b_i}{4}\, q\Big(\frac{w_i}{a_i b_i}\Big) - \frac{1}{4} \log \frac{a_i}{b_i}\, w_i \Big).$$

Which signs does $a_i > b_i$ favor, and what happens to the rich limit when $a_i b_i \to 0$ with a_i/b_i fixed? Compare with Azulay et al. (2021).

3.7 VARIATION. (The product parametrization.) Let $w = u \odot v$, $u(0) = c\mathbf{1}$ and $v(0) = 0$, with $c > 0$. Show that $u_i^2 - v_i^2 = c^2$ is conserved, that $w = -\frac{c^2}{2} \sinh(2G)$, and that the trajectory is that of (3.2) with $\alpha^2 = c^2/4$ run at half speed: $w_{\mathrm{prod}}(2t) = w(t)$.

3.8 VARIATION. ★ (Depth.) Let $w = u^{\odot L} - v^{\odot L}$ with $L \geq 3$ and $u(0) = v(0) = \alpha\mathbf{1}$. (a) Show that $u_i^{2-L} = \alpha^{2-L} + L(L-2)G_i$ and $v_i^{2-L} = \alpha^{2-L} - L(L-2)G_i$ as long as both are positive, and that $w_i = -\alpha^L \psi_L\big(L(L-2)\alpha^{L-2} G_i\big)$ with $\psi_L(z) = (1-z)^{-L/(L-2)} - (1+z)^{-L/(L-2)}$ on $(-1, 1)$. (b) Deduce a mirror flow with $\nabla\Phi(w)_i = \psi_L^{-1}(w_i/\alpha^L)/(L(L-2)\alpha^{L-2})$. (c) Show that $L(L-2)\alpha^{L-2}\,\Phi(w) \to \|w\|_1$ as $\alpha \to 0$, with a relative error that is a positive power of α for w in a compact set avoiding the coordinate hyperplanes, instead of the $1/\log(1/\alpha)$ of depth two.

3.9 VARIATION. (Unequal feature scales.) Replace Assumption 3.6 by $X^\top X/n = \mathrm{diag}(\lambda_1, \ldots, \lambda_d)$ with $\lambda_i > 0$. Solve the flow coordinate by coordinate and show that coordinate i is learned at accelerated time $1/(2\lambda_i |\bar{w}_i|) = n/(2|\langle X_{:i}, y\rangle|)$: the order of learning is the order of the correlations of the features with the labels, not of the coefficients.

3.10 SIMULATION. (The phase transition.) With $d = 100$ and $k = 5$, Gaussian designs and ± 1 targets, compute the limit w_α^Φ of the flow (by Theorem 3.4, through Newton's method on the dual, as in the script of Figure 3.1) for $\alpha \in \{10, 10^{-3}, 10^{-8}\}$, $n = 10, 15, \dots, 60$ and 20 random instances each, and plot the fraction of instances with $\|w - w_\star\| \le 10^{-2}\|w_\star\|$. The reader should observe that $\alpha = 10$ never succeeds; that for $\alpha = 10^{-3}$ the success rate rises from 0 at $n = 25$ to nearly 1 at $n = 50$; and that for $\alpha = 10^{-8}$ the rise comes earlier and is steeper, crossing one half between $n = 20$ and $n = 25$, close to the statistical dimension 20.4 of the ℓ_1 descent cone (Amelunxen et al. 2014). Explain the shift with (3.14): near the threshold the certificate margin $\gamma^\star$ is small.

3.11 SIMULATION. (Step size.) On the instance of Figure 3.1 with $\alpha = 0.1$, run gradient descent on (u, v) with the steps 0.0004, 0.0087, 0.0216 and 0.0346 until $\widehat{\mathcal{L}} < 10^{-11}$. The reader should observe errors $\|w - w_\star\|^2$ of 0.2284, 0.2289, 0.2297 and 0.2306: the limit moves away from w_α^Φ as η grows, toward denser solutions on this centered design, as Even et al. (2023) predict for full-batch gradient descent. Find the step at which gradient descent diverges, and explain the trend using the factor $1 - 4\eta^2 g_i^2$ by which a step multiplies $u_i v_i$.

3.12 SIMULATION. (Saddles.) With the script of Figure 3.3, change the seed, and plot the accelerated times at which the training loss of the flow crosses the midpoints of the predicted steps, for α from 10^{-4} down to 10^{-24}. The reader should observe convergence to the jump times of Definition 3.8, with an error decreasing roughly like $1/\log(1/\alpha)$. Show from Definition 3.8 that if a coordinate has opposite signs at two saddles, then it vanishes at some saddle in between.

3.13 RESEARCH. (Stochastic gradients with large steps.) Pesme et al. (2021) treat a stochastic gradient flow and Even et al. (2023) stochastic gradient descent with large steps, each with conditions on the data. A description of the limit of stochastic gradient descent on diagonal networks for general designs, as an explicit function of the step size, the batch size and α, uniform in the dimension, is open; so is the high-dimensional test error of that limit on Gaussian designs.

3.14 RESEARCH. (A potential for matrices.) For matrix sensing with a symmetric factorization $W = UU^\top$ from $U(0) = \alpha I$, the flow is a mirror flow when the measurement matrices commute (Gunasekar et al. 2017), and Li et al. (2021) describe the limit $\alpha \to 0$ as greedy low-rank learning under assumptions. Characterize the limit of the flow at fixed small α for general measurements, or prove that no function of the end-to-end matrix describes it.

Notes and References

The implicit-bias program grew out of the observations of Neyshabur et al. (2015) and Zhang et al. (2017). Gunasekar et al. (2018) showed that mirror descent in underdetermined regression converges to the interpolator closest to the initialization in the Bregman divergence of its potential, which is the template of Theorem 3.4. The diagonal network and the potential Φ_α are from Woodworth et al. (2020), first posted in 2019, who also treated depth L and the transition between the regimes; the equivalent product parametrization $w = u \odot v$ (Exercise 3.7) had been used in statistics by Hoff (2017), who showed that the Lasso can be computed by alternating ridge regressions in the factors, and Vaškevičius et al. (2019) and Zhao et al. (2022) used Hadamard-product parametrizations with early stopping for sparse regression. The potential is the hyperbolic entropy of Ghai et al. (2020), introduced to interpolate between gradient descent and the exponentiated gradient method of Kivinen and Warmuth (1997), which does well when few inputs are relevant.

The quantitative forms of Theorem 3.5 are ours; Woodworth and coauthors give the limits and discuss how small α must be. Theorem 3.7 is a direct integration in the spirit of Woodworth et al. (2020) and Gissin et al. (2020); the algorithm of Pesme and Flammarion (2023) is restated in our normalization and checked numerically in Figure 3.3.

Translating the literature. Woodworth et al. (2020) write β for our w and $Q_\alpha(\beta) = \alpha^2 \sum_i q(\beta_i/\alpha^2)$ with the same q, so their Q_α is our $4\Phi_\alpha$ and has the same minimizers; their depth is D, our L. Pesme and Flammarion (2023), Pesme et al. (2021) and Even et al. (2023) write β for our w; Pesme and Flammarion use $\beta = u \odot v$ from $u(0) = \sqrt{2}\,\alpha\mathbf{1}$, $v(0) = 0$, which by Exercise 3.7 is our network with $\alpha^2/2$ in place of α^2, run at half speed. Their normalizations of the loss and of accelerated time may differ from ours by further constant factors, which the reader should take from the paper at hand rather than from us. We have restated their saddle-to-saddle algorithm with our dual variable ζ, whose coordinates reach ± 1 at the jumps. Ghai et al. (2020) write $\phi_\beta(x) = x\,\mathrm{arcsinh}(x/\beta) - \sqrt{x^2+\beta^2}$ for the hypentropy; our $\Phi_\alpha(w)$ is $\frac{1}{4}\sum_i(\phi_\beta(w_i) - \phi_\beta(0))$ with $\beta = 2\alpha^2$. Chizat et al. (2019) write α for a multiplier of the model's output, which plays the role of our α^2: the output of a diagonal network scales as α^2 when its parameters scale as α.

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4 Training Past Zero Error

Logistic regression on separable data

THE MODEL — LINEAR CLASSIFIERS, EXPONENTIAL-TAILED LOSSES

IN THIS CHAPTER. Networks trained with the cross-entropy reach zero training error long before training stops, and what happens afterwards matters: the loss keeps falling like $1/t$, the weights grow like $\log t$, and the classifier keeps changing. The hydrogen atom of this regime is a linear classifier trained by gradient flow on the exponential loss over linearly separable data. We prove that the loss is at most $1/(1+\gamma^{\star 2}t)$, that the norm grows like $\log t/\gamma^{\star}$, that the normalized margin is within a factor $1-\log n/\log(\gamma^{\star 2}t/n)$ of the maximal margin $\gamma^{\star}$, and that the direction of the weights converges to the hard-margin support vector machine; the same holds for gradient descent, and a monotone smoothed margin carries the argument to homogeneous networks. The model is right about the direction and wrong about the speed: every rate is logarithmic, and for nonlinear networks the limit is a stationary point of the margin problem that need not be a maximizer.

4.1 The phenomenon: the loss keeps falling after the errors stop

A classifier trained with the cross-entropy never finishes. Once every training point is classified correctly, the loss is still positive, and it can be lowered further by scaling up the logits; no finite set of weights attains its infimum. Large networks reach this regime early: Zhang et al. (2017) showed that standard architectures for image classification drive the training error to zero even when the labels are replaced by random ones, and on real labels zero training error is the ordinary state of affairs long before the end of training. What does gradient descent do in the time that remains?

Soudry et al. (2018) gave a detailed answer for a convolutional network on CIFAR-10 trained by stochastic gradient descent with momentum at a constant learning rate. After the training error reached zero, the training loss decayed roughly like t^{-1} in the number of iterations, the Euclidean norm of the last weight layer grew roughly like $\log t$, the validation loss eventually started to *increase*, and the validation error, in contrast, kept slowly improving.[1] A year earlier Hoffer et al. (2017) had found, across several architectures, that the distance of the weights from their initialization grows logarithmically with the number of updates, and that the generalization gap of large-batch training closes when the number of updates is matched to that of small-batch training: train longer, generalize better.

[1] The opposite movements of validation loss and validation error matter in practice: early stopping on the validation loss would have stopped training while the error was still improving.

OBSERVATION 4.1 (Soudry et al. 2018; Hoffer et al. 2017). For networks trained with the cross-entropy past zero training error, at a constant learning rate: the training loss decays roughly like $1/t$ and the weight norm grows roughly like $\log t$; the classifier keeps changing; the test loss can increase while the test error decreases.

The status of the observation is uneven. The rates of the loss and of the norm have been seen repeatedly and are what the model below predicts. "The test error keeps improving" is weaker: it was reported for particular runs, it depends on the data, and it fails in simple settings. In a linear model trained on a high-dimensional Gaussian mixture we found the test error drifting slightly *upward* after the training error reached zero (Section 4.6). What is robust is that the training does not stop: the predictor keeps moving, slowly, and in a definite direction.

Boosting saw the same phenomenon first. Schapire et al. (1998) observed that AdaBoost, which greedily minimizes an exponential loss, keeps lowering its test error after the training error reaches zero, and explained it by the growth of the training margins.[2] The explanation was disputed for boosting, and margin-based explanations of deep networks are disputed today (Section 4.6). But the mechanism they point to is real, and in the model of this chapter it is exact.

Three questions follow. Where does the predictor go once the training set is fit, given that it never converges? How fast? And does the answer depend on the loss, the step size and the architecture?

[2] For a voting classifier the margin of a training example is the weighted vote for its correct label minus the largest weighted vote for an incorrect label, normalized by the total weight.

4.2 The model: a linear classifier and an exponential tail

The data are n pairs (x_i, y_i) with $x_i \in \mathbb{R}^d$ and labels $y_i \in \{-1, +1\}$. A linear classifier predicts $\operatorname{sgn}\langle w, x\rangle$, and the *margin* of the training point i is $q_i(w) = y_i\langle w, x_i\rangle$; the point is classified correctly when $q_i(w) > 0$. It is convenient to fold the labels into the data, $z_i = y_i x_i$, so that $q_i(w) = \langle w, z_i\rangle$. The training loss is the *exponential loss*

$$\widehat{\mathcal{L}}(w) = \frac{1}{n}\sum_{i=1}^{n} e^{-\langle w, z_i\rangle}, \qquad \nabla\widehat{\mathcal{L}}(w) = -\frac{1}{n}\sum_{i=1}^{n} e^{-\langle w, z_i\rangle} z_i, \tag{4.1}$$

and we train it by gradient flow, $\dot{w}(t) = -\nabla\widehat{\mathcal{L}}(w(t))$, from $w(0) = 0$; gradient descent comes in Section 4.4. The logistic loss $\frac{1}{n}\sum_i \log(1 + e^{-q_i})$ has the same tail, $\log(1 + e^{-q}) = e^{-q}(1 + o(1))$ as $q \to \infty$, and gives the same results with more bookkeeping (Exercise 4.2).[3]

Assumption 4.2 (Separable, bounded data). Every z_i satisfies $\|z_i\| \le 1$, and some $u \in \mathbb{R}^d$ satisfies $\langle u, z_i\rangle > 0$ for every i.

The bound on the norms only fixes units: rescaling the data by c rescales time by c^{-2} and margins by c. Separability is the substance.

[3] With K classes the cross-entropy of logits $f_1, \dots, f_K$ with correct class c is $\log\big(1 + \sum_{k\neq c} e^{-(f_c - f_k)}\big)$, again exponential in the logit gaps $f_c - f_k$; Soudry et al. (2018) treat the multiclass case.

Definition 4.3 (Normalized margin and maximal margin). The *normalized margin* of $w \neq 0$ is $\gamma(w) = \min_i q_i(w)/\|w\|$. The *maximal margin* is $\gamma^\star = \max_{\|u\|=1} \min_i \langle u, z_i \rangle$, and a unit vector attaining it is a *max-margin direction*. The *support vectors* are the points with $\langle u^\star, z_i \rangle = \gamma^\star$.

The normalized margin is invariant under $w \mapsto cw$ for $c > 0$, as the predictions are. The maximum exists because $u \mapsto \min_i \langle u, z_i \rangle$ is continuous on the compact unit sphere, and by Assumption 4.2 it is positive; by the Cauchy–Schwarz inequality $\gamma^\star \leq \min_i \|z_i\| \leq 1$. In the language of support vector machines, $\gamma^\star$ is the largest width of a slab around a hyperplane through the origin that separates the two classes, and $\hat{w} = u^\star/\gamma^\star$ is the solution of the hard-margin problem $\min \|w\|$ subject to $\langle w, z_i \rangle \geq 1$ for all i (Boser et al. 1992; Cortes and Vapnik 1995).

WHAT WAS THROWN AWAY. Everything below the last layer: the features x_i are fixed, and only a linear readout is trained. Stochastic gradients, momentum, adaptive methods, weight decay and normalization layers are gone, and the step size is infinitesimal. Two simplifications are deliberate and right to make first. Separability is the defining property of the regime "past zero error": a network that fits its training set has, in particular, last-layer features on which the training set is separable. And the exponential loss isolates the one property of the cross-entropy that matters here, the exponential tail, without the curvature near $q = 0$ that complicates the beginning of training and is irrelevant at its end.

DICTIONARY

IN THE MODEL	IN A TRAINED NETWORK
linear predictor $\langle w, x \rangle$ on fixed inputs x_i	the last layer of a network, on the features of the layer below
exponential loss e^{-q}	cross-entropy, whose tail is e^{-q} in the logit gap
separable data, $\gamma^\star > 0$	zero training error, reached early by overparameterized networks
$\widehat{\mathcal{L}}(w(t)) \approx 1/(\gamma^{\star 2} t)$	training loss decaying like $1/t$
$\|w(t)\| \approx \log t/\gamma^\star$	weight norms and logit scales growing like $\log t$
direction $w/\|w\|$, normalized margin $\gamma(w)$	the classifier itself, and the quantity it slowly improves
max-margin direction $u^\star$, support vectors	hard-margin SVM on the features; Karush–Kuhn–Tucker points for homogeneous networks
time t	learning rate times number of steps

WHAT ONE HAD TO NOTICE. That a loss without a minimizer still determines a limit, the limit of the *direction*; and that the exponential tail turns the gradient into a soft minimum. Write $w = \beta u$ with $\|u\| = 1$. Then

$$-\log\big(n\widehat{\mathcal{L}}(\beta u)\big) = -\log \sum_i e^{-\beta \langle u, z_i \rangle}$$

is a free energy at inverse temperature β, and as $\beta \to \infty$ it tends to the ground-state energy $\beta \min_i \langle u, z_i \rangle$: the Boltzmann weights $e^{-\beta\langle u,z_i\rangle}/\sum_j e^{-\beta\langle u,z_j\rangle}$, which are the weights of the points in the gradient (4.1), concentrate on the points of smallest margin. Gradient descent on the loss at large norm is gradient ascent on the smallest margin.[4] Training past zero error is a slow quench to zero temperature, and its ground state is the support vector machine.

[4] The function $-\log \sum_i e^{-q_i}$ is the log-sum-exp, or soft minimum, of the margins: it lies between $\min_i q_i - \log n$ and $\min_i q_i$.

4.3 The solution: the loss, the norm, and the margin

We first describe the target, the max-margin direction, by a lemma that collects the convex geometry we need. Write $\Delta_n = \{p \in \mathbb{R}^n : p_i \geq 0,\ \sum_i p_i = 1\}$ for the probability simplex and $C = \{\sum_i p_i z_i : p \in \Delta_n\}$ for the convex hull of the points z_i.

Lemma 4.4 (The max-margin direction). *Under Assumption 4.2:*

(i) The max-margin direction $u^\star$ is unique.

(ii) The point $z^\star$ of C closest to the origin is $z^\star = \gamma^\star u^\star$; equivalently, $\gamma^\star = \min_{p\in\Delta_n} \|\sum_i p_i z_i\|$.

(iii) Write $z^\star = \sum_i p_i^\star z_i$ with $p^\star \in \Delta_n$. Then $p_i^\star > 0$ only for support vectors, and $\hat{w} = u^\star/\gamma^\star = \sum_i \alpha_i z_i$ with $\alpha_i = p_i^\star/\gamma^{\star 2} \geq 0$ and $\sum_i \alpha_i = 1/\gamma^{\star 2}$.

(iv) If a unit vector u satisfies $\min_i \langle u, z_i\rangle \geq \gamma^\star - \delta$ with $\delta \geq 0$, then $\|u - u^\star\|^2 \leq 2\delta/\gamma^\star$.

Proof. *Part (i).* Suppose that unit vectors $u_1 \neq u_2$ both attain $\gamma^\star$. Their midpoint $v = \frac{1}{2}(u_1 + u_2)$ satisfies $\langle v, z_i\rangle \geq \gamma^\star$ for every i, and $\|v\| < 1$ by strict convexity of the Euclidean ball. Then $v \neq 0$, and $v/\|v\|$ has margin at least $\gamma^\star/\|v\| > \gamma^\star$, a contradiction.

Part (ii). The set C is compact and convex, so it has a unique point $z^\star$ closest to the origin, characterized by $\langle z - z^\star, z^\star\rangle \geq 0$ for every $z \in C$. By Assumption 4.2 every $z \in C$ has $\langle u, z\rangle > 0$ for the separating u, so $z^\star \neq 0$. For every unit vector u, the minimum of the linear function $\langle u, \cdot\rangle$ over C is attained at a vertex, so

$$\min_i \langle u, z_i\rangle = \min_{z\in C} \langle u, z\rangle \leq \langle u, z^\star\rangle \leq \|z^\star\|,$$

and hence $\gamma^\star \leq \|z^\star\|$. Conversely, for $u = z^\star/\|z^\star\|$ the characterization of $z^\star$ gives $\langle u, z\rangle \geq \langle u, z^\star\rangle = \|z^\star\|$ for every $z \in C$, in particular for every z_i. So $\gamma^\star = \|z^\star\|$, this u attains it, and by part (i) $u = u^\star$.

Part (iii). By part (ii), $\gamma^\star = \langle u^\star, z^\star\rangle = \sum_i p_i^\star \langle u^\star, z_i\rangle$, a convex combination of numbers that are all at least $\gamma^\star$. So every i with $p_i^\star > 0$ has $\langle u^\star, z_i\rangle = \gamma^\star$. Dividing $z^\star = \gamma^\star u^\star$ by $\gamma^{\star 2}$ gives the formula for $\hat{w}$, and $\sum_i \alpha_i = 1/\gamma^{\star 2}$ because $p^\star \in \Delta_n$.

Part (iv). Because $z^\star \in C$, the first display of part (ii) gives $\langle u, z^\star\rangle \geq \min_i \langle u, z_i\rangle \geq \gamma^\star - \delta$, that is, $\langle u, u^\star\rangle \geq 1 - \delta/\gamma^\star$. Since both vectors have norm one, $\|u - u^\star\|^2 = 2 - 2\langle u, u^\star\rangle \leq 2\delta/\gamma^\star$. ∎

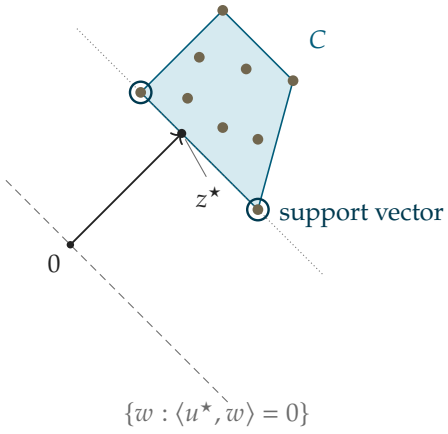


Figure 4.1. The geometry of Lemma 4.4 in the plane. The points are the $z_i = y_i x_i$; their convex hull C is shaded. The point $z^\star$ of C closest to the origin has norm $\gamma^\star$ and direction $u^\star$. The decision boundary (dashed) is orthogonal to $u^\star$, and the support vectors (circled) lie on the dotted line $\langle u^\star, z\rangle = \gamma^\star$; $z^\star$ is a convex combination of them.

Part (iii) is the Karush–Kuhn–Tucker condition of the hard-margin problem: $\hat{w}$ is a nonnegative combination of the support vectors. Part (ii) is the classical duality between the hard-margin support vector machine and the nearest point of a convex hull.[5] Part (iv) turns margins into angles: a direction whose margin is within δ of the best is within $\sqrt{2\delta/\gamma^\star}$ of $u^\star$. Figure 4.1 shows the picture.

[5] The weights $p_i^\star$ are the Lagrange multipliers of the hard-margin problem, normalized to sum to one.

Two inequalities connect the loss to the geometry. The first bounds the gradient in terms of the loss; the second bounds the smallest margin.

Lemma 4.5 (Gradient and soft minimum). *Under Assumption 4.2, every $w \in \mathbb{R}^d$ satisfies*

$$\gamma^\star \widehat{\mathcal{L}}(w) \le \langle -\nabla\widehat{\mathcal{L}}(w), u^\star\rangle \le \|\nabla\widehat{\mathcal{L}}(w)\| \le \widehat{\mathcal{L}}(w), \tag{4.2}$$

$$-\log\bigl(n\widehat{\mathcal{L}}(w)\bigr) \le \min_i q_i(w) \le \gamma^\star\|w\|. \tag{4.3}$$

Proof. For (4.2): by (4.1), $\langle -\nabla\widehat{\mathcal{L}}(w), u^\star\rangle = \frac{1}{n}\sum_i e^{-q_i(w)}\langle z_i, u^\star\rangle$, and every $\langle z_i, u^\star\rangle \ge \gamma^\star$; the middle inequality is the Cauchy–Schwarz inequality; and the triangle inequality with $\|z_i\| \le 1$ gives the last. For (4.3): the sum $n\widehat{\mathcal{L}}(w) = \sum_i e^{-q_i(w)}$ is at least its largest term $e^{-\min_i q_i(w)}$, which gives the left inequality; the right one holds for $w = 0$, and for $w \ne 0$ it is the definition of $\gamma^\star$ applied to $w/\|w\|$. ∎

The lower bound in (4.2) says that the loss has no flat directions while it is positive: the gradient always has a component of size at least $\gamma^\star\widehat{\mathcal{L}}$ along $u^\star$. This is what drives everything.

Theorem 4.6 (Loss, norm, and margin under gradient flow). *Under Assumption 4.2, gradient flow on (4.1) from $w(0) = 0$ exists for all $t \geq 0$, and for every $t \geq 0$:*

(i) $\dfrac{1}{1+t} \leq \widehat{\mathcal{L}}(w(t)) \leq \dfrac{1}{1+\gamma^{\star 2} t}$; *in particular every training point is classified correctly for $t > (n-1)/\gamma^{\star 2}$;*

(ii) $\dfrac{1}{\gamma^\star} \log \dfrac{1+\gamma^{\star 2} t}{n} \leq \|w(t)\| \leq \dfrac{1}{\gamma^{\star 2}} \log\bigl(1+\gamma^{\star 2} t\bigr)$;

(iii) $\min_i q_i(w(t)) \geq \gamma^\star \|w(t)\| - \log n$;

(iv) for $t > (n-1)/\gamma^{\star 2}$ the normalized margin satisfies

$$\gamma\bigl(w(t)\bigr) \geq \gamma^\star \Bigl(1 - \frac{\log n}{\log\bigl((1+\gamma^{\star 2} t)/n\bigr)}\Bigr). \tag{4.4}$$

Proof. The plan is to show that the loss decays at least like $1/t$, because the gradient is at least $\gamma^\star$ times the loss, and then to follow the free energy $N(t) = -\log(n\widehat{\mathcal{L}}(w(t)))$, which by (4.3) sits below the smallest margin and grows at least $\gamma^\star$ times as fast as the path length of w.

Existence. The loss (4.1) is infinitely differentiable, so the flow exists on a maximal interval $[0, T)$. Along it $\frac{\mathrm{d}}{\mathrm{d}t}\widehat{\mathcal{L}}(w(t)) = -\|\nabla\widehat{\mathcal{L}}(w(t))\|^2 \leq 0$, so $\widehat{\mathcal{L}}(w(t)) \leq \widehat{\mathcal{L}}(0) = 1$, and by (4.2) $\|\dot{w}\| \leq \widehat{\mathcal{L}} \leq 1$. Then $\|w(t)\| \leq t$ stays bounded on bounded intervals, the solution cannot blow up, and $T = \infty$. Write $\widehat{\mathcal{L}}(t)$ for $\widehat{\mathcal{L}}(w(t))$.

Part (i). By (4.2), $\gamma^{\star 2}\widehat{\mathcal{L}}^2 \leq \|\nabla\widehat{\mathcal{L}}\|^2 \leq \widehat{\mathcal{L}}^2$, so the derivative $-\|\nabla\widehat{\mathcal{L}}\|^2$ of $\widehat{\mathcal{L}}(t)$ lies between $-\widehat{\mathcal{L}}^2$ and $-\gamma^{\star 2}\widehat{\mathcal{L}}^2$. The loss is positive, so we may divide by $\widehat{\mathcal{L}}^2$: the derivative of $1/\widehat{\mathcal{L}}(t)$ lies between $\gamma^{\star 2}$ and 1. Integrating from 0, where $1/\widehat{\mathcal{L}} = 1$,

$$1 + \gamma^{\star 2} t \leq \frac{1}{\widehat{\mathcal{L}}(t)} \leq 1 + t.$$

If $1 + \gamma^{\star 2} t > n$, then $\widehat{\mathcal{L}}(t) < 1/n$, so every term e^{-q_i}/n is less than $1/n$ and every margin is positive.

Part (iii). Differentiate $N(t) = -\log(n\widehat{\mathcal{L}}(t))$ and use the lower bound of (4.2), $\|\nabla\widehat{\mathcal{L}}\| \geq \gamma^\star \widehat{\mathcal{L}}$, once:

$$\dot{N} = \frac{\|\nabla\widehat{\mathcal{L}}\|^2}{\widehat{\mathcal{L}}} \geq \gamma^\star \|\nabla\widehat{\mathcal{L}}\| = \gamma^\star \|\dot{w}\|.$$

Integrate from 0 to t. The length of the path is at least the distance between its endpoints, so with $N(0) = -\log n$ and $w(0) = 0$,

$$N(t) \geq -\log n + \gamma^\star \int_0^t \|\dot{w}(s)\| \, \mathrm{d}s \geq \gamma^\star \|w(t)\| - \log n. \tag{4.5}$$

The left inequality of (4.3) gives $\min_i q_i(w(t)) \geq N(t)$, which proves part (iii).

Part (ii). For the lower bound, chain the right inequality of (4.3), the left one, and part (i):

$$\gamma^\star \|w(t)\| \geq \min_i q_i(w(t)) \geq -\log\bigl(n\widehat{\mathcal{L}}(t)\bigr) \geq \log \frac{1+\gamma^{\star 2} t}{n}.$$

For the upper bound, $\|\dot{w}\| = \|\nabla \widehat{\mathcal{L}}\| \leq \widehat{\mathcal{L}}$ by (4.2), so by part (i)

$$\|w(t)\| \leq \int_0^t \widehat{\mathcal{L}}(s)\, \mathrm{d}s \leq \int_0^t \frac{\mathrm{d}s}{1+\gamma^{\star 2} s} = \frac{1}{\gamma^{\star 2}} \log\bigl(1+\gamma^{\star 2} t\bigr).$$

Part (iv). For $t > (n-1)/\gamma^{\star 2}$ the lower bound of part (ii) is positive, so $w(t) \neq 0$. Dividing part (iii) by $\|w(t)\|$ gives $\gamma(w(t)) \geq \gamma^\star - \log n / \|w(t)\|$, and the lower bound of part (ii) turns this into (4.4). ∎

The time $(n-1)/\gamma^{\star 2}$ to zero training error is a cousin of Novikoff's bound: the perceptron makes at most $1/\gamma^{\star 2}$ mistakes on data with $\|z_i\| \leq 1$ (Novikoff 1962).[6] The factor n comes from the normalization of the loss: zero training error needs $\widehat{\mathcal{L}} < 1/n$, not $\widehat{\mathcal{L}} < 1$.

[6] Novikoff presented the bound at a symposium held at the Polytechnic Institute of Brooklyn in April 1962; the paper is cited with the year 1962 or 1963.

Corollary 4.7 (Directional convergence). *Under the hypotheses of Theorem 4.6, for $t > (n-1)/\gamma^{\star 2}$,*

$$\left\| \frac{w(t)}{\|w(t)\|} - u^\star \right\|^2 \leq \frac{2 \log n}{\log\bigl((1+\gamma^{\star 2} t)/n\bigr)}.$$

In particular $w(t)/\|w(t)\| \to u^\star$ as $t \to \infty$.

Proof. By (4.4) the unit vector $u = w(t)/\|w(t)\|$ has $\min_i \langle u, z_i \rangle \geq \gamma^\star - \delta$ with $\delta = \gamma^\star \log n / \log((1+\gamma^{\star 2} t)/n)$. Part (iv) of Lemma 4.4 gives $\|u - u^\star\|^2 \leq 2\delta/\gamma^\star$. ∎

The rate of (4.4) is $O(\log n / \log t)$ in the margin, and Ji and Telgarsky (2021) prove a rate $O(\ln n / \ln t)$ of convergence to the max-margin direction for gradient descent with a constant step size and show that it is tight in both n and t. The rate of the corollary, $O((\log t)^{-1/2})$ in the angle, is not: the direction converges at the rate $O(1/\log t)$ for almost every data set, and at the rate $O(\log\log t / \log t)$ for some degenerate ones (Soudry et al. 2018). The heuristic derivation below shows where the faster rate comes from, and what its constant is.

Heuristic derivation of the asymptotic expansion. Guess that $w(t) = \hat{w} \log t + \tilde{w} + o(1)$ for fixed vectors $\hat{w}$ and $\tilde{w}$, and that the error term may be differentiated, so that $\dot{w}(t) = \hat{w}/t + o(1/t)$. Insert this into the flow. The point i contributes

$$\frac{1}{n} e^{-\langle w(t), z_i \rangle} z_i = \frac{1}{n} t^{-\langle \hat{w}, z_i \rangle} e^{-\langle \tilde{w}, z_i \rangle} z_i \bigl(1 + o(1)\bigr)$$

to $\dot{w}(t)$. A point with $\langle \hat{w}, z_i \rangle < 1$ would contribute a term decaying more slowly than $1/t$, which nothing could balance; so $\langle \hat{w}, z_i \rangle \geq 1$ for all i. The

points with $\langle \hat{w}, z_i \rangle > 1$ contribute $o(1/t)$, and matching the terms of order $1/t$ leaves

$$\hat{w} = \sum_{i \in \mathcal{S}} \alpha_i z_i, \qquad \alpha_i = \frac{1}{n} e^{-\langle \tilde{w}, z_i \rangle} > 0, \qquad \mathcal{S} = \{ i : \langle \hat{w}, z_i \rangle = 1 \}.$$

These are the Karush–Kuhn–Tucker conditions of the convex problem $\min \|w\|$ subject to $\langle w, z_i \rangle \geq 1$, which are sufficient for optimality; so $\hat{w} = u^\star / \gamma^\star$ is the hard-margin solution, and $\mathcal{S}$ is the set of support vectors. If the support vectors are linearly independent, then the coefficients α_i are determined by $\hat{w}$, and the equations

$$\langle \tilde{w}, z_i \rangle = -\log(n \alpha_i) \qquad \text{for } i \in \mathcal{S} \tag{4.6}$$

fix the component of $\tilde{w}$ in the span of the support vectors; the orthogonal component is set during the transient. Three predictions follow. The loss is dominated by the support vectors, and by part (iii) of Lemma 4.4,

$$\widehat{\mathcal{L}}(w(t)) \approx \frac{1}{t} \sum_{i \in \mathcal{S}} \alpha_i = \frac{1}{\gamma^{\star 2} t}.$$

The norm is $\|w(t)\| = \log t / \gamma^\star + \langle u^\star, \tilde{w} \rangle + o(1)$. And the normalized margin is

$$\frac{\gamma(w(t))}{\gamma^\star} = \frac{\log t + a}{\log t + b} + o\Big(\frac{1}{\log t}\Big), \qquad a = \min_{i \in \mathcal{S}} \langle \tilde{w}, z_i \rangle, \quad b = \gamma^\star \langle u^\star, \tilde{w} \rangle,$$

a gap of order $(b - a)/\log t$, while the angle to $u^\star$ is about $\gamma^\star \|P^\perp \tilde{w}\| / \log t$, with $P^\perp$ the projection orthogonal to $u^\star$. A point with $r_i = \langle \hat{w}, z_i \rangle > 1$ perturbs the expansion by a term of order $t^{1 - r_i}$, so the expansion is accurate once $t^{r_i - 1}$ is large for every non-support vector. ◇

Soudry et al. (2018) made this rigorous for gradient descent with small constant steps: for almost every separable data set the residual $w(t) - \hat{w} \log t$ stays bounded, and in degenerate cases it may grow like $\log \log t$ in directions orthogonal to the support vectors. The heuristic also explains the sharpness of part (i) of Theorem 4.6: the upper bound $1/(1 + \gamma^{\star 2} t)$ is attained asymptotically, while the lower bound $1/(1 + t)$ is off by the factor $\gamma^{\star 2}$.

4.4 The solution: steps and networks

Two extensions bring the model closer to practice: finite steps, and networks. The first is a short induction. The second needs one new idea, a smoothed margin that increases along the flow.

Proposition 4.8 (Gradient descent). *Under Assumption 4.2, let $w_{k+1} = w_k - \eta \nabla \widehat{\mathcal{L}}(w_k)$ with $w_0 = 0$ and $0 < \eta \leq \frac{1}{2}$, and write $\widehat{\mathcal{L}}_k = \widehat{\mathcal{L}}(w_k)$. For every $k \geq 0$:*

(i) $\widehat{\mathcal{L}}_k \leq 1/(1 + \frac{1}{2} \eta \gamma^{\star 2} k)$;

(ii) $\min_i q_i(w_k) \geq \gamma^\star \|w_k\| - \log n - \eta(\gamma^\star \eta + 2/\gamma^\star)$;
(iii) $\|w_k\| \geq \frac{1}{\gamma^\star} \log\big((1 + \frac{1}{2}\eta\gamma^{\star 2}k)/n\big)$.
In particular $\gamma(w_k) \to \gamma^\star$ *and* $w_k/\|w_k\| \to u^\star$ *as* $k \to \infty$.

The proof rests on a one-step inequality, which replaces the identity $\frac{\mathrm{d}}{\mathrm{d}t}\widehat{\mathcal{L}} = -\|\nabla\widehat{\mathcal{L}}\|^2$ of the flow.

Lemma 4.9 (One step). *Let* $w \in \mathbb{R}^d$, $g = \nabla\widehat{\mathcal{L}}(w)$ *and* $w^+ = w - \eta g$, *and suppose that* $\eta\widehat{\mathcal{L}}(w) \leq \frac{1}{2}$. *Then*

$$\widehat{\mathcal{L}}(w^+) \leq \widehat{\mathcal{L}}(w) - \eta\|g\|^2\big(1 - \eta\widehat{\mathcal{L}}(w)\big).$$

Proof. Put $v = -g = \frac{1}{n}\sum_i e^{-q_i(w)} z_i$ and $b_i = \eta\langle v, z_i\rangle$, so that $q_i(w^+) = q_i(w) + b_i$. By (4.2) and $\|z_i\| \leq 1$, $|b_i| \leq \eta\|v\| \leq \eta\widehat{\mathcal{L}}(w) \leq \frac{1}{2}$. On $[-1, 1]$ the function e^{-b} lies below $1 - b + b^2$: the difference $h(b) = 1 - b + b^2 - e^{-b}$ has $h(0) = h'(0) = 0$ and $h''(b) = 2 - e^{-b}$, which is positive for $b > -\log 2$, so $h \geq 0$ on $[-\log 2, 1]$; on $[-1, -\log 2]$ the function h is concave, with $h(-1) = 3 - e > 0$ and $h(-\log 2) = \log 2 + (\log 2)^2 - 1 > 0$, so $h > 0$ there too. Hence

$$\widehat{\mathcal{L}}(w^+) = \frac{1}{n}\sum_i e^{-q_i(w)} e^{-b_i} \leq \frac{1}{n}\sum_i e^{-q_i(w)}\big(1 - b_i + b_i^2\big).$$

Expand the right side term by term. The constant terms give $\widehat{\mathcal{L}}(w)$. The linear terms give $-\frac{\eta}{n}\sum_i e^{-q_i(w)}\langle v, z_i\rangle = -\eta\|v\|^2$, by the definition of v. In the quadratic terms $b_i^2 \leq \eta^2\|v\|^2$, so they give at most $\eta^2\widehat{\mathcal{L}}(w)\|v\|^2$. ∎

Proof of Proposition 4.8. The plan is to repeat the proof of Theorem 4.6 with sums instead of integrals, using Lemma 4.9 and absorbing the second-order terms into a constant. Write $g_k = \nabla\widehat{\mathcal{L}}(w_k)$.

Part (i). We show by induction that $\widehat{\mathcal{L}}_k$ is nonincreasing, so that $\eta\widehat{\mathcal{L}}_k \leq \eta\widehat{\mathcal{L}}_0 = \eta \leq \frac{1}{2}$ for all k. If $\eta\widehat{\mathcal{L}}_k \leq \frac{1}{2}$, then Lemma 4.9 and (4.2) give

$$\widehat{\mathcal{L}}_{k+1} \leq \widehat{\mathcal{L}}_k - \frac{\eta}{2}\|g_k\|^2 \leq \widehat{\mathcal{L}}_k\big(1 - c\widehat{\mathcal{L}}_k\big), \qquad c = \frac{\eta\gamma^{\star 2}}{2},$$

so $\widehat{\mathcal{L}}_{k+1} \leq \widehat{\mathcal{L}}_k$ and the induction proceeds. Since $0 \leq c\widehat{\mathcal{L}}_k < 1$ and $1/(1 - x) \geq 1 + x$ for $x \in [0, 1)$, inverting gives $1/\widehat{\mathcal{L}}_{k+1} \geq 1/\widehat{\mathcal{L}}_k + c$, and summing from $\widehat{\mathcal{L}}_0 = 1$ gives $1/\widehat{\mathcal{L}}_k \geq 1 + ck$.

Part (ii). Let $N_k = -\log(n\widehat{\mathcal{L}}_k)$ and $a_k = \eta\|g_k\|^2(1 - \eta\widehat{\mathcal{L}}_k)/\widehat{\mathcal{L}}_k$. Lemma 4.9 says $\widehat{\mathcal{L}}_{k+1} \leq \widehat{\mathcal{L}}_k(1 - a_k)$, and $a_k \leq \eta\widehat{\mathcal{L}}_k \leq \frac{1}{2}$ by (4.2); since $-\log(1 - x) \geq x$ for $x \in [0, 1)$, we get $N_{k+1} - N_k \geq a_k$. Now bound a_k below, using $\|g_k\| \geq \gamma^\star\widehat{\mathcal{L}}_k$ from (4.2) and then, with $\Delta w_k = w_{k+1} - w_k$, the bound $\|\Delta w_k\| = \eta\|g_k\| \leq \eta\widehat{\mathcal{L}}_k$:

$$a_k \geq \gamma^\star\eta\|g_k\|\big(1 - \eta\widehat{\mathcal{L}}_k\big) = \gamma^\star\|\Delta w_k\| - \gamma^\star\eta\widehat{\mathcal{L}}_k\|\Delta w_k\| \geq \gamma^\star\|\Delta w_k\| - \gamma^\star\eta^2\widehat{\mathcal{L}}_k^2.$$

Summing over $k < K$, using the triangle inequality $\sum_{k<K}\|\Delta w_k\| \geq \|w_K\|$ and $N_0 = -\log n$,

$$N_K \geq \gamma^\star \|w_K\| - \log n - \gamma^\star \eta^2 \sum_{k\geq 0} \widehat{\mathcal{L}}_k^2.$$

By part (i) and the comparison of a decreasing sum with an integral, $\sum_{k\geq 0}\widehat{\mathcal{L}}_k^2 \leq \sum_{k\geq 0}(1+ck)^{-2} \leq 1+\int_0^\infty (1+cs)^{-2}\,\mathrm{d}s = 1+1/c$, and $\gamma^\star\eta^2(1+1/c) = \gamma^\star\eta^2 + 2\eta/\gamma^\star$. Finally $\min_i q_i(w_K) \geq N_K$ by (4.3).

Part (iii). As in the proof of Theorem 4.6, (4.3) and part (i) give $\gamma^\star\|w_k\| \geq \min_i q_i(w_k) \geq N_k \geq \log((1+ck)/n)$.

The limits. Parts (ii) and (iii) give $\gamma(w_k) \geq \gamma^\star - (\log n + \gamma^\star\eta^2 + 2\eta/\gamma^\star)/\|w_k\|$ with $\|w_k\| \to \infty$, and part (iv) of Lemma 4.4 turns this into directional convergence. ∎

The constant $2\eta/\gamma^\star$ is the price of discreteness, and it is harmless: it enters only through $1/\|w_k\|$. The restriction $\eta \leq \frac{1}{2}$ comes from $\widehat{\mathcal{L}}_0 = 1$ and the descent lemma; larger steps work too, but by a different mechanism (Section 4.6).

HOMOGENEOUS NETWORKS. Let $f(x;\theta)$ be a network with parameters $\theta \in \mathbb{R}^p$, and suppose that it is *positively homogeneous* of degree $L \geq 1$ in θ: $f(x;c\theta) = c^L f(x;\theta)$ for every $c > 0$. A ReLU network without biases and with L layers is homogeneous of degree L; a linear classifier has $L = 1$.[7] Training uses the margins $q_i(\theta) = y_i f(x_i;\theta)$ in the loss $\widehat{\mathcal{L}}(\theta) = \frac{1}{n}\sum_i e^{-q_i(\theta)}$, and the normalized margin is now $\min_i q_i(\theta)/\|\theta\|^L$. The free energy of the proof of Theorem 4.6, divided by the matching power of the norm, becomes the *smoothed margin*

$$\tilde{\gamma}(\theta) = \frac{-\log\bigl(n\widehat{\mathcal{L}}(\theta)\bigr)}{\|\theta\|^L},$$

which by (4.3), whose proof uses only $n\widehat{\mathcal{L}} = \sum_i e^{-q_i}$, satisfies $\tilde{\gamma}(\theta) \leq \min_i q_i(\theta)/\|\theta\|^L \leq \tilde{\gamma}(\theta) + \log n/\|\theta\|^L$.

[7] A bias in the first layer can be absorbed by appending a constant coordinate to x, which keeps the network homogeneous; biases in later layers cannot.

Proposition 4.10 (The smoothed margin increases; Lyu and Li 2020). *Let $\theta \mapsto f(x_i;\theta)$ be continuously differentiable and positively homogeneous of degree $L \geq 1$ for every i, and let $\theta(t)$ solve $\dot{\theta} = -\nabla\widehat{\mathcal{L}}(\theta)$ on an interval $[t_0, T)$. If $\widehat{\mathcal{L}}(\theta(t_0)) < 1/n$, then on $[t_0, T)$ the loss stays below $1/n$, the norm $\|\theta(t)\|$ is strictly increasing, and $\tilde{\gamma}(\theta(t))$ is nondecreasing.*

PROOF. The plan is to compute the growth rates of the numerator $\nu = -\log(n\widehat{\mathcal{L}})$ and of the denominator ρ^L, with $\rho = \|\theta\|$, and to compare them through Euler's identity for homogeneous functions. Write $\widehat{\mathcal{L}}$, q_i, ν, ρ for their values at $\theta(t)$.

Preliminaries. Along the flow $\widehat{\mathcal{L}}$ is nonincreasing, so $\widehat{\mathcal{L}} < 1/n$ and $\nu > 0$ on $[t_0, T)$. Because $e^{-q_i} \leq n\widehat{\mathcal{L}}$ for each i, every margin satisfies $q_i \geq \nu > 0$. The parameter θ is never 0, because $f(x;0) = 0$ by homogeneity and $\widehat{\mathcal{L}}(0) = 1$. Differentiating $c \mapsto f(x;c\theta) = c^L f(x;\theta)$ at $c = 1$ gives Euler's identity $\langle\theta, \nabla_\theta q_i(\theta)\rangle = L q_i(\theta)$.

The norm. By Euler's identity and $q_i \geq \nu$,

$$\rho\dot{\rho} = \langle \theta, \dot{\theta} \rangle = \frac{1}{n} \sum_i e^{-q_i} \langle \theta, \nabla q_i \rangle = \frac{L}{n} \sum_i e^{-q_i} q_i \geq L\nu\widehat{\mathcal{L}} > 0. \tag{4.7}$$

The numerator. The flow gives $\dot{\nu} = -\dot{\widehat{\mathcal{L}}}/\widehat{\mathcal{L}} = \|\nabla\widehat{\mathcal{L}}\|^2/\widehat{\mathcal{L}}$. The component of $\nabla\widehat{\mathcal{L}}$ along θ is $\langle \nabla\widehat{\mathcal{L}}, \theta\rangle/\rho = -\dot{\rho}$, so $\|\nabla\widehat{\mathcal{L}}\| \geq \dot{\rho}$ and $\dot{\nu} \geq \dot{\rho}^2/\widehat{\mathcal{L}}$.

The ratio. Both ν and ρ are positive, so $\log\tilde{\gamma} = \log\nu - L\log\rho$ is differentiable, and

$$\frac{\mathrm{d}}{\mathrm{d}t}\log\tilde{\gamma} = \frac{\dot{\nu}}{\nu} - \frac{L\dot{\rho}}{\rho} \geq \frac{\dot{\rho}^2}{\widehat{\mathcal{L}}\nu} - \frac{L\dot{\rho}}{\rho} = \frac{\dot{\rho}}{\widehat{\mathcal{L}}\nu\rho}\left(\rho\dot{\rho} - L\nu\widehat{\mathcal{L}}\right) \geq 0,$$

where the last step uses $\dot{\rho} > 0$ and (4.7) twice. ∎

For $L = 1$ the proposition is a second proof of the margin part of Theorem 4.6, without its constants (Exercise 4.3). For networks it is the key step of a theory that we state without its full proof.

Theorem 4.11 (Lyu and Li 2020; Ji and Telgarsky 2020; stated with a sketch). *Let $f(x;\theta)$ be locally Lipschitz, admit a chain rule for Clarke subgradients, and be positively homogeneous of degree $L \geq 1$ in θ, such as a ReLU network without biases, possibly with max-pooling and convolutions; let $\theta(t)$ be a gradient flow of the exponential or logistic loss, in the sense of Clarke subgradients; and suppose that $\widehat{\mathcal{L}}(\theta(t_0)) < 1/n$ at some time t_0. Then:*

(i) $\widehat{\mathcal{L}}(\theta(t)) = \Theta\big(1/(t(\log t)^{2-2/L})\big)$ *and* $\|\theta(t)\| = \Theta\big((\log t)^{1/L}\big)$ *as* $t \to \infty$;

(ii) every limit point $\bar{\theta}$ of $\theta(t)/\|\theta(t)\|$ is, after rescaling, a Karush–Kuhn–Tucker point of

$$\min \tfrac{1}{2}\|\theta\|^2 \qquad \text{subject to} \quad y_i f(x_i;\theta) \geq 1 \quad \text{for all } i, \tag{4.8}$$

and the normalized margin $\min_i q_i(\theta(t))/\|\theta(t)\|^L$ converges, its limit being the normalized margin of every such limit point;

(iii) if the network is definable in an o-minimal structure, which covers ReLU networks with the exponential and logistic losses, then $\theta(t)/\|\theta(t)\|$ converges (Ji and Telgarsky 2020).

Proof sketch. Part (i) follows from Proposition 4.10 by comparing ν and ρ in the differential inequalities of its proof, as in Theorem 4.6; the logarithmic factor comes from $\nu \asymp \rho^L$. For part (ii), the Karush–Kuhn–Tucker conditions of (4.8) ask that θ be a nonnegative combination $\sum_i \lambda_i \nabla q_i(\theta)$ with $\lambda_i = 0$ whenever $q_i(\theta) > 1$. The negative gradient $\frac{1}{n}\sum_i e^{-q_i}\nabla q_i$ is such a combination with weights that, after normalization, vanish on the points of larger margin; what remains is to show that θ and $-\nabla\widehat{\mathcal{L}}$ become parallel along a sequence of times, which Lyu and Li (2020) deduce from the

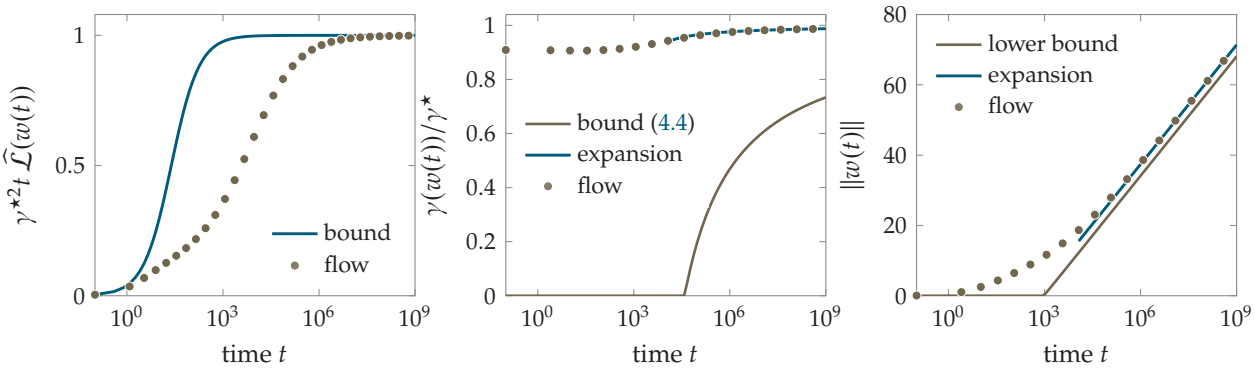


Figure 4.2. Gradient flow of the exponential loss on $n = 40$ points in the plane: the vectors $z_i = y_i x_i$ are drawn from $\mathcal{N}(\mu, 0.2^2 I_2)$ with $\mu = (0.55, 0.25)$ and rescaled to largest norm 1 (seed 31), so that $\gamma^\star = 0.2034$, with two support vectors and smallest non-support ratio $r_i = 1.49$. The flow is integrated with a stiff solver in the clock $\log(1 + t)$. Left: $\gamma^{\star 2} t\, \widehat{\mathcal{L}}$ against the upper bound $\gamma^{\star 2} t/(1 + \gamma^{\star 2} t)$ of Theorem 4.6; it reaches 0.999 at $t = 10^9$. Middle: the normalized margin, the bound (4.4) and the heuristic expansion $\hat{w} \log t + \tilde{w}$ with $\tilde{w}$ from (4.6), drawn for $t \geq 10^4$. Right: the norm, the lower bound of part (ii) and the expansion $\log t/\gamma^\star + \langle u^\star, \tilde{w} \rangle$.

integrability of $\frac{\mathrm{d}}{\mathrm{d}t} \log \tilde{\gamma}$ and a quantitative notion of approximate Karush–Kuhn–Tucker points. Part (iii) needs a Kurdyka–Łojasiewicz inequality for definable functions, unbounded and nonsmooth, which bounds the length of the curve $\theta/\|\theta\|$ in a logarithmic clock. Ji and Telgarsky (2020) also show that, when the network has locally Lipschitz gradients, $-\nabla\widehat{\mathcal{L}}(\theta(t))$ and $\theta(t)$ become aligned. The full proofs are in the two papers. ■

For a linear classifier (4.8) is the hard-margin problem, which is convex, so its Karush–Kuhn–Tucker points are its solutions and the theorem returns Corollary 4.7. For networks it is not convex, and part (ii) gives a stationary point, not a maximizer (Section 4.6).

4.5 Reading the solution

Theorem 4.6 and the heuristic expansion make four quantitative predictions. (i) The loss decays like $1/(\gamma^{\star 2} t)$, with the constant exactly $1/\gamma^{\star 2}$ at late times. (ii) The norm grows like $\log t/\gamma^\star$, with an offset $\langle u^\star, \tilde{w} \rangle$ fixed by the support vectors. (iii) The normalized margin approaches $\gamma^\star$ with a gap of order $1/\log t$: halving the gap takes squaring the training time.[8] (iv) The time scale is $1/\gamma^{\star 2}$: the data are fit by time $(n - 1)/\gamma^{\star 2}$, and the asymptotic regime begins once the non-support vectors are suppressed, when $t^{r_i - 1}$ is large for every $r_i = \langle \hat{w}, z_i \rangle > 1$.

Figure 4.2 tests these predictions on a data set small enough to see. In the left panel, $\gamma^{\star 2} t\, \widehat{\mathcal{L}}$ climbs to the bound and meets it: the upper bound of part (i) is the late-time law, not just a bound. The climb is slow because early on all forty points carry weight in the loss, and $\widehat{\mathcal{L}}$ exceeds the single-support-vector value. In the middle panel the normalized margin starts at $0.91\gamma^\star$, the margin of the direction $\sum_i z_i$ that the flow takes first, rises through the transient, and then creeps: from $0.975\gamma^\star$ at $t = 10^6$ to $0.988\gamma^\star$ at $t = 10^9$.

[8] A gap $c/\log t$ at time t becomes $c/(2 \log t)$ at time t^2.

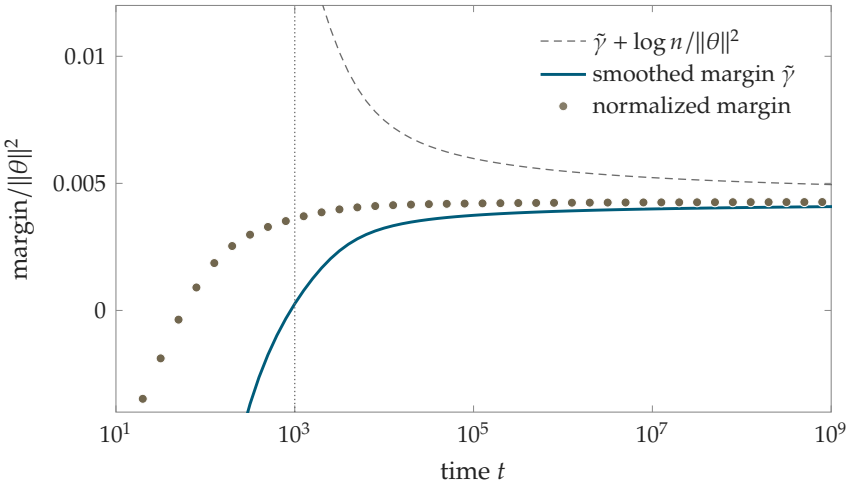


Figure 4.3. A ReLU network $f(x) = \sum_{r=1}^{50} a_r \,\mathrm{ReLU}(\langle w_r, x\rangle)$ without biases, homogeneous of degree $L = 2$, trained by gradient flow of the exponential loss on $n = 30$ points of the unit circle labeled by $\mathrm{sgn}(x_1 x_2)$, which no linear classifier separates; weights initialized $\mathcal{N}(0, 0.3^2)$, seed 0. The flow is integrated by the Euler method in the clock $\mathrm{d}\tau = \widehat{\mathcal{L}}\,\mathrm{d}t$ with $\mathrm{d}\tau = 10^{-3}$. The loss drops below $1/n$ at $t \approx 10^3$ (dotted line); from then on the smoothed margin never decreases, as Proposition 4.10 asserts, and the normalized margin lies between it and the dashed upper bound.

The expansion, which has no free parameter, matches it to within 0.003 from $t = 10^5$ on and to within 5×10^{-4} from $t = 10^7$ on. The rigorous bound (4.4) is valid but loose, at $0.73\gamma^\star$ at $t = 10^9$: it pays $\log n$ for not knowing which points are support vectors. In the right panel the norm is a straight line in $\log t$, with slope $1/\gamma^\star = 4.92$ per unit of $\log t$, as the lower bound of part (ii) and the expansion say. The angle to $u^\star$ is still 0.020 at $t = 10^9$; the expansion predicts $\|P^\perp \tilde w\|/\|w(t)\| = 1.452/71.4 = 0.020$.

Figure 4.3 shows the network version. Before the loss reaches $1/n$ the smoothed margin is negative and the normalized margin wanders; after, the smoothed margin increases monotonically, from 2.6×10^{-4} at $t = 10^3$ to 4.1×10^{-3} at $t = 10^9$, and the normalized margin settles near 4.27×10^{-3}, squeezed between the two curves as $\log n/\|\theta\|^2$ shrinks. The squared norm grows by about 600 per decade of t, linear in $\log t$, and $\widehat{\mathcal{L}}\, t \log t$ stays between 75 and 100 over the last three decades, as part (i) of Theorem 4.11 predicts for $L = 2$ (Exercise 4.11).

THE TEST LOSS RISES. Let (x, y) be a test point that the max-margin classifier gets wrong, $\langle \hat w, yx\rangle < 0$. Its exponential loss grows like $t^{|\langle \hat w, yx\rangle|}$ and its logistic loss like $|\langle \hat w, yx\rangle| \log t$; so the average test loss eventually increases, while the test error converges to that of the max-margin classifier. This is the decoupling of Observation 4.1, and Soudry et al. (2018) derived it in this form. The model says that the test loss is the wrong thing to monitor when training past zero error; it does not say that the test error improves.

WEIGHTS IN THE LOSS DO NOT MATTER IN THE LIMIT. If the loss weights the points, $\widehat{\mathcal{L}}(w) = \sum_i c_i e^{-\langle w, z_i\rangle}$ with $c_i > 0$ and $\sum_i c_i = 1$, then Theorem 4.6

holds with $\log n$ replaced by $\log(1/\min_i c_i)$ (Exercise 4.7): the limit direction is $u^\star$ whatever the weights. Importance weighting for class imbalance or distribution shift therefore has an effect that fades as training continues past zero error. Byrd and Lipton (2019) observed exactly this in deep networks: the effect of importance weights diminishes over training once the network separates the training data, and ℓ_2 regularization restores part of it. Here the model's prediction holds for networks.

LIMITS OF THE MODEL. Every statement is about $t \to \infty$ at fixed data. The parameter that sets the time scale is $\gamma^\star$, which can only decrease as points are added, and on hard data sets $1/\gamma^{\star 2}$ is large. The logarithm makes the asymptotic regime expensive even when $\gamma^\star$ is not small. In Figure 4.2 the transient lasts until about $t = 10^5$, some four thousand times the natural unit $1/\gamma^{\star 2} \approx 24$; with a non-support ratio $r_i = 1.15$ instead of 1.49, we found the expansion still 0.6% off at $t = 10^9$.

4.6 What the model gets wrong

WHAT WAS THROWN AWAY. *Fixed features.* The model trains a linear readout on frozen inputs; in a network the features move, and the geometry that defines the margin changes during training. Theorem 4.11 restores feature learning for homogeneous networks, but only at the level of stationary points, and biases, residual connections and normalization layers break homogeneity. *Separability.* Real training sets contain mislabeled and ambiguous points, which a network may or may not fit; for a linear model on nonseparable data the max-margin story applies only to a separable part (Exercise 4.8). *Infinite time.* The theorems are statements about $t \to \infty$, and the rates are logarithmic. *No regularization.* With weight decay λ the norm stops growing at about $\log(1/\lambda)/\gamma^\star$, and the limit becomes a regularized solution, which approaches the max-margin direction only as $\lambda \to 0$ (Rosset et al. 2003; Wei et al. 2019). *The optimizer.* Gradient flow, or gradient descent with $\eta \le \frac{1}{2}$; no stochasticity, momentum, adaptive preconditioning, or learning-rate schedules.

WHERE IT BREAKS. *The limit is never reached.* A gap of order $1/\log t$ means that the margin of a trained network reflects its training time and initialization as much as its limit: in Figure 4.2 the direction is still 0.02 away from $u^\star$ after 10^9 time units, on forty points in the plane. Statements about "the" implicit bias of a network trained for 10^5 steps should be read as statements about a point on a slow curve. *Stationary is not maximal.* Vardi et al. (2022) show that the Karush–Kuhn–Tucker points of Theorem 4.11 are in many cases not even local maximizers of the margin, for example for depth-two ReLU networks and depth-two diagonal linear networks, while for deep linear fully connected networks and several other settings they identify conditions that guarantee local or global optimality. The physicist's picture of a quench to the ground state fails: the network can freeze in a metastable state. *Margins do not predict generalization in deep networks.* Margin-based bounds such as the spectrally normalized bound of Bartlett et al. (2017) are far from tight; Nagarajan and Kolter (2019) found that such norm-based

bounds can grow with the training set size while the test error falls, and that uniform convergence itself can be vacuous for gradient descent; and in a large-scale comparison over thousands of networks trained on CIFAR-10 and SVHN, Jiang et al. (2020) found sharpness-based measures among the most promising predictors of generalization, while several norm-based measures, spectral ones included, correlated poorly or even negatively with it. The model explains the direction of the weights, not the quality of the classifier. *The test error need not improve.* In a linear model on a high-dimensional Gaussian mixture ($x_i = y_i\mu + g_i$ with $\mu = 0.25\,\mathbf{1}_d$, $g_i \sim \mathcal{N}(0, I_d)$, $d = 200$, $n = 100$) we found, for each of four seeds, the exact test error of the flow rising from about 0.05% at zero training error to about 0.13% at $t = 4 \times 10^6$: early-stopped gradient descent is close to the mean-difference classifier, which is near-optimal for isotropic noise, and the max-margin classifier is worse. When $d \gg n$ every training point becomes a support vector and the max-margin classifier coincides with the minimum-norm interpolator of the labels (Muthukumar et al. 2021), which brings the double descent of Chapter 1 to classification. *Large steps.* Proposition 4.8 needs $\eta \le \frac{1}{2}$ to make the loss decrease monotonically. Wu et al. (2024) show that gradient descent on logistic regression with separable data and a much larger constant step size η first goes through a phase in which the loss oscillates, leaves it within $O(\eta)$ steps, and then decreases at the rate $\tilde{O}(1/(\eta t))$; with η of the order of the step budget T, the loss after T steps is $\tilde{O}(1/T^2)$, much faster than the small-step analysis allows. The monotone picture of this chapter is the wrong picture of the beginning of training at large learning rates, the regime of Chapter 15. *Other optimizers, other norms.* The Euclidean norm is a property of gradient descent. Steepest descent with respect to another norm converges to the max-margin direction for that norm (Gunasekar et al. 2018a), and Zhang et al. (2024) prove that Adam, in linear logistic regression with diminishing learning rates, converges in direction to the maximum-ℓ_∞-margin classifier. Networks trained with Adam, which is most of them, are not described by the ℓ_2 geometry of this chapter. *Label noise.* With noisy labels, fitting the training set means fitting the noise. Whether the max-margin classifier still generalizes depends on the dimension: Chatterji and Long (2021) prove that, with sufficient overparameterization, the max-margin classifier trained on noisy data achieves nearly optimal population risk, and Frei et al. (2022) prove the analogue for two-layer networks trained by gradient descent on the logistic loss. The model, which has no noise and no population, is silent on this benign overfitting.

FINE STRUCTURE. *Architecture changes the norm.* For deep linear fully connected networks the end-to-end predictor still converges to the ℓ_2 max-margin direction, and the layers align (Ji and Telgarsky 2019a); for linear convolutional networks of depth L the limit is a stationary point of an $\ell_{2/L}$ bridge penalty on the Fourier coefficients of the predictor (Gunasekar et al. 2018b); for diagonal linear networks it is an ℓ_1-type margin, as the potential of Chapter 3 suggests (Exercise 4.9); and for infinitely wide two-layer networks Chizat and Bach (2020) characterize the limit as a max-margin classifier in a non-Hilbertian function space, which adapts to

low-dimensional structure. *Nonseparable data.* Ji and Telgarsky (2019b) show that on general data gradient descent on logistic regression converges in direction to the max-margin predictor of a maximal separable subset, at a rate of about $\log\log t/\log t$, and recovers the offset on the nonseparable part at a rate of about $(\log t)^2/\sqrt{t}$. *Faster rates.* The $1/\log t$ is a property of constant steps on a loss whose gradient vanishes. Gradient descent with aggressive step sizes that compensate for the vanishing gradient converges to the max-margin direction at the rate $O(\log t/\sqrt{t})$ for the exponential loss (Nacson et al. 2019), and Ji and Telgarsky (2021) reach $O(1/t)$ for the margin with an aggressive schedule, by reading gradient descent on the primal as mirror descent on the dual distribution $p \in \Delta_n$ of Lemma 4.4. *Boosting.* The same mathematics appeared first for boosting: Rosset et al. (2004) showed that boosting with infinitesimal steps follows a regularization path to an ℓ_1 max-margin classifier, and Telgarsky (2013) that shrinkage yields approximate max-margin classifiers. *Open.* Rates of directional convergence for nonlinear homogeneous networks are unknown in general, as is a description of which Karush–Kuhn–Tucker point gradient descent selects from a given initialization; and a theory of the margin at the finite training times used in practice, rather than in the limit, is missing.

4.7 Model card

MODEL CARD	
PHENOMENON	After zero training error, the cross-entropy keeps falling like $1/t$, the weights grow like $\log t$, and the classifier keeps changing; test loss and test error decouple.
MODEL	Linear classifier on fixed separable data with $\|z_i\| \le 1$, exponential loss, gradient flow or gradient descent with $\eta \le \frac{1}{2}$ from $w = 0$; homogeneous networks as an extension.
SOLUTION	$\widehat{\mathcal{L}} \le 1/(1+\gamma^{\star 2}t)$; $\|w\| \approx \log t/\gamma^\star$; normalized margin within $\gamma^\star \log n/\log(\gamma^{\star 2}t/n)$ of $\gamma^\star$; direction converges to the hard-margin SVM; the smoothed margin of a homogeneous network increases once $\widehat{\mathcal{L}} < 1/n$.
PREDICTS	$\gamma^{\star 2}t\widehat{\mathcal{L}} \to 1$; margin gap of order $1/\log t$; time scale $1/\gamma^{\star 2}$; importance weights fade; test loss of misclassified points grows like $\log t$.
GETS WRONG	Logarithmic rates never reached in practice; stationary points instead of maximizers for ReLU networks; margins do not predict deep-network generalization; large steps and Adam change the dynamics and the norm; label noise.
SUCCESSORS	Steepest descent and Adam (ℓ_∞ margins); linear convolutional and diagonal networks; Lyu–Li and Ji–Telgarsky for homogeneous networks; nonseparable data; normalized and primal–dual methods with fast rates.

The chapter is the hub of the thread on implicit bias. The diagonal

networks of Chapter 3 trained on classification data are homogeneous of degree two, and the potential that selects the sparse interpolator there becomes the ℓ_1 margin here (Exercise 4.9). The max-margin argument of Chapter 10 is Theorem 4.11 for a network homogeneous of degree three, with the margin problem solved in Fourier space; and the unconstrained-features model of neural collapse in Chapter 12 is a max-margin problem for the last layer and the features together, whose solution is a simplex. In all three, "max-margin solutions as attractors" means the limit of the dynamics studied here.

Two contrasts are instructive. In Chapter 1 the least-squares loss has a minimizer, and gradient descent from zero converges to the minimum-norm interpolator in finite effective time; here the loss has none, and the bias appears only in the limit of the direction. And the lazy regime of Chapter 6 is the case in which the features hardly move: then the network trained past zero error behaves like the linear model of this chapter on its tangent features, and converges, in the kernel sense, to a max-margin classifier of the neural tangent kernel.

Exercises

4.1 CONSOLIDATION. (Three points.) Let $z_1 = (1, 0)$, $z_2 = (0, 1)$ and $z_3 = (0.6, 0.6)$. (a) Show that $\gamma^\star = 1/\sqrt{2}$, $u^\star = (1, 1)/\sqrt{2}$, that the support vectors are z_1 and z_2, and that $\alpha = (1, 1, 0)$ in Lemma 4.4. (b) Show that gradient flow from 0 stays on the diagonal, $w(t) = s(t)(1, 1)$, with $\dot{s} = \frac{1}{3}(e^{-s} + 0.6e^{-1.2s})$, so that the normalized margin equals $\gamma^\star$ for every $t > 0$. (c) Show that $s(t) = \log(t/3) + O(t^{-1/5})$, and compare with (4.6). *(Hint: in (c), $u = e^s$ satisfies $\dot{u} = \frac{1}{3}(1 + 0.6u^{-1/5})$.)*

4.2 CONSOLIDATION. (The convex bound, and the logistic loss.) Let ℓ be the exponential or the logistic loss and $\widehat{\mathcal{L}}(w) = \frac{1}{n}\sum_i \ell(\langle w, z_i\rangle)$, which is convex. (a) Show that gradient flow from 0 satisfies $\widehat{\mathcal{L}}(w(t)) \le \widehat{\mathcal{L}}(v) + \|v\|^2/(2t)$ for every $v \in \mathbb{R}^d$ and $t > 0$. *(Hint: differentiate $\frac{1}{2}\|w(t) - v\|^2$ and use convexity and the monotonicity of $\widehat{\mathcal{L}}(w(t))$.)* (b) Taking $v = u^\star \log t/\gamma^\star$, deduce $\widehat{\mathcal{L}}(w(t)) \le \big(1 + (\log t)^2/(2\gamma^{\star 2})\big)/t$ for $t \ge 1$, for both losses. (c) For the logistic loss, show that $e^{-q} \le 2\ell(q)$ and $|\ell'(q)| \ge \ell(q)(1 - 2\ell(q))$ for $q \ge 0$. From the first time at which $\widehat{\mathcal{L}} < \frac{\log 2}{n}$, adapt the proof of part (iii) of Theorem 4.6 with $N = -\log(2n\widehat{\mathcal{L}})$, bounding the error terms by $\int \widehat{\mathcal{L}}^2 \, dt < \infty$ from (b), to prove that the normalized margin converges to $\gamma^\star$.

4.3 CONSOLIDATION. (The smoothed margin of a linear classifier.) For $L = 1$, Proposition 4.10 says that $\tilde{\gamma}(w(t)) = -\log(n\widehat{\mathcal{L}})/\|w\|$ is nondecreasing after the first time t_0 at which $\widehat{\mathcal{L}} < 1/n$. (a) Show that $\tilde{\gamma}(w(t)) \le \gamma^\star$ for all t, and that $\tilde{\gamma}(w(t)) \to \gamma^\star$. (b) Show that monotonicity alone does not give a rate, and recover one from (4.5). (c) Check the monotonicity numerically for gradient descent with $\eta = \frac{1}{2}$ on the data of Figure 4.2.

4.4 CONSOLIDATION. (The descent lemma, another way.) (a) Show that $\nabla^2\widehat{\mathcal{L}}(w) = \frac{1}{n}\sum_i e^{-\langle w,z_i\rangle} z_i z_i^\top$, so that $0 \preceq \nabla^2\widehat{\mathcal{L}}(w) \preceq \widehat{\mathcal{L}}(w) I_d$ under Assumption 4.2, and that $\widehat{\mathcal{L}}$ has no global Lipschitz gradient. (b) Show that $\widehat{\mathcal{L}}(w+\tau v) \le e^{\tau\|v\|}\widehat{\mathcal{L}}(w)$ for $\tau \ge 0$, and deduce from Taylor's formula a version of Lemma 4.9 with $1-\eta\widehat{\mathcal{L}}(w)$ replaced by $1-\frac{\eta}{2}e^{\eta\widehat{\mathcal{L}}(w)}\widehat{\mathcal{L}}(w)$. (c) For which step sizes does this version give a rate from $w_0 = 0$?

4.5 CONSOLIDATION. (Norm and loss from the expansion.) Assume the expansion of the heuristic derivation. (a) Show that $\gamma^\star\langle u^\star, \tilde{w}\rangle \ge \min_{i\in\mathcal{S}}\langle \tilde{w}, z_i\rangle$, so that the predicted gap $(b-a)/\log t$ is nonnegative. (b) Show that the angle between $w(t)$ and $u^\star$ is $\|P^\perp\tilde{w}\|/\|w(t)\| + o(1/\log t)$. (c) A misclassified test point has $\langle \hat{w}, z\rangle < 0$. Show that its logistic loss is $|\langle \hat{w}, z\rangle| \log t + O(1)$. *(Hint: in (a), $u^\star$ is a convex combination of the support vectors divided by $\gamma^\star$.)*

4.6 VARIATION. (Other norms.) Let $\|\cdot\|$ be a norm on $\mathbb{R}^d$ with dual norm $\|\cdot\|_*$, assume $\|z_i\|_* \le 1$, and let $\gamma^\star = \max_{\|u\|\le 1}\min_i\langle u, z_i\rangle > 0$. Consider an absolutely continuous solution of the steepest-descent flow, in which $\dot{w}(t)$ maximizes $\langle -\nabla\widehat{\mathcal{L}}(w), v\rangle$ over $\|v\| \le \|\nabla\widehat{\mathcal{L}}(w)\|_*$. (a) Show that $\frac{\mathrm{d}}{\mathrm{d}t}\widehat{\mathcal{L}} = -\|\nabla\widehat{\mathcal{L}}\|_*^2$ and $\|\nabla\widehat{\mathcal{L}}\|_* \ge \gamma^\star\widehat{\mathcal{L}}$. (b) Prove parts (i)–(iv) of Theorem 4.6 with $\|\cdot\|$ in place of the Euclidean norm. (c) For $\|\cdot\| = \|\cdot\|_\infty$, show that the flow is sign gradient descent rescaled by $\|\nabla\widehat{\mathcal{L}}\|_1$, and relate the result to Zhang et al. (2024). (d) Show by an example in $\mathbb{R}^2$ that the max-margin direction for $\|\cdot\|_\infty$ need not be unique, so that part (iv) of Lemma 4.4 fails.

4.7 VARIATION. (Weighted losses.) Let $\widehat{\mathcal{L}}(w) = \sum_i c_i e^{-\langle w,z_i\rangle}$ with $c_i > 0$ and $\sum_i c_i = 1$. Show that Theorem 4.6 holds with $\log n$ replaced by $\log(1/\min_i c_i)$ and n in part (i) replaced by $1/\min_i c_i$. Conclude that the limit direction does not depend on the weights, and that up-weighting a rare class only delays the approach to $u^\star$. Compare with Byrd and Lipton (2019).

4.8 VARIATION. (Nonseparable data.) Let $z_1 = (1,0)$, $z_2 = (-1,0)$ and $z_3 = (0.6, 0.8)$, so that no direction classifies all three points. (a) Show that $\inf\widehat{\mathcal{L}} = \frac{2}{3}$ and that it is not attained. (b) Show that gradient flow from 0 satisfies $w_1(t) \to 0$ and $w_2(t) - \frac{5}{4}\log t \to \frac{5}{4}\log\frac{16}{75}$, so that $w(t)/\|w(t)\| \to (0,1)$, the max-margin direction of the separable part restricted to the orthogonal complement of the span of z_1, z_2. Compare with Ji and Telgarsky (2019b). *(Hint: show first that $w_1 \ge 0$ and $w_1(t) = O(1/t)$, then solve for w_2 with $w_1 = 0$.)*

4.9 VARIATION. ★ (Diagonal networks and the ℓ_1 margin.) Let $f(x; u, v) = \langle u\odot u - v\odot v, x\rangle$, the diagonal network of Chapter 3, homogeneous of degree two. (a) Show that $\min\{\|u\|^2 + \|v\|^2 : u\odot u - v\odot v = \beta\} = \|\beta\|_1$. (b) Deduce that (u, v) is a global solution of (4.8) if and only if $\beta = u\odot u - v\odot v$ solves $\min\|\beta\|_1$ subject to $\langle \beta, z_i\rangle \ge 1$, the ℓ_1 max-margin problem. (c) For $z_1 = (1,0)$ and $z_2 = (0.6, 0.6)$, show that the ℓ_2 max-margin solution is $\hat{w} = (1, \frac{2}{3})$, and that the solutions of the ℓ_1 problem

form the segment $\{\beta_1 + \beta_2 = \frac{5}{3},\ 1 \le \beta_1 \le \frac{5}{3}\}$, which contains $\hat{w}$. The limit of the diagonal network is not determined by the ℓ_1 problem alone. (d) By Vardi et al. (2022), the Karush–Kuhn–Tucker points of this network need not be local maximizers. Explain why part (b) does not contradict this. *(Hint: in (a), $u_j^2 + v_j^2 \ge |u_j^2 - v_j^2|$ for each coordinate, with equality when $u_j = 0$ or $v_j = 0$.)*

4.10 SIMULATION. (Gradient descent and its bounds.) On the data of Figure 4.2, run gradient descent on the exponential loss with $\eta \in \{0.25, 0.5\}$ for 2×10^5 steps and check parts (i) and (ii) of Proposition 4.8 at every step. The reader should find both bounds satisfied, with $\min_i q_i(w_k) - \gamma^\star \|w_k\|$ ending near -0.2 for both step sizes: the constants $\log n = 3.7$ and $\eta(\gamma^\star \eta + 2/\gamma^\star)$ are almost entirely slack. Then try $\eta \in \{1, 2, 5\}$, outside the proposition: on these data the loss still decreases monotonically, and the normalized margin at time ηk is close to that of the flow.

4.11 SIMULATION. (Rates for a ReLU network.) Use the output of the script of Figure 4.3. (a) Plot $\|\theta\|^2$ against $\log t$ and check that it is asymptotically linear. (b) Assuming $\tilde{\gamma} \approx \bar{\gamma}$ constant and $\min_i q_i \approx \bar{\gamma}\|\theta\|^2$, use (4.7) to derive $u = \bar{\gamma}\|\theta\|^2$ from $e^u/u \approx 4\bar{\gamma}t/n$, and $\widehat{\mathcal{L}} \approx e^{-u}/n$. The reader should find that this predicts $\widehat{\mathcal{L}}(10^9) \approx 3.7 \times 10^{-9}$ with $\bar{\gamma} = 0.00427$, against 3.69×10^{-9} simulated, and that $\widehat{\mathcal{L}}\, t \log t$ varies slowly, as part (i) of Theorem 4.11 says for $L = 2$. (c) Repeat with several initialization seeds, keeping the data fixed, and record the normalized margin at $t = 10^7$; we found 0.00425 for five seeds, so on these data the flow appears to select the same stationary point.

4.12 RESEARCH. (Rates for networks.) For a two-layer ReLU network on separable data, bound the rate at which $\theta(t)/\|\theta(t)\|$ converges, or the normalized margin approaches its limit. Lyu and Li (2020) give rates for the loss and the norm but not for the direction; Ji and Telgarsky (2020) prove convergence without a rate. Even an example with a matching lower bound worse than $1/\log t$ would be new to us.

4.13 RESEARCH. (Which stationary point?) Characterize, for a two-layer ReLU network and a given data set, the Karush–Kuhn–Tucker points of (4.8) that gradient flow from small random initialization can reach, and decide when it reaches a global maximizer of the margin. Vardi et al. (2022) give settings with and without local optimality; a general criterion is open.

Notes and References

The implicit bias of gradient descent toward the max-margin direction was proved for linear models on separable data by Soudry et al. (2018), whose conference version appeared at ICLR 2018; they gave the expansion $\hat{w} \log t + \rho(t)$, the rates $1/\log t$ and $\log\log t/\log t$, and the CIFAR-10 experiments of Section 4.1. The ideas were older in boosting, where AdaBoost minimizes

an exponential loss by coordinate descent: Schapire et al. (1998) tied its test error to the margins, Rosset et al. (2003) showed that regularized solutions of losses with exponential tails approach margin-maximizing separators as the regularization vanishes, Rosset et al. (2004) connected boosting to an ℓ_1 max-margin classifier, and Telgarsky (2013) analyzed shrinkage. The max-margin classifier itself goes back to the support vector machines of Boser et al. (1992) and Cortes and Vapnik (1995), and the margin as the quantity that controls learning to the perceptron analysis of Novikoff (1962).

Our proof of Theorem 4.6 follows the free-energy argument used for homogeneous networks by Lyu and Li (2020), specialized to $L = 1$, with the convex geometry of Lemma 4.4 supplying the direction; the quantitative directional bound of Corollary 4.7 is a short consequence of part (iv) of that lemma. The primal–dual view, in which the normalized weights $e^{-q_i}/\sum_j e^{-q_j}$ are a dual variable converging to $p^\star$, is developed by Ji and Telgarsky (2021), who also show that the rate $\ln n/\ln t$ is tight. Proposition 4.8 is a standard descent argument; the constant $\frac{1}{2}$ in the step size is not optimized.

For networks, Lyu and Li (2020) proved the monotonicity of the smoothed margin, the rates of the loss and the norm, and convergence to Karush–Kuhn–Tucker points, for homogeneous networks with locally Lipschitz outputs; Ji and Telgarsky (2020) proved directional convergence and alignment using o-minimal geometry. Earlier, Ji and Telgarsky (2019a) treated deep linear networks and Gunasekar et al. (2018a,b) worked out the effect of the optimization geometry and of convolutional architectures. Wei et al. (2019) connected the regularized problem with small weight decay to the max-margin problem for networks, a link used in Chapter 10.

TRANSLATING THE LITERATURE. Soudry et al. (2018) absorb the labels into the inputs, writing $\mathbf{x}_n$ for our z_i and N for our n; their loss is the sum $\mathcal{L}(\mathbf{w}) = \sum_n \ell(\mathbf{w}^\top \mathbf{x}_n)$, which is n times our $\widehat{\mathcal{L}}$; they run gradient descent with step η, so that their iteration t corresponds to our time $n\eta t$; their $\hat{\mathbf{w}}$ is our $\hat{w} = u^\star/\gamma^\star$, and their residual $\boldsymbol{\rho}(t)$ is our $w(t) - \hat{w}\log t$ up to the constant $\hat{w}\log(n\eta)$ that the change of clock produces. Lyu and Li (2020) write $\Phi(\boldsymbol{\theta}; \mathbf{x})$ for our $f(x;\theta)$, q_n for our q_i, L for the degree of homogeneity, as we do, and ρ for $\|\theta\|$; their loss is again a sum, so for the exponential loss their smoothed margin $\log(1/\mathcal{L})/\rho^L$ is our $\tilde{\gamma}$. The dual variable of Ji and Telgarsky (2021) is our vector of normalized weights $e^{-q_i}/\sum_j e^{-q_j}$, whose limit is $p^\star$.

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Part II

Infinite Width

Two limits make wide networks simple. At initialization a wide network is a Gaussian process whose behaviour in depth has phases, as a magnet does. Under small updates it is a linear model in its parameters, a kernel machine. Both limits are exact, and both lose what makes networks interesting.

Order and Chaos at Initialization

5

Signal propagation in wide random networks

THE MODEL — INFINITELY WIDE RANDOM NETWORKS

IN THIS CHAPTER. A deep network trains only if its random initialization is scaled with care, and the deepest trainable network is set by a depth scale that diverges on a curve in the plane of weight and bias variances. Infinitely wide random networks explain this. We prove that their pre-activations are Gaussian processes whose covariance obeys a layer-to-layer recursion, and that the recursion has two phases: an ordered phase, where all inputs become perfectly correlated at a geometric rate χ_1 per layer, and a chaotic phase, where nearby inputs decorrelate; the depth scale is $-1/\log$ of the slope of the correlation map at its stable fixed point, and a heuristic expansion shows that it diverges like $1/|\chi_1 - 1|$. At He initialization ReLU networks sit exactly at the boundary and decorrelate only polynomially, $1 - c^\ell \sim 9\pi^2/(2\ell^2)$. The model is mean-field theory: it is exact when width is infinite at fixed depth, it has no training and no architecture, and at finite width its fluctuations grow with depth over width, so that criticality on average says little about the typical network once L is comparable to m.

5.1 The phenomenon: a narrow window of trainable initializations

Before a network learns anything, signals must pass through its random layers: forward, so that the output depends on the input, and backward, so that every layer receives a gradient. Each layer multiplies the signal by a random matrix and bends it through a nonlinearity, and a composition of L such maps generically shrinks or amplifies exponentially in L.

For gradients through time in recurrent networks the problem was identified early. Hochreiter (1991), in a diploma thesis written in German, analyzed how the error signal propagated backward through a recurrent network decays or blows up exponentially with the number of time steps, and Bengio et al. (1994) showed that a recurrent network that stores information robustly must have vanishing gradients over long time lags. These are the *vanishing and exploding gradients*.[1]

For depth, practice came first. Glorot and Bengio (2010) found that deep sigmoid networks with the then-standard initialization drove their top hidden layer into saturation early in training, and proposed to draw the weights of a layer with m_{in} inputs and m_{out} outputs with variance

[1] An unrolled recurrent network is a deep feedforward network whose layers share their weights, so the two problems are one problem, with depth playing the role of time.

$2/(m_{\text{in}}+m_{\text{out}})$, a compromise between keeping the variance of the activations and that of the back-propagated gradients constant in a linear network. He et al. (2015) redid the calculation for rectifiers: a ReLU passes on half of the second moment of its input, so the weight variance must be $2/m_{\text{in}}$ to keep the variance of the responses constant. With this initialization they trained a 30-layer rectifier network from scratch, while with the variance of Glorot and Bengio (2010) the same network failed to converge.

Poole et al. (2016) turned the variance-counting arguments into a theory. Following two inputs through a deep random network of large width, they found two phases. In the *ordered phase* every pair of inputs is mapped to nearly the same representation; in the *chaotic phase* nearby inputs are driven apart, and a smooth curve of inputs is mapped to a curve whose length and curvature grow exponentially with depth. Schoenholz et al. (2017) read off depth scales from the same theory: the number of layers over which the correlation between two inputs relaxes, and over which gradient norms grow or decay. The correlation depth scale diverges on the boundary between the phases, the *edge of chaos*, and their experiments put the limit of trainability at a fixed multiple of it.

Observation 5.1 (Schoenholz et al. 2017). Fully connected tanh networks with Gaussian weights of variance σ_w^2/m and biases of variance σ_b^2 were trained by stochastic gradient methods on MNIST and on CIFAR-10 for many values of the depth L and of (σ_w^2, σ_b^2). Networks trained when $L \lesssim 6\xi_c$ and failed when L was much larger, where ξ_c is the correlation depth scale of the infinitely wide random network with the same variances. Close to the critical line, where ξ_c diverges, the trainable depth grew accordingly.

How robust is this? The divergence of the trainable depth at the critical line has been reproduced many times; Xiao et al. (2018) trained vanilla convolutional networks with 10,000 layers, without residual connections or normalization, by initializing at criticality with orthogonal convolution kernels. The constant 6 is empirical and depends on the setting, the training budget, and the criterion for success.[2] Trainable is also not the same as trains well: Pennington et al. (2017) found that critical ReLU networks learn more slowly than critical sigmoidal networks with orthogonal weights, because criticality fixes the mean squared singular value of the input–output Jacobian and not its spread. Modern networks escape the constraint by architecture rather than by tuning (Section 5.6).

[2] The theory says nothing about the constant: it predicts a depth scale, not a training outcome.

The questions for the model are therefore: What computes ξ_c? Why is there a sharp boundary, and why does the depth scale diverge on it? Where does the ReLU network at He initialization sit?

5.2 The model: infinitely wide random networks

Assumption 5.2 (Random fully connected network). The input is $x \in \mathbb{R}^d$, the widths are $m_0 = d$ and $m_1, \dots, m_L$, and the activation $\phi\colon \mathbb{R} \to \mathbb{R}$ is continuous and polynomially bounded: $|\phi(t)| \le C_\phi(1 +$

$|t|)^{k_\phi}$ for constants C_ϕ, k_ϕ. The *pre-activations* are

$$h^1(x) = W^1 x + b^1, \qquad h^{\ell+1}(x) = W^{\ell+1}\phi\big(h^\ell(x)\big) + b^{\ell+1} \quad (1 \le \ell < L),$$

with ϕ applied coordinatewise, and all weights and biases are independent, with $W^\ell_{ij} \sim \mathcal{N}(0, \sigma_w^2/m_{\ell-1})$ and $b^\ell_i \sim \mathcal{N}(0, \sigma_b^2)$, where $\sigma_w^2 > 0$ and $\sigma_b^2 \ge 0$.

We write $a^0(x) = x$ and $a^\ell(x) = \phi(h^\ell(x))$ for the post-activations, so that $h^{\ell+1} = W^{\ell+1}a^\ell + b^{\ell+1}$ for $0 \le \ell < L$. The widths are the hidden widths m of the global notation, one per layer; the limit of the model sends them to infinity at fixed depth.

THE KERNEL RECURSION. The infinitely wide network is described by one deterministic function per layer, its *kernel*:

$$K^1(x,x') = \sigma_b^2 + \sigma_w^2 \frac{\langle x, x'\rangle}{d}, \qquad K^{\ell+1}(x,x') = \sigma_b^2 + \sigma_w^2\, \mathbb{E}\big[\phi(z)\phi(z')\big], \quad (5.1)$$

where (z, z') is a centered Gaussian pair with $\mathbb{E}z^2 = K^\ell(x,x)$, $\mathbb{E}z'^2 = K^\ell(x',x')$, and $\mathbb{E}zz' = K^\ell(x,x')$. The polynomial bound makes the expectation finite. The two *order parameters* are the length $q^\ell(x) = K^\ell(x,x)$, the limit of $\frac{1}{m_\ell}\|h^\ell(x)\|^2$, and the correlation

$$c^\ell(x,x') = \frac{K^\ell(x,x')}{\sqrt{q^\ell(x)\, q^\ell(x')}},$$

the limit of the cosine of the angle between $h^\ell(x)$ and $h^\ell(x')$. With $z \sim \mathcal{N}(0,1)$, and (u, v) standard Gaussians with correlation c, the length obeys the one-dimensional *length map*

$$q^{\ell+1} = \mathcal{V}(q^\ell), \qquad \mathcal{V}(q) = \sigma_b^2 + \sigma_w^2\, \mathbb{E}\big[\phi(\sqrt{q}\, z)^2\big], \quad (5.2)$$

and if both inputs sit at a fixed point $\bar{q} > 0$ of $\mathcal{V}$, then the correlation obeys the *correlation map*

$$c^{\ell+1} = \mathcal{C}(c^\ell), \qquad \mathcal{C}(c) = \frac{\sigma_b^2 + \sigma_w^2\, \mathbb{E}\big[\phi(\sqrt{\bar{q}}\, u)\phi(\sqrt{\bar{q}}\, v)\big]}{\bar{q}}. \quad (5.3)$$

The fixed point is written $\bar{q}$, not $q^\star$, because a superscript star marks an optimum in this book. The quantity that will decide everything is the slope

$$\chi_1 = \sigma_w^2\, \mathbb{E}\big[\phi'(\sqrt{\bar{q}}\, z)^2\big]. \quad (5.4)$$

DICTIONARY

IN THE MODEL	IN A TRAINED NETWORK
width $m_\ell \to \infty$ at fixed depth	width much larger than depth, $L/m \ll 1$
weight and bias variances σ_w^2, σ_b^2	the initialization scheme (LeCun, Glorot, He)
length q^ℓ	mean squared pre-activation of layer ℓ
correlation $c^\ell(x, x')$	cosine similarity of the representations of two inputs
fixed point $\bar{c}$ of C	how similar all inputs look to a deep layer
slope χ_1	growth factor per layer of squared gradient norms
depth scale ξ	depth beyond which training fails (about 6ξ for tanh in Observation 5.1)
critical line $\chi_1 = 1$	the family of initializations that train deep networks
kernel K^L	prior covariance of a wide Bayesian network; the NNGP kernel

WHAT WAS THROWN AWAY, AND WHY FIRST. The network is never trained: trainability is decided at initialization, at least in the sense of Observation 5.1, because a signal that does not reach the output cannot be learned from. The width is infinite, which turns sums over neurons into expectations and makes the theory deterministic, as the field of a magnet becomes deterministic when each spin interacts with infinitely many others. The architecture is a plain stack of dense layers, because it is the case in which depth is hardest. The data enter only through the inner products $\langle x, x' \rangle / d$, so two inputs are as good as a dataset.

WHAT ONE HAD TO NOTICE. Three things. Conditionally on one layer, the next is exactly Gaussian, because it is linear in Gaussian weights; only its covariance is random, and that covariance is an average over the neurons of the previous layer. Two inputs suffice: their lengths and their correlation are closed under the layer map, like the magnetization of a mean-field magnet. And the correlation map is a power series with nonnegative coefficients, so its behavior on $[0, 1]$ is decided by its slope at the fixed point $c = 1$.

5.3 The solution: Gaussian processes at infinite width

Let $\mathcal{X}$ be a finite set of inputs, and write $h_i^\ell(\mathcal{X}) = (h_i^\ell(x))_{x \in \mathcal{X}} \in \mathbb{R}^{\mathcal{X}}$ for the values of neuron i of layer ℓ on these inputs. Let $\mathcal{F}_\ell$ be the σ-algebra generated by $W^1, b^1, \ldots, W^\ell, b^\ell$, with $\mathcal{F}_0$ trivial.

Lemma 5.3 (Exact conditional Gaussianity). *Under Assumption 5.2, for $0 \le \ell < L$, conditionally on $\mathcal{F}_\ell$, the vectors $h_1^{\ell+1}(\mathcal{X}), \ldots, h_{m_{\ell+1}}^{\ell+1}(\mathcal{X})$ are*

independent with law $\mathcal{N}(0, \widehat{G}^\ell)$*, where*

$$\widehat{G}^\ell(x, x') = \sigma_b^2 + \frac{\sigma_w^2}{m_\ell} \sum_{j=1}^{m_\ell} a_j^\ell(x)\, a_j^\ell(x'), \qquad x, x' \in \mathcal{X}.$$

In particular $\widehat{G}^0 = K^1$ *on* $\mathcal{X}$*, and the vectors* $h_i^1(\mathcal{X})$ *are independent with law* $\mathcal{N}(0, K^1)$*.*

Proof. The post-activations $a^\ell(x)$ are $\mathcal{F}_\ell$-measurable, and the ith row of $W^{\ell+1}$, together with $b_i^{\ell+1}$, is a Gaussian vector independent of $\mathcal{F}_\ell$ and of the other rows. Conditionally on $\mathcal{F}_\ell$, the vector $h_i^{\ell+1}(\mathcal{X})$ is therefore a fixed linear image of this row, hence centered Gaussian, and the vectors for different i are independent. Since the entries of the row are independent with variances σ_w^2/m_ℓ and σ_b^2, the conditional covariance of $h_i^{\ell+1}(x) = \sum_j W_{ij}^{\ell+1} a_j^\ell(x) + b_i^{\ell+1}$ and $h_i^{\ell+1}(x')$ is $\frac{\sigma_w^2}{m_\ell} \sum_j a_j^\ell(x) a_j^\ell(x') + \sigma_b^2$. For $\ell = 0$, $a^0(x) = x$ and $m_0 = d$ give K^1. ∎

The lemma is exact at every width: a layer acts on a finite set of inputs through $\widehat{G}^\ell$ alone, and in law $h^{\ell+1}(\mathcal{X})$ is an $m_{\ell+1} \times |\mathcal{X}|$ matrix of independent standard Gaussians times $(\widehat{G}^\ell)^{1/2}$. On two inputs, a network of width m is therefore a Markov chain on $m \times 2$ matrices, which can be simulated in time $O(mL)$ instead of $O(m^2 L)$; all the finite-width simulations of this chapter use this reduction.[3]

[3] We checked the reduction against simulations with full $m \times m$ weight matrices at $m = 200$; the averaged correlations agree to within the Monte Carlo error.

Theorem 5.4 (Gaussian-process limit, sequential widths). *Under Assumption 5.2, let* $\mathcal{X}$ *be a finite set of inputs,* $1 \le \ell \le L$*, and* $J \ge 1$*, with* $m_\ell \ge J$*. In the iterated limit in which* $m_1 \to \infty$*, then* $m_2 \to \infty$*, and so on up to* $m_{\ell-1} \to \infty$*, the law of* $(h_1^\ell(\mathcal{X}), \dots, h_J^\ell(\mathcal{X}))$ *converges weakly to the law of* J *independent* $\mathcal{N}(0, K^\ell)$ *vectors, with* K^ℓ *restricted to* $\mathcal{X}$*. That is, for every bounded continuous* $F\colon (\mathbb{R}^{\mathcal{X}})^J \to \mathbb{R}$ *and* $z_1, \dots, z_J$ *i.i.d.* $\mathcal{N}(0, K^\ell)$*,*

$$\lim_{m_{\ell-1}\to\infty} \cdots \lim_{m_1\to\infty} \mathbb{E} F\big(h_1^\ell(\mathcal{X}), \dots, h_J^\ell(\mathcal{X})\big) = \mathbb{E} F(z_1, \dots, z_J).$$

For $\ell = 1$ *no limit is needed, and the law of* $(h_1^\ell(\mathcal{X}), \dots, h_J^\ell(\mathcal{X}))$ *does not depend on* $m_\ell, \dots, m_L$*.*

Proof. The plan is induction on ℓ: by Lemma 5.3, layer $\ell + 1$ is a Gaussian whose covariance is a continuous function of the neurons of layer ℓ; the induction hypothesis makes those neurons i.i.d. Gaussian, and the law of large numbers makes the covariance deterministic.

Two facts. First, centered Gaussian laws depend continuously on their covariance: if $\Sigma_n \to \Sigma$, then the characteristic functions $\exp(-\frac{1}{2} t^\top \Sigma_n t)$ converge to $\exp(-\frac{1}{2} t^\top \Sigma t)$, and Lévy's continuity theorem gives $\mathcal{N}(0, \Sigma_n) \to \mathcal{N}(0, \Sigma)$ weakly. Second, the last claim of the theorem holds because h^ℓ is a function of $W^1, b^1, \dots, W^\ell, b^\ell$, and the first J neurons of layer ℓ use only the first J rows of W^ℓ and b^ℓ.

Base case. By Lemma 5.3 with $\ell = 0$, the vectors $h_i^1(\mathcal{X})$ are i.i.d. $\mathcal{N}(0, K^1)$ at every width.

Induction step. Assume the statement for layer ℓ and every number of neurons, and fix J, F, and m_ℓ. For $y = (y_1, \dots, y_{m_\ell}) \in (\mathbb{R}^{\mathcal{X}})^{m_\ell}$ define the $\mathcal{X}\times\mathcal{X}$ matrix $G(y) = \sigma_b^2 \mathbf{1}\mathbf{1}^\top + \frac{\sigma_w^2}{m_\ell} \sum_j \phi(y_j)\phi(y_j)^\top$, and let $\Phi_F(y) = \mathbb{E}F(g_1, \dots, g_J)$ with $g_1, \dots, g_J$ i.i.d. $\mathcal{N}(0, G(y))$. The function Φ_F is bounded by $\sup|F|$, and it is continuous: ϕ is continuous, so G is continuous, and the first fact applies. By Lemma 5.3, conditionally on $\mathcal{F}_\ell$ the first J neurons of layer $\ell + 1$ are i.i.d. $\mathcal{N}(0, G(h_1^\ell(\mathcal{X}), \dots, h_{m_\ell}^\ell(\mathcal{X})))$, so

$$\mathbb{E}F\big(h_1^{\ell+1}(\mathcal{X}), \dots, h_J^{\ell+1}(\mathcal{X})\big) = \mathbb{E}\,\Phi_F\big(h_1^\ell(\mathcal{X}), \dots, h_{m_\ell}^\ell(\mathcal{X})\big).$$

The induction hypothesis with m_ℓ neurons in place of J, applied to the bounded continuous function Φ_F, shows that after the limits $m_1 \to \infty, \dots, m_{\ell-1} \to \infty$ the right side becomes $\mathbb{E}\,\Phi_F(z_1, \dots, z_{m_\ell})$ with z_j i.i.d. $\mathcal{N}(0, K^\ell)$ (for $\ell = 1$ this is the base case, with no limit).

Now let $m_\ell \to \infty$. Each entry of $\phi(z_j)\phi(z_j)^\top$ is integrable, since $|\phi(s)\phi(t)| \le \frac{1}{2}(\phi(s)^2 + \phi(t)^2)$ and $\phi(z_j(x))^2$ is bounded by a polynomial in a Gaussian variable. By the strong law of large numbers, $G(z_1, \dots, z_{m_\ell}) \to \sigma_b^2 \mathbf{1}\mathbf{1}^\top + \sigma_w^2 \mathbb{E}[\phi(z)\phi(z)^\top]$ almost surely, and by (5.1) the limit is $K^{\ell+1}$ on $\mathcal{X}$. By the first fact, $\Phi_F(z_1, \dots, z_{m_\ell}) \to \mathbb{E}F(z_1', \dots, z_J')$ almost surely, with z_i' i.i.d. $\mathcal{N}(0, K^{\ell+1})$, and bounded convergence gives the same limit for the expectation. This is the statement for layer $\ell + 1$. ∎

The proof shows more: in the same iterated limit, now including $m_\ell \to \infty$, the empirical kernel $\widehat{G}^\ell$ converges in probability to $K^{\ell+1}$, and in particular $\frac{1}{m_\ell}\|h^\ell(x)\|^2 \to q^\ell(x)$ (Exercise 5.1). This is the precise sense of *mean-field exactness*: each neuron of layer $\ell + 1$ feels the other neurons of layer ℓ only through an average over infinitely many of them, and the average does not fluctuate.

Theorem 5.5 (Simultaneous limits; Matthews et al. 2018, Hanin 2023). *The conclusion of Theorem 5.4 also holds when the widths $m_1, \dots, m_{\ell-1}$ tend to infinity simultaneously rather than one at a time, for activations satisfying a growth condition: Matthews et al. (2018) assume a linear envelope $|\phi(t)| \le c + M|t|$, and Hanin (2023) allows quite general nonlinearities and non-Gaussian weights under moment conditions.*

We do not prove Theorem 5.5; the exact hypotheses are in the two papers. For one hidden layer it is the central limit theorem used by Neal (1996).

THE LENGTH MAP. The kernel recursion needs a fixed length to define the correlation map. For odd activations such as tanh the length map has a unique attracting fixed point, and the proof is a Fourier argument.

Proposition 5.6 (Length map of tanh). *Let $\phi = \tanh$. The length map $\mathcal{V}$ of (5.2) is increasing and strictly concave on $[0, \infty)$, with $\mathcal{V}(0) = \sigma_b^2$, $\mathcal{V}'(0) = \sigma_w^2$, and $\mathcal{V} < \sigma_b^2 + \sigma_w^2$.*

(i) If $\sigma_b^2 > 0$, then $\mathcal{V}$ has a unique fixed point $\bar{q} > 0$, and $q^\ell \to \bar{q}$

monotonically from every $q^1 \geq 0$.

(ii) If $\sigma_b^2 = 0$ and $\sigma_w^2 \leq 1$, then $q^\ell \to \bar{q} = 0$ from every $q^1 \geq 0$.

(iii) If $\sigma_b^2 = 0$ and $\sigma_w^2 > 1$, then $\mathcal{V}$ has a unique positive fixed point $\bar{q}$, and $q^\ell \to \bar{q}$ monotonically from every $q^1 > 0$.

In cases (i) and (iii), $\mathcal{V}'(\bar{q}) < 1$, so the convergence is geometric.

PROOF. The plan is to write $1 - \tanh^2 = \operatorname{sech}^2$ as a mixture of cosines with positive weights, which turns $q \mapsto \mathbb{E}[\operatorname{sech}^2(\sqrt{q}z)]$ into a mixture of decaying exponentials.

The representation. The function sech^2 is integrable, with Fourier transform $\int_{\mathbb{R}} \operatorname{sech}^2(t)\cos(\omega t)\,\mathrm{d}t = g(\omega)$, where $g(\omega) = \pi\omega/\sinh(\pi\omega/2)$ for $\omega \neq 0$ and $g(0) = 2$ (Exercise 5.2). The function g is positive and decays exponentially, so Fourier inversion gives $\operatorname{sech}^2(t) = \frac{1}{2\pi}\int_{\mathbb{R}} g(\omega)\cos(\omega t)\,\mathrm{d}\omega$. Since $\mathbb{E}\cos(\omega\sqrt{q}z) = e^{-q\omega^2/2}$, Fubini's theorem gives

$$f(q) := \mathbb{E}\left[\tanh^2(\sqrt{q}\,z)\right] = 1 - \frac{1}{2\pi}\int_{\mathbb{R}} g(\omega)\, e^{-q\omega^2/2}\,\mathrm{d}\omega.$$

Monotonicity and concavity. The functions $\omega^2 g(\omega)$ and $\omega^4 g(\omega)$ are integrable, so we may differentiate under the integral for $q \geq 0$:

$$f'(q) = \frac{1}{4\pi}\int_{\mathbb{R}} \omega^2 g(\omega) e^{-q\omega^2/2}\,\mathrm{d}\omega > 0,$$
$$f''(q) = -\frac{1}{8\pi}\int_{\mathbb{R}} \omega^4 g(\omega) e^{-q\omega^2/2}\,\mathrm{d}\omega < 0.$$

At $q = 0$, differentiating the inversion formula twice at $t = 0$ gives $\frac{1}{2\pi}\int \omega^2 g = -(\operatorname{sech}^2)''(0) = 2$, because $\operatorname{sech}^2 t = 1 - t^2 + O(t^4)$; so $f'(0) = 1$. With $\mathcal{V} = \sigma_b^2 + \sigma_w^2 f$ and $0 \leq f < 1$, this proves the first sentence.

Case (ii). For $q > 0$, strict concavity gives $\mathcal{V}(q) < \mathcal{V}(0) + \mathcal{V}'(0)q = \sigma_w^2 q \leq q$. So q^ℓ decreases, its limit is a fixed point of the continuous map $\mathcal{V}$, and 0 is the only one.

Cases (i) and (iii). Let $g_0(q) = \mathcal{V}(q) - q$, strictly concave, with $g_0(q) \to -\infty$ because $\mathcal{V}$ is bounded. In case (i), $g_0(0) = \sigma_b^2 > 0$; in case (iii), $g_0(0) = 0$ and $g_0'(0) = \sigma_w^2 - 1 > 0$; in both cases g_0 is positive somewhere on $(0, \infty)$ and has a zero $\bar{q} > 0$ there by the intermediate value theorem. If $0 < q_1 < q_2$ were two zeros, then strict concavity on $[0, q_2]$ would give $g_0(q_1) > (1 - q_1/q_2)g_0(0) \geq 0$, a contradiction; so $\bar{q}$ is unique, $g_0 > 0$ on $(0, \bar{q})$ and $g_0 < 0$ on $(\bar{q}, \infty)$. For $0 < q < \bar{q}$, monotonicity gives $q < \mathcal{V}(q) < \mathcal{V}(\bar{q}) = \bar{q}$, and for $q > \bar{q}$ it gives $\bar{q} < \mathcal{V}(q) < q$: the iterates move monotonically toward $\bar{q}$, and their limit, a positive fixed point, is $\bar{q}$. In case (i) the start $q^1 = 0$ is mapped to $\sigma_b^2 > 0$. Finally, strict concavity gives $g_0'(\bar{q}) < \big(g_0(\bar{q}) - g_0(0)\big)/\bar{q} \leq 0$, that is, $\mathcal{V}'(\bar{q}) < 1$. ∎

In case (ii) the network forgets the scale of its input: every length decays to zero, slowly when $\sigma_w^2 = 1$, since $f(q) = q - 2q^2 + O(q^3)$ gives $q^\ell \sim 1/(2\ell)$.

5.4 The solution: order, chaos, and the critical line

Fix a fixed point $\bar{q} > 0$ of the length map and two inputs with $q^1(x) = q^1(x') = \bar{q}$, so that $q^\ell = \bar{q}$ in every layer. Write $\bar{\phi}(t) = \phi(\sqrt{\bar{q}}\, t)$ and expand it in the normalized Hermite polynomials, with coefficients $a_k = \mathbb{E}[\bar{\phi}(z)h_k(z)]$ (Lemma A.2). By Corollary A.4, the correlation map (5.3) is the power series

$$C(c) = \sum_{k\geq 0} \beta_k c^k, \qquad \beta_0 = \frac{\sigma_b^2 + \sigma_w^2 a_0^2}{\bar{q}}, \quad \beta_k = \frac{\sigma_w^2 a_k^2}{\bar{q}} \ (k \geq 1), \tag{5.5}$$

with nonnegative coefficients, convergent on $[-1, 1]$, with $\sum_k \beta_k = C(1) = \mathcal{V}(\bar{q})/\bar{q} = 1$.[4]

[4] The dual-activation view of Daniely et al. (2016) is exactly this: a layer acts on correlations through the power series of its activation's squared Hermite coefficients, and composition of layers is composition of power series.

Theorem 5.7 (Order and chaos). *Under Assumption 5.2, let ϕ be absolutely continuous and not constant, with ϕ' polynomially bounded, let $\bar{q} > 0$ be a fixed point of $\mathcal{V}$, and let χ_1 be given by (5.4).*

(i) The map C is continuous on $[-1, 1]$, maps $[-1, 1]$ into itself and $[0, 1]$ into $[0, 1]$, is nondecreasing and convex on $[0, 1]$, satisfies $C(1) = 1$, and has left derivative $C'(1) = \chi_1$ at 1. For every $c \in [-1, 1]$,

$$0 \leq 1 - C(c) \leq \chi_1 (1 - c). \tag{5.6}$$

(ii) Ordered phase, $\chi_1 < 1$. *For every $c^1 \in [-1, 1]$, $1 - c^\ell \leq 2\chi_1^{\ell-1}$, so $c^\ell \to 1$. If $c^1 \in (-1, 1)$, then $c^\ell < 1$ for every ℓ and $\frac{1}{\ell}\log(1 - c^\ell) \to \log \chi_1$.*

(iii) Critical line, $\chi_1 = 1$. *If ϕ is not affine and $c^1 \in [0, 1)$, then c^ℓ increases to 1, and $(1 - c^{\ell+1})/(1 - c^\ell) \to 1$: the decay is slower than every geometric rate.*

(iv) Chaotic phase, $\chi_1 > 1$. *The map C has a unique fixed point $\bar{c} \in [0, 1)$, and $C'(\bar{c}) < 1$. For every $c^1 \in [0, 1)$, $c^\ell \to \bar{c}$ monotonically, and $|c^\ell - \bar{c}| = O(\rho^\ell)$ for every $\rho \in (C'(\bar{c}), 1)$.*

Proof. The plan is to read every claim off the power series (5.5), using the identity

$$1 - C(c) = \sum_{k\geq 1} \beta_k (1 - c^k) = (1 - c) \sum_{k\geq 1} \beta_k \left(1 + c + \cdots + c^{k-1}\right), \tag{5.7}$$

which follows from $\sum_k \beta_k = 1$.

Part (i). The coefficients are nonnegative and summable, so the series converges uniformly on $[-1, 1]$ and C is continuous there; on $[0, 1]$ every term is nondecreasing and convex. Since $|c^k| \leq 1$, the first expression in (5.7) is nonnegative, and $C(c) \geq \beta_0 - \sum_{k\geq 1} \beta_k \geq -1$; for $c \in [0, 1]$ also $C(c) \geq \beta_0 \geq 0$. For the slope, the Hermite coefficients of $\bar{\phi}'$ are $\sqrt{k}\, a_k$ by Gaussian integration by parts (Lemma A.2, as in the proof of Lemma A.5), so Parseval's identity and $\bar{\phi}'(t) = \sqrt{\bar{q}}\, \phi'(\sqrt{\bar{q}}\, t)$ give

$$\sum_{k\geq 1} k\beta_k = \frac{\sigma_w^2}{\bar{q}} \sum_{k\geq 1} k a_k^2 = \frac{\sigma_w^2}{\bar{q}} \mathbb{E}\left[\bar{\phi}'(z)^2\right] = \sigma_w^2 \mathbb{E}\left[\phi'(\sqrt{\bar{q}}\, z)^2\right] = \chi_1,$$

which is finite because ϕ' is polynomially bounded. For $c \in [0,1)$ the sum in (5.7) increases to $\sum_k k\beta_k$ as $c \uparrow 1$, by monotone convergence, so the left derivative at 1 is χ_1. For $c \in [-1,1]$, $|1 + c + \cdots + c^{k-1}| \le k$ bounds the sum by χ_1, which proves (5.6).

Part (ii). Iterating (5.6) gives $1 - c^\ell \le \chi_1^{\ell-1}(1 - c^1) \le 2\chi_1^{\ell-1}$. Since ϕ is not constant, $\beta_k > 0$ for some $k \ge 1$, so for $c \in (-1,1)$ the first expression in (5.7) is positive and $C(c) \ge \beta_0 - \sum_{k\ge1} \beta_k |c|^k > -1$: the interval $(-1,1)$ is invariant, and $c^\ell < 1$ for every ℓ. The ratio $(1 - c^{\ell+1})/(1 - c^\ell)$ is the sum in (5.7) at $c = c^\ell$. As $c^\ell \to 1$, each term $\beta_k(1 + c^\ell + \cdots + (c^\ell)^{k-1})$ tends to $k\beta_k$ and is bounded by $k\beta_k$, which is summable; by dominated convergence the ratio tends to χ_1, and the logarithms of the ratios have Cesàro means converging to $\log \chi_1$, which is the claim.

Part (iii). If ϕ is not affine, then $\beta_k > 0$ for some $k \ge 2$ (if $a_k = 0$ for all $k \ge 2$, then $\bar\phi$ is affine), so for $c \in [0,1)$ the sum in (5.7) is strictly less than $\sum_k k\beta_k = 1$, and $C(c) > c$. Since C maps $[0,1]$ into itself, c^ℓ increases to a fixed point in $(c^1, 1]$, and $C(c) > c$ on $[0,1)$ leaves only 1. The ratio tends to $\chi_1 = 1$ by the argument of part (ii).

Part (iv). Let $g_1(c) = C(c) - c$, convex and continuous on $[0,1]$, with $g_1(0) = \beta_0 \ge 0$, $g_1(1) = 0$, and left derivative $\chi_1 - 1 > 0$ at 1; so $g_1 < 0$ on some interval $(1 - \delta, 1)$. By the intermediate value theorem g_1 has a zero in $[0, 1 - \delta]$; let $\bar c$ be the largest one. On $(\bar c, 1)$, g_1 has no zero and takes negative values, so $g_1 < 0$ there. If $0 \le c_0 < \bar c$ were another zero, then convexity on $[c_0, 1]$ with $g_1(c_0) = g_1(1) = 0$ would give $g_1 \le 0$ on $[c_0, 1]$, and the convex function g_1 would have the interior maximum $g_1(\bar c) = 0$ on this interval, forcing $g_1 \equiv 0$ on $[c_0, 1]$, which contradicts $g_1 < 0$ near 1. So $\bar c$ is the unique fixed point in $[0,1)$, $g_1 > 0$ on $[0, \bar c)$, and $g_1 < 0$ on $(\bar c, 1)$. Because C is nondecreasing on $[0,1]$, $c < \bar c$ implies $c < C(c) \le \bar c$, and $c > \bar c$ implies $\bar c \le C(c) < c$; so c^ℓ moves monotonically toward $\bar c$ and converges to a fixed point in $[0,1)$, which is $\bar c$. By convexity, $C'(\bar c) \le \big(C(c) - C(\bar c)\big)/(c - \bar c) < 1$ for each $c \in (\bar c, 1)$, where the last inequality is $g_1(c) < 0$. Finally, C is differentiable at $\bar c < 1$, so for $\rho \in (C'(\bar c), 1)$ there is $\eta > 0$ with $|C(c) - \bar c| \le \rho |c - \bar c|$ for $|c - \bar c| \le \eta$; once c^ℓ is within η of $\bar c$ the distance decreases by the factor ρ per layer. ∎

Theorem 5.7 says that a deep random network has two fates. In the ordered phase all inputs end up with the same representation at a geometric rate, and the deep network ignores its input. In the chaotic phase two inputs, however close, end up with correlation $\bar c < 1$, a correlation that the network assigns to every pair of distinct inputs: nearby points are pulled apart, and at depth the network behaves like a random function of its input. By Price's theorem (Lemma A.5), $C'(c) = \sigma_w^2 \mathbb{E}[\phi'(\sqrt{\bar q}u)\phi'(\sqrt{\bar q}v)]$, so the slope at $\bar c$ is a correlation of derivatives, as χ_1 is a mean square of derivatives.

The rate of approach to the stable fixed point defines the *depth scale*

$$\xi = -\frac{1}{\log C'(\bar c)}, \qquad \bar c = 1 \text{ in the ordered phase, so that } \xi = -\frac{1}{\log \chi_1} \text{ there.} \tag{5.8}$$

It is the ξ_c of Observation 5.1. The *critical line* is the curve $\chi_1 = 1$ in the plane (σ_w^2, σ_b^2); for tanh it starts at $(1, 0)$, where $\bar q = 0$, and χ_1 is evaluated

at the fixed length $\bar{q}(\sigma_w^2, \sigma_b^2)$ of Proposition 5.6. How does ξ diverge as the line is approached?

CRITICAL EXPONENT OF THE DEPTH SCALE. Let $C''(1) = \sum_k k(k-1)\beta_k = \sigma_w^2 \bar{q}\, \mathbb{E}[\phi''(\sqrt{\bar{q}}\, z)^2]$ be finite and positive, and expand near $c = 1$: $C(1-\varepsilon) = 1 - \chi_1 \varepsilon + \frac{1}{2} C''(1)\varepsilon^2 + O(\varepsilon^3)$. In the ordered phase $\xi = -1/\log \chi_1 \approx 1/(1-\chi_1)$ as $\chi_1 \uparrow 1$. In the chaotic phase, the fixed point $\bar{c} = 1 - \bar{\varepsilon}$ solves $\chi_1 \bar{\varepsilon} \approx \bar{\varepsilon} + \frac{1}{2} C''(1)\bar{\varepsilon}^2$, so $\bar{\varepsilon} \approx 2(\chi_1 - 1)/C''(1)$, and $C'(\bar{c}) \approx \chi_1 - C''(1)\bar{\varepsilon} \approx 2 - \chi_1$. Both sides give

$$\xi \approx \frac{1}{|\chi_1 - 1|} \qquad (\chi_1 \to 1),$$

a divergence with exponent 1, the value of mean-field theory. ◇

The derivation assumes that $\bar{c} \to 1$ as $\chi_1 \downarrow 1$, which fails at the end point $\sigma_b = 0$ of the tanh line, where $\bar{c} = 0$ throughout the chaotic phase (Exercise 5.3). The numerics of Figure 5.1(b) confirm the formula on both sides at $\sigma_b^2 = 0.05$: at $\chi_1 = 0.998$, $\xi = 496$ against $1/(1-\chi_1) = 497$, and at $\chi_1 = 1.0029$, $\xi = 349$ against 345.

RELU AT HE INITIALIZATION. For $\phi = \mathrm{ReLU}$ and $\sigma_b = 0$, the length map is $q \mapsto \frac{\sigma_w^2}{2} q$, so lengths neither explode nor vanish exactly when $\sigma_w^2 = 2$, the initialization of He et al. (2015). Then $\chi_1 = 2\,\mathbb{E}[\mathbf{1}\{z > 0\}] = 1$: He initialization is critical. The correlation map of the arc-cosine kernel of Cho and Saul (2009) decides how a critical ReLU network forgets.

Theorem 5.8 (ReLU at He initialization). *Let $\phi = \mathrm{ReLU}$, $\sigma_w^2 = 2$, and $\sigma_b = 0$. Then $\mathcal{V}(q) = q$ for every $q \geq 0$, and for every $\bar{q} > 0$ the correlation map is*

$$C(c) = \frac{\sqrt{1-c^2} + (\pi - \arccos c)\, c}{\pi}, \qquad c \in [-1, 1], \tag{5.9}$$

with $C'(c) = 1 - \frac{\arccos c}{\pi}$. For $\varepsilon \in [0, 1]$,

$$C(1-\varepsilon) = 1 - \varepsilon + \frac{2\sqrt{2}}{3\pi}\varepsilon^{3/2} + R(\varepsilon), \qquad 0 \leq R(\varepsilon) \leq \frac{\pi - 2}{5\sqrt{2}\,\pi}\varepsilon^{5/2}. \tag{5.10}$$

For every $c^1 \in [-1, 1)$, c^ℓ increases to 1 from $\ell = 2$ on, and

$$\lim_{\ell \to \infty} \ell^2 (1 - c^\ell) = \frac{9\pi^2}{2}. \tag{5.11}$$

PROOF. The plan is to compute C by integrating its derivative from $c = 0$, then to expand near $c = 1$, and finally to linearize the recursion in the variable $(1-c)^{-1/2}$.

Lengths. By symmetry of z, $\mathbb{E}[\mathrm{ReLU}(\sqrt{q}z)^2] = \frac{q}{2}\mathbb{E}z^2 = \frac{q}{2}$, so $\mathcal{V}(q) = 2 \cdot \frac{q}{2} = q$.

The map. By homogeneity $\mathbb{E}[\mathrm{ReLU}(\sqrt{\bar{q}}u)\,\mathrm{ReLU}(\sqrt{\bar{q}}v)] = \bar{q}\,\kappa(c)$ with $\kappa(c) = \mathbb{E}[\mathrm{ReLU}(u)\,\mathrm{ReLU}(v)]$, so $C = 2\kappa$. Price's theorem (Lemma A.5) applies to ReLU, which is absolutely continuous with derivative $\mathbf{1}\{t > 0\}$, and gives

$\kappa'(c) = \Pr[u > 0, v > 0]$ for $c \in (-1, 1)$. Write $u = \langle e, g\rangle$ and $v = \langle e', g\rangle$ for a standard Gaussian $g \in \mathbb{R}^2$ and unit vectors e, e' at angle $\theta = \arccos c$. The direction of g is uniform on the circle, and the set of directions with both inner products positive is an arc of length $\pi - \theta$, so $\kappa'(c) = (\pi - \arccos c)/(2\pi)$. At $c = 0$, independence gives $\kappa(0) = (\mathbb{E}\,\mathrm{ReLU}(u))^2 = 1/(2\pi)$. Integrating with the antiderivative $s \arccos s - \sqrt{1 - s^2}$ of $\arccos s$,

$$\kappa(c) = \frac{1}{2\pi} + \frac{1}{2\pi}\int_0^c (\pi - \arccos s)\,\mathrm{d}s = \frac{\sqrt{1 - c^2} + (\pi - \arccos c)\,c}{2\pi},$$

first for $c \in (-1, 1)$ and then on $[-1, 1]$ by continuity (Corollary A.4). This is (5.9), and $C' = 2\kappa'$.

The expansion. For $s \in [0, 1]$ the series $\arcsin s = s + \sum_{n\ge1} \gamma_n s^{2n+1}$ has nonnegative coefficients, so $(\arcsin s - s)/s^3$ is nondecreasing and $0 \le \arcsin s - s \le (\frac{\pi}{2} - 1)s^3$. Since $\arccos(1 - t) = 2\arcsin\sqrt{t/2}$ for $t \in [0, 2]$,

$$\arccos(1 - t) = \sqrt{2t} + \varrho(t), \qquad 0 \le \varrho(t) \le \frac{\pi - 2}{2\sqrt{2}}\,t^{3/2}.$$

Integrating C' from $1 - \varepsilon$ to 1 and substituting $s = 1 - t$,

$$1 - C(1 - \varepsilon) = \int_0^\varepsilon \left(1 - \frac{\arccos(1 - t)}{\pi}\right)\mathrm{d}t = \varepsilon - \frac{2\sqrt{2}}{3\pi}\varepsilon^{3/2} - \frac{1}{\pi}\int_0^\varepsilon \varrho(t)\,\mathrm{d}t,$$

which is (5.10) with $R(\varepsilon) = \frac{1}{\pi}\int_0^\varepsilon \varrho$, and the bound on ϱ integrates to the bound on R.

Convergence. For $c < 1$, $C' < 1$ on $(c, 1)$, so $1 - C(c) < 1 - c$, that is, $C(c) > c$; also $C \le 1$, and $C(c) \ge C(-1) = 0$ because C is nondecreasing. So $c^2 \in [0, 1)$, the sequence $(c^\ell)_{\ell\ge2}$ increases, and its limit is a fixed point of C, of which 1 is the only one.

The rate. Let $\varepsilon_\ell = 1 - c^\ell \to 0$, $\varepsilon_\ell > 0$, and $u_\ell = \varepsilon_\ell^{-1/2}$; write $a = \frac{2\sqrt{2}}{3\pi}$. By (5.10), $\varepsilon_{\ell+1} = \varepsilon_\ell(1 - \delta_\ell)$ with $\delta_\ell = a\varepsilon_\ell^{1/2} + R(\varepsilon_\ell)/\varepsilon_\ell$, and $0 \le \delta_\ell - a\varepsilon_\ell^{1/2} \le \varepsilon_\ell^{3/2}$. For $\delta \in [0, \frac{1}{2}]$, Taylor's theorem gives $(1 - \delta)^{-1/2} = 1 + \frac{\delta}{2} + r(\delta)$ with $0 \le r(\delta) \le 2\delta^2$. Once $\delta_\ell \le \frac{1}{2}$,

$$u_{\ell+1} - u_\ell = u_\ell\left(\frac{\delta_\ell}{2} + r(\delta_\ell)\right) = \frac{a}{2} + O\big(\varepsilon_\ell^{1/2}\big),$$

because $u_\ell\delta_\ell = a + O(\varepsilon_\ell)$ and $u_\ell\delta_\ell^2 = O(\varepsilon_\ell^{1/2})$. Hence $u_{\ell+1} - u_\ell \to a/2$, and by the Stolz–Cesàro theorem $u_\ell/\ell \to a/2$. So $\ell^2\varepsilon_\ell = (\ell/u_\ell)^2 \to 4/a^2 = \frac{4\cdot9\pi^2}{8} = \frac{9\pi^2}{2}$. ∎

ReLU networks at He initialization decorrelate polynomially: no depth scale, but a power law. The critical tanh network also decays as a power, but a different one: with $C''(1)$ finite, $\varepsilon_{\ell+1} = \varepsilon_\ell - \frac{1}{2}C''(1)\varepsilon_\ell^2 + O(\varepsilon_\ell^3)$ gives $1 - c^\ell \sim 2/(C''(1)\ell)$ (Exercise 5.4). The exponent at the critical point depends on the smoothness of ϕ at the origin, through the first nonlinear term of C at 1: ε^2 for tanh, $\varepsilon^{3/2}$ for ReLU.

GRADIENTS. Let $\mathcal{L}$ be a loss that depends on the network through h^L, and let $\delta^\ell = \partial\mathcal{L}/\partial h^\ell$. The chain rule gives $\delta^\ell = D^\ell (W^{\ell+1})^\top \delta^{\ell+1}$ with $D^\ell = \operatorname{diag}(\phi'(h^\ell))$.

GRADIENT NORMS. Assume the *gradient independence assumption*: in second moments, the matrix $W^{\ell+1}$ of the backward pass may be replaced by an independent copy, and $\delta^{\ell+1}$ is independent of h^ℓ. Then, conditionally on $\delta^{\ell+1}$ and h^ℓ, each coordinate $\sum_i W_{ij}^{\ell+1}\delta_i^{\ell+1}$ has variance $\frac{\sigma_w^2}{m_\ell}\|\delta^{\ell+1}\|^2$, and summing over j,

$$\mathbb{E}\|\delta^\ell\|^2 = \sigma_w^2\,\mathbb{E}\Big[\frac{1}{m_\ell}\sum_j \phi'(h_j^\ell)^2\Big]\,\mathbb{E}\|\delta^{\ell+1}\|^2 \;\longrightarrow\; \chi_1\,\mathbb{E}\|\delta^{\ell+1}\|^2$$

at infinite width and length $\bar{q}$, by the law of large numbers of Theorem 5.4. Iterating, $\mathbb{E}\|\delta^\ell\|^2 \approx \chi_1^{L-\ell}\,\mathbb{E}\|\delta^L\|^2$: gradients vanish geometrically in the ordered phase, explode in the chaotic phase, and keep their mean squared size on the critical line, with depth scale $\xi_\nabla = -1/\log\chi_1$. ◇

The assumption is false at finite width, where the forward and backward passes share the same matrices, but it gives the correct answer in the infinite-width limit for this network and many others: Yang (2020), in a preprint, gave a simple checkable condition under which computations with it are exact in the limit, and showed that they can be wrong when the condition fails.[5] Two cases are exactly solvable at every width, and they show what the mean hides.

Theorem 5.9 (Jacobian norms at finite width). *Let all widths equal m, $\sigma_b = 0$, and fix an input $x \neq 0$ and a direction $\dot{x} \neq 0$ in $\mathbb{R}^d$. Let $u^\ell = \partial_t h^\ell(x + t\dot{x})|_{t=0}$, so that $u^1 = W^1\dot{x}$ and $u^{\ell+1} = W^{\ell+1}D^\ell u^\ell$, and let $Y_\ell = \|D^\ell u^\ell\|^2$, the squared norm of the derivative of $a^\ell = \phi(h^\ell)$ along $\dot{x}$.*

(i) If $\phi(t) = t$ (a deep linear network), then $Y_L = Y_1\prod_{\ell=1}^{L-1} F_\ell$, where $F_1, \dots, F_{L-1}$ are i.i.d., independent of Y_1, with law $\frac{\sigma_w^2}{m}\chi_m^2$.

(ii) If $\phi = \mathrm{ReLU}$ with $D^\ell = \operatorname{diag}(\mathbf{1}\{h^\ell > 0\})$ and $\sigma_w^2 = 2$, then the same holds with F_ℓ of law $\frac{2}{m}\chi_B^2$, where $B \sim \mathrm{Bin}(m, \frac{1}{2})$ and $\chi_0^2 = 0$.

In both cases $\mathbb{E}Y_L = \chi_1^{L-1}\mathbb{E}Y_1$, and

$$\frac{\mathbb{E}Y_L^2}{(\mathbb{E}Y_L)^2} = \frac{\mathbb{E}Y_1^2}{(\mathbb{E}Y_1)^2}\Big(1 + \frac{\kappa}{m}\Big)^{L-1}, \qquad \kappa = 2 \text{ (linear)}, \quad \kappa = 5 \text{ (ReLU)}.$$

PROOF. The plan is to show that, conditionally on $\mathcal{F}_\ell$, the ratio $Y_{\ell+1}/Y_\ell$ has a fixed law that does not depend on $\mathcal{F}_\ell$; independence of the factors and the moment formulas follow.

Linear case. Here $D^\ell = I_m$ and $u^{\ell+1} = W^{\ell+1}u^\ell$. Conditionally on $\mathcal{F}_\ell$, the coordinates of $W^{\ell+1}u^\ell$ are independent $\mathcal{N}(0, \frac{\sigma_w^2}{m}\|u^\ell\|^2)$, so $Y_{\ell+1} = Y_\ell F_\ell$ with $F_\ell \sim \frac{\sigma_w^2}{m}\chi_m^2$ independent of $\mathcal{F}_\ell$ (if $Y_\ell = 0$, then $Y_{\ell+1} = 0$ and F_ℓ is drawn independently). Since $Y_1, F_1, \dots, F_{\ell-1}$ are $\mathcal{F}_\ell$-measurable, the factors are independent of each other and of Y_1.

[5] For tanh networks of depth 31 at $\sigma_b^2 = 0.05$, with h^1 at length $\bar{q}$, we measured the ratio $\|\delta^1\|^2/\|\delta^{31}\|^2$ over 30 backward layers. At criticality, averaged over 40 networks, it was 1.12 at width 800 and 1.03 at width 1600, against $\chi_1^{30} = 1$, with a spread across networks of 45% and 23% of the mean. Over 10 networks of width 800 it was 0.0060 against 0.0064 at $\sigma_w^2 = 1.2$, and 249 against 297 at $\sigma_w^2 = 3$.

ReLU case. Conditionally on $\mathcal{F}_\ell$, Lemma 5.3 (with $\mathcal{X} = \{x\}$, extended to the pair (h, u) by the same argument) shows that the pairs $(h_i^{\ell+1}, u_i^{\ell+1})$, $i \le m$, are i.i.d. centered Gaussian, with $\mathrm{Var}(u_i^{\ell+1}) = \frac{2}{m} Y_\ell$ and $\mathrm{Var}(h_i^{\ell+1}) = \frac{2}{m}\|a^\ell\|^2$. If $a^\ell = 0$, then $D^\ell = 0$, so $Y_\ell = 0$ and $Y_{\ell+1} = 0$. Otherwise $\mathrm{Var}(h_i^{\ell+1}) > 0$, and the law of a centered Gaussian pair is invariant under $(h, u) \mapsto (-h, -u)$, so for every Borel set A, $\Pr[h > 0, u^2 \in A] = \Pr[h < 0, u^2 \in A] = \frac{1}{2}\Pr[u^2 \in A]$: the sign of $h_i^{\ell+1}$ is a fair coin independent of $(u_i^{\ell+1})^2$. Hence $Y_{\ell+1} = \sum_i \mathbf{1}\{h_i^{\ell+1} > 0\}(u_i^{\ell+1})^2$ is, conditionally on $\mathcal{F}_\ell$, a sum of $B \sim \mathrm{Bin}(m, \frac{1}{2})$ independent squares of $\mathcal{N}(0, \frac{2}{m} Y_\ell)$ variables, that is, Y_ℓ times a variable of law $\frac{2}{m}\chi^2_B$. Independence follows as in the linear case. Finally, $x \neq 0$ and $\dot{x} \neq 0$ make $\mathbb{E}Y_1 > 0$, so the normalized moments are defined.

Moments. For the linear case $\mathbb{E}F = \sigma_w^2$ and $\mathbb{E}F^2 = \frac{\sigma_w^4}{m^2}(m^2 + 2m) = \sigma_w^4(1 + \frac{2}{m})$, and $\chi_1 = \sigma_w^2$. For ReLU, $\mathbb{E}[\chi^2_B \mid B] = B$ and $\mathbb{E}[(\chi^2_B)^2 \mid B] = B^2 + 2B$, with $\mathbb{E}B = \frac{m}{2}$ and $\mathbb{E}B^2 = \frac{m}{4} + \frac{m^2}{4}$; so $\mathbb{E}F = 1 = \chi_1$ and $\mathbb{E}F^2 = \frac{4}{m^2}\left(\frac{m^2}{4} + \frac{m}{4} + m\right) = 1 + \frac{5}{m}$. Independence gives $\mathbb{E}Y_L = \mathbb{E}Y_1(\mathbb{E}F)^{L-1}$ and $\mathbb{E}Y_L^2 = \mathbb{E}Y_1^2(\mathbb{E}F^2)^{L-1}$. ∎

At criticality the mean of Y_L/Y_1 is exactly 1 at every width, as the heuristic predicts, but $\log(Y_L/Y_1)$ is a sum of $L - 1$ independent terms, each with mean close to $-\kappa/(2m)$ and variance close to κ/m (for ReLU, conditionally on $B \ge 1$ in every layer, an event of probability at least $1 - (L-1)2^{-m}$): the typical network has $\log(Y_L/Y_1) \approx -\frac{\kappa}{2m}L \pm \sqrt{\kappa L/m}$. The mean is carried by rare networks with large gradients, and the parameter that controls the typical behavior is L/m, which the infinite-width limit sets to zero. For ReLU this is the result of Hanin (2018) and Hanin and Nica (2020), who prove the analogous statements for the input–output Jacobian with variable widths, where $\sum_\ell 1/m_\ell$ replaces L/m.

DYNAMICAL ISOMETRY. Criticality fixes the mean squared singular value of the input–output Jacobian, not its spread. A network has *dynamical isometry* if all these singular values concentrate near 1, as they do exactly for deep linear networks with orthogonal weights (Saxe et al. 2014). With free probability, Pennington et al. (2017) showed that ReLU networks cannot achieve it, whereas tanh networks can, with orthogonal weights on the critical line near $\bar{q} = 0$, and then learn faster.

5.5 Reading the solution

THE PHASE DIAGRAM. Figure 5.1(a) shows the critical line of tanh networks, computed by solving $\chi_1 = 1$ with $\bar{q}$ from Proposition 5.6. It starts at $(\sigma_w^2, \sigma_b^2) = (1, 0)$ and bends to larger weight variances as the bias variance grows: a bias adds a common component to all pre-activations, which raises $\bar{q}$, pushes the neurons into the saturated part of tanh, and lowers $\mathbb{E}[\phi'^2]$, so more weight variance is needed to keep $\chi_1 = 1$. Left of the line is order and right of it is chaos, and the contours of ξ crowd toward the line from both sides. Panel (b) shows ξ along the cut $\sigma_b^2 = 0.05$, where the critical point is $\sigma_w^2 = 1.761$: the divergence is the critical slowing down of a second-order transition, and the curve $1/|\chi_1 - 1|$ of the heuristic derivation is indistinguishable from it near the line. Combined with Observation 5.1,

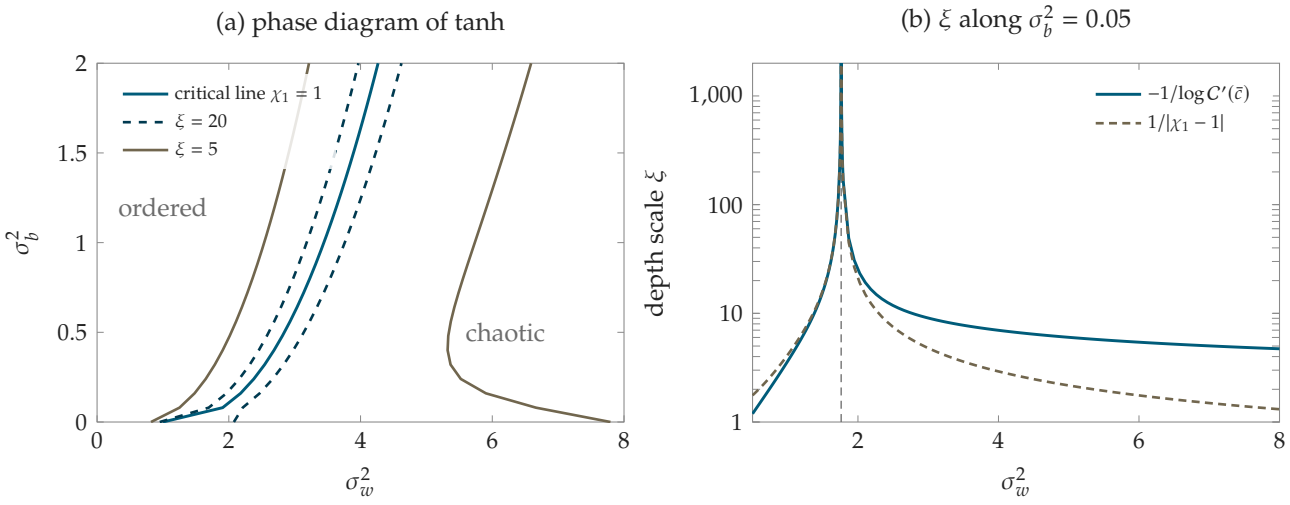


Figure 5.1. Infinitely wide tanh networks, computed by quadrature (theory only). (a) The critical line $\chi_1 = 1$ in the plane of weight and bias variances, and the contours $\xi = 5$ and $\xi = 20$ of the depth scale (5.8) on both sides of it. (b) The depth scale along $\sigma_b^2 = 0.05$ (critical point $\sigma_w^2 = 1.761$, dashed), with the approximation $1/|\chi_1 - 1|$ of the heuristic derivation.

the plot is a map of trainable depth: at $\sigma_w^2 = 1.2$, where $\xi = 5.9$, networks much deeper than $6\xi \approx 36$ layers should fail; at $\sigma_w^2 = 1.7$, where $\xi = 75$, several hundred layers should train.

Correlations, layer by layer. Figure 5.2 follows two inputs with initial correlation 0.6 through tanh networks of width 1000 in the three phases, against the iterated correlation map. In the ordered phase ($\sigma_w^2 = 1.2$, $\chi_1 = 0.845$) the correlation approaches 1 geometrically, the gap $1 - c^\ell$ shrinking by about χ_1 per layer; at criticality it approaches 1 like $6.2/\ell$, the power law of Exercise 5.4; in the chaotic phase ($\sigma_w^2 = 4$) it falls to $\bar{c} = 0.166$. Every average lies within about three standard errors of its curve (at most 3.1). The spread across networks, a finite-width effect, is smallest in the ordered phase and largest in the chaotic one.

The ReLU constant. Figure 5.3(a) shows $\ell^2(1 - c^\ell)$ for two orthogonal inputs of a ReLU network at He initialization. The limit $9\pi^2/2 = 44.41$ is approached slowly: the proof gives $u_{\ell+1} - u_\ell = \frac{a}{2} + \frac{3a^2}{8u_\ell} + O(u_\ell^{-2})$, hence $u_\ell = \frac{a}{2}\ell + \frac{3a}{4}\log\ell + O(1)$ and a relative correction of about $3\log\ell/\ell$ to $\ell^2(1 - c^\ell)$; at $\ell = 1000$ the exact iterate gives 43.2, at $\ell = 10^4$ it gives 44.26. Simulations at width 1000 agree with the iterated map up to depth 181, within two standard errors.

Depth over width. Figure 5.3(b) tests Theorem 5.9 at width $m = 50$ up to depth 151: the variance of $\log(Y_L/Y_1)$ grows linearly in L/m, with slope close to 5 for ReLU and 2 for linear networks. At $L = m$ for ReLU, the median of Y_L/Y_1 is about $e^{-5/2} \approx 0.08$ times its mean, and typical networks scatter around the median by a factor of order $e^{\sqrt{5}} \approx 9$ either way, although the network is critical on average. The same parameter controls the forward

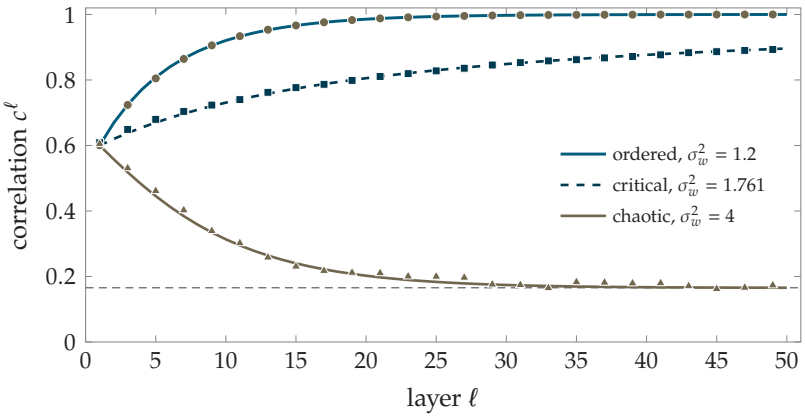


Figure 5.2. Correlation of the pre-activations of two inputs in tanh networks with $\sigma_b^2 = 0.05$, starting from $c^1 = 0.6$ at the fixed length $\bar{q}$. Curves: iterates of the correlation map (5.3). Marks: networks of width $m = 1000$, averages over 50 networks of the empirical cosine $\langle h^\ell(x), h^\ell(x')\rangle/(\|h^\ell(x)\|\|h^\ell(x')\|)$, every second layer (seed 20261017); the standard deviation across networks is about 0.03 at criticality and 0.07 in the chaotic phase. The dashed line is the chaotic fixed point $\bar{c} = 0.166$.

correlations: with the exact two-input chain at widths 25 to 200 and 1000 networks each, the mean of $1 - c^L$ at $L = m$ stays within 20% of the iterated map, a Monte Carlo error inflated by a heavy tail, but its median is only a quarter to a third of the mean (Exercise 5.10).

THE PHYSICS. The order parameters q and c play the role of the magnetization of a mean-field magnet, depth plays the role of time, and ξ is a correlation length. The chaotic phase has a positive Lyapunov exponent: by $1 - c^{\ell+1} \approx \chi_1(1 - c^\ell)$, infinitesimally close inputs separate by a factor $\sqrt{\chi_1}$ per layer. Mean-field theory is exact for the same reason as in the Curie–Weiss model: the interaction range, the width, is infinite.

5.6 What the model gets wrong

WHAT WAS THROWN AWAY. *Finite width.* The limit sets $L/m = 0$, and every fluctuation of order $\sqrt{L/m}$ disappears with it (Theorem 5.9). *Training.* The theory describes the network at time zero; it predicts where gradient descent can start, not what it finds, how fast, or whether the weights move far from their initial values (Chapter 6). *Architecture.* No residual connections, no normalization, no convolutions, no attention; each changes the recursion, and some remove the phase transition altogether. *Data.* Only pairwise inner products of inputs enter, so the theory cannot distinguish a dataset from a set of points with the same Gram matrix, and it says nothing about which functions of the data are easy to learn.

WHERE IT BREAKS. *Depth comparable to width.* Gradient statistics at finite width are controlled by $\sum_\ell 1/m_\ell$, which is L/m for constant widths: for ReLU networks the variance of the squared entries of the input–output

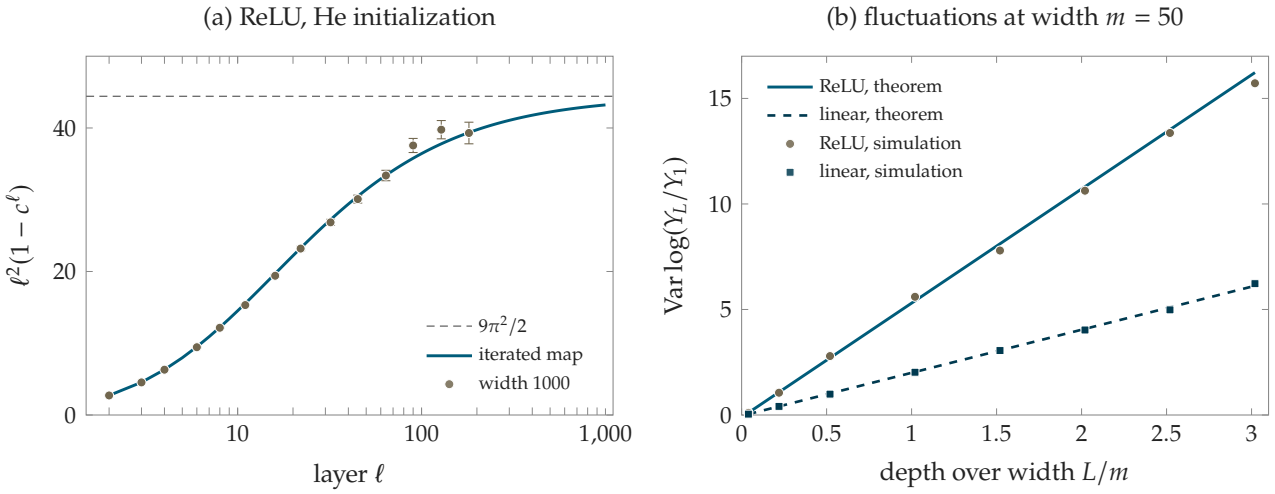


Figure 5.3. Critical networks at large and at finite width. (a) ReLU at $\sigma_w^2 = 2$, $\sigma_b = 0$, two orthogonal inputs: the iterated arc-cosine map (5.9) and its limit (5.11); points are averages over 400 networks of width 1000, with standard errors (seed 20261017). (b) Variance of the log squared Jacobian norm $\log(Y_L/Y_1)$ of Theorem 5.9. Lines: $(L-1)\,\mathrm{Var}\log F$, computed exactly from the laws of the factors ($\mathrm{Var}\log F = 0.108$ for ReLU, conditioned on $B \geq 1$, and 0.041 for linear networks at $m = 50$). Points: 2000 networks of width 50 with full Gaussian weight matrices.

Jacobian grows exponentially in this sum (Hanin 2018), and the logarithm of the squared norm of a Jacobian–vector product is approximately Gaussian, with mean and variance of order $\sum_\ell 1/m_\ell$, when depth and width grow together (Hanin and Nica 2020). Theorem 5.9 is the exactly solvable shadow of this result. When $L \gtrsim m$ the infinite-width kernel is the mean of a broad distribution and not the typical value: the network that is actually initialized is not the Gaussian process. Roberts et al. (2022) organize the finite-width corrections as an expansion in $1/m$, and find that criticality is necessary to keep the corrections from growing exponentially with depth, but that even at criticality the effective expansion parameter is the ratio L/m.

Criticality is necessary, not sufficient. Critical ReLU networks train more slowly than critical sigmoidal networks with orthogonal weights, because the spread of the Jacobian singular values, which the model does not compute, grows with depth for ReLU (Pennington et al. 2017). Inside the trainable region, which initialization trains fastest depends on the activation beyond χ_1 (Hayou et al. 2019).

Kernels are not networks. The kernel K^L is the covariance of the infinitely wide Bayesian network (Lee et al. 2018). In the large study of Lee et al. (2020), these kernels (and the tangent kernels of Chapter 6) outperformed finite fully connected networks trained in the standard way, but underperformed finite convolutional networks; large learning rates and weight decay, which help finite networks, break the correspondence.

Normalization changes the answer. The model suggests that a well-chosen activation and variance cure exploding gradients. Yang et al. (2019) showed that in fully connected networks with batch normalization and no skip connections, gradients explode exponentially with depth at initialization

whatever the activation and the weight variance: the normalization itself causes the explosion, and such networks are untrainable at large depth.

Attention collapses. A stack of self-attention layers without skip connections and feedforward blocks converges to a rank-one output doubly exponentially in depth (Dong et al. 2021), an ordered phase that variance tuning cannot leave. In transformers at initialization, this rank collapse makes the gradients of queries and keys vanish, and Noci et al. (2022) prevent it by scaling the residual branches with depth.

FINE STRUCTURE. The $1/m$ corrections are computed systematically by the effective theory of Roberts et al. (2022), in which the leading non-Gaussian correction to the law of the pre-activations is controlled at criticality by the ratio L/m. When depth and width go to infinity together with L/m fixed, the covariance of a network whose activation is shaped toward the identity at a rate tied to the width converges to the solution of a stochastic differential equation, the *neural covariance SDE* of Li et al. (2022): the deterministic recursion of this chapter becomes a diffusion, and L/m is its time. Residual networks do not have the exponential depth scales of plain networks, and their signal propagation is subexponential or polynomial (Yang and Schoenholz 2017); scaling the residual branches by $1/\sqrt{L}$ gives well-defined infinite-depth limits in which hyperparameters transfer across depth (Bordelon et al. 2024; Yang et al. 2024). Deep kernel shaping controls the whole correlation map, not only its slope at 1, and trains deep networks without skip connections or normalization (Martens et al. 2021). Open: a theory of trainability that predicts the constant in $L \lesssim 6\xi$, and the regime $L \sim m$ during training.

5.7 Model card

MODEL CARD	
PHENOMENON	Deep networks train only for well-scaled initializations; the maximal trainable depth is a multiple of a depth scale that diverges on an order–chaos boundary; He initialization for ReLU.
MODEL	Fully connected network with Gaussian weights $\mathcal{N}(0, \sigma_w^2/m)$ and biases $\mathcal{N}(0, \sigma_b^2)$, widths to infinity at fixed depth; order parameters q^ℓ and c^ℓ.
SOLUTION	Gaussian-process limit (Theorem 5.4); length map (Proposition 5.6); order and chaos from the dual power series (Theorem 5.7); ReLU at He initialization (Theorem 5.8); exact Jacobian fluctuations at finite width (Theorem 5.9).
PREDICTS	The critical line $\chi_1 = 1$; depth scales $\xi = -1/\log C'(\bar{c}) \approx 1/\lvert\chi_1 - 1\rvert$; gradient growth χ_1^L; $1 - c^\ell \sim 9\pi^2/(2\ell^2)$ for ReLU and $\sim 2/(C''(1)\ell)$ for smooth critical activations.
GETS WRONG	Finite width (fluctuations of order $\sqrt{L/m}$); training; residual connections, normalization, and attention; the spread of the Jacobian spectrum; the constant in $L \lesssim 6\xi$.

SUCCESSORS	$1/m$ effective field theory; neural covariance SDE at fixed L/m; dynamical isometry; residual and depth-μP scalings; kernel shaping.

The kernel K^L of this chapter is the starting point of Chapter 6: the neural tangent kernel is built from the same recursion together with the derivative map $c \mapsto C'(c)$, and the order–chaos analysis decides whether that kernel is degenerate at depth. The Hermite expansion of the activation, which turns a layer into a power series in the correlation, returns in Chapter 7, where the lowest nonzero Hermite coefficient of a link function sets the information exponent; in both chapters Gaussian inputs make a nonlinearity act diagonally on Hermite polynomials, as characters diagonalize convolution in Chapter 10. The spectrum of a product of random matrices in Theorem 5.9 is a relative of the Marchenko–Pastur law of Chapter 1.

Exercises

5.1 CONSOLIDATION. (The empirical kernel concentrates.) In the setting of Theorem 5.4, show that in the iterated limit $m_1 \to \infty, \dots, m_\ell \to \infty$ the empirical kernel $\widehat{G}^\ell(x, x')$ converges in probability to $K^{\ell+1}(x, x')$, and deduce that $\frac{1}{m_\ell}\|h^\ell(x)\|^2 \to q^\ell(x)$. *(Hint: apply the theorem to bounded continuous functions of $\frac{1}{J}\sum_{j\le J}\phi(h_j^\ell(x))\phi(h_j^\ell(x'))$, truncated, and use the exchangeability of the neurons.)*

5.2 CONSOLIDATION. (The Fourier transform of sech^2.) (a) Integrate $e^{i\omega t}\operatorname{sech}^2 t$, for $\omega > 0$, over the boundary of the rectangle with corners $\pm R$ and $\pm R + i\pi$, and show that $\int_{\mathbb{R}} \operatorname{sech}^2(t) e^{i\omega t}\,\mathrm{d}t = \pi\omega/\sinh(\pi\omega/2)$. (b) Deduce that $\mathbb{E}[\tanh^2(\sqrt{q}z)] = q - 2q^2 + O(q^3)$, and that for $\sigma_b = 0$ and $\sigma_w^2 = 1$ the length of a tanh network satisfies $\ell q^\ell \to \frac{1}{2}$. *(Hint: in (a), the only pole inside is the double pole at $i\pi/2$, where $\operatorname{sech}^2 t \approx -(t - i\pi/2)^{-2}$, and the top side of the rectangle is $-e^{-\pi\omega}$ times the bottom one.)*

5.3 CONSOLIDATION. ★ (The critical exponent.) Assume $C''(1) < \infty$. (a) Show that $\xi(1-\chi_1) \to 1$ as $\chi_1 \uparrow 1$. (b) For tanh with fixed $\sigma_b^2 > 0$, let σ_w^2 decrease to its critical value; show that the ratio of $1 - \bar{c}$ to $\chi_1 - 1$ tends to $2/C''(1)$, and that $\xi(\chi_1 - 1) \to 1$ as $\chi_1 \downarrow 1$. (c) For tanh with $\sigma_b = 0$, show that $\bar{c} = 0$ for every $\sigma_w^2 > 1$, and that $\chi_1 - 1$ and $1 - C'(0)$ are both of order $(\sigma_w^2 - 1)^2$. Why does (b) not apply?

5.4 CONSOLIDATION. (Smooth activations at criticality.) Let $\chi_1 = 1$ and $0 < C''(1) < \infty$, and let $C(1-\varepsilon) = 1 - \varepsilon + \frac{1}{2}C''(1)\varepsilon^2 + O(\varepsilon^3)$. Show that $\ell(1 - c^\ell) \to 2/C''(1)$, and that $C''(1) = \sigma_w^2 \bar{q}\,\mathbb{E}[\phi''(\sqrt{\bar{q}}z)^2]$. For tanh at $\sigma_b^2 = 0.05$, check numerically that $C''(1) = 0.324$, so that $1 - c^\ell \approx 6.17/\ell$, as in Figure 5.2. *(Hint: use $1/\varepsilon_{\ell+1} - 1/\varepsilon_\ell$.)*

5.5 VARIATION. (The error function.) Let $\phi = \operatorname{erf}$. (a) Show that for a centered Gaussian pair with variances q_1, q_2 and covariance q_{12}, $\mathbb{E}[\operatorname{erf}(z)\operatorname{erf}(z')] = \frac{2}{\pi}\arcsin\frac{2q_{12}}{\sqrt{(1+2q_1)(1+2q_2)}}$, the kernel of Williams (1997).

(b) Show that $\chi_1 = \frac{4\sigma_w^2}{\pi\sqrt{1+4\bar{q}}}$, that the critical point at $\sigma_b = 0$ is $\sigma_w^2 = \pi/4$, and that the critical line is the curve $\sigma_w^2 = \frac{\pi}{4}\sqrt{1+4\bar{q}}$, $\sigma_b^2 = \bar{q} - \frac{1}{2}\sqrt{1+4\bar{q}}\arcsin\frac{2\bar{q}}{1+2\bar{q}}$, $\bar{q} > 0$. *(Hint: in (a), use Price's theorem and* $\mathbb{E}[\phi'(z)\phi'(z')]$ *for Gaussian* ϕ'*.)*

5.6 VARIATION. (Leaky ReLU.) Let $\phi(t) = \max(t, \alpha t)$ with $\alpha \in [0, 1)$ and $\sigma_b = 0$. Show that lengths are preserved exactly when $\sigma_w^2 = 2/(1+\alpha^2)$, that this initialization is critical, and that

$$C(c) = \frac{(1+\alpha)^2 c + (1-\alpha)^2 \frac{2}{\pi}\left(\sqrt{1-c^2} + c\arcsin c\right)}{2(1+\alpha^2)}.$$

Deduce that $\ell^2(1-c^\ell) \to \frac{9\pi^2}{2}\left(\frac{1+\alpha^2}{(1-\alpha)^2}\right)^2$. Why does the constant blow up as $\alpha \to 1$?

5.7 VARIATION. (ReLU off the He point.) Let $\phi = \mathrm{ReLU}$. (a) For $\sigma_w^2 < 2$ and $\sigma_b^2 > 0$, show that $\bar{q} = \sigma_b^2/(1 - \sigma_w^2/2)$ and $\chi_1 = \sigma_w^2/2 < 1$: ReLU networks with a fixed length are never chaotic. (b) For $\sigma_w^2 > 2$, show that lengths grow geometrically, so that Theorem 5.7 does not apply; compute the correlation map at $\sigma_b = 0$ and show that it does not depend on σ_w^2.

5.8 VARIATION. (A residual network.) Let $h^{\ell+1} = h^\ell + \frac{\gamma}{\sqrt{L}} W^{\ell+1}\,\mathrm{ReLU}(h^\ell)$ with $W_{ij}^{\ell+1} \sim \mathcal{N}(0, 2/m)$ and no biases. (a) Derive the length recursion $q^{\ell+1} = (1 + \gamma^2/L)q^\ell$ at infinite width, and show that $q^L/q^1 \to e^{\gamma^2}$ as $L \to \infty$, while without the factor $1/\sqrt{L}$ the length grows like $(1+\gamma^2)^L$. (b) Derive the correlation recursion and show that, with the factor $1/\sqrt{L}$, the correlation of two inputs converges as $L \to \infty$ to the solution at time 1 of an ordinary differential equation in c; write it down.

5.9 SIMULATION. (Gradients in three phases.) Build tanh networks of width 800 and depth 31 with $\sigma_b^2 = 0.05$ and $\sigma_w^2 \in \{1.2, 1.761, 3\}$, start from first-layer pre-activations h^1 with i.i.d. $\mathcal{N}(0, \bar{q})$ coordinates, and backpropagate $\mathcal{L} = \langle v, h^{31}\rangle$ with $v \sim \mathcal{N}(0, I_m)$. Average $\|\delta^1\|^2/\|\delta^{31}\|^2$ over 10 networks and compare with χ_1^{30} (0.0064, 1, and 297). Observe agreement within the spread across networks, which is about 35–45% of the mean. Repeat at width 200 and observe the spread grow, as Theorem 5.9 suggests.

5.10 SIMULATION. (Typical versus mean.) Use the two-input Markov chain of Lemma 5.3 for ReLU at He initialization, with orthogonal inputs, widths $m \in \{25, 50, 100, 200\}$, depths $L \in \{m/4, m/2, m\}$, and 1000 networks each. (a) Observe that the mean of $1 - \hat{c}^L$ agrees with the iterated map up to Monte Carlo error, which is several percent at $L = m$ because the distribution has a heavy tail. (b) Observe that the ratio of the median of $1 - \hat{c}^L$ to its mean is about 0.75, 0.5, and 0.25 for $L/m = \frac{1}{4}, \frac{1}{2}, 1$, roughly independently of m, and that the relative standard deviation grows from about 0.8 to about 2. Explain why L/m controls both.

5.11 RESEARCH. (Trainability beyond signal propagation.) Observation 5.1 relates the trainable depth to 6ξ, with a constant fitted to experiments. Open: a solvable model of the first steps of training of a deep network near the critical line that predicts the trainable depth, including the constant and its dependence on the learning rate and the training budget.

5.12 RESEARCH. (The regime $L \sim m$.) The neural covariance SDE of Li et al. (2022) describes shaped networks at initialization when L/m is fixed. Open: an analogous limit for unshaped critical networks (ReLU at He initialization) for the law of the Jacobian spectrum and of c^L at fixed L/m, and its consequences for training.

Notes and References

Wide random networks became Gaussian processes with Neal (1996), for one hidden layer, and Williams (1997) computed the covariance functions for error-function and Gaussian units. Cho and Saul (2009) introduced the arc-cosine kernels as the covariances of infinitely wide rectifier layers and composed them into multilayer kernels, and Daniely et al. (2016) developed the duality between activations and their kernels, including the power-series form of the correlation map. The deep Gaussian-process limit was derived with sequential limits by Lee et al. (2018), who used it for Bayesian inference on MNIST and CIFAR-10, and proved for simultaneous limits by Matthews et al. (2018); the sequential proof given here follows the first route, made exact by Lemma 5.3. Hanin (2023) gives a new proof for general weight distributions, and Yang (2019) extends the limit to most modern architectures.

The order–chaos picture is due to Poole et al. (2016), who also showed that in the chaotic phase the length of a curve of inputs grows exponentially with depth, a measure of expressivity, and the depth scales and the link to trainability to Schoenholz et al. (2017), who also showed that dropout destroys the critical point. Pennington et al. (2017) and Xiao et al. (2018) added the spectrum of the Jacobian and orthogonal initializations, building on the orthogonal initializations of deep linear networks of Saxe et al. (2014).

The book of Roberts et al. (2022) develops the whole subject, criticality included, as a perturbation theory in $1/m$.[6]

TRANSLATING THE LITERATURE. Poole et al. (2016) and Schoenholz et al. (2017) write N_l for our width m_ℓ, q^l_{aa} and q^l_{ab} for our q^ℓ and $K^\ell(x,x')$, $\mathcal{V}$ for the length map, q^* and c^* for our fixed points $\bar{q}$ and $\bar{c}$, c^l_{ab} for our c^ℓ, and $\mathcal{D}z$ for the standard Gaussian measure; their χ_1 is ours, their χ_c is our $C'(\bar{c})$, their $\xi_c = -1/\log \chi_c$ is our ξ, and their ξ_∇ is our gradient depth scale $-1/\log \chi_1$. Lee et al. (2018) write $K^l(x,x')$ as we do, with N_l for the width. Roberts et al. (2022) write n_ℓ for the width, C_W and C_b for our σ_w^2 and σ_b^2, $K^{(\ell)}$ for the kernel, and use parallel and perpendicular susceptibilities $\chi_\parallel$ and $\chi_\perp$, which at a fixed point are our $\mathcal{V}'(\bar{q})$ and our χ_1, so that their criticality condition $\chi_\parallel = \chi_\perp = 1$ adds marginal stability of the length to our

[6] The Hermite form of the correlation map makes the order–chaos theorem a statement about power series with nonnegative coefficients summing to one, generating functions of probability distributions; Theorem 5.7(ii)–(iv) is then the extinction and survival dichotomy of a branching process, with χ_1 as the mean offspring number.

critical line. Hanin and Nica (2020) write n_j for the widths and $\beta = \sum_j 1/n_j$, which is L/m for constant widths.

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6 Lazy Training

The neural tangent kernel and what it cannot learn

THE MODEL — LINEARIZATION AROUND INITIALIZATION

IN THIS CHAPTER. Very wide networks, trained with small steps from a standard random initialization, fit their training data while each weight barely moves, and their predictions follow those of kernel regression with a fixed kernel, the neural tangent kernel. The model is the first-order Taylor expansion of the network around its initialization. We prove that a smooth model whose Jacobian is well conditioned at initialization and varies slowly converges to zero training loss inside a small ball, that a two-layer network of width of order n^2 satisfies these conditions, and that training then learns the eigenmodes of the kernel in order of their eigenvalues. We then prove the cost: no method that predicts from a fixed n-dimensional space of functions can learn a degree-k function of one unknown direction on the sphere from fewer than about $d^k/k!$ samples. Trained networks do far better because their features move.

6.1 The phenomenon: weights that do not move

A network with millions of parameters is trained by gradient descent on a loss that is not convex in them. For very wide networks trained with small steps, the optimization is nonetheless simple. The training loss decreases to zero at an exponential rate, and the weights at the end are close to the weights at the start: the *relative* change of each weight matrix decreases as the width grows. Du et al. (2019) proved that gradient descent drives the training loss of a two-layer ReLU network to zero at a linear rate when the width is a large polynomial in the number of examples and no two inputs are parallel, and Allen-Zhu et al. (2019) extended such guarantees to deep fully connected, convolutional, and residual networks.

Jacot et al. (2018) gave the phenomenon its name and its mathematics. During gradient descent the function computed by the network moves along the gradient of the loss in function space, preconditioned by a kernel, the *neural tangent kernel*

$$K_\theta(x, x') = \langle \nabla_\theta f(x;\theta), \nabla_\theta f(x';\theta) \rangle.$$

In a suitable scaling, as the widths of all hidden layers go to infinity the kernel at initialization converges to a deterministic limit K^∞, and it stays equal to K^∞ throughout training. The trained network is then kernel regression with K^∞. Lee et al. (2019) showed that a wide network trained by gradient descent stays close to its first-order Taylor expansion

around the initialization, in theory and in experiments with fully connected, convolutional, and residual networks. Because K^∞ is explicit, the infinitely wide network can be computed without training it; Arora et al. (2019a) computed the kernel of an 11-layer convolutional network with global average pooling and reached 77% test accuracy on CIFAR-10 with kernel regression, about ten points above the best result reported before for a Gaussian-process kernel.

The kernel picture also predicts *which* functions are learned fast: the error along each eigenfunction of the kernel decays at a rate equal to its eigenvalue, and these decrease with frequency. Rahaman et al. (2019) named the resulting tendency of networks to fit low frequencies first the *spectral bias*, and Basri et al. (2019) predicted the time needed to learn each frequency for data uniform on a sphere, where the eigenfunctions are spherical harmonics, and confirmed the prediction in shallow and deep networks.

Observation 6.1 (Basri et al. 2019; Du et al. 2019; Jacot et al. 2018; Lee et al. 2019). Train a network with m units per hidden layer, initialized with independent Gaussian weights of variance proportional to $1/(\text{fan-in})$, by gradient descent with a small step on the square loss. As m grows, the training loss converges to zero exponentially fast, the relative change $\|\theta_t-\theta_0\|/\|\theta_0\|$ of the parameters decreases to zero, the tangent kernel changes less and less during training, and the outputs on new inputs approach those of the network linearized at its initialization. Components of the target along eigenfunctions of the tangent kernel with larger eigenvalues are learned first.

The observation is robust in the regime it describes, and it is a regime, not a law of deep learning: convolutional networks forced to stay lazy perform worse than the same networks trained normally, and the tangent kernel of networks trained in the usual way changes fast early in training (Section 6.6).[1]

Two questions follow. Why does a nonlinear, nonconvex training problem behave like a linear one when the network is wide, and how wide is wide enough? And what does the linear behavior cost: which targets can a network that does not move its features never learn efficiently? The model of this chapter answers both.

[1] The term *lazy training* is due to Chizat et al. (2019), whose note first appeared in December 2018 under the title "A note on lazy training in supervised differentiable programming".

6.2 The model: linearization around initialization

Strip the network down to what gradient descent sees. Let $x_1, \dots, x_n$ be the training inputs and $y \in \mathbb{R}^n$ the labels, and collect the outputs of the network on the training set into one map

$$F\colon \mathbb{R}^P \to \mathbb{R}^n, \qquad F(\theta) = \big(f(x_1;\theta), \dots, f(x_n;\theta)\big),$$

with P the number of parameters. The training loss is $\widehat{\mathcal{L}}(\theta) = \frac{1}{2}\|F(\theta) - y\|^2$ (we do not divide by n, so that the kernel below has entries of order one), and we study its *gradient flow*

$$\dot\theta_t = -\nabla\widehat{\mathcal{L}}(\theta_t) = -J(\theta_t)^\top u_t, \qquad u_t = F(\theta_t) - y, \tag{6.1}$$

where $J(\theta) \in \mathbb{R}^{n \times P}$ is the Jacobian of F and u_t is the *residual*.[2] The chain rule gives the equation of motion of the outputs,

$$\dot{u}_t = J(\theta_t)\dot{\theta}_t = -K_t\, u_t, \qquad K_t = J(\theta_t)J(\theta_t)^\top \in \mathbb{R}^{n \times n}, \tag{6.2}$$

and the *tangent kernel* K_t is the matrix of the kernel K_{θ_t} of Section 6.1 on the training inputs. If K_t did not depend on t, then (6.2) would be a linear equation with the solution $u_t = \exp(-tK_0)u_0$, and every question about training would be a question about the spectrum of one fixed positive semidefinite matrix. This is exactly the dynamics of the linear model

$$f^{\text{lin}}(x;\theta) = f(x;\theta_0) + \langle \nabla_\theta f(x;\theta_0), \theta - \theta_0 \rangle, \tag{6.3}$$

the *linearized network*, whose tangent kernel is K_{θ_0} at every θ. Lazy training is the statement that the network and its linearization stay close.

The general theorem of Section 6.3 is about F; to see when its hypotheses hold we specialize to the simplest network.

Assumption 6.2 (Two-layer network). The inputs are unit vectors $x_1, \dots, x_n \in \mathbb{S}^{d-1}$, no two of which are parallel ($x_i \neq \pm x_j$ for $i \neq j$), and the labels satisfy $|y_i| \leq 1$. The network is

$$f(x;W) = \frac{1}{\sqrt{m}} \sum_{r=1}^{m} a_r\, \phi(\langle w_r, x \rangle),$$

with first-layer weights $w_1, \dots, w_m \in \mathbb{R}^d$, the rows of $W \in \mathbb{R}^{m \times d}$, and output weights $a_r \in \{-1, 1\}$ that are not trained. At initialization the w_r are independent $\mathcal{N}(0, I_d)$ and the a_r are independent uniform signs, independent of W. The activation ϕ is twice continuously differentiable, is not affine, and satisfies $|\phi'| \leq \beta$ and $|\phi''| \leq \beta$.

The activation tanh qualifies with $\beta = 1$; ReLU does not, because its derivative jumps, although the conclusions survive with more work (Du et al. 2019). The factor $m^{-1/2}$ is the *NTK parameterization*: with $\langle w_r, x \rangle \sim \mathcal{N}(0,1)$ each term is of order one, the output at initialization is a sum of m independent terms of mean zero, and the scale makes it of order one.[3] For this network $\partial f(x;W)/\partial w_r = m^{-1/2} a_r \phi'(\langle w_r, x \rangle)\, x$ and $a_r^2 = 1$, so the tangent kernel and its expectation at initialization are

$$\begin{aligned} K_W(x,x') &= \frac{\langle x, x' \rangle}{m} \sum_{r=1}^{m} \phi'(\langle w_r, x \rangle)\phi'(\langle w_r, x' \rangle), \\ K^\infty(x,x') &= \langle x, x' \rangle\, \kappa(\langle x, x' \rangle), \end{aligned} \tag{6.4}$$

where $\kappa(\rho) = \mathbb{E}[\phi'(g)\phi'(g')]$ for standard Gaussians g, g' with correlation ρ: for unit x, x' the pair $(\langle w, x \rangle, \langle w, x' \rangle)$ is such a pair with $\rho = \langle x, x' \rangle$. By the law of large numbers $K_{W_0} \to K^\infty$ as $m \to \infty$. We write K^∞ also for the $n \times n$ matrix $(K^\infty(x_i, x_j))_{ij}$, and $\lambda_{\min} = \lambda_{\min}(K^\infty)$.

[2] We write u for the residual because r is reserved for overlaps throughout the book.

[3] In the standard parameterization the factor $m^{-1/2}$ sits in the variance of the output weights instead. The two give the same network at initialization but gradients of different sizes, so the same step size means different things; Yang and Hu (2021) classify these choices.

DICTIONARY

IN THE MODEL	IN A TRAINED NETWORK
parameters $\theta \in \mathbb{R}^P$, map F	all weights; the outputs on the training set
Jacobian $J(\theta_0)$	per-example gradients at initialization
tangent kernel $K_t = J_t J_t^\top$	empirical neural tangent kernel on the training set
limit kernel K^∞	kernel of the infinitely wide architecture (fully connected, convolutional, ...)
$\lambda_{\min}(K^\infty)$	how distinguishable the training inputs are to the architecture
residual u_t	training errors
linearized model (6.3)	the network in the lazy regime
eigenvectors of K^∞	modes learned in order of their eigenvalues: spectral bias
width m, scale $m^{-1/2}$	hidden-layer width; NTK versus standard or mean-field parameterization
a fixed space of n functions	a kernel method, random features, or a lazy network

WHAT WAS THROWN AWAY, AND WHY FIRST. The features: the model is linear in the parameters, so the functions $x \mapsto \nabla_\theta f(x;\theta_0)$ are fixed before training, and learning is regression on them. Finite step sizes and stochastic gradients: we study the gradient flow. The second layer and depth: in Assumption 6.2 only one layer is trained (Exercise 6.5 trains both, and Theorem 6.11 states the kernel of deep networks). Each simplification moves toward a model in which the parameters move as little as the data allow; that is the right first model because there a nonconvex problem becomes solvable, and Section 6.4 computes the price.

WHAT ONE HAD TO NOTICE. Two facts, one about scale and one about geometry. The gradient of a wide network with respect to each single weight is small, of order $m^{-1/2}$, while the outputs depend on all m weights; so the outputs can change by order one while every weight changes by order $m^{-1/2}$. And the Hessian of the outputs, which is what makes K_t move, is block diagonal with one block of norm $O(m^{-1/2})$ per neuron, so it vanishes in the limit while the kernel, a sum of m terms of order $1/m$, does not. A third fact makes the model useful: the limit kernel is computable in closed form, by the Gaussian recursions of signal propagation (Chapter 5).

6.3 The solution: convergence without moving

The first theorem isolates what is needed of F: a Jacobian that is well conditioned at initialization and changes slowly over a ball whose radius is set by the initial error.

Theorem 6.3 (Lazy training). *Let $F\colon \mathbb{R}^P \to \mathbb{R}^n$ be continuously differentiable with Jacobian J, let $y \in \mathbb{R}^n$ and $\theta_0 \in \mathbb{R}^P$, and let $u_0 = F(\theta_0) - y$. Let*

$\mu > 0$, let $R = \|u_0\|/\mu$, and let B be the closed ball of radius R around θ_0. Assume

(i) $\lambda_{\min}\big(J(\theta_0)J(\theta_0)^\top\big) \geq 4\mu^2$;

(ii) $\|J(\theta) - J(\theta')\|_{\mathrm{op}} \leq L_J\|\theta - \theta'\|$ *for all* $\theta, \theta' \in B$;

(iii) $L_J R \leq \mu$.

Then the gradient flow (6.1) started at θ_0 exists for all $t \geq 0$, and for every $t \geq 0$,

$$\|u_t\| \leq e^{-\mu^2 t}\|u_0\|, \qquad \|\theta_t - \theta_0\| \leq \frac{\|u_0\| - \|u_t\|}{\mu}. \tag{6.5}$$

In particular θ_t stays in B, and it converges to a point $\theta_\infty \in B$ with $F(\theta_\infty) = y$.

Proof. The plan is to show that the kernel K_t stays at least μ^2 on the ball, which makes the residual decay, and that the path length is controlled by the decrease of the residual, which keeps the flow inside the ball.

The degenerate case. If $u_0 = 0$, then θ_0 is a stationary point of (6.1), the flow is constant, and (6.5) holds with $R = 0$. From now on $u_0 \neq 0$, so $R > 0$.

The Jacobian stays well conditioned on B. Let $\theta \in B$ and $v \in \mathbb{R}^n$. Hypothesis (i) gives $\|J(\theta_0)^\top v\|^2 = v^\top J(\theta_0)J(\theta_0)^\top v \geq 4\mu^2\|v\|^2$, and hypotheses (ii) and (iii) give $\|J(\theta) - J(\theta_0)\|_{\mathrm{op}} \leq L_J R \leq \mu$. By the triangle inequality,

$$\|J(\theta)^\top v\| \geq \|J(\theta_0)^\top v\| - \|(J(\theta) - J(\theta_0))^\top v\| \geq 2\mu\|v\| - \mu\|v\| = \mu\|v\|. \tag{6.6}$$

Local existence. On B the map J is Lipschitz by (ii), hence bounded, say $\|J(\theta)\|_{\mathrm{op}} \leq b$, and F is then Lipschitz with constant b. So the vector field $\theta \mapsto -J(\theta)^\top(F(\theta) - y)$ is Lipschitz on B, and by the Picard–Lindelöf theorem the flow has a unique solution on a maximal interval $[0, T^*)$ during which it stays in the open ball B°; if $T^* < \infty$, then θ_t leaves every compact subset of B° as $t \uparrow T^*$.

The residual decays. For $t < T^*$, (6.2) and (6.6) give

$$\frac{\mathrm{d}}{\mathrm{d}t}\|u_t\|^2 = -2u_t^\top K_t u_t = -2\|J(\theta_t)^\top u_t\|^2 \leq -2\mu^2\|u_t\|^2,$$

so $\|u_t\| \leq e^{-\mu^2 t}\|u_0\|$. The same identity and $\|J(\theta_t)^\top u_t\| \leq b\|u_t\|$ give $\frac{\mathrm{d}}{\mathrm{d}t}\|u_t\|^2 \geq -2b^2\|u_t\|^2$, hence $\|u_t\| \geq e^{-b^2 t}\|u_0\| > 0$.

The path length is bounded by the decrease of the residual. Since $u_t \neq 0$, the norm $\|u_t\|$ is differentiable, and (6.6) with $v = u_t$ gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\|u_t\| = -\frac{\|J(\theta_t)^\top u_t\|^2}{\|u_t\|} \leq -\mu\|J(\theta_t)^\top u_t\| = -\mu\|\dot\theta_t\|.$$

Integrating from 0 to t gives $\|\theta_t - \theta_0\| \leq \int_0^t \|\dot\theta_s\|\,\mathrm{d}s \leq (\|u_0\| - \|u_t\|)/\mu$, which is (6.5) on $[0, T^*)$.

The flow never leaves. Suppose $T^* < \infty$. By the last two steps, for $t < T^*$,

$$\|\theta_t - \theta_0\| \leq R - \frac{\|u_t\|}{\mu} \leq R - \frac{e^{-b^2 T^*}\|u_0\|}{\mu},$$

so θ_t stays in a closed ball of radius strictly less than R, a compact subset of B°, which contradicts the maximality of T^*. Hence $T^* = \infty$ and (6.5) holds for all t.

Convergence. The total path length $\int_0^\infty \|\dot\theta_s\|\,\mathrm{d}s$ is at most $\|u_0\|/\mu$, so θ_t is Cauchy as $t \to \infty$ and converges to some $\theta_\infty \in B$. By continuity of F, $F(\theta_\infty) - y = \lim_t u_t = 0$. ∎

The hypothesis is not convexity but $\|\nabla\widehat{\mathcal{L}}\|^2 \geq 2\mu^2\widehat{\mathcal{L}}$ on B, a Polyak–Łojasiewicz inequality, on which Liu et al. (2022) build a theory of overparameterized optimization. The radius $R = \|u_0\|/\mu$ does not involve L_J: the parameters travel a distance set by the initial error and the conditioning, while the curvature only has to be small *on that distance*. Wide networks will make R independent of the width and L_J of order $m^{-1/2}$.

Under the same hypotheses the network stays close to its linearization. Let θ_t^{lin} be the gradient flow of $\frac{1}{2}\|F^{\mathrm{lin}}(\theta)-y\|^2$ with $F^{\mathrm{lin}}(\theta) = F(\theta_0)+J(\theta_0)(\theta-\theta_0)$, from θ_0, and u_t^{lin} its residual. Its kernel is the constant $K_0 = J(\theta_0)J(\theta_0)^\top$, so $u_t^{\mathrm{lin}} = \exp(-tK_0)u_0$.

Corollary 6.4 (Closeness to the linearized model). *Under the hypotheses of Theorem 6.3, let $\beta_0 = \|J(\theta_0)\|_{\mathrm{op}}$. For every $t \geq 0$,*

$$\|u_t - u_t^{\mathrm{lin}}\| \leq \frac{\beta_0 L_J \|u_0\|^2}{\mu^3}\, e^{-\mu^2 t}, \qquad \|\theta_t - \theta_t^{\mathrm{lin}}\| \leq \frac{2\beta_0^2 L_J \|u_0\|^2}{\mu^5}.$$

If, moreover, a test input x has $\|\nabla_\theta f(x;\theta) - \nabla_\theta f(x;\theta')\| \leq L_x\|\theta-\theta'\|$ on B, then $|f(x;\theta_t) - f^{\mathrm{lin}}(x;\theta_t^{\mathrm{lin}})| \leq \frac{1}{2}L_x R^2 + \|\nabla_\theta f(x;\theta_0)\|\,\|\theta_t - \theta_t^{\mathrm{lin}}\|$.

Proof. The plan is to write the difference of the two residuals by variation of constants around the linear equation and to bound the forcing term by the motion of the kernel.

The kernel moves little. Hypothesis (i) gives $4\mu^2 \leq \lambda_{\min}(K_0) \leq \beta_0^2$, so $\mu \leq \beta_0/2$. By Theorem 6.3, $\|\theta_s - \theta_0\| \leq R$, so $\epsilon := \sup_s \|J(\theta_s) - J(\theta_0)\|_{\mathrm{op}} \leq L_J R \leq \mu$. Writing $J_s = J(\theta_s)$ and $J_0 = J(\theta_0)$,

$$\|K_s - K_0\|_{\mathrm{op}} = \|(J_s - J_0)J_s^\top + J_0(J_s - J_0)^\top\|_{\mathrm{op}} \leq \epsilon(2\beta_0 + \epsilon),$$

which is at most $3\beta_0 L_J R$.

Variation of constants. The difference $\delta_t = u_t - u_t^{\mathrm{lin}}$ satisfies $\delta_0 = 0$ and $\dot\delta_t = -K_0\delta_t - (K_t - K_0)u_t$, so $\delta_t = -\int_0^t \exp(-(t-s)K_0)(K_s - K_0)u_s\,\mathrm{d}s$. Since $K_0 \geq 4\mu^2 I_n$, $\|\exp(-(t-s)K_0)\|_{\mathrm{op}} \leq e^{-4\mu^2(t-s)}$, and with $\|u_s\| \leq e^{-\mu^2 s}\|u_0\|$,

$$\begin{aligned}\|\delta_t\| &\leq 3\beta_0 L_J R\|u_0\| \int_0^t e^{-4\mu^2(t-s)} e^{-\mu^2 s}\,\mathrm{d}s \\ &\leq 3\beta_0 L_J R\|u_0\|\,\frac{e^{-\mu^2 t}}{3\mu^2} = \frac{\beta_0 L_J\|u_0\|^2}{\mu^3} e^{-\mu^2 t}.\end{aligned}$$

Parameters. Integrating $\dot\theta_s - \dot\theta_s^{\mathrm{lin}} = -(J_s - J_0)^\top u_s - J_0^\top \delta_s$ and using the

bounds above,

$$\|\theta_t - \theta_t^{\mathrm{lin}}\| \le \int_0^\infty \Big(L_J R\|u_s\| + \beta_0\|\delta_s\|\Big)\,\mathrm{d}s \le \frac{L_J\|u_0\|^2}{\mu^3} + \frac{\beta_0^2 L_J\|u_0\|^2}{\mu^5} \le \frac{2\beta_0^2 L_J\|u_0\|^2}{\mu^5},$$

the last step by $\mu \le \beta_0$.

A test input. Taylor's theorem with a Lipschitz gradient bounds the difference between $f(x;\theta_t)$ and $f^{\mathrm{lin}}(x;\theta_t)$ by $\frac{1}{2}L_x\|\theta_t - \theta_0\|^2 \le \frac{1}{2}L_x R^2$. The linearized model changes by $\langle \nabla_\theta f(x;\theta_0), \theta_t - \theta_t^{\mathrm{lin}}\rangle$ between θ_t^{lin} and θ_t. ∎

The linearized model is solved in closed form, and its solution is kernel regression.

Proposition 6.5 (Lazy training is kernel regression). *Suppose $K_0 = J(\theta_0)J(\theta_0)^\top$ is invertible, and write $K_0(x,X)$ for the row vector $(K_{\theta_0}(x,x_i))_{i\le n}$. The linearized flow satisfies, for every $t \ge 0$ and every input x,*

$$f^{\mathrm{lin}}(x;\theta_t^{\mathrm{lin}}) = f(x;\theta_0) - K_0(x,X)\,K_0^{-1}\big(I_n - \exp(-tK_0)\big)u_0. \tag{6.7}$$

As $t \to \infty$ it converges to the kernel interpolant $f(x;\theta_0) + K_0(x,X)K_0^{-1}(y - F(\theta_0))$. The parameters move by $\|\theta_t^{\mathrm{lin}} - \theta_0\|^2 = u_0^\top K_0^{-1}(I_n - \exp(-tK_0))^2 u_0$, which increases to $u_0^\top K_0^{-1}u_0$, the smallest $\|\theta - \theta_0\|^2$ among all θ with $F^{\mathrm{lin}}(\theta) = y$.

Proof. With $J_0 = J(\theta_0)$, the linearized flow is $\dot\theta = -J_0^\top \exp(-tK_0)u_0$, so $\theta_t^{\mathrm{lin}} - \theta_0 = -J_0^\top \int_0^t \exp(-sK_0)\,\mathrm{d}s\,u_0 = -J_0^\top K_0^{-1}(I_n - \exp(-tK_0))u_0$. Since $\nabla_\theta f(x;\theta_0)^\top J_0^\top = K_0(x,X)$, inserting this in (6.3) gives (6.7). The squared norm is $v^\top K_0^{-1}J_0J_0^\top K_0^{-1}v = v^\top K_0^{-1}v$ with $v = (I_n - \exp(-tK_0))u_0$; in an eigenbasis of K_0 with eigenvalues $\lambda_k > 0$ it is $\sum_k (1 - e^{-\lambda_k t})^2 c_k^2/\lambda_k$, with c_k the coordinates of u_0, which increases in t. Finally every Δ with $J_0\Delta = -u_0$ splits as $\Delta = J_0^\top c + \Delta^\perp$ with $J_0\Delta^\perp = 0$; then $K_0 c = -u_0$, and $\|\Delta\|^2 = u_0^\top K_0^{-1}u_0 + \|\Delta^\perp\|^2$. ∎

The interpolant is the kernel analogue of the minimum-norm least-squares estimator of Chapter 1: among all functions in the span of the tangent features that fit the residual labels, gradient flow from θ_0 picks the one of smallest norm, here the norm of the reproducing kernel Hilbert space of K_0.

We now check the hypotheses of Theorem 6.3 for the two-layer network. Everything rests on one fact about the limit kernel.

Lemma 6.6 (The limit kernel is positive definite). *Under Assumption 6.2, the matrix $K^\infty = (K^\infty(x_i,x_j))_{i,j\le n}$ is positive definite.*

Proof. The plan is to expand K^∞ in tensor powers of the inputs and to show that a vector annihilated by infinitely many of them vanishes.

Expansion. Since ϕ' is bounded, it has a Hermite expansion $\phi' = \sum_k b_k h_k$ in $L^2(\mathcal{N}(0,1))$ with $\sum_k b_k^2 = \mathbb{E}\phi'(g)^2 \le \beta^2$, and Corollary A.4 gives $\kappa(\rho) = \sum_k b_k^2 \rho^k$. Hence $K^\infty_{ij} = \sum_k b_k^2 \langle x_i, x_j\rangle^{k+1}$, and since $\langle x_i, x_j\rangle^p = \langle x_i^{\otimes p}, x_j^{\otimes p}\rangle$, for every $v \in \mathbb{R}^n$

$$v^\top K^\infty v = \sum_{k\ge 0} b_k^2 \Big\| \sum_{i=1}^n v_i\, x_i^{\otimes(k+1)} \Big\|^2 \ge 0.$$

Infinitely many terms. If only finitely many b_k were nonzero, then ϕ' would agree almost everywhere, hence everywhere by continuity, with a polynomial; a bounded polynomial is constant, and ϕ would be affine, which Assumption 6.2 excludes. So the set $S = \{k : b_k \ne 0\}$ is infinite.

A vector in the kernel vanishes. Suppose $v^\top K^\infty v = 0$. Then $\sum_i v_i x_i^{\otimes(k+1)} = 0$ for each $k \in S$, and pairing with $z^{\otimes(k+1)}$ gives $\sum_i v_i \langle x_i, z\rangle^{k+1} = 0$ for every $z \in \mathbb{R}^d$. The vectors z with $\langle x_i, z\rangle = 0$ for some i, or with $\langle x_i, z\rangle = \pm\langle x_j, z\rangle$ for some $i \ne j$, lie in finitely many hyperplanes, the orthogonal complements of the nonzero vectors x_i and $x_i \mp x_j$ (nonzero because no two inputs are parallel); these do not cover $\mathbb{R}^d$. Choose z outside them and relabel the inputs so that $c_i = |\langle x_i, z\rangle|$ satisfies $c_1 > c_2 > \cdots > c_n > 0$, with signs $\epsilon_i = \operatorname{sgn}\langle x_i, z\rangle$. If $v \ne 0$, let i_0 be the first index with $v_{i_0} \ne 0$; dividing by $c_{i_0}^{k+1}$,

$$v_{i_0}\epsilon_{i_0}^{k+1} = -\sum_{i>i_0} v_i \epsilon_i^{k+1} \Big(\frac{c_i}{c_{i_0}}\Big)^{k+1} \qquad (k \in S),$$

and the right side tends to 0 as $k \to \infty$ in S, while the left side has absolute value $|v_{i_0}| > 0$. This contradiction shows $v = 0$. ∎

The hypothesis on parallel inputs is needed: for tanh the function ϕ' is even, $\kappa(-1) = \kappa(1)$, and an input x together with $-x$ gives two proportional rows of K^∞ (Exercise 6.3).[4] Positivity is qualitative. How large $\lambda_{\min}$ is depends on the data and the activation, and it can be very small: for 32 equally spaced points on a circle with a bias coordinate and $\phi = \tanh$, the setting of Figure 6.1, $\lambda_{\min} \approx 2.6 \times 10^{-8}$, because the eigenvalues of a smooth kernel decay fast with frequency.

[4] Du et al. (2019) prove positive definiteness for ReLU under the same hypothesis that no two inputs are parallel.

Lemma 6.7 (The network at initialization). *Under Assumption 6.2, let $J(W) \in \mathbb{R}^{n\times md}$ be the Jacobian of $W \mapsto (f(x_i; W))_{i\le n}$, let $c_\phi = \mathbb{E}\phi(g)^2$ for $g \sim \mathcal{N}(0,1)$, and let $\delta \in (0,1)$.*

(i) For all W, W', $\|J(W)\|_{\mathrm{op}} \le \beta\sqrt{n}$ and $\|J(W) - J(W')\|_{\mathrm{op}} \le \beta\sqrt{n/m}\,\|W - W'\|_F$.

(ii) If $m \ge \frac{8\beta^4 n^2}{\lambda_{\min}^2} \log\frac{4n^2}{\delta}$, then with probability at least $1 - \frac{\delta}{2}$, $\|K_0 - K^\infty\|_{\mathrm{op}} \le \frac{\lambda_{\min}}{2}$, and hence $\lambda_{\min}(K_0) \ge \frac{\lambda_{\min}}{2}$.

(iii) With probability at least $1 - \frac{\delta}{2}$, $\|u_0\|^2 \le 2n(c_\phi + 1)/\delta$.

Proof. *Part (i).* The entry of $J(W)$ in row i and in the block of coordinates of w_r is $m^{-1/2} a_r \phi'(\langle w_r, x_i\rangle) x_i^\top$. With $|\phi'| \le \beta$, $a_r^2 = 1$, and $\|x_i\| = 1$, the squared Frobenius norm is at most $\sum_i \sum_r \beta^2/m = n\beta^2$, which bounds the operator

norm. For the difference, $|\phi'(s)-\phi'(s')| \le \beta|s-s'|$ and the Cauchy–Schwarz inequality $|\langle w_r - w_r', x_i\rangle| \le \|w_r - w_r'\|$ give

$$\|J(W)-J(W')\|_F^2 = \frac{1}{m}\sum_{i=1}^{n}\sum_{r=1}^{m}\big(\phi'(\langle w_r,x_i\rangle)-\phi'(\langle w_r',x_i\rangle)\big)^2 \le \frac{n\beta^2}{m}\|W-W'\|_F^2.$$

Part (ii). By (6.4), $K_0 = \frac{1}{m}\sum_r Z_r$, where the matrix Z_r has the entries $(Z_r)_{ij} = \langle x_i, x_j\rangle \phi'(\langle w_r, x_i\rangle)\phi'(\langle w_r, x_j\rangle)$. The matrices Z_r are independent, have mean K^∞, and have entries in $[-\beta^2, \beta^2]$. Hoeffding's inequality (Theorem A.17) for the entry (i,j), with deviation ms and m summands of range $2\beta^2$, gives

$$\Pr\Big(|(K_0-K^\infty)_{ij}| \ge s\Big) \le 2\exp\Big(-\frac{2m^2s^2}{m(2\beta^2)^2}\Big) = 2\exp\Big(-\frac{ms^2}{2\beta^4}\Big).$$

Take $s = \lambda_{\min}/(2n)$ and a union bound over the n^2 entries: with probability at least $1 - 2n^2\exp(-m\lambda_{\min}^2/(8n^2\beta^4)) \ge 1-\frac{\delta}{2}$, every entry is below s, so $\|K_0-K^\infty\|_{\mathrm{op}} \le \|K_0-K^\infty\|_F \le ns = \lambda_{\min}/2$. Weyl's inequality then gives $\lambda_{\min}(K_0) \ge \lambda_{\min} - \lambda_{\min}/2$.

Part (iii). The output $f(x_i;W_0) = m^{-1/2}\sum_r a_r\phi(\langle w_r, x_i\rangle)$ is a normalized sum of m independent terms; each has mean zero, because a_r is a symmetric sign independent of w_r, and second moment c_ϕ, which is finite because $|\phi(s)| \le |\phi(0)| + \beta|s|$. Hence $\mathbb{E}\, f(x_i;W_0) = 0$, $\mathbb{E}\, f(x_i;W_0)^2 = c_\phi$, and $\mathbb{E}\|u_0\|^2 = \sum_i (c_\phi + y_i^2) \le n(c_\phi+1)$. Markov's inequality gives (iii). ∎

Theorem 6.8 (Wide two-layer networks train lazily). *Under Assumption 6.2, let $\delta \in (0,1)$ and suppose*

$$m \ge \frac{n^2}{\lambda_{\min}^2}\max\Big\{8\beta^4\log\frac{4n^2}{\delta},\ \frac{128\beta^2(c_\phi+1)}{\delta}\Big\}. \tag{6.8}$$

With probability at least $1-\delta$ over the initialization, the gradient flow of $\widehat{\mathcal{L}}(W) = \frac{1}{2}\sum_i (f(x_i;W)-y_i)^2$ from W_0 satisfies, for all $t \ge 0$,

$$\begin{aligned}
\|u_t\| &\le e^{-\lambda_{\min} t/8}\|u_0\|,\\
\|W_t - W_0\|_F &\le \sqrt{8/\lambda_{\min}}\,\|u_0\| \le 4\sqrt{n(c_\phi+1)/(\lambda_{\min}\delta)},\\
\max_{r\le m}\|w_r(t)-w_r(0)\| &\le 8\beta\sqrt{n}\,\|u_0\|/(\lambda_{\min}\sqrt{m}),\\
\|K_t - K_0\|_{\mathrm{op}} &\le 3\sqrt{8}\,\beta^2 n\|u_0\|/\sqrt{\lambda_{\min} m}.
\end{aligned}$$

Proof. The plan is to apply Theorem 6.3 with $\mu^2 = \lambda_{\min}/8$ on the event where parts (ii) and (iii) of Lemma 6.7 hold, which has probability at least $1-\delta$ by a union bound.

The hypotheses. The first term of (6.8) and Lemma 6.7(ii) give $\lambda_{\min}(K_0) \ge \lambda_{\min}/2 = 4\mu^2$, hypothesis (i). Lemma 6.7(i) gives hypothesis (ii) on all of $\mathbb{R}^{m\times d}$ with $L_J = \beta\sqrt{n/m}$. For hypothesis (iii), $L_J R = \beta\sqrt{n/m}\,\|u_0\|/\mu \le \mu$ is equivalent to $m \ge n\beta^2\|u_0\|^2/\mu^4 = 64n\beta^2\|u_0\|^2/\lambda_{\min}^2$, which holds by Lemma 6.7(iii) and the second term of (6.8).

The conclusions. Theorem 6.3 gives the decay of u_t and $\|W_t - W_0\|_F \le R = \|u_0\|/\mu$, and Lemma 6.7(iii) bounds $\|u_0\|$. For one neuron, $\dot w_r = -m^{-1/2} a_r \sum_i u_{t,i} \phi'(\langle w_r, x_i\rangle) x_i$, so $\|\dot w_r\| \le \beta m^{-1/2} \sum_i |u_{t,i}| \le \beta\sqrt{n/m}\,\|u_t\|$; integrating $e^{-\mu^2 t}\|u_0\|$ over $t \ge 0$ gives the factor $1/\mu^2 = 8/\lambda_{\min}$. For the kernel, the first step of the proof of Corollary 6.4 gives $\|K_t - K_0\|_{\mathrm{op}} \le 3\beta_0 L_J R$ with $\beta_0 \le \beta\sqrt{n}$ by Lemma 6.7(i). ∎

Read the theorem as a statement about scales. The displacement $\|W_t - W_0\|_F$ is bounded independently of m, while $\|W_0\|_F \approx \sqrt{md}$, so the relative change of the weights is of order $m^{-1/2}$; so is the largest move of a single neuron, and so is the change of the kernel. The exponents in the width $n^2/\lambda_{\min}^2$ are not sharp: Du et al. (2019) needed a larger power of n for ReLU, and Oymak and Soltanolkotabi (2020) reduced the requirement for smooth activations to a number of parameters of order n^2. Section 6.5 shows that the kernel in fact moves by order $1/m$, not $m^{-1/2}$.

6.4 The solution: what a kernel can and cannot learn

LEARNING BY MODES. On the training set the answer is in (6.7): if v_k is an eigenvector of K_0 with eigenvalue λ_k, then $\langle v_k, u_t^{\mathrm{lin}}\rangle = e^{-\lambda_k t}\langle v_k, u_0\rangle$, the normal modes of the quadratic of the prologue. For the population, let $x \sim \mathrm{Unif}(\mathbb{S}^{d-1})$, let $K(x,x') = \psi(\langle x, x'\rangle)$ be a bounded rotation-invariant kernel, and let T_K be its integral operator, $(T_K g)(x) = \mathbb{E}_{x'}[K(x,x')g(x')]$. The *population linearized flow* is the gradient flow of $\frac12\|f - f_\star\|^2$ in the geometry of the kernel,

$$\partial_t f_t = -T_K(f_t - f_\star), \tag{6.9}$$

the limit $n \to \infty$ of the function-space flow $\partial_t f_t(x) = -\frac1n \sum_i K(x, x_i)(f_t(x_i) - y_i)$ for the loss $\frac{1}{2n}\sum_i (f(x_i) - y_i)^2$ with $y_i = f_\star(x_i)$.

Proposition 6.9 (Spectral bias on the sphere). *Let Π_k be the orthogonal projection of $L^2(\mathbb{S}^{d-1})$ onto the spherical harmonics $\mathcal{H}_k$ of degree k, and let $\lambda_k = \frac{|\mathbb{S}^{d-2}|}{|\mathbb{S}^{d-1}|}\int_{-1}^{1} \psi(t) P_{d,k}(t)(1-t^2)^{(d-3)/2}\,\mathrm{d}t$, with $P_{d,k}$ the Gegenbauer polynomial normalized by $P_{d,k}(1) = 1$. The solution of (6.9) with $f_0 - f_\star \in L^2$ satisfies $\Pi_k(f_t - f_\star) = e^{-\lambda_k t}\,\Pi_k(f_0 - f_\star)$ for every $k \ge 0$ and $t \ge 0$.*

PROOF. The operator T_K is bounded on L^2, with norm at most $\sup|\psi|$, so (6.9) is a linear equation with a bounded operator and the unique solution $f_t - f_\star = \exp(-tT_K)(f_0 - f_\star)$. By the Funk–Hecke formula (Theorem A.6(iii)), $T_K Y = \lambda_k Y$ for every $Y \in \mathcal{H}_k$, so T_K commutes with Π_k and $\exp(-tT_K)\Pi_k = e^{-\lambda_k t}\Pi_k$. ∎

For the two-layer network, $K^\infty(x,x') = \rho\,\kappa(\rho)$ with $\rho = \langle x, x'\rangle$ and $\kappa(\rho) = \sum_j b_j^2 \rho^j$, so the eigenvalues are determined by the Hermite coefficients of ϕ', and their decay by the smoothness of ϕ. For ReLU, $\kappa(\rho) = (\pi - \arccos\rho)/(2\pi)$, a classical arc-cosine kernel (Cho and Saul 2009), and the eigenvalues decay like k^{-d} (Bietti and Mairal 2019);[5] for smooth activations they decay faster. The parity of ϕ' matters too. For ReLU, $\rho\kappa(\rho) = \frac{\rho}{4} + \frac{\rho \arcsin\rho}{2\pi}$ is a degree-one

[5] For $d = 3$ we computed $\lambda_4/\lambda_6 = 3.47$ by quadrature, against $(6/4)^3 = 3.375$.

term plus an even function, so $\lambda_k = 0$ for every odd $k \geq 3$ (Exercise 6.2): a two-layer ReLU network without bias cannot learn odd harmonics of degree three or more in the lazy regime at all (Basri et al. 2019). A bias, implemented as a constant input coordinate, removes these zeros.

WHAT NO FIXED SPACE OF FUNCTIONS CAN LEARN. High-degree harmonics are not only slow but expensive in samples, for every kernel, by a counting argument.

Theorem 6.10 (What a kernel cannot learn). *Let $d \geq 3$, $k \geq 0$, and $N = N_{d,k} = \dim \mathcal{H}_k$. For $w \in \mathbb{S}^{d-1}$ let*

$$f_w(x) = \frac{Z_k(x,w)}{\sqrt{N}} = \sqrt{N}\, P_{d,k}(\langle w, x\rangle),$$

with Z_k the reproducing kernel of $\mathcal{H}_k$ (Theorem A.6(ii)), so that $\|f_w\|_{L^2} = 1$. Let $w \sim \mathrm{Unif}(\mathbb{S}^{d-1})$, and let $V \subset L^2(\mathbb{S}^{d-1})$ be a random subspace of dimension at most n, spanned by a random orthonormal family independent of w. Then

$$\mathbb{E}\|f_w - \Pi_V f_w\|^2 = 1 - \frac{\mathbb{E}\,\mathrm{tr}(\Pi_V \Pi_k)}{N} \geq 1 - \frac{n}{N}.$$

Consequently, every estimator $\hat{f}$ that takes values in V, whatever it does with the labels, satisfies $\mathbb{E}\|\hat{f} - f_w\|^2 \geq 1 - n/N$.

PROOF. The plan is to compute the energy of f_w in one fixed direction, averaged over w, with the reproducing property, and to add over an orthonormal basis of V.

Normalization. The reproducing property gives $\|Z_k(\cdot, w)\|^2 = Z_k(w,w) = N$, so $\|f_w\| = 1$; the second expression for f_w is $Z_k(x,w) = N\, P_{d,k}(\langle x, w\rangle)$ from Theorem A.6(ii).

One direction. Let $e \in L^2$. Since Z_k is the kernel of Π_k, $\langle e, f_w\rangle = N^{-1/2} \int e(x) Z_k(x,w)\,\mathrm{d}x = N^{-1/2}(\Pi_k e)(w)$, the integral being against the uniform probability measure. Averaging over $w \sim \mathrm{Unif}(\mathbb{S}^{d-1})$,

$$\mathbb{E}_w \langle e, f_w\rangle^2 = \frac{1}{N}\|\Pi_k e\|^2 \leq \frac{\|e\|^2}{N}.$$

The subspace. Condition on V, which is independent of w, and let $e_1, \dots, e_{n'}$, $n' \leq n$, be its orthonormal basis. Then $\|f_w - \Pi_V f_w\|^2 = 1 - \sum_j \langle e_j, f_w\rangle^2$, and the previous step gives $\mathbb{E}_w \sum_j \langle e_j, f_w\rangle^2 = \frac{1}{N}\sum_j \|\Pi_k e_j\|^2 = \frac{1}{N}\,\mathrm{tr}(\Pi_V \Pi_k) \leq \frac{n'}{N}$. Take expectations over V.

Estimators. If $\hat{f} \in V$, then $\hat{f} - \Pi_V f_w \in V$ is orthogonal to $f_w - \Pi_V f_w$, so $\|\hat{f} - f_w\|^2 \geq \|f_w - \Pi_V f_w\|^2$. ∎

The theorem applies to every method that predicts from a fixed space of at most n functions. Kernel ridge regression with *every* bounded kernel predicts in the span of $K(\cdot, x_i)$, which depends on the inputs only; the linearized network of Proposition 6.5, started independently of w, predicts in $f(\cdot;\theta_0) + \mathrm{span}\{K_0(\cdot, x_i)\}$; a random-features model with p features predicts

in a space of dimension p whatever n is. Each therefore needs n (or p) of order $N_{d,k} = \frac{d^k}{k!}(1 + O(1/d))$ to reach error $\frac{1}{2}$ on a target that depends on one unknown direction through a polynomial of degree k. For $d = 30$ that is $N_{30,2} = 464$ for $k = 2$ and $N_{30,3} = 4930$ for $k = 3$.[6]

A network whose first layer moves can first find w and then fit a function of one variable. Chapter 7 shows that one neuron trained by online stochastic gradient descent finds the direction of a target of information exponent k from about d^{k-1} samples (up to logarithms), and Damian et al. (2022) prove that gradient descent on a two-layer network outside the kernel regime learns functions of r hidden directions of degree p from about $d^2 r + d r^p$ samples, against d^p for kernels. The kernel pays for not knowing the direction: it must be ready for all $N_{d,k}$ of them.

[6] For Gaussian inputs the same counting works with Hermite polynomials in place of spherical harmonics, up to one radial direction (Exercise 6.7).

THE KERNEL OF A DEEP NETWORK. The limit kernel of a deep fully connected network obeys the same Gaussian recursion as its forward signals (Chapter 5), with one more term.

Theorem 6.11 (Jacot et al. 2018; nonasymptotic form by Arora et al. 2019a). *Consider $h^1(x) = \sigma_w d^{-1/2} W^1 x + \sigma_b b^1$ and $h^{\ell+1} = \sigma_w m^{-1/2} W^{\ell+1} \phi(h^\ell) + \sigma_b b^{\ell+1}$ for $\ell < L$, with independent standard Gaussian entries in all W^ℓ and b^ℓ, hidden widths m, one output, and ϕ Lipschitz and twice differentiable with bounded second derivative. Define $\Sigma^1(x, x') = \sigma_w^2 \langle x, x' \rangle / d + \sigma_b^2$ and, with (g, g') centered Gaussian with covariance Σ^ℓ restricted to $\{x, x'\}$,*

$$\Sigma^{\ell+1} = \sigma_w^2 \mathbb{E}[\phi(g)\phi(g')] + \sigma_b^2, \qquad \dot{\Sigma}^{\ell+1} = \sigma_w^2 \mathbb{E}[\phi'(g)\phi'(g')],$$
$$\Theta^{\ell+1} = \Theta^\ell \, \dot{\Sigma}^{\ell+1} + \Sigma^{\ell+1},$$

with $\Theta^1 = \Sigma^1$. As the hidden widths tend to infinity one after the other, the tangent kernel of the output with respect to all weights and biases converges in probability to Θ^L at initialization, and, along gradient flow on the square loss over a fixed horizon, stays equal to Θ^L in the limit.

PROOF SKETCH. By the chain rule the tangent kernel of $h^{\ell+1}$ is the contribution $\Sigma^{\ell+1}$ of the weights and biases of layer $\ell + 1$, which converges by the law of large numbers as in Chapter 5, plus the tangent kernel Θ^ℓ of each unit of layer ℓ (by induction) multiplied by $\frac{\sigma_w^2}{m} \sum_j \phi'(h_j^\ell(x)) \phi'(h_j^\ell(x')) \to \dot{\Sigma}^{\ell+1}$; the cross terms average out. During training the weights of each layer move by $O(m^{-1/2})$ in operator norm, as in Theorem 6.3. The details are in Jacot et al. (2018, Theorems 1 and 2); Arora et al. (2019a) give explicit widths. ∎

For the two-layer network of Assumption 6.2 ($L = 2$, $\sigma_w = 1$, $\sigma_b = 0$, unit inputs scaled by $\sqrt{d}$) the part of Θ^2 coming from the first layer is $\Sigma^1 \dot{\Sigma}^2 = \langle x, x' \rangle \kappa(\langle x, x' \rangle)$, which is (6.4); training the second layer adds Σ^2 (Exercise 6.5).

THE OTHER INFINITE WIDTH. Laziness comes from the factor $m^{-1/2}$, not from the width itself (Chizat et al. 2019). The following derivation shows how.

LAZY AND MEAN-FIELD SCALINGS. Write $f(x) = m^{-\gamma}\sum_r a_r\phi(\langle w_r, x\rangle)$ with $\frac{1}{2} \le \gamma \le 1$, and train with the step (or time scale) η. The function moves at speed $\partial_t f = -\eta\sum_r \nabla_{w_r} f \cdot \nabla_{w_r}\widehat{\mathcal{L}}$, a sum of m terms each of order $m^{-2\gamma}$, so it changes by order one per unit time only if $\eta \asymp m^{2\gamma-1}$. Each neuron then moves at speed $\eta\|\nabla_{w_r}\widehat{\mathcal{L}}\| \asymp m^{2\gamma-1}m^{-\gamma} = m^{\gamma-1}$. For $\gamma = \frac{1}{2}$, the NTK parameterization with step of order one, each neuron moves by $O(m^{-1/2})$ while the function changes by $O(1)$: lazy. For $\gamma = 1$, the *mean-field parameterization* with step of order m, each neuron moves by $O(1)$, and the kernel $\eta\sum_r \nabla_{w_r} f(x)\cdot\nabla_{w_r} f(x')$, an average of m terms of order one, changes by order one: the features are learned. Between the two, the motion of one neuron is $m^{\gamma-1}$ and laziness sets in as $m \to \infty$ for every $\gamma < 1$. ◇

The mean-field limit is not a linear model but a nonlinear partial differential equation for the distribution of neurons. Let $\rho_t^m = \frac{1}{m}\sum_r \delta_{(a_r(t),w_r(t))}$ for a network $f(x) = \frac{1}{m}\sum_r a_r\phi(\langle w_r, x\rangle)$ trained in both layers, and write $f_\rho(x) = \int a\phi(\langle w, x\rangle)\,\mathrm{d}\rho(a,w)$.

Theorem 6.12 (Chizat and Bach 2018; Mei et al. 2018; Rotskoff and Vanden-Eijnden 2022; Sirignano and Spiliopoulos 2020; stated without proof). *Under regularity conditions on ϕ and on the data, and with initial neurons independent with law ρ_0, the empirical measure ρ_t^m of gradient flow on the population loss $\frac{1}{2}\mathbb{E}(f(x) - f_\star(x))^2$, run with time scale m, converges weakly as $m \to \infty$, uniformly on bounded time intervals, to the solution of*

$$\partial_t \rho_t = \nabla\cdot\big(\rho_t \nabla\Psi(\cdot;\rho_t)\big),$$

with $\Psi(a,w;\rho) = \mathbb{E}_x[(f_\rho(x) - f_\star(x))\,a\,\phi(\langle w, x\rangle)]$: a Wasserstein gradient flow of the loss as a function of ρ.

Chizat and Bach (2018) prove that, under homogeneity conditions and for an initialization of full support, if the flow converges, then its limit is a global minimizer. Yang and Hu (2021) classify the scalings of deep networks: each infinite-width limit either learns features or follows kernel gradient descent, and in their maximal-update parameterization (μP) every layer learns features.

6.5 Reading the solution

MODES. Figure 6.1 tests the linearized solution where it can be solved exactly. With equally spaced inputs on the circle the infinite-width kernel is circulant, its eigenvectors on the training set are the Fourier modes, and the linearized model predicts that the Fourier coefficient of frequency k of the residual is multiplied by $1 - \eta\lambda_k$ at each step of gradient descent. The eigenvalues $\lambda_1 = 3.62$, $\lambda_2 = 0.317$, and $\lambda_3 = 0.086$ of K^∞ decrease by factors of 11 and 4, and the three frequencies are learned one after the other on time scales separated by the same factors. A network of width 16384 follows the three predicted curves until each coefficient has decayed to about 10^{-2}; below that, the residual leaks between modes, because the kernel of a finite network is not exactly circulant and drifts slightly during

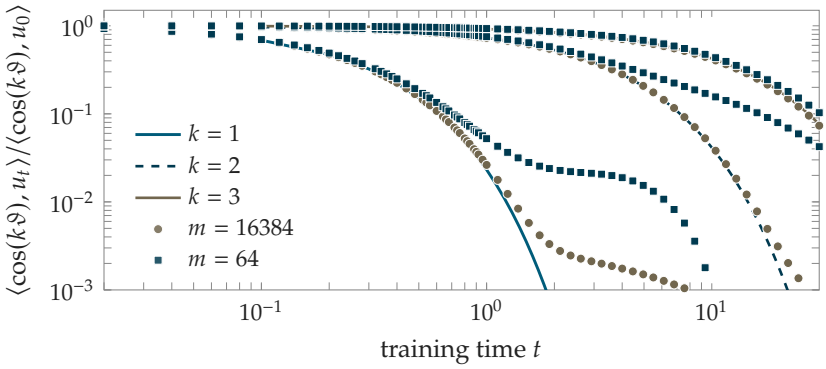


Figure 6.1. Learning by modes. Training set: $n = 32$ equally spaced angles ϑ_i on the circle, inputs $(\cos\vartheta_i, \sin\vartheta_i, 1)/\sqrt{2}$, labels $y_i = \cos\vartheta_i + \cos 2\vartheta_i + \cos 3\vartheta_i$. Network: Assumption 6.2 with $\phi = \tanh$ and an antisymmetric initialization $(w_{r+m/2} = w_r, a_{r+m/2} = -a_r)$, so that $f(\cdot; W_0) = 0$ and $u_0 = -y$; full-batch gradient descent with step $\eta = 0.02$ (seed 6). Curves: the linearized prediction $(1 - \eta\lambda_k)^{t/\eta}$ with the eigenvalues of K^∞, computed by Gauss–Hermite quadrature. Circles: width $m = 16384$; squares: $m = 64$. Ratios below 10^{-3} are not shown.

training. A network of width 64 learns all three frequencies more slowly than predicted, and its first coefficient stalls near 2×10^{-2}: at that width the kernel at initialization already differs from K^∞ by about 15% in operator norm.

How far things move. Figure 6.2 measures the motion of the parameters and of the kernel. Proposition 6.5 predicts the displacement exactly in the limit: after s steps the linearized model has moved by $D^2 = u_0^\top (K^\infty)^{-1}(I - (I - \eta K^\infty)^s)^2 u_0$, (here $D = 7.40$) whatever the width, while $\mathbb{E}\|W_0\|_F^2 = 3m$. The predicted relative change $D/\sqrt{3m}$ matches the simulation to within 2% from $m = 512$ on. The largest move of a single neuron also decreases like $m^{-1/2}$ (from 0.45 at $m = 1024$ to 0.11 at $m = 16384$). The kernel, however, moves by order $1/m$, not $m^{-1/2}$: its relative change falls by a factor of about 2 for each doubling of the width. Theorem 6.8 bounds it by $m^{-1/2}$, but the leading-order change cancels.

Why the kernel moves by $1/m$. Let $\chi(w) = \phi'(\langle w, x\rangle)\phi'(\langle w, x'\rangle)$ and expand the kernel to first order in the displacements:

$$K_t(x, x') - K_0(x, x') \approx \frac{\langle x, x'\rangle}{m} \sum_r \nabla\chi(w_r) \cdot (w_r(t) - w_r(0)).$$

The displacement of neuron r is $-m^{-1/2} a_r \int_0^t \sum_i u_{s,i} \phi'(\langle w_r(s), x_i\rangle) x_i \, \mathrm{d}s$, a sign a_r times a vector $m^{-1/2} g_r$ that depends on w_r and on the common residual. The first-order change is therefore $m^{-3/2} \sum_r a_r h_r$ with h_r of order one. The signs a_r are independent of w_r and enter the residual only through averages over all neurons, so the sum behaves like a sum of m terms of random sign, of order $\sqrt{m}$, and the first-order change is of order m^{-1}. The second-order terms are $\frac{1}{m}\sum_r O(\|w_r(t) - w_r(0)\|^2) = O(1/m)$. For the

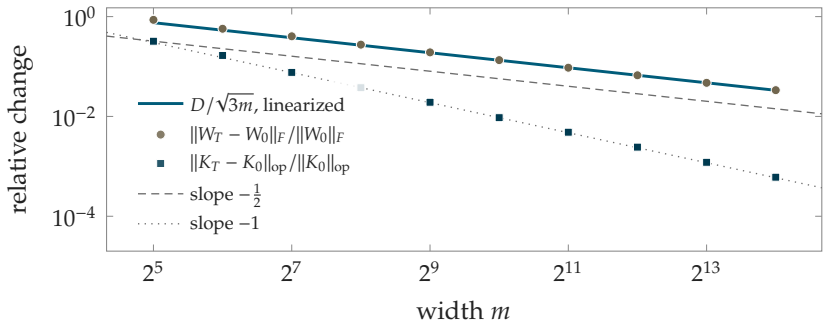


Figure 6.2. Motion of the weights and of the tangent kernel during training, against the width. Same training set, network, and initialization as Figure 6.1, labels $y_i = \cos\vartheta_i + \cos 2\vartheta_i$, gradient descent with $\eta = 0.02$ for time $T = 25$, at which the training residual is between 0.2% and 2.3% of $\|y\|$. Points: means over 4 seeds. Curve: the exact linearized displacement $D = 7.40$ of Proposition 6.5 divided by $\sqrt{3m}$. Dashed and dotted lines: reference slopes through the kernel change at $m = 32$ and $m = 1024$.

antisymmetric initialization of the figure, the first-order terms of each pair of neurons cancel to leading order. ◇

The $1/m$ rate agrees with the analysis of Dyer and Gur-Ari (2020), in which the tangent kernel is constant during training up to corrections of order $1/m$.[7] We checked that the same rate holds with independent random signs (Exercise 6.8).

[7] Dyer and Gur-Ari (2020) derive this from a conjecture on the scaling of correlation functions with width, which they analyze completely for deep linear networks and support with evidence for nonlinear ones.

The price. Figure 6.3 compares kernel ridge regression with a trained network on the target f_w of Theorem 6.10 with $k = 2$ in dimension $d = 30$, where $N_{30,2} = 464$. Kernel ridge regression with a two-layer ReLU tangent kernel, given the best ridge parameter in hindsight, learns nothing below $n = 100$, has error 0.6 at $n = 400$, and needs about $n = 1000$ samples to reach error 0.2; the theorem's bound $1 - n/464$ lies below it, as it must. A two-layer network with 256 units, trained in both layers from a small output scale,[8] reaches error 0.2 at $n = 240$ and 0.1 at $n = 480$. At this dimension the gap is a factor of four to five in samples, and it grows like d: by the theorem the kernel needs about $d^2/2$ samples, while a network that finds the direction needs order d up to logarithms (Chapter 7). The kernel curve also shows what the bound does not: $n \geq N$ samples are not enough, and even at $n = 600$ the error is 0.4; Ghorbani et al. (2021) prove that kernel ridge regression learns the degree-2 part only once $n \gg d^2$.

[8] No theorem of this chapter covers this network, trained by Adam with weight decay; it is there to show what moving features buy.

6.6 What the model gets wrong

What was thrown away. The features: in the lazy regime the representation is the random one at initialization, and everything learned is a linear combination of fixed tangent features. Width that is large compared with the data: Theorem 6.8 needs m of order $n^2/\lambda_{\min}^2$, and $\lambda_{\min}$ shrinks with n and is tiny for smooth activations (Section 6.3), so the theorem covers widths that no practical network has. Finite step sizes: gradient descent with a

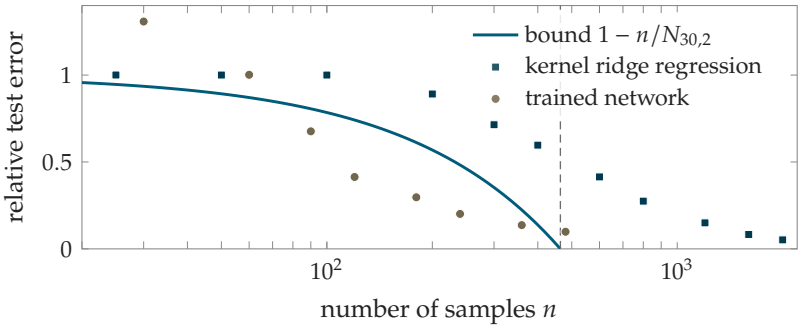


Figure 6.3. A kernel against a network that learns its features, on the target $f_w = \sqrt{N}\, P_{d,2}(\langle w, x\rangle)$ with $x \sim \mathrm{Unif}(\mathbb{S}^{29})$ and no noise; dashed line at $n = N_{30,2} = 464$. Squares: kernel ridge regression with the two-layer ReLU tangent kernel with a bias coordinate (both layers), ridge parameter chosen from $\{10^{-8}, \dots, 10\}$ by test error. Circles: a network $m^{-1/2} \sum_r a_r \,\mathrm{ReLU}(\sqrt{d}\langle w_r, x\rangle + b_r)$ with $m = 256$ trained in all parameters by full-batch Adam (step 0.02, 2000 steps, weight decay 10^{-4}), from $a_r \sim \mathcal{N}(0, 10^{-2})$. Relative errors $\|\hat{f} - f_w\|^2 / \|f_w\|^2$ on 4000 test points, averaged over 3 draws of w and of the data.

large step leaves the linear regime even at large width, and its catapult dynamics are the subject of Chapter 15. Depth and architecture enter only through the kernel, and the interaction of layers, in which one layer learns features that the next one uses, is absent. And the initialization is data independent, so the kernel knows nothing about the task.

WHERE IT BREAKS. *Performance.* The CNTK's 77% on CIFAR-10 (Arora et al. 2019a) is well below what the same kind of architecture reaches when trained in the usual way. In the comparisons of Lee et al. (2020) infinite-width kernels beat finite fully connected networks but lose to finite convolutional networks trained with large steps and weight decay, the very ingredients that break the correspondence. Deep convolutional networks forced into the lazy regime by a large output scale test worse (Chizat et al. 2019), and Geiger et al. (2020) found feature learning better for convolutional networks, though not for fully connected ones.

The kernel moves. In networks trained in the usual way the tangent kernel is not constant. Fort et al. (2020) found that it changes rapidly during the first few epochs and then slowly, and that kernel regression with the kernel reached partway through training comes close to the performance of the fully trained network, which the initial kernel does not. Atanasov et al. (2022) observed, in networks trained from small initializations, a *silent alignment*: early in training, while it is still small, the kernel rotates toward the task, and the network then behaves like kernel regression with the final kernel; the network is a kernel machine, but with a learned kernel. Hanin and Nica (2020) showed that the tangent kernel of a deep ReLU network at initialization is not deterministic when depth and width grow together, its fluctuations being exponential in depth over width, and that its first gradient step moves it by an amount set by the same ratio, so a wide network can learn features if it is deep enough.

Sample complexity. Theorem 6.10 is the cleanest instance of a family of separations. Ghorbani et al. (2021) show that kernel ridge regression with a rotation-invariant kernel learns, in the limit of infinite width, only the projection of the target on polynomials of degree ℓ when $d^\ell \ll n \ll d^{\ell+1}$, and that the linearized two-layer models with finite width are limited by the number of their features in the same way; random features cannot learn even a single ReLU neuron with Gaussian inputs unless their number or their weights are exponential in d (Yehudai and Shamir 2019); and three-layer residual networks learn some functions efficiently through hierarchical feature learning that kernels cannot (Allen-Zhu and Li 2019).

FINE STRUCTURE. *Corrections in* $1/m$. Dyer and Gur-Ari (2020) computed the corrections to the infinite-width dynamics order by order in $1/m$ with Feynman diagrams, Huang and Yau (2020) wrote the evolution of the kernel as an infinite hierarchy of equations whose truncations approximate it, and Bai and Lee (2020) replaced the linearization by a quadratic model that can escape the kernel regime and gains a factor of d in sample complexity. *The other limits.* The mean-field limit of Theorem 6.12 keeps feature learning at infinite width, and μP extends it to deep networks (Yang and Hu 2021). *Crossovers* are the subject of Exercise 6.10. *One step of feature learning.* Ba et al. (2022) and Damian et al. (2022) analyze networks whose first layer takes one or a few large gradient steps, enough to align the features with the target's directions and to beat every kernel. Open: a quantitative theory, for a realistic deep architecture and data, of how much the tangent kernel moves during training, in which directions, and how much the motion improves the test error.

6.7 Model card

MODEL CARD	
PHENOMENON	Very wide networks fit their training data while each weight barely moves; their predictions track kernel regression with the neural tangent kernel; low frequencies are learned first.
MODEL	A smooth map F from parameters to outputs, trained by gradient flow on the square loss; specialized to a two-layer network $m^{-1/2}\sum_r a_r\phi(\langle w_r, x\rangle)$ with Gaussian initialization.
SOLUTION	Convergence inside a ball of radius $\|u_0\|/\mu$ when $L_J\|u_0\| \le \mu^2$ (Theorem 6.3); closeness to the linearization (Corollary 6.4); width $n^2/\lambda_{\min}^2$ suffices for two-layer networks (Theorem 6.8); kernel lower bound $1 - n/N_{d,k}$ (Theorem 6.10).
PREDICTS	Exponential decay of each kernel mode at rate λ_k; relative weight change of order $m^{-1/2}$, with the exact displacement $\sqrt{u_0^\top K^{-1} u_0}$; kernel methods need about $d^k/k!$ samples for a degree-k function of one direction.

GETS WRONG	The features never move; the widths covered are unrealistic; kernels underperform trained networks; the kernel of practical networks changes early and aligns with the task.
SUCCESSORS	$1/m$ corrections and the tangent hierarchy; quadratic models; mean-field limits and μP; one-step feature learning; lazy-to-rich crossovers.

The chapter is the lazy end of a thread that runs through the book. Large initializations made deep linear networks learn all modes at once instead of in stages (Chapter 2) and made diagonal networks find the minimum-ℓ_2 interpolator instead of a sparse one (Chapter 3); grokking is read in Chapter 10 as a transition from an early lazy phase that memorizes to a late rich phase that generalizes; and the single-index model of Chapter 7 shows what the network gains by moving its features: d^{k-1} samples instead of the kernel's d^k. Spectra are a second thread: the time scale $1/\lambda_k$ of each mode here and the power-law spectra that turn kernel regression into scaling laws in Chapter 16 say that a linear model learns its target eigenvalue by eigenvalue. On the Boolean cube the same counting argument, with characters instead of spherical harmonics, shows why kernels cannot learn sparse parities (Chapter 9).

Exercises

6.1 CONSOLIDATION. (Gradient descent on the linearized model.) Run gradient descent $\theta_{s+1} = \theta_s - \eta J_0^\top (F^{\text{lin}}(\theta_s) - y)$ with $K_0 = J_0 J_0^\top$ invertible. (a) Show that $u_s = (I_n - \eta K_0)^s u_0$ and that $u_s \to 0$ for every u_0 if and only if $0 < \eta < 2/\lambda_{\max}(K_0)$. (b) Show that $\|\theta_s - \theta_0\|^2 = u_0^\top K_0^{-1}(I_n - (I_n - \eta K_0)^s)^2 u_0$, and that for $\eta \le 1/\lambda_{\max}(K_0)$ it increases with s to $u_0^\top K_0^{-1} u_0$.

6.2 CONSOLIDATION. (Arc-cosine kernels and odd harmonics.) Let g, g' be standard Gaussians with correlation ρ. (a) Show that $\Pr(g > 0, g' > 0) = (\pi - \arccos\rho)/(2\pi)$ and $\mathbb{E}[\mathrm{ReLU}(g)\,\mathrm{ReLU}(g')] = (\sqrt{1-\rho^2} + (\pi - \arccos\rho)\rho)/(2\pi)$. (b) For $\phi = \mathrm{ReLU}$ in (6.4), show that $K^\infty(x, x') = \frac{\rho}{4} + \frac{\rho \arcsin\rho}{2\pi}$ with $\rho = \langle x, x' \rangle$, and deduce from the parity of $P_{d,k}$ and the orthogonality of the Gegenbauer polynomials that $\lambda_k = 0$ for every odd $k \ge 3$. (c) On $\mathbb{S}^2$, where $P_{3,k}$ is the Legendre polynomial and the weight in Proposition 6.9 is $\frac{1}{2}$, show that $\lambda_0 = \frac{1}{16}$ and $\lambda_1 = \frac{1}{12}$, and check numerically that $\lambda_2 = \frac{7}{256}$. *(Hint: in (a), the event $\{g > 0, g' > 0\}$ is a wedge of angle $\pi - \arccos\rho$ in the plane; in (c), $\int_0^1 t \arcsin t \, \mathrm{d}t = \frac{\pi}{8}$.)*

6.3 CONSOLIDATION. (Parallel inputs.) (a) For $\phi = \tanh$ and inputs x_1 and $x_2 = -x_1$, show that the two rows of K^∞ are proportional, so $\lambda_{\min}(K^\infty) = 0$. (b) Show that appending a bias coordinate, $\tilde{x} = (x, 1)/\sqrt{2}$, makes distinct unit inputs pairwise non-parallel, so that Lemma 6.6 applies to every set of distinct inputs.

6.4 CONSOLIDATION. (Counting dimensions.) (a) Check that $N_{30,2} = 464$

and $N_{30,3} = 4930$. (b) A two-layer network of width m on inputs in $\mathbb{R}^d$ has md tangent features. Show that its linearization, trained on n examples from an initialization independent of w, has expected error at least $1 - (\min\{n, md\} + 1)/N_{d,k}$ on the target of Theorem 6.10.

6.5 VARIATION. ★ (Training both layers.) In Assumption 6.2 let the a_r be trained too, starting from ± 1. (a) Show that the tangent kernel becomes $K_W + K_a$ with $K_a(x, x') = \frac{1}{m}\sum_r \phi(\langle w_r, x\rangle)\phi(\langle w_r, x'\rangle)$, whose expectation is the Gaussian-process kernel $\mathbb{E}[\phi(g)\phi(g')]$, so that $\lambda_{\min}$ can only increase. (b) Show that on the ball of radius R around the initialization, $\|J(\theta) - J(\theta')\|_{\mathrm{op}} \le \beta\sqrt{n/m}\,\sqrt{1 + 2(1+R)^2}\,\|\theta - \theta'\|$. (c) Deduce a version of Theorem 6.8 with m of order $n^3/(\lambda_{\min}^3\delta^2)$, up to a factor depending on β and c_ϕ. Where does the extra factor $n/(\lambda_{\min}\delta)$ come from, and how would a bound on the motion of each a_r remove it? *(Hint: in (b), $|a_r| \le 1 + R$ on the ball, and $a\phi'(s) - a'\phi'(s') = (a - a')\phi'(s) + a'(\phi'(s) - \phi'(s'))$.)*

6.6 VARIATION. (Laziness is a matter of scale.) Let F satisfy hypotheses (i) and (ii) of Theorem 6.3 with constants μ and L_J on every ball around θ_0, with $F(\theta_0) = 0$, and train αF on $\frac{1}{2}\|\alpha F(\theta) - y\|^2$ for a scale $\alpha > 0$. (a) Show that the theorem applies to αF as soon as $\alpha \ge L_J\|y\|/\mu^2$. (b) Show that the parameters then move by at most $\|y\|/(\alpha\mu)$, which tends to 0 as $\alpha \to \infty$, and that the linearization error of Corollary 6.4 is of order $1/\alpha$. (c) Compare with the initialization scale α of diagonal linear networks in Chapter 3, where $F(\theta_0) \neq 0$ is avoided by the parameterization $u \odot u - v \odot v$.

6.7 VARIATION. ★ (Gaussian inputs.) Let $x \sim \mathcal{N}(0, I_d)$, let $w \sim \mathrm{Unif}(\mathbb{S}^{d-1})$, and let $f_w(x) = h_2(\langle w, x\rangle)$. (a) With $H_\alpha = \prod_i h_{\alpha_i}(x_i)$, show that

$$h_2(\langle w, x\rangle) = \sum_{|\alpha|=2} \sqrt{2/\alpha!}\, w^\alpha H_\alpha(x).$$

(b) For g in the span of the H_α with $|\alpha| = 2$, write $\langle g, f_w\rangle = w^\top A w$ with A symmetric and show that $\|g\| = \|A\|_F$. (c) Using $\mathbb{E}_w(w^\top A w)^2 = \frac{2\,\mathrm{tr}\,A^2 + (\mathrm{tr}\,A)^2}{d(d+2)}$, show that the operator $g \mapsto \mathbb{E}_w[\langle g, f_w\rangle f_w]$ has the eigenvalue $\frac{1}{d}$ once (on $g \propto \|x\|^2 - d$) and $\frac{2}{d(d+2)}$ on the rest of its range. (d) Deduce that every n-dimensional V independent of w has $\mathbb{E}\|f_w - \Pi_V f_w\|^2 \ge 1 - \frac{1}{d} - \frac{2(n-1)}{d(d+2)}$, and explain the term $\frac{1}{d}$, which the sphere does not have. *(Hint: for (d), the sum of $\langle e_j, Ce_j\rangle$ over an orthonormal family of size n is at most the sum of the n largest eigenvalues of C.)*

6.8 SIMULATION. (Weights and kernel against width.) Repeat the experiment of Figure 6.2 with independent uniform signs a_r and independent $w_r \sim \mathcal{N}(0, I_3)$, for $m \in \{256, 1024, 4096\}$ and 3 seeds. Observe a relative weight change of about 0.26, 0.14, and 0.07, and a relative kernel change of about 0.04, 0.011, and 0.003 (seeds 0 to 2; other seeds differ by up to 20%). (a) Which quantity follows $m^{-1/2}$ and which m^{-1}? (b) Explain both with Proposition 6.5 and the heuristic derivation of Section 6.5.

6.9 SIMULATION. (A kernel that cannot see odd harmonics.) On $\mathbb{S}^2$, fit kernel interpolation with $K(x,x') = \rho\,(\pi - \arccos\rho)/(2\pi)$, $\rho = \langle x, x'\rangle$ (a regularization of 10^{-10} suffices), to $n \in \{50, 200, 800, 2000\}$ uniform samples of the targets $\sqrt{5}P_2(\langle w, x\rangle)$ and $\sqrt{7}P_3(\langle w, x\rangle)$, with P_k the Legendre polynomials. (a) Observe that the relative test error for degree 2 falls below 10^{-4} by $n = 200$, and that for degree 3 it stays above 1 (between about 1.4 and 2.2 at every n, near 1.45 at $n = 2000$). (b) Explain both with Exercise 6.2. (c) Add a bias coordinate as in Figure 6.1 and observe that the degree-3 target is learned.

6.10 RESEARCH. (The lazy-to-rich crossover.) Train the network of Figure 6.3 with output $\alpha(f(x;\theta) - f(x;\theta_0))$ and loss $\frac{1}{\alpha^2}$ times the square loss, for a range of α and m. For large α it approaches kernel regression with its initial tangent kernel; for small α it learns w. Geiger et al. (2020) map such crossovers empirically, and Woodworth et al. (2020) solve one for diagonal linear networks. Open: for a nonlinear network on high-dimensional data, predict the scale $\alpha_c(m, n, d)$ at which the test error leaves the kernel curve, and the shape of the transition.

Notes and References

The neural tangent kernel was introduced by Jacot et al. (2018), first posted in June 2018; the plan of Theorem 6.8 follows Du et al. (2019), and Arora et al. (2019b) used the kernel to explain why random labels are fitted more slowly than true ones and to bound the generalization error. The abstract form of Theorem 6.3, with a condition on the Jacobian at initialization and its Lipschitz constant, follows Oymak and Soltanolkotabi (2020) and Chizat et al. (2019).

Theorem 6.10 is a simple instance of the lower bounds of Ghorbani et al. (2021), who also prove matching upper bounds; our proof uses only the addition theorem. The mean-field limit was found independently by Mei et al. (2018), Chizat and Bach (2018), Rotskoff and Vanden-Eijnden (2022), and Sirignano and Spiliopoulos (2020), all first posted in 2018.

TRANSLATING THE LITERATURE. Jacot et al. (2018) write Θ for our K^∞ and $\Theta^{(L)}$ for our Θ^L, n_ℓ for the widths, β for our σ_b, and $\Sigma^{(\ell)}$ and $\dot\Sigma^{(\ell)}$ as we do; they train with the loss normalized by the number of examples, which rescales time by n. Du et al. (2019) write $u(t)$ for our $F(\theta_t)$ (so $y - u(t)$ is our $-u_t$), $\mathbf{H}(t)$, $\mathbf{H}^\infty$, and λ_0 for our K_t, K^∞, and $\lambda_{\min}$, and σ for ϕ. Chizat et al. (2019) write h for our F, $Dh(w_0)$ for $J(\theta_0)$, α for the output scale of Exercise 6.6, and measure laziness by $\kappa_h = \|h(w_0) - y^\star\|\,\|D^2h(w_0)\|/\|Dh(w_0)\|^2$, the analogue of our $L_J\|u_0\|/\mu^2$.

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Part III

Feature Learning in High Dimensions

Real networks find the few directions that matter. In high dimension that search is the hard part, and its cost is set by how much signal the target leaks at low order. Plateaus, sudden drops and hidden progress are three faces of the same search.

7 The Search Phase

Single-index models and the information exponent

THE MODEL — SINGLE-INDEX MODEL, ONLINE SGD

IN THIS CHAPTER. When a network must find a few relevant directions in a high-dimensional input, its loss often stays flat for a long time and then drops. The single-index model, a target $g(\langle w_\star, x\rangle)$ of a Gaussian input learned by one neuron with online stochastic gradient descent on the sphere, isolates this search. Its state is one number, the overlap $r = \langle w, w_\star\rangle$, which leaves the equator $r = 0$ at a rate proportional to r^{k-1}, where k is the information exponent, the lowest Hermite degree that the target shares with the neuron. We prove that the population dynamics escapes in time of order $d^{(k-2)/2}$ and that online SGD, whose step size the noise at the equator limits to order $d^{-k/2}$, finds the direction with of order d^{k-1} samples when $k \geq 3$, with explicit constants; a theorem we quote shows that this is sharp up to logarithms. The model is wrong about the task itself: reusing samples, smoothing the loss, or transforming the labels finds the direction with far fewer samples, so the information exponent measures one algorithm, not the difficulty of the problem.

7.1 The phenomenon: plateaus whose length is set by the target

A trained network computes features, functions of the input that make the task easy, and it has to find them. When the task depends on a few directions of a high-dimensional input, finding them is a search over a sphere whose dimension is that of the input, and in the training curve the search shows up as a plateau: the loss stays near its initial value for a long time and then falls in a comparatively short one. Plateaus of this kind are among the oldest observations about gradient training. Committee machines trained online sit on plateaus while their hidden units are interchangeable (Chapter 8), networks trained on sparse parities show no progress in the loss for a long time and then learn abruptly (Chapter 9), and transformers acquire induction heads in a sudden transition (Chapter 14). This chapter isolates the simplest setting in which the length of the plateau can be computed, and the answer depends on the target through one integer.

The integer comes from the Hermite expansion (Lemma A.2).[1] A target $y = g(\langle w_\star, x\rangle)$ with $x \sim \mathcal{N}(0, I_d)$ and $\mathbb{E}g(z)^2 < \infty$ for $z \sim \mathcal{N}(0,1)$ expands as $g = \sum_{j\geq 0} \hat{g}_j h_j$, and the first index $j \geq 1$ with $\hat{g}_j \neq 0$ says how much of the target is visible to low-degree statistics of the input. The targets $\langle w_\star, x\rangle^3$ and $\mathrm{He}_3(\langle w_\star, x\rangle) = \langle w_\star, x\rangle^3 - 3\langle w_\star, x\rangle$ differ by a linear function. The first

[1] The polynomials carry the name of Charles Hermite, who studied them in 1864; Laplace and Chebyshev had used them earlier.

is correlated with $\langle w_\star, x\rangle$, so the empirical average of yx already points at $w_\star$; the second is uncorrelated with every polynomial of degree at most 2 in x, so no statistic of that kind sees it. Experiments show that this difference controls how long gradient training searches.

Observation 7.1 (Ben Arous et al. 2021; Abbe et al. 2023; Arnaboldi et al. 2023). Train a single neuron or a two-layer network by one-pass stochastic gradient descent on inputs with independent coordinates, Gaussian or uniform on $\{-1, 1\}^d$, with a target that depends on one or a few directions. The loss stays near its initial value for a stretch of training and then falls quickly; most of the samples are spent in the flat stretch, and the stretch is longer when the lowest-degree component of the target has higher degree. For sums of monomials of growing degree on the cube, the loss falls in several drops, one for each group of new coordinates found, and the plateau before a drop is longer when more coordinates enter at once. For targets with many flat directions at initialization, such as phase retrieval, two-layer networks escape after a number of samples of order $d \log d$, and overparameterization shortens the escape only by a constant factor.

These are controlled experiments on synthetic data, repeated with several architectures, losses, and step sizes. Most use one or two values of d, so the growth of the plateau as a power of d is read off from theory rather than measured. In practical networks plateaus are common, but we know of no case in which one has been traced to the Hermite degree of a target. The model of this chapter supplies the missing dimension sweep (Figure 7.1). The question it answers: how long does the search last, as a function of the dimension and of the target, and why does the answer depend on the target only through one integer?

7.2 The model: one neuron searching for one direction

Write $P_w^\perp = I_d - ww^\top$ for the orthogonal projection onto the tangent space of the sphere $\mathbb{S}^{d-1}$ at a unit vector w.

Definition 7.2 (The single-index model and spherical online SGD). Let $d \geq 2$ and $w_\star \in \mathbb{S}^{d-1}$, and let g and ϕ be polynomials. Inputs are $x \sim \mathcal{N}(0, I_d)$ and labels are $y = g(\langle w_\star, x\rangle)$. A neuron with weight w predicts $\phi(\langle w, x\rangle)$ and pays the *correlation loss* $\ell(w; x, y) = -y\,\phi(\langle w, x\rangle)$; its population loss is $\mathcal{L}(w) = \mathbb{E}\,\ell(w; x, y)$ for $w \in \mathbb{R}^d$. Given independent samples (x_t, y_t) for $t = 0, 1, 2, \ldots$, a start $w_0 \in \mathbb{S}^{d-1}$, and a step size $\eta > 0$, *spherical online SGD* is the recursion

$$w_{t+1} = \frac{w_t - \eta G_t}{\|w_t - \eta G_t\|}, \qquad G_t = P_{w_t}^\perp \nabla_w \ell(w_t; x_t, y_t) = -y_t\,\phi'(\langle w_t, x_t\rangle)\,P_{w_t}^\perp x_t. \tag{7.1}$$

The *overlap* is $r_t = \langle w_t, w_\star\rangle$.

Each step uses a new sample, so sample complexity and running time are the same number. The update moves along the tangent vector $-G_t$ and

returns to the sphere; because G_t is orthogonal to w_t, the normalization divides by $\sqrt{1+\eta^2\|G_t\|^2} \geq 1$, never by zero. By Lemma 7.4 below, the population loss on the sphere depends on w only through the overlap, $\mathcal{L}(w) = -\Psi(\langle w, w_\star\rangle)$, with

$$\Psi(r) = \sum_{j\geq 0} \hat{g}_j \hat{\phi}_j \, r^j, \qquad -1 \leq r \leq 1. \tag{7.2}$$

For the matched pure link $g = \phi = h_k$ this is $\Psi(r) = r^k$. That case is the hydrogen atom of the chapter, and the reader may keep it in mind throughout.

Definition 7.3 (Information exponent). The *information exponent* of a function g with $\mathbb{E}g(z)^2 < \infty$ is $k^\star(g) = \min\{j \geq 1 : \hat{g}_j \neq 0\}$. For a target g learned by a neuron ϕ on the correlation loss, the relevant integer is $k^\star = \min\{j \geq 1 : \hat{g}_j\hat{\phi}_j \neq 0\}$, the lowest degree at which Ψ is not constant. It is at least $k^\star(g)$, with equality when the Hermite coefficients of the student do not vanish.

The name is due to Ben Arous et al. (2021), who define it through the derivatives of the population loss at the equator; Dudeja and Hsu (2018) had already organized the learning of single-index models by the Hermite expansion of the link.[2] Linear functions, the ReLU, tanh, the sign, and z^3 have information exponent 1, because each is correlated with z; even functions such as z^2 and $|z|$ have information exponent 2; and He_k has information exponent k (Exercise 7.1). Near the equator $r = 0$ the population loss is $-\Psi(0) - \hat{g}_k\hat{\phi}_k r^k + O(r^{k+1})$ with $k = k^\star$, so for $k \geq 2$ the equator is a critical set of the loss, flat to order k. A random start lies almost on it, because a uniformly random unit vector has overlap of order $d^{-1/2}$.[3] The search phase is the escape from this degenerate saddle; for $k = 2$ it is the logistic escape of Chapter 2, and for larger k it is the higher-order saddle that recurs in Chapters 8 and 9.

Each simplification removes one difficulty so that the search can be seen alone. Gaussian inputs make the joint law of $\langle w_\star, x\rangle$ and $\langle w, x\rangle$ depend only on r, so the overlap is the only order parameter, and the Hermite basis diagonalizes the correlation of the two (Lemma 7.4), as the characters diagonalize noise on the cube in Chapter 9. This rotational invariance is the thread that also runs through Chapters 1, 6, 8 and 13. One direction and a known link separate the search for $w_\star$ from the fitting of a function. The sphere removes the dynamics of the norm, which in a network is set by the other layers and by weight decay; the price is the normalization step, which pulls the overlap toward the equator and is one of the two effects that limit the step size. The correlation loss changes nothing essential: on the sphere $\frac{1}{2}\mathbb{E}(y - \phi(\langle w, x\rangle))^2 = \frac{1}{2}\mathbb{E}g(z)^2 + \frac{1}{2}\mathbb{E}\phi(z)^2 - \Psi(r)$, so the square loss has the same landscape, and its stochastic gradient differs by a term of mean zero (Exercise 7.4). Fresh samples make the gradient noise a martingale difference sequence and make samples and steps the same thing.

To invent the model one has to notice two facts. Rotational invariance collapses the d-dimensional dynamics to one number, so that a high-

[2] Dudeja and Hsu designed a semiparametric estimator rather than analyzing SGD; their sample sizes are polynomial in d, with an exponent set by the smoothness of the link and the strength of its signal.

[3] Concentration of measure: all but a fraction $2e^{-d\varepsilon^2/2}$ of the sphere lies within distance ε of each equator.

dimensional search becomes a one-dimensional random walk with drift. And near the equator only the lowest common Hermite degree matters: the drift of the overlap is of order r^{k-1} while the noise is of order one, so the search is a contest between a signal that vanishes at the starting point and a noise that does not.

DICTIONARY

IN THE MODEL	IN A TRAINED NETWORK
weight $w \in \mathbb{S}^{d-1}$	incoming weights of one hidden neuron, normalized
direction $w_\star$	a feature on which the target depends
overlap $r = \langle w, w_\star \rangle$	alignment of a neuron with the feature: the progress of feature learning
equator $\lvert r \rvert \lesssim d^{-1/2}$	where every neuron starts under random initialization
Hermite coefficients $\hat{g}_j$	how the target depends on the feature, degree by degree
information exponent $k^\star$	how much of the target leaks to low-degree statistics
step size η	learning rate
one fresh sample per step	one pass over a large data set
normalization after each step	weight normalization, weight decay, or the scale set by other layers
search phase, r near 0	a loss plateau
descent phase, r near 1	the drop that ends the plateau

7.3 The solution: the overlap and its drift

Everything in this section rests on one identity for Gaussian vectors, which turns the d-dimensional problem into a problem about one number.

Lemma 7.4 (Hermite correlation). *Let U and V be jointly Gaussian standard variables with $\mathbb{E}[UV] = \rho$. Then $\mathbb{E}[h_j(U)h_k(V)] = \delta_{jk}\rho^k$ for all $j, k \geq 0$, and all functions f_1 and f_2 with $\mathbb{E} f_1(z)^2 < \infty$ and $\mathbb{E} f_2(z)^2 < \infty$ satisfy $\mathbb{E}[f_1(U) f_2(V)] = \sum_{k \geq 0} \hat{f}_{1,k} \hat{f}_{2,k} \rho^k$, with $\hat{f}_{i,k} = \mathbb{E}[f_i(z)h_k(z)]$. In particular $\mathcal{L}(w) = -\Psi(\langle w, w_\star \rangle)$ for every $w \in \mathbb{S}^{d-1}$.*

Proof. The first statement is Theorem A.3; we recall the proof because it takes three lines. The generating function $e^{sz - s^2/2} = \sum_{k \geq 0} s^k h_k(z)/\sqrt{k!}$ (Lemma A.2) and the Gaussian moment generating function $\mathbb{E} e^{sU + tV} = e^{(s^2 + 2\rho st + t^2)/2}$ give, for real s and t,

$$\sum_{j,k \geq 0} \frac{s^j t^k}{\sqrt{j!\, k!}} \mathbb{E}[h_j(U)h_k(V)] = \mathbb{E}\left[e^{sU - s^2/2} e^{tV - t^2/2}\right] = e^{\rho st} = \sum_{k \geq 0} \frac{(\rho st)^k}{k!}.$$

Taking the expectation term by term is legitimate because $\mathbb{E}\lvert h_j(U)h_k(V) \rvert \leq 1$ by the Cauchy–Schwarz inequality, so the double series converges absolutely; comparing the coefficients of $s^j t^k$ proves the first statement. For the second,

expand f_1 and f_2 in the orthonormal basis (h_k) and pass to the limit in the bilinear form $(f_1, f_2) \mapsto \mathbb{E}[f_1(U) f_2(V)]$, which the Cauchy–Schwarz inequality bounds by $(\mathbb{E} f_1(z)^2\, \mathbb{E} f_2(z)^2)^{1/2}$. For the last statement, $\langle w_\star, x\rangle$ and $\langle w, x\rangle$ are standard Gaussians with correlation $\langle w, w_\star\rangle$. ∎

For a unit vector w with overlap r and a fresh input x, write $u = \langle w_\star, x\rangle$, $v = \langle w, x\rangle$, and $F = g(u)\,\phi'(v)$, so that the spherical gradient is $G = -F\,P_w^\perp x$. Two scalars govern one step: the component of $-G$ along $w_\star$, which moves the overlap, and the squared length of G, which drives the normalization,

$$a = -\langle G, w_\star\rangle = F\,(u - rv), \qquad N = \|G\|^2 = F^2 \|P_w^\perp x\|^2.$$

One function of the overlap controls the size of both. For $-1 \le r \le 1$ let v and z be independent standard Gaussians, and put

$$\lambda(r) = \mathbb{E}\Big[g\big(rv + \sqrt{1-r^2}\, z\big)^2\, \phi'(v)^2\, (1 + z^2) \Big]. \tag{7.3}$$

Lemma 7.5 (One step in terms of the overlap). *Let $w \in \mathbb{S}^{d-1}$ have overlap r, let $x \sim \mathcal{N}(0, I_d)$, and define u, v, F, a, and N as above. Then*

(a) $\mathbb{E}[F\,P_w^\perp x] = \Psi'(r)\,P_w^\perp w_\star$, *and consequently* $\mathbb{E}a = (1-r^2)\,\Psi'(r)$;

(b) $\mathbb{E}a^2 \le \lambda(r)$ *and* $\mathbb{E}N \le d\,\lambda(r)$.

Proof. For (a), let e be a unit vector orthogonal to w, and let $c(\theta) = w\cos\theta + e\sin\theta$, a curve on the sphere through w with velocity e. By Lemma 7.4, $\mathbb{E}[g(u)\phi(\langle c(\theta), x\rangle)] = \Psi(\langle c(\theta), w_\star\rangle)$ for every θ. Differentiate at $\theta = 0$, under the expectation on the left, which dominated convergence allows because the integrand is a polynomial in Gaussian variables: $\mathbb{E}[F\langle e, x\rangle] = \Psi'(r)\langle e, w_\star\rangle$. The vector $m = \mathbb{E}[F\,P_w^\perp x]$ is orthogonal to w and satisfies $\langle m, e\rangle = \mathbb{E}[F\langle e, x\rangle]$ for every such e, so $m = \Psi'(r)\,P_w^\perp w_\star$. Since $a = F\langle P_w^\perp x, w_\star\rangle$ and $\langle P_w^\perp w_\star, w_\star\rangle = 1 - r^2$, the inner product of m with $w_\star$ is $\mathbb{E}a = (1-r^2)\Psi'(r)$.

For (b), suppose first that $|r| < 1$. Put $e = (w_\star - rw)/\sqrt{1-r^2}$ and $z = \langle e, x\rangle$, and let x' be the projection of x onto the orthogonal complement of w and $w_\star$. Then v, z, and x' are independent, v and z are standard Gaussians, $u = rv + \sqrt{1-r^2}\,z$, $P_w^\perp x = ze + x'$, and $\mathbb{E}\|x'\|^2 = d-2$. Hence $a = \sqrt{1-r^2}\,Fz$, so that $\mathbb{E}a^2 \le \mathbb{E}[F^2 z^2] \le \lambda(r)$, and

$$\mathbb{E}N = \mathbb{E}\big[F^2\big(z^2 + \|x'\|^2\big)\big] = \mathbb{E}[F^2 z^2] + (d-2)\,\mathbb{E}[F^2] \le d\,\lambda(r).$$

In the degenerate case $w = \pm w_\star$ we have $u = \pm v$, so $a = 0$; and $P_w^\perp x$ is independent of v with $\mathbb{E}\|P_w^\perp x\|^2 = d-1$, so $\mathbb{E}N = (d-1)\mathbb{E}[F^2] \le d\,\lambda(\pm 1)$. ∎

Remark 7.6 (The overlap is a Markov chain). The proof of (b) shows more. Given the past, the triple $(u_t, v_t, \|P_{w_t}^\perp x_t\|^2)$ that enters step t has the law of $(r_t v + \sqrt{1-r_t^2}\,z,\ v,\ z^2 + Q)$, with v and z standard Gaussians and Q a χ^2_{d-2} variable, all independent; and r_{t+1} is a function of r_t and this triple (Lemma 7.10 below). So (r_t) is a Markov chain on $[-1, 1]$ whose transition depends on d only through Q. Simulating it costs a few operations per step in every dimension, and we use it in Section 7.5 (Exercise 7.3).

The population flow is the limit of (7.1) as $\eta \to 0$ with ηt fixed.[4] Its overlap obeys a closed equation, and the time it needs to leave the equator already shows the role of k.

Theorem 7.7 (Population dynamics and escape times). *Suppose $\hat{g}_j\hat{\phi}_j \geq 0$ for every $j \geq 1$ and that Ψ is not constant, and let $k = k^\star$. The spherical gradient flow $\dot{w} = -P_w^\perp \nabla \mathcal{L}(w)$ started at $w_0 \in \mathbb{S}^{d-1}$ stays on the sphere, and its overlap solves*

$$\dot{r} = (1 - r^2)\,\Psi'(r). \tag{7.4}$$

If $0 < r_0 < \frac{1}{2}$, then r reaches $\frac{1}{2}$ at the time $t_{1/2} = \int_{r_0}^{1/2} \mathrm{d}s/((1-s^2)\Psi'(s))$, which satisfies

$$\frac{I_k(r_0)}{2^{k-1}\Psi'(\frac{1}{2})} \leq t_{1/2} \leq \frac{4\,I_k(r_0)}{3k\,\hat{g}_k\hat{\phi}_k}, \qquad I_k(r_0) = \int_{r_0}^{1/2} s^{1-k}\,\mathrm{d}s.$$

For $r_0 = d^{-1/2}$ with $d > 4$, the integral is $I_1 = \frac{1}{2} - d^{-1/2}$, $I_2 = \frac{1}{2}\log d - \log 2$, and $I_k = (d^{(k-2)/2} - 2^{k-2})/(k-2)$ for $k \geq 3$: the escape takes a time of order 1, $\log d$, and $d^{(k-2)/2}$ respectively.

[4] The reduction to one order parameter is exact here, in every dimension, because the student has one neuron; the committee machine of Chapter 8 needs a matrix of overlaps.

Proof. By Lemma 7.5(a), every unit vector w satisfies

$$-P_w^\perp \nabla \mathcal{L}(w) = \mathbb{E}[F\,P_w^\perp x] = \Psi'(\langle w, w_\star\rangle)\,(w_\star - \langle w, w_\star\rangle w).$$

Read the right side as a polynomial vector field on $\mathbb{R}^d$. Along its solutions $q = \|w\|^2 - 1$ satisfies the linear equation $\dot{q} = -2\Psi'(r)\,r\,q$, so q stays 0 when it starts at 0: the flow stays on the compact sphere and therefore exists for all time. The inner product of the flow with $w_\star$ is (7.4).

Nonnegative coefficients make $s \mapsto \Psi'(s)/s^{k-1} = \sum_{j\geq k} j\hat{g}_j\hat{\phi}_j s^{j-k}$ nondecreasing on $(0,1]$, so for $0 < s \leq \frac{1}{2}$

$$k\,\hat{g}_k\hat{\phi}_k\,s^{k-1} \leq \Psi'(s) \leq 2^{k-1}\Psi'(\tfrac{1}{2})\,s^{k-1},$$

and $\frac{3}{4} \leq 1 - s^2 \leq 1$. The right side of (7.4) is therefore bounded below by a positive number on $[r_0, \frac{1}{2}]$, so r increases and reaches $\frac{1}{2}$ in finite time, and the substitution $s = r(t)$ gives the formula for $t_{1/2}$. The two displayed bounds on Ψ' bound the integrand $1/((1-s^2)\Psi'(s))$ above and below, which gives the bounds on $t_{1/2}$; the values of I_k are elementary integrals. ∎

For $k \geq 2$ the flow is slow not because the landscape is rough (on the hemisphere $r > 0$ it has no critical points) but because it is flat where the search starts. Online SGD adds noise to this weak drift, and the noise sets the step size.

Why the step size must be of order $d^{-k/2}$. Take $\Psi(r) = r^k$ and expand one step of (7.1) to second order in η (the exact form is Lemma 7.10): at overlap $\rho \ll 1$ the overlap changes by a drift $\eta k \rho^{k-1}$, a centered noise $\eta\xi$ with $\mathbb{E}\xi^2$ of order one, and a normalization term $-\frac{1}{2}\eta^2 N\rho$ with $\mathbb{E}N$ of order d, by Lemma 7.5. To double the overlap, the drift needs $L \approx \rho^{2-k}/(\eta k)$ steps. Over these steps the noise accumulates to about $\eta\sqrt{L} \approx (\eta\rho^{2-k}/k)^{1/2}$, which

is small compared with ρ only if $\eta \ll \rho^k$; the normalization accumulates to about $\eta^2 d\rho L \approx \eta d\rho^{3-k}/k$, which is small compared with ρ only if $\eta \ll \rho^{k-2}/d$. For $k \geq 3$ both conditions are hardest at the start, $\rho = d^{-1/2}$, where both read $\eta \ll d^{-k/2}$. The doubling times form a geometric series dominated by its first term, about $d^{(k-2)/2}/(\eta k)$, which is of order d^{k-1} when η is of order $d^{-k/2}$. In words: the number of samples is the population escape time divided by the largest step that the noise at the equator allows. For $k = 1$ the binding condition is the normalization at $\rho \approx \frac{1}{2}$, which gives $\eta \approx 1/d$ and d samples; for $k = 2$ every doubling takes about $1/(2\eta)$ steps and both conditions read $\eta \ll 1/d$, which gives about $d \log d$ samples. ◇

7.4 The solution: online SGD needs d^{k-1} samples

We now make the heuristic rigorous for $k \geq 3$. The proof uses only the following properties of the pair (g, ϕ).

Assumption 7.8 (The search-phase hypotheses). The functions g and ϕ are polynomials, $k \geq 3$, and there are constants $c_0 > 0$ and $\Lambda < \infty$ such that $\Psi'(r) \geq c_0 r^{k-1}$ and $\lambda(r) \leq \Lambda$ for $0 \leq r \leq \frac{1}{2}$.

Example 7.9 (Matched Hermite links). For $g = \phi = h_k$ we have $\Psi(r) = r^k$, so the drift condition holds with $c_0 = k$. The function λ is a polynomial with $\lambda(0) = 2k(k+1)$ (Exercise 7.2). For $k = 3$ a direct expansion gives $\lambda(r) = 6(4 + 45r^2 + 12r^4 - 30r^6)$, which increases on $[0, \frac{1}{2}]$, so $\Lambda = \lambda(\frac{1}{2}) = 1491/16 < 94$; for $k = 4$ one finds $\Lambda = \lambda(\frac{1}{2}) = 3633/8 < 455$. More generally, every matched link $g = \phi$ with $\hat{g}_1 = \cdots = \hat{g}_{k-1} = 0$ satisfies the drift condition with $c_0 = k\hat{g}_k^2$.

The first step is an exact form of the expansion used in the heuristic. With $(r)_+ = \max(r, 0)$ and $\mathcal{F}_t$ the σ-algebra generated by $x_0, \dots, x_{t-1}$ (the start w_0 is fixed, so w_t and r_t are $\mathcal{F}_t$-measurable), write a_t and N_t for the quantities of Lemma 7.5 computed with $w = w_t$ and the sample x_t.

Lemma 7.10 (One step of SGD, bounded below). *For every t,*

$$r_{t+1} \geq r_t + \eta a_t - \eta^2 E_t, \qquad E_t = \tfrac{1}{2} N_t\,(r_t)_+ + |a_t|\sqrt{N_t/2}, \tag{7.5}$$

and $\mathbb{E}[E_t \mid \mathcal{F}_t] \leq \lambda(r_t)\big(\frac{d}{2}(r_t)_+ + \sqrt{d/2}\big)$.

Proof. Because G_t is orthogonal to w_t, $\|w_t - \eta G_t\|^2 = 1 + \eta^2 N_t$, and the inner product of (7.1) with $w_\star$ gives the exact recursion

$$r_{t+1} = \frac{r_t + \eta a_t}{\sqrt{1 + \eta^2 N_t}} = (r_t + \eta a_t)(1 - \theta_t), \qquad \theta_t = 1 - (1 + \eta^2 N_t)^{-1/2}.$$

Convexity of $q \mapsto (1+q)^{-1/2}$ gives $(1+q)^{-1/2} \geq 1 - q/2$ for $q \geq 0$, so $0 \leq \theta_t \leq \min(1, \eta^2 N_t/2)$; and $\min(1, p) \leq \sqrt{p}$ for $p \geq 0$ gives also $\theta_t \leq \eta\sqrt{N_t/2}$. Bound $r_t + \eta a_t$ above by $(r_t)_+ + \eta|a_t|$, then use the first bound on θ_t in the third term below and the second bound in the fourth:

$$r_{t+1} \geq r_t + \eta a_t - \theta_t (r_t)_+ - \theta_t\, \eta|a_t| \geq r_t + \eta a_t - \tfrac{1}{2}\eta^2 N_t (r_t)_+ - \eta^2 |a_t|\sqrt{N_t/2}.$$

For the conditional expectation, w_t is $\mathcal{F}_t$-measurable and x_t is independent of $\mathcal{F}_t$, so Lemma 7.5(b) applies conditionally with $w = w_t$: $\mathbb{E}[N_t \mid \mathcal{F}_t] \le d\lambda(r_t)$, and by the Cauchy–Schwarz inequality $\mathbb{E}[|a_t|\sqrt{N_t} \mid \mathcal{F}_t] \le (\lambda(r_t) \cdot d\lambda(r_t))^{1/2}$. ∎

The proof of the theorem cuts the search into epochs in which the overlap doubles. The next lemma treats one epoch; it is stated for a general band of overlaps because the descent phase will use it too. For a unit vector w and a fresh sample, E denotes $\frac{1}{2}N(r)_+ + |a|\sqrt{N/2}$, the quantity E_t of (7.5) at $w_t = w$.

Lemma 7.11 (One epoch). *Let $h > 0$, $\ell < \ell'$, $m \ge 0$, $v \ge 0$, and $e \ge 0$ be numbers such that every $w \in \mathbb{S}^{d-1}$ with $\ell - h \le \langle w, w_\star \rangle < \ell'$ satisfies $\mathbb{E}a \ge m$, $\mathbb{E}a^2 \le v$, and $\mathbb{E}E \le e$. Let τ be a stopping time with $r_\tau \ge \ell$ on the event $\{\tau < \infty\}$, let $L \ge 1$ be an integer, and let σ be the first time $t \ge \tau$ with $r_t \notin [\ell - h, \ell')$. Then, outside an event of probability at most*

$$\frac{16\eta^2 v L}{h^2} + \frac{2\eta^2 e L}{h}, \tag{7.6}$$

either $\tau = \infty$ or $r_t > \ell - h + \eta m(t - \tau)$ for every integer t with $\tau \le t \le \min(\sigma, \tau + L)$. Consequently, outside the same event and if $\tau < \infty$:

(i) *if $\eta m L \ge \ell' - \ell + h$, then $r_t \ge \ell'$ for some $t \le \tau + L$, and $r_s > \ell - h$ for $\tau \le s \le t$;*

(ii) *if $\ell' > 1$, then $r_t > \ell - h$ for $\tau \le t \le \tau + L$.*

Proof. The plan is to split the change of the overlap after τ into a drift, a martingale, and a nonnegative loss, and to bound the martingale by Doob's inequality and the loss by Markov's inequality. Let $\mu_s = \mathbb{E}[a_s \mid \mathcal{F}_s]$ and $\xi_s = a_s - \mu_s$, and let χ_s be the indicator of the event $\{\tau \le s < \min(\sigma, \tau + L)\}$. This event belongs to $\mathcal{F}_s$, because τ and σ are stopping times and $\{s < \tau + L\}$ is the complement of $\{\tau \le s - L\}$; on it $r_s \in [\ell - h, \ell')$, so that $\mu_s \ge m$, $\mathbb{E}[\xi_s^2 \mid \mathcal{F}_s] \le \mathbb{E}[a_s^2 \mid \mathcal{F}_s] \le v$, and $\mathbb{E}[E_s \mid \mathcal{F}_s] \le e$. Consider

$$M_t = \sum_{s<t} \chi_s \xi_s, \qquad S = \sum_{s \ge 0} \chi_s E_s.$$

The process (M_t) is a martingale with orthogonal increments, so $\mathbb{E}M_t^2 = \sum_{s<t} \mathbb{E}\big[\chi_s \mathbb{E}[\xi_s^2 \mid \mathcal{F}_s]\big] \le vL$ for every t, since at most L of the indicators equal 1. Doob's L^2 inequality (Lemma A.10) gives $\mathbb{E}\max_{t \le n} M_t^2 \le 4vL$ for every n, hence $\mathbb{E}\sup_t M_t^2 \le 4vL$ by monotone convergence, and Markov's inequality gives $\Pr(\sup_t |\eta M_t| \ge h/2) \le 16\eta^2 vL/h^2$. In the same way $\mathbb{E}S \le eL$, so $\Pr(\eta^2 S \ge h/2) \le 2\eta^2 eL/h$. Let $\mathcal{E}$ be the event that $\sup_t |\eta M_t| < h/2$ and $\eta^2 S < h/2$; its complement has probability at most (7.6).

On $\mathcal{E} \cap \{\tau < \infty\}$, let $\tau \le t \le \min(\sigma, \tau + L)$. Every s with $\tau \le s < t$ has $\chi_s = 1$ and no $s < \tau$ has, so summing (7.5) over $\tau \le s < t$ and using $\mu_s \ge m$ gives

$$r_t \ge r_\tau + \eta \sum_{s=\tau}^{t-1} \mu_s + \eta M_t - \eta^2 \sum_{s=\tau}^{t-1} E_s > \ell + \eta m(t - \tau) - \frac{h}{2} - \frac{h}{2},$$

which is the main claim. For (i), suppose $\sigma > \tau + L$; then $t = \tau + L$ is admissible, so $r_t > \ell - h + \eta m L \geq \ell'$, whereas $t < \sigma$ forces $r_t < \ell'$. Hence $\sigma \leq \tau + L$, and since $r_\sigma > \ell - h$, the overlap leaves the band through the top: $r_\sigma \geq \ell'$. For (ii), the overlap never reaches $\ell' > 1$, so an exit at a time $\sigma \leq \tau + L$ would have $r_\sigma < \ell - h$, contradicting the main claim; hence $\sigma > \tau + L$, and the main claim with $m \geq 0$ gives (ii). ∎

Theorem 7.12 (Online SGD finds the direction). *Suppose Assumption 7.8 holds, let $0 < \varepsilon \leq 1$, and let $r_0 \geq d^{-1/2}$. Run (7.1) with step size and horizon*

$$\eta = \frac{c_0\,\varepsilon}{2^{k+8}\Lambda}\,d^{-k/2}, \qquad T = \frac{2^{2k+10}\Lambda}{c_0^2\,\varepsilon}\,d^{k-1}.$$

Then with probability at least $1 - \varepsilon$ some $t \leq T$ has $r_t \geq \frac{1}{2}$.

Proof. The plan is to cut the climb from r_0 to $\frac{1}{2}$ into epochs in which the overlap doubles, to apply Lemma 7.11(i) to each, and to add up the failure probabilities, which form a geometric series because $k \geq 3$. If $r_0 \geq \frac{1}{2}$, there is nothing to prove. Otherwise let $J \geq 1$ be the least integer with $2^J r_0 \geq \frac{1}{2}$, put $\rho_j = 2^j r_0$ for $0 \leq j < J$ and $\rho_J = \frac{1}{2}$, so that $\rho_{j+1} = \min(2\rho_j, \frac{1}{2})$, and let τ_j be the first time t with $r_t \geq \rho_j$. In particular $\tau_0 = 0$.

The constants of an epoch. Fix $j < J$ and use Lemma 7.11 with $\tau = \tau_j$, $\ell = \rho_j$, $h = \rho_j/2$, and $\ell' = \rho_{j+1}$. Every overlap r in the band $[\rho_j/2, \rho_{j+1})$ lies in $(0, \frac{1}{2})$, so Lemma 7.5(a) and Assumption 7.8 give $\mathbb{E}a = (1 - r^2)\Psi'(r) \geq \frac{3}{4}c_0(\rho_j/2)^{k-1} =: m_j$, and Lemma 7.5(b) gives $\mathbb{E}a^2 \leq \Lambda$. Lemma 7.10 gives $\mathbb{E}E \leq \Lambda(\frac{d}{2}\cdot 2\rho_j + \sqrt{d/2}) \leq 2\Lambda d\rho_j$, because $r < 2\rho_j$ in the band and $\sqrt{d/2} \leq d\rho_j$ (here $\rho_j \geq r_0 \geq d^{-1/2}$ is used). Let L_j be the least integer at least

$$x_j = \frac{3\rho_j}{2\eta m_j} = \frac{2^k}{c_0\,\eta\,\rho_j^{k-2}}.$$

Since $\ell' - \ell + h \leq \frac{3}{2}\rho_j$, the hypothesis $\eta m_j L_j \geq \ell' - \ell + h$ of (i) holds. So outside an event $\mathcal{B}_j$ whose probability is at most (7.6), if $\tau_j < \infty$, then $\tau_{j+1} \leq \tau_j + L_j$.

The failure probability. The number x_j is at least 1, so $L_j \leq 2x_j$. To see this, compare the drift at overlap $\frac{1}{2}$ with its noise: $\frac{3}{4}c_0 2^{1-k} \leq \frac{3}{4}\Psi'(\frac{1}{2}) = \mathbb{E}a \leq (\mathbb{E}a^2)^{1/2} \leq \Lambda^{1/2}$, so $c_0^2 < 4^k\Lambda$ and $x_j \geq 2^k/(c_0\eta) \geq 2^{2k+8}\Lambda/c_0^2 > 1$. Inserting $v = \Lambda$, $e = 2\Lambda d\rho_j$, $h = \rho_j/2$, and $L_j \leq 2x_j$ into (7.6),

$$\Pr(\mathcal{B}_j) \leq \frac{64\eta^2\Lambda L_j}{\rho_j^2} + 8\eta^2\Lambda d L_j \leq \frac{2^{k+1}\Lambda\eta}{c_0}\Big(\frac{64}{\rho_j^k} + \frac{8d}{\rho_j^{k-2}}\Big).$$

Since $\rho_j = 2^j r_0$ and $r_0 \geq d^{-1/2}$, the geometric series give $\sum_{j<J}\rho_j^{-k} \leq \frac{8}{7}d^{k/2}$ and, because $k \geq 3$, $\sum_{j<J}\rho_j^{2-k} \leq 2d^{(k-2)/2}$. Hence

$$\sum_{j<J}\Pr(\mathcal{B}_j) \leq \frac{2^{k+1}\Lambda\eta\,d^{k/2}}{c_0}\Big(\frac{512}{7} + 16\Big) < \frac{2^{k+8}\Lambda\eta\,d^{k/2}}{c_0} = \varepsilon.$$

The time. Outside the union of the events $\mathcal{B}_j$, induction on j gives $\tau_j \le L_0 + \cdots + L_{j-1}$, and therefore

$$\tau_J \le \sum_{j<J} L_j \le \frac{2^{k+1}}{c_0 \eta} \sum_{j<J} \rho_j^{2-k} \le \frac{2^{k+2} d^{(k-2)/2}}{c_0\, \eta} = T. \qquad \blacksquare$$

Three remarks place the theorem. The constants are explicit but not good: for $k = 3$ and $\varepsilon = 0.1$ the step size is $1.6 \times 10^{-6} d^{-3/2}$, while the simulations of Section 7.5 escape reliably with steps about 10^4 times larger. The losses come from Chebyshev-type bounds, the margins of $\rho_j/4$, and the drift bound at the bottom of each band; the exponents are what the proof gets right. Second, the warm start $r_0 \ge d^{-1/2}$ holds with probability about 0.16 for a uniformly random w_0, because $\sqrt{d}\, r_0$ is approximately standard Gaussian; for even links the symmetry $w \mapsto -w$ makes $|r_0| \ge d^{-1/2}$ enough, and a start at $\gamma d^{-1/2}$ with $0 < \gamma < 1$ costs a factor γ^k in the step size (Exercise 7.5). For odd k a start with $r_0 < 0$ is genuinely bad: the population flow then tends to the equator from below and never crosses it (Exercise 7.8). Third, the failure probability enters as $1/\varepsilon$, the price of using only second moments; Ben Arous et al. (2021) obtain probability $1 - o(1)$ from random starts at the cost of factors $\log^2 d$, and Zweig et al. (2023) prove bounds of the same type for a class of non-Gaussian inputs.[5]

After the search, the overlap must still climb from $\frac{1}{2}$ to nearly 1. This descent phase is fast, and its accuracy is limited by the step size.

Proposition 7.13 (The descent phase). *Let $g = \phi = h_k$ with $k \ge 3$, let $d \ge 16$, let $\bar\Lambda = \max_{0 \le r \le 1} \lambda(r)$, and let η and T be as in Theorem 7.12 with $c_0 = k$ and $\Lambda = \bar\Lambda$. If $r_0 \ge d^{-1/2}$, then for every $\delta \in (0,1)$ and every integer t with $T \le t \le 2T$,*

$$\Pr\big(r_t < 1 - \delta\big) \le 2\varepsilon + \frac{1}{\delta}\Big(e^{-\kappa\eta(t-T)} + \frac{2^{k-10}\,\varepsilon}{d^{(k-2)/2}}\Big), \qquad \kappa = k\, 4^{1-k}.$$

Proof. The plan is to wait for the search to end, to show with Lemma 7.11(ii) that the overlap then stays above $\frac{1}{4}$, and to show that above $\frac{1}{4}$ the distance $D_t = 1 - r_t$ contracts in expectation. The hypotheses of Theorem 7.12 hold with $c_0 = k$ and the larger constant $\bar\Lambda$, so the first time τ with $r_\tau \ge \frac{1}{2}$ satisfies $\Pr(\tau > T) \le \varepsilon$.

The overlap stays above $\frac{1}{4}$. For overlaps $r \in [\frac{1}{4}, 1]$, Lemma 7.5 gives $\mathbb{E}a = k r^{k-1}(1-r^2) \ge 0$ and $\mathbb{E}a^2 \le \bar\Lambda$, and Lemma 7.10 gives $\mathbb{E}E \le \bar\Lambda(\frac{d}{2} + \sqrt{d/2}) \le \bar\Lambda d$. Apply Lemma 7.11(ii) with $\ell = \frac{1}{2}$, $h = \frac{1}{4}$, $\ell' = 2$, $m = 0$, and $L = \lfloor 2T \rfloor$: outside an event of probability at most $8\eta^2\bar\Lambda L(32 + d)$, if $\tau < \infty$, then $r_s > \frac{1}{4}$ for $\tau \le s \le \tau + L$, in particular for $\tau \le s \le 2T$. The definitions of η and T give $\eta^2\bar\Lambda T = \varepsilon/(64d)$, so this probability is at most $\varepsilon(32 + d)/(4d) \le \frac{3}{4}\varepsilon$ when $d \ge 16$.

Contraction. Let ν be the first time $s \ge \tau$ with $r_s < \frac{1}{4}$, and for $s \ge T$ let $V_s = D_s\, \mathbf{1}\{\tau \le T,\ s < \nu\}$. The event in the indicator belongs to $\mathcal{F}_s$, and on it $\frac{1}{4} \le r_s \le 1$, so that $\mathbb{E}[a_s \mid \mathcal{F}_s] = k r_s^{k-1}(1 + r_s)(1 - r_s) \ge \kappa D_s$ and $\mathbb{E}[E_s \mid \mathcal{F}_s] \le \bar\Lambda d$. By (7.5), $D_{s+1} \le D_s - \eta a_s + \eta^2 E_s$, and $D_{s+1} \ge 0$. The

[5] The doubling epochs, the martingale bound, and the treatment of the normalization are standard; we have arranged them so that every constant can be checked by hand.

indicator can only switch off as s grows, so

$$\mathbb{E}V_{s+1} \le \mathbb{E}\big[\mathbf{1}\{\tau \le T,\ s < \nu\}\,\mathbb{E}[D_{s+1} \mid \mathcal{F}_s]\big] \le (1-\kappa\eta)\,\mathbb{E}V_s + \eta^2\bar{\Lambda}d.$$

Starting from $\mathbb{E}V_{T'} \le \frac{3}{4}$ at $T' = \lceil T \rceil$ and iterating, then using $\frac{3}{4}e^{\kappa\eta} \le 1$ and inserting the value of η,

$$\mathbb{E}V_t \le \tfrac{3}{4}(1-\kappa\eta)^{t-T'} + \frac{\eta\bar{\Lambda}d}{\kappa} \le e^{-\kappa\eta(t-T)} + \frac{2^{k-10}\,\varepsilon}{d^{(k-2)/2}}.$$

Conclusion. If $\tau \le T$ and $\nu > t$, then $V_t = D_t$. So the event $\{r_t < 1-\delta\}$ is contained in the union of $\{\tau > T\}$, the event that the overlap falls below $\frac{1}{4}$ between τ and $2T$, and $\{V_t > \delta\}$, whose probabilities are at most ε, $\frac{3}{4}\varepsilon$, and $\mathbb{E}V_t/\delta$ by Markov's inequality. ∎

At $t = 2T$ the exponential term equals $\exp(-2^{4-k}d^{(k-2)/2})$, which is negligible: with probability at least $1-2\varepsilon$, up to terms that vanish as d grows, the overlap is within δ of 1 for every fixed $\delta > 0$. The descent takes about $\kappa^{-1}\eta^{-1}\log(1/(\varepsilon\delta))$ steps, of order $d^{k/2}\log(1/(\varepsilon\delta))/\varepsilon$, a vanishing fraction of T: the samples are spent in the search.

7.4.1 What is known to be sharp

Theorem 7.12 is an upper bound for one algorithm. Two kinds of lower bound complete the picture: one for online SGD, and one for every algorithm that learns from correlations. In the next statement, $a_d \ll b_d$ means $a_d/b_d \to 0$ as $d \to \infty$, and the square loss replaces the correlation loss; on the sphere the two have the same population landscape.

Theorem 7.14 (Sharp thresholds for online SGD; Ben Arous et al. 2021). *Let g be a polynomial with information exponent k, let $\phi = g$, and run (7.1) for the square loss $\frac{1}{2}(y-\phi(\langle w,x\rangle))^2$ from w_0 uniform on the hemisphere $\{w : \langle w, w_\star\rangle \ge 0\}$, with n samples and step size η depending on d, n at most polynomial in d.*

(a) *If $n \gg d$ for $k=1$, $n \gg d\log^2 d$ for $k=2$, or $n \gg d^{k-1}\log^2 d$ for $k \ge 3$, then for suitably chosen step sizes $r_n \to 1$ in probability.*

(b) *If $n \ll d$ for $k=1$, $n \ll d\log d$ for $k=2$, or $n \ll d^{k-1}$ for $k \ge 3$, then for small enough step sizes (of order at most $1/d$ for $k \le 2$, and at most $d^{-k/2}$ at the threshold for $k \ge 3$) $\max_{t\le n}|r_t| \to 0$ in probability.*

The thresholds d, $d\log d$, and d^{k-1} are sharp up to the factors $\log^2 d$. The lower bound for $k \ge 3$ is proved only for small step sizes; it says nothing about larger steps, which Section 7.5 shows to be harmful for a different reason. Barzilai and Shamir (2026) prove lower bounds for standard SGD, without the spherical constraint and without very small steps, by an argument outside the statistical-query framework described next. For networks that sum many single-index components with an even link of information exponent $k \ge 3$, Ren et al. (2025) locate the sharp transition times at which online SGD picks up each direction.

The second lower bound concerns *correlational statistical queries*: an algorithm chooses functions ψ of the input with $\mathbb{E}\psi(x)^2 \le 1$ and receives, for each, a number within a tolerance τ of $\mathbb{E}[y\,\psi(x)]$, with the errors chosen by an adversary. The model is the correlational restriction (Bshouty and Feldman 2002) of the statistical queries of Kearns (1998); a gradient of the square or correlation loss is such a query, estimated from samples.

Theorem 7.15 (Correlational lower bound, informal; Damian et al. 2022, Damian et al. 2023). *Let g satisfy $\mathbb{E}g(z) = 0$ and $\mathbb{E}g(z)^2 = 1$ and have information exponent $k \ge 1$. If an algorithm makes q correlational statistical queries of tolerance τ and recovers $w_\star$ to a fixed accuracy for every $w_\star \in \mathbb{S}^{d-1}$, then $\tau \le C_k(\log(qd)/d)^{k/4}$, with C_k depending only on k and the accuracy.*

An empirical average of n samples estimates $\mathbb{E}[y\psi(x)]$ to accuracy about $n^{-1/2}$. Reading $\tau \approx n^{-1/2}$ turns the theorem into the heuristic bound $n \gtrsim (d/\log(qd))^{k/2}$ for gradient methods; it is a heuristic, not a theorem about SGD, because the errors of SGD are random rather than adversarial, as both papers stress. The bound has the same form for odd and even k, and smoothed online SGD matches it up to logarithms for every k (Damian et al. 2023). For $k \le 2$ the correlational bound is weaker than the trivial requirement $n \gtrsim d$, and online SGD is optimal up to logarithms. For $k \ge 3$ it is $d^{k/2}$, and online SGD is off by the factor $d^{k/2-1}$. Smoothing the loss closes this gap (Section 7.6).

7.5 Reading the solution

The solution has three clocks. The population flow needs time of order $d^{(k-2)/2}$ to leave the equator; the equator limits the step size to order $d^{-k/2}$ for $k \ge 3$ and $1/d$ for $k \le 2$; and the number of samples is the first clock divided by the second: d^{k-1} for $k \ge 3$, $d \log d$ for $k = 2$, and d for $k = 1$, the thresholds of Theorem 7.14. The samples are spent almost entirely near the equator: the epoch lengths in the proof of Theorem 7.12 are proportional to ρ_j^{2-k}, so for $k \ge 3$ the first doubling of the overlap costs at least as much as all later ones together, and the descent costs a vanishing fraction (Proposition 7.13).

Figure 7.1 tests the first two clocks, with the step sizes of the heuristic. The slopes are 1 for $k = 1$, slightly more than 1 for $k = 2$ because of the logarithm, and 2 for $k = 3$. The medians agree with the curves within 1% for $k = 1$, within 7% for $k = 2$, and within 12% for $k = 3$, the worst cases at $d = 16$, where the start $r_0 = \frac{1}{4}$ is already halfway to the target. The spread is the other prediction. For $k = 1$ the quartiles narrow from ±7% of the median at $d = 16$ to ±0.3% at $d = 4096$, for $k = 2$ from ±16% to ±5%, while for $k = 3$ the interquartile range stays between 25% and 36% of the median for $d \ge 32$. At the step size $\eta \approx d^{-k/2}$ the noise is as large as it can be without stopping the search, so the escape time stays random as $d \to \infty$. We checked the one-dimensional chain against d-dimensional SGD for $k = 3$ at $d = 16$ and $d = 32$, with 1000 runs each: the medians agree within 2%.

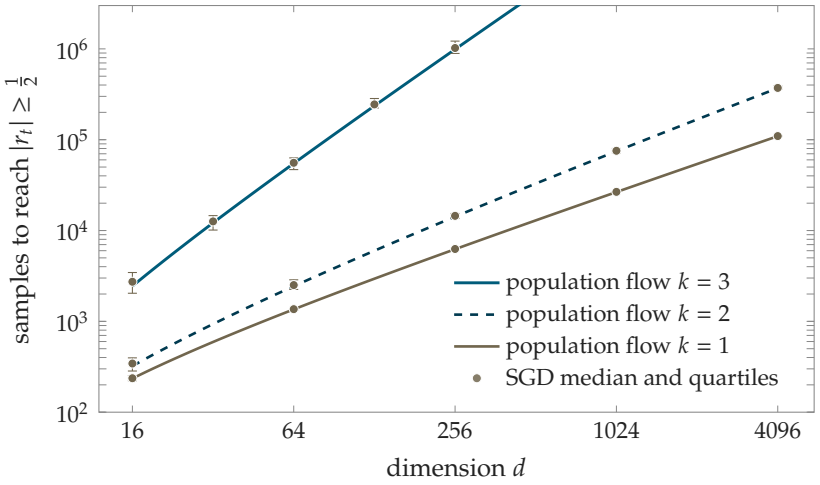


Figure 7.1. Escape times of spherical online SGD for the matched links $g = \phi = h_k$, from $r_0 = d^{-1/2}$, with $\eta = 0.02/d$ for $k \leq 2$ and $\eta = 0.02\, d^{-3/2}$ for $k = 3$. Curves: the population prediction $t_{1/2}/\eta$ of Theorem 7.7, with no fitted constant. Points: medians and quartiles of the first time t with $r_t \geq \frac{1}{2}$ (with $|r_t| \geq \frac{1}{2}$ for $k = 2$), over 200 runs for $k \leq 2$ and 100 runs for $k = 3$, simulated with the exact one-dimensional chain of Remark 7.6 (seed 20261017). One run in a hundred at $k = 3$, $d = 256$, had not escaped after three times the predicted time.

Figure 7.2 shows single runs. On a logarithmic scale the overlap grows from the start, as the population flow predicts; for $k = 3$ the flow makes $1/r_t$ decrease almost linearly in t while r_t is small. Yet the loss $1 - r_t^3$ stays within 10^{-2} of its initial value until the overlap passes 0.2: progress is hidden from the loss because the loss sees r^k, not r, the same mechanism as the hidden progress of Chapter 9. The runs escape at different times, between 0.8 and 1.4 times the population prediction, and then descend in a short and nearly deterministic time. They settle near $r \approx 0.985$, not at 1. Near $r = 1$ the normalization pulls the overlap down by about $\frac{1}{2}\eta^2 \mathbb{E}N \approx \frac{1}{2}\eta^2(d-1)\mathbb{E}[F^2]$ per step, and the drift pulls it up by about $\eta k r^{k-1}(1 - r^2) \approx 2k\eta(1 - r)$; they balance at $1 - r \approx \eta(d-1)\mathbb{E}[F^2]/(4k)$, which is 0.014 here, since $\mathbb{E}[F^2] = 93$ at $r = 1$ for $k = 3$. A smaller step after the search removes this floor, as in the two-phase schedules of Damian et al. (2023).

The equator also has a picture of its own. Near $r = 0$, Lemma 7.5 and Lemma 7.10 give, to leading order, $r_{t+1} - r_t \approx \eta k r_t^{k-1} - \frac{1}{2}\eta^2 k d\, r_t + \eta \xi_t$ with $\mathbb{E}\xi_t^2 \approx k(2k+1)$ for the matched link h_k. Without the drift this is an autoregressive process whose stationary variance is $(2k+1)/d$ (Exercise 7.11): SGD noise by itself keeps the overlap at the scale $d^{-1/2}$ of a random start. The drift beats the restoring force $\frac{1}{2}\eta^2 k d\, r$ only where $r^{k-2} > \eta d/2$, and this threshold lies inside the band of width $\sqrt{(2k+1)/d}$ exactly when $\eta \lesssim d^{-k/2}$. Larger steps do not stop the search outright but make it a rare event, a large deviation of the equatorial noise (Exercise 7.11); the lower bound of Theorem 7.14 does not cover them, and Rangriz (2026) proves limit theorems at the critical step size in which, near a fixed point, the rescaled summary statistics follow an Ornstein–Uhlenbeck process.[6]

[6] Ben Arous et al. (2022) prove limit theorems for summary statistics of SGD on time scales of order $1/\eta$. With $\eta = c/d$ and that time unit, the normalization term above becomes a restoring drift $-\frac{ck}{2} r$ of order one, which for $k \geq 3$ makes the equator a stable fixed point of the limiting

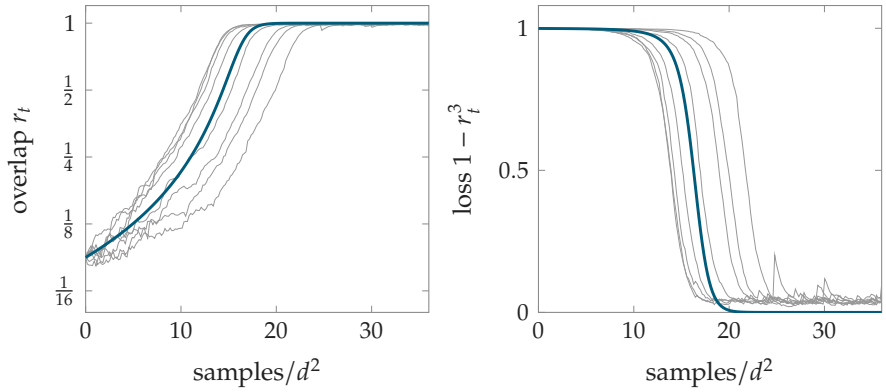


Figure 7.2. Eight runs of spherical online SGD in dimension $d = 128$ for $g = \phi = h_3$, with $\eta = 0.02\, d^{-3/2}$ and $r_0 = d^{-1/2}$ (grey), and the population flow (curve). Left: the overlap, on a logarithmic scale. Right: the population square loss $\frac{1}{2}\mathbb{E}(y - h_3(\langle w_t, x\rangle))^2 = 1 - r_t^3$. The runs simulate (7.1) in $\mathbb{R}^{128}$ directly (seed 7).

7.6 What the model gets wrong

WHAT WAS THROWN AWAY. Isotropic Gaussian inputs, which made the overlap a sufficient statistic and the Hermite polynomials the right basis; with structured or non-Gaussian inputs neither holds. One relevant direction and a known link: a network must find several directions and fit an unknown function of them, and the order in which directions are found becomes part of the problem. One neuron with nothing on top of it: width changes only the start, since the best of m random neurons has overlap about $\sqrt{2\log m/d}$, which shortens the search by a factor of order $(\log m)^{(k-2)/2}$ (Exercise 7.9). One pass over fresh samples. The sphere, which fixes the norm that real networks must learn. The correlation loss with a matched student, small constant steps, and no momentum or adaptive preconditioning. And a warm start.

WHERE IT BREAKS. The prediction that information exponent k costs about d^{k-1} samples concerns one algorithm; other gradient methods do much better (Table 7.1). *Smoothing the loss.* Damian et al. (2023) run online SGD on a loss averaged over a cap of radius λ around w; with $\lambda = d^{1/4}$ the signal near the equator grows faster than the noise, and $d^{k/2}$ samples suffice up to logarithms, which matches Theorem 7.15. So d^{k-1} is not a limit of correlational methods, only of plain online SGD. *Reusing samples.* Two steps on the same batch let the update depend nonlinearly on the label, which acts as an implicit transformation of y. Dandi et al. (2024) show that gradient descent that reuses a batch of $n \asymp d$ samples finds, within two steps, directions that one-pass SGD with as many samples cannot find, and Lee et al. (2024) show that a two-layer network trained by SGD with reused minibatches learns every polynomial single-index target with d samples up to polylogarithmic factors; Arnaboldi et al. (2024b), in a workshop paper, reach a similar conclusion for multi-index targets. *The information exponent is not the hardness of the problem.* Damian et al. (2024) define the generative exponent, the least information exponent of $T(y)$ over

Table 7.1. The cost, in samples and up to logarithmic factors, of finding $w_\star$ for $y = \mathrm{He}_k(\langle w_\star, x\rangle)$ with $k \geq 3$, by method. Statistical queries reach d because the generative exponent of every polynomial link is at most 2; the information-theoretic cost is of order d for every single-index model.

method	samples	source
kernels, random features	d^k	(Ghorbani et al. 2021); Chapter 6
online SGD, one neuron	d^{k-1}	Theorems 7.12 and 7.14
correlational queries	$d^{k/2}$	Theorem 7.15
smoothed online SGD	$d^{k/2}$	(Damian et al. 2023)
statistical queries	d	(Damian et al. 2024)
SGD with reused samples	d	(Lee et al. 2024)

all square-integrable transformations T of the label, prove that statistical-query and low-degree algorithms need $d^{k_g/2}$ samples when it equals k_g, and match this with a partial-trace algorithm. Every polynomial link has $k_g \leq 2$; for $y = \mathrm{He}_3(\langle w_\star, x\rangle)$ the generative exponent is 1, so the hydrogen atom of this chapter is easy for the right algorithm. *The loss and the student matter.* Joshi et al. (2024) show for sparse functions that gradient queries of the square loss are exactly as weak as correlational queries, while those of losses such as the absolute loss are as strong as general statistical queries; and a student whose Hermite coefficients vanish where the target's do not can lose the signal altogether (Exercise 7.1). *Anisotropy.* When the input covariance has a spike aligned with $w_\star$, the effective dimension drops and fewer samples suffice (Ba et al. 2023); but Mousavi-Hosseini et al. (2023) show that spherical gradient flow that ignores the covariance can then stay at the equator, while a normalization adapted to the covariance exploits the spike.

FINE STRUCTURE. The successors repair one omission each. For targets with several directions, the information exponent becomes the leap complexity: Abbe et al. (2022) characterize the sparse targets learnable with order d samples by a merged-staircase property, and Abbe et al. (2023) conjecture that online SGD needs about $d^{\max(L-1,1)}$ samples for a target of leap L, proving it in restricted Gaussian settings; the generative version of the leap, with a matching spectral algorithm that needs about $d^{\max(1,k/2)}$ samples for a leap k, is due to Damian et al. (2025). With large steps and $n \asymp d$ samples, a single gradient step already learns the linear part of a target (Ba et al. 2022). When the link is unknown, a shallow network can learn it together with the direction, with about d^k samples for gradient flow on the empirical loss (Bietti et al. 2022), and two-timescale gradient flow on multi-index targets passes a sequence of saddles predicted by the Hermite structure (Bietti et al. 2025). Beyond Gaussian inputs, a local growth condition on the population drift replaces the information exponent (Zweig et al. 2023), and for spherically symmetric inputs spherical harmonics replace Hermite polynomials, with one estimator that is optimal in samples and another, based on online SGD, that is optimal in running time (Joshi et al. 2025). The step size itself is a knob: for a class of gradient algorithms that covers

one-pass SGD and SGD with batch reuse, Tsiolis et al. (2025) show that in some cases a growing learning rate moves the learner from the regime governed by the information exponent to the one governed by the generative exponent. Open are whether plain minibatch SGD, with neither smoothing nor reuse, can reach $d^{k/2}$; the limit law of the escape time at the critical step size for odd k (Exercise 7.12); and lower bounds for online SGD that hold for every step size.

7.7 Model card

MODEL CARD	
PHENOMENON	A loss plateau while a network searches for the relevant directions of a high-dimensional input; longer for targets whose low-degree Hermite components vanish.
MODEL	$y = g(\langle w_\star, x\rangle)$, $x \sim \mathcal{N}(0, I_d)$; one neuron $\phi(\langle w, x\rangle)$ on the sphere; online SGD on the correlation loss; order parameter $r = \langle w, w_\star\rangle$.
SOLUTION	Population drift $(1 - r^2)\Psi'(r) \approx k\hat{g}_k\hat{\phi}_k r^{k-1}$; flow escapes in time of order $d^{(k-2)/2}$; with $\eta \asymp \varepsilon d^{-k/2}$, online SGD reaches $r = \frac{1}{2}$ within $O(d^{k-1}/\varepsilon)$ samples for $k \geq 3$ (proved), then descends in $O(d^{k/2}\log(1/\varepsilon\delta)/\varepsilon)$ samples.
PREDICTS	Sample complexities d, $d\log d$, d^{k-1} for $k = 1, 2, \geq 3$; a step size limited by the equator; flat loss with hidden progress in r; an accuracy floor proportional to ηd.
GETS WRONG	It measures one algorithm: smoothing reaches $d^{k/2}$, sample reuse and label transforms reach about d for polynomial links; anisotropic and non-Gaussian inputs change the exponent; one direction, known link, one neuron.
SUCCESSORS	Generative exponent; correlational and statistical-query lower bounds; smoothing and partial traces; leap complexity for multi-index targets; effective dynamics of SGD.

The search phase is the third appearance in this book of an escape from a saddle at small initialization. In Chapter 2 the escape is logistic, with time log(1/initialization)/strength, which is the case $k = 2$ here; in Chapter 3 the same mechanism orders the coordinates; here a degenerate saddle of order k turns the logarithm into the power $d^{(k-2)/2}$. The committee machine of Chapter 8 adds a symmetric saddle between students, analyzed with the same order parameters and the ODE method (Theorem A.13), and Chapter 9 is the Boolean version of this chapter: a parity on k coordinates has no correlation with low-degree functions, its hidden progress is the growth of $\log r$ under a flat loss, and its statistical-query bound is the counterpart of Theorem 7.15.

Against Chapter 6 the chapter gives the first quantitative separation between feature learning and kernels: a kernel method needs about d^k samples for a degree-k target, one neuron trained by online SGD needs d^{k-1}, and better gradient methods need $d^{k/2}$ or fewer.

Exercises

7.1 CONSOLIDATION. (Information exponents.) (a) Show that $z \mapsto \mathrm{ReLU}(z)$, $\tanh z$, $\operatorname{sgn} z$, and z^3 have information exponent 1, that z^2 and $|z|$ have information exponent 2 with $|\widehat{z}|_2 = 1/\sqrt{\pi}$, and that He_k has information exponent k. (b) Show that the Hermite coefficients of the ReLU vanish at every odd degree $j \geq 3$. Conclude that a ReLU neuron trained on the correlation loss for the target $h_3(\langle w_\star, x\rangle)$ has $\Psi \equiv 0$: it receives no signal at all, at every overlap. *(Hint:* $\mathrm{ReLU}(z) = \frac{z}{2} + \frac{|z|}{2}$.*)*

7.2 CONSOLIDATION. (Noise at the equator.) Let $g = \phi = h_k$. (a) Use $z h_k = \sqrt{k+1}\, h_{k+1} + \sqrt{k}\, h_{k-1}$ to show that $\lambda(0) = 2k(k+1)$, that $\mathbb{E}a^2 = k(2k+1)$ at overlap 0, and that $\mathbb{E}N = k(2k+1) + (d-2)k$ there. (b) Check the value $\lambda(\frac{1}{2}) = 1491/16$ of Example 7.9 by expanding $h_3(\frac{1}{2}v + \frac{\sqrt{3}}{2}z)$ in monomials.

7.3 CONSOLIDATION. (The overlap is a Markov chain.) Prove the claim of Remark 7.6: given $\mathcal{F}_t$, the law of r_{t+1} depends only on r_t. Write the transition as an explicit function of r_t and of independent $v, z \sim \mathcal{N}(0,1)$ and $Q \sim \chi^2_{d-2}$.

7.4 CONSOLIDATION. (Square loss.) For the square loss $\frac{1}{2}(y - \phi(\langle w, x\rangle))^2$, show that the spherical gradient is that of the correlation loss plus $\phi(v)\phi'(v)P^{\perp}_w x$, that this extra term has mean zero, and that Theorem 7.12 holds for the square loss with $g(u)$ replaced by $g(u) - \phi(v)$ in the definition (7.3) of λ.

7.5 CONSOLIDATION. ⋆ (Other exponents and colder starts.) (a) Adapt the proof of Theorem 7.12 to $k = 2$: show that η of order $\varepsilon/(d \log d)$ and $O(d \log^2 d/\varepsilon)$ samples suffice, and locate the logarithm that Theorem 7.14 suggests should be removable. (b) Do the same for $k = 1$, with η of order ε/d and $O(d/\varepsilon)$ samples. (c) For $k \geq 3$ and $r_0 \geq \gamma d^{-1/2}$ with $0 < \gamma < 1$, show that the conclusion holds with η multiplied by γ^k and T by γ^{2-2k}. *(Hint: in (a) every epoch contributes the same failure probability, and there are about* $\log_2 \sqrt{d}$ *epochs.)*

7.6 VARIATION. (Label noise.) Let $y = g(\langle w_\star, x\rangle) + \sigma\zeta$ with $\zeta \sim \mathcal{N}(0,1)$ independent of x. Show that the drift of Lemma 7.5(a) is unchanged and that $\lambda(r)$ increases by $2\sigma^2\mathbb{E}\phi'(z)^2$, which is $2k\sigma^2$ for $\phi = h_k$. By what factor does the bound of Theorem 7.12 grow?

7.7 VARIATION. (A little leakage.) Let $g = \phi = h_3 + \beta h_1$ with $0 < \beta < 1$, so that $\Psi(r) = r^3 + \beta^2 r$ and the information exponent is 1. Solve the population flow near the equator and show that the escape from $r_0 = d^{-1/2}$ takes time of order $\min(\sqrt{d}, 1/\beta)$. For which β does the degree-1 leakage dominate the search?

7.8 VARIATION. (Starting below the equator.) Let $g = \phi = h_k$ and $-1 < r_0 < 0$. (a) For even k, show that the population flow converges to $-w_\star$, and explain why that is as good as $w_\star$. (b) For odd k, show that

$r(t)$ increases to 0 and never becomes positive. What must the noise of SGD do in that case?

7.9 VARIATION. (Width.) Start m neurons independently and uniformly on the sphere, and train each by (7.1). Show that the largest initial overlap is about $\sqrt{2\log m/d}$ when $\log m \ll d$, and use Theorem 7.7 to show that the fastest neuron escapes earlier than a typical one by a factor of order $(2\log m)^{(k-2)/2}$.

7.10 SIMULATION. (Escape times for $k = 4$.) Using the chain of Remark 7.6 with $g = \phi = h_4$, $r_0 = d^{-1/2}$, and $\eta = 0.02\, d^{-2}$, measure the median escape time to $|r| \geq \frac{1}{2}$ over 50 runs for $d = 16, 32, 64, 128$. You should observe a slope close to 3 on a log-log plot, and medians about 30% above $t_{1/2}/\eta$ from Theorem 7.7 at $d = 16$ but within 10% for $d \geq 32$; explain why the agreement is worse at small d.

7.11 SIMULATION. (The equator and the step size.) (a) Show that the recursion $r_{t+1} = (1 - \frac{1}{2}\eta^2 kd)r_t + \eta\xi_t$ with independent ξ_t of variance $k(2k+1)$ has stationary variance close to $(2k+1)/d$ when $\eta^2 kd \ll 1$. (b) For $k = 3$ and $d = 256$, simulate the chain of Remark 7.6 with $\eta = c\, d^{-3/2}$ for c between 0.25 and 16, and plot the median escape time divided by d^2 against c. You should observe a decrease like $0.3/c$, the population prediction, for small c, a minimum for c between 2 and 4, and a steep increase beyond (at $c = 16$ almost no run escapes within $30d^2$ steps). Relate the increase to the equatorial band of (a).

7.12 RESEARCH. (The law of the search time.) Let $g = \phi = h_k$ with $k \geq 3$, $\eta = c\, d^{-k/2}$, and $r_0 = d^{-1/2}$. Show heuristically that $\bar{r}_s = \sqrt{d}\, r_{\lfloor s d^{k-1} \rfloor}$ approximately solves $\mathrm{d}\bar{r} = c\big(k\bar{r}^{k-1} - \frac{ck}{2}\bar{r}\big)\,\mathrm{d}s + c\sqrt{k(2k+1)}\,\mathrm{d}W_s$, so that the escape time divided by d^{k-1} should converge in law to the blow-up time of this equation started at $\bar{r} = 1$. A proof is open as far as we know: the limit theorems of Ben Arous et al. (2022) cover time scales of order $1/\eta$, shorter by the factor $d^{(k-2)/2}$.

7.13 RESEARCH. (Plain SGD at the correlational rate.) Can minibatch SGD without explicit smoothing or sample reuse, with suitably tuned batch size and step size, learn targets of information exponent $k \geq 3$ with about $d^{k/2}$ samples? Test this numerically for h_3. Open as far as we know; Arnaboldi et al. (2024a) analyze the effect of the batch size on online SGD, and Damian et al. (2023) the effect of explicit smoothing.

Notes and References

The rigorous theory of the search phase starts with Ben Arous et al. (2021), who named the information exponent, proved the thresholds of Theorem 7.14 for a general class of losses, and described the search and descent phases; a precursor is their analysis of Langevin dynamics and gradient descent for spiked tensor models (Ben Arous et al. 2020), where the order of the tensor plays the role of k and sets the signal strength needed to leave

the equator.[7] Before them, Tan and Vershynin (2023) had shown, in work first posted in 2019, that online SGD solves phase retrieval, an example with $k = 2$, from an arbitrary start.[8] Mondelli and Montanari (2019) located the information-theoretic weak-recovery threshold at order d samples.

The correlational lower bound is due to Damian et al. (2022), who also showed that two-layer networks trained by gradient descent beat kernel methods on such targets; Damian et al. (2023) extended it to every link and matched it with smoothed online SGD. Abbe et al. (2023) stated the corresponding bounds for multi-index targets in terms of the leap. The generative exponent (Damian et al. 2024) and the sample-reuse papers (Arnaboldi et al. 2024b; Dandi et al. 2024; Lee et al. 2024) showed that the correlational picture is not the whole story. Bruna and Hsu (2025) survey algorithms for single- and multi-index models.

The effective-dynamics theorems of Ben Arous et al. (2022) turn the heuristic expansion of Section 7.3 into limit theorems for summary statistics on time scales of order $1/\eta$, and the ODE method of Theorem A.13 is the special case in which the noise vanishes in the limit and the summary statistics follow a deterministic equation.

TRANSLATING THE LITERATURE. Ben Arous et al. (2021) write N for our d, $M = \alpha N$ for our n, δ/N for our η, and the latitude m, the inner product of the parameter with the planted direction, for our r. Their population loss is our $-\Psi(r)$ up to a constant, and their information exponent, the order of the first nonvanishing derivative of the population loss at $m = 0$, agrees with our $k^\star$ for matched links. Damian et al. (2023) write σ for the known link, our $g = \phi$; $w^\star$ for $w_\star$; $\alpha = w \cdot w^\star$ for r; He_k for the monic Hermite polynomials, with coefficients $c_k = \mathbb{E}[\sigma\,\mathrm{He}_k] = \sqrt{k!}\,\hat{g}_k$; $k^\star$ for the information exponent; and $L(w; x; y) = 1 - \sigma(w \cdot x)y$ for our $\ell + 1$, with λ the smoothing radius, unrelated to our $\lambda(r)$. In Damian et al. (2024) the letter $k^\star$ changes role and denotes the generative exponent, our k_g. Dudeja and Hsu (2018) write p for our d.

[7] In tensor PCA the equator is the set of vectors uncorrelated with the planted spike, and a random start lies on it for the same reason as here.

[8] Phase retrieval, $y = |\langle w_\star, x\rangle|$, is even, so its information exponent is 2, and the escape is the logistic escape of Chapter 2.

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8 Plateaus and Specialization

The committee machine and the ODE method

THE MODEL — SOFT COMMITTEE MACHINE, TEACHER AND STUDENT

IN THIS CHAPTER. A network with a few hidden units, trained online from small weights on a target computed by a network of the same form, often learns in two steps: the error drops to an intermediate value, stays there while the hidden units are interchangeable copies of one another, and drops again when they specialize. The soft committee machine, a sum of erf units learning from a teacher of the same kind on Gaussian inputs, reproduces this. We prove that as the dimension grows the overlaps between student and teacher weights follow a deterministic system of ordinary differential equations, with closed-form coefficients and a term of order η^2 that comes from the gradient noise. From it we derive a learning-rate threshold $\pi\sqrt{5/3}$ for a single unit, and, for K students and K orthonormal teachers, an exact symmetric fixed point with an explicit unstable eigenvalue, which gives a plateau whose length grows like $\log d$. The model is silent about why real networks seldom show such plateaus: structured data, overparameterization, preconditioned gradients, and large steps all change the picture, and the plateau's length is set by fluctuations that the limiting equations leave out.

8.1 The phenomenon: redundant units and a sudden division of labor

A hidden layer has a symmetry that a single neuron lacks. Permuting the hidden units of a two-layer network, together with their outgoing weights, does not change the function it computes, so every function the network can represent is represented many times, and the loss landscape repeats itself under the permutation group. Points fixed by a permutation, where two units carry the same weights, are special: the gradient there respects the symmetry, so two identical units receive identical updates and stay identical. Small random initializations drive the network toward such points, because at the start every unit sees roughly the same average target and moves the same way.

The consequence shows up in training curves as a plateau that has nothing to do with a search for a hidden direction. The chapter on single-index models (Chapter 7) produced plateaus because the target was invisible to low-degree statistics; here every hidden unit sees its signal at once, and the delay comes instead from the need to break a symmetry. Physicists who analyzed online learning in small sigmoidal networks during the

1990s called the redundant stretch the *symmetric phase* and the end of it *specialization*.

Observation 8.1 (Saad and Solla 1995b; Biehl et al. 1996; Fukumizu and Amari 2000). Train a two-layer network with a few sigmoidal hidden units, by online gradient descent from small random weights, on inputs with independent coordinates and labels produced by a network of the same form. The error falls quickly to an intermediate value and then stays nearly constant for a long stretch, during which the hidden units have nearly the same incoming weights, each a blend of all the teacher's units. The stretch ends with a second drop, in which the units differentiate and each aligns with one teacher unit. With several teacher units the dynamics can pass through several such plateaus in a cascade. The plateaus are associated with critical points at which units are duplicated, which exist for every target, loss, and activation function.

The evidence is of an unusual kind. The first two sources combine an exact theory, the one this chapter develops, with simulations of the stochastic algorithm; the third is a structural statement about the loss landscape: critical points of a network with fewer hidden units reappear in a larger network as lines of critical points, along which training can stall (Fukumizu and Amari 2000).[1] Plateaus of this kind were a recognized problem of backpropagation: Amari (1998) suggested that the natural gradient would make it less serious. In current practice they are seldom seen, a gap that Yoshida and Okada (2019) trace in part to the input statistics. Section 8.6 returns to the reasons.

The question the model answers: what sets the level of the plateau, why does it last as long as it does, and what in the dynamics decides when it ends?

[1] Fukumizu and Amari show that such a line can contain local minima as well as saddles, and that the conclusion does not depend on the target, the loss, or the activation, only on the hierarchical structure of the model.

8.2 The model: the soft committee machine

Write $g(z) = \operatorname{erf}(z/\sqrt{2}) = 2\Pr(\mathcal{N}(0,1) \le z) - 1$.[2] This is an odd sigmoid with values in $(-1, 1)$, and

$$g'(z) = \sqrt{\tfrac{2}{\pi}}\, e^{-z^2/2}, \qquad g''(z) = -z\, g'(z). \tag{8.1}$$

Every derivative of g is a polynomial times $e^{-z^2/2}$, so all derivatives are bounded.

[2] So g is the Gaussian distribution function rescaled to $(-1, 1)$, the probit counterpart of tanh.

Definition 8.2 (Soft committee machine, teacher and student). Let $d \ge m + m_\star$. A *teacher* has weights $w_{\star 1}, \dots, w_{\star m_\star} \in \mathbb{R}^d$ with Gram matrix $T = (\langle w_{\star k}, w_{\star k'} \rangle)$, and a *student* has weights $W = (w_1, \dots, w_m)$. For an input $x \sim \mathcal{N}(0, I_d)$ put $z_j = \langle w_j, x \rangle$ and $z_{\star k} = \langle w_{\star k}, x \rangle$; the label and the prediction are

$$y = \sum_{k=1}^{m_\star} g(z_{\star k}), \qquad f(x; W) = \sum_{j=1}^{m} g(z_j),$$

and the population loss is $\mathcal{L}(W) = \tfrac{1}{2}\mathbb{E}\big(y - f(x;W)\big)^2$. Online SGD

with step size η/d draws a fresh input x^s at step $s = 0, 1, 2, \dots$ and updates

$$w_j^{s+1} = w_j^s + \frac{\eta}{d}\,\delta_j^s\,x^s, \qquad \delta_j^s = g'(z_j^s)\big(y^s - f(x^s; W^s)\big), \tag{8.2}$$

which is a gradient step on $\frac{1}{2}(y^s - f(x^s; W^s))^2$. The *order parameters* are the overlaps

$$Q_{jl} = \langle w_j, w_l\rangle, \qquad R_{jk} = \langle w_j, w_{\star k}\rangle, \qquad T_{kk'} = \langle w_{\star k}, w_{\star k'}\rangle.$$

This is the *soft committee machine*: a two-layer network whose second layer is fixed to ones, so that the units vote with equal weight, and whose units are soft versions of the sign. Its weights have norm of order one, so the preactivations z_j are of order one, and the gradient step $\frac{\eta}{d}\delta_j x$ has norm of order $\eta d^{-1/2}$, which is the scaling at which each step changes the overlaps by $O(1/d)$. Time is measured in samples per dimension, $t = s/d$.

Rotational invariance does the first reduction. Given the current weights, the vector $(z_1, \dots, z_m, z_{\star 1}, \dots, z_{\star m_\star})$ is a centered Gaussian with covariance

$$\mathsf{M} = \begin{pmatrix} Q & R \\ R^\top & T \end{pmatrix}, \tag{8.3}$$

so the population loss, and the mean of every update of the overlaps, depend on the d-dimensional weights only through Q and R. The order parameters are to this model what the single overlap r was to the single-index model of Chapter 7: the reason the high-dimensional dynamics collapses to a few numbers. The difference is that now there are several of them, and permutations of the student units act on them.

The model freezes the readout, takes isotropic Gaussian inputs, which make the overlaps sufficient statistics, keeps the width fixed while $d \to \infty$ (the opposite of the infinite-width limits of Chapters 5 and 6), and uses noiseless labels, each sample once. Each simplification isolates the mechanism of interest, the interplay between the permutation symmetry and the gradient noise, and the activation g is chosen because every Gaussian integral the dynamics needs is then elementary (Lemma 8.5). The target has information exponent 1 in each direction, since $\hat{g}_1 = \mathbb{E}[z\,g(z)] = \mathbb{E}\,g'(z) = 1/\sqrt{\pi} \neq 0$, so no unit has to search; whatever delay we find is due to symmetry, not to the flatness studied in Chapter 7.

To invent the model one had to notice that two facts compose. Rotational invariance reduces the state to the Gram matrix of the student and teacher weights; and the gradient noise of online learning in high dimension averages out over d steps, except for a deterministic contribution of order η^2 that a naive gradient-flow limit misses. Together they turn a stochastic process in $\mathbb{R}^{md}$ into an autonomous ODE in $m(m+1)/2 + mm_\star$ variables.

DICTIONARY

IN THE MODEL	IN A TRAINED NETWORK
student weights w_j	incoming weights of hidden unit j
teacher weights $w_{\star k}$	the features the target is built from
fixed readout $f = \sum_j g(z_j)$	second layer, frozen at equal weights
overlap R_{jk}	how much unit j has learned feature k
overlap Q_{jl}, $j \neq l$	redundancy between units j and l
symmetric fixed point	a plateau: units are copies, each a blend of all features
specialization	the drop that ends the plateau: each unit takes one feature
unstable eigenvalue λ_{u}	the rate at which redundancy breaks
asymmetry of order $d^{-1/2}$	accidents of a random initialization
η^2 term	gradient noise, which limits the step size

8.3 The solution: Gaussian integrals and the ODE limit

Three lemmas prepare the main theorem. The first is the computational engine: a Gaussian weight $e^{-z^2/2}$, which is what g' contributes, can be absorbed into the covariance.

Lemma 8.3 (Gaussian tilting). *Let $X \sim \mathcal{N}(0, C)$ in $\mathbb{R}^n$ with C positive semidefinite, let $S = (s_1, \dots, s_r)$ be a list of indices in $\{1, \dots, n\}$, repetitions allowed, and write $X_S = (X_{s_1}, \dots, X_{s_r})$, $C_{SS} = (C_{s_i s_{i'}})$, and $C_{\cdot S}$ for the $n \times r$ matrix with columns $C_{\cdot s_i}$. For every measurable H of polynomial growth,*

$$\mathbb{E}\left[e^{-\|X_S\|^2/2} H(X)\right] = \det(I_r + C_{SS})^{-1/2}\, \mathbb{E} H(\tilde{X}),$$

where $\tilde{X} \sim \mathcal{N}(0, \tilde{C})$ with $\tilde{C} = C - C_{\cdot S}(I_r + C_{SS})^{-1} C_{\cdot S}^{\top}$.

Proof. Write $C = LL^{\top}$ with $L \in \mathbb{R}^{n \times n}$ and $X = LZ$ with $Z \sim \mathcal{N}(0, I_n)$, and let L_S be the $r \times n$ matrix whose rows are the rows $s_1, \dots, s_r$ of L, so that $X_S = L_S Z$ and $C_{SS} = L_S L_S^{\top}$. The weight times the standard Gaussian density is

$$e^{-\|L_S z\|^2/2} (2\pi)^{-n/2} e^{-\|z\|^2/2} = (2\pi)^{-n/2} \exp\!\left(-\tfrac{1}{2} z^{\top} (I_n + L_S^{\top} L_S) z\right),$$

an unnormalized Gaussian density with covariance $\Sigma = (I_n + L_S^{\top} L_S)^{-1}$ and total mass $\det(I_n + L_S^{\top} L_S)^{-1/2} = \det(I_r + L_S L_S^{\top})^{-1/2}$, by Sylvester's determinant identity. Hence the left side equals $\det(I_r + C_{SS})^{-1/2} \mathbb{E} H(L\tilde{Z})$ with $\tilde{Z} \sim \mathcal{N}(0, \Sigma)$. The Woodbury identity $\Sigma = I_n - L_S^{\top}(I_r + L_S L_S^{\top})^{-1} L_S$ and $L L_S^{\top} = C_{\cdot S}$ give $\mathrm{Cov}(L\tilde{Z}) = L \Sigma L^{\top} = C - C_{\cdot S}(I_r + C_{SS})^{-1} C_{\cdot S}^{\top}$. ∎

The covariance of $\tilde{X}$ is a Schur complement of $I + C$ minus the identity, hence positive semidefinite. The second lemma says how a Gaussian expectation changes with the covariance; it contains Price's theorem (Lemma A.5) for variables that are not standardized, and it will also supply the Lipschitz bounds that the ODE method needs.

Lemma 8.4 (Gaussian interpolation). *Let $\psi\colon \mathbb{R}^n \to \mathbb{R}$ be twice continuously differentiable with $|\psi|$, $|\partial_a \psi|$, and $|\partial_a \partial_b \psi|$ bounded by $\kappa(1 + \|\xi\|^p)$. For positive semidefinite C write $\mathbb{E}_C \psi = \mathbb{E}\psi(X)$ with $X \sim \mathcal{N}(0, C)$. For positive semidefinite C, C', with $C_\theta = (1-\theta)C + \theta C'$,*

$$\mathbb{E}_{C'}\psi - \mathbb{E}_C \psi = \frac{1}{2}\int_0^1 \sum_{a,b=1}^{n} (C' - C)_{ab}\, \mathbb{E}_{C_\theta}[\partial_a \partial_b \psi]\, \mathrm{d}\theta .$$

Consequently, if $\|C\|_F, \|C'\|_F \le B$, then $|\mathbb{E}_{C'}\psi - \mathbb{E}_C\psi| \le L_B \|C' - C\|_F$ with $L_B = \frac{1}{2} n \kappa (1 + \mu_p B^{p/2})$, where $\mu_p = \mathbb{E}\|\mathcal{N}(0, I_n)\|^p$.

Proof. The plan is to join the two laws by a path of Gaussians and to differentiate along it, converting the result with Stein's lemma. Let $X \sim \mathcal{N}(0, C)$ and $X' \sim \mathcal{N}(0, C')$ be independent and $X_\theta = \sqrt{1-\theta}\, X + \sqrt{\theta}\, X'$, which has law $\mathcal{N}(0, C_\theta)$; put $\varphi(\theta) = \mathbb{E}\psi(X_\theta)$. For $0 < \theta < 1$ the growth bounds justify differentiating under the expectation:

$$\varphi'(\theta) = \sum_a \mathbb{E}\Big[\partial_a \psi(X_\theta)\Big(\frac{X'_a}{2\sqrt{\theta}} - \frac{X_a}{2\sqrt{1-\theta}}\Big)\Big].$$

Condition on X and apply Stein's lemma (Theorem A.1) to X' and the function $h(X') = \partial_a \psi(\sqrt{1-\theta}X + \sqrt{\theta}X')$, whose derivatives are $\partial_b h = \sqrt{\theta}\, \partial_b \partial_a \psi$: this gives $\mathbb{E}[X'_a h(X')] = \sqrt{\theta}\, \sum_b C'_{ab} \mathbb{E}[\partial_a \partial_b \psi(X_\theta)]$. The same computation for X gives the second term, and

$$\varphi'(\theta) = \frac{1}{2}\sum_{a,b} (C'_{ab} - C_{ab})\, \mathbb{E}[\partial_a \partial_b \psi(X_\theta)].$$

The right side is bounded on $(0,1)$, and φ is continuous on $[0,1]$ by dominated convergence, so integrating proves the identity. For the bound, $\|C_\theta\|_F \le B$ gives $\mathbb{E}\|X_\theta\|^p \le \|C_\theta\|_{\mathrm{op}}^{p/2} \mu_p \le B^{p/2}\mu_p$, so every $|\mathbb{E}_{C_\theta}\partial_a\partial_b\psi|$ is at most $\kappa(1 + \mu_p B^{p/2})$, and the Cauchy–Schwarz inequality over the n^2 pairs (a,b) finishes the proof. ∎

Taking C' near C shows that $C \mapsto \mathbb{E}_C\psi$ is continuously differentiable on the cone of positive semidefinite matrices, with partial derivative $\mathbb{E}_C[\partial_a\partial_b\psi]$ with respect to an off-diagonal pair $C_{ab} = C_{ba}$ and $\frac{1}{2}\mathbb{E}_C[\partial_a^2\psi]$ with respect to a diagonal entry. The third lemma evaluates every integral the committee machine needs.[3]

Lemma 8.5 (The erf integrals). *Let $X \sim \mathcal{N}(0, C)$ with C positive semidefi-*

[3] These are the integrals I_2, I_3, and I_4 of Saad and Solla (1995b); the matrix form of I_4 through a Schur complement is our arrangement of their formula.

nite, and let a, b, c, c' be indices, not necessarily distinct. Then

$$I_2(a,b) := \mathbb{E}[g(X_a)g(X_b)] = \frac{2}{\pi}\arcsin\frac{C_{ab}}{\sqrt{(1+C_{aa})(1+C_{bb})}},$$

$$\begin{aligned} I_3(a,b,c) &:= \mathbb{E}[g'(X_a)\,X_b\,g(X_c)] \\ &= \frac{2}{\pi}\,\frac{C_{bc}(1+C_{aa}) - C_{ab}C_{ac}}{(1+C_{aa})\sqrt{(1+C_{aa})(1+C_{cc}) - C_{ac}^2}}, \end{aligned}$$

$$\begin{aligned} I_4(a,b;c,c') &:= \mathbb{E}[g'(X_a)g'(X_b)g(X_c)g(X_{c'})] \\ &= \frac{4}{\pi^2}\,\frac{1}{\sqrt{\det(I_2 + C_{SS})}}\arcsin\frac{\tilde{C}_{cc'}}{\sqrt{(1+\tilde{C}_{cc})(1+\tilde{C}_{c'c'})}}, \end{aligned}$$

where $S = (a,b)$ and $\tilde{C} = C - C_{\cdot S}(I_2 + C_{SS})^{-1}C_{S\cdot}^{\top}$. All denominators are at least 1, and all arguments of arcsin *lie in* $(-1,1)$.

Proof. For I_2, only the covariance of the pair (X_a, X_b) matters; write $\alpha = C_{aa}$, $\beta = C_{bb}$, and $\gamma = C_{ab}$, so that $|\gamma| \le \sqrt{\alpha\beta}$. Move the off-diagonal entry from 0 to γ with the diagonal fixed; every matrix on the way is positive semidefinite. By Lemma 8.4 with $\psi(u,v) = g(u)g(v)$, whose derivatives are bounded, the derivative with respect to the off-diagonal entry c is $\mathbb{E}[g'(X_a)g'(X_b)]$, which is $\frac{2}{\pi}\mathbb{E}[e^{-(X_a^2+X_b^2)/2}] = \frac{2}{\pi}((1+\alpha)(1+\beta) - c^2)^{-1/2}$ by Lemma 8.3 with $H = 1$. At $c = 0$ the two variables are independent and g is odd, so the expectation vanishes. Integrating,

$$I_2(a,b) = \frac{2}{\pi}\int_0^{\gamma}\frac{\mathrm{d}c}{\sqrt{(1+\alpha)(1+\beta) - c^2}} = \frac{2}{\pi}\arcsin\frac{\gamma}{\sqrt{(1+\alpha)(1+\beta)}}.$$

For I_3, apply Lemma 8.3 with $S = (a)$: $I_3 = \sqrt{2/\pi}\,(1+C_{aa})^{-1/2}\mathbb{E}[\tilde{X}_b g(\tilde{X}_c)]$. Stein's lemma (Theorem A.1) gives $\mathbb{E}[\tilde{X}_b g(\tilde{X}_c)] = \tilde{C}_{bc}\mathbb{E}g'(\tilde{X}_c)$, and Lemma 8.3 with $H = 1$ gives $\mathbb{E}g'(\tilde{X}_c) = \sqrt{2/\pi}\,(1+\tilde{C}_{cc})^{-1/2}$. Substituting $\tilde{C}_{bc} = C_{bc} - C_{ab}C_{ac}/(1+C_{aa})$ and $1 + \tilde{C}_{cc} = ((1+C_{aa})(1+C_{cc}) - C_{ac}^2)/(1+C_{aa})$ gives the formula. For I_4, apply Lemma 8.3 with $S = (a,b)$ and then the formula for I_2 to the Gaussian vector $\tilde{X}$.

For the last sentence: $(1+C_{aa})(1+C_{cc}) - C_{ac}^2 \ge 1 + C_{aa} + C_{cc} \ge 1$ because $C_{ac}^2 \le C_{aa}C_{cc}$; $\det(I_2 + C_{SS}) \ge 1$ because C_{SS} is positive semidefinite; and for a positive semidefinite matrix D, $|D_{ab}| \le \sqrt{D_{aa}D_{bb}} < \sqrt{(1+D_{aa})(1+D_{bb})}$, which applies to C and to $\tilde{C}$. ∎

With the lemma, the population loss is explicit. Expanding the square and using I_2 with the covariance (8.3),

$$\begin{aligned} \pi\mathcal{L} = &\sum_{k,k'}\arcsin\frac{T_{kk'}}{\sqrt{(1+T_{kk})(1+T_{k'k'})}} + \sum_{j,l}\arcsin\frac{Q_{jl}}{\sqrt{(1+Q_{jj})(1+Q_{ll})}} \\ &- 2\sum_{j,k}\arcsin\frac{R_{jk}}{\sqrt{(1+Q_{jj})(1+T_{kk})}}. \end{aligned} \tag{8.4}$$

For an orthonormal teacher, $T = I_{m_\star}$, the loss at $W = 0$ is $\frac{1}{\pi} m_\star \arcsin\frac{1}{2} = m_\star/6$.

8.3.1 The order parameters follow an ODE

Collect the order parameters in a vector $\omega = ((Q_{jl})_{j\le l}, (R_{jk})) \in \mathbb{R}^D$ with $D = m(m+1)/2 + m m_\star$, and write $\mathsf{M}(\omega)$ for the matrix (8.3), with the teacher Gram matrix T fixed. Call ω *realizable* if $\mathsf{M}(\omega)$ is positive semidefinite; these are exactly the order parameters of some weights, in every dimension $d \ge m + m_\star$. Taking the inner product of (8.2) with $w_{\star k}$ and with w_l gives the exact recursion

$$\begin{aligned} R_{jk}^{s+1} &= R_{jk}^s + \frac{\eta}{d}\,\delta_j^s z_{\star k}^s, \\ Q_{jl}^{s+1} &= Q_{jl}^s + \frac{\eta}{d}\left(\delta_j^s z_l^s + \delta_l^s z_j^s\right) + \frac{\eta^2}{d^2}\,\delta_j^s \delta_l^s \,\|x^s\|^2. \end{aligned} \tag{8.5}$$

The last term is the one that a gradient-flow limit would drop. It carries a factor $\|x^s\|^2/d^2 \approx 1/d$, the same order as the others, because the step moves the weights along x^s, a vector of length $\sqrt{d}$. For realizable ω define $F_\eta(\omega) \in \mathbb{R}^D$ by Gaussian expectations under $(z, z_\star) \sim \mathcal{N}(0, \mathsf{M}(\omega))$, with $\delta_j = g'(z_j)(\sum_k g(z_{\star k}) - \sum_l g(z_l))$:

$$\begin{aligned} F_\eta(\omega)_{R,jk} &= \eta\,\mathbb{E}[\delta_j z_{\star k}], \\ F_\eta(\omega)_{Q,jl} &= \eta\,\mathbb{E}[\delta_j z_l + \delta_l z_j] + \eta^2\,\mathbb{E}[\delta_j \delta_l]. \end{aligned} \tag{8.6}$$

Expanding δ_j and applying Lemma 8.5 with the covariance $\mathsf{M}(\omega)$ makes the field explicit:

$$\begin{aligned} \mathbb{E}[\delta_j z_{\star k}] &= \sum_{k'} I_3(z_j, z_{\star k}, z_{\star k'}) - \sum_{l'} I_3(z_j, z_{\star k}, z_{l'}), \\ \mathbb{E}[\delta_j z_l] &= \sum_{k'} I_3(z_j, z_l, z_{\star k'}) - \sum_{l'} I_3(z_j, z_l, z_{l'}), \\ \mathbb{E}[\delta_j \delta_l] &= \sum_{k,k'} I_4(z_j, z_l; z_{\star k}, z_{\star k'}) - 2\sum_{k,l'} I_4(z_j, z_l; z_{\star k}, z_{l'}) \\ &\quad + \sum_{l',l''} I_4(z_j, z_l; z_{l'}, z_{l''}), \end{aligned}$$

where each I is evaluated by reading the entries of its arguments' covariance from Q, R, and T. This is the *Saad–Solla system*. To apply the ODE method we extend F_η to all of $\mathbb{R}^D$: let Π be the Frobenius-norm projection onto the closed convex cone of positive semidefinite matrices of size $m + m_\star$, and put $\bar F_\eta(\omega) = F_\eta(\Pi \mathsf{M}(\omega))$, where $F_\eta(\mathsf{M})$ means the expectations (8.6) under $\mathcal{N}(0, \mathsf{M})$. On realizable points $\bar F_\eta = F_\eta$.

Theorem 8.6 (The ODE limit). *Fix m, $m_\star$, $\eta > 0$, a positive semidefinite teacher Gram matrix T, a realizable starting point $\bar\omega_0$, and a horizon $t_1 > 0$. The equation $\dot{\bar\omega} = \bar F_\eta(\bar\omega)$, $\bar\omega(0) = \bar\omega_0$, has a unique solution on $[0, t_1]$, the solution is realizable for every t, and so it solves $\dot{\bar\omega} = F_\eta(\bar\omega)$ with the explicit*

field above. There is a constant C, depending only on $m, m_\star, \eta, T, \bar\omega_0, t_1$, such that the following holds. Let $d \geq m + m_\star$, let the teacher have Gram matrix T, run (8.2) from deterministic weights with order parameters ω_0, and let ω_s be the order parameters after s steps. For every $a \in (0, 1]$ with $\Delta := C(\|\omega_0 - \bar\omega_0\| + a + d^{-1/2}) < 1$,

$$\Pr\Big(\max_{0\leq s\leq t_1 d} \|\omega_s - \bar\omega(s/d)\| > \Delta\Big) \leq \frac{C}{a^2 d}.$$

With $\omega_0 \to \bar\omega_0$ and $a = d^{-1/4}$, the order parameters stay within $O(\|\omega_0 - \bar\omega_0\| + d^{-1/4})$ of the ODE over $t_1 d$ steps, with probability $1 - O(d^{-1/2})$. A random initialization $w_j^0 \sim \mathcal{N}(0, \frac{q_0}{d} I_d)$ has $\|\omega_0 - \bar\omega_0\| = O(d^{-1/2})$ with high probability for $\bar\omega_0 = (q_0 I_m, 0)$, and the theorem applies conditionally on the initialization.

Proof. The plan is to verify the hypotheses of the ODE method (Theorem A.13): a field that is Lipschitz and bounded on a neighborhood of the path, increments whose noise has bounded conditional variance, and a small bias.

Bounds on the integrands. Since $|g| < 1$ and $0 < g' \leq \sqrt{2/\pi}$, every δ_j satisfies $|\delta_j| \leq \bar\delta := \sqrt{2/\pi}\,(m + m_\star)$. The integrands $\delta_j z_{\star k}$, $\delta_j z_l$, and $\delta_j \delta_l$ are smooth functions of $(z, z_\star)$ whose derivatives up to order two are bounded by $\kappa(1 + \|(z, z_\star)\|)$ for a constant κ depending on m and $m_\star$, by (8.1).

The field is Lipschitz on bounded sets and grows sublinearly. The map $\omega \mapsto \mathsf{M}(\omega)$ is affine with $\|\mathsf{M}(\omega) - \mathsf{M}(\omega')\|_F \leq \sqrt{2}\|\omega - \omega'\|$, and Π is nonexpansive with $\Pi 0 = 0$, so $\|\Pi\mathsf{M}(\omega)\|_F \leq \|\mathsf{M}(\omega)\|_F$. By Lemma 8.4, $\bar F_\eta$ is Lipschitz on every ball. By the Cauchy–Schwarz inequality, $|\mathbb{E}[\delta_j z_l]| \leq \bar\delta\,(\Pi\mathsf{M})_{ll}^{1/2} \leq \bar\delta \|\mathsf{M}(\omega)\|_F^{1/2}$, and similarly for the other entries, so $\|\bar F_\eta(\omega)\| \leq c_0(1 + \|\omega\|)$. A locally Lipschitz field with linear growth has a unique solution on $[0, t_1]$; call it $\bar\omega$, let $r_1 = \max_t \|\bar\omega(t)\|$, and let U be the open ball of radius $r_1 + 2$. On U the field is bounded by some B and L-Lipschitz, and every point within distance $\rho = 1$ of the path lies in U.

Noise and bias. Let $\mathcal{F}_s$ be generated by $x^0, \dots, x^{s-1}$. Given $\mathcal{F}_s$, the vector $(z^s, z_\star^s)$ is $\mathcal{N}(0, \mathsf{M}(\omega_s))$, and $\mathsf{M}(\omega_s)$ is positive semidefinite, so $\Pi\mathsf{M}(\omega_s) = \mathsf{M}(\omega_s)$. Write $d(\omega_{s+1} - \omega_s) = \bar F_\eta(\omega_s) + \xi_{s+1} + \epsilon_{s+1}$ with $\epsilon_{s+1} = \mathbb{E}[d(\omega_{s+1} - \omega_s) \mid \mathcal{F}_s] - \bar F_\eta(\omega_s)$ and ξ_{s+1} the centered remainder. By (8.5), the R-entries of ϵ_{s+1} vanish and its Q-entries are $\eta^2 \mathbb{E}[\delta_j \delta_l (\|x^s\|^2/d - 1) \mid \mathcal{F}_s]$, which are at most $\eta^2 \bar\delta^2 \mathbb{E}|\|x\|^2/d - 1| \leq \eta^2 \bar\delta^2 \sqrt{2/d}$ in absolute value, because $\mathrm{Var}\|x\|^2 = 2d$. So $\|\epsilon_{s+1}\| \leq \epsilon := \sqrt{D}\,\eta^2 \bar\delta^2 \sqrt{2/d}$. The conditional second moment of ξ_{s+1} is at most that of $d(\omega_{s+1} - \omega_s)$, whose entries are bounded in mean square by $\eta^2 \bar\delta^2 T_{kk}$, by $3\eta^2 \bar\delta^2 (Q_{jj} + Q_{ll}) + 3\eta^4 \bar\delta^4 \mathbb{E}\|x\|^4/d^2$, and $\mathbb{E}\|x\|^4 = d^2 + 2d \leq 3d^2$. On the event $\{\omega_s \in U\}$ these are bounded by a constant ς^2.

Conclusion. Theorem A.13 applies with these constants: with $\Delta' = (\|\omega_0 - \bar\omega_0\| + a + t_1\epsilon + t_1 LB/(2d))e^{Lt_1}$, if $\Delta' < 1$, the probability that $\|\omega_s - \bar\omega(s/d)\| > \Delta'$ for some $s \leq t_1 d$ is at most $\varsigma^2 t_1/(a^2 d)$. Since ϵ and $1/d$ are at most constants times $d^{-1/2}$, $\Delta' \leq \Delta$ for a suitable C, which proves the bound.

Realizability. Fix $t \in [0, t_1]$. For each $d \geq m + m_\star$ choose initial weights

with $\omega_0 = \bar{\omega}_0$ exactly, which is possible because $\bar{\omega}_0$ is realizable, and take $a = d^{-1/4}$. For large d the bound shows that with positive probability $\|\omega_{\lfloor td \rfloor} - \bar{\omega}(\lfloor td \rfloor/d)\| \le C(2d^{-1/4})$, so some realizable point lies within $2Cd^{-1/4} + Bd^{-1}$ of $\bar{\omega}(t)$. The realizable set is closed, so $\bar{\omega}(t)$ is realizable, and there $\bar{F}_\eta = F_\eta$. ∎

REMARK 8.7 (The overlaps form a Markov chain). The proof shows more. Given $\mathcal{F}_s$, the update (8.5) is a function of ω_s and of the triple $(z^s, z^s_\star, \|x^s\|^2)$, whose conditional law depends only on ω_s: if P is the orthogonal projection onto the span of the current student and teacher weights, of dimension ϱ, then $\|x\|^2 = \|Px\|^2 + \chi^2_{d-\varrho}$ with an independent chi-square variable, and $\|Px\|^2$ is a function of $(z, z_\star)$. So (ω_s) is a Markov chain whose law can be simulated exactly in every dimension at a cost independent of d per step. All simulations in this chapter use it; for $m = m_\star = 2$, $d = 100$, and 400 runs, its median loss agrees with that of SGD in $\mathbb{R}^{100}$ within 1% at $t = 10, 20, 40, 80$ (Exercise 8.3).

REMARK 8.8 (Two time scales). The η-term of F_η is the image of gradient flow on the population loss: $-\nabla_{w_j}\mathcal{L}(W) = \mathbb{E}[\delta_j x]$, and its inner products with $w_{\star k}$ and w_l are the entries of F_η/η without the η^2 term. So for small η, in the time $\tau = \eta t$, the order parameters follow

$$\frac{\mathrm{d}\omega}{\mathrm{d}\tau} = G(\omega), \qquad G(\omega)_{R,jk} = \mathbb{E}[\delta_j z_{\star k}], \quad G(\omega)_{Q,jl} = \mathbb{E}[\delta_j z_l + \delta_l z_j], \tag{8.7}$$

the order-parameter shadow of $\dot{w}_j = -\nabla_{w_j}\mathcal{L}(W)$, and the η^2 term is a correction of relative size η.

8.3.2 One unit: the learning-rate threshold

The simplest case already shows what the η^2 term does. Let $m = m_\star = 1$ and $T = 1$, write $R = R_{11}$ and $Q = Q_{11}$, and use the coordinates $u = R - 1$ and $v = Q - R^2$, the squared distance of w from the line through $w_\star$; the realizable set is $v \ge 0$, and $\|w - w_\star\|^2 = u^2 + v$. Put $c_1 = \mathbb{E}[g'(z)^2 z^2]$, $c_2 = \mathbb{E}[g'(z)^2]$, and $c_3 = \mathbb{E}[g'(z)^4]$ for $z \sim \mathcal{N}(0,1)$, so that

$$c_1 = \frac{2}{3\sqrt{3}\,\pi}, \qquad c_2 = \frac{2}{\sqrt{3}\,\pi}, \qquad c_3 = \frac{4}{\sqrt{5}\,\pi^2};$$

these values follow from $g'(z)^2 = \frac{2}{\pi}e^{-z^2}$, $\mathbb{E}e^{-z^2} = 3^{-1/2}$, $\mathbb{E}z^2e^{-z^2} = 3^{-3/2}$, and $\mathbb{E}e^{-2z^2} = 5^{-1/2}$.

Proposition 8.9 (A single unit and its learning-rate threshold). *For every $\eta > 0$ the point $w = w_\star$, that is $(u, v) = (0, 0)$, is a fixed point of the ODE of Theorem 8.6. In the coordinates (u, v) the Jacobian of the field there is upper triangular, with diagonal entries $-\eta c_1$ and $-(2\eta c_2 - \eta^2 c_3)$. Consequently, with*

$$\eta_{\max} = \frac{2c_2}{c_3} = \pi\sqrt{5/3} \approx 4.056,$$

the fixed point is linearly stable for $0 < \eta < \eta_{\max}$: solutions starting near it

converge exponentially, and the loss decays at the rate $\min(2\eta c_1,\, 2\eta c_2 - \eta^2 c_3)$, *which is largest,* $8\sqrt{5}/27 \approx 0.663$, *at* $\eta = \frac{2}{3}\eta_{\max}$. *For* $\eta > \eta_{\max}$ *the fixed point is unstable, and an eigenvector of the positive eigenvalue points into the realizable region* $v > 0$.

PROOF. The plan is to write the ODE in the coordinates (u, v), to compute its derivatives at $(0, 0)$ with the interpolation lemma, and to read off the eigenvalues. Represent $z = Rz_\star + \sqrt{v}\,\zeta$ with $z_\star, \zeta$ independent standard Gaussians, and let $\delta = g'(z)(g(z_\star) - g(z))$. At $w = w_\star$ we have $z = z_\star$ and $\delta = 0$ identically, so every term of (8.6) vanishes: the point is fixed for every η.

The equation for v. By the chain rule and (8.6),

$$\dot v = \dot Q - 2R\dot R = 2\eta\,\mathbb{E}[\delta(z - Rz_\star)] + \eta^2\mathbb{E}[\delta^2] = 2\eta\sqrt{v}\,\mathbb{E}[\delta\zeta] + \eta^2\mathbb{E}[\delta^2].$$

Stein's lemma in ζ gives $\mathbb{E}[\delta\zeta] = \sqrt{v}\,\mathbb{E}[\partial_z\delta]$, so $\dot v = 2\eta v\,A + \eta^2 B$ with $A = \mathbb{E}[\partial_z\delta]$ and $B = \mathbb{E}[\delta^2]$, both continuously differentiable functions of (R, Q) by Lemma 8.4. At the fixed point, $\partial_z\delta = g''(z)(g(z_\star) - g(z)) - g'(z)^2 = -g'(z_\star)^2$, so $A = -c_2$. For B, use the derivative rules after Lemma 8.4, with Q the variance of z and R its covariance with $z_\star$. At $z = z_\star$, $\delta = 0$ and $\partial_z\delta = -\partial_{z_\star}\delta = -g'^2$, so $\partial_Q B = \frac{1}{2}\mathbb{E}[\partial_z^2\delta^2] = \mathbb{E}[(\partial_z\delta)^2] = c_3$ and $\partial_R B = \mathbb{E}[\partial_z\partial_{z_\star}\delta^2] = 2\mathbb{E}[\partial_z\delta\,\partial_{z_\star}\delta] = -2c_3$. Since $\mathrm{d}Q = \mathrm{d}v + 2R\,\mathrm{d}R$ and $\mathrm{d}R = \mathrm{d}u$, at $R = 1$ this reads $\mathrm{d}B = c_3\,\mathrm{d}v$. Hence $\partial_u\dot v = 0$ and $\partial_v\dot v = 2\eta A + \eta^2 c_3 = -(2\eta c_2 - \eta^2 c_3)$ at the fixed point.

The equation for u. Here $\dot u = \eta\phi(R, Q)$ with $\phi = \mathbb{E}[\delta z_\star]$. At the fixed point, writing $z_\star = z$ after differentiating,

$$\partial_R\phi = \mathbb{E}[\partial_z\partial_{z_\star}(\delta z_\star)] = \mathbb{E}[-g'^2 - z_\star^2 g'^2] = -c_2 - c_1,$$
$$\partial_Q\phi = \tfrac{1}{2}\mathbb{E}[z_\star\partial_z^2\delta] = \tfrac{1}{2}\mathbb{E}[-3z_\star g' g''] = \tfrac{3}{2}c_1,$$

using $\partial_z^2\delta = -3g'g''$ at $z = z_\star$ and $g'' = -zg'$. Moving along u at fixed v changes Q by $2\,\mathrm{d}u$, so $\partial_u\dot u = \eta(\partial_R\phi + 2\partial_Q\phi) = \eta(2c_1 - c_2) = -\eta c_1$, because $c_2 = 3c_1$.

Conclusion. The Jacobian is $\begin{pmatrix} -\eta c_1 & * \\ 0 & -(2\eta c_2 - \eta^2 c_3) \end{pmatrix}$. For $\eta < \eta_{\max}$ both eigenvalues are negative and the linearization theorem gives local exponential convergence; it applies at the boundary point $(0, 0)$ of the realizable set because the closed-form field of Lemma 8.5 is real-analytic on a neighborhood of that set, and solutions that start realizable stay realizable (Theorem 8.6). The loss is $\frac{1}{2}\mathbb{E}[g'(z_\star)^2(z - z_\star)^2]$ to second order, that is $\frac{1}{2}(c_1u^2 + c_2v)$, and u^2 decays at rate $2\eta c_1$ while v, forced by terms of order u^2, decays at rate $\min(2\eta c_1, 2\eta c_2 - \eta^2 c_3)$; so does the loss. The two rates are equal at $\eta = 2(c_2 - c_1)/c_3 = \frac{2}{3}\eta_{\max}$, where the first is increasing and the second decreasing in η, which gives the maximum $\frac{4}{3}\eta_{\max}c_1 = 8\sqrt{5}/27$. For $\eta > \eta_{\max}$ the eigenvalue $-(2\eta c_2 - \eta^2 c_3)$ is positive, and its eigenvector in a triangular matrix has a nonzero second coordinate, which can be taken positive. ∎

The threshold is a stability condition of the kind met in the prologue and in Chapter 15: gradient descent on a quadratic with curvature λ_1 diverges

for steps above $2/\lambda_1$. Here the curvature in the directions orthogonal to $w_\star$ is c_2 per unit time, and the noise term $\eta^2 c_3 v$ plays the role of the overshoot. The variance of the gradient, not the curvature alone, sets the threshold. Beyond it the ODE converges to a different fixed point with positive error: numerically, $\mathcal{L} \approx 0.0056$ at $\eta = 4.2$ and 0.036 at $\eta = 5$, with $Q > 1$; the student overshoots in norm.

8.4 The solution: the symmetric plateau

Now take $K \geq 2$ students and K orthonormal teachers: $m = m_\star = K$ and $T = I_K$. The permutations of the students, and separately of the teachers, leave the dynamics unchanged, and they single out two linear subspaces of order parameters,

$$\mathcal{S} = \{\omega : \; R_{jk} = R \text{ for all } j,k;\; Q_{jj} = Q \text{ for all } j;\; Q_{jl} = C \text{ for all } j \neq l\},$$
$$\mathcal{S}_0 = \mathcal{S} \cap \{C = Q\}.$$

Points of $\mathcal{S}_0$ are the order parameters of identical students, $w_1 = \cdots = w_K$, since then $\|w_j - w_l\|^2 = 2Q - 2C = 0$.

Theorem 8.10 (The symmetric plateau). *Let $m = m_\star = K \geq 2$ and $T = I_K$.*

(a) *For every $\eta > 0$, solutions of the ODE of Theorem 8.6 that start in $\mathcal{S}$ stay in $\mathcal{S}$, and solutions that start in $\mathcal{S}_0$ stay in $\mathcal{S}_0$. For SGD itself, students that start identical stay identical.*

(b) *The point $P_\star \in \mathcal{S}_0$ with*

$$R = a := \frac{1}{\sqrt{K(2K-1)}}, \qquad Q = C = \frac{1}{2K-1},$$

the order parameters of $w_j = a \sum_k w_{\star k}$ for every j, is a fixed point of the gradient-flow field G of (8.7), and the only one of the form $w_j = a' \sum_k w_{\star k}$ for all j with $a' > 0$. Its loss is

$$\mathcal{L}_\star = \frac{K}{6} - \frac{K^2}{\pi} \arcsin \frac{1}{2K}.$$

(c) *Let Ξ be the direction with $\Xi_{R,11} = \Xi_{R,22} = 1$, $\Xi_{R,12} = \Xi_{R,21} = -1$, and all other entries 0. Then Ξ is an eigenvector of the Jacobian of G at $P_\star$ with eigenvalue*

$$\lambda_{\mathrm{u}} = \frac{2K}{\pi\,(2K+1)^{3/2}(2K-1)^{1/2}} > 0.$$

So $P_\star$ is a saddle, and its unstable direction makes student 1 prefer teacher 1 and student 2 prefer teacher 2. For $K = 2$, $\lambda_{\mathrm{u}} = 4\sqrt{15}/(75\pi) \approx 0.0657$ and $\mathcal{L}_\star = \frac{1}{3} - \frac{4}{\pi} \arcsin \frac{1}{4} \approx 0.01161$.

(d) *Perturbations of $P_\star$ in which the students split into directions orthogonal to all teachers, $w_j = w + \varepsilon u_j$ with $\sum_j u_j = 0$ and each u_j orthogonal to $w_{\star 1}, \ldots, w_{\star K}$, have velocity $O(\varepsilon^2)$ under the gradient flow: these directions are marginal.*

The plateau is the time spent near $P_\star$: the dynamics is drawn into $\mathcal{S}_0$ toward $P_\star$ along stable directions and leaves it along Ξ. In $\mathcal{S}_0$ itself and for $K = 2$, the two eigenvalues at $P_\star$ are -0.592 and -1.972 (computed numerically), so $P_\star$ attracts identical students exponentially. The full 7×7 Jacobian of G at $P_\star$ for $K = 2$ has the eigenvalues

$$0.0657, \quad 0, \quad 0, \quad -0.592, \quad -0.920, \quad -0.986, \quad -1.972,$$

where the zeros belong to directions that are of second order in the weights, among them the marginal directions of (d), and -0.920 to identical students turning toward one teacher (Exercise 8.4). Within $\mathcal{S}$ the fixed point therefore attracts only algebraically along the coordinate $Q - C$; numerically $Q - C$ decays like $5/\tau$ along the flow.

Proof. The plan is to prove (a) by checking that the field is tangent to the two subspaces, (b) with the integrals of Lemma 8.5, and (c) and (d) in weight space, where the gradient flow is a gradient flow and the Hessian of the loss can be computed with Stein's lemma.

(a) Invariance. We use one fact about ODEs: if a field F is Lipschitz on bounded sets and $F(V) \subseteq V$ for a linear subspace V, then solutions starting in V stay in V, because the ODE restricted to V has a solution, which solves the full equation, and solutions are unique. For $\omega \in \mathcal{S}$, the covariance $\mathsf{M}(\omega)$ is unchanged when the students are permuted, and also when the teachers are permuted, so the joint law of $(z, z_\star)$ is invariant under both kinds of relabeling, while the label and the prediction are symmetric functions. Hence $\mathbb{E}[\delta_j z_{\star k}]$ does not depend on (j, k), $\mathbb{E}[\delta_j z_j]$ and $\mathbb{E}[\delta_j^2]$ do not depend on j, and $\mathbb{E}[\delta_j z_l + \delta_l z_j]$ and $\mathbb{E}[\delta_j \delta_l]$ for $j \neq l$ do not depend on the pair. So $F_\eta(\omega) \in \mathcal{S}$, and the same holds for $\bar{F}_\eta$, because the projection Π commutes with simultaneous permutations of rows and columns. For $\omega \in \mathcal{S}_0$ the variables $z_1, \dots, z_K$ coincide almost surely, because $\mathbb{E}(z_j - z_l)^2 = 2Q - 2C = 0$; so are the δ_j, and $F_\eta(\omega)_{Q,jl}$ does not depend on whether $j = l$: $F_\eta(\omega) \in \mathcal{S}_0$. For SGD, (8.2) gives identical students identical updates.

(b) The fixed point. At $w_j = w = a' \sum_k w_{\star k}$, all preactivations equal $z = a' \sum_k z_{\star k}$, with $\operatorname{Var} z = Q = Ka'^2$ and $\operatorname{Cov}(z, z_{\star k}) = a'$. Put $\Lambda = \sqrt{2 + (2K-1)a'^2}$. By Lemma 8.5(ii), the three integrals $\mathbb{E}[g'(z) z_{\star 1} g(z_{\star 1})]$, $\mathbb{E}[g'(z) z_{\star 1} g(z_{\star 2})]$, and $\mathbb{E}[g'(z) z_{\star 1} g(z)]$ are $\frac{2}{\pi}(1+Q)^{-1}$ times, respectively,

$$\frac{1 + Q - a'^2}{\Lambda}, \qquad \frac{-a'^2}{\Lambda}, \qquad \frac{a'}{\sqrt{1 + 2Q}},$$

where we used $2(1+Q) - a'^2 = \Lambda^2$, $(1+Q)^2 - Q^2 = 1 + 2Q$, and $C_{bc}(1 + C_{aa}) - C_{ab}C_{ac} = a'(1+Q) - a'Q$ in the third. Summing with weights 1, $K-1$, and $-K$, and using $1 + Q - Ka'^2 = 1$,

$$G_{R,jk} = \frac{2}{\pi(1+Q)} \Big(\frac{1}{\sqrt{2 + (2K-1)a'^2}} - \frac{Ka'}{\sqrt{1 + 2Ka'^2}} \Big).$$

The bracket vanishes exactly when $a' > 0$ and $1 + 2Ka'^2 = K^2 a'^2 (2 + (2K - 1)a'^2)$, a quadratic equation in a'^2 with product of roots $-1/(K^2(2K-1)) < 0$,

hence with exactly one positive root, and $a'^2 = 1/(K(2K-1))$ satisfies it: $\frac{1}{2K-1} + \frac{2(K-1)}{2K-1} = 1$. The Q-entries of G vanish too, because $z = a\sum_k z_{\star k}$ makes $\mathbb{E}[\delta_j z] = a\sum_k \mathbb{E}[\delta_j z_{\star k}] = 0$. For the loss, (8.4) with $Q/(1+Q) = \frac{1}{2K}$ and $a/\sqrt{2(1+Q)} = \frac{1}{2K}$ gives $\mathcal{L}_\star = \frac{1}{\pi}(K \arcsin\frac{1}{2} + K^2 \arcsin\frac{1}{2K} - 2K^2 \arcsin\frac{1}{2K})$.

The Hessian at $P_\star$. Let $W_\star = (w, \dots, w)$ with $w = a\sum_k w_{\star k}$, and let $z = \langle w, x\rangle$, $\delta = g'(z)(y - Kg(z))$. The point $W_\star$ is a critical point of $\mathcal{L}$: the gradient $-\mathbb{E}[\delta x]$ lies in the span of the teachers, because δ depends on x only through $z_{\star 1}, \dots, z_{\star K}$, and its inner products with the teachers vanish by (b). Differentiating $\nabla_{w_j}\mathcal{L} = -\mathbb{E}[g'(z_j)(y - f)x]$ once more at $W_\star$, the blocks of the Hessian are $H_{jl} = \mathcal{B} + \mathbf{1}\{j = l\}\mathcal{A}$ with

$$\mathcal{B} = \mathbb{E}[g'(z)^2 x x^\top], \qquad \mathcal{A} = -\mathbb{E}[(y - Kg(z))g''(z)\, x x^\top] = \mathbb{E}[\delta z\, x x^\top],$$

using $g''(z) = -z g'(z)$. For a perturbation $W = W_\star + \varepsilon(p_1, \dots, p_K)$ with $\sum_j p_j = 0$, the sum $\sum_l H_{jl} p_l$ equals $\mathcal{A} p_j$, so the gradient flow gives $\dot{p}_j = -\mathcal{A} p_j + O(\varepsilon)$.

(d) The marginal directions. Write $x = \sum_k z_{\star k} w_{\star k} + x_\perp$, with $x_\perp$ independent of the $z_{\star k}$. Since δz depends only on the $z_{\star k}$, $\mathcal{A}u = \mathbb{E}[\delta z]\, u$ for u orthogonal to the teachers, and $\mathbb{E}[\delta z] = 0$ at $P_\star$ by (b). So $\mathcal{A}u_j = 0$, which is (d).

(c) The specialization eigenvalue. Let $v = (w_{\star 1} - w_{\star 2})/\sqrt{2}$ and $V = \langle v, x\rangle = (z_{\star 1} - z_{\star 2})/\sqrt{2}$, a standard Gaussian independent of z and of $z_{\star 3}, \dots, z_{\star K}$. Exchanging teachers 1 and 2 preserves the law of x and the function δz but changes the sign of V; so $\langle w_{\star k}, \mathcal{A}v\rangle = \mathbb{E}[\delta z\, z_{\star k} V]$ vanishes for $k \geq 3$ and is antisymmetric in $k \in \{1, 2\}$, and with $\mathcal{A}$ vanishing off the teacher span this means $\mathcal{A}v = \mu v$ with $\mu = \mathbb{E}[\delta z V^2]$. For a standard Gaussian V and a smooth h of polynomial growth, two applications of Stein's lemma give $\mathbb{E}[V^2 h(V)] = \mathbb{E}h(V) + \mathbb{E}h''(V)$. Write $z_{\star 1} = U + V/\sqrt{2}$ and $z_{\star 2} = U - V/\sqrt{2}$ with $U = (z_{\star 1} + z_{\star 2})/2$; then V is independent of $(U, z_{\star 3}, \dots, z_{\star K})$, and z is a function of these variables. Apply the identity conditionally on them. Only $g(z_{\star 1}) + g(z_{\star 2})$ depends on V, with second derivative $\frac{1}{2}(g''(z_{\star 1}) + g''(z_{\star 2}))$, so

$$\mu = \mathbb{E}[\delta z] + \tfrac{1}{2}\mathbb{E}\big[g'(z) z\,(g''(z_{\star 1}) + g''(z_{\star 2}))\big] = -\mathbb{E}\big[g'(z) g'(z_{\star 1})\, z\, z_{\star 1}\big],$$

using $\mathbb{E}[\delta z] = 0$, the symmetry between teachers 1 and 2, and $g'' = -z g'$. The pair $(z, z_{\star 1})$ has covariance $C_2 = \left(\begin{smallmatrix} Q & a \\ a & 1 \end{smallmatrix}\right)$, with $\det(I_2 + C_2) = 2(1+Q) - a^2 = (2K+1)/K$. Lemma 8.3 with S the pair gives

$$\mathbb{E}[g'(z) g'(z_{\star 1}) z z_{\star 1}] = \frac{2}{\pi}\frac{\big(C_2(I_2 + C_2)^{-1}\big)_{12}}{\det(I_2 + C_2)^{1/2}} = \frac{2}{\pi}\frac{a}{\det(I_2 + C_2)^{3/2}},$$

because $C_2 - C_2(I_2 + C_2)^{-1}C_2 = C_2(I_2 + C_2)^{-1}$, whose off-diagonal entry is $(a(1+Q) - aQ)/\det(I_2 + C_2)$. Inserting a and the determinant gives $-\mu = \lambda_{\mathrm{u}}$.

From weights to order parameters. Take $p_1 = v$, $p_2 = -v$, and $p_j = 0$ for $j \geq 3$. Along $W_\star + \varepsilon(p_j)$ the order parameters are $\omega(\varepsilon) = P_\star + \frac{\varepsilon}{\sqrt{2}}\Xi + O(\varepsilon^2)$, because $\langle v, w_{\star 1}\rangle = -\langle v, w_{\star 2}\rangle = 1/\sqrt{2}$ and $\langle v, w\rangle = 0$. By Remark 8.8, $G(\omega(\varepsilon))$ is the time derivative of the order parameters under the gradient flow, whose first-order part is $\dot{w}_j = \varepsilon\lambda_{\mathrm{u}} p_j$. Its R-entries are $\varepsilon\lambda_{\mathrm{u}}\langle p_j, w_{\star k}\rangle = \frac{\varepsilon}{\sqrt{2}}\lambda_{\mathrm{u}}\Xi_{R,jk}$,

and its Q-entries are $\varepsilon\lambda_u(\langle p_j, w\rangle + \langle w, p_l\rangle) = 0$. Differentiating at $\varepsilon = 0$ gives $\frac{1}{\sqrt{2}} J\Xi = \frac{1}{\sqrt{2}}\lambda_u \Xi$ for the Jacobian J of G, which is continuously differentiable near $P_\star$ by Lemma 8.4. ■

The proof of (d) uses the identity $g'' = -zg'$, and it is special to this activation: for a general odd g the orthogonal split has curvature $-\mathbb{E}[(y - Kg(z))g''(z)]$, which need not vanish (Exercise 8.7).

THE PLATEAU LASTS $\frac{1}{2\lambda_u(\eta)} \log d$. Let $\lambda_u(\eta)$ be the eigenvalue of the full field F_η in the direction Ξ, at the symmetric fixed point $P_\star(\eta)$ of F_η in $\mathcal{S}_0$; by (c), $\lambda_u(\eta) = \eta\lambda_u + O(\eta^2)$. Start from small random weights. The initial order parameters are symmetric up to fluctuations of order $d^{-1/2}$, and in time $O(1)$ the ODE carries them close to $P_\star(\eta)$ while the component along Ξ, call it S, stays of order $d^{-1/2}$. Near $P_\star(\eta)$ the recursion (8.5) for S reads $S_{s+1} = S_s + \frac{1}{d}(\lambda_u(\eta) S_s + \xi_s)$ to first order, with centered noise ξ_s of variance of order one, so in time t,

$$S(t) \approx e^{\lambda_u(\eta) t} \Big(S(0) + \frac{1}{d} \sum_{s < td} e^{-\lambda_u(\eta) s/d} \xi_s \Big),$$

and the bracket converges to a random variable $\Gamma d^{-1/2}$ with Γ of order one: the initial asymmetry and the asymmetry injected by the noise are of the same order. The plateau ends when S becomes of order one, at

$$t_{\text{esc}} \approx \frac{1}{2\lambda_u(\eta)} \log d + \frac{1}{\lambda_u(\eta)} \log \frac{1}{|\Gamma|} + O(1),$$

a time of order $\log d$ with fluctuations of order one that do not shrink as d grows. In steps, the plateau costs about $\frac{d}{2\lambda_u(\eta)} \log d$ samples. ◇

The heuristic leaves the range of Theorem 8.6, which controls the dynamics only up to a fixed horizon t_1, while t_{esc} grows with d; and the escape is decided by fluctuations of order $d^{-1/2}$, exactly the scale the theorem neglects. Without them the ODE started on $\mathcal{S}$ never leaves it, and SGD started from identical students never specializes, at every d.

8.5 Reading the solution

The solution makes four quantitative predictions. The plateau height $\mathcal{L}_\star = K/6 - (K^2/\pi) \arcsin \frac{1}{2K}$ is 0.0116, 0.0203, and 0.0284 for $K = 2, 3, 4$, a small fraction of the initial loss $K/6$: the symmetric student, which sends every unit to the average teacher direction, already captures most of the variance of the label, and what remains is the part that requires a division of labor. The plateau length is $\frac{1}{2\lambda_u} \log d$ in units of $\tau = \eta t$, and since $\lambda_u \approx \frac{1}{2\pi K}$ for large K, it grows linearly in the number of units as well as logarithmically in the dimension.[4] The level and the escape rate depend on η through the η^2 term: at $\eta = 0.5$ and $K = 2$ the plateau rises from 0.0116 to 0.0147, and $\lambda_u(\eta)/\eta$ is 0.0662, 0.0670, 0.0687, and 0.0748 for $\eta = 0.1$, 0.25, 0.5, and 1. Finally, every unit's step size is limited by the threshold of Proposition 8.9,

[4] The exact values are $\lambda_u = 0.0657$, 0.0461, and 0.0356 for $K = 2, 3, 4$, against $1/(2\pi K) = 0.0796$, 0.0531, and 0.0398.

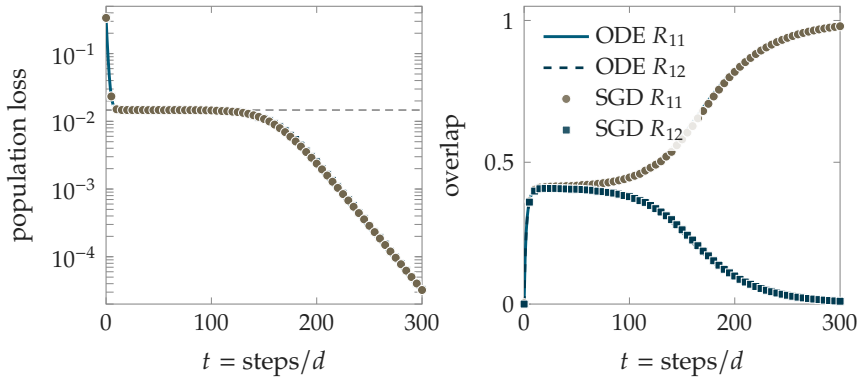


Figure 8.1. Plateau and specialization for $K = 2$ students and orthonormal teachers, $\eta = 0.5$. Left: the population loss (8.4) along the ODE of Theorem 8.6 (curve), and medians over 16 runs of SGD in dimension $d = 300$ (squares) and $d = 3000$ (dots); the dashed line is the plateau level $\mathcal{L}_\star(\eta) = 0.0147$ of the full field. Right: the overlaps of student 1 with teachers 1 and 2, ODE (curves) and medians at $d = 3000$ (points). All runs start from the same order parameters, $R = 0.003\, I$ and $Q - RR^\top = 0.01\, I$, and are simulated with the exact chain of Remark 8.7 (seed 8).

and the committee's threshold is lower: for $K = 2$ the specialized fixed point $R = Q = I$ loses stability at $\eta \approx 2.31$, against 4.06 for one unit (Exercise 8.9).

Figure 8.1 tests the ODE limit across a plateau. The runs start with a slight preference of each student for its own teacher, $R(0) = 0.003\, I$, so that the ODE itself escapes; its loss falls below half the plateau level at $t = 166$. In the first 20 time units the loss falls by a factor 23 to the plateau, and both students move to the blend $R_{jk} \approx 0.41$, close to $a = 1/\sqrt{6} \approx 0.408$ (the full field at $\eta = 0.5$ puts the fixed point at 0.410). On the plateau the medians agree with the ODE to within 1% in loss, for both dimensions. The escape is where the noise matters: the medians cross the half-plateau level at $t = 170$ in both dimensions, about 4 time units after the ODE; the noise injects asymmetry of order $d^{-1/2}$ in both directions, which shortens some plateaus and lengthens others. After the escape the medians trail the ODE by a few time units: R_{11} rises to 0.98 and R_{12} falls to 0.01 by $t = 300$, with the loss decaying exponentially at the rate of the specialized fixed point.

Figure 8.2 tests the heuristic from random initialization, where the only asymmetry is accidental. The predicted slope is $1/(2\lambda_{\mathrm{u}}(\eta)) = 14.6$ per unit of $\log d$; a least-squares fit to the medians gives 13.7, and over the factor 32 in dimension the medians rise from 145 to 193. The slope is a parameter-free prediction; the intercept is not, since it depends on the law of Γ, which the heuristic does not compute. The quartiles show the other prediction: the interquartile range stays between 30 and 40 time units at every d, as it must if the escape time is $\frac{1}{2\lambda_{\mathrm{u}}} \log d$ plus a fluctuation of order one.

Figure 8.3 tests Proposition 8.9. The measured rates agree with the predicted rate within 2% for $\eta \le 3.75$ and within 4% at $\eta = 4$, where the decay is slow and the fitting window short; the largest deviations elsewhere are near the kink at $\eta = 2.70$, where the two modes have nearly equal rates. For small η the slow mode is the component along $w_\star$; for large η it is the

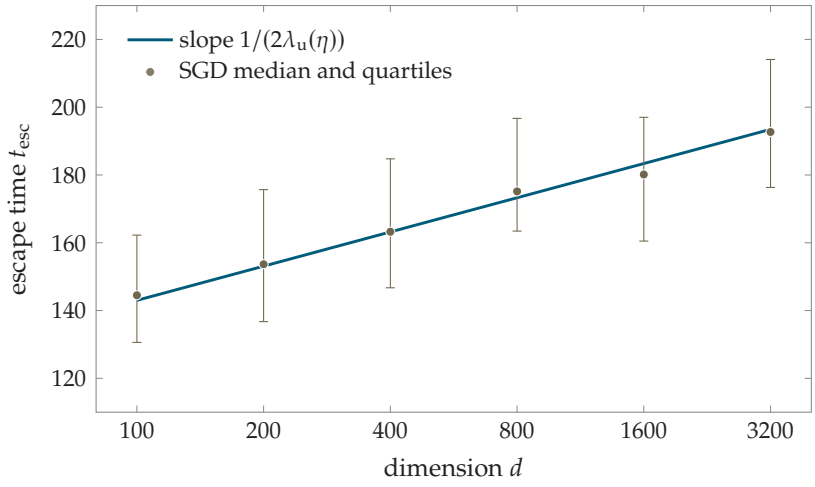


Figure 8.2. Length of the symmetric plateau from random initialization, $K = 2$, $\eta = 0.5$, $w_j(0) \sim \mathcal{N}(0, \frac{0.01}{d} I_d)$. Points: median and quartiles over 40 runs of the first time at which the loss falls below half the plateau level, simulated with the exact chain of Remark 8.7 (seed 9). Line: slope $1/(2\lambda_u(\eta)) = 14.6$ per unit of $\log d$, computed from the Jacobian of the full field at $P_\star(\eta)$, with the intercept fitted to the medians.

orthogonal component, whose contraction $2\eta c_2$ is fought by the noise $\eta^2 c_3$. At $\eta = 4.25$ and 4.5 the runs settle at a nonzero error, as the bifurcation in Section 8.3.2 predicts.

8.6 What the model gets wrong

WHAT WAS THROWN AWAY. Isotropic Gaussian inputs, which made the overlaps sufficient statistics and every integral explicit. A fixed second layer, which forbids reweighting units instead of moving them. A width fixed while $d \to \infty$, and a teacher of the same form and width. One pass over fresh samples. Steps of size η/d, which make the dynamics deterministic at the scale of the theorem and push the decisive fluctuations below it. And small initialization, which places the trajectory near the symmetric saddle; a student started far from $\mathcal{S}$ need not visit it.

WHERE IT BREAKS. The first failure is internal: the plateau length is set by fluctuations of order $d^{-1/2}$ that Theorem 8.6 discards, and we do not know a proof of the $\frac{1}{2\lambda_u} \log d$ law for this model. The other failures are external. *Overparameterization.* With more students than teachers the picture changes qualitatively: Goldt et al. (2019) prove that the ODE description remains exact and find that the final error of the student grows with the width when only the first layer is trained but stays constant or decreases when both layers are trained; and Richert et al. (2022) show that in the over-realizable case the approach to perfect learning follows a power law, not the exponential convergence of the matched case. *Data.* Yoshida and Okada (2019) show that the plateau depends on the input statistics, and becomes less noticeable when the covariance has small and dispersed eigenvalues;

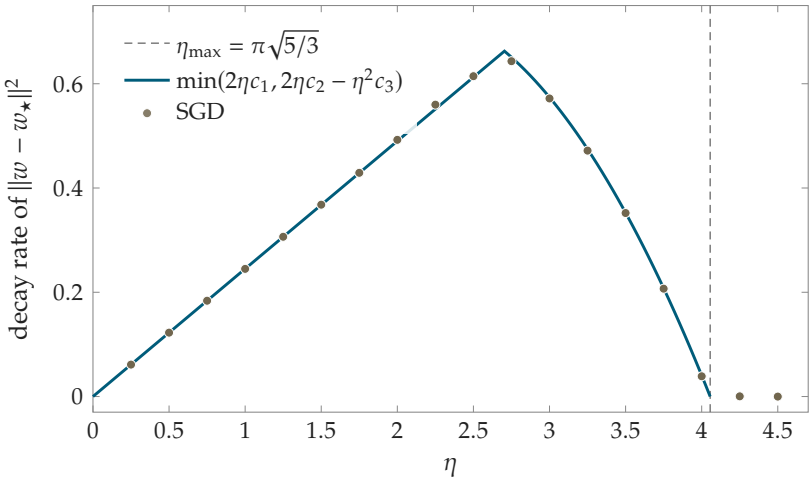


Figure 8.3. The learning-rate threshold of a single erf unit (Proposition 8.9). Curve: the asymptotic decay rate predicted by the linearization, which vanishes at $\eta_{\max} \approx 4.056$ (dashed). Points: minus the slope of the median of $\log\|w - w_\star\|^2$ over 8 runs of SGD with $d = 1000$, fitted where $10^{-40} < \|w - w_\star\|^2 < 10^{-6}$, from $R = 0$ and $Q = 0.25$ (exact chain, seed 10). Beyond $\eta_{\max}$ the runs do not converge and the fitted rate is zero.

Goldt et al. (2020) replace isotropic inputs by the hidden manifold model, inputs that lie near a low-dimensional manifold with labels that depend on the position on it, and describe one-pass SGD on such data through a Gaussian equivalence property, which brings in data parameters, such as the dimension of the manifold, that isotropic inputs do not have. *Optimizers.* Natural gradient descent, which preconditions by the Fisher information, shortens or removes the plateaus (Amari 1998; Rattray and Saad 1999), so the plateau is a property of plain gradient descent as much as of the network. *Steps and width.* Veiga et al. (2022) map the interplay of the learning rate, the time scale, and the number of hidden units, and show that the regime of this chapter, small steps with m fixed, is one corner of a phase diagram whose other corners include the mean-field limit of wide networks and regimes in which the noise dominates the dynamics. *The readout.* Riegler and Biehl (1995) train the hidden-to-output weights as well and compare with fixed readouts.

Fine structure. The repairs come in three kinds. *Exact descriptions of more general dynamics.* Arnaboldi et al. (2023) derive one low-dimensional deterministic description that connects the small-step, high-dimensional, and mean-field limits of one-pass SGD in two-layer networks, and Collins-Woodfin et al. (2023), in an arXiv preprint, give ODE deterministic equivalents for streaming SGD on generalized linear and multi-index models, with learning-rate thresholds for stability that generalize Proposition 8.9; Ben Arous et al. (2022) prove limit theorems for summary statistics of SGD, find a critical step size at which a correction term, the counterpart of our η^2 term, enters the effective dynamics, and show that diffusive limits near fixed points, where escapes are decided, can be complex. *Targets*

that require several escapes. For multi-index targets on Gaussian or Boolean inputs, Abbe et al. (2023) introduce the leap complexity, conjecture that SGD on two-layer networks needs total time about $d^{\max(\mathrm{Leap},2)}$, that is about $d^{\max(\mathrm{Leap}-1,1)}$ samples, prove a version for Gaussian inputs under additional assumptions on how SGD is run, and show that the support of the target is learned in a sequence of saddle-to-saddle transitions; the cascade of plateaus of Observation 8.1 is the matched-teacher ancestor of this picture, and Chapters 7 and 9 develop the single-direction case. Berthier et al. (2023), in a preprint and partly non-rigorously, find separated time scales for wide networks on single-index targets. *Landscape and symmetry.* Fukumizu and Amari (2000) and Amari et al. (2006) relate plateaus to the singular regions created by duplicated units, and Cousseau et al. (2008) analyze the dynamics near them; Şimşek et al. (2021) count the symmetry-induced critical points of overparameterized networks and show that one extra neuron per layer connects the permutation-equivalent minima into a single manifold. Open: a proof of the logarithmic plateau law and of the law of the escape time; the plateau length when $m > m_\star$; and a description of plateaus with data that are neither Gaussian nor isotropic.

8.7 Model card

MODEL CARD	
PHENOMENON	Plateaus in online training of small sigmoidal networks from small weights: hidden units are redundant copies, then specialize.
MODEL	Teacher and student soft committee machines with $g(z) = \mathrm{erf}(z/\sqrt{2})$, Gaussian inputs, online SGD with step η/d; order parameters Q, R, T.
SOLUTION	As $d \to \infty$ the overlaps follow explicit ODEs with an η^2 noise term (proved); one unit converges for $\eta < \pi\sqrt{5/3}$ (proved); for K students and teachers, a symmetric saddle with loss $\frac{K}{6} - \frac{K^2}{\pi}\arcsin\frac{1}{2K}$ and unstable eigenvalue $\frac{2K}{\pi(2K+1)^{3/2}(2K-1)^{1/2}}$ (proved).
PREDICTS	Plateau level; plateau length $\frac{1}{2\lambda_u}\log d$ in units of ηt, growing linearly in K (derived); optimal and maximal learning rates.
GETS WRONG	The plateau length is decided by fluctuations the ODE omits; overparameterization, structured data, natural gradients, large steps, and trained readouts change or remove the plateau.
SUCCESSORS	Overparameterized students; hidden manifold and Gaussian equivalence; dimensionless ODEs and phase diagrams of SGD; leap complexity and saddle-to-saddle dynamics; effective dynamics of summary statistics.

The committee machine is the matrix version of the single-index model of Chapter 7: Gaussian rotational invariance again reduces the state to overlaps, and the ODE method of Theorem A.13 turns online SGD into a

deterministic flow on them. The two chapters produce plateaus for different reasons. In Chapter 7 the target is invisible at low degree, and the plateau is a search whose length is a power of d; here every unit sees its signal at once, and the plateau is a symmetric saddle whose length is a logarithm. Both are escapes from small initialization, the thread that runs from the deep linear networks of Chapter 2 and the diagonal networks of Chapter 3 to the hidden progress of Chapter 9 and the induction heads of Chapter 14; the escape here has the logistic form of Chapter 2, with initialization $d^{-1/2}$.

The threshold $\pi\sqrt{5/3}$ is a stability bound of the kind studied in the prologue and Chapter 15, with the variance of the stochastic gradient in the role of the top curvature. The plateau is one instance of the abrupt transitions collected in Chapters 9, 10, 14 and 16, where a smooth underlying dynamics produces a sudden drop in the loss.

Exercises

8.1 CONSOLIDATION. (Facts about the erf unit.) (a) Prove (8.1), $\mathbb{E}g(z)^2 = \frac{1}{3}$, and $\hat{g}_1 = 1/\sqrt{\pi}$ for $z \sim \mathcal{N}(0,1)$. (b) Show that the loss at $W = 0$ equals $m_\star/6$ for an orthonormal teacher and that the information exponent of g is 1. (c) Use Lemma 8.3 with $S = (a, a)$ to show $\mathbb{E}g'(X_a)^2 = \frac{2}{\pi}(1 + 2C_{aa})^{-1/2}$.

8.2 CONSOLIDATION. (One unit, explicitly.) For $m = m_\star = 1$ and $T = 1$, show that the gradient-flow part of the ODE is

$$\frac{\mathrm{d}R}{\mathrm{d}\tau} = \frac{2}{\pi(1+Q)}\Big(\frac{1+Q-R^2}{\sqrt{2(1+Q)-R^2}} - \frac{R}{\sqrt{1+2Q}}\Big),$$

derive the corresponding equation for Q, check that $(R, Q) = (1, 1)$ is a fixed point, and compute the Jacobian there numerically for $\eta = 1$ to confirm the eigenvalues $-c_1$ and $-(2c_2 - c_3)$ of Proposition 8.9.

8.3 CONSOLIDATION. (The exact chain.) Prove Remark 8.7: write $\|Px\|^2$ as a quadratic form in $(z, z_\star)$ using the pseudoinverse of $\mathsf{M}(\omega)$, and show that, given $\mathcal{F}_s$, the law of ω_{s+1} depends only on ω_s. For $m = m_\star = 2$ and $T = I$, give an explicit sampler that uses five scalar random variables per step.

8.4 CONSOLIDATION. (The other modes at $P_\star$.) Let $p_j = v$ for every j, with $v = (w_{\star 1} - w_{\star 2})/\sqrt{2}$: identical students turning toward teacher 1. Show that the gradient flow gives $\dot{p} = -(K\mathcal{B} + \mathcal{A})p$ to first order, and that the eigenvalue is

$$-\frac{2K}{\pi}\sqrt{\frac{2K-1}{2K+1}} + \lambda_{\mathrm{u}},$$

which is -0.920 for $K = 2$.

8.5 VARIATION. (Label noise.) Let $y = \sum_k g(z_{\star k}) + \sigma\zeta$ with $\zeta \sim \mathcal{N}(0,1)$ independent. Show that the ODE acquires the term $\eta^2\sigma^2\mathbb{E}[g'(z_j)g'(z_l)]$ in $\dot{Q}_{jl}$ and nothing else. For one unit, show that $w_\star$ is no longer a fixed

point, and that to leading order the excess loss at the new fixed point is $\frac{\eta\sigma^2 c_2^2}{2(2c_2-\eta c_3)}$, which is about $\eta\sigma^2/(2\sqrt{3}\pi)$ for small η and diverges at $\eta_{\max}$.

8.6 VARIATION. (Unequal teachers.) Let $K = 2$ and $T = \mathrm{diag}(1, \vartheta^2)$ with $\vartheta \neq 1$. Show that $\mathcal{S}$ is no longer invariant, so that the plateau of Theorem 8.10 disappears as an exact object. Integrate the ODE from a symmetric start for $\vartheta = 0.9$ and 0.5: which teacher is learned first, and how does the delay depend on $|1 - \vartheta|$?

8.7 VARIATION. (Other activations.) For an odd activation ϕ in place of g, with the corresponding symmetric fixed point, show that the Hessian block $\mathcal{A}$ acts on directions orthogonal to the teachers as multiplication by $-\mathbb{E}[(y - K\phi(z))\phi''(z)]$. Show that this vanishes for every activation with $\phi''(z) = -c\, z\, \phi'(z)$ for a constant c, and give an odd activation for which it does not vanish. What does this change in Theorem 8.10(d)?

8.8 VARIATION. (Specialization for general K.) Prove that for $K \geq 3$ every vector v in the teacher span with $\langle v, \sum_k w_{\star k}\rangle = 0$ satisfies $\mathcal{A}v = -\lambda_{\mathrm{u}} v$, so that the unstable eigenvalue of $P_\star$ has multiplicity at least $K - 1$ in weight space. How many distinct ways to break the symmetry does this give, and how does the plateau length depend on K?

8.9 SIMULATION. (The committee's learning-rate threshold.) For $K = 2$, $T = I$, compute numerically the Jacobian of the full field F_η at the specialized fixed point $R = Q = I$ for η between 1 and 3. You should find that its largest eigenvalue crosses zero at $\eta \approx 2.31$, well below the single-unit threshold 4.06; identify the unstable eigenvector and explain the role of the cross terms $\mathbb{E}[\delta_1\delta_2]$.

8.10 SIMULATION. (Plateau length and the number of units.) Integrate the ODE with $\eta = 0.1$ from $Q(0) = 0.01\, I$ and $R(0) = \epsilon I$ for $\epsilon = 10^{-3}$ and 10^{-5}, for $K = 2$ and $K = 3$, and record the first time after the approach at which the loss falls below $\mathcal{L}_\star/2$. You should find that the escape time grows by about 153 (for $K = 2$) and 217 (for $K = 3$) per factor e decrease of ϵ, within 1% of $1/(\eta\lambda_{\mathrm{u}})$ from Theorem 8.10.

8.11 RESEARCH. (Plateaus with more students than teachers.) For $m_\star = 2$ orthonormal teachers and $m = 3$ or 4 students, find the symmetric fixed points of the gradient-flow field, their unstable eigenvalues, and the predicted plateau length, and compare with simulations. How the plateau depends on $m/m_\star$ is, as far as we know, not settled.

8.12 RESEARCH. (A theorem for the escape.) Prove that for $K = 2$ and fixed $\eta < \eta_{\max}$, the escape time from random initialization divided by $\log d$ converges in probability to $1/(2\lambda_{\mathrm{u}}(\eta))$, and identify the limit law of $t_{\mathrm{esc}} - \frac{1}{2\lambda_{\mathrm{u}}(\eta)} \log d$. We do not know of a proof for this model.

Notes and References

The committee machine belongs to the statistical mechanics of learning, surveyed by Engel and Van den Broeck (2001). The exact dynamics of online learning came in 1995. Saad and Solla (1995a) announced the exact solution for soft committee machines with arbitrary numbers of hidden units in student and teacher, and Saad and Solla (1995b) gave the full derivation and treated the realizable, over-realizable, and unrealizable cases and the role of the learning rate. Biehl and Schwarze (1995) analyzed the same dynamics independently, Riegler and Biehl (1995) trained the readout too, and Biehl et al. (1996), in a preprint, found several fixed points and cascades of plateaus between them.

The plateau problem shaped the next decade of work on optimization. It led to the natural gradient (Amari 1998), analyzed in the committee machine with the methods of this chapter by Rattray and Saad (1999), and to the geometry of the singular regions created by duplicated units (Amari et al. 2006; Cousseau et al. 2008; Fukumizu and Amari 2000); Section 8.6 describes both.[5]

[5] The geometric view and the ODE view describe the same object: the set $\mathcal{S}_0$ of identical students is the image of a network with one hidden unit inside the network with K.

The modern revival began with Goldt et al. (2019), who proved that the ODE description is asymptotically exact and studied overparameterized students, and Goldt et al. (2020), who treated structured inputs. Veiga et al. (2022) drew the phase diagram of one-pass SGD with rigorous convergence rates, and Arnaboldi et al. (2023) unified the high-dimensional and mean-field limits. The weight-space derivation of the unstable eigenvalue in Theorem 8.10 is our arrangement.

Translating the literature. Saad and Solla (1995b) write N for our d, ξ for our input x, J_i for our student weights w_j, B_n for our teacher weights $w_{\star k}$, K and M for our m and $m_\star$, and μ for our step s, so that their update is (8.2); their time $\alpha = \mu/N$ is our t, their generalization error ϵ_g is our $\mathcal{L}$, and the overlaps Q, R, T keep their names, with student indices i, k and teacher indices n, m. Their integrals I_2, I_3, I_4 are those of Lemma 8.5. Goldt et al. (2019) write N for d and K, M for the numbers of student and teacher units, keep the entries of the weights of order one and divide the preactivations by $\sqrt{N}$, so that their order parameters, normalized by N, are ours; they also study training the second layer, which our model freezes at one.

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9 Hidden Progress

Sparse parities and the limits of gradient methods

THE MODEL — SPARSE PARITY ON THE HYPERCUBE

IN THIS CHAPTER. Networks trained by stochastic gradient descent on a sparse parity, the product of k unknown coordinates of a uniformly random sign vector, show no improvement in loss or error for a long time and then learn abruptly, while measurements of the weights reveal progress during the flat stretch. The model of this chapter is the parity itself, learned by one neuron with the activation $z \mapsto z^k$. Its correlation with the target is exactly $k!$ times the product of the k relevant weights, so the population gradient is supported on the relevant coordinates; we prove that one gradient step estimated from about $d^{k-1} \log d$ samples finds them, that every statistical-query algorithm needs a number of queries times an inverse squared tolerance of order d^k, and that the population flow spends a time of order $d^{(k-2)/2}$ on a loss plateau while the logarithm of the relevant weights grows steadily from the start. The model is wrong about how networks actually find the coordinates: with random initialization the progress sits in a few lucky neurons, and natural tasks offer low-degree footholds that parities deny.

9.1 The phenomenon: abrupt learning with steady progress

Parity is the oldest hard case of neural network theory. Minsky and Papert (1969) showed that a perceptron that combines predicates on a bounded number of inputs cannot compute the parity of all its inputs; the parity of n bits needs predicates of order n.[1] Modern networks represent parities without difficulty: a single hidden layer of $k+1$ rectified linear units computes the parity of k given bits (Corollary 9.11). The difficulty that remains is learning: when the k bits are unknown, nothing in the data points to them until they have been found.

Fix a dimension d and an unknown set $S \subseteq [d] = \{1, \dots, d\}$ of size k. Inputs are uniform on the cube $\{-1, 1\}^d$, and the label is the *sparse parity*

$$y = \chi_S(x) = \prod_{i \in S} x_i .$$

Statistically the problem is easy: there are $\binom{d}{k}$ candidate sets, so about $k \log d$ samples identify S by exhaustive search, and Gaussian elimination over the field with two elements solves it in polynomial time from about d samples.[2]

[1] Their *order* of a perceptron is the largest number of inputs that one of its predicates reads.

[2] Write $x_i = (-1)^{z_i}$ with $z_i \in \{0, 1\}$; then $\chi_S(x) = (-1)^{\sum_{i \in S} z_i}$, and each sample is a linear equation modulo 2 for the indicator vector of S. About d random equations determine it.

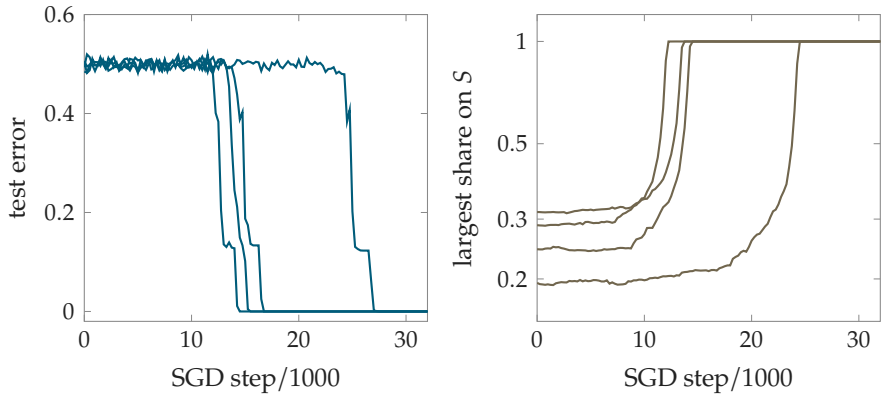


Figure 9.1. Four runs of SGD for a two-layer ReLU network of width 256 on the parity $x_1x_2x_3$ of $d = 50$ uniform bits: hinge loss, step size 0.02, fresh minibatches of 32 samples, Gaussian initialization, all layers trained (seeds 0 to 3). Left: error on 4000 test points. Right: the progress measure $\max_j \sum_{i\in S} W_{ji}^2/\|W_j\|^2$ over the hidden units j, on a logarithmic scale; a unit with no preference has share $k/d = 0.06$.

Neither algorithm resembles gradient descent. For a gradient method the task is hard for a reason that the Fourier–Walsh expansion makes precise (Theorem A.15): the label χ_S is orthogonal to every function of the input that does not depend on all k coordinates of S, and in particular to every polynomial of degree less than k. No statistic of degree below k sees the target.

Observation 9.1 (Barak et al. 2022). Train a network by stochastic gradient descent on fresh samples of a k-sparse parity of d bits. Across a range of architectures and initializations, the loss and the error stay at their chance values for a long stretch of training and then fall abruptly; on small instances the number of iterations before the drop grows like $d^{O(k)}$, close to the statistical-query lower bound of Theorem 9.7 below. During the flat stretch the training makes continual progress that the loss and the error do not show: the authors argue that SGD amplifies the sparse solution gradually, through a Fourier gap in the population gradient, rather than stumbling on it by chance.

Figure 9.1 reproduces the observation at a size that runs in a minute. A two-layer network of 256 rectified linear units, with standard Gaussian initialization, learns $x_1x_2x_3$ in dimension $d = 50$ by plain SGD on the hinge loss. In each of four runs the test error sits at $\frac{1}{2}$ for between 12,000 and 25,000 steps and then falls to zero within about 3,000 steps. The progress measure in the right panel, the largest share of a neuron's squared input weight carried by the three relevant coordinates, starts at a value set by the luck of the initialization and begins to grow several thousand steps before the drop, in every run.

How firm is the observation? The flat-then-sudden curve is easy to reproduce, and it was seen long before the name: Shalev-Shwartz et al. (2017) used parities as their first example of a failure of gradient-based

learning, arguing that when the gradient carries little information about the target, training by gradients cannot succeed. The scaling $d^{O(k)}$ is measured only on small instances, and the exponent depends on architecture and hyperparameters. The statement about hidden progress depends on the measure: the share of the best neuron in Figure 9.1 grows during the last third of the plateau, not from the first step, while averages over all neurons barely move until the drop.

The observation matters beyond parities because it is the cleanest instance of an *emergent* capability: a skill that is absent for a long time and then present. Wei et al. (2022) catalogued abilities of large language models that appear abruptly as models are scaled, and Schaeffer et al. (2023) argued that many such jumps are produced by the choice of metric, nonlinear or discontinuous metrics turning smooth progress into apparent jumps. Parities sit between these positions: the loss is flat for a real reason, yet a different measurement sees continuous progress. The questions for the model are therefore three. How much signal does a gradient carry about S? How many samples, or how much computation, must every gradient method spend? And what progresses while the loss stands still?

9.2 The model: one monomial neuron on the cube

We use the Fourier–Walsh expansion of Theorem A.15: every $f : \{-1,1\}^d \to \mathbb{R}$ is $\sum_{T\subseteq[d]} \hat{f}(T)\chi_T$ with $\hat{f}(T) = \mathbb{E}[f(x)\chi_T(x)]$, and $\mathbb{E}f(x)^2 = \sum_T \hat{f}(T)^2$. Throughout, x is uniform on $\{-1,1\}^d$.

Definition 9.2 (The sparse parity model). Let $1 \le k \le d$ and let $S \subseteq [d]$ with $|S| = k$. Inputs are $x \sim \mathrm{Unif}\{-1,1\}^d$ and labels are $y = \chi_S(x)$. The *monomial neuron* with weight $w \in \mathbb{R}^d$ predicts $\langle w, x\rangle^k$ and pays the correlation loss $-y\langle w, x\rangle^k$. Its population correlation and loss are

$$\Psi(w) = \mathbb{E}\left[\chi_S(x)\langle w, x\rangle^k\right], \qquad \mathcal{L}(w) = -\Psi(w). \tag{9.1}$$

For a two-layer network $f(x;\theta) = \sum_{j=1}^m a_j\phi(\langle w_j, x\rangle + b_j)$ the correlation is $\mathbb{E}[y\,f(x;\theta)]$.

The activation $z \mapsto z^k$ is chosen because it is the lowest-degree neuron that can see the target at all: a polynomial of degree less than k in x is orthogonal to χ_S, so a neuron with a polynomial activation of degree less than k has $\Psi \equiv 0$ and receives no gradient. The correlation loss replaces the hinge and square losses of practice for the same reason as in Chapter 7: it makes the population landscape an explicit polynomial, and the square loss $(y - f)^2$ equals twice the correlation loss plus terms that do not involve the label. On the sphere $\|w\| = 1$, where we study the dynamics, the neuron is the Boolean counterpart of the single-index student of Chapter 7 with information exponent k; the difference is that the target depends on k coordinates rather than one direction, and that the cube, unlike the Gaussian, is not rotationally invariant, so the coordinates themselves are the natural basis.

Three algorithms will be studied, in increasing realism. First, *statistical-query algorithms*, the abstraction of every method that learns only from averages over the data, including gradient descent with estimated gradients; this gives the lower bound. Second, *one gradient step* from the symmetric initialization $w_0 \propto \mathbf{1}_d$, estimated from n samples, followed by a small readout; this gives a matching upper bound. Third, the *population gradient flow* on the sphere from a small start, and its noisy version online SGD; this gives the plateau and the hidden progress.

Each simplification has a reason. Uniform inputs make the characters orthonormal and independent of S, so that all $\binom{d}{k}$ targets look alike to every algorithm, which is the source of the lower bound.[3] Exact parity labels remove every low-degree correlation: the target has Fourier weight only at the single set S. The monomial activation makes the correlation of one neuron with the target an explicit polynomial, Lemma 9.3 below. The symmetric initialization $w_0 \propto \mathbf{1}_d$ removes the luck of a random start, so that what remains is the deterministic signal. What a reader would have had to notice to invent the model is that the population gradient of a neuron is itself a Fourier coefficient: by the chain rule, $\partial_{w_i} \mathbb{E}[y\phi(\langle w, x\rangle)] = \mathbb{E}[\chi_{S\triangle\{i\}}(x)\,\phi'(\langle w, x\rangle)]$, the Fourier coefficient of $\phi'(\langle w, x\rangle)$ at the set S with i removed or added. The question of how much a gradient knows becomes a question about the Fourier spectrum of one function, a theme that Chapter 10 develops on the group $\mathbb{Z}_p$.

[3] Uniformity is what makes the $\binom{d}{k}$ targets an orthonormal family; under a biased product distribution the characters are no longer orthogonal, and Exercise 9.7 shows how much this helps.

DICTIONARY

IN THE MODEL	IN A TRAINED NETWORK
the set S, $\lvert S\rvert = k$	the relevant features of the input
parity χ_S	a target with no low-degree footprint: every statistic of degree $< k$ is blind to it
monomial neuron $\langle w, x\rangle^k$	one hidden unit; its degree is the lowest that can see the target
correlation $\Psi(w)$	minus the loss; the network's agreement with the labels
weights w_i, $i \in S$	the progress measure: how much the unit attends to the relevant inputs
effective overlap ρ	the alignment of the unit with the target, on a scale where the loss is $1 - \rho^k$
symmetric start $w_0 \propto \mathbf{1}_d$	an initialization with no luck in it
statistical query of tolerance τ	an average over a minibatch of about τ^{-2} samples
loss plateau, ρ near 0	the long flat stretch of Figure 9.1
growth of $\log \rho$ during the plateau	hidden progress

9.3 The solution: what a gradient can see

Everything rests on one computation, which says that the correlation of a monomial neuron with a parity counts the ways to match the factors of the monomial with the coordinates of S.

Lemma 9.3 (Monomials against characters). *Let $T \subseteq [d]$, let $j \geq 0$ be an integer with $j \leq |T|$, and let $w \in \mathbb{R}^d$. Then*

$$\mathbb{E}\left[\chi_T(x)\langle w, x\rangle^j\right] = \begin{cases} j! \prod_{i\in T} w_i, & |T| = j, \\ 0, & |T| > j. \end{cases}$$

Proof. Expand the power: $\langle w, x\rangle^j = \sum w_{i_1} \cdots w_{i_j}\, x_{i_1} \cdots x_{i_j}$, the sum over all sequences $(i_1, \ldots, i_j)$ in $[d]^j$. Because $x_i^2 = 1$, the product $\chi_T(x) x_{i_1} \cdots x_{i_j}$ equals $\chi_U(x)$, where U is the set of indices that occur an odd number of times in the list made of the elements of T and of $i_1, \ldots, i_j$; and $\mathbb{E}\chi_U(x) = 1$ if $U = \emptyset$ and 0 otherwise, by Theorem A.15. So a sequence contributes only if every element of T occurs in it an odd number of times, and every other index an even number of times. Such a sequence has at least $|T|$ terms. If $|T| > j$, no sequence qualifies and the expectation is 0. If $|T| = j$, every element of T occurs exactly once and nothing else occurs, so the qualifying sequences are the $j!$ orderings of T, each contributing $\prod_{i\in T} w_i$. ∎

Theorem 9.4 (The exact signal of a monomial neuron). *For every $w \in \mathbb{R}^d$,*

$$\Psi(w) = k! \prod_{i\in S} w_i, \qquad \partial_{w_i}\Psi(w) = \begin{cases} k! \prod_{j\in S\setminus\{i\}} w_j, & i \in S, \\ 0, & i \notin S. \end{cases}$$

In particular, at the symmetric start $w_0 = d^{-1/2}\mathbf{1}_d$ the population gradient equals $k!\, d^{-(k-1)/2}$ on every coordinate of S and exactly 0 on every other coordinate.

Proof. The formula for Ψ is Lemma 9.3 with $T = S$ and $j = k$. Differentiating the product gives the gradient. For the second statement differentiate under the expectation, which is a finite sum: $\partial_{w_i}\Psi(w) = k\,\mathbb{E}[\chi_S(x) x_i \langle w, x\rangle^{k-1}] = k\,\mathbb{E}[\chi_{S\triangle\{i\}}(x)\langle w, x\rangle^{k-1}]$, where $\triangle$ is the symmetric difference. For $i \notin S$ the set $S\triangle\{i\} = S \cup \{i\}$ has $k+1 > k-1$ elements, so Lemma 9.3 gives 0; for $i \in S$ it has $k-1$ elements and gives $k\,(k-1)! \prod_{j\in S\setminus\{i\}} w_j$, consistent with the first formula. At w_0 every product of $k-1$ coordinates is $d^{-(k-1)/2}$. ∎

Two features of the theorem drive the rest of the chapter. The gradient off S vanishes identically, not only at w_0: in the population, the irrelevant weights of a monomial neuron never move. And the gradient on S is small, of order $d^{-(k-1)/2}$ at a start of unit norm, because it is a product of $k-1$ small weights. The first feature says that the population gradient identifies S perfectly; the second says that a sample estimate of it is drowned in noise

unless the sample is large. Before quantifying the second, we check that the first is not an artifact of the monomial.

Proposition 9.5 (The Fourier gap of a ReLU neuron). *Let d be odd, let k be even with $2 \le k < d$, and let $\alpha > 0$. The correlation $\Phi(w) = \mathbb{E}[\chi_S(x)\,\mathrm{ReLU}(\langle w, x\rangle)]$ is differentiable at $w = \alpha\mathbf{1}_d$, and its partial derivative there is $\frac{1}{2}\widehat{\mathrm{Maj}}_d(S \setminus \{i\})$ for $i \in S$ and $\frac{1}{2}\widehat{\mathrm{Maj}}_d(S \cup \{i\})$ for $i \notin S$, where $\mathrm{Maj}_d(x) = \mathrm{sgn}(x_1 + \cdots + x_d)$. These numbers have opposite signs, and the ratio of their absolute values, off S to on S, is exactly $(k-1)/(d-k)$. For odd k both vanish.*

Proof. Since d is odd, $\langle \alpha\mathbf{1}_d, x\rangle = \alpha(x_1 + \cdots + x_d)$ is never 0, so near $\alpha\mathbf{1}_d$ each of the finitely many terms of the expectation is a linear function of w, and $\partial_{w_i}\Phi(\alpha\mathbf{1}_d) = \mathbb{E}[\chi_S(x)x_i\mathbf{1}\{\langle\mathbf{1}_d, x\rangle > 0\}]$. Write $\mathbf{1}\{\langle\mathbf{1}_d, x\rangle > 0\} = \frac{1}{2}(1+\mathrm{Maj}_d(x))$; the set $S\triangle\{i\}$ is not empty, so the constant contributes nothing and the derivative is $\frac{1}{2}\widehat{\mathrm{Maj}}_d(S\triangle\{i\})$. For the values we use the formula for the Fourier coefficients of majority (O'Donnell 2014, Chapter 5): they vanish on sets of even size, and on a set of odd size j,

$$\widehat{\mathrm{Maj}}_d(T) = (-1)^{(j-1)/2}\,\frac{\binom{(d-1)/2}{(j-1)/2}}{\binom{d-1}{j-1}}\cdot\frac{2}{2^d}\binom{d-1}{(d-1)/2}.$$

For odd k the sets $S \setminus \{i\}$ and $S \cup \{i\}$ have even sizes, so both derivatives vanish. For even k the sizes are $k-1$ and $k+1$, consecutive odd numbers, so the signs differ. With $\nu = (d-1)/2$, the ratio of absolute values is

$$\frac{\binom{\nu}{k/2}}{\binom{\nu}{k/2-1}}\cdot\frac{\binom{d-1}{k-2}}{\binom{d-1}{k}} = \frac{\nu - k/2 + 1}{k/2}\cdot\frac{k(k-1)}{(d-k+1)(d-k)} = \frac{k-1}{d-k},$$

because $\nu - k/2 + 1 = (d-k+1)/2$. ∎

The ReLU neuron at a symmetric start therefore also separates S from its complement, but with a finite *Fourier gap*: its gradient off S is smaller by the factor $(k-1)/(d-k)$, not zero. By Stirling's formula the coefficient at level $k-1$ is of order $d^{-(k-1)/2}$, the same order as the monomial's signal. The gradient of a ReLU neuron, read as a Fourier coefficient of the threshold function $\mathbf{1}\{\langle w, x\rangle > 0\}$, is the mechanism behind Barak et al. (2022)'s explanation of hidden progress; the parity of k and the bias matter, since for odd k the weights of a symmetric unit without bias receive no signal at all (Exercise 9.2).[4]

[4] The bias does receive signal for odd k: its derivative is $\frac{1}{2}\widehat{\mathrm{Maj}}_d(S)$, nonzero because $|S|$ is odd. A symmetric unit learns that the parity is correlated with its threshold, not which coordinates matter.

9.3.1 The statistical-query lower bound

A gradient estimated from data is a list of averages. The model of computation that keeps only this feature is due to Kearns (1998).

Definition 9.6 (Statistical queries). A *statistical query* is a function $q\colon \{-1,1\}^d \times \{-1,1\} \to \mathbb{R}$ with $\mathbb{E}[q(x,1)^2] \le 1$ and $\mathbb{E}[q(x,-1)^2] \le 1$; every q with values in $[-1,1]$ qualifies. An *oracle of tolerance* $\tau > 0$

for the target χ_S answers each query with some number v satisfying $|v - \mathbb{E}q(x, \chi_S(x))| \le \tau$. A *statistical-query learner* chooses its queries one at a time, each depending on the previous answers and on internal randomness, and finally outputs a set $\hat{S}$.

An average of $q(x_s, y_s)$ over a minibatch of b fresh samples is within about $b^{-1/2}$ of $\mathbb{E}q(x, y)$; a query of tolerance τ is the idealization of a minibatch of size τ^{-2}, with the sampling error replaced by an adversarial one.

Theorem 9.7 (Statistical-query lower bound for sparse parities). *Let $1 \le k \le d$, $M = \binom{d}{k}$, and $0 < \tau \le 1$, and let S be uniformly random among the subsets of $[d]$ of size k, independent of the learner's internal randomness. There is an oracle of tolerance τ such that every statistical-query learner that makes at most Q queries outputs $\hat{S} = S$ with probability at most*

$$\frac{Q\tau^{-2} + 1}{M}.$$

In particular, a learner that identifies every S with certainty against every oracle of tolerance τ makes at least $(M-1)\tau^2$ queries.

Proof. The plan is to let the oracle answer every query by its label-independent part, which is correct for all but a few targets, and to count the targets for which it is not. For a query q put

$$q_0(x) = \tfrac{1}{2}\big(q(x,1) + q(x,-1)\big), \qquad q_1(x) = \tfrac{1}{2}\big(q(x,1) - q(x,-1)\big).$$

Since $\chi_S(x) \in \{-1, 1\}$, checking the two cases gives $q(x, \chi_S(x)) = q_0(x) + \chi_S(x) q_1(x)$, and therefore

$$\mathbb{E}q(x, \chi_S(x)) = \mathbb{E}q_0(x) + \hat{q}_1(S).$$

The oracle. Given the target S and a query q, the oracle answers $\mathbb{E}q_0(x)$ if $|\hat{q}_1(S)| \le \tau$, and the exact value $\mathbb{E}q(x, \chi_S(x))$ otherwise. In both cases the answer is within τ of the truth.

Few targets are exposed by one query. Let $B(q)$ be the set of sets T of size k with $|\hat{q}_1(T)| > \tau$. Parseval's identity (Theorem A.15) and the inequality $(a-b)^2 \le 2a^2 + 2b^2$ give

$$|B(q)|\,\tau^2 \le \sum_T \hat{q}_1(T)^2 = \mathbb{E}q_1(x)^2 \le \tfrac{1}{2}\big(\mathbb{E}q(x,1)^2 + \mathbb{E}q(x,-1)^2\big) \le 1,$$

so $|B(q)| \le \tau^{-2}$.

Counting. Fix the learner's internal randomness. Run the learner against the responder that answers $\mathbb{E}q_0(x)$ to every query q; this responder ignores S, so the run produces queries $q^{(1)}, \dots, q^{(Q')}$ with $Q' \le Q$ and an output $\hat{S}_0$ that do not depend on S. Let $B = B(q^{(1)}) \cup \cdots \cup B(q^{(Q')})$, a set of at most $Q\tau^{-2}$ targets. If $S \notin B$, the oracle for S gives the same answer as the responder at each of these queries, so by induction on the number of queries the learner, facing the oracle, asks the same queries and outputs $\hat{S}_0$. Hence $\hat{S} = S$ forces

$S \in B$ or $S = \hat{S}_0$, an event of probability at most $(|B|+1)/M \le (Q\tau^{-2}+1)/M$ for uniformly random S. Averaging over the internal randomness proves the bound. A learner that succeeds with certainty has $1 \le (Q\tau^{-2}+1)/M$, which is the last statement. ∎

The argument goes back to the beginnings of the model: Kearns observed that parities cannot be learned from statistical queries in polynomial time, and Blum et al. (1994) turned the counting of pairwise orthogonal functions into a characterization of weak statistical-query learning by what is now called the statistical-query dimension.[5] The same counting bounds learners that only need a weak hypothesis (Exercise 9.4).

[5] For the class of all 2^d parities the counting gives $Q\tau^{-2} \gtrsim 2^d$; restricting to sets of size k replaces 2^d by $\binom{d}{k}$.

Remark 9.8 (Gradient methods are statistical algorithms). For a model $f(x;\theta)$ with p parameters, the population gradient of the correlation loss has coordinates $-\mathbb{E}[y\,\partial_{\theta_j} f(x;\theta)]$, and each is a statistical query after division by $B_j = (\mathbb{E}[\partial_{\theta_j} f(x;\theta)^2])^{1/2}$; for the square loss the gradient differs by $\mathbb{E}[f\,\partial_{\theta_j} f]$, which does not involve the label. So t steps of gradient descent whose gradient coordinates are accurate to τB_j are a statistical-query algorithm with $Q = tp$ queries, and if the errors are adversarial, Theorem 9.7 requires $tp\,\tau^{-2} \gtrsim \binom{d}{k}$. With $\tau \approx b^{-1/2}$ for minibatches of size b, this reads $tpb \gtrsim \binom{d}{k}$: the total computation of a gradient method, parameters times samples, is at least of order $d^k/k!$. This translation is a heuristic, not a theorem about SGD, because sampling errors are random rather than adversarial. The difference can be decisive: Abbe and Sandon (2020) showed that SGD with small batches on suitably constructed networks can emulate every polynomial-time learning algorithm, including those for parities that no statistical-query algorithm matches, and proved that this power disappears when gradients are averaged over large batches.

9.4 The solution: one step, and a plateau with progress

We now show that a single gradient step meets the lower bound up to a constant and a logarithm. The noise of the empirical gradient is controlled by the moments of a sum of independent signs.

Lemma 9.9 (Moments of a sum of signs). *Let $s = x_1 + \cdots + x_d$. For every integer $j \ge 1$, $\mathbb{E}s^{2j} \le (2j-1)!!\, d^j$, where $(2j-1)!! = 1 \cdot 3 \cdots (2j-1)$.*

Proof. Expanding, $\mathbb{E}s^{2j}$ is the number of sequences $(i_1, \dots, i_{2j}) \in [d]^{2j}$ in which every index occurs an even number of times, because the expectation of a product of signs is 1 for such sequences and 0 for the others. Each such sequence arises from at least one pair of a perfect matching of the $2j$ positions and an assignment of one index to each of the j matched pairs: match the occurrences of each index among themselves. There are $(2j-1)!!$ perfect matchings and d^j assignments, which bounds the number of sequences. ∎

The bound is the corresponding Gaussian moment; for $j = 2$ the exact value is $\mathbb{E}s^4 = 3d^2 - 2d$. Write the empirical gradient of the correlation at the start $w_0 = \mathbf{1}_d$, scaled for convenience (the gradient is homogeneous of degree $k-1$ in w, so the scale of the start does not matter):

$$\hat{G}_i = \frac{1}{n}\sum_{s=1}^{n} Z_i(x_s), \qquad Z_i(x) = k\,\chi_S(x)\,\langle \mathbf{1}_d, x\rangle^{k-1} x_i, \qquad i \in [d]. \tag{9.2}$$

Theorem 9.10 (One gradient step finds the support). *Let $2 \le k \le d$ and $0 < \delta < 1$, and let $x_1, \dots, x_n$ be independent and uniform on $\{-1,1\}^d$ with labels $\chi_S(x_s)$. If*

$$n \ge C_k\, d^{k-1} \log \frac{2d}{\delta}, \qquad C_k = \frac{8k\big(k\,(2k-3)!! + k!/3\big)}{(k!)^2},$$

then with probability at least $1-\delta$ every $i \in S$ has $\hat{G}_i > k!/2$ and every $i \notin S$ has $\hat{G}_i < k!/2$. In particular the k largest coordinates of the empirical gradient are exactly the coordinates of S.

Proof. The plan is to apply Bernstein's inequality to each coordinate and a union bound over the d coordinates. Fix i. By Theorem 9.4 with $w = \mathbf{1}_d$, $\mathbb{E}Z_i = \partial_{w_i}\Psi(\mathbf{1}_d)$ equals $k!$ for $i \in S$ and 0 for $i \notin S$. Three bounds on Z_i are needed.

- *Range.* $|\langle \mathbf{1}_d, x\rangle| \le d$, so $|Z_i| \le k d^{k-1}$; also $k! = k\,(k-1)! \le k d^{k-1}$ because $k \le d$. Hence $|Z_i - \mathbb{E}Z_i| \le 2k d^{k-1}$.
- *Variance.* Since $\chi_S^2 = x_i^2 = 1$, Lemma 9.9 with $j = k-1 \ge 1$ gives $\operatorname{Var} Z_i \le \mathbb{E}Z_i^2 = k^2\,\mathbb{E}\langle \mathbf{1}_d, x\rangle^{2k-2} \le k^2(2k-3)!!\,d^{k-1}$.
- *Deviation.* Bernstein's inequality (Theorem A.18) for the n independent centered variables $Z_i(x_s) - \mathbb{E}Z_i$, with $c = 2k d^{k-1}$, $v = nk^2(2k-3)!!\,d^{k-1}$, and threshold $nk!/2$, gives

$$\Pr\Big(|\hat{G}_i - \mathbb{E}Z_i| \ge \frac{k!}{2}\Big) \le 2\exp\Big(-\frac{n^2(k!)^2/4}{2\big(v + c\,nk!/6\big)}\Big),$$

  and inserting v and c, the exponent equals $-n(k!)^2/\big(8d^{k-1}(k^2(2k-3)!! + k\,k!/3)\big) = -n/(C_k d^{k-1})$.

By the hypothesis on n, the right side is at most δ/d. By the union bound, with probability at least $1-\delta$ every coordinate satisfies $|\hat{G}_i - \mathbb{E}Z_i| < k!/2$, which is the claim. ∎

For $k = 3$ the constant is $C_3 = 22/3 \approx 7.3$.[6] The theorem uses a nonlinear step after the gradient, a threshold or a choice of the k largest coordinates. A plain gradient step followed by a linear readout would need more samples: the noise of about $(d^{k-1}/n)^{1/2}$ on each of the $d-k$ irrelevant coordinates adds up, in $\langle w_1, x\rangle$, to a perturbation of order $(d^k/n)^{1/2}$, so about d^k samples would be needed for the noise to be small compared with the signal (Exercise 9.10). With the support found, a small second layer finishes the job.

[6] The constant is not optimized. Bernstein's inequality is used because the range of Z_i and its standard deviation times $\sqrt{n}$ are of the same order d^{k-1} at the sample size of the theorem; Chebyshev's inequality would cost a factor d.

Corollary 9.11 (A network that computes the parity). *Let $\hat{S} \subseteq [d]$ have k elements and $s(x) = \sum_{i\in\hat{S}} x_i$. Then for every $x \in \{-1,1\}^d$,*

$$\chi_{\hat{S}}(x) = (-1)^k \Big[\mathrm{ReLU}\big(s(x)+k+1\big) + \sum_{j=1}^{k} (-1)^j\, 4j\ \mathrm{ReLU}\big(s(x)+k+1-2j\big)\Big],$$

a network of $k+1$ ReLU units with input weights $\mathbf{1}_{\hat{S}}$. Consequently, under the hypotheses of Theorem 9.10, one gradient step at $w_0 = \mathbf{1}_d$, the choice of the k largest coordinates, and this readout output a network equal to χ_S with probability at least $1-\delta$.

Proof. The value $s(x)$ lies in $\{-k+2i : 0 \le i \le k\}$, and at $s = -k+2i$ the parity is $(-1)^{k-i}$, since i coordinates equal $+1$ and $k-i$ equal -1. Put $c_0 = 1$ and $c_j = (-1)^j 4j$ for $j \ge 1$. At $s = -k+2i$ the unit $\mathrm{ReLU}(s+k+1-2j)$ equals $2(i-j)+1$ for $j \le i$ and 0 for $j > i$, so the bracket equals $A_i = \sum_{j=0}^{i} c_j\,(2(i-j)+1)$, and we must show $A_i = (-1)^i$. The partial sums of the coefficients are $P_i = \sum_{j=0}^{i} c_j = 1 + 4\sum_{j=1}^{i}(-1)^j j = (-1)^i(2i+1)$, as one checks separately for even and odd i. Hence, for $i \ge 1$,

$$A_i - A_{i-1} = 2P_{i-1} + c_i = 2(-1)^{i-1}(2i-1) + (-1)^i 4i = 2(-1)^i,$$

and since $A_0 = 1$, $A_i = 1 + 2\sum_{l=1}^{i}(-1)^l = (-1)^i$. The last statement combines this identity with Theorem 9.10. ∎

How close to the limit? Read the d coordinates of the empirical gradient as statistical queries. Normalized to unit second moment, the query for coordinate i is $q_i(x,y) = y\,k\langle \mathbf{1}_d, x\rangle^{k-1} x_i/\beta$ with $\beta^2 = k^2(2k-3)!!\,d^{k-1}$, its expectation is $k!/\beta$ on S and 0 off S, and answers of every tolerance below $k!/(2\beta)$ suffice to find S. So the one-step algorithm is a statistical-query algorithm with $Q = d$ and, as τ approaches $k!/(2\beta)$,

$$Q\tau^{-2} \to \frac{4k^2(2k-3)!!}{(k!)^2}\, d^k = \frac{4k\,(2k-3)!!}{(k-1)!}\cdot\frac{d^k}{k!},$$

while Theorem 9.7 requires $Q\tau^{-2} \ge \binom{d}{k} - 1 \approx d^k/k!$. The algorithm is within the factor $4k(2k-3)!!/(k-1)!$ of the lower bound, 18 for $k=3$, and its $n \approx d^{k-1}\log d$ samples times d coordinates is the computation $d^k \log d$. This is the precise sense in which gradient methods can learn parities "near the computational limit": the limit is a product of samples and queries, and a single step on the right neuron saturates it.

9.4.1 The plateau and the hidden progress

The one-step algorithm compresses the search into one large average. Training by small steps spreads it over time, and the loss curve acquires a plateau. We study the population flow of the monomial neuron on the unit sphere, where the loss is bounded. Two constants recur:

$$c_k = \max_{\|w\|=1} \Psi(w) = \frac{k!}{k^{k/2}}, \qquad \kappa_k = k\,c_k. \tag{9.3}$$

The maximum is attained when $w_i^2 = 1/k$ on S and $w_i = 0$ off S, by the inequality of arithmetic and geometric means: $\prod_{i\in S} w_i^2 \le (\frac{1}{k}\sum_{i\in S} w_i^2)^k \le k^{-k}$. For $\Psi(w) > 0$ define the *effective overlap*

$$\rho(w) = \big(\Psi(w)/c_k\big)^{1/k} \in (0,1],$$

the overlap with $\mathbf{1}_S/\sqrt{k}$ of the vector with equal weights on S that has the same correlation as w. In these units the normalized loss $(\mathcal{L}(w) - \min\mathcal{L})/(0 - \min\mathcal{L})$ is $1 - \rho^k$, so the loss sees ρ^k while the weights carry ρ. A weight vector is *balanced* if $|w_i|$ is the same for all $i \in S$. The symmetric start $w_0 = d^{-1/2}\mathbf{1}_d$ is balanced, with $\rho_0 = \sqrt{k/d}$; a uniformly random start has $\Psi(w_0) > 0$ with probability $\frac{1}{2}$, and then ρ_0 is of order $d^{-1/2}$.

Theorem 9.12 (Hidden progress of the population flow). *Let $k \ge 3$ and $w_0 \in \mathbb{S}^{d-1}$ with $\Psi(w_0) > 0$, and let $w(t)$ solve the spherical gradient flow of the correlation,*

$$\dot w = \nabla\Psi(w) - \langle w, \nabla\Psi(w)\rangle\, w, \qquad w(0) = w_0. \tag{9.4}$$

(a) *The solution exists for all $t \ge 0$ and stays on the sphere, $\Psi(w(t))$ is nondecreasing, and*

$$\frac{\mathrm{d}}{\mathrm{d}t} w_i^2 = 2\Psi\,(1 - k w_i^2) \quad (i \in S), \qquad \frac{\mathrm{d}}{\mathrm{d}t} w_i^2 = -2k\Psi\, w_i^2 \quad (i \notin S).$$

(b) *The effective overlap satisfies*

$$\dot\rho \ge \kappa_k\, \rho^{k-1}(1 - \rho^2), \tag{9.5}$$

with equality at all times if w_0 is balanced.

(c) *Let t_a be the first time with $\rho \ge a$. If $\rho_0 \le a < b \le \frac{1}{2}$, then*

$$t_b - t_a \le \frac{4\, a^{2-k}}{3(k-2)\kappa_k},$$

and if w_0 is balanced, then also $t_b - t_a \ge (a^{2-k} - b^{2-k})/((k-2)\kappa_k)$.

(d) *On $[0, t_{1/2}]$ the logarithm of the effective overlap grows at least linearly, $\frac{\mathrm{d}}{\mathrm{d}t}\log\rho \ge \frac{3}{4}\kappa_k \rho_0^{k-2}$.*

Proof. The plan is to compute the evolution of each squared weight, to derive from it a closed differential inequality for Ψ, and to integrate that inequality. Write $g = \nabla\Psi$. By Theorem 9.4, $g_i = \Psi/w_i$ for $i \in S$ whenever $\Psi \ne 0$, and $g_i = 0$ for $i \notin S$; and since Ψ is homogeneous of degree k, Euler's identity gives $\langle w, g\rangle = k\Psi$.

(a) Read the right side of (9.4) as a polynomial vector field on $\mathbb{R}^d$. Along its solutions $q = \|w\|^2 - 1$ satisfies $\dot q = 2\langle w, g\rangle(1 - \|w\|^2) = -2k\Psi q$, a linear equation, so q stays 0: the solution stays on the compact sphere and therefore exists for all time. On the sphere, $\dot\Psi = \langle g, \dot w\rangle = \|g\|^2 - \langle w, g\rangle^2 \ge 0$ by the Cauchy–Schwarz inequality, so $\Psi \ge \Psi(w_0) > 0$ for all t, and no weight in

S vanishes. For $i \in S$, $\frac{\mathrm{d}}{\mathrm{d}t} w_i^2 = 2w_i(g_i - k\Psi w_i) = 2\Psi(1 - kw_i^2)$; for $i \notin S$, $\frac{\mathrm{d}}{\mathrm{d}t} w_i^2 = -2k\Psi w_i^2$.

(b) By (a) and the formula for g,

$$\dot{\Psi} = \|g\|^2 - \langle w, g\rangle^2 = \Psi^2 \sum_{i \in S} w_i^{-2} - k^2\Psi^2.$$

The inequality of arithmetic and geometric means bounds the sum below: $\sum_{i\in S} w_i^{-2} \geq k\big(\prod_{i\in S} w_i^{-2}\big)^{1/k} = k\,(k!/\Psi)^{2/k}$, using $\prod_{i\in S} w_i^2 = (\Psi/k!)^2$. Substitute $\Psi = c_k\rho^k$, so that $\dot{\Psi} = kc_k\rho^{k-1}\dot{\rho}$, and use $(k!)^{2/k}c_k^{-2/k} = k$:

$$kc_k\rho^{k-1}\dot{\rho} \geq k\,(k!)^{2/k}(c_k\rho^k)^{2-2/k} - k^2c_k^2\rho^{2k} = k^2c_k^2\rho^{2k-2}(1-\rho^2).$$

Dividing by $kc_k\rho^{k-1} > 0$ gives (9.5). Equality holds in the inequality of means exactly when all w_i^2, $i \in S$, are equal. By (a), $\frac{\mathrm{d}}{\mathrm{d}t}(w_i^2 - w_j^2) = -2k\Psi(w_i^2 - w_j^2)$ for $i, j \in S$, a linear equation, so a balanced start stays balanced and (9.5) is then an equality.

(c) Let $h(s) = \kappa_k s^{k-1}(1-s^2)$, which is positive on $(0,1)$. By (b), ρ is nondecreasing, and it is strictly increasing while $\rho < 1$. Put $H(r) = \int_a^r \mathrm{d}s/h(s)$. Suppose first that t_a is finite, which holds for $a = \rho_0$ since then $t_a = 0$. On $[t_a, t_b]$ the chain rule and (b) give $\frac{\mathrm{d}}{\mathrm{d}t} H(\rho(t)) = \dot{\rho}/h(\rho) \geq 1$, so $H(\rho(t)) \geq t - t_a$ there; since $H(\rho(t)) < H(b)$ for $t < t_b$, the time t_b is finite and $t_b - t_a \leq H(b)$. Applied with $a = \rho_0$, this shows that t_c is finite for every $c < 1$, so the supposition holds for every a. On $[a, b] \subseteq (0, \frac{1}{2}]$ we have $1 - s^2 \geq \frac{3}{4}$, hence

$$H(b) \leq \frac{4}{3\kappa_k}\int_a^b s^{1-k}\,\mathrm{d}s \leq \frac{4\,a^{2-k}}{3(k-2)\kappa_k}.$$

For a balanced start $\frac{\mathrm{d}}{\mathrm{d}t} H(\rho(t)) = 1$, so $t_b - t_a = H(b) \geq \kappa_k^{-1}\int_a^b s^{1-k}\,\mathrm{d}s = (a^{2-k} - b^{2-k})/((k-2)\kappa_k)$, using $1 - s^2 \leq 1$.

(d) By (b), $\frac{\mathrm{d}}{\mathrm{d}t}\log\rho = \dot{\rho}/\rho \geq \kappa_k\rho^{k-2}(1-\rho^2)$, and on $[0, t_{1/2}]$ we have $\rho \geq \rho_0$ and $1 - \rho^2 \geq \frac{3}{4}$. ∎

The theorem is the Boolean version of the search phase of Chapter 7: for a balanced start, ρ obeys exactly the equation $\dot{r} = (1-r^2)\Psi'(r)$ of that chapter with $\Psi(r) = c_k r^k$. It adds two things. An unbalanced start is never slower than a balanced one with the same correlation, and the imbalance between the relevant weights decays; the irrelevant weights decay at the rate $2k\Psi \geq 2k\Psi(w_0)$, so the neuron forgets the coordinates outside S exponentially fast once the correlation is positive. Without the sphere, (a) becomes the statement that $w_i^2 - w_j^2$ is conserved for $i, j \in S$, the balancedness of the deep linear networks of Chapter 2: the monomial neuron on a parity is a depth-k product of scalars (Exercise 9.6).

Corollary 9.13 (Plateau and drop). *Let $k \geq 3$, let $w_0 = d^{-1/2}\mathbf{1}_d$, and let $0 < \varepsilon < 2^{-k}$ with $\varepsilon^{1/k} > \rho_0 = \sqrt{k/d}$. The normalized loss $1 - \rho^k$ stays*

above $1-\varepsilon$ *until the time*

$$t_{\varepsilon^{1/k}} \geq \frac{(d/k)^{(k-2)/2} - \varepsilon^{-(k-2)/k}}{(k-2)\kappa_k},$$

and it reaches $1-2^{-k}$ *at most* $\frac{4}{3(k-2)\kappa_k}\varepsilon^{-(k-2)/k}$ *later, a time that does not depend on* d*. During the plateau* $[0, t_{\varepsilon^{1/k}}]$ *the effective overlap grows by the factor* $\varepsilon^{1/k}\sqrt{d/k}$*.*

Proof. Apply Theorem 9.12(c) twice: with $a = \rho_0$ and $b = \varepsilon^{1/k}$ for the lower bound, using $t_{\rho_0} = 0$ and $\rho_0^{2-k} = (d/k)^{(k-2)/2}$; and with $a = \varepsilon^{1/k}$ and $b = \frac{1}{2}$ for the upper bound. The last statement is the definition of $t_{\varepsilon^{1/k}}$. ■

As $d \to \infty$ with k and ε fixed, the plateau lasts a time of order $d^{(k-2)/2}$ and the drop a time of order one: the loss curve becomes a step. Yet nothing abrupt happens to the weights. The relevant weights are proportional to ρ, and by (d) the logarithm of ρ climbs at a rate bounded below from the first instant; most of the climb in $\log \rho$, a fraction $1 - O(1/\log d)$ of it, happens while the loss is within ε of its initial value. The loss is flat because it sees ρ^k, the kth power of a quantity that grows steadily. This is the mechanism of Schaeffer et al. (2023) in miniature: a nonlinear readout of continuous progress produces a jump.

Online SGD needs about d^{k-1} samples. Replace the population gradient by the gradient of one fresh sample, $G_t = -y_t\, k\langle w_t, x_t\rangle^{k-1} P_{w_t}^{\perp} x_t$ with $P_w^{\perp} = I_d - ww^{\top}$, and renormalize after each step, as in Chapter 7. For a unit vector w, $\langle w, x\rangle$ has bounded moments by Theorem A.16, so the component of G_t along $\mathbf{1}_S/\sqrt{k}$ has variance of order one and $\|G_t\|^2$ has mean of order d. These are the two noise terms of the step-size heuristic of Chapter 7, with the same orders, and the drift is the right side of (9.5). The heuristic of that chapter then applies verbatim: the step size must be of order $d^{-k/2}$ to keep the noise below the drift at $\rho \approx d^{-1/2}$, and the number of samples is the escape time $d^{(k-2)/2}$ divided by the step, of order d^{k-1}. This matches the one-step count of Theorem 9.10 up to the logarithm: spreading the average over many small steps costs no extra samples. ◇

9.5 Reading the solution

The solution has three numbers. The *signal*: the population gradient at a start of unit norm is $k!\, d^{-(k-1)/2}$ on S and zero elsewhere, while each coordinate of a single-sample gradient has standard deviation of order d^0 in the same units; one sample therefore carries a signal-to-noise ratio of order $d^{-(k-1)/2}$ per coordinate. The *sample size*: averaging n samples multiplies the ratio by $\sqrt{n}$, and finding the largest of d coordinates needs a ratio of order $\sqrt{\log d}$, hence $n \asymp d^{k-1}\log d$. The *product*: d queries at this precision cost $d^k \log d$, against the lower bound $\binom{d}{k}$ of Theorem 9.7. Kernel methods, which cannot adapt to S, pay instead with the number of features needed

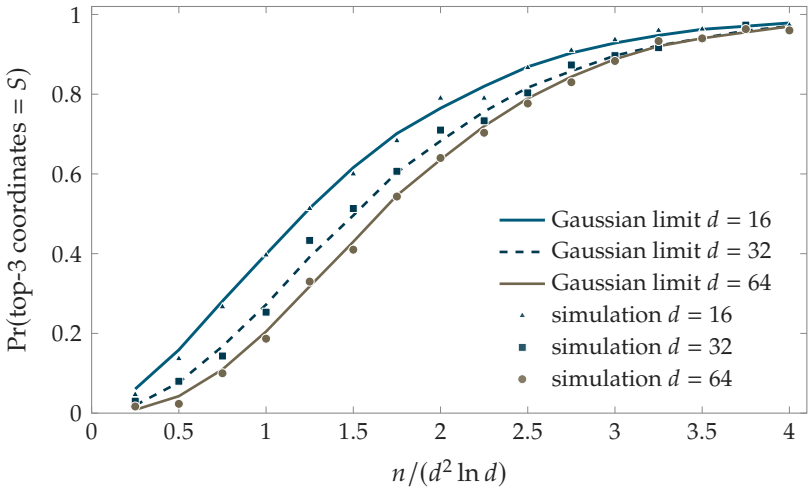


Figure 9.2. One gradient step of the monomial neuron $\langle w, x\rangle^3$ at $w_0 = \mathbf{1}_d$ on the parity $x_1x_2x_3$: the probability that the three largest coordinates of the empirical gradient (9.2) are the relevant ones, against the number of samples in units of $d^2 \ln d$. Points: 300 independent data sets for each (d, n) (seed 20261017). Curves: the central-limit prediction, a Gaussian vector with the exact mean and covariance of $\hat{G}$, estimated with 20,000 draws. Theorem 9.10 guarantees success probability $1 - \delta$ from $n = 7.3\, d^2 \ln(2d/\delta)$.

to represent all degree-k monomials, also of order d^k (Chapter 6); the gain of gradient training over kernels is in memory, not in computation.

Figure 9.2 tests the sample size. The points are simulations of the one-step algorithm for $k = 3$ and $d = 16$, 32, and 64; the curves are the Gaussian approximation with the exact mean and covariance of $\hat{G}$, which uses the exact moments $\mathbb{E}\langle \mathbf{1}_d, x\rangle^4 = 3d^2 - 2d$ and $\mathbb{E}[\langle \mathbf{1}_d, x\rangle^4 x_1x_2] = 12d - 16$. The two agree to within about two Monte Carlo standard errors (one standard error is about 0.03), and the curves for the three dimensions are close when n is measured in units of $d^2 \ln d$, as the signal-to-noise count predicts. That count also gives the location: the largest of $d - k$ roughly independent centered Gaussians with standard deviation $\sigma = (k^2(2k-3)!!\, d^{k-1}/n)^{1/2}$ is about $\sigma\sqrt{2 \ln d}$, and the smallest relevant coordinate is about $k!$, so success becomes likely near $n \approx 2k^2(2k-3)!!\, d^{k-1} \ln d/(k!)^2$, which is $1.5\, d^2 \ln d$ for $k = 3$. The simulated success probability crosses $\frac{1}{2}$ between about 1.2 (for $d = 16$) and 1.7 (for $d = 64$) in these units. The guarantee of Theorem 9.10 at $\delta = 0.05$ and $d = 64$ asks for about $14\, d^2 \ln d$ samples, four times the $3.5\, d^2 \ln d$ at which the simulations succeed about 95% of the time. Most of the factor comes from asking every coordinate to stay within $k!/2$ of its mean, half the gap, which costs a factor 4 in the variance term; the range term of Bernstein's inequality adds about 20% for $k = 3$, and $\ln(2d/\delta)$ exceeds $\ln d$.

Figure 9.3 tests the plateau. On the logarithmic scale of the left panel the effective overlap rises from the first step, along the population curve, while the loss in the right panel stays above 0.99 until ρ has doubled from its start $\rho_0 = \sqrt{3/256} \approx 0.108$; on the flow this takes 61% of the time

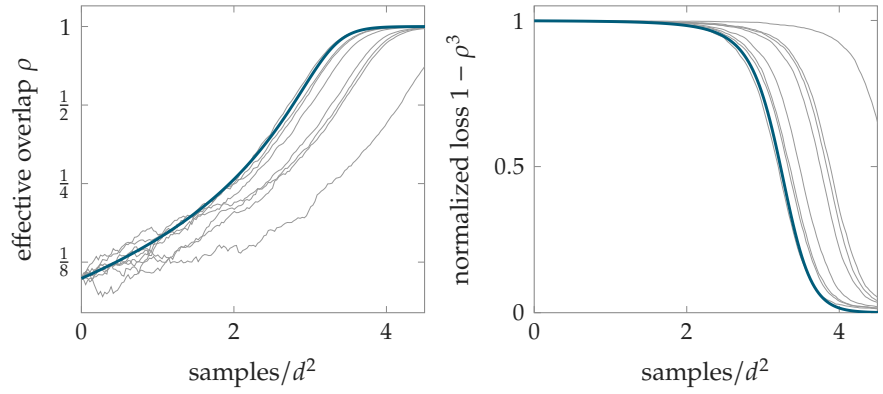


Figure 9.3. Eight runs of spherical online SGD for the monomial neuron $\langle w, x\rangle^3$ on the parity $x_1x_2x_3$ in dimension $d = 256$, from the symmetric start $w_0 = d^{-1/2}\mathbf{1}_d$, with step size $\eta = 0.05\,d^{-3/2}$ and one fresh sample per step (grey; seed 11), and the population flow of Theorem 9.12, $\dot\rho = \eta\kappa_3\rho^2(1-\rho^2)$ per step (curve). Left: the effective overlap $\rho = (\Psi(w)/c_3)^{1/3}$, on a logarithmic scale. Right: the normalized population loss $1-\rho^3$.

to reach $\rho = \frac{1}{2}$, and the fraction tends to 1 as d grows. The prediction $t_{1/2} = \kappa_3^{-1}\big(\rho_0^{-1} - 2 + \operatorname{artanh}\frac{1}{2} - \operatorname{artanh}\rho_0\big)$ (Exercise 9.5) gives $2.77\,d^2$ samples at this step size, with no fitted constant. The SGD runs reach $\rho = \frac{1}{2}$ between 0.99 and 1.51 times the prediction, with median 1.14: the noise delays the escape slightly at this step size, as in Chapter 7, and the delay shrinks at smaller steps (Exercise 9.12). After the escape every run descends in a short, nearly deterministic time.

Placed side by side, Figures 9.1 and 9.3 show what the model explains and what it does not. The plateau, its abrupt end, and the steady growth of a weight-based measure underneath a flat loss appear in both. But in the network the progress measure is flat for the first half of the plateau and grows only in the best few neurons, and the length of the plateau varies by a factor of two between seeds. Both features come from the random initialization, which the symmetric start of the model removes.

9.6 What the model gets wrong

WHAT WAS THROWN AWAY. Uniform inputs, which make every target look like every other and every character orthogonal to every other; a real input distribution is neither uniform nor a product. Exact parity labels, with all Fourier weight on one set of size k and none below; natural targets have weight spread over many degrees. The monomial activation, chosen so that one neuron's correlation is an exact product; a ReLU unit has the finite Fourier gap of Proposition 9.5 and, for odd k without bias, no gap at all. The symmetric initialization, which removes luck; with a random start the progress of Figure 9.1 concentrates in the neurons whose initial weights happen to favor S. Population dynamics or a single large step, which removes the interaction between noise and the slow escape; only a heuristic connects the theorems to online SGD. One neuron, so that width

plays no role. And the correlation loss, which is unbounded below off the sphere.

WHERE IT BREAKS. *Natural tasks have footholds.* A parity is hard because it has no low-degree component, and this is the exception. Abbe et al. (2021) define the staircase property, under which each high-degree Fourier coefficient is reachable from lower-degree ones along a nested chain, and show that such functions are learned in polynomial time by layerwise training of suitable networks, with experiments showing that ResNets trained by SGD learn them too; Abbe et al. (2022) prove that a merged-staircase property is necessary and nearly sufficient for two-layer networks in the mean-field regime to learn sparse functions with a number of samples of order d. For such targets there is no long plateau before every step, and the lower bound of Theorem 9.7 does not apply. *Non-uniform inputs.* Under a product distribution with $\mathbb{E}x_i = \mu \neq 0$, the parity correlates with single coordinates, $\mathbb{E}[\chi_S(x)x_i] = \mu^{k-1}$ for $i \in S$ against μ^{k+1} for $i \notin S$, and a linear statistic finds S from a number of samples that grows only logarithmically in d (Exercise 9.7). Cornacchia and Mossel (2023) turn this into a model of curriculum learning, showing that drawing the training samples from two or more product distributions, rather than from the uniform one alone, substantially lowers the computational cost of learning k-parities by SGD, and Daniely and Malach (2020) use non-uniform input distributions to separate two-layer networks trained by gradient descent from linear methods. *Luck, width, and sparsity.* Edelman et al. (2023) study the trade-offs between data, computation, width, and the luck of the initialization for offline sparse parity learning; they find that wide networks and sparse initialization improve sample efficiency, and they describe width as parallel search for neurons whose initialization is close to S. The symmetric model has no luck in it and cannot express these trade-offs. *Statistical queries are not the end.* Remark 9.8 lists the gap between adversarial and random errors: SGD on specially built networks can emulate Gaussian elimination (Abbe and Sandon 2020). With label noise, on the other hand, parity learning is believed to be hard for every algorithm, and the best general algorithms, such as that of Blum et al. (2003), which runs in polynomial time when the parity depends on the first $O(\log d \log\log d)$ bits, are far from gradient methods. *Architecture.* For transformers, Hahn and Rofin (2024) argue that functions whose output is sensitive to many input bits, parity foremost, occupy isolated regions of parameter space, which biases training toward low-sensitivity functions; and Kim and Suzuki (2025) prove that a one-layer transformer learns parity in a single gradient update when trained with intermediate parities as supervision and teacher forcing, while without intermediate supervision finite-precision gradient methods need many iterations. Chain of thought turns one parity of degree k into $k-1$ parities of degree two; the statistical-query cost of Theorem 9.7 applies to each link, not to the chain.

FINE STRUCTURE. The successors repair one omission at a time. For two-layer networks, Barak et al. (2022) prove that a network of width $2^{\Theta(k)}$, trained by SGD with batch size $d^{O(k)}$, learns k-sparse parities to error

ε in $2^{O(k)}\,\mathrm{poly}(1/\varepsilon)$ iterations, using the Fourier gap of the population gradient. Kou et al. (2024) show that sign SGD on a two-layer network with $2^{\Theta(k)}$ neurons solves k-sparse parity with $\tilde{O}(d^{k-1})$ samples, matching the statistical-query lower bound in the sense of the product computed above. For $k = 2$, the XOR problem, Glasgow (2024) proves that minibatch SGD on a two-layer ReLU network with logistic loss reaches small error with $d\,\mathrm{polylog}(d)$ samples, after a signal-finding phase in which neurons evolve independently, the random-start analogue of our hidden progress. Mean-field Langevin dynamics achieves a sample complexity in which k is decoupled from the exponent of d (Suzuki et al. 2023). For sparse targets that combine monomials of several degrees, the leap complexity of Abbe et al. (2023) predicts a sequence of plateaus, one for each new set of coordinates, with lengths set by the size of the leap; this is the multi-index form of Theorem 9.12 and the Boolean counterpart of Chapter 7. Open are sharp lower bounds for plain SGD on standard networks beyond the statistical-query heuristic, and a description of hidden progress from a random start: which quantity, computed from the weights at a given time, predicts when the drop will come.

9.7 Model card

MODEL CARD	
PHENOMENON	Loss and error flat for a long time, then an abrupt drop, with progress in the weights during the plateau; training time growing like $d^{O(k)}$ for k-sparse parities.
MODEL	$x \sim \mathrm{Unif}\{-1,1\}^d$, $y = \chi_S(x)$ with $\lvert S\rvert = k$; one neuron $\langle w, x\rangle^k$ on the correlation loss; statistical queries, one gradient step from $w_0 \propto \mathbf{1}_d$, and the spherical population flow.
SOLUTION	$\Psi(w) = k!\prod_{i\in S} w_i$, gradient supported on S (proved); every statistical-query learner needs $Q\tau^{-2} \geq \binom{d}{k} - 1$ (proved); one step with $O(d^{k-1}\log(d/\delta))$ samples finds S and $k+1$ ReLUs finish (proved); $\dot\rho \geq \kappa_k \rho^{k-1}(1-\rho^2)$ with equality for balanced starts (proved).
PREDICTS	Sample size $d^{k-1}\log d$ for one step, about $1.5\,d^2 \ln d$ at $k = 3$; computation d^k; plateau of length $d^{(k-2)/2}$ and drop of length $O(1)$; steady growth of $\log\rho$ under a flat loss.
GETS WRONG	No luck: random starts concentrate progress in few neurons; no low-degree structure, which natural tasks have; uniform inputs; statistical queries can be beaten by SGD on special networks; architecture-specific effects for transformers.
SUCCESSORS	Staircase and leap complexity; biased inputs and curricula; width, sparsity, and luck; sign SGD matching the SQ bound; chain of thought.

This chapter is the Boolean partner of Chapter 7. There a single direction hides behind a link whose low Hermite coefficients vanish, here k

coordinates hide behind a character whose low-degree Fourier coefficients vanish; the information exponent becomes the degree k, the Hermite basis becomes the characters, and the escape equations coincide for balanced starts. Both chapters belong to the thread of escapes from small initialization that starts with the logistic escape of Chapter 2: without the sphere, the monomial neuron is a depth-k product of scalars whose relevant weights stay balanced, and the plateau of length $d^{(k-2)/2}$ is the higher-order saddle that also produces the plateaus of the committee machine (Chapter 8) and the sudden formation of induction heads (Chapter 14).

The statistical-query argument is Fourier analysis on the group $\{-1,1\}^d$; Chapter 10 uses Fourier analysis on $\mathbb{Z}_p$ to describe what a network computes after an equally abrupt transition, and its progress measures play the role of $\log \rho$ here. Against Chapter 6, the chapter gives a second separation between feature learning and kernels: both pay about d^k in computation, but a gradient method stores d weights where a kernel method stores d^k features.

Exercises

9.1 CONSOLIDATION. (Sums of signs.) (a) Compute $\mathbb{E}s^4 = 3d^2 - 2d$ for $s = x_1 + \cdots + x_d$, and check Lemma 9.9 for $j = 2$. (b) Show that $\mathbb{E}[s^4 x_1 x_2] = 12d - 16$, and conclude that two coordinates of the empirical gradient (9.2) for $k = 3$ have correlation of order $1/d$.

9.2 CONSOLIDATION. (Majority.) (a) Verify $\mathrm{Maj}_3(x) = \frac{1}{2}(x_1 + x_2 + x_3) - \frac{1}{2}x_1x_2x_3$. (b) For $d = 5$ and $k = 2$, compute the partial derivatives of Proposition 9.5 directly from the 32 points of the cube and check the ratio $(k-1)/(d-k) = \frac{1}{3}$. (c) For odd $k \geq 3$, show that the weights of the symmetric unit $\mathrm{ReLU}(\alpha\langle \mathbf{1}_d, x\rangle + b)$ with $b = 0$ receive zero population gradient, while its bias receives $\frac{1}{2}\widehat{\mathrm{Maj}}_d(S) \neq 0$.

9.3 CONSOLIDATION. (Fewer units.) Show that k ReLU units with input weights $\mathbf{1}_S$ and an output bias suffice to compute χ_S. *(Hint: interpolate the values $(-1)^{k-i}$ at $s = -k + 2i$ by a zigzag with breakpoints at the interior nodes, and write the affine part with the unit* $\mathrm{ReLU}(s + k)$*.)*

9.4 CONSOLIDATION. (Weak learners.) A weak learner outputs a function $h\colon \{-1,1\}^d \to [-1,1]$ and succeeds if $\mathbb{E}[h(x)\chi_S(x)] \geq \gamma$. Adapt the proof of Theorem 9.7 to show that for uniformly random S the success probability is at most $(Q\tau^{-2} + \gamma^{-2})/\binom{d}{k}$.

9.5 CONSOLIDATION. (The escape time for $k = 3$.) For a balanced start with $\rho_0 < \frac{1}{2}$, solve (9.5) with equality and show $t_{1/2} = \kappa_3^{-1}\big(\rho_0^{-1} - 2 + \operatorname{artanh}\frac{1}{2} - \operatorname{artanh}\rho_0\big)$. Evaluate it for $d = 256$ and $\rho_0 = \sqrt{3/d}$, and check the value $2.77\,d^2$ samples quoted in Section 9.5 for $\eta = 0.05\,d^{-3/2}$.

9.6 VARIATION. (Without the sphere.) Let $\dot{w} = \nabla\Psi(w)$ on $\mathbb{R}^d$. (a) Show that $w_i^2 - w_j^2$ is conserved for $i, j \in S$ and that w_i is constant for $i \notin S$. (b) For $w_i(0) = u_0 > 0$ on S and $k \geq 3$, show that $u(t) =$

$\big(u_0^{2-k} - (k-2)k!\,t\big)^{-1/(k-2)}$, which blows up at $t^\star = u_0^{2-k}/((k-2)k!)$, and that the correlation $k!u^k$ stays below ε until a time $t^\star - O(\varepsilon^{-(k-2)/k})$. (c) For $k = 2$ show exponential growth, and compare with the logistic escape of Chapter 2.

9.7 VARIATION. (Biased inputs.) Let the coordinates of $x \in \{-1, 1\}^d$ be independent with $\mathbb{E}x_i = \mu \in (0, 1)$. (a) Show $\mathbb{E}[\chi_S(x)x_i] = \mu^{k-1}$ for $i \in S$ and μ^{k+1} for $i \notin S$. (b) Using Hoeffding's inequality (Theorem A.17), show that the k largest coordinates of $\frac{1}{n}\sum_s y_s x_s$ are S with probability at least $1 - \delta$ once $n \geq 8\log(2d/\delta)/(\mu^{2k-2}(1-\mu^2)^2)$. Why does Theorem 9.7 not apply?

9.8 VARIATION. (Label noise.) Flip each label independently with probability $p < \frac{1}{2}$. Show that every correlation in this chapter is multiplied by $1 - 2p$, that the sample size of Theorem 9.10 grows by the factor $(1-2p)^{-2}$ up to the change in the range term, and that the statistical-query bound keeps its form with τ replaced by $\tau/(1-2p)$.

9.9 VARIATION. (Gaussian inputs.) Let $x \sim \mathcal{N}(0, I_d)$ and $y = \prod_{i\in S} x_i$. Use Isserlis' theorem to show that $\mathbb{E}[y\langle w, x\rangle^k] = k!\prod_{i\in S} w_i$ and that the gradient vanishes off S, as on the cube, and that $\mathbb{E}\langle \mathbf{1}_d, x\rangle^{2k-2} = (2k-3)!!\, d^{k-1}$ exactly. Which conclusions of the chapter survive, and how does the target relate to the multi-index models of Chapter 7?

9.10 VARIATION. ★ (A plain step.) Let $w_1 = \hat{G}$ be the empirical gradient (9.2) itself, used without thresholding, and let $f(x) = \langle w_1, x\rangle/k!$. Show that $\mathbb{E}[(f(x) - s_S(x))^2]$, where $s_S = \sum_{i\in S} x_i$, is of order d^k/n, so that a readout of f by a few ReLU units needs about d^k samples, a factor $d/\log d$ more than Theorem 9.10. *(Hint: write $\hat{G} = k!\,\mathbf{1}_S + \xi$; for a fresh input x, independent of the data, $\mathbb{E}\langle \xi, x\rangle^2 = \mathbb{E}\|\xi\|^2$, and each $\mathbb{E}\xi_i^2$ is the variance bound in the proof of Theorem 9.10 divided by n.)*

9.11 SIMULATION. (One step for $k = 4$.) Repeat the simulation of Figure 9.2 for $k = 4$ and $d = 12, 16, 24$, with $n = a\,d^3 \ln d$ and 100 data sets per point. You should observe a success probability near $\frac{1}{2}$ for a between 0.5 and 0.8, larger for larger d,, and above 0.9 at $a = 1.6$, close to the prediction $a \approx 2k^2(2k-3)!!/(k!)^2 = 0.83$ of Section 9.5.

9.12 SIMULATION. (Step size and random starts.) (a) Rerun the simulation of Figure 9.3 with $\eta = c\,d^{-3/2}$ for $c = 0.02$ and $c = 0.1$. You should observe a median escape time within about 15% of the population prediction for $c = 0.02$, and a wider spread for $c = 0.1$, from about 0.7 to 1.7 times the prediction, with the median delayed by about 20%; relate this to the noise at the start, as in Chapter 7. (b) Integrate the population flow (9.4) for $d = 50$, $k = 3$ from several uniformly random starts with $\Psi(w_0) > 0$. You should find that $\rho(t)$ always stays above the balanced solution with the same ρ_0, as Theorem 9.12(b) asserts, and that the time to $\rho = \frac{1}{2}$ is shorter than the balanced one, typically by a factor between 0.5 and 0.95, and by more for starts with very small ρ_0.

9.13 RESEARCH. (Lower bounds for plain SGD.) Prove that online SGD on a two-layer ReLU network of polynomial width with Gaussian initialization needs at least $d^{k-1-o(1)}$ samples to learn a k-sparse parity, for every step size. Theorem 9.7 gives this only for adversarial gradient errors, and Abbe and Sandon (2020) show that random errors can be exploited by specially constructed networks; whether standard architectures can exploit them is open as far as we know.

9.14 RESEARCH. (A progress measure from a random start.) In the setting of Figure 9.1, find a quantity computed from the weights at time t that predicts the step of the drop with an error that shrinks as d grows, from early in the plateau. The largest share of a neuron on S moves only late; candidates include products of the relevant weights of each neuron, the analogue of Ψ, and the population Fourier gap of each neuron. No such predictor with guarantees is known to us.

Notes and References

Parities entered the theory of neural networks with Minsky and Papert (1969), whose limitations of bounded-order perceptrons included the parity of all inputs. The statistical-query model was introduced by Kearns in 1993; the journal version (Kearns 1998) shows that statistical-query algorithms tolerate random classification noise and that parities cannot be learned from statistical queries in polynomial time. Blum et al. (1994) characterized weak statistical-query learning through Fourier analysis by the number of nearly orthogonal functions in a class, and Blum et al. (2003) gave a noise-tolerant algorithm for parities on $O(\log d \log\log d)$ bits that no statistical-query algorithm matches, separating the two models. The Fourier analysis used here is in O'Donnell (2014), including the Fourier coefficients of majority.

Shalev-Shwartz et al. (2017) put parities at the head of their list of failures of gradient-based learning, and Abbe and Sandon (2020) delineated what SGD can and cannot do beyond statistical queries. Daniely and Malach (2020) proved that gradient descent on two-layer networks learns sparse parities under distributions where linear methods fail. The phenomenon of this chapter, abrupt learning with hidden progress near the statistical-query limit, is the subject of Barak et al. (2022); its follow-up on data, compute, width, and luck is Edelman et al. (2023), and the matching upper bound for sign SGD is Kou et al. (2024). The monomial neuron, the one-step theorem, and the comparison argument for the spherical flow are presented here in the form that makes every constant explicit; the doubling-epoch analysis of online SGD that would turn the heuristic of Section 9.4.1 into a theorem is the one carried out for single-index models in Chapter 7, following Ben Arous et al. (2021).[7]

For structure that rescues gradient methods see the staircase papers (Abbe et al. 2021, 2022), the leap complexity (Abbe et al. 2023), and curricula with biased inputs (Cornacchia and Mossel 2023). For the debate on emergence see Wei et al. (2022) and Schaeffer et al. (2023); for transformers, Hahn and Rofin (2024) and Kim and Suzuki (2025).

[7] The equality case of Theorem 9.12(b) is exactly the population equation of Chapter 7 with $\Psi(r) = c_k r^k$, so the proof of the online SGD theorem there transfers to balanced starts once the noise bounds are redone on the cube.

TRANSLATING THE LITERATURE. Barak et al. (2022) write n for our dimension d, keep k for the sparsity and S for the support, and call the task the (n, k)-sparse parity; their labels are in $\{\pm 1\}$, as ours. Kearns (1998) pairs a query with its tolerance τ, as we do, but his original formulation takes labels and query values in $\{0, 1\}$, and later work mostly uses queries with values in $[-1, 1]$; the decomposition $q = q_0 + yq_1$ in the proof of Theorem 9.7 is the $\{\pm 1\}$ form of the split into a label-independent and a correlational part. O'Donnell (2014) writes n for our d, χ_S and $\hat{f}(S)$ as we do, and Maj_n for our Maj_d. Kou et al. (2024) write d for the dimension and k for the sparsity, as we do. Our effective overlap ρ is the overlap r of Chapter 7 for balanced weights, and our c_k is the coefficient of r^k in that chapter's Ψ.

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Part IV

What Networks Represent

Trained networks build structures that can be read off their weights: Fourier features for modular arithmetic, overcomplete codes for sparse features, a regular simplex for classes. Each structure is the optimum of a small problem that can be solved by hand.

10 Grokking

Modular addition and its Fourier circuit

THE MODEL — MODULAR ADDITION, QUADRATIC TWO-LAYER NETWORK

IN THIS CHAPTER. Networks trained on part of the addition table of $\mathbb{Z}_p$ fit their training data early and predict the rest of the table much later: this is grokking. In a two-layer network with quadratic activation the generalizing solution can be written down: each neuron carries one Fourier frequency, and trigonometric identities turn the neurons into clocks $\cos(\omega(a+b-c))$ that peak at the correct answer. We prove that margin maximization forces this circuit (single-frequency neurons whose readout phase is the sum of their input phases, every frequency present, at least six neurons per frequency), and that on a diagonal linear network weight decay acts exactly as an initialization scale shrinking like $e^{-\lambda t}$, which predicts a delay $\lambda^{-1}\log(\alpha/\alpha_c)$ that simulations confirm. The model is silent on transformers that use few frequencies or another algorithm, on grokking without weight decay, and on networks far from the margin limit.

10.1 The phenomenon: memorize now, generalize later

In 2021 a group at OpenAI trained small transformers on tables of binary operations: addition, subtraction and division modulo 97, polynomials such as x^2+xy+y^2, and composition of permutations in S_5. Each network saw a random part of a table and had to fill in the rest (Power et al. 2022).[1] The networks were two-layer transformers of width 128 with about 4×10^5 parameters, trained with AdamW at learning rate 10^{-3} and weight decay 1. On division modulo 97 with half of the table for training, training accuracy was nearly perfect after about 10^3 steps, while validation accuracy stayed near chance and approached 100% only after close to 10^6 steps. Power and coauthors called this *grokking*. The delay grew steeply as the training set shrank: at 25 to 30% of the table, removing 1% of the data raised the median time to generalization by 40 to 50%, and some operations never generalized. Of the interventions they tried, weight decay improved data efficiency the most.

[1] The work was presented at an ICLR 2021 workshop on mathematical reasoning and posted to arXiv in January 2022. The verb *grok*, to understand something completely, comes from Robert Heinlein's novel *Stranger in a Strange Land* (1961).

OBSERVATION 10.1 (Power et al. 2022; Nanda et al. 2023). A small network trained with AdamW and weight decay on a random fraction of the addition table of $\mathbb{Z}_p$ (between about 25% and 50%, for p near 100) reaches 100% training accuracy early, stays near chance on the rest of the table for one to three orders of magnitude longer, and then reaches about 100% test accuracy within a comparatively short interval. The delay grows steeply as the training fraction decreases, and weight decay shortens it.

What does the network compute once it generalizes? Nanda et al. (2023) reverse-engineered a one-layer transformer trained on 30% of the addition table modulo 113. Its embedding of a number a is concentrated on five "key frequencies" $\omega_k = 2\pi k/113$, $k \in \{14, 35, 41, 42, 52\}$, as $\cos(\omega_k a)$ and $\sin(\omega_k a)$. Attention and the MLP combine these into $\cos(\omega_k(a+b))$ and $\sin(\omega_k(a+b))$ by the addition formulas, and the unembedding produces logits close to $\sum_k \cos(\omega_k(a+b-c))$, largest at $c = a+b$. Progress measures built from these Fourier components showed three phases: memorization (epochs 0 to 1.4 thousand); circuit formation (to 9.4 thousand), while the test loss stays flat; and cleanup (to 14 thousand), when weight decay removes the memorizing components and the test loss drops. The jump in test accuracy ends a gradual process, the kind of hidden progress met in Chapter 9.

Grokking on group operations is reproducible across transformers and multilayer perceptrons. Liu et al. (2022) mapped phase diagrams over learning rate and weight decay with four phases (comprehension, grokking, memorization, confusion). Liu et al. (2023) induced grokking on MNIST, with 1000 training images and an enlarged initialization, and argued that weight decay λ shrinks the weight norm like $e^{-\lambda t}$, so that generalization waits until the norm reaches a critical value, after a time of about $\lambda^{-1}\log(w_0/w_c)$; on modular addition, constraining the weight norm to a smaller value largely removed the delay. Delayed generalization has also been reported for Gaussian processes and linear regression (Miller et al. 2024), and a delayed onset of adversarial robustness for convolutional networks on CIFAR-10 (Humayun et al. 2024). Grokking is thus a property of a training configuration (initialization scale, weight decay, data fraction) as much as of a task, and on group operations the generalizing networks compute with characters of the group.

Two questions follow. *Which* computation does a generalizing network perform, and why that one among the many functions that fit the training data? And *why so late*: what sets the delay between fitting the training set and generalizing?

10.2 The model: a quadratic network on the addition table

In this chapter $p \geq 3$ is an odd prime, the modulus; this overrides, locally, the global use of p for a number of features. The elements of $\mathbb{Z}_p = \{0, 1, \ldots, p-1\}$ are written a, b, c, and arithmetic on them is modulo p.[2]

[2] The proofs use only that p is odd. Primality matters elsewhere: for composite p, a frequency that shares a factor with p cannot separate all the classes on its own.

THE TASK. The data are the p^2 correct triples $(a, b, a+b)$. A training set is a random set of pairs (a, b), labeled by $a+b$; the test error is the error on the remaining pairs. The input is one-hot, $x = (e_a, e_b) \in \mathbb{R}^{2p}$, and says nothing about the arithmetic.

THE NETWORK. A two-layer network with m hidden neurons, quadratic activation, and p outputs, the logits. The first-layer weight vector $w_r \in \mathbb{R}^{2p}$ of neuron r splits into two functions $u_r, u'_r \colon \mathbb{Z}_p \to \mathbb{R}$, so that $\langle w_r, x\rangle = u_r(a) + u'_r(b)$, and its readout is a third function $v_r \colon \mathbb{Z}_p \to \mathbb{R}$. The logit of

class c is

$$f_c(a,b;\theta) = \sum_{r=1}^{m} v_r(c)\big(u_r(a) + u'_r(b)\big)^2. \tag{10.1}$$

The parameters are $\theta = (\theta_1, \dots, \theta_m)$ with $\theta_r = (u_r, u'_r, v_r) \in (\mathbb{R}^p)^3$; we write $\|u\|^2 = \sum_a u(a)^2$ and $\|\theta_r\|^2 = \|u_r\|^2 + \|u'_r\|^2 + \|v_r\|^2$. The network predicts the class with the largest logit. Training minimizes the cross-entropy over the training set plus the weight decay $\frac{\lambda}{2}\|\theta\|^2$, by gradient flow in the analysis and by AdamW in the experiments.

TWO STRUCTURAL FACTS. The network is *homogeneous of degree three* neuron by neuron: replacing θ_r by $t\theta_r$ with $t > 0$ multiplies the contribution of neuron r to every logit by t^3. Homogeneity makes margins the right currency, as in Chapter 4. And the task is invariant under translations: if (a, b, c) is a correct triple, then so is $(a + g, b + h, c + g + h)$ for all $g, h \in \mathbb{Z}_p$. The characters $a \mapsto e^{2\pi i k a/p}$ of $\mathbb{Z}_p$ diagonalize this symmetry, so we expect Fourier analysis.

WHAT WAS THROWN AWAY. Attention, ReLU, normalization and Adam. One-hot inputs with free first-layer weights are a learned embedding, and the quadratic activation is the simplest nonlinearity that multiplies, $(x+y)^2 = x^2 + 2xy + y^2$; it makes every logit a polynomial in the weights, so the Fourier analysis is exact. The margin analysis uses the full table. For the delay we use the diagonal linear network of Chapter 3 with weight decay, in which the passage from the lazy to the rich regime can be computed.

DICTIONARY

IN THE MODEL	IN A TRAINED NETWORK
one-hot (e_a, e_b); first-layer functions u_r, u'_r	token embeddings of a and b, read by an MLP neuron
quadratic activation; the cross term $2u_r(a)u'_r(b)$	the multiplication done by attention and by the MLP nonlinearity
frequency ζ and phases φ, ψ of a Fourier neuron	a key frequency, and the cosine and sine directions of the embedding (Nanda et al. 2023)
readout $v_r(c) \propto \cos(\omega_\zeta c + \varphi + \psi)$	rows of the unembedding; logits $\propto \sum_\zeta \cos(\omega_\zeta(a + b - c))$
normalized margin, norm $\|\theta\|_{2,3}$	the long-time limit of cross-entropy training with small weight decay
initialization scale α; effective scale $\alpha e^{-\lambda t}$	parameter norm, decaying under weight decay after memorization
critical scale α_c	the norm below which training leaves the lazy regime

WHAT ONE HAD TO NOTICE. That generalizing on part of a group's table means finding the group structure, carried for a cyclic group by characters; that one multiplication suffices, since $\cos(\omega(a + b))$ is a sum of products of a function of a and a function of b; and that training selects among the networks that fit by a norm, so "which circuit?" becomes an extremal

problem that the same characters diagonalize.

10.3 The solution: a Fourier circuit for addition

FOURIER ANALYSIS ON $\mathbb{Z}_p$. For $u\colon \mathbb{Z}_p \to \mathbb{R}$ let $\hat{u}(k) = \sum_{a\in\mathbb{Z}_p} u(a)e^{-2\pi i ka/p}$ for $k \in \mathbb{Z}_p$. In this normalization inversion, Plancherel's identity, and the symmetry of real functions read

$$u(a) = \frac{1}{p}\sum_{k\in\mathbb{Z}_p} \hat{u}(k)e^{2\pi i ka/p}, \qquad \sum_a u(a)^2 = \frac{1}{p}\sum_k |\hat{u}(k)|^2, \qquad \hat{u}(-k) = \overline{\hat{u}(k)}$$

(Theorem A.14). All three follow from the orthogonality relation

$$\sum_{a\in\mathbb{Z}_p} e^{2\pi i qa/p} = p\,\mathbf{1}\{q \equiv 0 \pmod p\} \qquad \text{for every integer } q. \tag{10.2}$$

Because p is odd, the nonzero frequencies pair up as $\pm\zeta$ with $\zeta \in \{1, \dots, \frac{p-1}{2}\}$; we write $\omega_\zeta = 2\pi\zeta/p$. A real function whose transform vanishes outside $\pm\zeta$ is $a \mapsto \rho\cos(\omega_\zeta a + \varphi)$ with $\rho = \frac{2}{p}|\hat{u}(\zeta)|$ and $\varphi = \arg\hat{u}(\zeta)$.

Lemma 10.2 (Sum of cosines). *For every integer q,* $\displaystyle\sum_{\zeta=1}^{(p-1)/2} \cos\frac{2\pi\zeta q}{p} = \frac{p\,\mathbf{1}\{q\equiv 0\} - 1}{2}$.

PROOF. In (10.2) the term $a = 0$ equals 1, and the terms $a = \zeta$ and $a = p - \zeta$ are complex conjugates with real part $\cos(2\pi\zeta q/p)$. Hence (10.2) reads $1 + 2\sum_{\zeta=1}^{(p-1)/2}\cos(2\pi\zeta q/p) = p\,\mathbf{1}\{q \equiv 0\}$. ∎

With $q = a + b - c$ the lemma says that the logits

$$f_c(a,b) = \sum_{\zeta=1}^{(p-1)/2} \cos\big(\omega_\zeta(a+b-c)\big) = \frac{p\,\mathbf{1}\{c = a+b\} - 1}{2} \tag{10.3}$$

are $\frac{p-1}{2}$ on the correct class and $-\frac{1}{2}$ on every other class. Each term is a *clock*: $\cos(\omega_\zeta(a+b-c))$ is largest when the angle $\omega_\zeta c$ equals $\omega_\zeta a + \omega_\zeta b$, so it compares the sum of two rotations with a third. One clock already peaks only at the correct class, because p is prime; the sum of all of them makes every wrong class tie. How does a sum of squares of first-layer activations produce one clock?

ONE FREQUENCY AT A TIME. Fix a frequency ζ and write $\omega = \omega_\zeta$. Call the neuron with $u(a) = \cos(\omega a + \varphi)$, $u'(b) = \cos(\omega b + \psi)$ and $v(c) = \cos(\omega c + \varphi + \psi)$ the *Fourier neuron* of frequency ζ and phases (φ, ψ), and write

$$n_{\varphi,\psi}(a,b,c) = \big(\cos(\omega a + \varphi) + \cos(\omega b + \psi)\big)^2 \cos(\omega c + \varphi + \psi)$$

for its contribution to the logits. Its readout phase is the sum of its input phases; Theorem 10.9 will explain why. We use the readout phase $\chi = \varphi + \psi$

and the difference $\delta = \varphi - \psi$ as coordinates; the function $n_{\varphi,\psi}$ depends on (φ, ψ) only through χ and δ modulo 2π.

Lemma 10.3 (Cancellation within one frequency). *Let $\beta_1, \dots, \beta_K$ be real, let (φ_r, ψ_r) be phases, and let $\chi_r = \varphi_r + \psi_r$ and $\delta_r = \varphi_r - \psi_r$. Then*

$$\sum_{r=1}^{K} \beta_r \, n_{\varphi_r,\psi_r}(a,b,c) = \frac{1}{2}\Big(\sum_{r=1}^{K} \beta_r\Big) \cos\big(\omega(a+b-c)\big) \qquad \text{for all } a,b,c \in \mathbb{Z}_p \tag{10.4}$$

if and only if, for all integers j and s with $|j| \le 2$, $|s| \le 1$ and $(j,s) \neq (0,0)$,

$$M_{j,s} := \sum_{r=1}^{K} \beta_r e^{\mathrm{i}(j\chi_r + s\delta_r)} = 0. \tag{10.5}$$

Proof. The plan is to expand one neuron into cosines of linear forms in (a,b,c) and to read off the coefficient of each. Put $A = \omega a + \varphi$, $B = \omega b + \psi$ and $C = \omega c + \chi$. Using $\cos^2 x = \frac{1}{2}(1 + \cos 2x)$ and $4\cos A \cos B \cos C = \cos(A+B-C) + \cos(A+B+C) + \cos(A-B+C) + \cos(A-B-C)$,

$$\begin{aligned} n_{\varphi,\psi} = {} & \tfrac{1}{2}\cos(A+B-C) + \cos C + \tfrac{1}{4}\big[\cos(2A \pm C) + \cos(2B \pm C)\big] \\ & + \tfrac{1}{2}\big[\cos(A+B+C) + \cos(A-B \pm C)\big], \end{aligned}$$

each $\pm$ standing for the sum of the two terms. Because $\chi = \varphi + \psi$, the first term is $\frac{1}{2}\cos(\omega(a+b-c))$ for every neuron. Every other term is $\cos(\omega\ell + j\chi + s\delta)$ with a linear form ℓ in (a,b,c) and integers (j,s): C has $(\ell; j, s) = (c; 1, 0)$; $2A + C$ has $(2a + c; 2, 1)$; $2A - C$ has $(2a - c; 0, 1)$; $2B + C$ has $(2b + c; 2, -1)$; $2B - C$ has $(2b - c; 0, -1)$; $A + B + C$ has $(a + b + c; 2, 0)$; $A - B + C$ has $(a - b + c; 1, 1)$; and $A - B - C$ has $(a - b - c; -1, 1)$. Up to sign, these pairs (j,s) are the seven nonzero pairs with $|j| \le 2$ and $|s| \le 1$. Weighting by β_r and summing over r, the term with $(\ell; j, s)$ contributes its coefficient times $\mathrm{Re}\big(e^{\mathrm{i}\omega\ell} M_{j,s}\big)$, and $M_{-j,-s} = \overline{M_{j,s}}$ because the weights are real. If (10.5) holds, then every such contribution vanishes, and (10.4) follows.

Conversely, the eighteen functions $e^{\pm\mathrm{i}\omega(a+b-c)}$ and $e^{\pm\mathrm{i}\omega\ell}$ are characters of $\mathbb{Z}_p^3$ with distinct frequency vectors: two of these vectors differ in some coordinate by 1, 2 or 4, which multiplied by ζ is not divisible by p. Characters with distinct frequency vectors are orthogonal by (10.2). So if (10.4) holds, then the coefficient of each $e^{\mathrm{i}\omega\ell}$ on its left side vanishes, and that coefficient is a nonzero multiple of the corresponding $M_{j,s}$. ∎

In the language of numerical integration, (10.5) says that the points (χ_r, δ_r) with weights β_r integrate every trigonometric polynomial of degree at most 2 in χ and at most 1 in δ exactly, as the uniform measure of total mass $\sum_r \beta_r$ does. Such weighted points form a *cubature rule* on the torus, and every cubature rule of this kind gives a circuit.

Theorem 10.4 (A Fourier network for modular addition). *For each frequency $\zeta \in \{1, \dots, \frac{p-1}{2}\}$ and each pair of phases $(\varphi, \psi) \in \{0, \frac{2\pi}{3}, \frac{4\pi}{3}\}^2$,*

take the neuron with $u(a) = \cos(\omega_\zeta a + \varphi)$, $u'(b) = \cos(\omega_\zeta b + \psi)$ and $v(c) = \frac{2}{9}\cos(\omega_\zeta c + \varphi + \psi)$. The resulting network of width $\frac{9}{2}(p-1)$ has logits

$$f_c(a,b) = \frac{p\,\mathbf{1}\{c = a+b\} - 1}{2} \qquad \text{for all } a,b,c \in \mathbb{Z}_p.$$

In particular it classifies the whole table correctly, and every margin $f_{a+b}(a,b) - f_c(a,b)$ with $c \neq a+b$ equals $\frac{p}{2}$.

PROOF. Fix ζ. Its nine neurons are the Fourier neurons $n_{\varphi,\psi}$ with weights $\beta = \frac{2}{9}$. Writing $j\chi + s\delta = (j+s)\varphi + (j-s)\psi$, the moments factorize over the grid,

$$M_{j,s} = \frac{2}{9}\Big(\sum_{\varphi} e^{\mathrm{i}(j+s)\varphi}\Big)\Big(\sum_{\psi} e^{\mathrm{i}(j-s)\psi}\Big),$$

and by (10.2) with modulus 3 the first factor vanishes unless 3 divides $j+s$, the second unless 3 divides $j-s$. If 3 divides both, then it divides $2j$ and $2s$, hence j and s, and $|j| \le 2$, $|s| \le 1$ force $j = s = 0$. So (10.5) holds, and Lemma 10.3 shows that the nine neurons of frequency ζ contribute $\frac{1}{2}\cdot 9 \cdot \frac{2}{9}\cos(\omega_\zeta(a+b-c)) = \cos(\omega_\zeta(a+b-c))$. Summing over ζ gives (10.3). ∎

Three phases per input turn every unwanted term into a sum of three unit vectors spaced by $2\pi/3$. We checked the identity numerically for every odd $p \le 23$.

REMARK 10.5 (Other designs). A pair of Fourier neurons with phases (φ,ψ) and $(\varphi,\psi+\pi)$ turns squares into a product, $n_{\varphi,\psi} + n_{\varphi,\psi+\pi} = \big[(x+y)^2 - (x-y)^2\big]z = 4xyz$, with $x = \cos(\omega a + \varphi)$, $y = \cos(\omega b + \psi)$ and $z = \cos(\omega c + \varphi + \psi)$. Morwani et al. (2024) use four such pairs, one for each product in $\cos(A+B-C) = \cos A\cos B\cos C - \sin A\sin B\cos C + \sin A\cos B\sin C + \cos A\sin B\sin C$: eight neurons per frequency. Three pairs suffice. For $\theta \in \{0, \frac{\pi}{3}, \frac{2\pi}{3}\}$ take $(\varphi,\psi) = (\theta,\theta)$ and $(\theta,\theta+\pi)$; expanding $4xyz$, the three unwanted terms carry the phases 4θ, 2θ and -2θ, whose exponentials sum to zero over θ, so

$$\begin{aligned}\frac{1}{3}\sum_{\theta}\big(n_{\theta,\theta} + n_{\theta,\theta+\pi}\big) &= \frac{4}{3}\sum_{\theta}\cos(\omega a+\theta)\cos(\omega b+\theta)\cos(\omega c+2\theta)\\ &= \cos\big(\omega(a+b-c)\big).\end{aligned}$$

These six neurons per frequency give width $3(p-1)$; small blocks of this kind are the building material of Tian (2025), who composes global minimizers of a squared loss from them. Gromov (2023) uses independent uniformly random phases with $\chi = \varphi + \psi$ and N neurons per frequency; with readout weight $2/N$ in our normalization, each $M_{j,s}$ is then of size about $2/\sqrt{N}$, and the unwanted terms cancel only as the width grows.

The six-neuron identity writes $\cos(A+B-C) = \mathrm{Re}(e^{\mathrm{i}A}e^{\mathrm{i}B}e^{-\mathrm{i}C})$ with three products, as few as multiplying two complex numbers needs.[3] Can

[3] Gauss's trick multiplies $x_1 + \mathrm{i}x_2$ by $y_1 + \mathrm{i}y_2$ with the three real products x_1y_1, x_2y_2 and $(x_1+x_2)(y_1+y_2)$; over the reals three is the minimum.

a frequency do with fewer than six Fourier neurons? Not with positive weights, the case that margins select.

Lemma 10.6 (Six neurons per frequency). *If $\beta_1, \dots, \beta_K > 0$ and the moment conditions (10.5) hold, then the points (χ_r, δ_r), taken modulo 2π, include at least six distinct points. In particular $K \geq 6$.*

PROOF. Neurons with the same point (χ_r, δ_r) modulo 2π contribute the same summand to every moment, so merging them and adding their weights changes neither the hypothesis nor the positivity; assume that the K points are distinct. Consider the six-dimensional space of trigonometric polynomials

$$P(\chi, \delta) = \sum_{|j|\leq 1} \sum_{s\in\{0,1\}} c_{js} e^{\mathrm{i}(j\chi + s\delta)}.$$

If $K < 6$, then the K linear conditions $P(\chi_r, \delta_r) = 0$ on six unknowns have a nonzero solution (c_{js}). Expanding $|P|^2$, every product of two of its exponentials has frequency $(j - j', s - s')$ with $|j - j'| \leq 2$ and $|s - s'| \leq 1$, so the moment conditions kill every term with $(j, s) \neq (j', s')$:

$$0 = \sum_{r=1}^{K} \beta_r |P(\chi_r, \delta_r)|^2 = \sum_{j,s,j',s'} c_{js}\overline{c_{j's'}}\, M_{j-j',s-s'} = M_{0,0} \sum_{j,s} |c_{js}|^2.$$

But $M_{0,0} = \sum_r \beta_r > 0$ and the coefficients are not all zero, a contradiction. ∎

This is the classical lower bound for cubature rules with positive weights: a rule exact on all products $P\bar{Q}$ from a space of dimension N needs N points.[4] Without positivity the bound fails: four neurons of one frequency that are not Fourier neurons (a numerical search finds them) produce the clock up to a term depending on c alone, and Tian (2025) composes exact solutions from such non-Fourier blocks. Margins exclude them.

[4] The six-point rule of Remark 10.5 attains the bound; such a tight rule has equal weights (Exercise 10.2).

10.4 The solution: margins select the circuit

Why would training find this circuit? Morwani et al. (2024) proposed that margin maximization alone determines it. Recall from Chapter 4 that for a homogeneous network trained by cross-entropy on data it can classify, the regularized minimizers converge in direction, as the weight decay tends to zero, to a network of maximal normalized margin (Wei et al. 2019), and gradient flow without regularization converges in direction to a Karush–Kuhn–Tucker point of the margin problem (Lyu and Li 2020).[5] Weight decay penalizes $\|\theta\|^2 = \sum_r \|\theta_r\|^2$. Morwani and coauthors normalize instead by

$$\|\theta\|_{2,3} = \Big(\sum_{r=1}^{m} \|\theta_r\|^3\Big)^{1/3},$$

the norm matched to neurons homogeneous of degree three, with which the margin problem decouples into one problem per neuron. With the Euclidean norm it does not (Exercise 10.7), and we know of no complete solution then.

[5] The first statement is about global minimizers, for the norm that is penalized; the second gives a stationary point of the margin problem, not necessarily a global maximizer.

Definition 10.7 (Margins). The *margin* of θ at (a,b) against $c \neq a+b$ is $f_{a+b}(a,b;\theta) - f_c(a,b;\theta)$. The margin $\gamma(\theta)$ of the network is the smallest of these $p^2(p-1)$ numbers, and its *average margin* $\bar{\gamma}(\theta)$ is their average. The *normalized margin* of $\theta \neq 0$ is $\gamma(\theta)/\|\theta\|_{2,3}^3$, which is invariant under $\theta \mapsto t\theta$ for $t > 0$. A network of width m is a *max-margin network* if it maximizes the normalized margin among networks of width m.

The maximum exists, because γ is continuous on the compact set $\{\|\theta\|_{2,3} = 1\}$. The average margin is the class-weighted margin of Morwani and coauthors, the margin averaged over a uniform pair (a,b) and a uniform wrong class. It is linear in the neurons: $\bar{\gamma}(\theta) = \sum_r \bar{\gamma}(\theta_r)$, the sum of the average margins of the one-neuron networks θ_r.

Lemma 10.8 (Average margin of one neuron). *For one neuron (u,u',v), with $\bar{v} = \frac{1}{p}\sum_c v(c)$ and (a,b) uniform on $\mathbb{Z}_p^2$,*

$$\begin{aligned}\bar{\gamma}(u,u',v) &= \frac{2p}{p-1}\,\mathbb{E}_{a,b}\big[u(a)u'(b)\big(v(a+b)-\bar{v}\big)\big] \\ &= \frac{2}{(p-1)p^2}\sum_{k\neq 0}\hat{u}(k)\,\hat{u}'(k)\,\overline{\hat{v}(k)}. \qquad (10.6)\end{aligned}$$

Proof. For fixed (a,b) the average of $v(c)$ over the $p-1$ wrong classes is $(p\bar{v}-v(a+b))/(p-1)$, so the average margin at (a,b) is $\frac{p}{p-1}\big(u(a)+u'(b)\big)^2\big(v(a+b)-\bar{v}\big)$. Expand the square. The term $u(a)^2$ averages to zero, because for fixed a the class $a+b$ is uniform when b is, so that $\mathbb{E}_b[v(a+b)] = \bar{v}$; the term $u'(b)^2$ vanishes in the same way. The cross term $2u(a)u'(b)$ remains, which proves the first equality. For the second, insert $v(c) = \frac{1}{p}\sum_k \hat{v}(k)e^{2\pi ikc/p}$ and sum over a and b separately:

$$\sum_{a,b} u(a)u'(b)v(a+b) = \frac{1}{p}\sum_k \hat{v}(k)\,\hat{u}(-k)\,\hat{u}'(-k).$$

The term $k=0$ equals $\hat{u}(0)\hat{u}'(0)\hat{v}(0)/p = \sum_{a,b} u(a)u'(b)\bar{v}$, which the subtraction of $\bar{v}$ removes. Divide by p^2, multiply by $\frac{2p}{p-1}$, replace k by $-k$, and use $\hat{v}(-k) = \overline{\hat{v}(k)}$. ∎

The squares cancel because the wrong classes are averaged uniformly: the terms $u(a)^2v(c)$ and $u'(b)^2v(c)$, which no single neuron can avoid, carry no information about $a+b$ and contribute equally to the correct logit and to the average wrong logit. What remains is a cubic form that the Fourier transform diagonalizes.

Theorem 10.9 (The best single neuron). *Every neuron satisfies $\bar{\gamma}(\theta_r) \leq \gamma^\star\|\theta_r\|^3$, with*

$$\gamma^\star = \sqrt{\frac{2}{27}}\,\frac{1}{\sqrt{p}\,(p-1)}.$$

A neuron $\theta_r \neq 0$ attains equality if and only if it is a positive multiple of

a Fourier neuron: $u(a) = \rho\cos(\omega_\zeta a + \varphi)$, $u'(b) = \rho\cos(\omega_\zeta b + \psi)$ *and* $v(c) = \rho\cos(\omega_\zeta c + \varphi + \psi)$ *for some* $\rho > 0$, *frequency* $\zeta \in \{1, \dots, \frac{p-1}{2}\}$, *and phases* φ, ψ.

PROOF. The plan is to pair conjugate frequencies, to bound each pair by the arithmetic–geometric mean inequality, and then to use that $s \mapsto s^{3/2}$ grows faster than linearly to put all the weight on one pair. Both sides are homogeneous of degree three, so assume $\|\theta_r\| = 1$.

Pairing. Because u, u' and v are real, the terms k and $-k$ of (10.6) are complex conjugates, and $k \neq -k$ because p is odd. With $x_\zeta = |\hat{u}(\zeta)|$, $y_\zeta = |\hat{u}'(\zeta)|$ and $z_\zeta = |\hat{v}(\zeta)|$,

$$\bar{\gamma}(\theta_r) = \frac{4}{(p-1)p^2} \sum_{\zeta=1}^{(p-1)/2} \operatorname{Re}\Big(\hat{u}(\zeta)\hat{u}'(\zeta)\overline{\hat{v}(\zeta)}\Big) \le \frac{4}{(p-1)p^2} \sum_{\zeta=1}^{(p-1)/2} x_\zeta y_\zeta z_\zeta.$$

The constraint. Plancherel's identity, with the terms $\pm\zeta$ grouped and the term $k = 0$ dropped, gives $1 = \|u\|^2 + \|u'\|^2 + \|v\|^2 \ge \frac{2}{p}\sum_\zeta s_\zeta$ with $s_\zeta = x_\zeta^2 + y_\zeta^2 + z_\zeta^2$; equality requires $\hat{u}(0) = \hat{u}'(0) = \hat{v}(0) = 0$.

One frequency. By the arithmetic–geometric mean inequality $x_\zeta y_\zeta z_\zeta \le (s_\zeta/3)^{3/2}$, with equality if and only if $x_\zeta = y_\zeta = z_\zeta$. And $\sum_\zeta s_\zeta^{3/2} \le (\max_\zeta s_\zeta)^{1/2}\sum_\zeta s_\zeta \le \big(\sum_\zeta s_\zeta\big)^{3/2}$, with equality if and only if at most one s_ζ is nonzero. With $\sum_\zeta s_\zeta \le p/2$ these give

$$\bar{\gamma}(\theta_r) \le \frac{4}{(p-1)p^2}\Big(\frac{p}{6}\Big)^{3/2} = \sqrt{\frac{2}{27}}\,\frac{1}{\sqrt{p}\,(p-1)}.$$

Equality. Equality holds throughout if and only if the transforms vanish at 0 and at every frequency except one pair $\pm\zeta$, the three moduli at ζ are equal (to $\sqrt{p/6}$), and $\hat{u}(\zeta)\hat{u}'(\zeta)\overline{\hat{v}(\zeta)}$ is a positive real, that is, $\arg\hat{v}(\zeta) = \arg\hat{u}(\zeta) + \arg\hat{u}'(\zeta)$. By the remark before Lemma 10.2, these are exactly the functions $\rho\cos(\omega_\zeta a + \varphi)$, $\rho\cos(\omega_\zeta b + \psi)$ and $\rho\cos(\omega_\zeta c + \varphi + \psi)$ with $\rho = \frac{2}{p}\sqrt{p/6} = \sqrt{2/(3p)}$. Conversely, such a neuron makes every inequality of the chain an equality. ∎

The best neuron is *single-frequency*, because a cubic objective under a quadratic constraint gains more from concentration than it loses; and its phases *add*, so that the term $\frac{1}{2}\cos(\omega(a+b)+\chi)$ of the cross term meets the readout $\cos(\omega c + \chi)$ and leaves the clock $\frac{1}{4}\cos(\omega(a+b-c))$. One neuron cannot be the whole network: its logits peak at the correct class only for some pairs. The next theorem assembles optimal neurons into a network whose margins are all equal.

Theorem 10.10 (Max-margin networks; Morwani et al. 2024, with sharp width).

(i) Every network of width m *satisfies* $\gamma(\theta) \le \gamma^\star \|\theta\|_{2,3}^3$.

(ii) A network $\theta \neq 0$ attains equality if and only if its logits are

$$f_c(a,b;\theta) = \gamma^\star \|\theta\|_{2,3}^3 \Big(\mathbf{1}\{c = a+b\} - \frac{1}{p}\Big) \qquad \text{for all } a,b,c.$$

Then every nonzero neuron is a positive multiple of a Fourier neuron, and each frequency $\zeta \in \{1,\dots,\frac{p-1}{2}\}$ is carried by at least six neurons.

(iii) Equality is attained at width m if and only if $m \geq 3(p-1)$.

Proof. The plan is to compare the minimum margin with the average margin, which decomposes over neurons, and then to read the equality cases in Fourier space; part (iii) rests on the cubature lemmas of Section 10.3.

The bound. A minimum is at most an average, so by Theorem 10.9,

$$\gamma(\theta) \leq \bar\gamma(\theta) = \sum_{r=1}^m \bar\gamma(\theta_r) \leq \gamma^\star \sum_{r=1}^m \|\theta_r\|^3 = \gamma^\star \|\theta\|_{2,3}^3. \tag{10.7}$$

Equality. If the logits have the stated form, then every margin equals $\gamma^\star\|\theta\|_{2,3}^3$, and equality holds. Conversely, suppose that equality holds, and write $\gamma' = \gamma(\theta)$. The first inequality in (10.7) is then an equality, so all margins equal γ': for every (a,b) the wrong logits share a common value $g(a,b)$, and $f_c(a,b) = g(a,b) + \gamma'\mathbf{1}\{c = a+b\}$. The second inequality is an equality term by term, so by Theorem 10.9 every nonzero neuron is a positive multiple of a Fourier neuron. A Fourier readout has mean zero over c, so $\sum_c f_c(a,b) = 0$, which gives $g = -\gamma'/p$.

Every frequency, six times. Group the neurons by frequency. The neurons of frequency ζ are ρ_r times Fourier neurons with phases (φ_r, ψ_r), and they contribute $B_\zeta = \sum_r \rho_r^3 n_{\varphi_r,\psi_r}$, whose Fourier transform in c, for fixed (a,b), lives on $\pm\zeta$. By Lemma 10.2 the logits are $\gamma'(\mathbf{1}\{c = a+b\} - \frac{1}{p}) = \frac{2\gamma'}{p}\sum_\zeta \cos(\omega_\zeta(a+b-c))$, and the transform in c of this sum at $\pm\zeta$ is that of its ζ term alone. Here $\gamma' = \gamma^\star\|\theta\|_{2,3}^3 > 0$. Comparing transforms in c, frequency by frequency,

$$B_\zeta(a,b,c) = \frac{2\gamma'}{p}\cos\big(\omega_\zeta(a+b-c)\big) \qquad \text{for every } \zeta.$$

The right side is not zero, so some neuron carries frequency ζ. Expand B_ζ as in the proof of Lemma 10.3, with the positive weights $\beta_r = \rho_r^3$: the character $e^{\mathrm{i}\omega_\zeta(a+b-c)}$ occurs only in the clock term, so comparing its coefficients gives $\frac12\sum_r \beta_r = \frac{2\gamma'}{p}$, and B_ζ has exactly the form (10.4). By Lemma 10.3 the moment conditions hold; and by Lemma 10.6 at least six neurons carry ζ.

Width. Six neurons for each of the $\frac{p-1}{2}$ frequencies make $m \geq 3(p-1)$ necessary. For $m \geq 3(p-1)$, use the six-neuron blocks of Remark 10.5 at every frequency and set the other neurons to zero. Multiplying u and u' by t and v by t^{-2} does not change a neuron's function, so the neuron $(\cos(\omega a+\varphi), \cos(\omega b+\psi), \frac13\cos(\omega c+\chi))$ may be replaced by the Fourier neuron $3^{-1/3}(\cos(\omega a+\varphi), \cos(\omega b+\psi), \cos(\omega c+\chi))$. The logits are (10.3), so all margins are equal and the first inequality in (10.7) is an equality; every neuron is a Fourier neuron, so the second is one as well. ∎

The max-margin networks thus form a large family, one positive cubature rule per frequency, but they all compute the clock sum (10.3) up to scale and share three visible features: single-frequency neurons, additive phases, and all $\frac{p-1}{2}$ frequencies present. Morwani et al. (2024) prove parts (i) and (ii) for width $m \geq 4(p-1)$ with their eight-neuron blocks; the six-neuron blocks and Lemma 10.6 make the threshold exact. They also treat sparse parities and finite groups, where irreducible representations replace characters.

10.5 The solution: why the circuit comes late

The max-margin theorem describes the end of training. Why is the end reached so long after the training set is fit? Two groups gave the same answer in 2023 (Kumar et al. 2024; Lyu et al. 2024): training starts in the lazy regime of Chapter 6, where the network fits the training set with its initial tangent kernel and memorizes, and reaches the rich regime only later.

Theorem 10.11 (Lyu et al. 2024; stated without proof). *Let $f(\theta; x)$ be L-homogeneous and twice continuously differentiable in θ, let $f(\bar{\theta}; x) = 0$ for every input x, and run gradient flow on $\widehat{\mathcal{L}}(\theta) + \frac{\lambda}{2}\|\theta\|^2$ from $\theta(0) = \alpha\bar{\theta}$, with $\lambda = \lambda(\alpha)$ of order α^{-q} for a constant $q > 0$. Fix a constant $c \in (0, 1)$ and let $\alpha \to \infty$.*

(i) At time $(1-c)\lambda^{-1}\log\alpha$ the predictor converges pointwise, after normalization for classification, to the kernel predictor of the tangent features $\nabla_\theta f(\bar{\theta}; x)$: the maximum-$\ell_2$-margin classifier if these features separate the data, or the minimum-norm interpolant for regression with linearly independent features.

(ii) At suitable times between $\lambda^{-1}\log\alpha$ and $(1+c)\lambda^{-1}\log\alpha$, every limit point of $\theta/\|\theta\|$ (classification) or of θ (regression) is a Karush–Kuhn–Tucker point of $\min \frac{1}{2}\|\theta\|^2$ subject to $y_i f(\theta; x_i) \geq 1$, respectively $f(\theta; x_i) = y_i$, for all i.

For diagonal linear networks these are the maximum-ℓ_2-margin and maximum-ℓ_1-margin classifiers, and the second needs far fewer samples for a sparse target: that gap is grokking. Kumar et al. (2024) observe grokking without weight decay, in a two-layer network learning the square of a linear function of Gaussian inputs by gradient descent, controlled by an output scale that sets the laziness. For modular addition, Mohamadi et al. (2024) prove that permutation-equivariant kernel methods, a model of the lazy phase, need a constant fraction of the table to generalize.

The diagonal network of Chapter 3 with weight decay makes the two phases explicit. Let $X \in \mathbb{R}^{n\times d}$ with $n < d$, let $y = Xw_\star$ with a k-sparse target $w_\star$, let $w = u \odot u - v \odot v$ with $u(0) = v(0) = \alpha\mathbf{1}_d$, and run gradient flow on

$$\widehat{\mathcal{L}}(w) + \frac{\lambda}{2}\left(\|u\|^2 + \|v\|^2\right), \qquad \widehat{\mathcal{L}}(w) = \frac{1}{2n}\|Xw - y\|^2.$$

Recall from Chapter 3 the potential $\Phi_\alpha(w) = \frac{\alpha^2}{4}\sum_i q(w_i/\alpha^2)$ with $q(z) = 2 - \sqrt{4+z^2} + z\,\operatorname{arcsinh}(z/2)$, whose gradient is $\nabla\Phi_\alpha(w)_i = \frac{1}{4}\operatorname{arcsinh}(w_i/2\alpha^2)$.

Without weight decay the flow converges to $w_\alpha^\Phi = \arg\min\{\Phi_\alpha(w) : Xw = y\}$, which is close to the minimum-ℓ_2-norm interpolator when α is large and to the minimum-ℓ_1-norm interpolator when α is small.

Proposition 10.12 (Weight decay shrinks the initialization). *Let $\alpha_t = \alpha e^{-\lambda t}$ and $g(t) = \nabla\widehat{\mathcal{L}}(w(t))$. For every $t \geq 0$ and every coordinate i:*

(i) $u_i(t)v_i(t) = \alpha_t^2$;

(ii) $\frac{\mathrm{d}}{\mathrm{d}t}\nabla\Phi_{\alpha_t}(w(t)) = -g(t)$;

(iii) $w(t) = \arg\min\{\Phi_{\alpha_t}(w') : Xw' = Xw(t)\}$.

Proof. The flow is $\dot{u} = -2u \odot g - \lambda u$ and $\dot{v} = 2v \odot g - \lambda v$. It exists for all time, because the objective decreases along it and is at least $\frac{\lambda}{2}(\|u\|^2 + \|v\|^2)$, so the trajectory stays bounded.

Part (i). The product rule gives $\frac{\mathrm{d}}{\mathrm{d}t}(u_iv_i) = (-2g_i - \lambda)u_iv_i + (2g_i - \lambda)u_iv_i = -2\lambda u_iv_i$, and $u_i(0)v_i(0) = \alpha^2$.

Part (ii). Let $\sigma_i = u_i^2 + v_i^2$. By part (i), $\sigma_i^2 - w_i^2 = 4u_i^2v_i^2 = 4\alpha_t^4$, so $\sigma_i = \sqrt{w_i^2 + 4\alpha_t^4}$, and the flow gives $\dot{w}_i = 2u_i\dot{u}_i - 2v_i\dot{v}_i = -4\sigma_ig_i - 2\lambda w_i$. Let $z_i = w_i/(2\alpha_t^2)$. Since $\frac{\mathrm{d}}{\mathrm{d}t}\alpha_t^2 = -2\lambda\alpha_t^2$, the quotient rule gives $\dot{z}_i = (\dot{w}_i + 2\lambda w_i)/(2\alpha_t^2) = -2\sigma_ig_i/\alpha_t^2$, while $\sqrt{1 + z_i^2} = \sigma_i/(2\alpha_t^2)$. Hence

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{1}{4}\operatorname{arcsinh} z_i = \frac{\dot{z}_i}{4\sqrt{1 + z_i^2}} = -g_i.$$

Part (iii). At $t = 0$, $\nabla\Phi_\alpha(w(0)) = 0$, so by part (ii) $\nabla\Phi_{\alpha_t}(w(t)) = -\int_0^t g = -\frac{1}{n}X^\top\int_0^t(Xw - y)$ lies in the row space of X. The potential Φ_{α_t} is strictly convex, because $q'' > 0$, so a point w' minimizes it on the affine set $\{Xw' = Xw(t)\}$ if and only if $\nabla\Phi_{\alpha_t}(w')$ lies in the row space; $w(t)$ is such a point, and the minimizer is unique. ∎

On a diagonal network, then, weight decay acts exactly as an initialization scale shrinking like $e^{-\lambda t}$: at every moment the network sits where the flow of Chapter 3 from scale α_t would have put it, among predictors with its own training outputs. Nothing here is asymptotic; the rest needs one approximation.

Heuristic derivation of the grokking time. Assume that the training residual stays negligible after an initial transient, so that $Xw(t) \approx y$. Then Proposition 10.12 gives $w(t) \approx w_{\alpha_t}^\Phi$, and the test error at time t is a fixed function of $\log\alpha - \lambda t$, read off from the static family $a \mapsto w_a^\Phi$.

Memorization. At first $\sigma_i \approx 2\alpha^2$, so $\dot{w} \approx -8\alpha^2\nabla\widehat{\mathcal{L}}(w)$, and the residual $Xw - y$ decays at rate at least $8\alpha^2\lambda_{\min}(XX^\top/n)$. The training set is fit by the time $t_{\mathrm{mem}} \approx \log(1/\varepsilon)/\big(8\alpha^2\lambda_{\min}(XX^\top/n)\big)$, while $w \approx w_\alpha^\Phi$ is close to the minimum-norm interpolator, whose test error for a sparse target is near the null risk.

Generalization. The static family passes from ℓ_2-like to ℓ_1-like behavior when a crosses a critical scale α_c, so the test error drops when $\alpha e^{-\lambda t} = \alpha_c$, that is, at

$$t_{\text{grok}} \approx \frac{1}{\lambda} \log \frac{\alpha}{\alpha_c}. \tag{10.8}$$

The critical scale. For $|w_i| \ll a^2$ the potential is quadratic, $\Phi_a(w) \approx \sum_i w_i^2/(16a^2)$; for $|w_i| \gg a^2$ it is $\Phi_a(w) \approx \frac{1}{4}\sum_i |w_i|(\log(|w_i|/a^2) - 1)$. The minimum-norm interpolator $w_2 = X^+ y$ is spread over all coordinates, with $\|w_2\|^2 \approx \frac{n}{d}\|w_\star\|^2$; the target is concentrated on k of them; and the minimizer switches from one to the other when their potentials are comparable. For a target with entries ± 1 this gives, to logarithmic accuracy, $\|w_2\|^2/(16a^2) \approx \frac{1}{4}\|w_\star\|_1 \log(1/a^2)$, that is,

$$\alpha_c^2 \log \frac{1}{\alpha_c^2} \approx \frac{\|w_2\|^2}{4\|w_\star\|_1} \approx \frac{n}{4d}. \quad \diamond$$

The ratio $t_{\text{grok}}/t_{\text{mem}}$ grows like $\alpha^2 \log(\alpha/\alpha_c)/\lambda$, so large initialization and small weight decay make grokking dramatic. The same formula, with the weight norm in place of α, is the estimate of Liu et al. (2023); here α_c has a definition, and (10.8) refines the $\log \alpha$ of Theorem 10.11 to $\log(\alpha/\alpha_c)$, which matters at moderate α. The estimate of α_c is the crudest step: on random instances it is within a factor of about 1.5 of the scale at which the static test error halves, an error that enters t_{grok} through a logarithm.

10.6 Reading the solution

The model makes four quantitative predictions. (i) Every max-margin network has the logits $\gamma(\mathbf{1}\{c = a + b\} - \frac{1}{p})$, in which the wrong classes tie; as vectors in $\mathbb{R}^p$, the logit vectors of the p possible sums are the vertices of a regular simplex centered at the origin, the geometry of neural collapse (Chapter 12). (ii) Each of its neurons has a single frequency, equal norms $\|u\| = \|u'\| = \|v\|$, and phase defect $\chi - \varphi - \psi = 0$, where φ, ψ, χ are the phases of u, u', v at their common frequency. (iii) All $\frac{p-1}{2}$ frequencies appear, and the maximal normalized margin is $\gamma^\star = \sqrt{2/27}/(\sqrt{p}(p-1))$, for instance 6.1×10^{-4} for $p = 59$. (iv) In the diagonal caricature the training set is fit at $t_{\text{mem}} \propto \alpha^{-2}$, the test error drops at $t_{\text{grok}} \approx \lambda^{-1} \log(\alpha/\alpha_c)$, and the test error depends on α and t only through $\log \alpha - \lambda t$.

Figure 10.1 tests the delay. The left panel is grokking in miniature: the training error falls near $t = 10^{-3}$, the test error near $t = 4 \times 10^2$, and in between the test error sits at 0.72 for five decades. After the fast transient the simulation is indistinguishable from the quasi-static curve, computed from the static interpolators alone. The right panel shows the predicted collapse: $\lambda t_{1/2}$ depends on α only through $\log(\alpha/\alpha_c)$. The small offset grows with λ, from 0.005 to 0.05; it is the training residual, of order λ, that the quasi-static assumption neglects. The heuristic equation for α_c gives 0.131 here, and $t_{\text{grok}} = 434$.

Figure 10.2 tests the features. The network groks, with a delay factor of about 34, and ends with the neurons that Theorem 10.9 prescribes: the large

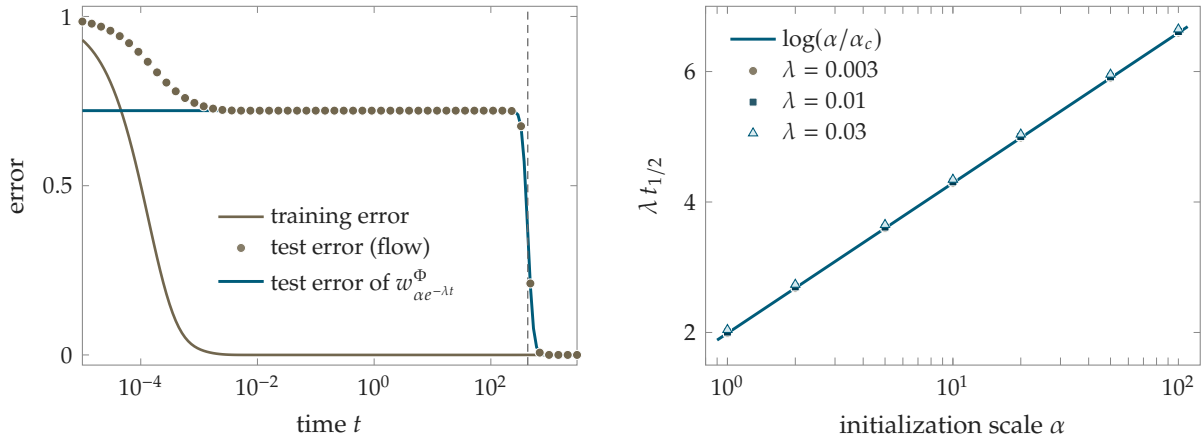


Figure 10.1. Grokking in a diagonal linear network with weight decay ($d = 100$, $n = 30$, a 3-sparse target with entries 1, Gaussian design, seed 0); the gradient flow of (u, v) is integrated with a stiff solver. Left: $\alpha = 10$, $\lambda = 0.01$. The training error $\|Xw - y\|^2/\|y\|^2$ falls below 10^{-4} at $t \approx 0.005$; the relative test error $\|w - w_\star\|^2/\|w_\star\|^2$ stays at 0.72, its value for the minimum-norm interpolator, until $t \approx 300$ and falls below 0.01 by $t \approx 690$. The curve is the quasi-static prediction of Proposition 10.12, the test error of $\arg\min\{\Phi_{\alpha_t}(w) : Xw = y\}$; the dashed line is $\lambda^{-1}\log(\alpha/\alpha_c) = 429$, where $\alpha_c = 0.137$ is the scale at which that static test error halves. Right: the time $t_{1/2}$ at which the simulated test error halves, for three weight decays and seven initialization scales; all 21 points lie within 0.06 of the line $\log(\alpha/\alpha_c)$.

ones have single frequencies, and their readout phases equal the sums of their input phases to within a few hundredths of a radian, although nothing in training refers to phases. All 29 frequencies are present, as Theorem 10.10 predicts and as Morwani et al. (2024) observed. The Fourier concentration rises long before the test accuracy, a progress measure in the sense of Nanda et al. (2023).

The margin predictions fail, informatively. At step 10^4 the smallest margin is 0.23 times the average margin, where a max-margin network has all margins equal; the normalized margin is $0.15\,\gamma^\star$; and the clocks carry 59% of the variance of the logits (centered over c), against 100% at max margin, so the unwanted terms of Lemma 10.3 are far from cancelled. Classification needs only the largest logit to be right. The features of the max-margin solution are in place long before its margin is.

The delay roughly follows $t_{\text{grok}} \propto \lambda^{-1}$: with weight decay 0.5, 1 and 2 the test accuracy reaches 99% after about 14000, 6000 and 4000 steps. The law fails at both ends (Exercise 10.9): with weight decay 0.3 the test accuracy is still 98% after 40000 steps, and with weight decay 4 it stalls near 94% from step 20000 to 40000. Too strong a weight decay stops the circuit short of completion, an effect that the diagonal model, in which weight decay only shrinks the effective scale, cannot see.

10.7 What the model gets wrong

WHAT WAS THROWN AWAY. *The task* is an abelian group, whose characters are known in advance and diagonalize everything; natural data have no such basis. *The architecture* is a two-layer network with a polynomial

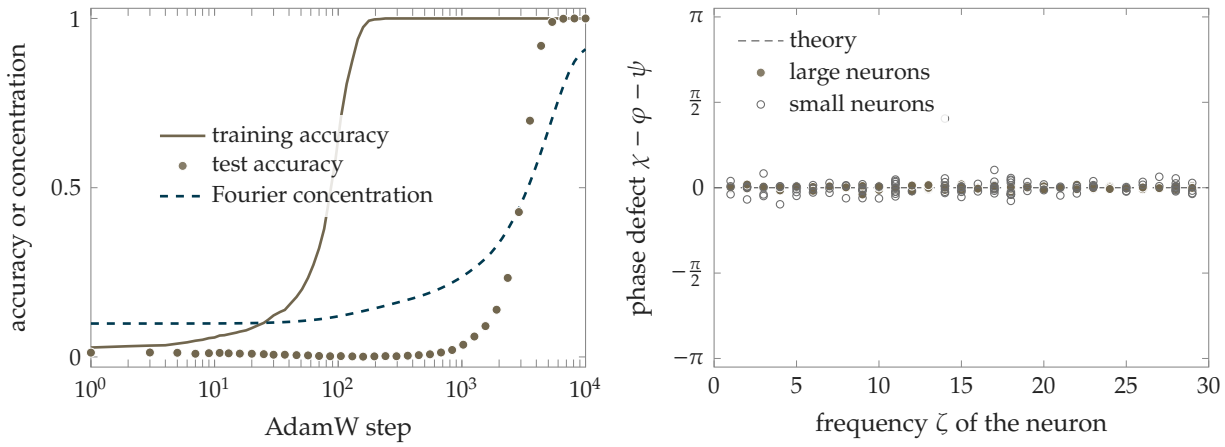


Figure 10.2. The quadratic network (10.1) trained on modular addition: $p = 59$, width $m = 256$, 30% of the table for training, full-batch AdamW (learning rate 10^{-3}, $\beta = (0.9, 0.98)$, weight decay 1), cross-entropy, seed 0. Left: training accuracy reaches 99% at step 177, test accuracy at step 5960. The Fourier concentration is the share of a neuron's first-layer energy at its dominant frequency $\pm\zeta$, averaged with weights $\|\theta_r\|^3$; it rises steadily from 0.10 before the test accuracy moves. Right: after 10^4 steps, the phase defect of every neuron at its dominant frequency. The 139 large neurons (norm above 0.3 times the largest) have median concentration 0.91 and defects below 0.13 in absolute value (median 0.015), and they cover all 29 frequencies.

activation, which makes the Fourier analysis exact; there is no attention, no factorization into embedding and unembedding, no normalization, and no ReLU. *The margin theory* describes the end of training on the full table, in the limit of vanishing weight decay and with the $\ell_{2,3}$ norm instead of the ℓ_2 norm that weight decay penalizes: it explains which features a generalizing network uses, not why training on part of the table generalizes, which is the phenomenon. *The delay model* is a reparameterized linear model whose only "feature" is a choice of coordinates; it has no circuit to form.

WHERE IT BREAKS. *Different networks find different algorithms.* Zhong et al. (2023) found that transformers with weak attention learn a "pizza" algorithm, which averages the embeddings of a and b instead of multiplying them, so that all pairs with the same sum lie on a line through the origin; its logits carry an extra factor $|\cos(\omega(a-b)/2)|$ and are not clocks. Which algorithm appears depends on the architecture, and margin maximization within one architecture does not predict it.[6] *Transformers use few frequencies.* The transformer of Nanda et al. (2023) uses five key frequencies, and their other seeds three or four, while every max-margin network of Theorem 10.10 at $p = 113$ uses all 56; our MLP agrees with the theory, the transformers do not. *Grokking happens without weight decay.* Kumar et al. (2024) observe it with plain gradient descent, controlled by laziness; Thilak et al. (2024) with Adam and no regularization, at the onset of cyclic instabilities in which the norm of the last layer jumps; and Prieto et al. (2025) show that without regularization the logits grow until floating-point absorption saturates the softmax and the gradient vanishes exactly, halting learning, while the gradient aligns with a direction that scales the logits without changing

[6] Zhong and coauthors vary an attention rate between constant attention, under which the pizza appears, and standard attention, under which the clock appears.

predictions. *Grokking happens without circuits.* Levi et al. (2024) show that a linear estimator on Gaussian data groks in thresholded accuracy, with a delay computed by random-matrix methods of the kind used in Chapter 1, while the losses evolve smoothly; and Varma et al. (2023) (unrefereed) explain grokking by the relative efficiency of memorizing and generalizing circuits, which predicts a critical dataset size and "ungrokking" below it, an axis that the initialization scale does not capture. *Non-abelian groups.* For finite groups, irreducible representations replace characters, and Chughtai et al. (2023) found networks composing representation matrices, with the representations learned varying from seed to seed; but Stander et al. (2024), reverse-engineering networks trained on S_5 and S_6, found neurons that detect membership in cosets of subgroups and little evidence that the embeddings store representation matrices.

FINE STRUCTURE. *Representation theory* replaces Fourier analysis for finite groups: Morwani et al. (2024) characterize max-margin networks for general group operations, and Chughtai et al. (2023) describe the algorithm by matrix products. *Algebraic structure*: Tian (2025) decomposes the squared loss of quadratic networks on abelian groups frequency by frequency and composes global minimizers from small blocks, and reports that about 95% of the solutions found by gradient descent match these constructions. *Phase transitions*: Rubin et al. (2024) treat grokking in two-layer teacher–student models, with a cubic teacher and with modular addition, as a first-order transition in an adaptive-kernel theory of the Bayesian posterior; Žunkovič and Ilievski (2024) compute exact critical exponents in teacher–student models with local rules; and Beck et al. (2025) place grokking in logistic regression at the edge of linear separability. *Fourier features elsewhere*: recursive feature machines, kernel machines with learned feature metrics, grok modular arithmetic with block-circulant features that implement the same Fourier algorithm (Mallinar et al. 2025); GPT-2-XL fine-tuned on addition represents numbers by sparse Fourier features, low frequencies in the MLPs for magnitude and high frequencies in attention for the residue (Zhou et al. 2024); and language models place days of the week and months of the year on circles (Engels et al. 2025), a cross-link to Chapter 11. Open: a proof that gradient descent with weight decay on a fraction of the table finds the circuit, with the time it takes and the frequencies it selects.

10.8 Model card

MODEL CARD	
PHENOMENON	A network fits part of a group's table early and generalizes much later; then it computes with characters of the group.
MODEL	Two-layer network with quadratic activation and one-hot inputs, on $a + b \bmod p$; for the delay, a diagonal linear network with weight decay and large initialization.
SOLUTION	An explicit Fourier circuit; max-$\ell_{2,3}$-margin networks consist of Fourier neurons with additive phases and use every frequency, with at least six neurons each, which requires

	width $3(p-1)$; weight decay acts as an initialization scale $\alpha e^{-\lambda t}$.
PREDICTS	Logits $\propto \sum_\zeta \cos(\omega_\zeta(a+b-c))$; $\gamma^\star = \sqrt{2/27}/(\sqrt{p}(p-1))$; phase defect 0; $t_{\text{grok}} \approx \lambda^{-1}\log(\alpha/\alpha_c)$, $t_{\text{mem}} \propto \alpha^{-2}$.
GETS WRONG	Transformers use few frequencies or the pizza algorithm; trained networks are far from max margin; grokking without weight decay or without circuits; coset circuits for permutation groups.
SUCCESSORS	Representation theory of finite groups; algebraic composition of optima; lazy-to-rich dynamics; phase-transition theories; Fourier features in language models.

Two threads of the book meet here. Implicit bias: the margin argument is that of Chapter 4 for a network homogeneous of degree three, and Chapter 3 supplies the potential and the exact effect of weight decay. Lazy versus rich: memorization is the kernel regime of Chapter 6, and grokking is the crossing into the rich regime, which weight decay forces at a predictable time; the search phase of Chapter 7 and the hidden progress of Chapter 9 tell the same story from small initialization, with the plateau before fitting rather than after. Fourier analysis on groups links the chapter to the parities of Chapter 9 and the circular features of Chapter 11, and the equal-margin logits form the simplex of Chapter 12.

Exercises

10.1 CONSOLIDATION. (The normalized margin of the circuit.) (a) Show that the network of Theorem 10.4 has normalized margin $0.39\,\gamma^\star$, to two digits. (b) Multiply u and u' by $(2/9)^{1/3}$ and v by $(2/9)^{-2/3}$ in every neuron; show that the logits do not change and the normalized margin becomes $\gamma^\star$. (c) Show that the max-margin logit vectors $f(a,b) \in \mathbb{R}^p$ of the p possible sums form a regular simplex, as in Chapter 12.

10.2 CONSOLIDATION. ★ (Six phases suffice, and tight rules have equal weights.) (a) Verify the six-neuron identity of Remark 10.5, and check that its points (χ_r, δ_r) are $(\frac{\pi r}{3}, \pi r)$, $r = 0, \dots, 5$, up to order. (b) Suppose that six points with positive weights satisfy (10.5). Show that every weight equals $\frac{1}{6}\sum_r \beta_r$. (c) Conclude that the nine phases of Theorem 10.4 are not minimal, and that six is the least number of Fourier neurons per frequency. *(Hint: in (b), the evaluation map from the space of Lemma 10.6 to the six points is a bijection, so for each point there is a P_r equal to 1 there and 0 at the others; compute $\sum_s \beta_s |P_r(\chi_s, \delta_s)|^2$ in two ways and use $1 = P_r(\chi_r, \delta_r) \le \sqrt{6}\,(\sum_{j,s} |c_{js}|^2)^{1/2}$.)*

10.3 CONSOLIDATION. (Why the full table.) (a) Redo Lemma 10.8 with a non-uniform weighting of the wrong classes, and show that the terms $u(a)^2$ and $u'(b)^2$ no longer cancel in general. (b) Redo it with (a,b) uniform on a training set D, and show that the squares cancel if D is a union of rows $\{a\} \times \mathbb{Z}_p$, but not for a typical random D. What

does this say about explaining generalization from part of the table by margins?

10.4 CONSOLIDATION. (Weight decay and the Lasso.) (a) Show that the minimum of $\|u\|^2+\|v\|^2$ subject to $u\odot u-v\odot v=w$ is $\|w\|_1$; deduce that the regularized objective of Section 10.5 and the Lasso $\widehat{\mathcal{L}}(w)+\frac{\lambda}{2}\|w\|_1$ have the same minimum value, and that global minimizers correspond. (b) For gradient descent with step η, show that u_iv_i is multiplied by $(1-\eta\lambda)^2-4\eta^2g_i^2$ at each step, so part (i) of Proposition 10.12 holds up to a relative error of order η^2 per step. (c) Evaluate t_{mem} for the run of Figure 10.1 and compare with the figure.

10.5 VARIATION. ★ (Multiplication and even moduli.) (a) Let g generate the cyclic group $\mathbb{Z}_p^\times$ of order $p-1$. Show that for $a,b,c\in\mathbb{Z}_p^\times$, $c=ab$ if and only if $\log_g c=\log_g a+\log_g b$ in $\mathbb{Z}_{p-1}$, and build a Fourier network for multiplication. (b) The modulus $N=p-1$ is even, so the frequency $N/2$, whose character $(-1)^x$ becomes the Legendre symbol, is real and has no conjugate partner. Redo Theorem 10.9 on $\mathbb{Z}_N$ and show that the best single neuron sits at frequency $N/2$, with average margin $2/(3\sqrt{3}(N-1)\sqrt{N})$, a factor $\sqrt{2}$ above the best paired frequency. (c) Show that if all neurons sit at frequency $N/2$, then every wrong class of the same parity as $a+b$ has margin zero, and conclude that the proof of Theorem 10.10 fails for even moduli.

10.6 VARIATION. (A non-abelian group.) Let G be a finite group with irreducible characters χ_ρ of dimensions d_ρ. (a) Use the orthogonality of characters to show that $\sum_\rho d_\rho\chi_\rho(g)=|G|\,\mathbf{1}\{g=e\}$. (b) Write the character table of S_3 and check (a). (c) Deduce that the logits $f_c(a,b)=\sum_\rho d_\rho\chi_\rho(c^{-1}ab)-1$ give every margin $|G|$. (d) Write $\chi_\rho(c^{-1}ab)$ as the trace of a product of three matrices, and explain why a network now needs products of matrix entries; compare with the algorithm of Chughtai et al. (2023).

10.7 VARIATION. (The Euclidean norm.) (a) Show that $\min_{t>0}\big(t^2(\|u\|^2+\|u'\|^2)+t^{-4}\|v\|^2\big)=3\cdot2^{-2/3}(\|u\|^2+\|u'\|^2)^{2/3}\|v\|^{2/3}$: the weight-decay cost of a neuron grows like the power $\frac{2}{3}$ of the size of its contribution to the logits. (b) Show that under $\|\theta\|_2\le1$ the average margin is maximized only by networks with a single nonzero neuron, and that such a network does not have equal margins (Theorem 10.10). Conclude that the proof of Theorem 10.10 does not apply to the norm that weight decay penalizes.

10.8 VARIATION. (Unbalanced initialization.) In the diagonal network start from $u(0)=\alpha\mathbf{1}_d$ and $v(0)=\alpha'\mathbf{1}_d$ with $\alpha\neq\alpha'$. Show that $u_iv_i=\alpha\alpha'e^{-2\lambda t}$, that part (ii) of Proposition 10.12 holds with $\alpha_t^2=\alpha\alpha'e^{-2\lambda t}$, and that $w(t)$ minimizes $\Phi_{\alpha_t}(w')-\frac{1}{4}\log(\alpha/\alpha')\sum_i w'_i$ on $\{Xw'=Xw(t)\}$. Which sign of target coordinates does $\alpha>\alpha'$ favor?

10.9 SIMULATION. (Weight decay and the delay in the network.) Run the script of Figure 10.2 with weight decay 0.3, 0.5, 1, 2 and 4 for 40000

steps, and record the first step at which the test accuracy reaches 99%. The reader should observe about 14000, 6000 and 4000 steps for 0.5, 1 and 2, consistent with $t_{\text{grok}} \propto \lambda^{-1}$ up to a slowly varying logarithm; for 0.3, a test accuracy still below 99% (about 98%) at step 40000; and for 4, a test accuracy that stalls near 94%. Plot the parameter norm against time and compare with the effective scale $\alpha e^{-\lambda t}$ of the diagonal model.

10.10 SIMULATION. (The critical scale.) With the script of Figure 10.1: (a) set $\lambda = 0$ and check that the test error stays at that of w_α^Φ; (b) for $n/d \in \{0.2, 0.3, 0.4\}$ and several random designs, compare the scale $\alpha_{1/2}$ at which the static test error halves with the root of $\alpha_c^2 \log(1/\alpha_c^2) = \|w_2\|^2/(4\|w_\star\|_1)$. The reader should find agreement within a factor of about 1.5, an error of about $\log 1.5 \approx 0.4$ in λt_{grok}, small against $\log(\alpha/\alpha_c)$ when α is large.

10.11 RESEARCH. (Dynamics on part of the table.) Prove that gradient descent with small weight decay, from a small random initialization on a random fraction of the table, converges to a network of approximately Fourier neurons, and estimate the time. Mohamadi et al. (2024) show that quadratic networks fitting the training set with bounded ℓ_∞ norm generalize from far fewer points than kernel methods, and that gradient descent with a small ℓ_∞ penalty finds such networks; Tian (2025) describes global minimizers of a squared loss. The dynamics with weight decay, and the selection of frequencies, are open.

10.12 RESEARCH. (The narrowest circuit.) What is the smallest width of a network (10.1) whose logits equal the clock sum (10.3) exactly? Fourier neurons need $3(p-1)$, and four neurons of one frequency give its clock up to a function of c alone (see the discussion after Lemma 10.6), which further neurons must cancel. A matching lower bound, and the analogous question for the ℓ_2 margin problem of Exercise 10.7, are open.

Notes and References

Grokking was reported by Power et al. (2022), and its Fourier explanation comes from Nanda et al. (2023). Gromov (2023) trained the quadratic network of this chapter with a squared loss and wrote down an approximate solution with random phases. Liu et al. (2022, 2023) first tied the delay to hyperparameters, and Liu et al. (2023) gave the estimate $\lambda^{-1}\log(w_0/w_c)$ that Section 10.5 derives for the diagonal model. The margin analysis is that of Morwani et al. (2024), resting on Wei et al. (2019) and Lyu and Li (2020); Tian (2025) composes solutions from small blocks, and the cubature view of Lemmas 10.3 and 10.6 organizes all such designs.

For the delay, Lyu et al. (2024) prove the dichotomy of early kernel and late margin biases, with diagonal networks and matrix completion as examples, and also exhibit "misgrokking", in which the early bias generalizes and the late one does not. Kumar et al. (2024) build on the lazy regime of

Chizat et al. (2019) and the kernel and rich regimes of Woodworth et al. (2020). Proposition 10.12 is a short computation in the setting of Chapter 3. AdamW (Loshchilov and Hutter 2019) multiplies the parameters by $1 - \eta\lambda$ at each step, so $\eta\lambda$ is the decay per step in our experiments.

TRANSLATING THE LITERATURE. Morwani et al. (2024) write a neuron as $(u_a + v_b)^2 w$: their u, v, w are our u, u', v; their frequency ζ and phases θ_u^*, θ_v^*, θ_w^* are our ζ, φ, ψ, χ; their class-weighted margin with uniform weighting is our average margin $\bar{\gamma}$, their $L_{2,3}$ norm our $\|\cdot\|_{2,3}$, and their γ^* our $\gamma^\star$; their Fourier transform is unnormalized, like ours. Nanda et al. (2023) write $w_k = 2\pi k/p$ for our ω_ζ with $\zeta = k$. Gromov (2023) writes N for the width (our m), n and m for the inputs (our a and b), q for the output (our c), and $\phi^{(1)}$, $\phi^{(2)}$, $\phi^{(3)}$ for our φ, ψ, χ. Tian (2025) writes d for the order of the group (our p); the order of a block is our number of neurons per frequency. Lyu et al. (2024) use our α and λ, write $\bar{\theta}_{\text{init}}$ for our $\bar{\theta}$, and write $T_c^{\pm}(\alpha) = (1 \pm c)\lambda^{-1} \log \alpha$ for the two times of Theorem 10.11.

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11 Superposition

More features than dimensions

THE MODEL — RELU TOY AUTOENCODER

IN THIS CHAPTER. Neurons of trained networks often respond to several unrelated things, and the activations of language models decompose into sparse combinations of many more directions than there are dimensions. The toy model of this chapter stores p sparse features in $m < p$ dimensions with a linear map and reads them out with a ReLU. We prove that without the ReLU the optimum is principal component analysis at every sparsity, so a linear network never stores more features than dimensions; that with it, two features share one dimension as an antipodal pair once they are sparse enough, with exact losses and a provable region of the phase diagram; that sparse features fit without first-order interference exactly when their directions have pairwise nonpositive inner products, which is possible for at most $2m$ of them; and that random directions and a thresholded readout store e^{cm} features if few are active at a time. The model knows nothing of correlated or multidimensional features, of computation, or of how sparse autoencoders choose among decompositions.

11.1 The phenomenon: polysemantic neurons and sparse dictionaries

Look at what makes a single neuron of a trained vision network fire, and the answer is often a list of things with nothing in common. In their survey of the circuits of InceptionV1, Olah et al. (2020) found many *polysemantic neurons*, neurons that respond to several unrelated inputs, and named them a main obstacle to reading networks neuron by neuron.[1] Word embeddings tell the same story in a form one can compute with. Arora et al. (2018) showed that the embedding of a polysemous word such as *tie* lies close to a linear combination of vectors for its senses, and that sparse coding of the embeddings against a dictionary of about 2000 "discourse atoms" recovers those senses.

Elhage et al. (2022) turned these observations into a hypothesis. A network may represent more *features*, properties of the input that it uses, than it has dimensions, by assigning each feature a direction, with the directions almost but not exactly orthogonal. The cost is *interference*: reading one feature picks up a little of every other. If features are *sparse*, so that few are present in a given input, the interference is usually small, and a nonlinearity can remove what remains. They called this *superposition*, studied it in the toy model of this chapter, and found that whether a feature is stored in

[1] Neuroscience met the same fact earlier under the name *mixed selectivity*: Rigotti et al. (2013) found that neurons in monkey prefrontal cortex respond to combinations of task variables, and that every variable can be decoded from the population although single cells mix them.

superposition changes abruptly with its sparsity and importance, and that stored features arrange themselves in regular polytopes.

If the hypothesis is right, then the right units for reading a network are directions, not neurons, and they can be found by sparse dictionary learning: write each activation vector $h \in \mathbb{R}^m$ as $h \approx \sum_i a_i(h)\, d_i$ with many more dictionary vectors d_i than dimensions and few nonzero coefficients $a_i(h) \geq 0$, as Olshausen and Field (1997) did for natural images. A *sparse autoencoder* computes the coefficients with one ReLU layer and learns the dictionary. Cunningham et al. (2024) trained such autoencoders on the residual stream of small language models of the Pythia family and found directions more interpretable than neurons. Bricken, Templeton, et al. (2023) decomposed the 512 neurons of a one-layer transformer's MLP into more than 4000 features, among them features for DNA sequences, legal language, HTTP requests, and Hebrew text, invisible in single neurons. Templeton, Conerly, et al. (2024) scaled the method to a production model, Claude 3 Sonnet, with dictionaries of up to 34 million features, many of them multilingual, multimodal, and usable to steer the model, and Gao et al. (2025) trained a 16-million-latent autoencoder on GPT-4 activations and measured scaling laws in the dictionary size.

OBSERVATION 11.1 (Olah et al. 2020; Elhage et al. 2022; Bricken, Templeton, et al. 2023; Templeton, Conerly, et al. 2024). Individual neurons of trained networks are often polysemantic. The activation vectors of a layer of width m are approximated, with a reconstruction error that is small but not negligible, by sparse nonnegative combinations of $p \gg m$ learned directions, and many of these directions are interpretable.

The status of the observation should be stated carefully. Polysemanticity is robust and old. That activations *are* sparse combinations of a dictionary is a modeling claim, supported by the interpretability of the recovered directions, and challenged by three facts we return to in Section 11.6: the reconstructions leave a structured error, the dictionaries depend on the random seed, and some features are not one-dimensional. The question for this chapter is the one that comes before all of this. *When and why does a network store more features than it has dimensions, how many can it store, and at what cost?*

11.2 The model: a ReLU autoencoder for sparse features

THE FEATURES. The input is a vector $x \in [0,1]^p$ of p features with independent coordinates, each equal to 0 with probability S and uniform on $[0,1]$ otherwise. The number $S \in [0,1)$ is the *sparsity*; it is local notation, as is the name p for the number of features, which matches the global use of p for the number of features of a linear model. A feature is *active* when it is nonzero. One feature has mean, second moment, and variance

$$\mathbb{E}x_i = \frac{1-S}{2}, \qquad \mathbb{E}x_i^2 = \frac{1-S}{3}, \qquad v(S) := \operatorname{Var} x_i = \frac{(1-S)(1+3S)}{12}. \tag{11.1}$$

Each feature carries an *importance* $\omega_i > 0$, and $\Omega = \sum_i \omega_i$.

THE NETWORK. A matrix $W \in \mathbb{R}^{m \times p}$ with $m < p$ compresses the features to $h = Wx \in \mathbb{R}^m$, and the transpose decodes them through a ReLU with bias $b \in \mathbb{R}^p$:

$$x' = \mathrm{ReLU}\big(W^\top W x + b\big), \qquad \mathcal{L}(W, b) = \sum_{i=1}^{p} \omega_i \, \mathbb{E}\big(x_i - x_i'\big)^2. \tag{11.2}$$

We write $w_i \in \mathbb{R}^m$ for the i-th column of W, the direction of feature i, and $G = W^\top W$ for the Gram matrix of the directions. Coordinate i of the output reads

$$x_i' = \mathrm{ReLU}\Big(\|w_i\|^2 x_i + \sum_{j \neq i} \langle w_i, w_j \rangle \, x_j + b_i\Big) :$$

the signal, the *interference* from the other features, and a bias. The *linear model* is the same network without the ReLU, $x' = W^\top W x + b$.

Definition 11.2 (Representation and superposition). A configuration (W, b) *represents* feature i if $w_i \neq 0$, and it *drops* feature i if $w_i = 0$. It is in *superposition* if it represents more than m features.

Represented directions in superposition cannot all be orthogonal, so superposition means interference. Elhage et al. (2022) also measure superposition by $\|W\|_F^2 / m$, the number of features per dimension, which equals 1 when $W^\top W$ is an orthogonal projection of rank m; we use it in Section 11.5.

WHAT WAS THROWN AWAY. The features are given rather than learned from data, independent, identically distributed except for their importances, and of one kind (a magnitude in $[0, 1]$). The task is to reconstruct them, not to compute with them. The decoder is tied to the encoder, there is one layer, and the loss is the population loss. Each simplification removes something that would hide the trade-off we want to see: with independent features, interference is pure noise; with reconstruction, the target is the identity and every error is visible; tied weights make the geometry of W the whole story.

DICTIONARY

IN THE MODEL	IN A TRAINED NETWORK
features x_i, sparsity S, importance ω_i	properties of the input a network uses (a word sense, a language, a concept), present rarely, of unequal value to the loss
directions $w_i \in \mathbb{R}^m$, $m < p$	directions in a layer's activation space: the rows of a sparse autoencoder's decoder
$h = Wx$	the activation vector of the layer, of width m
interference $\langle w_i, w_j \rangle x_j$	polysemanticity: a neuron or direction picks up unrelated features
ReLU with bias $b_i < 0$	a nonlinearity that thresholds out small interference; the encoder of a sparse autoencoder
loss $\mathcal{L}(W, b)$	what the layer must preserve for the rest of the network

WHAT ONE HAD TO NOTICE. That a sparse feature is usually absent, so that interference with it costs only when it is present together with another one; that a ReLU with a slightly negative bias discards small positive interference for free and that negative interference costs nothing at all, because the ReLU clips it; and that in high dimension there are exponentially many almost orthogonal directions. The rest is computation.

11.3 The solution: what the ReLU buys

NO SUPERPOSITION WITHOUT A NONLINEARITY. The linear model is principal component analysis, and so it never superposes. The theorem holds for every distribution with uncorrelated coordinates; we use only the mean $\mu \in \mathbb{R}^p$ and the diagonal covariance $\operatorname{diag}(v_1, \dots, v_p)$ of x.

Theorem 11.3 (The linear model keeps the top m features). *Let x have mean μ and covariance $\operatorname{diag}(v_1, \dots, v_p)$ with all $v_i > 0$, let $m < p$, and let $c_{(1)} \geq \cdots \geq c_{(p)}$ be the numbers $c_i = \omega_i v_i$ in decreasing order. For every $W \in \mathbb{R}^{m \times p}$ and $b \in \mathbb{R}^p$,*

$$\mathcal{L}_{\text{lin}}(W, b) := \sum_{i=1}^{p} \omega_i \mathbb{E}\big(x_i - (W^\top W x + b)_i\big)^2 \;\geq\; \sum_{k=m+1}^{p} c_{(k)}.$$

Equality holds when, for a set K of m indices carrying the m largest values c_i, the columns $(w_i)_{i \in K}$ are orthonormal, $w_i = 0$ for $i \notin K$, and $b = (I_p - W^\top W)\mu$. If $c_{(m)} > c_{(m+1)}$, then these are the only minimizers; in particular every minimizer drops exactly $p - m$ features, whatever the sparsity.

PROOF. The plan is to remove the bias, to rewrite the loss as the distance from a diagonal matrix to a matrix of rank at most m, and to bound that distance by a linear program on the diagonal of a projection. Put $P = W^\top W$, $D = \operatorname{diag}(\omega_i)$ and $\Sigma = \operatorname{diag}(v_i)$.

The bias. Split the error as $x - Px - b = (I_p - P)(x - \mu) + \big((I_p - P)\mu - b\big)$. The first part has mean zero, so the cross term vanishes in expectation, and

$$\mathcal{L}_{\text{lin}}(W, b) = \sum_i \omega_i \big[(I_p - P)\Sigma(I_p - P)\big]_{ii} + \sum_i \omega_i \big((I_p - P)\mu - b\big)_i^2.$$

The second sum is nonnegative and vanishes exactly for $b = (I_p - P)\mu$. The first equals $\|A - B\|_F^2$ with $A = D^{1/2}\Sigma^{1/2} = \operatorname{diag}(\sqrt{c_i})$ and $B = D^{1/2}P\Sigma^{1/2}$, a matrix of rank at most m.

A projection. Because $\operatorname{rank} B \le m < p$, there is an orthogonal projection Q of rank m whose range contains the range of B, so that $(I_p - Q)B = 0$. The matrices $(I_p - Q)A$ and $Q(A - B)$ have columns in orthogonal subspaces and sum to $A - B$, so

$$\|A - B\|_F^2 = \|(I_p - Q)A\|_F^2 + \|Q(A - B)\|_F^2 \;\ge\; \sum_i c_i \|(I_p - Q)e_i\|^2 = \sum_i c_i(1 - q_i), \tag{11.3}$$

with $q_i = \|Qe_i\|^2 = Q_{ii}$. The numbers q_i lie in $[0, 1]$ and add up to $\operatorname{tr} Q = m$.

The linear program. For every real t, $\sum_i c_i q_i = tm + \sum_i (c_i - t)q_i$. Take $t = c_{(m)}$: the terms with $c_i < t$ are at most 0, and those with $c_i \ge t$ are at most $c_i - t$, since $q_i \le 1$; the indices with $c_i > t$ are among the top m. Hence $\sum_i c_i q_i \le tm + \sum_{k \le m}(c_{(k)} - t) = \sum_{k \le m} c_{(k)}$, and (11.3) gives the bound.

Attainment. For the configuration in the statement, P is the diagonal projection Π_K onto the coordinates in K, so $B = \operatorname{diag}(\sqrt{c_i}\,\mathbf{1}\{i \in K\})$ and $\|A - B\|_F^2 = \sum_{i \notin K} c_i = \sum_{k > m} c_{(k)}$; the bias term vanishes by the choice of b.

Uniqueness. Let $c_{(m)} > c_{(m+1)}$, so that K is unique, and take t strictly between the two values. Every term $(c_i - t)q_i$ is then at most $c_i - t$ for $i \in K$ and at most 0 for $i \notin K$, with equality only if $q_i = 1$, respectively $q_i = 0$. So equality forces $\|(I_p - Q)e_i\| = 0$ for $i \in K$, and since $\operatorname{rank} Q = m = |K|$, $Q = \Pi_K$. Equality in (11.3) also forces $Q(A - B) = 0$, that is $B = \Pi_K A$, and then $P = D^{-1/2}B\Sigma^{-1/2} = \Pi_K$. The identity $W^\top W = \Pi_K$ says $\langle w_i, w_j \rangle = \mathbf{1}\{i = j \in K\}$, and the bias must be $(I_p - \Pi_K)\mu$. ∎

Sparsity enters the linear model only through the variances, so it cannot make superposition pay. The reason is visible in the proof: a linear decoder pays for interference in every input, whether the interfering feature is present or not, through the variance it adds. When importances tie the minimizer is not unique, and rotations of W within a tied block may represent more than m features; but $W^\top W$ is still an orthogonal projection, one feature per dimension, and the slightest difference in importances removes the rotation (Exercise 11.2).[2]

[2] That a linear autoencoder trained by least squares recovers the principal subspace goes back to Bourlard and Kamp (1988) and Baldi and Hornik (1989), who also showed that its other critical points are saddles. The low-rank approximation behind it is that of Eckart and Young (1936).

TWO FACTS ABOUT THE RELU. Everything that follows uses two inequalities: for all real z, z' and every $y \ge 0$,

$$|\mathrm{ReLU}(z) - \mathrm{ReLU}(z')| \le |z - z'|, \qquad |y - \mathrm{ReLU}(z)| \le |y - z|. \tag{11.4}$$

The first says that the ReLU is 1-Lipschitz. For the second, if $z \ge 0$ both sides agree, and if $z < 0$, then $|y - 0| = y \le y - z$. The ReLU is the projection onto $[0, \infty)$, and projecting onto a convex set that contains the target never increases the distance to it. This is the whole advantage of the nonlinearity: negative interference is free.

Lemma 11.4 (The cost of dropping a feature). *If $w_i = 0$, then the term i of $\mathcal{L}(W, b)$ is at least $\omega_i\, v(S)$, with equality if and only if $b_i = \frac{1-S}{2}$.*

Proof. If $w_i = 0$, then $x'_i = \mathrm{ReLU}(b_i)$ is a constant $c \geq 0$, and $\mathbb{E}(x_i - c)^2 = v(S) + (\mathbb{E}x_i - c)^2$ is smallest exactly at $c = \mathbb{E}x_i = \frac{1-S}{2} > 0$, that is, at $b_i = \frac{1-S}{2}$. ∎

A dropped feature is replaced by its mean, and the bias makes this possible; without a bias the cost of dropping would be $\omega_i \mathbb{E}x_i^2$, which changes the comparisons below (Exercise 11.6).

Two features in one dimension. Take $p = 2$, $m = 1$, so that $W = (w_1, w_2)$ is a row of two numbers, and importances $\omega = (1, r)$ with $0 < r \leq 1$. The loss is a function of four numbers, and we compare three explicit configurations: the *dedicated* configuration $w = (1, 0)$, $b = (0, \frac{1-S}{2})$, which keeps feature 1 exactly and replaces feature 2 by its mean; the *antipodal pair* $w = (1, -1)$, $b = (0, 0)$; and the antipodal tilt $w = (1, -c)$ of the dedicated configuration. The tilt needs one more function. For $\tau \in \mathbb{R}$ and a feature x, let

$$f_S(\tau) = S\,\mathbb{E}\,\mathrm{ReLU}(\tau - x)^2 + (1 - S)\,\mathbb{E}(\tau - x)^2, \qquad \kappa(S) = \min_{\tau \in \mathbb{R}} f_S(\tau), \quad (11.5)$$

and let τ_S be the minimizer (Exercise 11.3 shows that it exists, is unique, lies in $(0, 1)$, and solves a quadratic equation). Numerically, $\kappa(S)/v(S)$ equals 1 at $S = 0$, 0.60 at $S = \frac{1}{2}$, and 0.11 at $S = 0.9$, and $\kappa(S) \leq f_S(0) = \frac{(1-S)^2}{3}$.

Theorem 11.5 (Two features in one dimension). *Let $p = 2$, $m = 1$, $\omega = (1, r)$ with $0 < r \leq 1$, and $0 \leq S < 1$.*

(i) Every configuration that drops a feature has $\mathcal{L} \geq r\, v(S)$. The dedicated configuration attains $r\, v(S)$.

(ii) The antipodal pair has $\mathcal{L} = \frac{1+r}{6}(1 - S)^2$.

(iii) For every $c > 0$, the configuration $w = (1, -c)$, $b = \big(c\tau_S, (1 + c - c^2)\frac{1-S}{2}\big)$ has

$$\mathcal{L} \leq r\, v(S) - c^2\big(r\, v(S) - \kappa(S)\big) + c^4 r\, v(S).$$

(iv) Consequently, if $r\, v(S) > \min\big\{\kappa(S), \frac{1+r}{6}(1 - S)^2\big\}$, then the infimum of $\mathcal{L}$ is strictly smaller than the loss of every configuration that drops a feature: no optimal or nearly optimal configuration drops one. This holds in particular whenever $S > \frac{2+r}{2+5r}$.

Proof. The plan is to evaluate the antipodal pair exactly and to bound the tilt with (11.4); part (iv) then compares numbers.

Part (i). If $w_2 = 0$, then Lemma 11.4 bounds the second term by $r\, v(S)$, and the first is nonnegative. If $w_1 = 0$, then the first term alone is at least $v(S) \geq r\, v(S)$. For the dedicated configuration, $x'_1 = \mathrm{ReLU}(x_1) = x_1$, and the second term equals $r\, v(S)$ by Lemma 11.4.

Part (ii). Here $x'_1 = \mathrm{ReLU}(x_1 - x_2)$. If $x_2 = 0$, then $x'_1 = x_1$. If $x_1 = 0$, then $x'_1 = \mathrm{ReLU}(-x_2) = 0 = x_1$: the interference is negative and the ReLU

clips it. If both are active, which happens with probability $(1-S)^2$, then $x_1 - \mathrm{ReLU}(x_1 - x_2) = \min(x_1, x_2)$, and for independent uniform U, V the minimum has density $2(1-t)$ on $[0,1]$, so $\mathbb{E}\min(U,V)^2 = \int_0^1 2t^2(1-t)\,\mathrm{d}t = \frac{1}{6}$. The first term is therefore $\frac{(1-S)^2}{6}$, and by symmetry the second is r times the same.

Part (iii), feature 1. Now $x_1' = \mathrm{ReLU}(x_1 + c(\tau_S - x_2))$. If $x_1 > 0$, then $x_1 = \mathrm{ReLU}(x_1)$ and the Lipschitz bound in (11.4) gives $|x_1 - x_1'| \le c|\tau_S - x_2|$. If $x_1 = 0$, then $(x_1 - x_1')^2 = c^2\,\mathrm{ReLU}(\tau_S - x_2)^2$, because $c > 0$. Since x_1 and x_2 are independent, the first term is at most $c^2 f_S(\tau_S) = c^2\kappa(S)$.

Part (iii), feature 2. Now $x_2' = \mathrm{ReLU}(z)$ with $z = -cx_1 + c^2x_2 + b_2$. By the second inequality in (11.4), $(x_2 - x_2')^2 \le (x_2 - z)^2 = \big((1-c^2)x_2 + cx_1 - b_2\big)^2$. Our b_2 is the mean of $(1-c^2)x_2 + cx_1$, so the expectation is its variance, $(1-c^2)^2 v(S) + c^2 v(S) = v(S)(1 - c^2 + c^4)$. Adding r times this to the bound for feature 1 gives (iii).

Part (iv). If $r\,v(S) > \kappa(S)$, take $c^2 = (rv(S) - \kappa(S))/(2rv(S))$ in (iii); the loss is then at most

$$r\,v(S) - \frac{\big(r\,v(S) - \kappa(S)\big)^2}{4r\,v(S)} < r\,v(S).$$

If $r\,v(S) > \frac{1+r}{6}(1-S)^2$, then (ii) gives a loss below $r\,v(S)$. In both cases (i) shows that every configuration dropping a feature is worse than one of these. Finally, by (11.1) and because $1 - S > 0$, the inequality

$$\frac{1+r}{6}(1-S)^2 < r\,\frac{(1-S)(1+3S)}{12}$$

is equivalent to $2(1+r)(1-S) < r(1+3S)$, that is, to $(2+5r)S > 2 + r$. ∎

Both constructions superpose by making the interference negative. The antipodal pair is the extreme case: each feature is read out exactly whenever it is alone, and the error $\min(x_1, x_2)$ appears only when both are present, an event of probability $(1-S)^2$, while dropping a feature costs order $1 - S$. The tilt is the infinitesimal version. Feature 2 gains $c^2 rv(S)$ because its readout now sees a little of x_2; feature 1 pays $c^2\kappa(S)$, and since negative interference is clipped when $x_1 = 0$, $\kappa(S)$ is smaller than the variance $v(S)$ that a linear decoder would pay, which is what makes the trade profitable. At $S = 0$ the two coincide, $\kappa(0) = \frac{1}{12} = v(0)$, and at $r = 1$ the tilt is then neutral to second order in c, the rotational degeneracy of Theorem 11.3; at higher order the ReLU, which clips the rare negative outputs of a rotated readout, breaks the tie in favor of superposition.

Part (iii) also shows that for $r > \kappa(S)/v(S)$ the dedicated configuration is not a local minimum: the loss decreases along the tilt. The converse is what a second-order computation gives.

Heuristic derivation of the spinodal. Perturb the dedicated configuration to $w = (1+\delta, \pm c)$ and $b = (b_1, \frac{1-S}{2} + \beta)$ with all four perturbations small. For feature 2, whose ReLU stays in its linear region, the best readout of x_2 from $\pm c(1+\delta)x_1 + c^2x_2$ plus a constant gains at most $c^2 rv(S) + O(c^4)$. For feature 1, the error is $-(2\delta x_1 \pm cx_2 + b_1)$ when $x_1 > 0$, up to a set of

measure $O(c)$ near $x_1 = 0$, and $\mathrm{ReLU}(\pm c x_2 + b_1)$ when $x_1 = 0$; the term in δ only adds a positive quadratic, the sign + is worse than − (it is not clipped), and with $b_1 = c\tau$ the cost is $c^2 f_S(\tau) \geq c^2 \kappa(S)$. So the second variation is $c^2(\kappa(S) - r v(S))$ to leading order: the dedicated configuration is a strict local minimum for $r < \kappa(S)/v(S)$ and a saddle for $r > \kappa(S)/v(S)$. ◇

We checked the heuristic numerically: from small perturbations of the dedicated configuration with a negative tilt $w_2 < 0$, gradient descent on the exact loss returns to it at $(S, r) = (0.7, 0.3)$, $(0.5, 0.55)$ and $(0.3, 0.75)$, just below the curve $r = \kappa(S)/v(S)$, and leaves it at $(0.7, 0.4)$, $(0.5, 0.65)$ and $(0.3, 0.85)$, just above. In the language of phase transitions, the curve $r = \kappa(S)/v(S)$ is a *spinodal*, the edge of metastability of the dedicated phase. Section 11.5 shows that the true transition comes before it and is discontinuous.

11.4 The solution: geometry and capacity

THE SPARSE LIMIT. With many features, the cleanest statements are about $S \to 1$. Condition on the set $A \subseteq \{1, \dots, p\}$ of active features, which has probability $(1-S)^{|A|} S^{p-|A|}$, and let $\ell_A(W, b)$ be the conditional expected loss. For $A = \emptyset$ and $A = \{i\}$, with U uniform on $[0, 1]$,

$$\ell_\emptyset = \sum_j \omega_j \,\mathrm{ReLU}(b_j)^2,$$

$$\ell_{\{i\}} = \omega_i \mathbb{E}\big(U - \mathrm{ReLU}(\|w_i\|^2 U + b_i)\big)^2 + \sum_{j \neq i} \omega_j \mathbb{E}\,\mathrm{ReLU}\big(\langle w_j, w_i\rangle U + b_j\big)^2.$$

Since $x \in [0, 1]^p$, every ℓ_A is bounded by a constant $C_{W,b}$ that depends on the configuration, and the probability that two or more features are active is at most $\binom{p}{2}(1-S)^2$. Hence

$$\mathcal{L}(W, b) = S^p \ell_\emptyset + (1-S) S^{p-1} \sum_{i=1}^{p} \ell_{\{i\}} + R, \qquad 0 \leq R \leq \tbinom{p}{2} C_{W,b} (1-S)^2. \tag{11.6}$$

To first order in the density $1 - S$, the loss sees only inputs with one active feature: the problem becomes one of *one-hot* inputs.

Definition 11.6 (Obtuse configurations). A configuration is *obtuse* if its directions are unit vectors with pairwise nonpositive inner products, $\|w_i\| = 1$ and $\langle w_i, w_j\rangle \leq 0$ for $i \neq j$, and its bias is $b = 0$.

Lemma 11.7 (No first-order interference). *A configuration has $\ell_\emptyset = 0$ and $\ell_{\{i\}} = 0$ for every i if and only if it is obtuse.*

PROOF. The term $\omega_i \mathbb{E}(U - \mathrm{ReLU}(\|w_i\|^2 U + b_i))^2$ vanishes if and only if $\mathrm{ReLU}(\|w_i\|^2 u + b_i) = u$ for almost every $u \in [0, 1]$; the left side is then a linear function equal to u on an interval, so $\|w_i\|^2 = 1$ and $b_i = 0$. Given $b = 0$, the term $\mathbb{E}\,\mathrm{ReLU}(\langle w_j, w_i\rangle U)^2$ vanishes if and only if $\langle w_j, w_i\rangle \leq 0$, and $\ell_\emptyset = 0$ holds. Conversely, an obtuse configuration makes every one of these terms zero. ■

How many directions can be pairwise obtuse? The answer is classical.

Lemma 11.8 (At most $2m$ obtuse directions). *If $u_1, \dots, u_k \in \mathbb{R}^m$ are nonzero and $\langle u_i, u_j \rangle \le 0$ for all $i \neq j$, then $k \le 2m$. The $2m$ vectors $\pm e_1, \dots, \pm e_m$ attain the bound.*

Proof. We use induction on m. For $m = 1$ the vectors are nonzero numbers, and two of the same sign have a positive product, so $k \le 2$. Let $m \ge 2$, and project $u_1, \dots, u_{k-1}$ onto the hyperplane orthogonal to u_k: $u_i' = u_i - \frac{\langle u_i, u_k \rangle}{\|u_k\|^2} u_k$. For $i \neq j$,

$$\langle u_i', u_j' \rangle = \langle u_i, u_j \rangle - \frac{\langle u_i, u_k \rangle \langle u_j, u_k \rangle}{\|u_k\|^2} \le 0,$$

because the subtracted product of two nonpositive numbers is nonnegative. A projection u_i' vanishes only if u_i is a multiple of u_k, necessarily a negative one; two such u_i would have a positive inner product, so at most one u_i' vanishes. The remaining $k - 2$ or more nonzero vectors lie in an $(m-1)$-dimensional space with pairwise nonpositive inner products, so $k - 2 \le 2(m-1)$ by the induction hypothesis. ∎

Theorem 11.9 (Sparse features are superposed). *Let $m < p$ and $0 \le S < 1$, and let $\omega_{\min}$ be the smallest importance.*

(i) A configuration (W, b) satisfies $\mathcal{L}(W, b) = O\big((1-S)^2\big)$ as $S \to 1$ if and only if it is obtuse. If it is not, then $\mathcal{L}(W, b) \ge 2^{1-p}(1-S)\big(\ell_\emptyset + \sum_i \ell_{\{i\}}\big) > 0$ for $S \ge \frac{1}{2}$.

(ii) Obtuse configurations exist if and only if $p \le 2m$. Every obtuse configuration has

$$\mathcal{L}(W, 0) \le \frac{\Omega}{3}(1-S)\big(1 - S^{p-1}\big) \le \frac{(p-1)\Omega}{3}(1-S)^2.$$

(iii) If $p \le 2m$ and $4(1 - S^{p-1})\Omega < \omega_{\min}(1+3S)$, then the loss of every configuration that drops a feature exceeds the loss of every obtuse configuration. In particular, for S close enough to 1, no optimal or nearly optimal configuration drops a feature, and all are in superposition.

Proof. The plan is to read part (i) from the expansion (11.6), part (ii) from Lemma 11.8 and the clipping of negative interference, and part (iii) from Lemma 11.4.

Part (i). By (11.6), $\mathcal{L}(W, b) = O((1-S)^2)$ if and only if $S^p \ell_\emptyset + (1 - S)S^{p-1} \sum_i \ell_{\{i\}} = O((1-S)^2)$, which forces $\ell_\emptyset = 0$ (the coefficient S^p tends to 1) and then $\sum_i \ell_{\{i\}} = 0$; by Lemma 11.7 this means obtuse. Conversely, for an obtuse configuration the first two terms vanish and $\mathcal{L} = R = O((1-S)^2)$. For the lower bound, drop $R \ge 0$ and use $S^p \ge (1-S)S^{p-1}$ and $S^{p-1} \ge 2^{1-p}$, both valid for $S \ge \frac{1}{2}$.

Part (ii), existence. By Lemma 11.8, p unit vectors with pairwise nonpositive inner products exist only if $p \le 2m$. If $p \le 2m$, then $\lceil p/2 \rceil \le m$, and the

directions $w_{2k-1} = e_k$, $w_{2k} = -e_k$ for $k \le \lfloor p/2 \rfloor$, together with $w_p = e_{(p+1)/2}$ when p is odd, are obtuse.

Part (ii), the bound. Fix i and an obtuse configuration. If $x_i = 0$, then $x_i' = \mathrm{ReLU}(\sum_{j\neq i}\langle w_i, w_j\rangle x_j) = 0 = x_i$, since every term of the sum is at most 0. If $x_i > 0$, then $x_i' = \mathrm{ReLU}(x_i + \sum_{j\neq i}\langle w_i, w_j\rangle x_j)$ lies in $[0, x_i]$, so $|x_i - x_i'| \le x_i$, and $x_i' = x_i$ when no other feature is active. By independence, $\mathbb{E}(x_i - x_i')^2 \le \mathbb{E}x_i^2 \cdot \Pr\{\text{some } x_j \neq 0,\ j \neq i\} = \frac{1-S}{3}(1 - S^{p-1})$. Summing with weights ω_i gives the first bound, and $1 - S^{p-1} \le (p-1)(1-S)$ gives the second.

Part (iii). By Lemma 11.4 and (11.1), a configuration that drops a feature has loss at least $\omega_{\min}\frac{(1-S)(1+3S)}{12}$, and this exceeds the bound of part (ii) exactly under the stated inequality. As $S \to 1$ its left side tends to 0 and its right side to $4\omega_{\min}$. A configuration that drops no feature represents $p > m$ of them. ∎

For $p = 2$ and $m = 1$ the obtuse configuration is the antipodal pair, and part (ii) overestimates its exact loss of Theorem 11.5 by a factor of two. For $p \le 2m$ in general, sparse features pair up into antipodal pairs in orthogonal planes, the cross-polytope, or form other obtuse sets such as a regular simplex of $m + 1$ directions with pairwise inner products $-\frac{1}{m}$; at first order all of them are free. With $p > 2m$ features first-order interference cannot be avoided. Then some inner products are positive, and a negative bias must cut off the interference, at the price of also cutting off the bottom of every true signal; part (i) says that the loss of every configuration is then of order $1 - S$. Its constant is positive configuration by configuration; that it is bounded below uniformly is the content of the next remark.

Remark 11.10 (More than $2m$ features). For $p > 2m$ the infimum of $\ell_\emptyset + \sum_i \ell_{\{i\}}$ over all configurations is positive, so the optimal loss is at least a constant times $1 - S$ for $S \ge \frac{1}{2}$.

Proof sketch. Along a sequence of configurations on which this sum tends to 0, $\mathrm{ReLU}(b_j) \to 0$, and the functions $u \mapsto \mathrm{ReLU}(\|w_i\|^2 u + b_i)$ tend to $u \mapsto u$ in $L^2[0,1]$; one checks that this forces $\|w_i\| \to 1$ and $b_i \to 0$, so a subsequence of W converges, and its limit is obtuse, which Lemma 11.8 forbids. We omit the routine verification of the second step; the lower bound then follows from the inequality of part (i) of Theorem 11.9. ∎

The shape of the solutions. In the configurations that training finds (Figure 11.2), the represented directions form regular polygons and polytopes, and the share of a dimension that a feature receives takes simple fractional values. Elhage et al. (2022) define the *feature dimensionality* of a represented feature,

$$\mathrm{D}_i = \frac{\|w_i\|^4}{\sum_j \langle w_i, w_j\rangle^2},$$

which is 1 for a feature with a dimension of its own and smaller under interference; the sum runs over all j, including $j = i$.

Proposition 11.11 (Tight frames share dimensions equally). *Let $w_1, \dots, w_N$ have equal norms and span a subspace V of dimension k, and suppose that they form a* tight frame *of V: $\sum_j w_j w_j^\top = \gamma \Pi_V$ for a number γ, with Π_V the orthogonal projection onto V. Then $\mathrm{D}_i = k/N$ for every i.*

Proof. Taking traces, $N\|w_1\|^2 = \gamma k$. For each i, $w_i \in V$ gives $\sum_j \langle w_i, w_j\rangle^2 = w_i^\top \big(\sum_j w_j w_j^\top\big) w_i = \gamma \|w_i\|^2 = \frac{N}{k}\|w_i\|^4$, so $\mathrm{D}_i = k/N$. ∎

An antipodal pair is a tight frame of a line, so each of its features has dimensionality $\frac{1}{2}$; the vertices of a regular N-gon with $N \geq 3$ are a tight frame of the plane (Exercise 11.5), with dimensionality $\frac{2}{N}$: $\frac{2}{3}$ for a triangle, $\frac{2}{5}$ for a pentagon; a regular tetrahedron in $\mathbb{R}^3$ gives $\frac{3}{4}$. When the represented features split into tight frames of mutually orthogonal subspaces, each frame keeps its own value.

Exponentially many almost orthogonal directions.

Obtuse configurations stop at $2m$ features. Beyond that, interference is unavoidable but can be made small, and in high dimension small costs little room.

Theorem 11.12 (Packing almost orthogonal directions). *For every $\varepsilon \in (0,1)$ and every integer $m \geq 1$, there are $N = \lfloor e^{\varepsilon^2 m/4} \rfloor$ unit vectors $u_1, \dots, u_N \in \mathbb{R}^m$ with $|\langle u_i, u_j\rangle| \leq \varepsilon$ for all $i \neq j$.*

Proof. The plan is to choose the vectors at random and to show that the expected number of bad pairs is less than one. Let $u_i = s_i/\sqrt{m}$ with independent uniformly random sign vectors $s_i \in \{\pm 1\}^m$; each u_i is a unit vector. For $i \neq j$, $m\langle u_i, u_j\rangle = \sum_{k=1}^m s_{ik}s_{jk}$ is a sum of m independent uniform signs, because the products $s_{ik}s_{jk}$ are independent and uniform on $\{\pm 1\}$. Hoeffding's inequality (Theorem A.17) for m variables with values in $[-1,1]$ gives

$$\Pr\big\{|\langle u_i, u_j\rangle| > \varepsilon\big\} \leq \Pr\Big\{\Big|\sum_{k=1}^m s_{ik}s_{jk}\Big| \geq \varepsilon m\Big\} \leq 2\exp\Big(-\frac{2(\varepsilon m)^2}{4m}\Big) = 2e^{-\varepsilon^2 m/2}.$$

Summing over the $\binom{N}{2}$ pairs, the expected number of bad pairs is at most $N(N-1)e^{-\varepsilon^2 m/2} < N^2 e^{-\varepsilon^2 m/2} \leq 1$, where the last step uses $N \leq e^{\varepsilon^2 m/4}$. A nonnegative integer random variable with mean less than 1 is 0 with positive probability, so some choice of signs has no bad pair. ∎

The theorem is a form of the lemma of Johnson and Lindenstrauss (1984), and its exponent is the right order up to a logarithm.[3] For $m = 1000$ and $\varepsilon = 0.25$ it guarantees 6.1 million directions with all cosines at most 0.25 in absolute value; in two samples of 3000 random sign vectors the largest cosines were 0.16 and 0.18.

Reading sparse features through a ReLU.

Almost orthogonal directions are useful only if the features can be read back. For the toy model with unit directions this is one ReLU with a negative bias.

[3] The Welch bound, Exercise 11.4, shows that $N > m$ unit vectors always have some $|\langle u_i, u_j\rangle| \geq \sqrt{(N-m)/(m(N-1))}$ (Welch 1974), so ε cannot go below about $(2m)^{-1/2}$ once $N \geq 2m$. That $\log N$ can be at most of order $\varepsilon^2 m \log(1/\varepsilon)$ is a theorem of Alon (2003).

Theorem 11.13 (Thresholded readout). *Let the directions $w_1, \dots, w_p \in \mathbb{R}^m$ be unit vectors with* coherence $\mu = \max_{i \neq j} |\langle w_i, w_j \rangle|$, *let $s \geq 1$ be an integer, let $\beta \geq s\mu$, and set $b = -\beta \mathbf{1}_p$ in (11.2). If $x \in [0,1]^p$ has at most s nonzero coordinates, then for every i:*

(i) if $x_i = 0$, then $x'_i = 0$;

(ii) if $x_i > 0$, then $x_i - \beta - (s-1)\mu \leq x'_i \leq x_i$; with $\beta = s\mu$, the error is at most $(2s-1)\mu$.

Proof. Write $x'_i = \mathrm{ReLU}(x_i + \iota_i - \beta)$ with the interference $\iota_i = \sum_{j \neq i} \langle w_i, w_j \rangle x_j$, which has at most as many nonzero terms as there are active features $j \neq i$, each of absolute value at most μ. If $x_i = 0$, there are at most s such terms, so $\iota_i - \beta \leq s\mu - \beta \leq 0$ and $x'_i = 0$. If $x_i > 0$, there are at most $s-1$, so $|\iota_i| \leq (s-1)\mu$. The ReLU is nondecreasing, hence $x'_i \leq \mathrm{ReLU}(x_i + (s-1)\mu - \beta) \leq \mathrm{ReLU}(x_i - \mu) \leq x_i$, using $\beta \geq s\mu$ and $x_i \geq 0$; and $x'_i \geq x_i + \iota_i - \beta \geq x_i - \beta - (s-1)\mu$. ∎

Corollary 11.14 (Capacity of the toy model). *Let $p \geq 2$, $s \geq 1$ and $\delta \in (0,1)$. If $m \geq 4(2s-1)^2 \log p / \delta^2$, then there is a configuration of (11.2) that outputs $x'_i = 0$ for every inactive feature and $|x'_i - x_i| \leq \delta$ for every active one, for every $x \in [0,1]^p$ with at most s active features.*

Proof. Let $\varepsilon = \delta/(2s-1) \in (0,1)$. The hypothesis gives $\varepsilon^2 m/4 \geq \log p$, so Theorem 11.12 provides at least p unit vectors with coherence at most ε; take p of them as columns of W. Apply Theorem 11.13 with $\mu \leq \varepsilon$ and $\beta = s\varepsilon \geq s\mu$: the error on active features is at most $\beta + (s-1)\mu \leq (2s-1)\varepsilon = \delta$. ∎

The number of features that fit in m dimensions is therefore exponential in $m\delta^2/s^2$. This is compressed sensing done by a single layer.[4] The guarantee is worst case: it holds for every input with at most s active features, while for random sparse features the interference is a sum of terms with random signs and is typically of order $\mu\sqrt{s}$ rather than μs.

[4] Compressed sensing recovers every s-sparse vector from m of order $s \log(p/s)$ random linear measurements by ℓ_1 minimization (Candès et al. 2006; Donoho 2006). A one-step thresholded readout pays a factor of about s in the number of dimensions, and the Welch bound shows that no guarantee based on coherence alone can do better (Exercise 11.4).

11.5 Reading the solution

The theorems make four kinds of prediction. (i) A linear network never superposes, at every sparsity. (ii) Two features share one dimension when they are sparse and nearly equally important; superposition provably wins for $S > \frac{2+r}{2+5r}$ and for $r > \kappa(S)/v(S)$, and the dedicated configuration is a saddle in the second region. (iii) Features fit in m dimensions without first-order interference exactly up to $p = 2m$, in obtuse configurations with zero bias; beyond $2m$, positive interference appears and must be cut off by a negative bias, and the loss jumps from order $(1-S)^2$ to order $1-S$. (iv) With few active features, the number of storable features is exponential in the dimension.

Figure 11.1 tests prediction (ii). On the left, at $r = \frac{1}{2}$, the optimal loss follows the dedicated curve $rv(S)$ exactly up to $S \approx 0.52$, then breaks away,

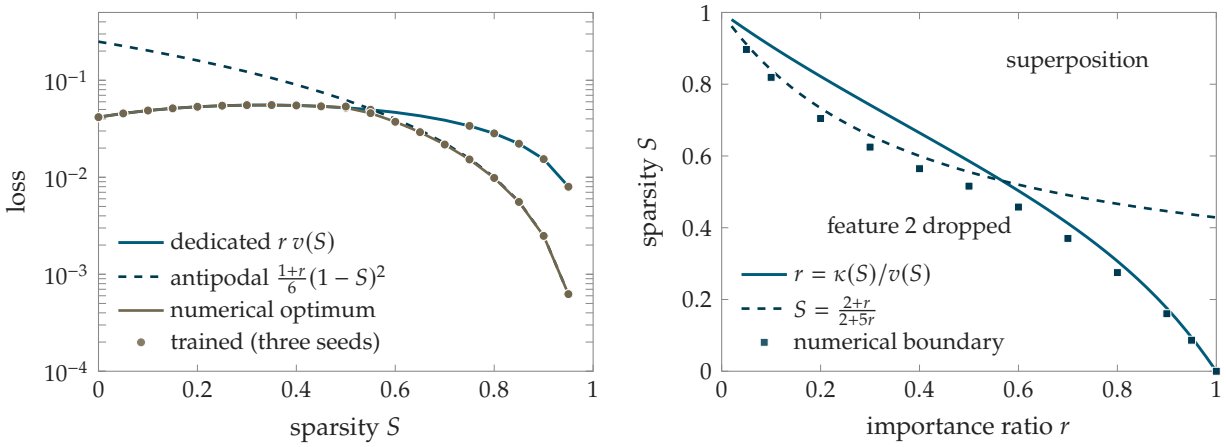


Figure 11.1. Two features in one dimension. Left: importance ratio $r = \frac{1}{2}$. The curves are the exact losses of the dedicated configuration and of the antipodal pair (Theorem 11.5) and the minimum of the exact loss over (w_1, w_2, b_1, b_2), computed by L-BFGS from three starting points. The points are the exact loss of parameters trained by Adam on fresh batches of 2048 samples (1500 steps, learning rate 10^{-2}, then 2×10^{-3} after step 1000; initialization $\mathcal{N}(0, \frac{1}{4})$; seeds 0, 1, 2). Right: in the region above the dots the numerical optimum represents both features, and below them it drops feature 2; superposition is proved above each of the two curves.

and above $S \approx 0.7$ it is within 4% of the antipodal curve, which it approaches as $S \to 1$. Training finds the optimum to within 0.1% whenever it does not stall, and the stalls are informative. At $S = 0.5$ one seed ends in a superposed configuration 3% worse than the dedicated optimum: below the transition the superposed branch is metastable. At $S = 0.55$ two of the three seeds end at the dedicated configuration, which there is a strict local minimum: 0.55 lies between the true transition, $S_c = 0.516$, and the spinodal, $S = 0.586$, at which $\kappa(S)/v(S) = \frac{1}{2}$. For $S \geq 0.75$ one seed still sits near the dedicated configuration, now a saddle, but a flat one: the negative curvature $2(rv(S) - \kappa(S))$ is about 0.02 at $S = 0.9$, and the escape from a saddle with a small starting tilt takes a time of order $\log(1/c_0)$ divided by the curvature, the logistic escape met in Chapters 2 and 7.

On the right, the numerical boundary $S_c(r)$ lies below both proven curves for every r. The transition is first order: along the left panel, the optimal configuration jumps at $S_c = 0.516$ from $w = (1, 0)$ to $w \approx (1.04, -0.77)$, with $\|W\|_F^2$ jumping from 1 to 1.67, and then moves toward the antipodal pair, reaching $w \approx (1.04, -0.91)$ at $S = 0.6$; it does not grow continuously from the dedicated one. Near $r = 1$ the boundary falls to $S = 0$ and meets the spinodal, and at $r = 1$ every $S \geq 0$ superposes, by a small rotation when S is small (at $S = 0$ the optimal loss is 0.08326, against $\frac{1}{12} \approx 0.08333$ for the dedicated configuration). For $r \to 0$ both proven curves and the numerical boundary tend to $S = 1$; a feature much less important than the other is superposed only when it is very sparse. The proven curve $S = \frac{2+r}{2+5r}$ is close to the true boundary at small r, because the transition goes to a nearly antipodal configuration with nearly zero bias, and the spinodal is close to it at large r.

Figure 11.2 tests prediction (iii) with $p = 5$ and $m = 2$, so that $2m = 4$. In

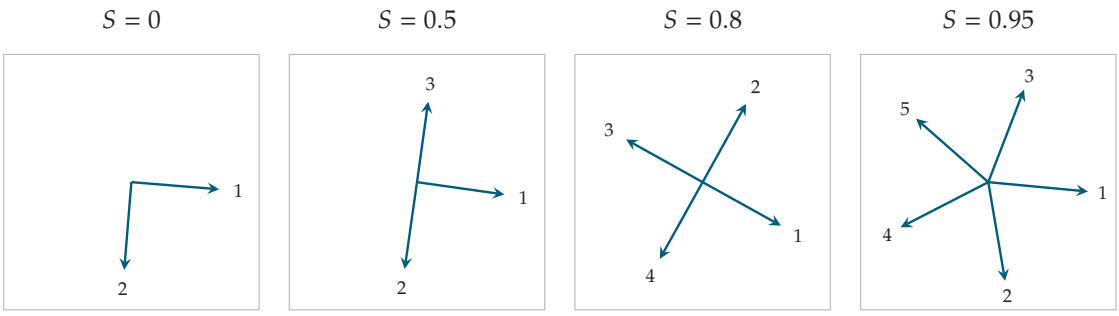


Figure 11.2. Five features in two dimensions: the directions w_i of norm above 0.1 in trained models (importances $\omega_i = 0.8^{i-1}$; Adam, 3000 steps on fresh batches of 4096 samples, learning rate 10^{-2}, then 2×10^{-3} after step 2000; initialization $\mathcal{N}(0, 0.09)$; the better of seeds 0 and 1). Labels are feature indices. Dense: the two most important features, orthogonal. $S = 0.5$: feature 1 alone and features 2, 3 as an antipodal pair. $S = 0.8$: two antipodal pairs, biases below 0.01. $S = 0.95$: all five, nearly a regular pentagon, with norms 1.09 to 1.14 and biases -0.22 to -0.24.

the dense model the two most important features get orthogonal directions and the other three are dropped, with biases within 0.001 of their means $\frac{1-S}{2}$, as Theorem 11.3 and Lemma 11.4 predict. As S grows, features enter in order of importance: three features at $S = 0.5$, as a dedicated direction plus an antipodal pair orthogonal to it, and four at $S = 0.8$, as two antipodal pairs, the obtuse configuration of Theorem 11.9 with zero bias. The fifth feature cannot join without positive interference, and at $S = 0.95$ it has joined: the five directions form a nearly regular pentagon, with consecutive angles between 68.5° and 75°, neighbors have inner products between 0.34 and 0.45, and every bias is about -0.23, the negative threshold that Theorem 11.13 uses. The features per dimension, $\|W\|_F^2/m$, are 1.0, 1.4, 2.0 and 3.2, and the dimensionalities are close to those of Proposition 11.11: 1, 0.54 and 0.46 at $S = 0.5$ (a dedicated feature and an unequal antipodal pair), 0.49 to 0.51 at $S = 0.8$, and 0.38 to 0.41 at $S = 0.95$, against $\frac{2}{5}$ for a regular pentagon.

LIMITS OF THE PREDICTIONS. The phase boundary of the two-feature model is numerical: we prove which side of each curve superposes, not where the true transition is, and we know no closed form for it. The capacity bound is worst case and loose for random inputs. And all predictions concern optimal configurations; the stalls in Figure 11.1 show that training can miss them.

11.6 What the model gets wrong

WHAT WAS THROWN AWAY. *The features are given.* In a network, features are not inputs but functions of the input, learned along with the weights that store them; the model says how a fixed set would be stored, not which set would be learned, and so it cannot say what a polysemantic neuron means. *The features are independent* and identically distributed apart from importance. Correlations change everything the geometry depends on: features that never co-occur can be stored antipodally for free (Exercise 11.8), and features that always co-occur interfere in every input. *The values are*

magnitudes in $[0, 1]$; with binary features the thresholds move (Exercise 11.7), and features with signs cannot use the clipping of negative interference. *The task is reconstruction* through tied weights and one layer, so the model has nothing to say about computing with features stored in superposition. *The loss is the population loss*, so there is no memorization and no finite-sample effect.

WHERE IT BREAKS. *Computation in superposition.* Hänni et al. (2024) (unrefereed) construct one-layer networks that compute pairwise ANDs of many sparse Boolean features, with the inputs themselves in superposition, using far fewer neurons than there are features; there superposition is a resource for computation, not only a way through a bottleneck, and the reconstruction model cannot express this. *Features that are not directions.* Engels et al. (2025a) found, with sparse autoencoders, multidimensional features in language models: days of the week and months of the year lie on circles in GPT-2 and Mistral 7B, and the circles are used to compute with modular arithmetic. A circular feature is a two-dimensional irreducible object, the Fourier feature of Chapter 10, and the toy model, in which every feature is a single direction, cannot produce it. *The geometry is not Euclidean.* Park et al. (2024) formalize the hypothesis that concepts are linear directions, and argue that the inner product in which concept directions are orthogonal is a "causal" inner product determined by the model, not the Euclidean one; the toy model's choice of $\langle \cdot, \cdot \rangle$ is a convention. *The decomposition is not unique.* Sparse autoencoders do not recover a unique dictionary: Paulo and Belrose (2026) find that autoencoders differing only in their random seed share only about 30% of their latents in one large configuration, and Engels et al. (2025b) show that the reconstruction error, the "dark matter", is structured, with about half of the error vector linearly predictable from the input activation. In the toy model the true features are given and every optimal configuration stores them; in a network one can find many sparse dictionaries, and the model offers no criterion for choosing among them. *Data, not features.* With a finite training set, Henighan et al. (2023) found that the same toy model stores individual training examples in superposition when the data set is small and features when it is large, with double descent in between (Chapter 1); the infinite-data model sees only the second regime.

FINE STRUCTURE. The geometry of optimal configurations with $p > 2m$ is richer than Theorem 11.9 suggests: Elhage et al. (2022) report that the directions organize into polytopes whose dimensionalities cluster at fractions such as $\frac{3}{4}$, $\frac{2}{3}$ and $\frac{1}{2}$, that several configurations match solutions of the Thomson problem of placing charges on a sphere, and that the transition of a feature into superposition is abrupt. Scherlis et al. (2022) (unrefereed) recast superposition in a related toy model as an allocation of capacity, the dimensionality D_i, among features: the most important features get dimensions of their own, less important ones share, the least important are dropped, and sharing is more common for sparser and heavier-tailed inputs. Liu et al. (2025) vary the strength of superposition in the toy model with weight decay and find that in the strong-superposition regime the loss scales inversely with the dimension m for a broad class of feature frequencies, a

mechanism for the scaling laws of Chapter 16. On the side of dictionaries, Gao et al. (2025) find clean scaling laws for k-sparse autoencoders. What is open: the dynamics by which gradient descent reaches these polytopes, the theory of computation in superposition beyond constructions, and a criterion that selects one sparse dictionary among many.

11.7 Model card

MODEL CARD	
PHENOMENON	Polysemantic neurons; activations that decompose into sparse combinations of many more directions than dimensions.
MODEL	p independent sparse features in $[0, 1]$, stored in $m < p$ dimensions by W and read out by $\mathrm{ReLU}(W^\top W x + b)$, importance-weighted squared loss.
SOLUTION	Linear readout: principal components at every sparsity. ReLU: exact losses for two features in one dimension, superposition provable for $S > \frac{2+r}{2+5r}$ or $r > \kappa(S)/v(S)$; zero first-order interference if and only if obtuse, possible if and only if $p \leq 2m$; $e^{\varepsilon^2 m/4}$ directions with coherence ε and thresholded readout.
PREDICTS	A first-order transition into superposition with a metastable region; antipodal pairs, cross-polytopes, then polygons with negative bias; loss $O((1-S)^2)$ up to $2m$ features and order $1-S$ beyond; capacity exponential in $m\delta^2/s^2$.
GETS WRONG	Features are given, independent, one-dimensional; reconstruction instead of computation; Euclidean geometry; unique true dictionary.
SUCCESSORS	Computation in superposition; multidimensional and circular features; causal inner products; memorization in superposition; scaling laws from superposition.

The chapter stands at a junction of the book. Its linear half is Chapter 2: a linear autoencoder learns principal components, and in stages, one per singular value, so that under gradient flow features enter one at a time, as they do here when the sparsity grows. Its nonlinear half is a phase transition with metastability, and the stalls of Figure 11.1 are escapes from a saddle, as in Chapters 7 and 8. The almost orthogonal directions of Theorem 11.12 are the random features of Chapter 6 seen from the other side: there random directions are fixed and only a readout is learned; here learned directions are as orthogonal as random ones would be.

The circular features of Engels et al. (2025a) are the Fourier features of Chapter 10, now found in language models: a two-dimensional irreducible representation where this chapter has one direction. The many directions sharing few dimensions are the opposite of the simplex of Chapter 12, in which K class means sit in $K-1$ dimensions, as far apart as possible, because classes are never present together; sparse features are rarely present

together, and that is why superposition works. Finally, the dependence of the loss on the dimension through the number of features stored links superposition to the scaling laws of Chapter 16.

Exercises

11.1 CONSOLIDATION. (Moments and the antipodal pair with a bias.) (a) Derive (11.1). (b) For the antipodal pair with $b = (\beta, \beta)$ and $r = 1$, write the loss of feature 1 as $S^2\beta^2 + S(1-S)(\beta^2 + \frac{\beta^3}{3}) + (1-S)^2 C(\beta)$ for $0 \le \beta \le 1$, with $C(\beta) = \mathbb{E}\big(U - \mathrm{ReLU}(U - V + \beta)\big)^2$. (c) Show that $C'(0) = -\frac{1}{3}$, and conclude that a small positive bias improves the antipodal pair at every $S < 1$. Why does a positive bias help, although it produces false positives?

11.2 CONSOLIDATION. (Ties in the linear model.) (a) With equal importances and equal variances, show that every W with $W^\top W$ an orthogonal projection of rank m minimizes $\mathcal{L}_{\mathrm{lin}}$, so that for $m = 1$, $p = 2$ every unit vector $w = (\cos\theta, \sin\theta)$ is optimal, and represents both features when $\sin 2\theta \neq 0$. (b) Show that $\|W\|_F^2 = m$ for all these minimizers. (c) Show that if $\omega_1 > \omega_2$, then only $\theta \in \{0, \pi\}$ remains.

11.3 CONSOLIDATION. (The function κ.) (a) Show that for $\tau \in [0,1]$, $f_S(\tau) = S^2\tau^2 + \frac{S(1-S)}{3}\tau^3 + (1-S)\big(\tau^2 - (1-S)\tau + \frac{1-S}{3}\big)$, that f_S is decreasing on $(-\infty, 0]$ and increasing on $[1, \infty)$, and that f_S is strictly convex on $[0,1]$. (b) Conclude that for $0 < S < 1$ the minimizer is the root in $(0,1)$ of $S(1-S)\tau^2 + 2(S^2 + 1 - S)\tau - (1-S)^2 = 0$, and that $\tau_0 = \frac{1}{2}$, $\kappa(0) = \frac{1}{12}$. (c) Compute $\kappa(S)/v(S)$ at $S = \frac{1}{2}$ and find the S at which it equals $\frac{1}{2}$; compare with the spinodal in Figure 11.1.

11.4 CONSOLIDATION. ★ (The Welch bound.) Let $u_1, \dots, u_N \in \mathbb{R}^m$ be unit vectors with $N > m$, and let $\mu = \max_{i \neq j} |\langle u_i, u_j \rangle|$. (a) Let G be their Gram matrix. Show that $\|G\|_F^2 \ge (\mathrm{tr}\, G)^2 / \mathrm{rank}\, G \ge N^2/m$, and deduce $\mu^2 \ge \frac{N-m}{m(N-1)}$. (b) Conclude that if $N \ge 2m$, then $(2s-1)\mu < 1$ requires $m > (2s-1)^2/2$, so that Theorem 11.13 needs m of order s^2 to say anything. *(Hint: in (a), the nonzero eigenvalues of G add up to N, and there are at most m of them; use the Cauchy–Schwarz inequality.)*

11.5 CONSOLIDATION. (Tight frames.) (a) Show that the vertices $w_j = (\cos\frac{2\pi j}{N}, \sin\frac{2\pi j}{N})$ of a regular N-gon, $N \ge 3$, satisfy $\sum_j w_j w_j^\top = \frac{N}{2} I_2$, and that this fails for $N = 2$. (b) Compute D_i for the configurations of Figure 11.2 from the data file, and compare with Proposition 11.11.

11.6 VARIATION. (No bias.) Force $b = 0$ in the two-feature model. (a) Show that the best configuration that drops feature 2 has loss $r\frac{1-S}{3}$, and that the antipodal pair still has loss $\frac{1+r}{6}(1-S)^2$. (b) Conclude that the antipodal pair beats every configuration dropping a feature when $S > \frac{1-r}{1+r}$, so for $r = 1$ at every $S > 0$, and that at $S = 0$, $r = 1$ the two tie. Explain why the bias, which lets a dropped feature be replaced by its mean, makes dropping more attractive.

11.7 VARIATION. (Binary features.) Let each feature equal 1 with probability $1-S$ and 0 otherwise. (a) Show that dropping feature 2 costs at least $rS(1-S)$ and that the antipodal pair costs $(1+r)(1-S)^2$. (b) Conclude that superposition beats dropping when $S > \frac{1+r}{1+2r}$, for instance $S > \frac{2}{3}$ at $r = 1$, and compare with $\frac{3}{7}$ for uniform values. Why do binary features superpose less readily?

11.8 VARIATION. (Anticorrelated features.) Let x_1 and x_2 never be active together: $x = (U, 0)$, $(0, U)$, or $(0, 0)$ with probabilities $\frac{1-S}{2}$, $\frac{1-S}{2}$, S. (a) Show that the antipodal pair has loss 0 for every S. (b) More generally, show that if at most one of p features is active at a time, then every obtuse configuration has loss 0, so $p \le 2m$ features fit exactly at every sparsity. (c) Let instead $x_2 = x_1$ always. Show that $w = (1, 1)/\sqrt{2}$ with $b = 0$ has loss 0. Correlations decide the geometry.

11.9 VARIATION. ★ (Random sparsity.) Use the configuration of Corollary 11.14 with $\varepsilon = \delta/(2s-1)$ on the random features of Section 11.2, and let K be the number of active features, a binomial variable with parameters p and $1-S$. (a) Show that, for every x, every coordinate has error at most $\max(K-s, 0)\varepsilon$ if it is inactive and at most $(s+K-1)\varepsilon$ if it is active. (b) Deduce $\mathcal{L} \le \Omega\big[(1-S)\delta^2 + \mathbb{E}\big((s+K)^2\varepsilon^2 \mathbf{1}\{K > s\}\big)\big]$, and choose s for $p(1-S) = 10$ and $p = 10^6$.

11.10 SIMULATION. (The pentagon and its successors.) Run the script of Figure 11.2 for $S \in \{0, 0.1, \dots, 0.9, 0.95, 0.98\}$ and record $\|W\|_F^2$ and the number of directions of norm above 0.1. With the better of seeds 0 and 1, the reader should observe two features up to $S = 0.3$, three at $S = 0.4$ and 0.5, four from $S = 0.6$ to $S = 0.8$, and five from $S = 0.9$ on; the counts depend on the seed (seed 0 gives four at $S = 0.5$, seed 1 only four at $S = 0.9$: training ends in different local optima). The biases of represented features should be above -0.05 with up to four features, and near -0.2 with five. Repeat with equal importances and describe what changes.

11.11 SIMULATION. (Escape from the dedicated saddle.) In the two-feature model with $r = \frac{1}{2}$ and $S = 0.9$, start gradient descent on the exact loss (the function `loss` of the script of Figure 11.1) from $w = (1, -c_0)$, $b = (0, \frac{1-S}{2})$ with $c_0 \in \{10^{-2}, 10^{-4}, 10^{-6}\}$ and step size $\eta = 0.1$. Record the number of steps until $|w_2| > \frac{1}{2}$. The reader should observe about 1200, 3100 and 5000 steps: increments of about 1900 steps per factor 100 in c_0, as predicted by the linearization, $\log(100)/(2\eta(rv(S) - \kappa(S))) \approx 1900$, the logistic escape of Chapter 7. Explain why the linearization predicts the increments but not the counts themselves.

11.12 RESEARCH. (The phase boundary.) Prove that the dedicated configuration is a global minimizer of the two-feature loss for $S < S_c(r)$ for some explicit function S_c, and find the exact transition. The optimum on the superposed side has nonzero biases and unequal norms, and we know of no closed form for it; part (iii) of Theorem 11.5 and the heuristic spinodal give one side only.

11.13 RESEARCH. (Computation in superposition.) Replace reconstruction by a computation, for instance the outputs $|x_i - x_j|$ or $x_i \wedge x_j$ for selected pairs, with a hidden ReLU layer of width m. How many pairs can a width-m network compute when the inputs are s-sparse? Hänni et al. (2024) give constructions; matching lower bounds, the role of training, and the geometry of learned solutions are open.

Notes and References

Overcomplete sparse representations are older than deep learning: Olshausen and Field (1997) learned dictionaries with more atoms than dimensions for natural images and argued that cortex uses such codes, and compressed sensing (Candès et al. 2006; Donoho 2006) made precise when a sparse vector can be recovered from few linear measurements. The packing argument of Theorem 11.12 is the probabilistic method in the form used for the lemma of Johnson and Lindenstrauss (1984), with Hoeffding's inequality (Hoeffding 1963), and Lemma 11.8 is a classical fact about obtuse sets of vectors. Theorem 11.3 is the Eckart–Young theorem (Eckart and Young 1936) specialized to a diagonal matrix, with the bias handled first; its proof by a linear program on the diagonal of a projection is Ky Fan's maximum principle in disguise.

The superposition hypothesis grew out of the circuits program (Olah et al. 2020) and was made concrete by Elhage et al. (2022), whose toy model this chapter solves in its simplest cases; they also studied the phase diagram of two features in one dimension and the polytope geometry in larger cases numerically, and compared the former with the losses of a few configurations. Henighan et al. (2023) extended the toy model to finite data, and Scherlis et al. (2022) to an allocation of capacity. Word senses as sparse combinations of directions are due to Arora et al. (2018). The dictionary-learning program followed: Cunningham et al. (2024) and Bricken, Templeton, et al. (2023) at small scale, Templeton, Conerly, et al. (2024) and Gao et al. (2025) at large scale.[5]

The two-feature analysis of Theorem 11.5, the spinodal, and Theorem 11.9 were derived for this chapter. The comparison of explicit configurations is in the spirit of Elhage et al. (2022), but we have not compared our formulas with theirs line by line, and the bounds on the side of superposition are new as stated here as far as we know.

[5] The Transformer Circuits Thread is the web publication of Anthropic's interpretability team; its articles are not refereed. Elhage et al. (2022) is also on arXiv.

TRANSLATING THE LITERATURE. Elhage et al. (2022) write n for the number of features (our p), m for the number of dimensions (our m), $W \in \mathbb{R}^{m \times n}$ with columns W_i (our w_i), S for the sparsity, as here, I_i for the importance (our ω_i; we keep I for the identity), and $x' = \mathrm{ReLU}(W^\top W x + b)$ for the reconstruction, as here; their feature dimensionality $D_i = \|W_i\|^2 / \sum_j (\hat{W}_i \cdot W_j)^2$, with $\hat{W}_i$ the unit vector, equals our D_i, and their dimensions per feature $m/\|W\|_F^2$ is the inverse of our features per dimension. In Bricken, Templeton, et al. (2023) and Templeton, Conerly, et al. (2024) the activation vector of the layer being decomposed plays the role of our h, the number of dictionary features plays the role of our p, and the rows of the decoder play the role of our directions

w_i; the encoder's ReLU with bias is our thresholded readout.

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12 Neural Collapse

The geometry of the last layer

THE MODEL — UNCONSTRAINED FEATURES

IN THIS CHAPTER. Train a deep classifier long past zero training error and its last layer becomes rigid: the features of each class shrink to their mean, the K class means spread out into a regular simplex, the classifier vectors line up with the means, and the network classifies by the nearest class mean. The hydrogen atom of this neural collapse treats the last-layer features as free variables, constrained only in size. We prove that every global minimizer of the cross-entropy with norm constraints is a simplex, solve the weight-decay and square-loss versions in closed form, find a transition at a single dimensionless strength $\kappa = K\sqrt{n\lambda_W\lambda_H}$ of the regularization, and show that the max-margin problem reached by unregularized gradient descent has the same solution. The model explains the geometry and predicts how it breaks under class imbalance; it is silent on test data, on depth, and on whether collapse is good for anything.

12.1 The phenomenon: four regularities at the end of training

Chapter 4 followed a linear classifier on fixed features into the regime past zero training error, where the cross-entropy keeps falling and the weights keep growing. In a network the features below the last layer keep moving as well. What do they converge to?

Papyan et al. (2020) measured it. They trained networks from three families (VGG, ResNet and DenseNet) on seven image classification data sets (MNIST, FashionMNIST, CIFAR10, CIFAR100, SVHN, STL10 and ImageNet), by stochastic gradient descent, far into what they called the *terminal phase of training*: the epochs after the training error first reaches zero. Throughout that phase they recorded the penultimate-layer features $h \in \mathbb{R}^m$ of the training examples and the last-layer weights. Write K for the number of classes, $h_{k,i}$ for the features of the ith training example of class k, $\bar{h}_k$ for the mean of class k, h_G for the global mean, and w_k for the kth row of the last weight matrix $W \in \mathbb{R}^{K\times m}$, so that the logits of an input with features h are $Wh + b$.

OBSERVATION 12.1 (Papyan et al. 2020). During the terminal phase of training, with the cross-entropy loss and weight decay, four quantities tend toward zero together:

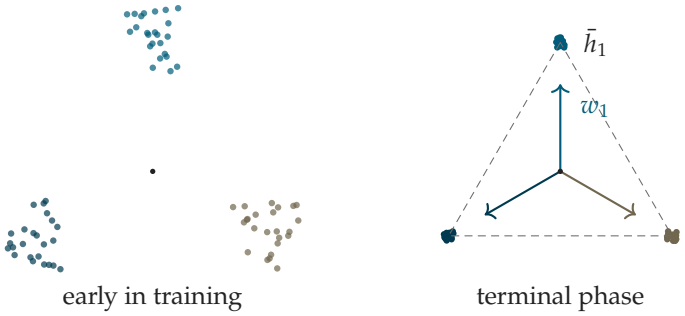


Figure 12.1. Neural collapse for $K = 3$ classes, drawn in the plane of the class means. Left: the penultimate features of the training examples (dots, one color per class) form clouds. Right: in the terminal phase the clouds shrink to their means (NC1), the means form an equilateral triangle centered at the global mean, which is the simplex of NC2, and the classifier vectors w_k (arrows) point at the means (NC3).

(NC1) variability collapse: the within-class scatter, relative to the between-class scatter of the means;

(NC2) convergence to a simplex: the spread of the norms $\|\bar{h}_k - h_G\|$, and the deviation of every cosine between two centered means from $-1/(K-1)$;

(NC3) self-duality: the distance between $W/\|W\|_F$ and the matrix of centered means, normalized the same way;

(NC4) nearest class center: the fraction of training points on which the network's prediction differs from the class of the nearest mean.

The name *neural collapse* covers all four. NC1 is measured by $\mathrm{tr}(\Sigma_W \Sigma_B^+)/K$, with Σ_W the average within-class covariance, Σ_B the covariance of the class means and Σ_B^+ its pseudo-inverse. NC2 says that the centered means become the vertices of a regular simplex centered at the origin, the most spread-out configuration of K points of equal norm; NC3 says that the classifier is that same simplex, rescaled; NC4 follows from the other three (Exercise 12.2). Papyan et al. (2020) argued that the collapsed geometry brings benefits in generalization, robustness and interpretability, and their paper started a literature: Han et al. (2022) reported the same four regularities for networks trained with the square loss, across three networks and five data sets, and He and Su (2023) found a regularity one level down, a *law of equi-separation*: in a trained network each layer improves a measure of class separation by roughly the same factor, so that collapse builds up geometrically with depth.[1]

The status of Observation 12.1 is good on training data and weaker elsewhere. On the training set the four metrics decrease steadily in a wide range of architectures and data sets, but they do not reach zero in finite time, and how close they get depends on the training time, the weight decay and the learning-rate schedule. In a small experiment of our own (Figure 12.3) NC1 fell by a factor of thirty and then stalled. On test data collapse is much weaker (Hui et al. 2022), and NC2 needs at least $K - 1$ feature dimensions,

[1] Rangamani et al. (2023) report a related "intermediate neural collapse": the within-class covariance, relative to the between-class one, decreases with depth.

so it cannot hold literally when the classes outnumber the dimensions, as they do for the vocabulary of a language model (Section 12.6).

The regularity is still striking. A network has millions of parameters and its features are produced by a long nonlinear computation, yet at the end of training the last layer settles into the most symmetric configuration available, the same for every architecture. Why a simplex? And why does the classifier copy it?

12.2 The model: free features of bounded size

The data are $K \geq 2$ classes with n training examples each; deviating from the global convention, n counts the examples per class, so the training set has Kn points. The model keeps only the last layer. The classifier is $W \in \mathbb{R}^{K\times m}$ with rows $w_1, \dots, w_K$, and the features are Kn vectors $h_{k,i} \in \mathbb{R}^m$, collected as the columns of $H \in \mathbb{R}^{m\times Kn}$. The features are *free variables*: they are optimized directly, as if the network below the last layer could produce whatever features it likes. This is the *unconstrained features model*, introduced under that name by Mixon et al. (2020) and as the *layer-peeled model* by Fang et al. (2021). The loss of an example with logits $z \in \mathbb{R}^K$ and class k is the cross-entropy

$$\ell(z, k) = \log \sum_{j=1}^{K} e^{z_j - z_k},$$

and the training loss is $\widehat{\mathcal{L}}(W, H) = \frac{1}{Kn} \sum_{k=1}^{K} \sum_{i=1}^{n} \ell(W h_{k,i}, k)$, without a bias; adding one changes nothing in the main theorem (Exercise 12.3). Write $\bar{w} = \frac{1}{K} \sum_k w_k$ for the mean classifier and $\bar{h}_k = \frac{1}{n} \sum_i h_{k,i}$ for the class means.

The loss alone has no minimizer: scaling up W and H in a separating configuration drives it to zero. Something must bound the size, and the literature uses three devices, each a stand-in for a feature of real training.

Assumption 12.2 (Size of the last layer). One of the following holds.

(i) (Norm constraints.) For given $E_W, E_H > 0$, the parameters satisfy $\frac{1}{K}\|W\|_F^2 \leq E_W$ and $\frac{1}{Kn}\|H\|_F^2 \leq E_H$.

(ii) (Weight decay.) The objective is $\widehat{\mathcal{L}}(W, H) + \frac{\lambda_W}{2}\|W\|_F^2 + \frac{\lambda_H}{2}\|H\|_F^2$ with $\lambda_W, \lambda_H > 0$.

(iii) (Margins.) No regularization; the object of study is the direction of (W, H) along gradient descent, as in Chapter 4.

Norm constraints are the cleanest for proofs; weight decay is what practice uses, on all layers, and a penalty on the features stands in for the penalty on the layers that produce them; and without regularization the margin story of Chapter 4 applies, because the logits Wh are homogeneous of degree two in (W, H). All three give the same geometry.

DICTIONARY

IN THE MODEL	IN A TRAINED NETWORK
classifier W, rows w_k	last weight matrix
free features $h_{k,i} \in \mathbb{R}^m$	penultimate-layer activations of the training examples
bound on $\|H\|_F$	weight decay on all the layers that produce the features
$m \geq K-1$	penultimate width at least the number of classes minus one
class means $\bar{h}_k$, $h_{k,i} = \bar{h}_k$	NC1
simplex equiangular tight frame	NC2
$w_k \propto \bar{h}_k$	NC3
global minimizer, or max-margin direction	the end of the terminal phase of training
regularization strength $\kappa = K\sqrt{n\lambda_W\lambda_H}$	weight decay times the square root of the data per class

WHAT WAS THROWN AWAY. Everything that makes the features depend on the inputs. The model assumes that the network can realize every configuration of last-layer features at a cost that depends only on their norms. Gone too are the test set, the layers below the last, the optimizer, the bias, and every asymmetry between classes. Expressiveness is the right assumption to make first, because a network that drives the training error to zero has already shown that it can place its features almost where it likes; and the question of Observation 12.1 is about the training features only.

WHAT ONE HAD TO NOTICE. That NC3 and NC4 make the last layer look like the solution of an optimization problem in which the features and the classifier play symmetric roles; and that a problem with that much symmetry should have a symmetric ground state. The problem is invariant under every rotation $Q \in O(m)$ acting on both sides, $(W, H) \mapsto (WQ^\top, QH)$, and under every relabeling of the classes. The simplex is the configuration of K points that relabelings map to itself by rotations. Its minimizers will form a single orbit of the rotation group: the permutation symmetry is preserved, and the rotational symmetry is broken, as in a ferromagnet below its Curie point, where the magnetization picks a direction.

12.3 The solution: the simplex is the ground state

We first describe the simplex and the property that makes it optimal.

Definition 12.3 (Simplex equiangular tight frame). Vectors $v_1, \dots, v_K \in \mathbb{R}^m$ form a *simplex equiangular tight frame* of squared norm $r > 0$ if

$$\langle v_j, v_k \rangle = r\left(\delta_{jk} - \frac{1-\delta_{jk}}{K-1}\right) \qquad \text{for all } j, k,$$

that is, if their Gram matrix is $\frac{rK}{K-1}(I_K - \frac{1}{K}\mathbf{1}_K\mathbf{1}_K^\top)$.

The Gram matrix is $\frac{rK}{K-1}$ times the orthogonal projection onto the complement of $\mathbf{1}_K$, so it is positive semidefinite of rank $K-1$, and a simplex equiangular tight frame exists in $\mathbb{R}^m$ if and only if $m \geq K-1$. The vectors sum to zero, because $\|\sum_k v_k\|^2 = \mathbf{1}^\top G \mathbf{1} = 0$ for the Gram matrix G, and every two frames with the same r differ by a rotation of $\mathbb{R}^m$. For $K=2$ the frame is a pair of antipodal points, for $K=3$ an equilateral triangle, for $K=4$ a regular tetrahedron. "Tight frame" refers to $\sum_k v_k v_k^\top$ being a multiple of the projection onto the span of the v_k; "equiangular" to the equal angles.[2]

[2] Equiangular tight frames are a classical subject in frame theory and coding, where they solve packing problems for lines and points on spheres (Strohmer and Heath 2003).

Lemma 12.4 (The simplex is the best code). *Let $u_1, \dots, u_K$ be unit vectors in $\mathbb{R}^m$ with $K \geq 2$. Then*

$$\max_{j \neq k} \langle u_j, u_k \rangle \geq -\frac{1}{K-1},$$

with equality if and only if $u_1, \dots, u_K$ form a simplex equiangular tight frame of squared norm 1.

Proof. Write $\beta = \max_{j \neq k} \langle u_j, u_k \rangle$. Expanding a square,

$$0 \leq \Big\| \sum_k u_k \Big\|^2 = K + \sum_{j \neq k} \langle u_j, u_k \rangle \leq K + K(K-1)\beta,$$

which gives the inequality. Equality holds if and only if both steps are equalities: every $\langle u_j, u_k \rangle$ with $j \neq k$ equals $\beta = -1/(K-1)$, which is the definition of the frame, and then $\sum_k u_k = 0$ holds automatically. ∎

Among K points on a sphere, the simplex makes the largest pairwise correlation as small as possible, and therefore the smallest pairwise distance as large as possible. That is already the shape of the answer. What remains is to show that the loss and the classifier force the features onto a sphere and push them apart, and this is the job of one inequality.

Lemma 12.5 (Alignment). *For every $W \in \mathbb{R}^{K \times m}$ and $H \in \mathbb{R}^{m \times Kn}$,*

$$\frac{1}{Kn} \sum_{k,i} \langle w_k - \bar{w}, h_{k,i} \rangle \leq \Big(\frac{1}{K} \|W\|_F^2 \Big)^{1/2} \Big(\frac{1}{Kn} \|H\|_F^2 \Big)^{1/2}. \tag{12.1}$$

If equality holds and the left side is positive, then $\bar{w} = 0$, $h_{k,i} = \bar{h}_k$ for all k and i, and $w_k = c\,\bar{h}_k$ for all k, with $c = \sqrt{n}\|W\|_F / \|H\|_F > 0$.

Proof. The plan is to average over the examples of each class, apply the Cauchy–Schwarz inequality in $\mathbb{R}^{K \times m}$, and then drop two nonnegative terms. Averaging over i turns the left side of (12.1) into $\frac{1}{K}\sum_k \langle w_k - \bar{w}, \bar{h}_k \rangle$, and the Cauchy–Schwarz inequality bounds this by

$$\Big(\frac{1}{K} \sum_k \|w_k - \bar{w}\|^2 \Big)^{1/2} \Big(\frac{1}{K} \sum_k \|\bar{h}_k\|^2 \Big)^{1/2}.$$

Two variance identities finish the proof:

$$\sum_k \|w_k - \bar{w}\|^2 = \|W\|_F^2 - K\|\bar{w}\|^2, \qquad \|\bar{h}_k\|^2 = \frac{1}{n}\sum_i \|h_{k,i}\|^2 - \frac{1}{n}\sum_i \|h_{k,i} - \bar{h}_k\|^2.$$

Suppose now that equality holds in (12.1) with a positive left side. Then both dropped terms vanish, so $\bar{w} = 0$ and $h_{k,i} = \bar{h}_k$, and equality holds in the Cauchy–Schwarz inequality with a positive inner product, so $w_k - \bar{w} = c\,\bar{h}_k$ for a constant $c > 0$. With $\bar{w} = 0$ this reads $w_k = c\,\bar{h}_k$, and then $\|W\|_F^2 = c^2 \sum_k \|\bar{h}_k\|^2 = c^2\|H\|_F^2/n$ determines c. ∎

The left side of (12.1) is the average over examples of the gap between the mean logit and the logit of the correct class, with the sign reversed. The lemma says that this average gap is at most the product of the root-mean-square sizes of the classifier and the features, and that it can reach that bound only in a configuration that already satisfies NC1 and NC3. The theorem adds NC2.

Theorem 12.6 (Neural collapse under norm constraints). *Let $K \geq 2$, $m \geq K - 1$, and $E_W, E_H > 0$, and minimize $\widehat{\mathcal{L}}(W, H)$ subject to $\frac{1}{K}\|W\|_F^2 \leq E_W$ and $\frac{1}{Kn}\|H\|_F^2 \leq E_H$. The minimum value is*

$$\widehat{\mathcal{L}}^\star = \log\Bigl(1 + (K-1)\exp\Bigl(-\frac{K}{K-1}\sqrt{E_W E_H}\Bigr)\Bigr), \tag{12.2}$$

and (W, H) is a minimizer if and only if

(i) (NC1) $h_{k,i} = \bar{h}_k$ for all k and i;

(ii) (NC2) $\bar{h}_1, \ldots, \bar{h}_K$ form a simplex equiangular tight frame of squared norm E_H;

(iii) (NC3) $w_k = \sqrt{E_W/E_H}\,\bar{h}_k$ for all k.

Proof. The plan is to bound the loss below in three steps, by the arithmetic–geometric mean inequality inside each example, by Jensen's inequality across examples, and by Lemma 12.5, and then to read off the configuration from the cases of equality. Fix a feasible (W, H) and write $z^{k,i} = W h_{k,i}$ for the logits of example (k, i).

Step 1: within an example. The $K - 1$ numbers $e^{z_j - z_k}$ with $j \neq k$ have geometric mean $\exp\bigl(\frac{1}{K-1}\sum_{j\neq k}(z_j - z_k)\bigr)$, and $\frac{1}{K-1}\sum_{j\neq k}(z_j - z_k) = \frac{K}{K-1}(\bar{z} - z_k)$ with $\bar{z} = \frac{1}{K}\sum_j z_j$. Put $\psi(u) = \log(1 + (K-1)e^u)$. The arithmetic–geometric mean inequality gives

$$\ell(z^{k,i}, k) = \log\Bigl(1 + \sum_{j\neq k} e^{z_j^{k,i} - z_k^{k,i}}\Bigr) \geq \psi(u_{k,i}), \qquad u_{k,i} = \frac{K}{K-1}\langle \bar{w} - w_k, h_{k,i}\rangle,$$

because $\bar{z} - z_k = \langle \bar{w} - w_k, h_{k,i}\rangle$ and $\log(1 + x)$ is increasing.

Step 2: across examples. The function ψ is strictly convex and strictly increasing, since $\psi'(u) = (K-1)e^u/(1 + (K-1)e^u)$ is positive and strictly increasing. By Jensen's inequality, $\widehat{\mathcal{L}}(W, H) \geq \psi(\bar{u})$ with $\bar{u} = \frac{1}{Kn}\sum_{k,i} u_{k,i}$.

Step 3: the alignment bound. By Lemma 12.5 and the constraints, $\bar{u} \geq -\frac{K}{K-1}\sqrt{E_W E_H}$, and since ψ is increasing, $\widehat{\mathcal{L}}(W,H) \geq \psi(-\frac{K}{K-1}\sqrt{E_W E_H})$, which is the right side of (12.2).

Attainment. Let (W,H) satisfy (i)–(iii); such configurations exist because $m \geq K-1$, and they are feasible with equality. Put $c = \sqrt{E_W/E_H}$. For $j \neq k$, $z_j^{k,i} - z_k^{k,i} = c(\langle \bar{h}_j, \bar{h}_k \rangle - \|\bar{h}_k\|^2) = -cE_H\frac{K}{K-1} = -\frac{K}{K-1}\sqrt{E_W E_H}$, so every example has loss equal to the right side of (12.2). This proves (12.2) and that (i)–(iii) describe minimizers.

Equality. Conversely, let (W,H) be a minimizer, so that all three steps hold with equality. In Step 3, since ψ is strictly increasing, the left side of (12.1) equals $\sqrt{E_W E_H}$, which is positive and at least the right side of (12.1) by the constraints. So equality holds in Lemma 12.5 and in both constraints: $\bar{w} = 0$, NC1 holds, and $w_k = c\bar{h}_k$ with $c = \sqrt{n}\|W\|_F/\|H\|_F = \sqrt{E_W/E_H}$, which is NC3. In Step 2, strict convexity forces all $u_{k,i}$ to be equal; here $u_{k,i} = -\frac{K}{K-1}c\|\bar{h}_k\|^2$, so all the means have the same squared norm r. In Step 1, equality in the arithmetic–geometric mean inequality forces the numbers $z_j - z_k = c(\langle \bar{h}_j, \bar{h}_k \rangle - r)$, $j \neq k$, to be equal for each k, so $\langle \bar{h}_j, \bar{h}_k \rangle = \beta_k$ for all $j \neq k$; by symmetry of the inner product $\beta_k = \beta_j$, so all the off-diagonal inner products equal one number β. Finally $\sum_k \bar{h}_k = K\bar{w}/c = 0$, and

$$0 = \Big\|\sum_k \bar{h}_k\Big\|^2 = Kr + K(K-1)\beta$$

gives $\beta = -r/(K-1)$. The constraint on H holds with equality and $h_{k,i} = \bar{h}_k$, so $r = \frac{1}{K}\sum_k \|\bar{h}_k\|^2 = E_H$, which is NC2. ∎

The proof follows Fang et al. (2021) and Lu and Steinerberger (2022) in outline, with the equality cases made explicit; Graf et al. (2021) proved a version with spherical constraints at about the same time. Each inequality produces one collapse property: the variance identity for the features gives NC1, the Cauchy–Schwarz step gives NC3, Jensen's inequality across examples gives the means equal norms, and the arithmetic–geometric mean inequality inside each example gives them equal angles, which completes NC2. The minimizer is unique up to rotations of $\mathbb{R}^m$, the orbit predicted in Section 12.2, and every logit gap of every example equals $\frac{K}{K-1}\sqrt{E_W E_H}$: the configuration of Lemma 12.4.

WEIGHT DECAY. The penalized problem reduces to the constrained one at the norms the minimizer chooses, and the choice of norms is a problem in one variable. Write

$$\kappa = K\sqrt{n\lambda_W\lambda_H}, \qquad \rho(W,H) = \Big(\frac{1}{K}\|W\|_F^2 \cdot \frac{1}{Kn}\|H\|_F^2\Big)^{1/2} = \frac{\|W\|_F\|H\|_F}{K\sqrt{n}}.$$

The number κ is the only combination of λ_W, λ_H and n that matters, and ρ is the root-mean-square size of the logits that the last layer can produce.

Corollary 12.7 (Neural collapse under weight decay). *Let $K \geq 2$, $m \geq K-1$ and $\lambda_W, \lambda_H > 0$, and minimize $F(W,H) = \widehat{\mathcal{L}}(W,H) +$*

$\frac{\lambda_W}{2}\|W\|_F^2 + \frac{\lambda_H}{2}\|H\|_F^2$.

(i) If $\kappa \geq 1$, the unique minimizer is $W = 0$, $H = 0$.

(ii) If $\kappa < 1$, the minimizers are exactly the configurations of Theorem 12.6 with $\lambda_W\|W\|_F^2 = \lambda_H\|H\|_F^2$ and

$$\rho = \rho^\star = \frac{K-1}{K}\log\Big(1 + \frac{K(1-\kappa)}{\kappa}\Big). \tag{12.3}$$

At every minimizer the training loss is $\log\frac{K}{K-(K-1)\kappa}$, and every training example is assigned probability $1 - \frac{K-1}{K}\kappa$ to its class.

Proof. The plan is to bound F below by a convex function g of ρ alone, using Theorem 12.6 for the loss and the arithmetic–geometric mean inequality for the penalty, and then to minimize g. Put

$$g(\rho) = \psi\Big(-\frac{K}{K-1}\rho\Big) + \kappa\rho, \qquad \rho \geq 0,$$

with ψ as in the proof of Theorem 12.6. A minimizer of F exists, because F is continuous and $F(W,H) \geq \frac{\lambda_W}{2}\|W\|_F^2 + \frac{\lambda_H}{2}\|H\|_F^2$ tends to infinity with (W,H).

The lower bound. Fix (W,H) and set $E_W = \frac{1}{K}\|W\|_F^2$, $E_H = \frac{1}{Kn}\|H\|_F^2$, so that $\rho = \sqrt{E_W E_H}$. If $\rho > 0$, then (W,H) is feasible for Theorem 12.6 with these budgets, and $\widehat{\mathcal{L}}(W,H) \geq \psi(-\frac{K}{K-1}\rho)$. If $\rho = 0$, then $W = 0$ or $H = 0$, all logits vanish, and $\widehat{\mathcal{L}} = \log K = \psi(0)$. For the penalty, the arithmetic–geometric mean inequality gives

$$\frac{\lambda_W}{2}\|W\|_F^2 + \frac{\lambda_H}{2}\|H\|_F^2 = \frac{K}{2}\big(\lambda_W E_W + n\lambda_H E_H\big) \geq K\sqrt{n\lambda_W\lambda_H E_W E_H} = \kappa\rho,$$

with equality if and only if $\lambda_W E_W = n\lambda_H E_H$, that is, $\lambda_W\|W\|_F^2 = \lambda_H\|H\|_F^2$. Hence $F(W,H) \geq g(\rho(W,H))$.

The function g. It is strictly convex, as a strictly convex function of a linear function plus a linear function. Its derivative is $g'(\rho) = \kappa - \frac{K}{K-1}\psi'(-\frac{K}{K-1}\rho)$, and $\psi'(0) = \frac{K-1}{K}$ gives $g'(0) = \kappa - 1$. If $\kappa \geq 1$, then $g' \geq g'(0) \geq 0$ on $[0,\infty)$, and g is uniquely minimized at $\rho = 0$. If $\kappa < 1$, then $g'(0) < 0$ and $g'(\rho) \to \kappa > 0$ as $\rho \to \infty$, so g is uniquely minimized at the root of g'. With $s = \frac{K}{K-1}\rho$ the equation $g' = 0$ reads $Ke^{-s} = \kappa\big(1 + (K-1)e^{-s}\big)$, so $e^{-s} = \kappa/(K-(K-1)\kappa)$, which is (12.3).

Minimizers. The bound $F \geq g(\rho) \geq \min g$ is attained: for $\kappa \geq 1$ by $(0,0)$, and for $\kappa < 1$ by a configuration of Theorem 12.6 with $\rho = \rho^\star$ and balanced norms, which exists because $m \geq K-1$. So $\min F = \min g$, and (W,H) is a minimizer if and only if $\rho(W,H)$ minimizes g and both inequalities of the lower bound are equalities. For $\kappa \geq 1$ this means $\rho = 0$ and $\lambda_W\|W\|_F^2 = \lambda_H\|H\|_F^2$, so $W = 0$ and $H = 0$. For $\kappa < 1$ it means $\rho = \rho^\star > 0$, balanced norms, and $\widehat{\mathcal{L}}(W,H)$ equal to the minimum of Theorem 12.6 at the budgets E_W, E_H, which by that theorem holds exactly for the configurations it describes. The loss is then $\psi(-s) = \log\big(1 + (K-1)e^{-s}\big) = \log\frac{K}{K-(K-1)\kappa}$, and each example's probability of its class is $1/(1 + (K-1)e^{-s}) = 1 - \frac{K-1}{K}\kappa$. ∎

THE SQUARE LOSS. Han et al. (2022) found neural collapse with the square loss, and in the unconstrained features model the square loss can be solved in closed form. Let $Y \in \mathbb{R}^{K \times Kn}$ be the matrix of one-hot labels, whose column (k, i) is e_k, and add an unpenalized bias $b \in \mathbb{R}^K$:

$$F_{\mathrm{sq}}(W, H, b) = \frac{1}{2Kn}\left\|WH + b\mathbf{1}_{Kn}^\top - Y\right\|_F^2 + \frac{\lambda_W}{2}\|W\|_F^2 + \frac{\lambda_H}{2}\|H\|_F^2. \qquad (12.4)$$

We use the nuclear norm $\|Z\|_*$, the sum of the singular values of Z, and two facts about it: it is the dual of the operator norm, $\|Z\|_* = \max\{\langle G, Z\rangle : \|G\|_{\mathrm{op}} \le 1\}$ with $\langle G, Z\rangle = \mathrm{tr}(G^\top Z)$; and consequently $\|WX\|_* \le \|W\|_F \|X\|_F$, because $\langle G, WX\rangle = \langle W^\top G, X\rangle \le \|W^\top G\|_F \|X\|_F \le \|W\|_F \|G\|_{\mathrm{op}} \|X\|_F$.

Proposition 12.8 (Square loss: closed form). *Let $K \ge 2$, $m \ge K - 1$ and $\lambda_W, \lambda_H > 0$, and let $\kappa = K\sqrt{n\lambda_W\lambda_H}$.*

(i) If $\kappa \ge 1$, the unique minimizer of (12.4) is $W = 0$, $H = 0$, $b = \frac{1}{K}\mathbf{1}_K$.

(ii) If $\kappa < 1$, every minimizer satisfies NC1, $\sum_k \bar{h}_k = 0$, $w_k = \frac{\kappa}{K\lambda_W}\bar{h}_k$ (NC3), $b = \frac{1}{K}\mathbf{1}_K$, and

$$W\bar{h}_k + b = (1 - \kappa)e_k + \frac{\kappa}{K}\mathbf{1}_K \qquad \text{for every } k,$$

so that the means form a simplex equiangular tight frame (NC2) and the correct-class output is $1 - \frac{K-1}{K}\kappa$. The minimum value is $\frac{K-1}{2K}\kappa(2 - \kappa)$.

PROOF. The plan is to eliminate the bias, to bound F_{sq} below by a convex function of $Z = WH$, to solve that convex problem with a dual certificate, and then to read off the configuration from the stationarity conditions. Put $\mu = \sqrt{\lambda_W\lambda_H}$, let $P = I_{Kn} - \frac{1}{Kn}\mathbf{1}\mathbf{1}^\top$ be the projection that centers the columns, and let $\widetilde{Y} = YP$, whose column (k, i) is $e_k - \frac{1}{K}\mathbf{1}_K$.

The bias. For fixed $A = WH - Y$, the minimum of $\|A + b\mathbf{1}^\top\|_F^2$ over b is $\|AP\|_F^2$, attained only at $b = -\frac{1}{Kn}A\mathbf{1}$, and $AP = WHP - \widetilde{Y}$.

The lower bound. By the arithmetic–geometric mean inequality, $\|HP\|_F \le \|H\|_F$, and the second fact about the nuclear norm,

$$\frac{\lambda_W}{2}\|W\|_F^2 + \frac{\lambda_H}{2}\|H\|_F^2 \ge \mu\|W\|_F\|H\|_F \ge \mu\|W\|_F\|HP\|_F \ge \mu\|WHP\|_*. \qquad (12.5)$$

Hence $F_{\mathrm{sq}}(W, H, b) \ge \Phi(WHP)$ with $\Phi(Z) = \frac{1}{2Kn}\|Z - \widetilde{Y}\|_F^2 + \mu\|Z\|_*$.

The convex problem. Let $Q = \widetilde{Y}/\sqrt{n}$. A direct count gives $QQ^\top = I_K - \frac{1}{K}\mathbf{1}\mathbf{1}^\top$, an orthogonal projection of rank $K - 1$, so all the nonzero singular values of Q equal 1: $\|Q\|_{\mathrm{op}} = 1$ and $\|Q\|_* = \langle Q, Q\rangle = K - 1$. Let $G = Q/\max(1, \kappa)$, so that $\|G\|_{\mathrm{op}} \le 1$ and $\|Z\|_* \ge \langle G, Z\rangle$ for every Z. Completing the square,

$$\Phi(Z) \ge \frac{1}{2Kn}\left\|Z - \widetilde{Y} + Kn\mu G\right\|_F^2 + \text{const},$$

a strictly convex function of Z minimized only at $Z_0 = \widetilde{Y} - Kn\mu G = \sqrt{n}\big(1 - \kappa/\max(1, \kappa)\big)Q$, because $Kn\mu = \sqrt{n}\kappa$. At Z_0 the bound is an equality: Z_0 is a nonnegative multiple of Q, so $\|Z_0\|_* = \langle G, Z_0\rangle$ when $\kappa < 1$, and $Z_0 = 0$ when

$\kappa \geq 1$. So Z_0 is the unique minimizer of Φ, with $Z_0 = \sqrt{n}(1-\kappa)Q = (1-\kappa)\widetilde{Y}$ if $\kappa < 1$ and $Z_0 = 0$ otherwise.

Minimizers. A minimizer of F_{sq} exists by coercivity in (W, H) and by the formula for the optimal bias. The configurations described below attain $\Phi(Z_0)$ (Exercise 12.5), so $\min F_{\mathrm{sq}} = \Phi(Z_0)$ and every minimizer has $WHP = Z_0$, equality throughout (12.5), and the optimal bias. If $\kappa \geq 1$, equality in (12.5) with $\|Z_0\|_* = 0$ forces $W = 0$ and $H = 0$, and then $b = \frac{1}{K}\mathbf{1}$; this is part (i). If $\kappa < 1$, the residual is

$$R = WH + b\mathbf{1}^\top - Y = WHP - \widetilde{Y} = -\kappa\widetilde{Y}.$$

The objective is differentiable, so at a minimizer its gradients in H and W vanish:

$$\frac{1}{Kn}W^\top R + \lambda_H H = 0, \qquad \frac{1}{Kn}RH^\top + \lambda_W W = 0.$$

The first gives $h_{k,i} = \frac{\kappa}{Kn\lambda_H}W^\top(e_k - \frac{1}{K}\mathbf{1}) = \frac{\kappa}{Kn\lambda_H}(w_k - \bar{w})$, which depends only on k (NC1) and sums to zero over k. The second, with $\sum_{k,i} h_{k,i} = 0$, gives $W = \frac{\kappa}{Kn\lambda_W}\sum_{k,i}(e_k - \frac{1}{K}\mathbf{1})h_{k,i}^\top = \frac{\kappa}{K\lambda_W}\sum_k e_k \bar{h}_k^\top$, which is NC3. Since the features are centered, the optimal bias is $b = \frac{1}{K}\mathbf{1} - \frac{1}{Kn}WH\mathbf{1} = \frac{1}{K}\mathbf{1}$, and the column of $WH = WHP = Z_0$ for class k is $(1-\kappa)(e_k - \frac{1}{K}\mathbf{1})$; adding b gives the displayed logits. Reading the jth entry with NC3, $\frac{\kappa}{K\lambda_W}\langle \bar{h}_j, \bar{h}_k\rangle = (1-\kappa)(\delta_{jk} - \frac{1}{K})$, so the Gram matrix of the means is a positive multiple of $I_K - \frac{1}{K}\mathbf{1}\mathbf{1}^\top$, which is NC2. The minimum value is $\Phi(Z_0) = \frac{\kappa^2}{2Kn}\|\widetilde{Y}\|_F^2 + \mu(1-\kappa)\sqrt{n}(K-1) = \frac{K-1}{2K}\kappa(2-\kappa)$, using $\|\widetilde{Y}\|_F^2 = n(K-1)$ and $\mu\sqrt{n} = \kappa/K$. ∎

The bias matters. Without it the target is Y itself, whose K singular values all equal $\sqrt{n}$, and the same argument gives class means that are orthogonal rather than a simplex, and needs $m \geq K$ (Exercise 12.6). The cross-entropy needs no bias, because the softmax ignores a common shift of the logits.

12.4 The solution: margins, landscapes, and imbalance

Practice does not minimize a constrained problem; it runs gradient descent, often with little weight decay. Three questions remain: where gradient descent goes without regularization, whether it can get stuck, and what happens when the classes are not balanced.

THE MARGIN PROBLEM. Without regularization the logits Wh are homogeneous of degree two in $\theta = (W, H)$, and the cross-entropy has the exponential tail of Chapter 4 in every logit gap. By the results described in that chapter for homogeneous models, once the training loss is small the direction of θ along gradient flow is governed by the max-margin problem

$$\min \frac{1}{2}\left(\|W\|_F^2 + \|H\|_F^2\right) \quad \text{subject to} \quad \langle w_k - w_j, h_{k,i}\rangle \geq 1 \ \text{ for all } k, i, \ j \neq k. \tag{12.6}$$

Theorem 12.9 (The max-margin solution is neural collapse). *Let $K \geq 2$ and $m \geq K - 1$. The minimum value of (12.6) is $\sqrt{n}(K-1)$, and (W, H) is a minimizer if and only if NC1 holds, the means form a simplex equiangular*

tight frame of squared norm $\frac{K-1}{K\sqrt{n}}$*, and* $w_k = \sqrt{n}\,\bar{h}_k$ *for every* k*.*

PROOF. The plan is to average the constraints into a single inequality for the left side of (12.1), and then to apply Lemma 12.5. Let (W, H) be feasible. Summing the constraints of example (k, i) over $j \neq k$ gives $\langle Kw_k - K\bar{w}, h_{k,i}\rangle \geq K - 1$, that is,

$$\langle w_k - \bar{w}, h_{k,i}\rangle \geq \frac{K-1}{K} \qquad \text{for every } (k, i). \tag{12.7}$$

Averaging over the Kn examples and applying Lemma 12.5, $\frac{K-1}{K} \leq \frac{\|W\|_F\|H\|_F}{K\sqrt{n}}$, so $\|W\|_F\|H\|_F \geq \sqrt{n}(K-1)$, and

$$\frac{1}{2}\left(\|W\|_F^2 + \|H\|_F^2\right) \geq \|W\|_F\|H\|_F \geq \sqrt{n}(K-1).$$

Now let (W, H) attain the bound. Then $\|W\|_F = \|H\|_F$, equality holds in Lemma 12.5 with a positive left side, and every inequality (12.7) is an equality. The lemma gives $\bar{w} = 0$, NC1, and $w_k = c\bar{h}_k$ with $c = \sqrt{n}\|W\|_F/\|H\|_F = \sqrt{n}$. Equality in (12.7) means that the $K-1$ constraints of each example, each at least 1, sum to $K-1$, so each equals 1: $c(\|\bar{h}_k\|^2 - \langle \bar{h}_j, \bar{h}_k\rangle) = 1$ for all $j \neq k$. Exchanging j and k shows that all the means have the same squared norm r, and then all the off-diagonal inner products equal $r - 1/c$. As in the proof of Theorem 12.6, $\sum_k \bar{h}_k = 0$ forces $r - 1/c = -r/(K-1)$, so $r = \frac{K-1}{Kc} = \frac{K-1}{K\sqrt{n}}$. Conversely, a configuration of this form exists because $m \geq K-1$; every constraint is active, since $\sqrt{n}(\|\bar{h}_k\|^2 - \langle \bar{h}_j, \bar{h}_k\rangle) = \sqrt{n}\, r \frac{K}{K-1} = 1$; and $\|H\|_F^2 = nKr = \sqrt{n}(K-1) = \|W\|_F^2$, so its value is $\sqrt{n}(K-1)$. ∎

Theorem 12.9 identifies the global minimizers of (12.6). The problem is not convex, and the theory of Chapter 4 for homogeneous models gives convergence to its Karush–Kuhn–Tucker points, not to its minimizers. For the unconstrained features model, the landscape closes this gap.

Theorem 12.10 (Benign landscapes; stated with a sketch). *(i) (Ji et al. 2022) For the cross-entropy without regularization, gradient flow on* (W, H) *converges in direction to Karush–Kuhn–Tucker points of* (12.6)*, and the unregularized loss has a benign landscape: its critical points that are not global minimizers are strict saddles, in a sense they make precise.*

(ii) (Zhu et al. 2021) For the cross-entropy with weight decay on W, H *and a bias, and with a feature dimension large enough relative to* K*, every critical point is either a global minimizer, which is neural collapse, or a strict saddle, at which the Hessian has a negative eigenvalue.*

(iii) (Zhou et al. 2022) The same holds for the square loss with weight decay in the unconstrained features model.

PROOF SKETCH. We describe a mechanism common to parts (ii) and (iii), and omit the case analysis, for which we refer to the three papers. The

objective depends on W and H through $Z = WH$ and through the penalty, and by (12.5) and its equality case (Exercise 12.5), minimizing the penalty over factorizations of a fixed Z gives $\mu\|Z\|_*$. So the problem in (W, H) is a factorized form of the convex problem $\min_Z \widehat{\mathcal{L}}(Z) + \mu\|Z\|_*$. A critical point of the factorized problem that is not a global minimizer either has an unbalanced factorization or a product Z that is not optimal for the convex problem; when m exceeds the rank of the optimal Z, there is room in $\mathbb{R}^m$ to move W and H in a new direction along which the objective decreases to second order, which is the negative curvature. For part (i), the convergence to Karush–Kuhn–Tucker points is the homogeneous-model theory of Lyu and Li (2020) described in Chapter 4, and Ji et al. (2022) analyze the landscape of the unregularized loss directly. ■

Strict saddles do not attract gradient descent from random initialization, so the theorem predicts that gradient methods find neural collapse in the model, and they do: in all our runs of Figure 12.2, quasi-Newton descent from random starting points reached the predicted minimum to six digits.

IMBALANCED CLASSES. Suppose now that K_A majority classes have n_A examples each and $K_B \geq 2$ minority classes have n_B each, with imbalance ratio $R = n_A/n_B$, $K = K_A + K_B$, and loss averaged over all examples. Following Fang et al. (2021), constrain $\frac{1}{K}\sum_k \|w_k\|^2 \leq E_W$ and $\frac{1}{K}\sum_k \frac{1}{n_k}\sum_i \|h_{k,i}\|^2 \leq E_H$, where n_k is the size of class k. Imbalance breaks the permutation symmetry, and with it the simplex. Fang et al. (2021) proved that it breaks in a definite way.

Theorem 12.11 (Minority collapse; Fang et al. 2021; stated with a sketch). *Let $m \geq K$. With K_A, K_B, E_W, E_H fixed, let (W^R, H^R) be a minimizer of the imbalanced problem for each R. Then $\lim_{R\to\infty}(w_k^R - w_{k'}^R) = 0$ for every two minority classes k and k'.*

The minority classifiers become identical, so that the network cannot tell the minority classes apart, however their features are arranged. The key step is an averaging lemma.

Lemma 12.12 (Averaging the minority classifiers). *Fix the features and the majority classifiers, and replace each minority classifier w_j, $j \in B$, by their mean $\bar{w}_B = \frac{1}{K_B}\sum_{j\in B} w_j$. Then the loss of every majority example does not increase, and $\sum_{j\in B}\|w_j\|^2$ strictly decreases unless the w_j, $j \in B$, are already equal.*

PROOF. Let (k, i) be an example of a majority class, with logits $z = Wh_{k,i}$. Its loss is

$$\ell(z, k) = \log\Big(\sum_{j\notin B} e^{z_j} + \sum_{j\in B} e^{\langle w_j, h_{k,i}\rangle}\Big) - z_k,$$

an increasing function of the sum over B, and the averaging changes nothing else. By convexity of the exponential, $\sum_{j\in B} e^{\langle w_j, h\rangle} \geq K_B e^{\langle \bar{w}_B, h\rangle}$, which is the value after averaging. The norm statement is strict convexity of $\|\cdot\|^2$: $\sum_{j\in B}\|w_j\|^2 = K_B\|\bar{w}_B\|^2 + \sum_{j\in B}\|w_j - \bar{w}_B\|^2$. ■

WHY THE MINORITY CLASSIFIERS MERGE. Take $K_A = K_B = 2$ for definiteness. The minority examples carry a fraction $1/(1+R)$ of the loss, so as $R \to \infty$ the problem tends to that of minimizing the majority loss alone. In that limit the minority features cost budget and earn nothing, so they shrink to zero, and the minority classifiers matter only through the logits they give to majority examples, which they should make as small as possible. By Lemma 12.12, merging the two minority classifiers into their mean never hurts the majority examples and frees classifier budget, which can be spent on the majority margins; so at the limiting optimum the two minority classifiers coincide, both pointing away from the majority means. What this argument does not show is that the freed budget strictly lowers the loss at every limiting optimum; Fang et al. (2021) give the full proof. ◇

The numerical solutions of Figure 12.4 show something stronger than the limit: the two minority classifiers become exactly equal at a finite imbalance ratio, which grows quickly with the budget. With the square loss and no bias the threshold can be computed: a class whose size n_k satisfies $n_k \le N^2 \lambda_W \lambda_H$, where N is the total number of examples, gets the classifier $w_k = 0$ and zero features, so all such minority classes collapse to the same point (Exercise 12.7).

12.5 Reading the solution

The solution makes four predictions about the last layer of a classifier trained to convergence.

The geometry is universal. With norm constraints, weight decay, or no regularization at all (in the max-margin limit), with the cross-entropy or with the square loss and a bias, the class means form a simplex, the classifier copies it, and the features collapse. Only the scale depends on the details.

A transition at $\kappa = 1$. Corollary 12.7 and Proposition 12.8 have the same threshold. For $\kappa = K\sqrt{n\lambda_W\lambda_H} \ge 1$ the regularization wins and the last layer is zero; for $\kappa < 1$ it is a simplex whose scale grows as κ decreases. Near the threshold both orders of the transition can be read off: $\rho^\star \approx (K-1)(1-\kappa)$ for the cross-entropy and $\rho = \frac{K-1}{K}(1-\kappa)$ for the square loss, and since ρ is a product of two norms, $\|W\|_F$ and $\|H\|_F$ vanish like $(1-\kappa)^{1/2}$: a continuous transition with the mean-field exponent $\frac{1}{2}$ of Landau theory.[3] With two or more linear layers on top of the features the transition becomes discontinuous (Exercise 12.8).

The confidence is set by κ *alone.* The probability that the cross-entropy solution assigns to the correct class and the correct-class output of the square-loss solution are the same number, $1 - \frac{K-1}{K}\kappa$. Small weight decay gives confident training predictions, and the training loss of the cross-entropy solution, $\log \frac{K}{K-(K-1)\kappa} \approx \frac{K-1}{K}\kappa$ for small κ, is proportional to κ.

Logits grow like $\log(1/\lambda)$. As $\kappa \to 0$, $\rho^\star = \frac{K-1}{K}\log\frac{1}{\kappa} + O(1)$: the logits grow logarithmically in the inverse weight decay. This is the regularized counterpart of the $\log t$ growth of Chapter 4, where time plays the role of $1/\lambda$.

[3] In Landau's theory of a continuous transition the order parameter vanishes like the square root of the distance to the critical point; here $\|W\|_F^2 \propto \rho$ plays the role of the squared order parameter.

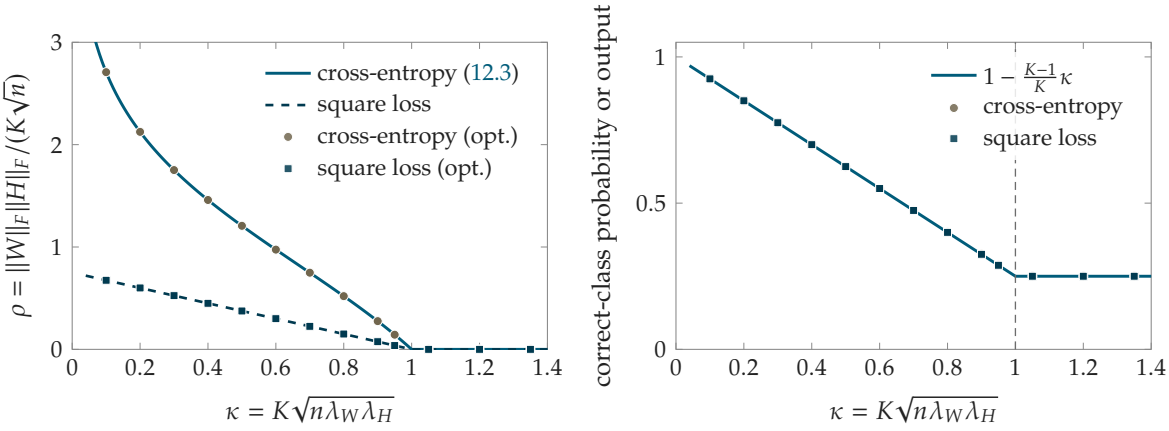


Figure 12.2. The transition at $\kappa = 1$ in the unconstrained features model with weight decay, $K = 4$, $n = 5$, $m = 6$, $\lambda_W = \lambda_H$. Curves: Corollary 12.7 for the cross-entropy and Proposition 12.8 for the square loss with a bias. Points: the regularized objective minimized numerically by L-BFGS from three random initializations (seeds 0, 1, 2), keeping the best. Left: the scale ρ of the last layer. Right: the probability of the correct class (cross-entropy) and the correct-class output (square loss), which the theory predicts to be equal.

Figure 12.2 tests the first three predictions. The points are numerical minimizers of the regularized objectives from random initializations, and they fall on the curves to the precision of the optimizer: at $\kappa \in \{0.1, 0.5, 0.9, 0.95\}$ the discrepancy in ρ is below 10^{-7} for both losses, the squared within-class scatter is at most about 10^{-13} of the squared size of the means, and the cosines between the centered means are within 10^{-6} of $-\frac{1}{3}$. For $\kappa > 1$ the optimizer returns $W = 0$, $H = 0$, as Corollary 12.7 and Proposition 12.8 predict. In the left panel the cross-entropy scale diverges logarithmically as $\kappa \to 0$, while the square-loss scale saturates at $\frac{K-1}{K}$, where the outputs equal their targets. The model is exactly solvable, and a figure of this kind tests the algebra and the landscape (the optimizer was never trapped), not the model's relevance to networks.

Figure 12.3 turns to a network. The training error reaches zero near step 35; at that point NC1 has fallen from 2.3 to 0.2, mostly as a by-product of learning to classify. In the terminal phase the metrics decrease again, slowly: NC2 from 0.35 to 0.09 and NC3 from 0.32 to 0.08 over the next twenty thousand steps, and NC4 is exactly zero throughout. The prediction of the model is that they tend to zero. They do not, within our budget: NC1 stalls near 0.08 after step 2000, and the test NC1 stalls near 0.21, more than twice the training value; the test error wanders between 7.5% and 8.7% during the terminal phase, ending near its highest value. The geometry of the model is the direction in which the network moves; the network stops short of it.

Figure 12.4 shows Theorem 12.11 at work. With budget $E_W = E_H = 1$ the two minority classifiers are identical from $R \approx 5$ on; with budget 3, from $R \approx 300$; with budget 10 their cosine has only risen from -0.33 to -0.17 at $R = 1000$. A larger budget means a smaller effective regularization, and the merging is a consequence of the regularization: with a small budget the

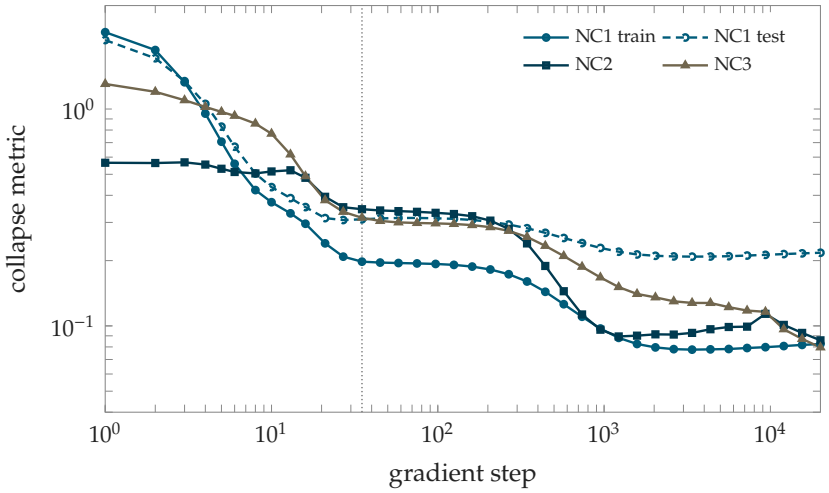


Figure 12.3. Collapse metrics of a ReLU network $32 \to 256 \to 256 \to 32 \to 4$ trained by full-batch gradient descent (step 0.1, momentum 0.9, weight decay 10^{-3} on all parameters, cross-entropy, seed 0) on a Gaussian mixture with $K = 4$ classes in $\mathbb{R}^{32}$ (class means $\mathcal{N}(0, \frac{1}{4}I)$, unit noise, 64 training examples per class). NC1 is $\operatorname{tr}(\Sigma_W \Sigma_B^+)/K$ on the training and on 2000 test examples; NC2 is the Frobenius distance between the normalized Gram matrix of the centered means and that of a simplex; NC3 is the distance between $W/\|W\|_F$ and the centered means normalized the same way. The training error reaches zero near step 35 (dotted line). The unconstrained features model predicts that all the metrics tend to zero.

classifier norm is precious, and the minority classes are not worth it.

12.6 What the model gets wrong

WHAT WAS THROWN AWAY. *The inputs.* The features are free, so the model cannot distinguish a training set that is easy to collapse from one that is hard, and it cannot predict how fast collapse happens or whether it happens at all for a network of given width and depth. *The test set.* With no population, there is nothing to say about the features of unseen examples, which is where collapse would matter for generalization. *The layers below.* A penalty on $\|H\|_F$ replaces the weight decay on all the layers that produce the features; the replacement is exact for no architecture. *Balance and room.* The simplex needs equal class sizes and $m \geq K - 1$; both fail in common settings. *The optimizer.* The theorems concern minimizers and limits; the time scale of collapse is absent.

WHERE IT BREAKS. *Test features do not collapse.* Hui et al. (2022), in an unrefereed preprint, separate collapse on training data, which they find, from collapse on test data, which they find much weaker, and conclude that neural collapse is primarily an optimization phenomenon whose relation to generalization is unclear; in Figure 12.3 the test NC1 stalls at more than twice the training value. *Collapse can hurt transfer.* Kornblith et al. (2021) compared nine training objectives on ImageNet and found that those producing more class separation in the penultimate layer gave higher accuracy but frozen

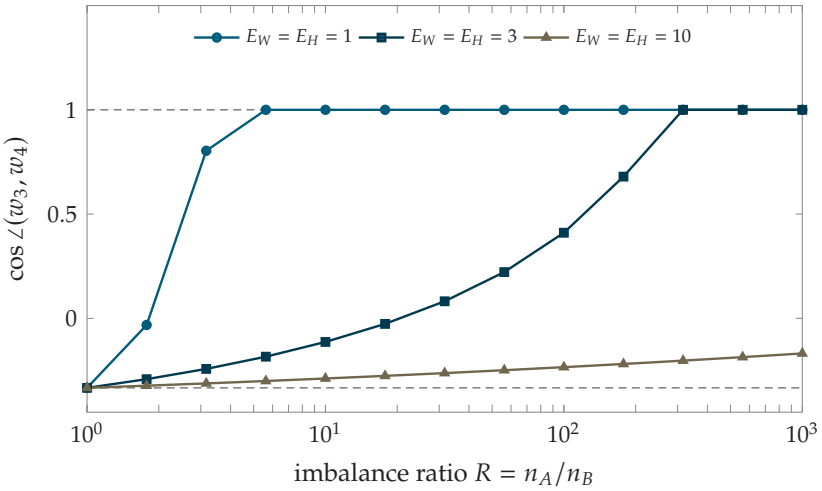


Figure 12.4. Minority collapse in the norm-constrained model with $K_A = 2$ majority and $K_B = 2$ minority classes, $m = 4$: the cosine between the two minority classifiers at the numerical optimum (SLSQP from four random starts, seeds 0 to 3, best kept) against the imbalance ratio. At $R = 1$ the classifiers form a simplex (cosine $-\frac{1}{3}$, lower dashed line); as R grows they merge (cosine 1, upper dashed line), at a ratio that grows quickly with the budget.

features that transferred worse; Hui et al. (2022) report that training longer can likewise hurt transfer, while Galanti et al. (2022) argue that collapse helps few-shot learning on new classes. The model, which knows only the training classes, can decide neither. *Depth changes the optimum.* With several free layers between the features and the classifier, a deep unconstrained features model, the global optimum exhibits neural collapse layer after layer for binary classification (Súkeník et al. 2023); but beyond two layers or two classes, solutions of lower rank have lower cost, because a product of penalized layers favors low rank (Súkeník et al. 2024). Exercise 12.8 shows the mechanism: depth makes the penalty concave in the singular values. *More classes than dimensions.* The classes of a language model are tokens, and the vocabulary exceeds the embedding dimension, so no simplex fits: the geometry is frustrated, as for many charges on a sphere. Jiang et al. (2024) show that the minimizers of a spherically constrained model then maximize the smallest one-versus-rest margin, under conditions on m and K; Wu and Papyan (2024) find that in causal language models, where the vocabulary is also imbalanced and training lasts a few epochs, collapse properties develop with scale and are linked to generalization; and Zhao et al. (2024) argue, with experiments on synthetic and small real data sets, that next-token prediction drives the logits toward a sparse-plus-low-rank structure in which contexts with the same set of possible successors collapse once projected onto a suitable subspace. *Imbalance.* For moderate imbalance the geometry is not a simplex: Thrampoulidis et al. (2022) characterize it, for vanishing regularization, by the singular value decomposition of a simplex-encoded label matrix, and find that ridge regularization matters for the minority classes, as the dependence on the budget in Figure 12.4

suggests.

FINE STRUCTURE. Tirer and Bruna (2022) show that for the regularized square loss the minimizers of the model can have a more delicate structure than for the cross-entropy, and extend their results to a model with a layer of weights and a ReLU below the free features. Jacot et al. (2025) remove the free features: for networks ending in at least two linear layers, trained with weight decay and with a wide first layer, they prove approximate collapse at solutions with low training error, balanced linear layers and well-conditioned features, and give conditions under which gradient descent with weight decay finds such solutions. For dynamics, Han et al. (2022) observe that the classifier stays close to the least-squares classifier of the current features, and analyze a renormalized gradient flow on the resulting central path, which predicts collapse. E and Wojtowytsch (2022) explain the simplex through toy models of very expressive networks, and He and Su (2023) and Rangamani et al. (2023) document partial collapse that builds up with depth. *Open:* the rate of collapse in a network of given architecture; a model with data in which test features collapse partially; and the geometry that gradient descent selects when $K > m + 1$.

12.7 Model card

MODEL CARD	
PHENOMENON	In the terminal phase of training, last-layer features collapse to their class means (NC1), the means form a simplex equiangular tight frame (NC2), the classifier aligns with them (NC3), and prediction becomes nearest class center (NC4).
MODEL	Unconstrained features: free $h_{k,i} \in \mathbb{R}^m$, classifier W, K balanced classes, $m \geq K - 1$; cross-entropy or square loss with norm constraints, weight decay, or no regularization (max-margin limit).
SOLUTION	Every minimizer is neural collapse, with closed-form scale; a transition at $\kappa = K\sqrt{n\lambda_W\lambda_H} = 1$; the max-margin problem has the same solution; benign landscapes (sketched); minority collapse under imbalance (sketched).
PREDICTS	The simplex geometry; correct-class confidence $1 - \frac{K-1}{K}\kappa$; logits growing like $\log(1/\kappa)$; minority classifiers merging at large imbalance, at a threshold that grows with the budget.
GETS WRONG	Silent on test data and transfer; partial collapse in networks; deeper free models prefer lower rank; no simplex when $K > m + 1$ (language models); the geometry under moderate imbalance is not a simplex.
SUCCESSORS	Deep and extended unconstrained features models; generalized neural collapse; SELI geometry; central-path dynamics; collapse in end-to-end training of wide networks.

The chapter ends the thread on implicit bias that starts in Chapter 3 and runs through Chapter 4. In Chapter 4 the features were fixed and gradient descent found the max-margin classifier; here the features are free as well, the margin problem (12.6) is solved over both, and the solution is the simplex: the max-margin attractor of a model that is homogeneous of degree two. The Fourier features of Chapter 10 are another solution of a margin problem for a homogeneous network, found the same way, by symmetry. The superposition of Chapter 11 is the opposite regime: there more features must be stored than there are dimensions, and the geometry is a compromise; here there is room, and the geometry is perfect.

The unconstrained features model is a mean-field theory: it replaces the layers below by a single effective cost on the features, and it gets the symmetry of the ground state right while missing the test set and the finite training time.

Exercises

12.1 CONSOLIDATION. (The simplex.) Let $U \in \mathbb{R}^{m\times K}$ have orthonormal columns, $m \geq K$, and put $M = \sqrt{\frac{rK}{K-1}}\, U(I_K - \frac{1}{K}\mathbf{1}\mathbf{1}^\top)$. (a) Show that the columns of M form a simplex equiangular tight frame of squared norm r. (b) Show that $MM^\top$ is $\frac{rK}{K-1}$ times an orthogonal projection of rank $K-1$, which is the tight-frame property. (c) For $m = K-1$, construct a frame explicitly by projecting the standard basis vectors of $\mathbb{R}^K$ onto the hyperplane orthogonal to $\mathbf{1}_K$ and choosing an orthonormal basis of that hyperplane.

12.2 CONSOLIDATION. (NC4 from NC1–NC3.) Suppose that the training features satisfy NC1–NC3 with means of equal norm centered at the origin and zero bias. Show that $\arg\max_k \langle w_k, h\rangle = \arg\min_k \|h - \bar{h}_k\|$ for every $h \in \mathbb{R}^m$. Show by an example with $K = 2$ that the conclusion fails if the means have unequal norms.

12.3 CONSOLIDATION. (A bias does not help.) Add a bias $b \in \mathbb{R}^K$ to the logits in Theorem 12.6, unconstrained. Show that the lower bound of the proof still holds, because $\frac{1}{Kn}\sum_{k,i}(\bar{b} - b_k) = 0$ for balanced classes, and deduce that the minimum value is unchanged and that at every minimizer b is a multiple of $\mathbf{1}_K$.

12.4 CONSOLIDATION. (Checking Corollary 12.7.) (a) Show that at a minimizer with $\kappa < 1$, $\|W\|_F^2 = K\rho^\star\sqrt{n\lambda_H/\lambda_W}$ and $\|H\|_F^2 = K\rho^\star\sqrt{n\lambda_W/\lambda_H}$. (b) Show that $\rho^\star = (K-1)(1-\kappa) + O((1-\kappa)^2)$ as $\kappa \uparrow 1$ and $\rho^\star = \frac{K-1}{K}\log\frac{1}{\kappa} + \frac{K-1}{K}\log K + o(1)$ as $\kappa \to 0$.

12.5 CONSOLIDATION. (Attaining the square-loss bound.) Let $\kappa < 1$. Take a simplex frame $\bar{h}_1, \dots, \bar{h}_K$ of squared norm r, put $h_{k,i} = \bar{h}_k$, $w_k = c\,\bar{h}_k$ and $b = \frac{1}{K}\mathbf{1}$. (a) Find c and r such that $\lambda_W\|W\|_F^2 = \lambda_H\|H\|_F^2$ and $WH = (1-\kappa)\widetilde{Y}$. (b) Show that then $\|W\|_F\|H\|_F = \|WH\|_*$, so that every inequality in (12.5) is an equality and $F_{\mathrm{sq}} = \Phi(Z_0)$.

12.6 VARIATION. (The square loss without a bias.) Drop b from (12.4) and assume $m \geq K$. Show that the unique minimizer of the analogue of Φ is $(1-\kappa)_+ Y$, and that for $\kappa < 1$ every minimizer has collapsed features whose means are orthogonal with equal norms: an orthogonal frame, not a simplex. What does NC4 look like for this geometry?

12.7 VARIATION. (Minority collapse, exactly solvable.) In the square loss without a bias, let class k have n_k examples, $N = \sum_k n_k$, and minimize $\frac{1}{2N}\|WH - Y\|_F^2 + \frac{\lambda_W}{2}\|W\|_F^2 + \frac{\lambda_H}{2}\|H\|_F^2$ with $m \geq K$. (a) Show that $YY^\top = \operatorname{diag}(n_1, \ldots, n_K)$ and that the analogue of Φ is minimized by $Z_0 = \sum_k (1 - N\mu/\sqrt{n_k})_+ e_k \mathbf{1}_{(k)}^\top$, where $\mathbf{1}_{(k)}$ is the indicator of the examples of class k and $\mu = \sqrt{\lambda_W \lambda_H}$. *(Hint: the singular value decomposition of Y is explicit, and the dual certificate of Proposition 12.8 is replaced by a different multiple of each singular pair.)* (b) Show that a class with $n_k \leq N^2 \lambda_W \lambda_H$ has $w_k = 0$ and $h_{k,i} = 0$ at every minimizer, so that all such classes become indistinguishable. (c) With $n = (20, 20, 3)$ and $\lambda_W = \lambda_H = \sqrt{8}/43$, check that the correct-class output is 0.3675 for the two large classes and 0 for the small one.

12.8 VARIATION. ★★ (Depth makes the transition discontinuous.) Replace WH in (12.4) by $W_2 W_1 H$ with $W_1 \in \mathbb{R}^{m \times m}$, penalize $\frac{\lambda}{2}(\|W_2\|_F^2 + \|W_1\|_F^2 + \|H\|_F^2)$, and you may use the fact that the minimum of $\frac{1}{3}(\|A\|_F^2 + \|B\|_F^2 + \|C\|_F^2)$ over factorizations $Z = ABC$ with large enough inner dimensions is $\sum_i \sigma_i(Z)^{2/3}$. (a) Assuming that the optimal Z shares its singular vectors with $\widetilde{Y}$, reduce the problem to minimizing $(t-1)^2 + at^{2/3}$ over $t \geq 0$ for each nonzero singular value $\sqrt{n}\,t$ of Z, with $a = 3\lambda K n^{1/3}$. (b) Show that the minimizer jumps from $t = \frac{1}{2}$ to $t = 0$ at $a_c = \frac{3}{2} \cdot 2^{-1/3}$. (c) Compare with the continuous transition of Proposition 12.8, and relate to Súkeník et al. (2024). *(Hint: at the jump, the value at the nonzero critical point equals the value 1 at $t = 0$, and the critical point satisfies $a = 3(1-t)t^{1/3}$.)*

12.9 SIMULATION. (Collapse in the model under gradient descent.) Run gradient descent with step 0.5 on the unregularized cross-entropy of the unconstrained features model with $K = 4$, $n = 5$, $m = 6$, from small random initialization ($\mathcal{N}(0, 0.1^2)$ entries, seed 0) for 2×10^5 steps. Record the within-class scatter $\sum_{k,i}\|h_{k,i} - \bar{h}_k\|^2 / \sum_k \|\bar{h}_k\|^2$, the largest deviation of the cosines between centered means from $-\frac{1}{3}$, and the normalized margin $\min_{k,i,j\neq k}\langle w_k - w_j, h_{k,i}\rangle / \big(\frac{1}{2}(\|W\|_F^2 + \|H\|_F^2)\big)$. The reader should observe that all three quantities converge slowly: after 2×10^5 steps the normalized margin is still about 3% below the value $1/(\sqrt{n}(K-1)) = 0.1491$ of Theorem 12.9, and the scatter and the cosine deviation are still about 0.03 and 0.05, falling by a quarter to a third per decade of steps. Explain the slowness with the logarithmic rate of directional convergence for exponentially tailed losses in Chapter 4.

12.10 SIMULATION. (Thresholds of minority collapse.) With the script of Figure 12.4, locate the smallest ratio R at which the two minority classifiers coincide, for $E_W = E_H \in \{0.5, 1, 2, 3\}$, and plot it on a

logarithmic scale. The reader should find thresholds near $R = 2, 4, 30$ and 300, growing roughly exponentially in E over this range. Explain qualitatively why a larger budget delays the merging.

12.11 RESEARCH. (Dynamics of collapse.) In a network trained past zero error, NC1 decreases slowly and may stall (Figure 12.3). Find a solvable model with data, for example a two-layer network on a Gaussian mixture, in which the rate of decrease of NC1 and its limiting value can be computed. Han et al. (2022) and Ji et al. (2022) analyze dynamics with free features; a model in which the features are produced from inputs is open, beyond the results of Jacot et al. (2025).

12.12 RESEARCH. (More classes than dimensions.) For $K > m + 1$ the minimizers of Theorem 12.6 are no longer simplices. Characterize them, for example for $m = 2$ and all K, where a regular K-gon is the natural guess, and decide when gradient descent finds them. Lu and Steinerberger (2022) study the limit of many points, and Jiang et al. (2024) prove one-versus-rest margin maximization under conditions; a general description is open.

Notes and References

Neural collapse was named and measured by Papyan et al. (2020), whose paper appeared on arXiv in August 2020 and in PNAS that October; Han et al. (2022) extended it to the square loss, an ICLR 2022 outstanding paper, and introduced the central path. The unconstrained features model appeared within months in several groups: Mixon et al. (2020) gave it its name and studied its loss landscape; Lu and Steinerberger (2022) solved the cross-entropy on the sphere and studied the limit of many classes; E and Wojtowytsch (2022) proved simplex symmetry in toy models of expressive networks; Fang et al. (2021) introduced the layer-peeled model, proved Theorem 12.6 in their formulation, and discovered minority collapse; and Graf et al. (2021) proved that both the cross-entropy and the supervised contrastive loss are minimized by simplex configurations. Zhu et al. (2021) proved the benign landscape with weight decay and used it to fix the last layer to a simplex in practice, saving memory; Zhou et al. (2022) did the same for the square loss; and Ji et al. (2022) connected the unregularized model to the margin theory of Chapter 4. Kothapalli (2023) reviews this literature.

We isolated the alignment inequality of Lemma 12.5 because Theorem 12.6, Corollary 12.7 and Theorem 12.9 all reduce to it. The closed form of Corollary 12.7 in the single parameter κ, the dual-certificate proof of Proposition 12.8, and the confidence $1 - \frac{K-1}{K}\kappa$ shared by both losses are our own presentation; we have not found them stated in this form.

Equiangular tight frames, and the bound of Lemma 12.4, belong to frame theory and coding (Strohmer and Heath 2003); the margin view of collapse comes from Soudry et al. (2018) and Lyu and Li (2020), through Ji et al. (2022). Language models have made the regime $K > m + 1$ the practically important one.

TRANSLATING THE LITERATURE. Papyan et al. (2020) write C for our number of classes K, N for our examples per class n, p for our feature dimension m, $\boldsymbol{h}_{i,c}$ for our $h_{c,i}$, $\boldsymbol{\mu}_c$ and $\boldsymbol{\mu}_G$ for our $\bar{h}_c$ and h_G, and Σ_W, Σ_B as we do; their NC1 metric is $\operatorname{tr}(\Sigma_W \Sigma_B^{\dagger})/C$, our $\operatorname{tr}(\Sigma_W \Sigma_B^{+})/K$. Fang et al. (2021) write p for our feature dimension m and use K, n_k, $\boldsymbol{h}_{k,i}$ and the budgets E_W, E_H as we do, write the classifier as $\boldsymbol{W} = [\boldsymbol{w}_1, \dots, \boldsymbol{w}_K]^{\top}$, and denote the majority and minority counts by K_A, K_B and the sizes by n_A, n_B, with ratio $R = n_A/n_B$. Zhu et al. (2021) and Zhou et al. (2022) write d for our feature dimension m (we keep d for the input dimension), $\boldsymbol{H} \in \mathbb{R}^{d \times N}$ with $N = nK$ as we do, and $\lambda_{\boldsymbol{W}}, \lambda_{\boldsymbol{H}}, \lambda_{\boldsymbol{b}}$ for the weight decays; our κ is not used there, and their thresholds can be rewritten in terms of it. Several papers, including Fang et al. (2021), call the configuration of Definition 12.3 a simplex ETF and normalize it with $\sqrt{K/(K-1)}$ as in Exercise 12.1.

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Part V

Transformers

Two solvable models of learning in context: a linear attention layer that performs one step of gradient descent, and a two-layer circuit that counts bigrams. Each is exact in its own world and partial in ours.

13 Learning in Context

Linear attention as one step of gradient descent

THE MODEL — LINEAR SELF-ATTENTION ON REGRESSION TASKS

IN THIS CHAPTER. Large language models learn a task from a few examples placed in the prompt, without changing a weight, and transformers trained on prompts of linear regression problems predict nearly as well as least squares. A single layer of linear self-attention, trained on such prompts, is the solvable model: its prediction is one step of preconditioned gradient descent on the prompt, and we prove that the best such layer, among all its parameters, takes the step $\Gamma^\star = N\big((N+1)\Sigma + (\operatorname{tr}\Sigma + \sigma^2)I_d\big)^{-1}$, that gradient flow finds it, and that for isotropic inputs and a small balanced initialization it does so after a plateau, along a logistic curve. The model gets the mechanism of one trained layer right and the rate wrong: one step loses a factor $(d+1+\sigma^2)/\sigma^2$ against least squares, deeper trained transformers do much better, softmax attention computes different estimators, and in language models in-context learning is in part the recognition of a task already learned.

13.1 The phenomenon: learning from the prompt

A trained network is a fixed function. Yet Brown et al. (2020) found that GPT-3, a transformer language model with 175 billion parameters, could be taught a new task by its input alone: shown a few demonstrations of a task, written as text (pairs of English and French words, or arithmetic problems with their answers), followed by a new instance, it often completed the instance correctly, and more demonstrations usually helped. They called this *in-context learning*, distinguished zero-, one- and few-shot prompts, and stressed that no gradient update takes place: whatever learning happens is computed by the forward pass, from the prompt.[1]

Language is a poor laboratory: the tasks are vague, and one cannot tell whether a model learns from the demonstrations or recognizes a task it has seen in training. Garg et al. (2022) replaced language with a well-posed problem. Each training sequence is a prompt $(x_1, f(x_1), \ldots, x_N, f(x_N), x_q)$ with a function f drawn afresh from a class, and the model is trained to predict $f(x_q)$. For linear functions $f(x) = \langle w_\star, x\rangle$ with x and $w_\star$ isotropic Gaussian in dimension 20, a transformer trained from scratch reached an error comparable to that of least squares on the same prompt, and it did so on functions it had never seen. Sparse linear functions, decision trees and two-layer networks could be learned in context too.

What does the trained transformer compute? Akyürek et al. (2023)

[1] The idea that a network can learn to learn inside its activations is older: Hochreiter et al. (2001) trained LSTM networks whose recurrent dynamics, with fixed weights, learned new functions from the examples presented to them.

showed by construction that transformers can implement gradient descent and closed-form ridge regression, and found that trained models track the predictions of least squares and ridge. von Oswald et al. (2023) showed that one layer of linear self-attention can implement exactly one step of gradient descent on the squared loss of the prompt, and found that a trained single layer matches that step closely, in prediction and in its weights. Raventós et al. (2023) varied the number of distinct tasks $w_\star$ seen in training and found a threshold: below it, the transformer behaves like the Bayes estimator for the finite set of training tasks and fails on new ones; above it, it behaves like ridge regression, the Bayes estimator for a Gaussian prior over all tasks.

Observation 13.1 (Garg et al. 2022; von Oswald et al. 2023; Akyürek et al. 2023). Transformers trained on prompts of noisy linear regression problems with random coefficients, to predict the label of a query input from the labeled examples before it, learn to predict nearly as well as least squares or ridge regression on the prompt, for coefficients never seen in training. A single trained layer of linear self-attention predicts like one step of gradient descent on the squared loss of the prompt.

How robust is this? On synthetic regression the observation has been reproduced many times, but it describes models trained on nothing else, with many distinct tasks. In language models the evidence is mixed. Min et al. (2022) found that replacing the labels of the demonstrations by random ones barely hurts many classification tasks, and Pan et al. (2023) separated *task recognition*, which works without correct labels and does not improve much with scale, from *task learning*, which emerges with scale; Wei et al. (2023) (a preprint) found that large models, unlike small ones, can follow demonstrations whose labels are flipped. In-context learning also depends on the training data (Chan et al. 2022) and can be transient, appearing and then fading during training (Singh et al. 2023).

The question for this chapter: when a single attention layer is trained on regression prompts, which algorithm does it learn, how well does that algorithm predict, and how does training find it?

13.2 The model: one layer of linear attention

Prompts. A task is a coefficient vector $w_\star \in \mathbb{R}^d$; a prompt shows N labeled examples of the task and one query.

Assumption 13.2 (Linear regression prompts). For each prompt, independently: the task is $w_\star \sim \mathcal{N}(0, I_d)$; the inputs $x_1, \dots, x_N, x_q$ are independent with law $\mathcal{N}(0, \Sigma)$, Σ positive definite; the labels are $y_i = \langle w_\star, x_i \rangle + \varepsilon_i$ and $y_q = \langle w_\star, x_q \rangle + \varepsilon_q$ with noises $\varepsilon_1, \dots, \varepsilon_N, \varepsilon_q \sim \mathcal{N}(0, \sigma^2)$; all these variables are independent.

The prompt is stored as a matrix whose columns are tokens, one per example, with the query's label left blank:

$$Z = \begin{pmatrix} x_1 & x_2 & \cdots & x_N & x_q \\ y_1 & y_2 & \cdots & y_N & 0 \end{pmatrix} \in \mathbb{R}^{(d+1)\times(N+1)}. \tag{13.1}$$

We write $z_i = (x_i, y_i)$ for the ith column and $z_q = (x_q, 0)$ for the last. Three statistics of the prompt will appear everywhere: the sample covariance, the label-weighted input mean, and the mean square label,

$$\hat{\Sigma} = \frac{1}{N}\sum_{i=1}^{N} x_i x_i^\top, \qquad s = \frac{1}{N}\sum_{i=1}^{N} y_i x_i, \qquad \hat{c} = \frac{1}{N}\sum_{i=1}^{N} y_i^2.$$

Their relevance is no accident: the empirical loss of the prompt, $\widehat{\mathcal{L}}(w) = \frac{1}{2N}\sum_i (y_i - \langle w, x_i\rangle)^2$, has gradient $\nabla\widehat{\mathcal{L}}(w) = \hat{\Sigma}w - s$, so $\hat{\Sigma}$ and s are the sufficient statistics of least squares, and $\nabla\widehat{\mathcal{L}}(0) = -s$.

LINEAR SELF-ATTENTION. A softmax attention head (Vaswani et al. 2017) maps Z to $Z + W^P W^V Z \,\mathrm{softmax}\big((W^K Z)^\top W^Q Z\big)$, the softmax acting on each column. Removing the softmax and merging the products of matrices that appear only together gives *linear self-attention* (Schlag et al. 2021; von Oswald et al. 2023), with parameters $\theta = (W^{KQ}, W^{PV})$, two $(d+1)\times(d+1)$ matrices:

$$f(Z;\theta) = Z + \frac{1}{N} W^{PV} Z\, M\, Z^\top W^{KQ} Z, \qquad M = \begin{pmatrix} I_N & 0 \\ 0 & 0 \end{pmatrix} \in \mathbb{R}^{(N+1)\times(N+1)}. \tag{13.2}$$

The mask M lets every token attend to the N examples but not to the query, whose blank label carries no information. The prediction is read from the bottom entry of the query column, $\hat{y}_q = f(Z;\theta)_{d+1,N+1}$; since $Z_{d+1,N+1} = 0$, the residual term does not contribute. The model is trained on the *pretraining loss*, the population risk over prompts,

$$\mathcal{L}(\theta) = \mathbb{E}\big(y_q - \hat{y}_q(\theta)\big)^2,$$

with the expectation over Assumption 13.2. A good parameter is a learning algorithm: it must predict well for every task, using only the prompt.

BLOCKS. Only some blocks of θ matter. Write

$$W^{KQ} = \begin{pmatrix} W_{11} & w_{12} \\ w_{21}^\top & w_{22} \end{pmatrix}, \qquad W^{PV} = \begin{pmatrix} V_{11} & v_{12} \\ v_{21}^\top & v_{22} \end{pmatrix},$$

with $W_{11}, V_{11} \in \mathbb{R}^{d\times d}$, column vectors $w_{12}, w_{21}, v_{12}, v_{21} \in \mathbb{R}^d$, and scalars w_{22}, v_{22}. The prediction is the last row of W^{PV} applied to the last column of $\frac{1}{N} Z M Z^\top W^{KQ} Z$, and $\frac{1}{N} Z M Z^\top = \frac{1}{N}\sum_{i\le N} z_i z_i^\top$ has blocks $\hat{\Sigma}$, s, $s^\top$, $\hat{c}$. So V_{11}, v_{12} (the rows that write into the inputs) and w_{12}, w_{22} (which multiply the blank label) never affect $\hat{y}_q$; they matter only in deeper models.

DICTIONARY

IN THE MODEL	IN A TRAINED NETWORK
prompt of N pairs (x_i, y_i) and a query	few-shot prompt: demonstrations followed by a new input
task $w_\star$, drawn afresh for each prompt	the task the demonstrations specify
law of $w_\star$	the diversity of tasks implicit in the pretraining corpus
one token $z_i = (x_i, y_i)$ per example	an input and its label in separate tokens, interleaved
W^{KQ}, W^{PV}	key–query and value–output products of one head
linear attention, one layer	softmax attention, many heads and layers, MLPs, normalization
preconditioner Γ	the learning rule implemented by the forward pass
pretraining loss $\mathcal{L}(\theta)$	next-token loss on documents
Bayes estimator (ridge)	the best a learner could do with the prompt

WHAT WAS THROWN AWAY, AND WHY FIRST. The softmax goes, because without it attention is a cubic polynomial in the prompt and every expectation is a Gaussian moment. Depth, MLPs and normalization go, because one layer already learns a nontrivial algorithm. Language goes, replaced by a task family where the optimal learner is known, so that "learning in context" can be measured against ridge regression. Each example is one token, rather than two, which spares the model the job of pairing inputs with labels, the job of the induction heads of Chapter 14. Section 13.6 says what each choice costs.

WHAT ONE HAD TO NOTICE. Attention computes $\sum_i z_i z_i^\top$, a sum of outer products over the prompt, and so does the gradient of the squared loss: $-\nabla\widehat{\mathcal{L}}(0) = \frac{1}{N}\sum_i y_i x_i$ is a block of $\frac{1}{N}\sum_i z_i z_i^\top$. A layer whose update is a sum over the examples can therefore take a gradient step on them.[2]

[2] Schlag et al. (2021) had shown that linear attention is a *fast weight programmer*: it writes the outer products $v_i k_i^\top$ of values and keys into a weight matrix, which then multiplies the query.

13.3 The solution: the optimal single layer

The first result reads off what one layer computes; the second finds the best learner of that form; the third shows that nothing in the full parameter space does better.

Proposition 13.3 (What one layer computes). *For every θ, with $\Gamma(\theta) = v_{22}W_{11}^\top + w_{21}v_{21}^\top$ and $b(\theta) = W_{11}^\top\hat{\Sigma}\,v_{21} + \hat{c}\,v_{22}w_{21}$,*

$$\hat{y}_q = x_q^\top\Gamma(\theta)\,s + x_q^\top b(\theta). \tag{13.3}$$

In particular, if $v_{21} = 0$ and $w_{21} = 0$, then $\hat{y}_q = x_q^\top\Gamma s$ with $\Gamma = v_{22}W_{11}^\top$, which is the prediction $\langle w_1, x_q\rangle$ of one step of preconditioned gradient descent $w_1 = w_0 - \Gamma\nabla\widehat{\mathcal{L}}(w_0)$ from $w_0 = 0$. Every $\Gamma \in \mathbb{R}^{d\times d}$ arises this way, for instance with $v_{22} = 1$ and $W_{11} = \Gamma^\top$.

PROOF. The query column of $Z^\top W^{KQ} Z$ is $Z^\top W^{KQ} z_q$. The last entry of z_q is 0, so $W^{KQ} z_q$ has blocks $W_{11} x_q$ and $w_{21}^\top x_q$. Applying $\frac{1}{N} Z M Z^\top$ and then the last row $(v_{21}^\top, v_{22})$ of W^{PV} gives

$$\begin{aligned}\hat{y}_q &= (v_{21}^\top,\ v_{22}) \begin{pmatrix} \hat{\Sigma} & s \\ s^\top & \hat{c} \end{pmatrix} \begin{pmatrix} W_{11} x_q \\ w_{21}^\top x_q \end{pmatrix} \\ &= v_{21}^\top \hat{\Sigma} W_{11} x_q + (v_{21}^\top s)(w_{21}^\top x_q) + v_{22}\, s^\top W_{11} x_q + v_{22}\, \hat{c}\, w_{21}^\top x_q .\end{aligned}$$

Each term is a scalar; transposing the first and third and regrouping gives (13.3). If $v_{21} = w_{21} = 0$, then $b = 0$ and $\Gamma = v_{22} W_{11}^\top$, and $w_1 = 0 - \Gamma(\hat{\Sigma} \cdot 0 - s) = \Gamma s$. ∎

The odd and even parts of (13.3) will be the key. Flipping the sign of the task and of all the noises maps $y_i \mapsto -y_i$, so $s \mapsto -s$ while $\hat{\Sigma}$ and $\hat{c}$ are unchanged: the first term of $\hat{y}_q$ is odd in the labels and the second, $x_q^\top b$, is even. The target y_q is odd. An even term can only add error, as the proof of Theorem 13.6 will show.

The risk of the one-step learner needs the fourth moments of a Gaussian vector.

Lemma 13.4 (Gaussian fourth moments). *Let $x \sim \mathcal{N}(0, \Sigma)$ in $\mathbb{R}^d$. For every $A \in \mathbb{R}^{d \times d}$,*

$$\mathbb{E}\left[x x^\top A x x^\top\right] = \Sigma A \Sigma + \Sigma A^\top \Sigma + \operatorname{tr}(A\Sigma)\, \Sigma .$$

Consequently, if $x_1, \dots, x_N$ are independent $\mathcal{N}(0, \Sigma)$ and $\hat{\Sigma} = \frac{1}{N} \sum_i x_i x_i^\top$, then $\mathbb{E}[\hat{\Sigma}^2] = \frac{1}{N}\big((N+1)\Sigma^2 + \operatorname{tr}(\Sigma)\Sigma\big)$.

PROOF. *The standard case.* Let $g \sim \mathcal{N}(0, I_d)$. Its coordinates are independent with $\mathbb{E} g_i^2 = 1$, $\mathbb{E} g_i^4 = 3$, and vanishing odd moments, so $\mathbb{E}[g_i g_j g_k g_l]$ is 1 if the indices form two equal pairs and differ across pairs, 3 if all four are equal, and 0 otherwise; in every case $\mathbb{E}[g_i g_j g_k g_l] = \delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}$. For $B \in \mathbb{R}^{d \times d}$, the (i, l) entry of $g g^\top B g g^\top$ is $\sum_{j,k} g_i g_j B_{jk} g_k g_l$, so

$$\mathbb{E}\left[g g^\top B g g^\top\right]_{il} = \sum_{j,k} B_{jk}\big(\delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}\big) = B_{il} + B_{li} + \delta_{il} \operatorname{tr} B .$$

General Σ. Write $x = \Sigma^{1/2} g$ and $B = \Sigma^{1/2} A \Sigma^{1/2}$. Then $x x^\top A x x^\top = \Sigma^{1/2} g g^\top B g g^\top \Sigma^{1/2}$, whose mean is $\Sigma^{1/2}(B + B^\top + \operatorname{tr}(B) I)\Sigma^{1/2} = \Sigma A \Sigma + \Sigma A^\top \Sigma + \operatorname{tr}(A\Sigma)\Sigma$.

The sample covariance. Expanding $\hat{\Sigma}^2 = \frac{1}{N^2} \sum_{i,j} x_i x_i^\top x_j x_j^\top$, the $N(N-1)$ terms with $i \neq j$ have mean Σ^2 by independence, and the N terms with $i = j$ have mean $2\Sigma^2 + \operatorname{tr}(\Sigma)\Sigma$ by the first part with $A = I_d$. Adding, $\mathbb{E}[\hat{\Sigma}^2] = \frac{1}{N^2}\big(N(N+1)\Sigma^2 + N \operatorname{tr}(\Sigma)\Sigma\big)$. ∎

Theorem 13.5 (The optimal one-step learner). *Under Assumption 13.2,*

let $\mathcal{R}(\Gamma) = \mathbb{E}\big(y_q - x_q^\top \Gamma s\big)^2$ *for* $\Gamma \in \mathbb{R}^{d\times d}$, *and let*

$$H = \frac{1}{N}\Big((N+1)\Sigma^2 + (\operatorname{tr}\Sigma + \sigma^2)\Sigma\Big).$$

Then

$$\mathcal{R}(\Gamma) = \sigma^2 + \operatorname{tr}\Sigma - 2\operatorname{tr}(\Sigma\Gamma\Sigma) + \operatorname{tr}(\Sigma\Gamma H\Gamma^\top), \tag{13.4}$$

and $\mathcal{R}$ *has the unique minimizer*

$$\Gamma^\star = N\Big((N+1)\Sigma + (\operatorname{tr}\Sigma + \sigma^2)I_d\Big)^{-1}, \tag{13.5}$$

with $\mathcal{R}(\Gamma) - \mathcal{R}(\Gamma^\star) = \operatorname{tr}\big(\Sigma(\Gamma-\Gamma^\star)H(\Gamma-\Gamma^\star)^\top\big)$. *If* Σ *has eigenvalues* $\lambda_1, \dots, \lambda_d$, *the minimal risk is*

$$\mathcal{R}^\star = \sigma^2 + \sum_{k=1}^{d} \frac{\lambda_k(\lambda_k + \operatorname{tr}\Sigma + \sigma^2)}{(N+1)\lambda_k + \operatorname{tr}\Sigma + \sigma^2}. \tag{13.6}$$

For $\Sigma = I_d$: $\Gamma^\star = \gamma^\star I_d$ *with* $\gamma^\star = \frac{N}{N+d+1+\sigma^2}$, *and* $\mathcal{R}^\star = \sigma^2 + \frac{d(d+1+\sigma^2)}{N+d+1+\sigma^2}$.

Proof. The plan is to average over the query first, then over the prompt with Lemma 13.4, and to complete the square.

The query. The pair (x_q, ε_q) is independent of $(w_\star, s)$, with $\mathbb{E}x_q x_q^\top = \Sigma$ and $\mathbb{E}\varepsilon_q^2 = \sigma^2$. Since $y_q - x_q^\top \Gamma s = \varepsilon_q + x_q^\top(w_\star - \Gamma s)$ is a sum of two uncorrelated terms given $(w_\star, s)$,

$$\mathcal{R}(\Gamma) = \sigma^2 + \mathbb{E}\big[(w_\star - \Gamma s)^\top \Sigma (w_\star - \Gamma s)\big].$$

Two moments of s. Write $s = \hat{\Sigma} w_\star + \bar{\xi}$ with $\bar{\xi} = \frac{1}{N}\sum_i \varepsilon_i x_i$. The task is independent of the inputs and noises, with $\mathbb{E}w_\star w_\star^\top = I_d$, and $\mathbb{E}[\bar{\xi} \mid X] = 0$, so $\mathbb{E}[s w_\star^\top] = \mathbb{E}[\hat{\Sigma}] = \Sigma$. For the second moment, the cross terms vanish because the noises have mean zero given everything else; conditioning on the inputs gives $\mathbb{E}[\hat{\Sigma} w_\star w_\star^\top \hat{\Sigma}] = \mathbb{E}[\hat{\Sigma}^2]$; and independence of the noises gives $\mathbb{E}[\bar{\xi}\bar{\xi}^\top] = \frac{\sigma^2}{N^2}\sum_i \mathbb{E}[x_i x_i^\top] = \frac{\sigma^2}{N}\Sigma$. With Lemma 13.4,

$$\mathbb{E}[s s^\top] = \frac{1}{N}\big((N+1)\Sigma^2 + \operatorname{tr}(\Sigma)\Sigma\big) + \frac{\sigma^2}{N}\Sigma = H.$$

The risk. Expanding the quadratic form, $\mathbb{E}[w_\star^\top \Sigma w_\star] = \operatorname{tr}\Sigma$, $\mathbb{E}[w_\star^\top \Sigma\Gamma s] = \operatorname{tr}(\Sigma\Gamma\,\mathbb{E}[s w_\star^\top]) = \operatorname{tr}(\Sigma\Gamma\Sigma)$, and $\mathbb{E}[s^\top\Gamma^\top\Sigma\Gamma s] = \operatorname{tr}(\Sigma\Gamma H\Gamma^\top)$, which is (13.4).

Completing the square. Let $K = \frac{1}{N}\big((N+1)\Sigma + (\operatorname{tr}\Sigma + \sigma^2)I_d\big)$, so that $H = \Sigma K = K\Sigma$ and $\Gamma^\star = K^{-1}$, a symmetric matrix with $H\Gamma^\star = \Sigma$. Expanding $\operatorname{tr}\big(\Sigma(\Gamma-\Gamma^\star)H(\Gamma-\Gamma^\star)^\top\big)$ and using $H\Gamma^{\star\top} = \Sigma$ turns its cross term into $-2\operatorname{tr}(\Sigma\Gamma\Sigma)$ and its last term into $\operatorname{tr}(\Sigma\Gamma^\star\Sigma)$; comparing with (13.4),

$$\mathcal{R}(\Gamma) = \sigma^2 + \operatorname{tr}\Sigma - \operatorname{tr}(\Sigma\Gamma^\star\Sigma) + \operatorname{tr}\big(\Sigma(\Gamma-\Gamma^\star)H(\Gamma-\Gamma^\star)^\top\big).$$

The matrices Σ and H are positive definite (H is a product of two commuting positive definite matrices), so the last term equals $\|\Sigma^{1/2}(\Gamma-\Gamma^\star)H^{1/2}\|_F^2$,

which vanishes only at $\Gamma = \Gamma^\star$. Hence $\Gamma^\star$ is the unique minimizer and $\mathcal{R}^\star = \sigma^2 + \operatorname{tr}\Sigma - \operatorname{tr}(\Sigma^2\Gamma^\star)$. In an eigenbasis of Σ, the matrix $\Gamma^\star$ is diagonal with entries $N/\big((N+1)\lambda_k + \operatorname{tr}\Sigma + \sigma^2\big)$, and $\lambda_k - N\lambda_k^2/\big((N+1)\lambda_k + \operatorname{tr}\Sigma + \sigma^2\big)$ simplifies to the kth term of (13.6). For $\Sigma = I_d$, set $\lambda_k = 1$ and $\operatorname{tr}\Sigma = d$. ∎

The proof used only $\mathbb{E}w_\star w_\star^\top = I_d$, independence, and Gaussian inputs: the law of the task enters only through its second moment.[3] The one-step learner is restricted, but the restriction is not ours: the full layer can do no better.

[3] We checked (13.4) and (13.6) by Monte Carlo with 5×10^5 prompts for $d = 5$, $N = 12$, $\sigma^2 = 0.7$, and $\Sigma = \operatorname{diag}(2, 1, 0.5, 0.3, 0.2)$, at $\Gamma^\star$ and at a random Γ; the deviations were within 2.5 standard errors.

Theorem 13.6 (The global minimum of one layer). *Under Assumption 13.2, every $\theta = (W^{KQ}, W^{PV})$ satisfies*

$$\mathcal{L}(\theta) = \mathcal{R}\big(\Gamma(\theta)\big) + \mathbb{E}\big(x_q^\top b(\theta)\big)^2 \;\geq\; \mathcal{R}^\star,$$

with $\Gamma(\theta)$ and $b(\theta)$ as in Proposition 13.3. Equality holds if and only if $\Gamma(\theta) = \Gamma^\star$ and $b(\theta) = 0$ almost surely. In particular, $\min_\theta \mathcal{L}(\theta) = \mathcal{R}^\star$, and every global minimizer predicts $\hat{y}_q = x_q^\top \Gamma^\star s$.

Proof. The plan is to split the prediction (13.3) into its odd and even parts and to show that they do not interact.

The symmetry. Let τ map $(w_\star, \varepsilon_1, \dots, \varepsilon_N, \varepsilon_q)$ to its negative and leave the inputs unchanged. Under Assumption 13.2 these variables are independent of the inputs and have centered Gaussian laws, so τ preserves the joint law of all the variables. It maps every label to its negative, hence $s \mapsto -s$ and $y_q \mapsto -y_q$, while $\hat{\Sigma}$, $\hat{c}$, and x_q are unchanged.

No interaction. Write $\hat{y}_q = x_q^\top \Gamma s + e$ with $e = x_q^\top b(\theta)$, and let $F = (y_q - x_q^\top \Gamma s)\, e$. All the variables are polynomials in Gaussian variables, so F is integrable. By the symmetry, $F \circ \tau = -F$ and $F \circ \tau$ has the law of F, so $\mathbb{E}F = -\mathbb{E}F = 0$. Expanding the square,

$$\mathcal{L}(\theta) = \mathbb{E}\big(y_q - x_q^\top \Gamma s\big)^2 - 2\mathbb{E}F + \mathbb{E}e^2 = \mathcal{R}(\Gamma) + \mathbb{E}e^2.$$

The bound. Theorem 13.5 gives $\mathcal{R}(\Gamma) \geq \mathcal{R}^\star$, with equality only at $\Gamma = \Gamma^\star$, and $\mathbb{E}e^2 \geq 0$, with equality only if $e = 0$ almost surely. Since x_q is independent of $b(\theta)$ and has a positive definite covariance, $\mathbb{E}e^2 = \mathbb{E}\big[b^\top \Sigma b\big]$ vanishes exactly when $b(\theta) = 0$ almost surely. The parameter with $v_{22} = 1$, $W_{11} = \Gamma^{\star\top}$, and all other blocks zero attains $\mathcal{R}^\star$ by Proposition 13.3. ∎

The theorem says more than that the optimum is attained: *whatever* global minimum training reaches, the trained layer runs one step of gradient descent with preconditioner $\Gamma^\star$. The minimizers themselves are not unique (rescaling v_{22} by $c \neq 0$ and W_{11} by $1/c$ changes nothing), but their predictions are.

How good is this learner? The best predictor of y_q from the prompt and x_q is the posterior mean, which, for Gaussian prior and noise, is ridge regression.

Corollary 13.7 (One step against the Bayes estimator). *Let $\Sigma = I_d$ in Assumption 13.2, and let $X \in \mathbb{R}^{N\times d}$ have rows x_i. Then:*

(i) the Bayes-optimal prediction is $x_q^\top \hat{w}$ with $\hat{w} = (X^\top X + \sigma^2 I_d)^{-1} X^\top y$, and its risk is $\sigma^2 + \mathbb{E}\,\mathrm{tr}\big(\sigma^2 (X^\top X + \sigma^2 I_d)^{-1}\big)$;

(ii) if $d, N \to \infty$ with $d/N \to \gamma \in (0,\infty)$ and $\sigma^2/d \to \varsigma > 0$, then the excess risks per dimension converge:

$$\frac{\mathcal{R}^\star - \sigma^2}{d} \to \frac{\gamma(1+\varsigma)}{1+\gamma(1+\varsigma)}, \qquad \frac{\mathcal{R}_{\mathrm{B}} - \sigma^2}{d} \to \frac{\sqrt{b^2+4\varsigma} - b}{2}, \tag{13.7}$$

where $\mathcal{R}_{\mathrm{B}}$ is the Bayes risk and $b = \frac{1}{\gamma} - 1 + \varsigma$;

(iii) as $\gamma \to 0$ (many examples per dimension), the ratio of the two limits tends to $(1+\varsigma)/\varsigma$.

Proof. *Part (i).* Given (X, y), the density of $w_\star$ at w is proportional to

$$\exp\Big(-\frac{1}{2\sigma^2}\|y - Xw\|^2 - \frac{1}{2}\|w\|^2\Big),$$

a Gaussian whose precision matrix is $(X^\top X + \sigma^2 I_d)/\sigma^2$ and whose mean $\hat{w}$ is the stationary point $X^\top(y - Xw) = \sigma^2 w$. The squared loss is minimized by the conditional mean of y_q given the prompt and x_q, which is $x_q^\top \hat{w}$, because x_q and ε_q are independent of the rest. Its risk is $\sigma^2 + \mathbb{E}\|w_\star - \hat{w}\|^2$, and the conditional mean of $\|w_\star - \hat{w}\|^2$ is the trace of the posterior covariance $\sigma^2(X^\top X + \sigma^2 I_d)^{-1}$.

Part (ii). For the one-step learner, Theorem 13.5 gives $(\mathcal{R}^\star - \sigma^2)/d = (d + 1 + \sigma^2)/(N + d + 1 + \sigma^2)$; divide numerator and denominator by N. For the Bayes estimator, write $\hat{\Sigma}_X = X^\top X/N$ and $\lambda = \sigma^2/N$; then $\frac{1}{d}\,\mathrm{tr}\big(\sigma^2(X^\top X + \sigma^2 I)^{-1}\big) = \frac{1}{d}\,\mathrm{tr}\big(\lambda(\hat{\Sigma}_X + \lambda I)^{-1}\big)$, which is the function $v_p(\lambda)$ of Chapter 1 with $p = d$ features and $n = N$ samples. Here $\lambda = \lambda_N = \varsigma_N\, d/N \to \varsigma\gamma$ with $\varsigma_N = \sigma^2/d$. The proof of Theorem 1.11 shows that $\mathbb{E}v_p(\lambda) \to v(\lambda)$ at each fixed $\lambda > 0$; since $|\partial_\lambda v_p(\lambda)| \le 1/(4\lambda)$, the convergence also holds along $\lambda_N \to \varsigma\gamma > 0$. The limit $v(\varsigma\gamma)$ is the positive root of $\gamma v^2 + (1 - \gamma + \varsigma\gamma)v - \varsigma\gamma = 0$, that is, of $v^2 + bv - \varsigma = 0$, which is the stated limit.

Part (iii). As $\gamma \to 0$, the first limit is $\gamma(1+\varsigma) + O(\gamma^2)$, and the second is $\frac{2\varsigma}{\sqrt{b^2+4\varsigma}+b} = \varsigma\gamma + O(\gamma^2)$, because $b = 1/\gamma + O(1)$. ∎

13.4 The solution: how training finds the step

Theorem 13.6 is about the landscape; does gradient descent reach its minimum? Two facts make the dynamics solvable. First, the set

$$\mathcal{M} = \{\theta : v_{21} = 0,\ w_{21} = 0\}$$

is invariant: on it the odd–even symmetry kills the gradient in the off-manifold directions. Second, on $\mathcal{M}$ the model is a two-layer linear network, $\Gamma = v_{22} W_{11}^\top$, a scalar times a matrix, with the conservation law of Chapter 2. Gradient flow is $\dot{\theta} = -\nabla\mathcal{L}(\theta)$.

Theorem 13.8 (Gradient flow finds the optimal step). *Under Assumption 13.2, let $\theta(0) \in \mathcal{M}$ with $v_{22}(0) > \|W_{11}(0)\|_F$. Then the gradient flow exists for all $t \geq 0$, stays in $\mathcal{M}$, and conserves $c = v_{22}^2 - \|W_{11}\|_F^2 > 0$; and as $t \to \infty$, $\Gamma(\theta(t)) \to \Gamma^\star$ and $\mathcal{L}(\theta(t)) \to \mathcal{R}^\star$.*

PROOF. The plan is to show invariance by the odd–even symmetry, to obtain the conservation law and boundedness on $\mathcal{M}$, and then to identify every limit point as a global minimum.

Invariance. The loss does not depend on V_{11}, v_{12}, w_{12}, w_{22}, so their gradients vanish everywhere. By Proposition 13.3, $\partial \hat{y}_q / \partial v_{21} = \hat{\Sigma} W_{11} x_q + (w_{21}^\top x_q) s$ and $\partial \hat{y}_q / \partial w_{21} = (v_{21}^\top s + v_{22} \hat{c}) x_q$. At a point of $\mathcal{M}$ these are $\hat{\Sigma} W_{11} x_q$ and $v_{22} \hat{c}\, x_q$, which are even under the map τ of the proof of Theorem 13.6, while $y_q - \hat{y}_q = y_q - x_q^\top \Gamma s$ is odd. So $\partial \mathcal{L} / \partial v_{21} = -2\mathbb{E}[(y_q - \hat{y}_q)\, \partial \hat{y}_q / \partial v_{21}] = 0$ on $\mathcal{M}$, and likewise for w_{21}. The vector field is polynomial, hence locally Lipschitz: the solution of the flow restricted to $\mathcal{M}$, in the variables (W_{11}, v_{22}), solves the full equation, and by uniqueness it is the full solution, as long as it exists.

The flow on $\mathcal{M}$. On $\mathcal{M}$, $\mathcal{L}(\theta) = \mathcal{R}(\Gamma)$ with $\Gamma = v_{22} A$ and $A = W_{11}^\top$. Let $G = \nabla \mathcal{R}(\Gamma) = 2(\Sigma \Gamma H - \Sigma^2)$, from (13.4). The chain rule gives

$$\dot{A} = -v_{22}\, G, \qquad \dot{v}_{22} = -\langle G, A \rangle_F,$$

so $\frac{\mathrm{d}}{\mathrm{d}t} v_{22}^2 = -2 v_{22} \langle G, A \rangle_F = \frac{\mathrm{d}}{\mathrm{d}t} \|A\|_F^2$, and $c = v_{22}^2 - \|A\|_F^2$ is constant. In particular $v_{22}^2 \geq c > 0$, so v_{22} never vanishes and stays positive.

Boundedness. The loss does not increase along the flow, and by Theorem 13.5, $\mathcal{R}(\Gamma) \geq \mathcal{R}^\star + \mu \|\Gamma - \Gamma^\star\|_F^2$ with $\mu = \lambda_{\min}(\Sigma) \lambda_{\min}(H) > 0$. So $\|\Gamma(t)\|_F \leq B$ for a constant B. Since $\|\Gamma\|_F^2 = v_{22}^2 \|A\|_F^2 = (c + \|A\|_F^2)\|A\|_F^2 \geq \|A\|_F^4$, also $\|A\|_F^2 \leq B$ and $v_{22}^2 \leq c + B$. A bounded solution of a locally Lipschitz equation exists for all time.

Limit points. Integrating $\frac{\mathrm{d}}{\mathrm{d}t} \mathcal{L} = -\|\nabla \mathcal{L}\|^2$ gives $\int_0^\infty \|\nabla \mathcal{L}(\theta(t))\|^2 \,\mathrm{d}t \leq \mathcal{L}(\theta(0)) - \mathcal{R}^\star < \infty$, so $\nabla \mathcal{L}(\theta(t_k)) \to 0$ along some sequence $t_k \to \infty$. By boundedness, a subsequence of $\theta(t_k)$ converges to some $\bar{\theta} \in \mathcal{M}$, where $\nabla \mathcal{L}(\bar{\theta}) = 0$ by continuity and $\bar{v}_{22}^2 \geq c > 0$. The A-component of the gradient, $\bar{v}_{22}\, G(\bar{\Gamma})$, vanishes, so $\nabla \mathcal{R}(\bar{\Gamma}) = 0$, and $\bar{\Gamma} = \Gamma^\star$, the only critical point of the strictly convex $\mathcal{R}$. Hence $\mathcal{L}(\theta(t_k)) \to \mathcal{R}^\star$ along the subsequence; since $\mathcal{L}(\theta(t))$ is nonincreasing and bounded below by $\mathcal{R}^\star$, $\mathcal{L}(\theta(t)) \to \mathcal{R}^\star$. Finally $\mu \|\Gamma(t) - \Gamma^\star\|_F^2 \leq \mathcal{R}(\Gamma(t)) - \mathcal{R}^\star \to 0$. ∎

The hypothesis $v_{22}(0) > \|W_{11}(0)\|_F$ keeps the flow away from the saddle at $\Gamma = 0$. From a small, balanced initialization, $c = 0$, and the flow must escape that saddle. For isotropic inputs this escape can be solved in closed form.

Proposition 13.9 (Escape from the saddle: a logistic curve). *Let $\Sigma = I_d$, and start the gradient flow on $\mathcal{M}$ at $W_{11}(0) = a_0 I_d$ and $v_{22}(0) = \sqrt{d}\, a_0$ with $a_0 > 0$, so that $\Gamma(0) = \gamma_0 I_d$ with $\gamma_0 = \sqrt{d}\, a_0^2$. Then $\Gamma(t) = \gamma(t) I_d$ for all t,*

where

$$\dot\gamma = 4\sqrt{d}\,\gamma\Big(1-\frac{\gamma}{\gamma^\star}\Big), \qquad \gamma(t) = \frac{\gamma^\star}{1+(\gamma^\star/\gamma_0-1)e^{-4\sqrt{d}\,t}}, \tag{13.8}$$

with $\gamma^\star = N/(N+d+1+\sigma^2)$. *The step reaches half its final value at* $t_{1/2} = \log(\gamma^\star/\gamma_0 - 1)/(4\sqrt{d})$.

PROOF. With $\Sigma = I_d$ we have $H = hI_d$ with $h = (N+d+1+\sigma^2)/N = 1/\gamma^\star$, and $G = 2(h\Gamma - I_d)$. Try $A = a(t)I_d$ and $v_{22} = \sqrt{d}\,a(t)$, so $\Gamma = \gamma I_d$ with $\gamma = \sqrt{d}\,a^2$ and $G = 2(h\gamma - 1)I_d$. The flow equations of the proof of Theorem 13.8 become

$$\dot a\, I_d = -2\sqrt{d}\,a(h\gamma-1)I_d, \qquad \dot v_{22} = -\langle G, aI_d\rangle_F = -2d\,a(h\gamma-1) = \sqrt{d}\,\dot a,$$

so the ansatz is consistent, with the single equation $\dot a = -2\sqrt{d}\,a(h\gamma-1)$; by uniqueness it gives the solution. Then $\dot\gamma = 2\sqrt{d}\,a\dot a = -4d\,a^2(h\gamma-1) = 4\sqrt{d}\,\gamma(1-h\gamma)$, a logistic equation with carrying capacity $1/h = \gamma^\star$, whose solution with $\gamma(0) = \gamma_0$ is (13.8). Setting $\gamma = \gamma^\star/2$ gives $e^{4\sqrt{d}\,t} = \gamma^\star/\gamma_0 - 1$. ∎

For general Σ, Zhang et al. (2024) prove that gradient flow on the population loss, from a suitable random initialization, converges to a global minimum; by Theorem 13.6, the limit predicts with $\Gamma^\star$.

PROOF SKETCH. What is omitted is the case $c = 0$ for a nonisotropic Σ: one must show that a balanced initialization whose matrix part is positive semidefinite moves away from the saddle at $\Gamma = 0$, where $\nabla\mathcal{R}(0) = -2\Sigma^2$ pulls Γ toward positive definite matrices, and that it is not attracted to the critical set $\{v_{22} = 0,\ \mathrm{tr}(\Sigma^2 W_{11}^\top) = 0\}$. Once Γ is bounded away from 0, the argument of Theorem 13.8 applies. Zhang et al. (2024) carry out this step for their initialization, whose exact form we do not reproduce; we have not checked their argument line by line. ∎

TASK DIVERSITY. Assumption 13.2 draws a fresh task for every prompt. If instead the pretraining tasks are a finite set $\{w_1, \dots, w_M\}$, then the derivation of Theorem 13.5 goes through with the identity replaced by the task second-moment matrix $\frac{1}{M}\sum_k w_k w_k^\top$ (Exercise 13.8): a single linear layer cannot memorize tasks, it can only adapt its preconditioner to their span. Lu et al. (2025) solve a linear-attention model of this kind in a proportional limit in which N and the number of tasks grow linearly with d and the number of pretraining prompts quadratically; they find a double descent in the number of pretraining prompts and a transition from memorizing the pretraining tasks to generalizing beyond them as the diversity grows, and they report the same behavior in experiments with linear attention and with nonlinear transformers.

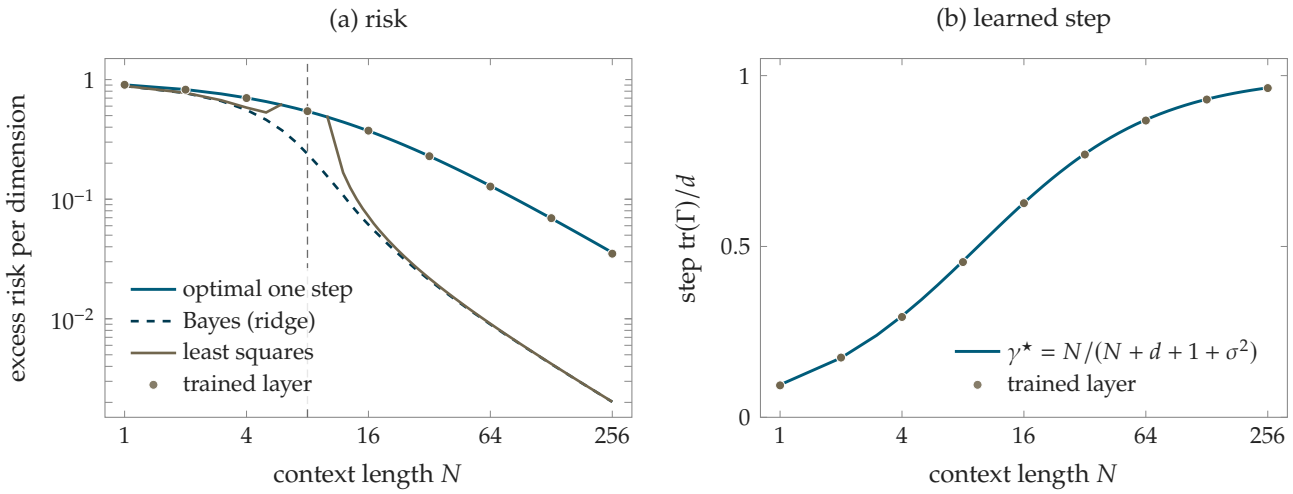


Figure 13.1. In-context regression with $d = 8$, $\Sigma = I_d$, and $\sigma^2 = 0.5$; excess risk is (risk $- \sigma^2)/d$. (a) Curves: the optimal one-step learner (Theorem 13.5), the Bayes estimator (Corollary 13.7(i), Monte Carlo over 4000 designs per N), and minimum-norm least squares (Theorem 1.6 with $\mathbb{E}\|w_\star\|^2 = d$; infinite for $|N - d| \le 1$; the dashed line marks $N = d$). Points: one layer of linear self-attention with all parameters trained by Adam (1500 steps, batches of 1024 fresh prompts, cosine-decayed rate from 10^{-2}, initialization $\mathcal{N}(0, 0.01)$), risk estimated on 10^5 fresh prompts (seed 20261017). (b) The learned step $\mathrm{tr}\,\Gamma(\theta)/d$ of the same layers.

13.5 Reading the solution

Figure 13.1 compares the theory with trained layers. Each point is a layer with all $2(d+1)^2 = 162$ parameters trained from a random initialization; its risk lies within 1.8 standard errors of $\mathcal{R}^\star$ at every N, its learned step $\mathrm{tr}(\Gamma)/d$ within 0.004 of $\gamma^\star$, the off-diagonal part of Γ is below 0.011 in normalized Frobenius norm, and the even part $b(\theta)$ has coefficients of order 10^{-2}, the level of the optimizer's noise. Training did not get stuck in this experiment, although Theorem 13.8 covers only initializations on $\mathcal{M}$.

THE LEARNED STEP. The optimal preconditioner (13.5) is a step size that knows how much to trust the prompt. For $\Sigma = I_d$ it is $\gamma^\star = N/(N+d+1+\sigma^2)$: a short or noisy prompt gives a short step. The denominator adds N (the signal in s, whose mean is $w_\star$) to $d + 1 + \sigma^2$, the fluctuation of s around $w_\star$: $\mathbb{E}\|s - w_\star\|^2 = \frac{d(d+1+\sigma^2)}{N}$, of which $\sigma^2 d/N$ comes from the noise and $d(d+1)/N$ from the fluctuation of $\hat{\Sigma}$ around I_d. As $N \to \infty$, $\Gamma^\star \to \Sigma^{-1}$, so for long prompts the one step becomes a Newton step, and for short ones it is a ridge-like shrinkage, $\Gamma^\star \approx \frac{N}{\mathrm{tr}\,\Sigma + \sigma^2} I_d$, which does not depend on the shape of Σ at all.

THE RATE. At fixed d the excess risk of the optimal step is $d(d+1+\sigma^2)/N + O(N^{-2})$, and that of least squares is $\sigma^2 d/(N-d-1)$ (Theorem 1.6); the one-step learner is consistent but pays a factor $(d+1+\sigma^2)/\sigma^2$ for not inverting $\hat{\Sigma}$. Without noise least squares is exact as soon as $N \ge d$, while one step never is: its excess risk is $d(d+1)/(N+d+1)$. In Figure 13.1(a) the gap at $N = 32$ is a factor of about 10.5, against the asymptotic $1 + (d+1)/\sigma^2 = 19$. Near $N = d$ the one-step learner beats the interpolating least-squares estimator,

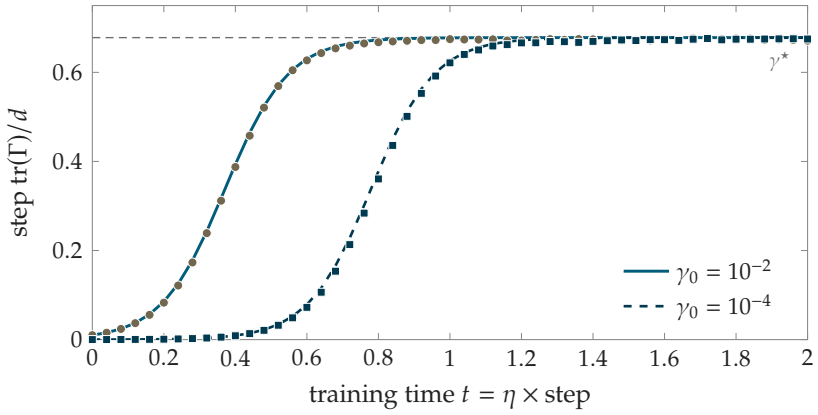


Figure 13.2. Escape from the saddle. Curves: the logistic solution (13.8) for $d = 8$, $N = 20$, $\sigma^2 = 0.5$ ($\gamma^\star = 0.678$, dashed line) and two initial steps γ_0. Points: gradient descent with step $\eta = 0.005$ on all 162 parameters, each step on a batch of 4096 fresh prompts, from the balanced initialization of Proposition 13.9 plus independent $\mathcal{N}(0, 10^{-8})$ perturbations of every parameter (seed 1017). The largest deviation from the curves is 0.017, on the steep part of the second curve.

whose variance explodes, but not the Bayes estimator, which is ridge; in the proportional limit at $\gamma = d/N = 1$ its excess risk per dimension is $(1 + \varsigma)/(2 + \varsigma)$, against $\frac{1}{2}\big(\sqrt{\varsigma^2 + 4\varsigma} - \varsigma\big)$ for Bayes (Corollary 13.7): 0.52 against 0.27 at $\varsigma = 0.1$.[4]

[4] We checked both limits in (13.7) at $d = 200$ for $(N, \varsigma) \in \{(200, 0.1), (400, 0.1), (100, 0.5)\}$: the finite-size values differ from the limits by less than 0.002.

PLATEAU, THEN A LOGISTIC JUMP. Figure 13.2 shows the dynamics of Proposition 13.9. The origin $\Gamma = 0$ is a saddle of the loss in the parameters, as in the deep linear networks of Chapter 2: the product $v_{22} W_{11}^\top$ has a vanishing gradient when both factors are small, so a small initialization spends a time $\log(\gamma^\star/\gamma_0)/(4\sqrt{d})$ on a plateau before the step jumps to $\gamma^\star$, the time growing like the logarithm of the inverse initialization. In-context learning appears abruptly in training, but in this model the abruptness is the generic escape from a saddle, not a new mechanism; the induction heads of Chapter 14 give a second instance.

WHAT SHIFTS IT TOLERATES. The prediction $x_q^\top \Gamma^\star s$ is linear in the labels, and its risk depends on the law of the task only through $\mathbb{E} w_\star w_\star^\top$; for a fixed task it depends on $w_\star$ only through $\|w_\star\|$ when $\Sigma = I_d$ (Exercise 13.4). So the learner transfers to new task distributions with the same second moment, including tasks never seen in training. It does not transfer across input distributions: $\Gamma^\star$ depends on Σ, and the prediction is cubic in the inputs. If the inputs of a test prompt are scaled by a, the noiseless part of the target scales by a and that of the prediction by a^3. If the input covariance is multiplied by $c = 2$, with the parameters of Figure 13.2, the excess risk rises from 2.6 to 15.6, nearly the null risk 16, while least squares does not notice the scaling (Exercise 13.6). Zhang et al. (2024) found the trained layer brittle under covariate shift in this sense, and reported that larger nonlinear transformers are more robust.

13.6 What the model gets wrong

WHAT WAS THROWN AWAY. Without the softmax, attention cannot select examples: every example enters the prediction with a weight that is a quadratic form, so the layer cannot implement nearest neighbors, sparse regression, or a data-dependent step. With one layer, the forward pass can take one step, and the learner cannot be consistent at the rate of least squares. Putting x_i and y_i in one token removes the hardest part of in-context learning in language, finding which earlier token goes with which, the job of induction heads (Chapter 14). Linear tasks with a Gaussian prior make the Bayes estimator ridge regression, a linear function of the labels; for other priors the Bayes estimator is nonlinear in the labels, and no single linear layer can follow it. Drawing a fresh task for every prompt hides the memorization regime of Raventós et al. (2023), and training on the population loss hides the number of pretraining prompts needed (Lu et al. 2025; Wu et al. 2024).

WHERE IT BREAKS. *Deeper models are not one step.* The transformers of Garg et al. (2022) reach the error of least squares, far below the one-step risk $d(d+1)/(N+d+1)$ of the noiseless model, so they cannot be computing one gradient step. Ahn et al. (2023) proved that for L layers of linear attention certain critical points of the loss implement L steps of preconditioned gradient descent, but not that training finds them; Fu et al. (2024) found that the predictions of successive layers of trained transformers track the iterates of an iterative Newton method, which converges exponentially faster than gradient descent, and proved that k Newton iterations can be implemented with $k + O(1)$ layers; von Oswald et al. (2023) themselves reported that deeper trained models improve on plain gradient descent with a learned curvature correction. "One layer is one gradient step" survives; "in-context learning is gradient descent" does not follow from it.

Softmax attention computes something else. Collins et al. (2024) showed that a softmax attention layer trained on in-context regression learns a nearest-neighbor-like predictor whose window adapts to the Lipschitz constant and the noise of the pretraining functions, which linear attention cannot do. A softmax layer is not a gradient step on the squared loss, and its optimum depends on the function class in a way the linear model does not see.

Language models are not trained on regression prompts. Shen et al. (2024) compared in-context learning in pretrained language models (LLaMa-7B among them) with gradient descent fine-tuning on the same demonstrations and found that the two respond differently, for instance to the order of the demonstrations; they argued that the evidence for the equivalence came from models trained specifically for in-context regression, with hand-built or specially trained weights. Much of in-context learning in language models is task recognition rather than learning (Min et al. 2022; Pan et al. 2023), a retrieval of something learned in the weights, which no model with a fresh task per prompt can represent. And in-context learning in transformers can be transient, giving way to in-weights learning as training continues (Singh et al. 2023), while in this model the global minimum is

stable.

The data decide whether there is in-context learning at all. Chan et al. (2022) found that in-context learning emerges only for training data with particular properties, bursty sequences and many rare classes, and that it trades off against in-weights learning. In Assumption 13.2 the task changes with every prompt, so in-context learning is the only option.

FINE STRUCTURE. The Bayesian view of Xie et al. (2022) explains in-context learning in language as inference of a latent concept shared by a document, of which the Gaussian posterior of Corollary 13.7 is the simplest case. Task diversity was quantified by Raventós et al. (2023) empirically and by Wu et al. (2024) and Lu et al. (2025) in solvable linear-attention models: Wu et al. (2024) bound the number of independent pretraining tasks needed and show that the pretrained model is then nearly Bayes optimal at a fixed context length. Transformers can select among algorithms in context, for instance between ridge with different penalties, and reach near-Bayes risk on mixtures of noise levels (Bai et al. 2023). Looped transformers, which apply one block repeatedly, learn in-context regression and other function classes about as well as standard transformers with under a tenth of the parameters (Yang et al. 2024). For softmax attention, Huang et al. (2024) proved convergence of gradient descent for a one-layer transformer on in-context linear regression with features drawn from a finite set, with stage-wise dynamics when the features are imbalanced. The solvable model of how a multi-layer softmax transformer comes to learn in context, and of which algorithm it then runs, is open; the induction-head models of Chapter 14 solve one piece of it.

13.7 Model card

MODEL CARD	
PHENOMENON	Transformers learn a task from the examples in the prompt, without weight updates; trained on regression prompts, they predict nearly as well as least squares.
MODEL	One layer of linear self-attention on prompts $(x_1, y_1, \dots, x_N, y_N, x_q)$ with Gaussian inputs, a Gaussian task drawn per prompt, and Gaussian noise; population pretraining loss.
SOLUTION	The layer computes $x_q^\top \Gamma s + x_q^\top b$; the global optimum is $b = 0$, $\Gamma^\star = N((N+1)\Sigma + (\operatorname{tr}\Sigma + \sigma^2)I)^{-1}$ (Theorems 13.5 and 13.6); gradient flow finds it (Theorem 13.8), along a logistic curve from a small initialization (Proposition 13.9).
PREDICTS	One preconditioned gradient step, a Newton step for long prompts; excess risk $d(d+1+\sigma^2)/N$, a factor $(d+1+\sigma^2)/\sigma^2$ above least squares; a plateau of length $\log(1/\gamma_0)/(4\sqrt{d})$; tolerance of task shift, brittleness under covariate shift.
GETS WRONG	Deeper and softmax transformers run other algorithms (Newton-like iterations, adaptive nearest neighbors); language models also recognize tasks; data properties and

	transience; no memorization regime.
SUCCESSORS	Multi-layer linear attention; task diversity in proportional limits; algorithm selection; looped transformers; softmax dynamics; induction heads.

The Gaussian design returns from Chapter 1: the Bayes estimator of Corollary 13.7 is the optimally tuned ridge of Proposition 1.12, and its proportional limit comes from the Marchenko–Pastur law. Rotational and sign symmetries of Gaussian data did the work here as in Chapters 7 and 8. The escape from the saddle at $\Gamma = 0$ is that of the deep linear networks of Chapter 2, with the same conservation law and the same logistic curve, and it recurs in the induction heads of Chapter 14, where a second layer must compose with a first before either can learn.

Exercises

13.1 CONSOLIDATION. (Gaussian fourth moments.) (a) Prove $\mathbb{E}[g_i g_j g_k g_l] = \delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}$ for $g \sim \mathcal{N}(0, I_d)$ by counting cases. (b) Deduce from Lemma 13.4 that $\mathbb{E}[\hat{\Sigma}A\hat{\Sigma}] = \frac{1}{N}\big((N-1)\Sigma A\Sigma + \Sigma A\Sigma + \Sigma A^\top\Sigma + \operatorname{tr}(A\Sigma)\Sigma\big)$. (c) Show that $\mathbb{E}\|s - w_\star\|^2 = d(d+1+\sigma^2)/N$ when $\Sigma = I_d$.

13.2 CONSOLIDATION. (The best scalar step.) Under Assumption 13.2, restrict to $\Gamma = \eta I_d$. Show that the best η is $\eta^\star = N\operatorname{tr}(\Sigma^2)/\big((N+1)\operatorname{tr}(\Sigma^3) + (\operatorname{tr}\Sigma + \sigma^2)\operatorname{tr}(\Sigma^2)\big)$, and that it coincides with $\Gamma^\star$ exactly when Σ is a multiple of I_d. For $\Sigma = \operatorname{diag}(1, \epsilon)$ and $\sigma = 0$, show that as $N \to \infty$ the risk of $\Gamma^\star$ tends to 0, while that of $\eta^\star I_2$ tends to $\epsilon(1-\epsilon)^2/(1+\epsilon^3)$.

13.3 CONSOLIDATION. (Logistic escape.) (a) For the setting of Figure 13.2, compute $t_{1/2}$ for $\gamma_0 = 10^{-2}$ and $\gamma_0 = 10^{-4}$ (answers: 0.37 and 0.78). (b) Show that halving γ_0 delays the escape by $\log 2/(4\sqrt{d}) + O(\gamma_0/\gamma^\star)$ as $\gamma_0 \to 0$, and that the time to reach $(1-\delta)\gamma^\star$ is $\log\big((\gamma^\star/\gamma_0 - 1)(1-\delta)/\delta\big)/(4\sqrt{d})$. (c) Show that along the flow of Proposition 13.9 the loss is $\sigma^2 + d - 2d\gamma + d\gamma^2/\gamma^\star$, and sketch it against t.

13.4 CONSOLIDATION. (Task shift.) Let $\Sigma = I_d$ and fix $w_\star$. Show that the risk of the predictor $x_q^\top \gamma s$, averaged over the inputs and noises only, is $\sigma^2 + \|w_\star\|^2\big(1 - 2\gamma + \gamma^2\frac{N+d+1}{N}\big) + \gamma^2\frac{\sigma^2 d}{N}$. Deduce that the trained layer has the same risk on every task distribution with $\mathbb{E}\|w_\star\|^2 = d$, for instance tasks uniform on the sphere of radius $\sqrt{d}$, or a single fixed task of that norm.

13.5 CONSOLIDATION. (The proportional limit.) (a) Check part (iii) of Corollary 13.7, and show that at $\gamma = 1$ the limits are $(1+\varsigma)/(2+\varsigma)$ and $\frac{1}{2}(\sqrt{\varsigma^2+4\varsigma} - \varsigma)$. (b) Show that at $\varsigma \to 0$ the Bayes limit is $(1 - 1/\gamma)_+$, the fraction of directions the prompt does not see, while the one-step limit is $\gamma/(1+\gamma)$.

13.6 VARIATION. (Covariate shift.) Train at $\Sigma = I_d$, so that $\Gamma = \gamma^\star I_d$, and test on prompts whose inputs (context and query) have law $\mathcal{N}(0, cI_d)$. (a)

Show that the excess risk is $dc - 2d\gamma^\star c^2 + d\gamma^{\star 2}c^2\big((N+1+d)c+\sigma^2\big)/N$, which grows like c^3. (b) With $d = 8$, $N = 20$, $\sigma^2 = 0.5$, evaluate it at $c \in \{\frac{1}{2}, 1, 2\}$ and compare with the null risk dc. (c) Show that for $N \geq d+2$ the risk of least squares does not depend on c.

13.7 VARIATION. (Two layers, two steps.) Stack L layers (13.2), each acting on all columns of the output of the previous one, with $W^{KQ} = \begin{pmatrix} \eta I_d & 0 \\ 0 & 0 \end{pmatrix}$ and $W^{PV} = \begin{pmatrix} 0 & 0 \\ 0 & -1 \end{pmatrix}$ in every layer, and read the prediction as minus the bottom entry of the query column. Show by induction that after ℓ layers the label entry of column $i \leq N$ is $y_i - \langle w_\ell, x_i\rangle$ and that of the query column is $-\langle w_\ell, x_q\rangle$, where w_ℓ are the gradient descent iterates $w_{\ell+1} = w_\ell - \eta\nabla\widehat{\mathcal{L}}(w_\ell)$ from $w_0 = 0$ (von Oswald et al. 2023). Replace ηI_d by $\Gamma^\top$ to get preconditioned steps.

13.8 VARIATION. (Finite task diversity.) Let the task be $\pm w_k$, with k uniform on $\{1, \dots, M\}$ and an independent uniform sign, let $\Sigma = I_d$, and let $W = \frac{1}{M}\sum_k w_k w_k^\top$. (a) Show that Theorems 13.5 and 13.6 hold with

$$\Gamma^\star = NW\big((N+1)W + (\operatorname{tr} W + \sigma^2)I_d\big)^{-1}.$$

(b) If $M < d$ and the w_k are linearly independent, show that as $N \to \infty$ the trained layer tested on fresh tasks $w_\star \sim \mathcal{N}(0, I_d)$ has excess risk tending to $d - M$: it has learned the span of the training tasks, not the tasks themselves. *(Hint: the sign flips keep the law symmetric, which is what Theorem 13.6 needs.)*

13.9 VARIATION. (Other symmetric laws.) (a) Show that Theorem 13.5 holds for every task law with $\mathbb{E}w_\star w_\star^\top = I_d$ independent of the inputs and noises. (b) Show that Theorem 13.6 holds whenever, in addition, the laws of $w_\star$ and of the noises are symmetric and have finite fourth moments. (c) Let the query attend to itself, replacing M by I_{N+1} in (13.2); show that the extra terms are even in the labels, so that the global minimum is unchanged.

13.10 SIMULATION. (Anisotropic inputs.) Take $d = 4$, $\Sigma = \operatorname{diag}(4, 2, 1, \frac{1}{2})$, $N = 10$, and $\sigma^2 = \frac{1}{4}$. Train all 50 parameters of the layer with Adam on fresh prompts, as in Figure 13.1. Observe that the learned $\Gamma(\theta)$ is close to $\operatorname{diag}(0.193, 0.336, 0.533, 0.755)$, that the coefficients of $b(\theta)$ are near zero, and that the risk approaches (13.6). Then test the trained layer with $\Sigma' = \operatorname{diag}(\frac{1}{2}, 1, 2, 4)$ and explain the result with (13.4).

13.11 SIMULATION. (A softmax head.) Replace linear attention by one softmax head with the same embedding (13.1), train it on the prompts of Figure 13.1 with $d = 8$ and $N \in \{8, 32\}$, and compare its risk with $\mathcal{R}^\star$. Then train both models on prompts with labels $y_i = \operatorname{sgn}\langle w_\star, x_i\rangle$ and compare again. Report which model wins in each case; Theorem 13.6 predicts only the linear model's risk, and the outcome for the softmax head is the point of the exercise.

13.12 RESEARCH. ★★ (Global minima of deeper linear transformers.) For $L \geq 2$ layers of linear self-attention on the prompts of Assumption 13.2, Ahn et al. (2023) identify critical points that implement L steps of preconditioned gradient descent. Open, as far as we know: whether these are global minima for general L and Σ, what the minimal risk is as a function of (N, d, σ^2, L), and whether gradient flow from small initialization reaches it, with plateaus as in Proposition 13.9 between steps.

13.13 RESEARCH. (A solvable softmax model.) Find a task family and a softmax-attention model for which the global minimum of the pre-training loss and the gradient-flow dynamics can be computed as explicitly as in Theorem 13.6 and Proposition 13.9, and which reproduces the adaptive window of Collins et al. (2024). Results exist for special data (Huang et al. 2024); a model as complete as the linear one is open.

Notes and References

Brown et al. (2020) described in-context learning in GPT-3 and measured it against the number of demonstrations; the idea of a network that learns in its activations goes back at least to the meta-learning experiments of Hochreiter et al. (2001). Garg et al. (2022) turned the phenomenon into a controlled experiment on function classes, and within a few months Akyürek et al. (2023) and von Oswald et al. (2023), first posted in November and December 2022, connected trained transformers with gradient descent and ridge regression. Linear attention had been introduced for efficiency (Katharopoulos et al. 2020) and connected with fast weights (Schlag et al. 2021).

The landscape of one layer was analyzed in 2023 by three groups. Ahn et al. (2023) showed that the global minimum of a single layer of linear attention implements one step of preconditioned gradient descent, with a preconditioner that adapts to the input distribution and to the finite number of examples, and studied critical points of deeper models. Mahankali et al. (2024) proved that one step of gradient descent is the optimal one-layer learner for isotropic Gaussian inputs, preconditioned for nonisotropic ones, and that it remains one step of gradient descent on the least-squares loss even when the responses are nonlinear functions. Zhang et al. (2024) proved convergence of gradient flow and computed the limit; they analyzed robustness to distribution shift. The odd–even argument of Theorem 13.6 is a short route to the global minimum for the masked layer; we do not know whether it appears in this form in the literature.

The connection with ridge regression and the Bayes estimator is due to Akyürek et al. (2023) and Raventós et al. (2023); asymptotic analyses of task diversity are in Wu et al. (2024) and Lu et al. (2025). Algorithm selection in context is analyzed by Bai et al. (2023). The mechanistic study of in-context learning in language models began with induction heads (Olsson et al. 2022), the subject of Chapter 14.

TRANSLATING THE LITERATURE. Zhang et al. (2024) write N, as we do, for the number of examples in a training prompt, and M for the length of a test prompt, which may differ from N; their trained layer predicts with our $\Gamma^\star$ for their label model and training length N, applied to test prompts of length M. Ahn et al. (2023) write n for our N and P and Q for the two matrices of a layer, our W^{PV} and W^{KQ}, and call the matrix that multiplies the gradient the preconditioner, our Γ. von Oswald et al. (2023) write $e_j = (x_j, y_j)$ for our tokens z_j and keep separate key, query, value, and projection matrices, whose products are our W^{KQ} and W^{PV}; their construction of gradient steps is Exercise 13.7. Garg et al. (2022) write k for the number of in-context examples, our N, and report squared errors divided by the dimension, as in Figure 13.1.

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14 Induction Heads

In-context Markov chains and a two-layer circuit

THE MODEL — ATTENTION-ONLY TRANSFORMER ON RANDOM MARKOV CHAINS

IN THIS CHAPTER. Small transformers trained on text acquire, in a short burst early in training, attention heads that complete the pattern $[A][B]\dots[A] \to [B]$, and their ability to use long contexts jumps at the same moment. We study the phenomenon on sequences drawn from a random Markov chain, a new chain for every sequence, so that next-token prediction can only be learned in context. The Bayes-optimal predictor is an add-κ bigram estimator, and we prove that a two-layer attention-only circuit (a previous-token head, an induction head, and an attention sink that carries the prior) computes it with an error exponentially small in the attention scales. In a three-parameter caricature with linear attention we prove that gradient flow first learns the in-context unigram statistics, sits on a plateau, and leaves it after a time $\log(1/\alpha)/(2G)$ set by the initialization scale α and an explicit induction gain G; a small softmax transformer, with a simplified readout, shows a plateau whose length is consistent with the same logarithmic dependence. The model is silent on semantics, on heads that copy abstract patterns, and on why induction heads in language models coincide with other changes.

14.1 The phenomenon: a phase change in context use

In December 2021 a group at Anthropic published a study of transformers stripped down to attention: no MLP layers, and at most two layers (Elhage et al. 2021). A one-layer model of this kind is a sum of terms that each read one earlier token, and it learns what the authors called skip-trigrams. Two-layer models did something new. Some heads in the second layer attended, from the current token A, to the position *just after* an earlier occurrence of A, and copied the token found there: on a sequence $\dots[A][B]\dots[A]$ they predicted $[B]$. Elhage and coauthors named these *induction heads*. Such a head cannot work alone. Its key at position s must contain the token at position $s-1$, which a *previous-token head* in the first layer writes there; the second head's keys read the first head's output, a mechanism they called *K-composition*.[1]

Olsson et al. (2022) followed the heads through training. They measured context use by an *in-context learning score*: the loss on the 500th token of a context minus the loss on the 50th token, averaged over text. In models of many sizes, with and without MLPs, induction heads appear in a short window early in training, the score improves abruptly in the same window,

[1] In their language a head is described by two low-rank matrices, $W_{QK} = W_Q^\top W_K$, which decides where to attend, and $W_{OV} = W_O W_V$, which decides what to copy. K-composition means that W_{QK} of a later head reads a subspace written by W_{OV} of an earlier one.

and the training loss shows a visible bump there. They argued, from several lines of evidence, that induction heads may be the mechanism behind most in-context learning; the evidence is causal for small attention-only models, where removing the heads removes the effect, and correlational for large models with MLPs.

OBSERVATION 14.1 (Elhage et al. 2021; Olsson et al. 2022). In transformers with at least two layers trained on text, induction heads form during a short period early in training. During it the in-context learning score improves abruptly, and the training loss curve shows a bump. One-layer attention-only models do not form induction heads.

The observation has been reproduced in controlled settings, which also show that it is fragile. Chan et al. (2022) trained transformers on sequences of image–label pairs and found that in-context learning emerges when the training data are *bursty*, with classes appearing in clusters within a sequence, and contain many rare classes; with a Zipfian class distribution in-context learning and learning in the weights coexist. Singh et al. (2023) found that in-context learning can be *transient*: on such data it emerges and then fades as in-weights learning takes over, although the training loss keeps decreasing. Reddy (2024) reduced an in-context classification task to a two-parameter model of an induction head, in which the abrupt transition comes from nested nonlinearities learned in sequence. Singh et al. (2024) dissected the formation of induction heads in a synthetic task and found that several subcircuits must form and interact before the head works.[2]

The cleanest setting is the one we adopt. Edelman et al. (2024) trained small transformers on sequences, each generated by its own random Markov chain, so that the next token can be predicted well only from the statistics of the current sequence. They observed training in stages: uniform predictions, then a plateau at a suboptimal strategy that uses only the in-context frequencies of single tokens, then a rapid jump to predictions that use the in-context frequencies of pairs, computed by what they called *statistical induction heads*. They also gave evidence that the simpler unigram solution delays the bigram one.

Three questions follow. *What* should an induction head compute, statistically? *How* does a two-layer network compute it, and why two layers? And *why* does it form late and abruptly, after a plateau, rather than gradually from the start?

[2] Their tool is a causal intervention on the training dynamics, modifying activations throughout training, rather than on the trained network.

14.2 The model: Markov chains drawn afresh

THE DATA. We study *in-context Markov chains*. Tokens take values in $[K] = \{1, \dots, K\}$; we use K, the global symbol for a number of classes, because next-token prediction is a K-class problem. Fix a concentration $\kappa > 0$.[3] Each sequence is generated in three steps.

Assumption 14.2 (In-context Markov chains). (i) A transition matrix P is drawn with independent rows $P_{i\cdot} \sim \mathrm{Dir}(\kappa \mathbf{1}_K)$, $i \in [K]$. (ii) The first token x_1 is drawn from a law μ on $[K]$ that does not depend on P. (iii)

[3] Papers on this task often write α for the Dirichlet parameter; we reserve α for the initialization scale, as everywhere in the book.

Given P and $x_1, \dots, x_s$, the next token is $x_{s+1} = j$ with probability $P_{x_s j}$.

Recall that the Dirichlet law $\mathrm{Dir}(\kappa_1, \dots, \kappa_K)$ on the simplex has density proportional to $\prod_j q_j^{\kappa_j - 1}$ and mean $\kappa_j / \sum_l \kappa_l$. With probability one every entry of P is positive, so P has a unique stationary law $\pi = \pi(P)$, with $\pi P = \pi$. A model sees a context $x_{1:t} = (x_1, \dots, x_t)$ and outputs a probability vector $q_t \in \mathbb{R}^K$ for x_{t+1}; it is trained on the cross-entropy $\mathbb{E}[-\log q_t(x_{t+1})]$ averaged over the positions $t = 1, \dots, N$ of contexts of length N. A fresh P for every sequence makes all in-weights knowledge about transitions useless: averaged over P, every row is uniform.[4] The counts in the context are everything,

$$v_{ij}(t) = \#\{s < t : \ x_s = i, \ x_{s+1} = j\}, \qquad v_i(t) = \sum_j v_{ij}(t),$$
$$v_j^U(t) = \#\{s \le t : \ x_s = j\},$$

the transition counts out of state i and the unigram counts.

[4] In the language of Chapter 13, the chain P is the task and the context is its training set; the network must learn a learning algorithm.

THE NETWORK. We use an attention-only transformer with two layers, one head each, in which every layer writes its output to a fresh subspace of the residual stream, so that later layers can read each part separately.[5] Position $s \ge 1$ carries the one-hot vector $e_{x_s} \in \mathbb{R}^K$ of its token; an extra position $s = 0$, the *attention sink*, carries $e_{x_0} := 0$. Layer 1 is a previous-token head of inverse temperature β_1: position $s \ge 1$ attends to $s' \in \{0, \dots, s\}$ with weights

$$a^{(1)}_{ss'} = \frac{e^{\beta_1 \mathbf{1}\{s' = s-1\}}}{\sum_{r=0}^{s} e^{\beta_1 \mathbf{1}\{r = s-1\}}}, \qquad h_s = \sum_{s'=0}^{s} a^{(1)}_{ss'} \, e_{x_{s'}} \in \mathbb{R}^K. \tag{14.1}$$

[5] This "disentangled" form is used by Nichani et al. (2024); a standard transformer with a wide residual stream can embed it, by placing token, position and head outputs in orthogonal subspaces.

Its logits depend on positions only (one-hot positions and a shift matrix as W_{QK}), and its output h_s is written next to e_{x_s}. Layer 2 is an induction head of inverse temperature β_2 with a sink. From the last position t it attends to $s \in \{0, \dots, t\}$ with logits

$$\ell_0 = \beta_2 + \log(K\kappa), \qquad \ell_s = \beta_2 \langle e_{x_t}, h_s \rangle \quad (1 \le s \le t), \tag{14.2}$$

and weights $a^{(2)}_s = e^{\ell_s} / \sum_r e^{\ell_r}$; the query reads the token at t and the key reads the subspace written by layer 1, which is K-composition. Its values are e_{x_s} for $s \ge 1$ and the uniform vector $\frac{1}{K}\mathbf{1}_K$ for the sink, so the output

$$\hat{y}_t = a^{(2)}_0 \, \tfrac{1}{K}\mathbf{1}_K + \sum_{s=1}^{t} a^{(2)}_s \, e_{x_s} \tag{14.3}$$

is a probability vector with positive entries, and the network predicts $q_t = \hat{y}_t$. Reading the output in this way, as a mixture of the values, replaces the unembedding and MLP of a real transformer, which would have to compute logarithms of the mixture; in the experiments of Section 14.5 the logits are $\gamma \log \hat{y}_t$ with a learned γ.

THE CARICATURE FOR THE DYNAMICS. Gradient flow on softmax attention is not solvable in closed form. For the dynamics we keep the composition and drop everything else: linear attention, square loss, three scalar parameters. Layer 1 has positional score $a\mathbf{1}\{s' = s-1\}$, so that $h_s = a\, e_{x_{s-1}}$ (and $h_1 = 0$); layer 2 has score $c + b\langle e_{x_t}, h_s\rangle$, values e_{x_s}, and normalization $1/t$. Its output is

$$\hat{y}_t = c\,\varphi_U + ab\,\varphi_B, \qquad \varphi_U = \frac{1}{t}\sum_{s=1}^{t} e_{x_s}, \qquad \varphi_B = \frac{1}{t}\sum_{s=2}^{t} \mathbf{1}\{x_{s-1} = x_t\}\, e_{x_s}, \tag{14.4}$$

that is, $\varphi_U(j) = v_j^U(t)/t$ and $\varphi_B(j) = v_{x_t j}(t)/t$. The parameter c is a unigram strength, a the strength of the previous-token head, and b that of the induction head; the bigram feature enters through the product ab because it needs both layers. We train at one position $t = T$ on the square loss $\mathcal{L}(c, a, b) = \mathbb{E}\|\hat{y}_T - e_{x_{T+1}}\|^2$ by gradient flow from $c = 0$ and $a = b = \alpha$, with a small initialization scale α.

DICTIONARY

IN THE MODEL	IN A TRAINED NETWORK
random transition matrix P, fresh for each sequence	a document with its own statistics; patterns that recur within it
transition counts $v_{ij}(t)$	repeated bigrams in a context, $[A][B]\ldots[A][B]$
layer 1, logits $\beta_1\mathbf{1}\{s' = s-1\}$	a previous-token head
layer 2, logits $\beta_2\langle e_{x_t}, h_s\rangle$	an induction head, K-composed with the previous-token head
sink of logit $\beta_2 + \log(K\kappa)$, value $\frac{1}{K}\mathbf{1}_K$	an attention sink, here carrying the Dirichlet prior
unigram strength c	uniform attention over the context, the first thing learned
product ab of layer strengths	the induction circuit, useful only when both layers are in place
induction gain G	how much the context's bigram statistics improve on its unigram statistics

WHAT WAS THROWN AWAY. Semantics, MLPs, layer normalization, multiple heads, learned embeddings, and every in-weights regularity: the data are designed so that only in-context statistics help. The circuit model fixes the form of both heads and leaves two scales; the caricature fixes the form of the heads and leaves three scalars, linear attention and a square loss. These are the right first simplifications because the phenomenon is a fact about *composition* (a computation that needs two heads, one feeding the other) and about *statistics* (what an optimal predictor should do with a context), and both survive them.

WHAT ONE HAD TO NOTICE. That an induction head is a statistical estimator: copying the token after an earlier match, averaged over all matches, is counting bigrams. That a sequence model trained on a mixture of tasks predicts with the posterior of the task, so that the right benchmark is a

Bayes predictor. And that a product of two small strengths is a saddle, the object of Chapter 2, so that composition alone predicts a plateau whose length grows like the logarithm of the inverse initialization.

14.3 The solution: what an induction head computes

14.3.1 The Bayes predictor

The *Bayes predictor* of the next token is its conditional law given the context.

Theorem 14.3 (Bayes-optimal next-token prediction). *Under Assumption 14.2, for every $t \geq 1$ and every context $x_{1:t}$ of positive probability, the rows of P are conditionally independent given $x_{1:t}$, with $P_{i\cdot} \sim \mathrm{Dir}(\kappa \mathbf{1}_K + \nu_{i\cdot}(t))$, and*

$$\Pr(x_{t+1} = j \mid x_{1:t}) = p_t^{\mathrm{B}}(j) := \frac{\nu_{x_t j}(t) + \kappa}{\nu_{x_t}(t) + K\kappa} \qquad \text{for every } j \in [K]. \quad (14.5)$$

Every predictor q_t, a function of $x_{1:t}$ with values in the simplex, satisfies $\mathbb{E}[-\log q_t(x_{t+1})] \geq \mathbb{E}[-\log p_t^{\mathrm{B}}(x_{t+1})]$, with equality if and only if $q_t = p_t^{\mathrm{B}}$ almost surely.

Proof. The plan is to write the joint density of P and the context, recognize a product of Dirichlet densities, and take the posterior mean. Let $Z = \Gamma(\kappa)^K / \Gamma(K\kappa)$ be the normalizing constant of $\mathrm{Dir}(\kappa \mathbf{1}_K)$. By Assumption 14.2, the joint density of P and $x_{1:t}$ is

$$\mu(x_1) \prod_{s=1}^{t-1} P_{x_s x_{s+1}} \prod_{i=1}^{K} \frac{1}{Z} \prod_{j=1}^{K} P_{ij}^{\kappa - 1} = \frac{\mu(x_1)}{Z^K} \prod_{i=1}^{K} \prod_{j=1}^{K} P_{ij}^{\kappa + \nu_{ij}(t) - 1},$$

because the product over s contains the factor P_{ij} exactly $\nu_{ij}(t)$ times. Here we used (ii): the law of x_1 does not involve P. As a function of P, the right side is a product over rows of unnormalized $\mathrm{Dir}(\kappa \mathbf{1}_K + \nu_{i\cdot}(t))$ densities, which proves the first claim. By (iii), $\Pr(x_{t+1} = j \mid x_{1:t}, P) = P_{x_t j}$, so $\Pr(x_{t+1} = j \mid x_{1:t}) = \mathbb{E}[P_{x_t j} \mid x_{1:t}]$, the mean of the jth coordinate of $\mathrm{Dir}(\kappa \mathbf{1}_K + \nu_{x_t \cdot}(t))$, which is (14.5).

For the second claim, condition on $x_{1:t}$. The difference of the conditional cross-entropies of q_t and p_t^{B} is $\sum_j p_t^{\mathrm{B}}(j) \log(p_t^{\mathrm{B}}(j)/q_t(j))$, the Kullback–Leibler divergence of q_t from p_t^{B}, which is nonnegative and vanishes only when $q_t = p_t^{\mathrm{B}}$ (Gibbs' inequality). Take expectations. ∎

Two cases deserve a word. At $t = 1$, and whenever the current token has not been seen before position t, all counts out of x_t vanish and the prediction is uniform. Hypothesis (ii) is used only in the first display; if the first token is drawn from the stationary law $\pi(P)$, the formula fails, because the first token then carries information about P (Exercise 14.1). The estimator (14.5) adds κ pseudo-counts to each transition: $\kappa = 1$ is Laplace's rule of succession and $\kappa = \frac{1}{2}$ the estimator of Krichevsky and Trofimov (1981).[6]

[6] Laplace derived his rule for two outcomes: after s successes in n trials, the probability of a success next is $(s+1)/(n+2)$ under a uniform prior. Krichevsky and Trofimov showed that $\kappa = \frac{1}{2}$ is nearly minimax for the cumulative log loss of universal coding.

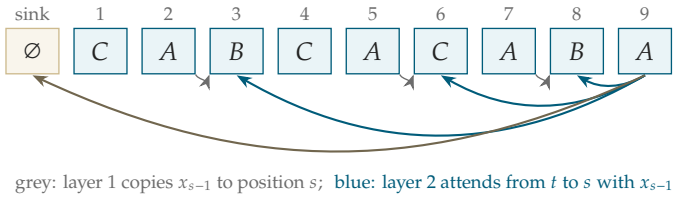


Figure 14.1. The circuit of Theorem 14.6 on the context $CABCACABA$, with $t = 9$ and $x_t = A$. Layer 1 writes x_{s-1} at each position s (three of the grey arrows drawn). Layer 2 attends from position 9 to the positions $3, 6, 8$ that follow an earlier A, whose tokens are B, C, B, and to the sink. With $\kappa = 1$ and $K = 3$ the output is $\left(\frac{0+1}{3+3}, \frac{2+1}{3+3}, \frac{1+1}{3+3}\right)$ for (A, B, C), the Bayes predictor.

Two cruder predictors serve as baselines: the *uniform* predictor $\frac{1}{K}\mathbf{1}_K$, with loss $\log K$ at every position, and the *unigram* predictors $(\nu_j^U(t)+\kappa')/(t+K\kappa')$, which use the frequencies of single tokens in the context but not the current token. For long contexts their losses have simple limits. Write $\mathrm{H}(q) = -\sum_j q_j \log q_j$ for entropy.

Proposition 14.4 (Long-context losses). *Under Assumption 14.2, as $t \to \infty$, the expected loss of p_t^{B} converges to the mean entropy rate $\mathrm{H}_B = \mathbb{E}\sum_i \pi_i \mathrm{H}(P_{i\cdot})$, and that of the unigram predictor, for every fixed $\kappa' > 0$, to the mean stationary entropy $\mathrm{H}_U = \mathbb{E}\,\mathrm{H}(\pi)$. Moreover $\mathrm{H}_B \le \mathrm{H}_U \le \log K$.*

Proof sketch. Fix P; its entries are positive, so the chain is ergodic, and by the ergodic theorem $\nu_j^U(t)/t \to \pi_j$ and $\nu_{ij}(t)/t \to \pi_i P_{ij}$ almost surely. Hence $p_t^{\mathrm{B}}(j) \to P_{x_t j}$ and the unigram predictor tends to π, uniformly over the next token. The law of (x_t, x_{t+1}) given P converges to $\pi_i P_{ij}$. The losses are bounded by $\log((t + K\kappa)/\kappa)$, which grows too slowly to matter after a truncation argument, and the limits follow by dominated convergence; we omit the routine details. For the inequalities: $\mathrm{H}(\pi) = \mathrm{H}(\sum_i \pi_i P_{i\cdot}) \ge \sum_i \pi_i \mathrm{H}(P_{i\cdot})$ by concavity of entropy and stationarity, and $\mathrm{H}(\pi) \le \log K$. ∎

For $K = 2$ the rows are $(1-u, u)$ and $(w, 1-w)$ with u, w independent uniform, and the three limits are $\log 2 = 0.693$, $\mathrm{H}_U = 0.564$ and $\mathrm{H}_B = 0.481$ (two-dimensional quadrature, Exercise 14.2). For $K = 3$, Monte Carlo over 2×10^5 matrices gives $\log 3 = 1.099$, $\mathrm{H}_U = 0.979$ and $\mathrm{H}_B = 0.819$. These are the three levels at which training will pause.

14.3.2 A two-layer circuit

The network (14.1)–(14.3) computes p_t^{B} in the limit of large inverse temperatures (Figure 14.1). We need one fact about softmax.

Lemma 14.5 (Softmax is Lipschitz in ratio). *Let ℓ, ℓ' be real vectors indexed by the same finite set, with $|\ell_s - \ell'_s| \le \varepsilon$ for every s, and let w and w' be their softmax weights. Then $e^{-2\varepsilon} \le w_s/w'_s \le e^{2\varepsilon}$ for every s.*

PROOF. Write $w_s/w'_s = e^{\ell_s-\ell'_s}\cdot\sum_r e^{\ell'_r}/\sum_r e^{\ell_r}$. The first factor lies in $[e^{-\varepsilon}, e^{\varepsilon}]$, and so does the second, because each term e^{ℓ_r} lies between $e^{-\varepsilon}e^{\ell'_r}$ and $e^{\varepsilon}e^{\ell'_r}$. ∎

Theorem 14.6 (A two-layer circuit computes the Bayes predictor). *Let $\kappa > 0$, $\beta_1 \in \mathbb{R}$ and $\beta_2 \geq 0$. For every $t \geq 1$, every sequence $x_1, \dots, x_t \in [K]$, and every $j \in [K]$, the output (14.3) satisfies*

$$|\log \hat{y}_t(j) - \log p^{\mathrm{B}}_t(j)| \leq 2\beta_2 t e^{-\beta_1} + \frac{t}{\kappa}e^{-\beta_2}. \tag{14.6}$$

In particular, if $0 < \varepsilon < 2t/\kappa$, $\beta_2 = \log(2t/(\kappa\varepsilon))$ and $\beta_1 \geq \log(4\beta_2 t/\varepsilon)$, then for every law of the sequence the cross-entropy of $\hat{y}_t$ exceeds that of p^{B}_t by at most ε; under Assumption 14.2, p^{B}_t is the Bayes predictor.

PROOF. The plan is to compare the network with an ideal one whose first layer is exact, using Lemma 14.5, and then to compare the ideal network at finite β_2 with its limit. Write $i = x_t$ and set $\mathbf{1}\{x_0 = i\} := 0$ for the sink.

Layer 1. For $s \geq 1$ the softmax (14.1) puts weight $e^{\beta_1}/(e^{\beta_1}+s)$ on $s-1$, so the remaining weight is $\delta_s = s/(e^{\beta_1}+s) \leq te^{-\beta_1}$. Since $\langle e_i, e_{x_{s'}}\rangle = \mathbf{1}\{x_{s'} = i\} \in \{0,1\}$,

$$\begin{aligned}\langle e_i, h_s\rangle &= (1-\delta_s)\mathbf{1}\{x_{s-1}=i\} + \sum_{s'\neq s-1} a^{(1)}_{ss'}\mathbf{1}\{x_{s'}=i\}\\ &\in \big[(1-\delta_s)\mathbf{1}\{x_{s-1}=i\},\ (1-\delta_s)\mathbf{1}\{x_{s-1}=i\}+\delta_s\big],\end{aligned}$$

so $|\langle e_i, h_s\rangle - \mathbf{1}\{x_{s-1}=i\}| \leq \delta_s \leq te^{-\beta_1}$.

An ideal second layer. Let $\ell^*_0 = \ell_0$ and $\ell^*_s = \beta_2\mathbf{1}\{x_{s-1}=i\}$ for $1 \leq s \leq t$, and let w be their softmax weights. By the bound on layer 1, $|\ell_s - \ell^*_s| \leq \beta_2 te^{-\beta_1}$ for every s, so by Lemma 14.5 each weight $a^{(2)}_s$ lies within a factor $e^{\pm 2\beta_2 te^{-\beta_1}}$ of w_s. The values in (14.3) have nonnegative coordinates, so the same factor bounds $\hat{y}_t(j)$ against $\tilde{y}(j) := w_0/K + \sum_{s\geq 1} w_s\mathbf{1}\{x_s = j\}$:

$$|\log \hat{y}_t(j) - \log \tilde{y}(j)| \leq 2\beta_2 te^{-\beta_1}. \tag{14.7}$$

The ideal network against the Bayes predictor. The positions $s \in \{2,\dots,t\}$ with $x_{s-1} = i$ are exactly the ends of the transitions out of i within $x_{1:t}$: there are $\nu_i(t)$ of them, and $\nu_{ij}(t)$ of them carry the token j. Each has unnormalized weight e^{β_2}; the sink has $K\kappa e^{\beta_2}$; the other $M = t - \nu_i(t)$ positions have weight 1, and $r_j \in [0, M]$ of them carry j. Hence

$$\tilde{y}(j) = \frac{e^{\beta_2}(\nu_{ij}(t)+\kappa) + r_j}{e^{\beta_2}(\nu_i(t)+K\kappa) + M}.$$

Dividing by $p^{\mathrm{B}}_t(j) = (\nu_{ij}(t)+\kappa)/(\nu_i(t)+K\kappa)$ and using $0 \leq r_j \leq M \leq t$, $\nu_{ij}(t)+\kappa \geq \kappa$ and $\nu_i(t)+K\kappa \geq \kappa$,

$$\Big(1+\frac{t}{\kappa}e^{-\beta_2}\Big)^{-1} \leq \frac{\tilde{y}(j)}{p^{\mathrm{B}}_t(j)} \leq 1+\frac{t}{\kappa}e^{-\beta_2},$$

and $\log(1+u) \le u$ gives $|\log \tilde{y}(j) - \log p_t^{\mathrm{B}}(j)| \le te^{-\beta_2}/\kappa$. Together with (14.7) this proves (14.6).

The choice of scales. With the stated β_2, which is positive because $\varepsilon < 2t/\kappa$, the second term of (14.6) is $\varepsilon/2$, and with the stated β_1 the first is at most $\varepsilon/2$. The cross-entropies of $\hat{y}_t$ and p_t^{B} are expectations of $-\log \hat{y}_t(x_{t+1})$ and $-\log p_t^{\mathrm{B}}(x_{t+1})$, whose difference is at most ε pointwise. ■

The bound holds sequence by sequence: the circuit is an exact algorithm for the add-κ bigram estimator, and the Markov assumption is needed only to make that estimator optimal. We checked (14.6) on 300 random sequences and scales; the observed error never exceeded 0.65 times the bound. Three features of the construction will matter later. The *sink* carries the prior: its logit offset $\log(K\kappa)$ adds κ pseudo-counts to every transition, and without it the circuit outputs the empirical bigram distribution, whose cross-entropy is infinite whenever a transition is unseen (Exercise 14.5). When the current token has not occurred before, every non-sink logit is near 0 against $\beta_2 + \log(K\kappa)$ at the sink, and the output is nearly uniform, as the Bayes predictor requires. And when $\beta_2 = 0$ the second layer ignores the first and returns a mixture of the in-context unigram frequencies and the uniform vector: an untrained induction head is a unigram head.[7]

Why two layers? The key at position s must depend on x_{s-1}, information that sits at another position, and in a one-layer transformer keys are computed from one position's embedding alone. Sanford et al. (2024) make this precise: by a communication-complexity argument, a one-layer transformer that solves an induction task must be much larger than a two-layer one. Higher-order chains need more composition: Rajaraman et al. (2024) show that transformers with $O(\log k)$ layers can represent the in-context conditional distribution of kth-order Markov sources, and that layer normalization brings this to a constant number of layers.

[7] The same remark explains the first stage of training in Figure 14.3: uniform attention is already a unigram estimator, and only its readout must be learned.

14.4 The solution: why the circuit forms late

We now turn to the caricature (14.4). Its loss depends on (a, b) only through the product $v = ab$, and it is a quadratic in (c, v).

Lemma 14.7 (The loss is quadratic in (c, ab)). *Let $A_{UU} = \mathbb{E}\|\varphi_U\|^2$, $A_{UB} = \mathbb{E}\langle\varphi_U, \varphi_B\rangle$, $A_{BB} = \mathbb{E}\|\varphi_B\|^2$, $g_U = \mathbb{E}\langle\varphi_U, e_{x_{T+1}}\rangle$ and $g_B = \mathbb{E}\langle\varphi_B, e_{x_{T+1}}\rangle$, at the training position $T \ge 2$. Then*

$$\mathcal{L}(c, a, b) = 1 - 2g_U c - 2g_B v + A_{UU}c^2 + 2A_{UB}cv + A_{BB}v^2, \qquad v = ab.$$

The moments satisfy $A_{UB} \ge 0$, $g_U > 0$, and $A_{UU}A_{BB} \ge A_{UB}^2$ with $A_{UU}, A_{BB} > 0$.

Proof. Expand $\|c\varphi_U + v\varphi_B - e_{x_{T+1}}\|^2$ and use $\|e_{x_{T+1}}\|^2 = 1$. The features have nonnegative coordinates, so $A_{UB} \ge 0$; the inequality is Cauchy–Schwarz in L^2. Finally $g_U > 0$ and $A_{UU} > 0$ because $\varphi_U \ne 0$, and $A_{BB} > 0$ because $\varphi_B \ne 0$ with positive probability (for instance when $x_{T-1} = x_T$, which needs $T \ge 2$). ■

The gradient flow $\dot c = -\partial_c \mathcal{L}$, $\dot a = -\partial_a \mathcal{L}$, $\dot b = -\partial_b \mathcal{L}$ reads

$$\begin{aligned} \dot c &= 2\big(g_U - A_{UU}c - A_{UB}v\big), \qquad \dot a = b\,R, \qquad \dot b = a\,R, \\ R &= 2\big(g_B - A_{UB}c - A_{BB}v\big). \end{aligned} \tag{14.8}$$

At the origin of the (a, b) plane the gradients in a and b vanish, while that in c is $-2g_U \neq 0$: the unigram direction is learned at once, and the induction directions sit at a saddle, the saddle of a product ab met in Chapter 2.[8] On the unigram manifold $v = 0$ the loss is minimized at $c^\star = g_U/A_{UU}$, with value $\mathcal{L}_U = 1 - g_U^2/A_{UU}$. The rate at which the induction circuit grows there is governed by the *induction gain*

$$G \coloneqq g_B - A_{UB}c^\star = -\tfrac{1}{2}\,\partial_v \mathcal{L}(c^\star, 0), \tag{14.9}$$

the correlation of the bigram feature with the target that the unigram feature does not already explain. If $G > 0$ and $A_{UU}A_{BB} > A_{UB}^2$, the quadratic has the unique minimizer

$$v^{\star\star} = \frac{A_{UU}G}{A_{UU}A_{BB} - A_{UB}^2} > 0, \qquad c^{\star\star} = \frac{g_U A_{BB} - g_B A_{UB}}{A_{UU}A_{BB} - A_{UB}^2},$$

with value $\mathcal{L}^{\star\star}$ below the plateau by $\mathcal{L}_U - \mathcal{L}^{\star\star} = A_{UU}G^2/(A_{UU}A_{BB} - A_{UB}^2)$.

[8] There the product of the two layers of a linear network multiplies a singular value of the data; here the product of two attention strengths multiplies a feature of the context. With $a(0) \neq b(0)$ the conserved quantity $a^2 - b^2$ plays the role of the balancedness of Chapter 2 (Exercise 14.7).

Lemma 14.8 (Long-context moments). *Under Assumption 14.2, as $T \to \infty$, A_{UU} and g_U both converge to $\mathbb{E}\|\pi\|^2$, so that $c^\star \to 1$, and*

$$A_{UB} \to \mathbb{E}\sum_i \pi_i^2 \langle \pi, P_{i\cdot}\rangle, \qquad A_{BB} \to \mathbb{E}\sum_i \pi_i^3 \|P_{i\cdot}\|^2,$$
$$G \to \mathbb{E}\sum_i \pi_i^2 \langle P_{i\cdot} - \pi, P_{i\cdot}\rangle.$$

Proof sketch. By the ergodic theorem, $\varphi_U \to \pi$ and $\frac{1}{T}\nu_{i\cdot}(T) \to \pi_i P_{i\cdot}$ almost surely given P, so $\varphi_B = \sum_i \mathbf{1}\{x_T = i\}\frac{1}{T}\nu_{i\cdot}(T)$ is $\pi_{x_T}P_{x_T\cdot} + o(1)$. The law of (x_T, x_{T+1}) given P tends to $\pi_i P_{ij}$, and all features are bounded by 1, so bounded convergence gives the limits; for instance $g_U \to \mathbb{E}\sum_i \pi_i \langle \pi, P_{i\cdot}\rangle = \mathbb{E}\|\pi\|^2$ by stationarity, $\sum_i \pi_i P_{i\cdot} = \pi$. ∎

In the long-context limit the plateau predictor is $\hat y = \varphi_U$, the in-context unigram frequencies, and $G = \mathbb{E}\sum_i \pi_i^2 \langle P_{i\cdot} - \pi, P_{i\cdot}\rangle$ measures how far the rows of the chain are from its stationary law. If all rows are equal, the tokens are independent, $G = 0$, and there is nothing for an induction head to learn. For $\kappa = 1$ Monte Carlo gives $G = 0.034$ for $K = 2$, 0.037 for $K = 3$ and 0.030 for $K = 4$, against $A_{UU} = 0.61$, 0.41 and 0.30: the unigram coordinate relaxes at rate $2A_{UU}$, five to nine times the growth rate $4G$ of the induction circuit.

Theorem 14.9 (Unigram plateau and escape time). *Assume the conclusions of Lemma 14.7, $A_{UU}A_{BB} > A_{UB}^2$, and $G > 0$. Run (14.8) from*

$c(0) = 0$ *and* $a(0) = b(0) = \alpha > 0$. *Let*

$$\vartheta_0 = \min\Big\{\frac{G}{2A_{BB}},\ \frac{GA_{UU}}{4A_{UB}^2}\Big\}$$

(the second term is $+\infty$ *if* $A_{UB} = 0$*), fix* $\vartheta \in (0, \vartheta_0]$ *with* $\alpha^2 < \vartheta$*, and let* $\tau_\vartheta = \inf\{t \ge 0 :\ v(t) \ge \vartheta\}$*. Then:*

(i) (The plateau.) *For* $0 \le t \le \tau_\vartheta$, $|c(t) - c^\star(1 - e^{-2A_{UU}t})| \le A_{UB}\vartheta/A_{UU}$ *and* $|\mathcal{L}(t) - \mathcal{L}_U| \le 2A_{UU}c^{\star 2}e^{-4A_{UU}t} + 4g_B\vartheta$.

(ii) (The escape time.) τ_ϑ *is finite, and*

$$\frac{\log(\vartheta/\alpha^2) - 2A_{UB}c^\star/A_{UU}}{4\big(G + A_{UB}^2\vartheta/A_{UU}\big)} \le \tau_\vartheta \le \frac{\log(\vartheta/\alpha^2)}{4\big(G - A_{BB}\vartheta\big)}.$$

(iii) (Convergence.) *If moreover* $2A_{UU}c^{\star 2}e^{-4A_{UU}\tau_\vartheta} < G\vartheta$*, which holds for every small enough* α *by (ii), then* $(c, a, b) \to (c^{\star\star}, \sqrt{v^{\star\star}}, \sqrt{v^{\star\star}})$ *and* $\mathcal{L}(t) \to \mathcal{L}^{\star\star} < \mathcal{L}_U$.

Proof. The plan is to reduce the flow to two scalar equations, solve the linear one for c in terms of v, and then integrate the growth rate of v between two comparison bounds. Write $A = A_{UU}$, $B = A_{UB}$, $C = A_{BB}$ for brevity.

Reduction. The flow (14.8) has polynomial right side, so a solution exists on a maximal interval. Along it $\mathcal{L}$ is nonincreasing, since $\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{L} = -\|\nabla\mathcal{L}\|^2$. The difference $a - b$ satisfies $\frac{\mathrm{d}}{\mathrm{d}t}(a - b) = -R\,(a - b)$ and vanishes at $t = 0$, so $a = b$ for all t, and $v = a^2$. The quadratic form of Lemma 14.7 is positive definite because $AC > B^2$, so $\mathcal{L}$ tends to infinity as $|c| + v \to \infty$; since $\mathcal{L}$ is nonincreasing, c and a stay bounded and the solution exists for all $t \ge 0$. Finally $\dot a = aR$ gives $a(t) = \alpha \exp\int_0^t R > 0$ and

$$v(t) = \alpha^2 \exp\Big(\int_0^t 2R(s)\,\mathrm{d}s\Big), \qquad 2R = 4\big(g_B - Bc - Cv\big). \tag{14.10}$$

The unigram coordinate. The equation for c is linear with source $-2Bv$. Variation of constants, with $c(0) = 0$ and $g_U = Ac^\star$, gives

$$c(t) = c^\star\big(1 - e^{-2At}\big) - 2B\int_0^t e^{-2A(t-s)}v(s)\,\mathrm{d}s. \tag{14.11}$$

For $t \le \tau_\vartheta$ we have $0 < v \le \vartheta$, so the integral term lies between 0 and $2B\vartheta\int_0^\infty e^{-2As}\mathrm{d}s = B\vartheta/A$. This is the first claim of (i), and it gives in particular

$$c^\star\big(1 - e^{-2At}\big) - \frac{B\vartheta}{A} \le c(t) \le c^\star \qquad (0 \le t \le \tau_\vartheta). \tag{14.12}$$

Upper bound on τ_ϑ. For $t < \tau_\vartheta$, (14.12) and $v < \vartheta$ give $2R \ge 4(g_B - Bc^\star - C\vartheta) = 4(G - C\vartheta)$, and $G - C\vartheta \ge G/2 > 0$ because $\vartheta \le G/(2C)$. By (14.10), $v(t) \ge \alpha^2 e^{4(G - C\vartheta)t}$ for $t < \tau_\vartheta$. The right side reaches ϑ at the time on the right of (ii), so τ_ϑ is finite and at most that time.

Lower bound on τ_ϑ. For $t \le \tau_\vartheta$, drop the term $-Cv \le 0$ and use the lower bound of (14.12):

$$2R \le 4\big(g_B - Bc\big) \le 4\Big(G + Bc^\star e^{-2At} + \frac{B^2\vartheta}{A}\Big).$$

Integrating from 0 to τ_ϑ, with $\int_0^\infty 4Bc^\star e^{-2As}\mathrm{d}s = 2Bc^\star/A$, (14.10) at $t = \tau_\vartheta$, where $v = \vartheta$ by continuity, gives $\log(\vartheta/\alpha^2) \le 4(G + B^2\vartheta/A)\tau_\vartheta + 2Bc^\star/A$, which is the left inequality of (ii).

The loss on the plateau. Completing the square in c and then expanding in v,

$$\mathcal{L} = \mathcal{L}_U + A(c - c^\star)^2 - 2v\big(g_B - Bc\big) + Cv^2. \tag{14.13}$$

For $t \le \tau_\vartheta$, (14.12) gives $|c - c^\star| \le c^\star e^{-2At} + B\vartheta/A$, hence $A(c - c^\star)^2 \le 2Ac^{\star 2}e^{-4At} + 2B^2\vartheta^2/A$, and $0 \le g_B - Bc \le g_B + B^2\vartheta/A$. With $v \le \vartheta$, $C\vartheta \le G/2$, $4B^2\vartheta/A \le G$ (both from $\vartheta \le \vartheta_0$) and $G \le g_B$,

$$|\mathcal{L} - \mathcal{L}_U| \le 2Ac^{\star 2}e^{-4At} + 2g_B\vartheta + \vartheta^2\Big(\frac{4B^2}{A} + C\Big) \le 2Ac^{\star 2}e^{-4At} + 2g_B\vartheta + \tfrac{3}{2}G\vartheta,$$

which is at most the bound in (i).

Convergence. At $t = \tau_\vartheta$ we have $v = \vartheta$ and, by (14.12), $g_B - Bc \ge G$; so (14.13) and the estimates of the previous step give

$$\mathcal{L}(\tau_\vartheta) - \mathcal{L}_U \le 2Ac^{\star 2}e^{-4A\tau_\vartheta} + \frac{2B^2\vartheta^2}{A} - 2G\vartheta + C\vartheta^2 \le 2Ac^{\star 2}e^{-4A\tau_\vartheta} - G\vartheta < 0$$

under the hypothesis of (iii). Since $\int_0^\infty \|\nabla\mathcal{L}\|^2\mathrm{d}t = \mathcal{L}(0) - \lim_t \mathcal{L}(t) < \infty$ and $\|\nabla\mathcal{L}\|^2$ has a bounded time derivative on the bounded trajectory, $\nabla\mathcal{L} \to 0$ along the flow. Every limit point is therefore a critical point with $\mathcal{L} \le \mathcal{L}(\tau_\vartheta) < \mathcal{L}_U$ and $a = b \ge 0$. A critical point with $a = b$ has $\partial_c\mathcal{L} = 0$ and either $a = 0$ or $\partial_v\mathcal{L} = 0$. If $a = 0$, then $v = 0$, $c = c^\star$, and $\mathcal{L} = \mathcal{L}_U$, which is excluded. Otherwise (c, v) is the unique minimizer $(c^{\star\star}, v^{\star\star})$ of the quadratic, and $a = b = \sqrt{v^{\star\star}}$. The limit point is unique, so the flow converges to it. ∎

Corollary 14.10 (Logarithmic escape). *Under the hypotheses of Theorem 14.9, for every fixed $\vartheta \in (0, \vartheta_0]$,*

$$\frac{1}{2(G + A_{UB}^2\vartheta/A_{UU})} \le \liminf_{\alpha\to 0} \frac{\tau_\vartheta}{\log(1/\alpha)}, \qquad \limsup_{\alpha\to 0} \frac{\tau_\vartheta}{\log(1/\alpha)} \le \frac{1}{2(G - A_{BB}\vartheta)}.$$

Proof. Divide the bounds of (ii) by $\log(1/\alpha)$, note $\log(\vartheta/\alpha^2)/\log(1/\alpha) \to 2$, and let $\alpha \to 0$. ∎

The plateau thus lasts $\log(1/\alpha)/(2G)$, up to a factor $1 + O(\vartheta)$ and an additive constant, while the unigram coordinate settles in time $O(1/A_{UU})$. The escape is the logistic escape of Chapter 2: two factors that each need

the other grow together as $v \propto e^{4Gt}$ while small, the composite strength v at rate $4G$, until the quadratic term $A_{BB}v^2$ stops them. The induction gain G plays the role of a singular value of the input–output correlation, and the unigram feature the role of a mode that has already been learned and must be projected out of the bigram signal; this projection is the competition that Edelman et al. (2024) observed.

Rigorous results for softmax models. Nichani et al. (2024) analyze gradient descent on a simplified two-layer transformer for in-context learning of a latent causal structure; they show that the first attention layer learns the causal graph because the gradient of its attention matrix reflects mutual information between tokens, and for Markov chains the trained network implements an induction head. Bietti et al. (2023) study a two-layer transformer on data mixing global bigrams with context-specific ones, describe weight matrices as associative memories, and show that global bigrams are learned quickly and the induction head more slowly, with theoretical insight into how gradient steps build these memories. Neither result gives the plateau length as a function of the initialization; that is the piece our caricature supplies, at the cost of linear attention.

14.5 Reading the solution

The model makes five quantitative predictions. (i) Loss levels: a trained network should pass through the uniform level $\log K$, then the unigram level, then the Bayes level of Theorem 14.3, whose long-context values are 1.099, 0.979 and 0.819 for $K = 3$, $\kappa = 1$. (ii) The Bayes predictor is computable by two heads and a sink, with error $O(\beta_2 t e^{-\beta_1} + t e^{-\beta_2}/\kappa)$, so a two-layer network can reach the Bayes level exactly. (iii) After the unigram coordinate settles, the loss sits at $\mathcal{L}_U$ for a time $\log(1/\alpha)/(2G)$: each factor 10 in the initialization adds $\log(10)/(2G)$ to the plateau, and the drop from it is a logistic curve of rate $4G$. (iv) Weaker rows mean longer plateaus: G falls as κ grows, like $(K-1)^2/(K^3(K\kappa+1))$ for large κ (Exercise 14.9); for $K = 3$ and $\kappa = \frac{1}{4}, \frac{1}{2}, 1, 2, 4$ Monte Carlo gives $G = 0.070, 0.055, 0.037, 0.021$ and 0.0115. (v) Even the Bayes predictor learns slowly from a finite context. The add-κ estimator of one row from n transitions has expected excess loss $(K-1)/(2n)$ to leading order; with $n \approx \pi_i t$ transitions out of state i and the current state distributed as π, the excess loss at position t is

$$\mathbb{E}\big[-\log p_t^{\mathrm{B}}(x_{t+1})\big] - \mathrm{H}_B \approx \sum_i \pi_i \frac{K-1}{2\pi_i t} = \frac{K(K-1)}{2t}, \tag{14.14}$$

a heuristic that Monte Carlo confirms slowly (0.048, 0.027 and 0.0141 at $t = 50, 100, 200$ for $K = 3$, against 0.060, 0.030 and 0.015). This is why the Bayes level averaged over a short context lies well above H_B.

Figure 14.2 tests (iii). The plateau is flat to four digits, at the value $\mathcal{L}_U$ predicted from the moments, and its length grows by 34.2 time units per decade of α, the slope $\log(100)/(4G)$ of Corollary 14.10; the curves sit about 5 units below the asymptote, the additive constant of (ii). The bounds of (ii) bracket the flow: at $\vartheta = \vartheta_0 = 0.107$ and $\alpha = 10^{-3}$ they give $64 \le \tau_\vartheta \le 135$,

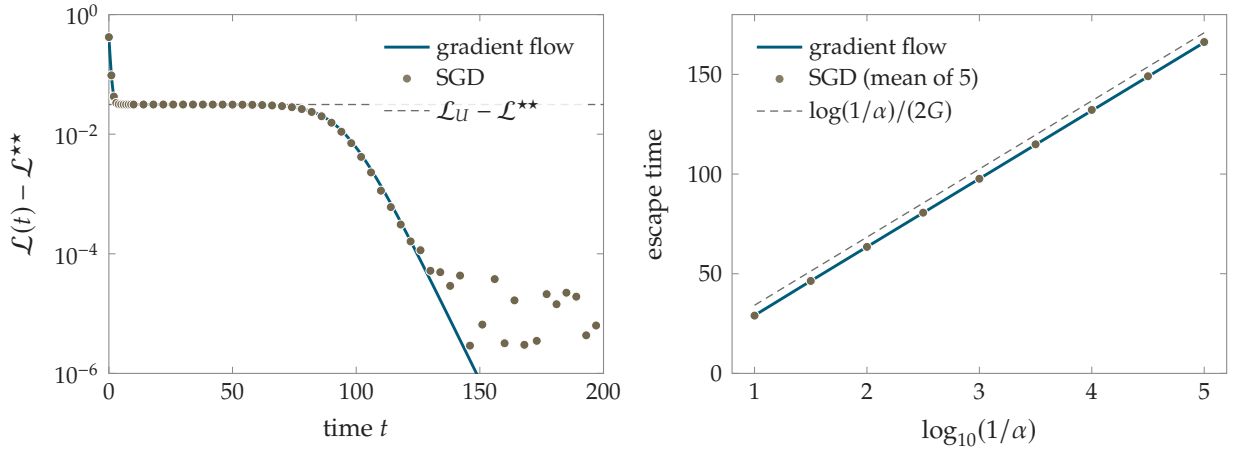


Figure 14.2. The caricature (14.4) with $K = 3$, $\kappa = 1$, uniform first token and training position $T = 24$. The moments are computed on a pool of 4×10^5 sequences (seed 0): $A_{UU} = 0.433$, $A_{UB} = 0.185$, $A_{BB} = 0.115$, $g_U = 0.410$, $g_B = 0.209$, hence $c^\star = 0.947$, $G = 0.0337$, $\mathcal{L}_U = 0.6116$ and $\mathcal{L}^{\star\star} = 0.5802$. Curves: the gradient flow (14.8), integrated numerically. Points: minibatch SGD on (c, a, b) with batches of 256 sequences from the pool and step 0.05, plotted against time 0.05×step. Left: excess loss for $\alpha = 10^{-3}$; after the unigram coordinate settles ($t \approx 5$) the loss stays on the plateau until $t \approx 70$ and then drops logistically; the SGD points level off near 10^{-5}, the noise floor of the step size. Right: the time at which $v = ab$ reaches $v^{\star\star}/2$; flow and SGD (five seeds, spread at most 1.3) agree, and both are parallel to $\log(1/\alpha)/(2G)$, with slope 34.2 per decade.

and the flow takes 81. The second drop is small, $\mathcal{L}_U - \mathcal{L}^{\star\star} = 0.031$ against the first drop 0.39: the square loss of the caricature rewards the bigram feature less than cross-entropy does, and only the time scale, not the size of the gain, should be compared with experiments.

Figure 14.3 tests (i) and the qualitative content of (iii) on a softmax network that can represent the circuit of Theorem 14.6. The three stages appear: the readout γ, whose gradient is of order one, turns the uniform attention into a unigram estimator within about 300 steps; the loss then sits at about 1.032, the level of the best add-κ' unigram estimator; and it then drops to within 0.006 of the Bayes level. The plateau grows by about 530 steps per decade of initialization, consistent with the linear dependence on $\log(1/\alpha)$ of Corollary 14.10, although three initializations cannot establish linearity and the corollary is proved only for the caricature. The slope itself is not predicted: the softmax network has more than two factors in its induction path (relative-position logits, a value matrix, a bilinear score and the readout), cross-entropy instead of square loss, and stochastic gradients. Nor does the figure test a standard transformer: the readout $\gamma \log \hat{y}_t$ hands the network the logarithm that a real unembedding and MLP would have to learn, and the sink value is fixed to be uniform. With the previous-token head frozen in place the plateau disappears, and the loss crosses the midpoint within 100 steps (Exercise 14.10), which supports, in this architecture, the view that composition causes the delay.

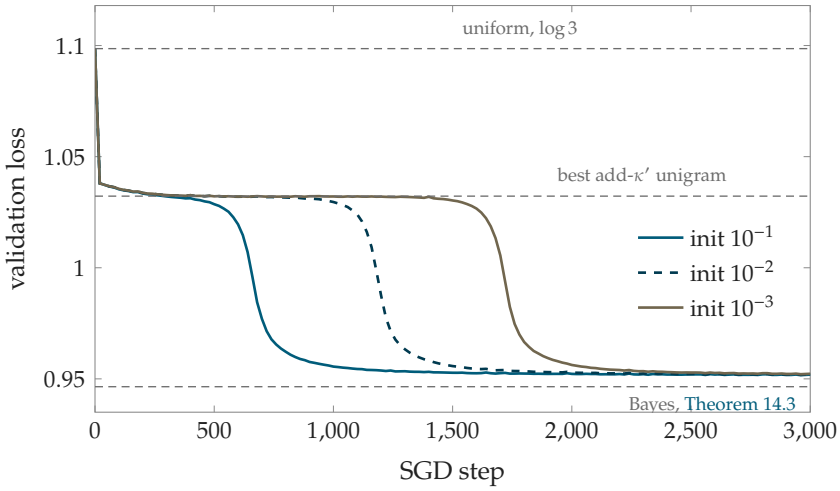


Figure 14.3. A softmax transformer on in-context Markov chains ($K = 3$, $\kappa = 1$, context $N = 32$, loss averaged over positions, validation set of 4000 sequences, seed 0). Layer 1: one head with learned relative-position logits and value matrix; layer 2: one head with a learned bilinear score on (token, layer-1 output) and a sink with learned logit; logits $\gamma \log \hat{y}_t$ with learned γ, a nonstandard readout (Section 14.2); all weights initialized as the stated scale times a standard Gaussian, γ and the sink logit at 0. Plain SGD, learning rate 1, batches of 64 fresh sequences. Dashed lines: the three levels, computed on the validation set. The loss crosses the midpoint between the unigram and Bayes levels at steps 680, 1200 and 1740, and ends at 0.952; the previous-token weight $a^{(1)}_{t,t-1}$ ends near 0.9.

14.6 What the model gets wrong

WHAT WAS THROWN AWAY. *The data* are designed so that in-context statistics are the only signal: a fresh chain per sequence, a symmetric prior, no vocabulary structure, no semantics. Real text mixes in-context and in-weights regularities, and their competition is a large part of the phenomenon. *The circuit* is fixed by hand, with one head per layer and the positional pattern of a previous-token head built in; real induction heads match on fuzzy, learned features and copy through learned subspaces. *The caricature* keeps one scalar per layer, linear attention and a square loss; it captures the saddle created by composition and nothing else. In particular it does not explain why gradient descent finds the *previous-token* pattern among all positional patterns, a symmetry-breaking question that Nichani et al. (2024) answer with mutual information and that our three-parameter ansatz assumes away. *The readout* of our experiments is a logarithm of the attention output, which spares the network the MLP a real transformer needs to turn mixtures of one-hot vectors into calibrated logits.

WHERE IT BREAKS. *Language models may do more at the phase change.* For large models with MLPs the evidence of Olsson et al. (2022) that induction heads cause in-context learning is correlational: the heads and the jump in context use appear in the same window of training, but other mechanisms may form in that window too, and the abrupt improvement is not cleanly attributable

to a two-head circuit. *In-context learning in large models goes beyond copying.* Todd et al. (2024) find a small set of heads whose outputs, averaged over demonstrations, form *function vectors* that trigger a task even in zero-shot contexts; Hendel et al. (2023) find that in-context learning compresses the demonstrations into a single *task vector*. These are abstractions of a task, not counts of repeated tokens. *Data control whether the head forms at all.* Chan et al. (2022) show that in-context learning requires bursty data with many rare classes, and Singh et al. (2023) that it can disappear with further training as in-weights learning takes over; in our model $G = 0$ for data without in-context structure (Exercise 14.8), which captures the first finding, but nothing in it can capture the second, because in-weights knowledge is useless by construction. *One layer gets stuck.* Makkuva et al. (2025) characterize the loss landscape of single-layer transformers on Markov data and find bad, unigram-like local minima whose existence depends on the data and the architecture; our caricature has no local minima at all, only a saddle. *The plateau is not always one plateau.* Singh et al. (2024) find that induction-head formation depends on several interacting subcircuits, and Reddy (2024) that the abrupt transition comes from a sequence of nested nonlinearities; the caricature, with one product of two strengths, has a single escape.

FINE STRUCTURE. *Rigorous dynamics.* Nichani et al. (2024) prove that gradient descent on a simplified two-layer transformer learns latent causal graphs, and induction heads for Markov chains, and Bietti et al. (2023) explain the order in which global and in-context bigrams are learned through associative memories. *Higher-order chains.* For kth-order Markov sources the Bayes predictor counts $(k+1)$-grams; Edelman et al. (2024) extend their experiments to such sources, Akyürek et al. (2024) find "n-gram heads", higher-order induction heads, in transformers trained on in-context language learning tasks, and Rajaraman et al. (2024) show that depth logarithmic in k suffices, and constant depth with layer normalization. *Lower bounds.* Sanford et al. (2024) prove that one-layer transformers need much larger size than two-layer ones for induction tasks. *Attention sinks.* Xiao et al. (2024) found that language models put large attention on the first tokens regardless of content and that keeping these "sinks" stabilizes streaming inference; in our circuit a sink has a statistical role, carrying the prior. Open: a theorem for the softmax network of Figure 14.3 that predicts the slope of the plateau length against $\log(1/\alpha)$, and an account of how the unigram stage, which our proof treats as fast, changes the escape when it is not.

14.7 Model card

MODEL CARD	
PHENOMENON	Induction heads form abruptly after a plateau, and context use jumps at the same time.
MODEL	Sequences from a random Markov chain with $\mathrm{Dir}(\kappa\mathbf{1}_K)$ rows, fresh per sequence; a two-layer attention-only circuit with a sink; for the dynamics, linear attention with three scalar strengths c, a, b.
SOLUTION	Bayes predictor $(\nu_{x_t j} + \kappa)/(\nu_{x_t} + K\kappa)$; a two-layer circuit computes it with error $O(\beta_2 t e^{-\beta_1} + t e^{-\beta_2}/\kappa)$; gradient flow learns the unigram first and escapes the induction saddle at time $\log(1/\alpha)/(2G)$.
PREDICTS	Plateaus at $\log K$, H_U and H_B; plateau length linear in $\log(1/\alpha)$; G and escape rate set by the spread of the rows, $G \sim (K-1)^2/(K^3(K\kappa+1))$ as $\kappa \to \infty$.
GETS WRONG	No semantics or in-weights competition; no transience; language-model phase changes involve more than one circuit; function and task vectors; bad local minima of one-layer models.
SUCCESSORS	Causal-structure dynamics; associative memories; n-gram heads and higher-order chains; depth lower bounds; attention sinks.

The chapter closes the book's thread on escape from saddles. The escape time $\log(1/\alpha)/(2G)$ is that of a deep linear network (Chapter 2), with G in the role of a singular value; it is a cousin of the search phase of single-index models (Chapter 7), of the plateaus of the committee machine (Chapter 8), where the student must break a symmetry, and of the hidden progress of sparse parities (Chapter 9). As in grokking (Chapter 10), a circuit can be written down exactly and the hard part is the dynamics. The comparison with Chapter 13 is instructive: there, linear attention on regression tasks implements one step of gradient descent with a single layer; here, the in-context task is discrete and the optimal algorithm needs two composed heads, which is where the plateau comes from. The question whether such plateaus add up to the smooth loss curves of large models is taken up in Chapter 16.

Exercises

14.1 CONSOLIDATION. (When the first token knows the chain.) (a) Redo Theorem 14.3 with rows $P_{i\cdot} \sim \mathrm{Dir}(\kappa_{i1}, \dots, \kappa_{iK})$ and show $\Pr(x_{t+1} = j \mid x_{1:t}) = (\nu_{x_t j} + \kappa_{x_t j})/(\nu_{x_t} + \sum_l \kappa_{x_t l})$. (b) Let $K = 2$, $\kappa = 1$, and draw x_1 from the stationary law $\pi(P)$. Show that $\Pr(x_2 = x_1 \mid x_1) = 2\mathbb{E}[w(1-u)/(u+w)]$ with u, w independent uniform on $[0,1]$, and evaluate it numerically (about 0.591), while (14.5) gives $\frac{1}{2}$. Explain which step of the proof fails.

14.2 CONSOLIDATION. (Three levels for two states.) For $K = 2$ and $\kappa = 1$,

write the stationary law and the entropy rate of $P = \left(\begin{smallmatrix} 1-u & u \\ w & 1-w \end{smallmatrix}\right)$, and compute $\mathrm{H}_U = \mathbb{E}\,\mathrm{H}(\pi)$ and H_B by two-dimensional quadrature. You should find 0.564 and 0.481. Show that $\mathrm{H}_B < \mathrm{H}_U$ unless $u + w = 1$, and interpret $u + w = 1$.

14.3 CONSOLIDATION. (The $1/t$ excess.) Let n draws from $q \in \mathbb{R}^K$, with all $q_j > 0$, give counts n_j, and let $\hat{q}_j = (n_j + \kappa)/(n + K\kappa)$. Show that $\mathbb{E}\,\mathrm{KL}(q \,\|\, \hat{q}) = (K-1)/(2n) + O(n^{-2})$ by expanding $\log \hat{q}_j$ to second order around q_j. Derive (14.14), and compare with the Monte Carlo values in Section 14.5.

14.4 CONSOLIDATION. (The quadratic.) (a) Verify the formulas for $v^{\star\star}$, $c^{\star\star}$ and $\mathcal{L}_U - \mathcal{L}^{\star\star}$ after (14.9). (b) Check them against the moments in the caption of Figure 14.2. (c) Show that $c^\star - c^{\star\star} = A_{UB}G/(A_{UU}A_{BB} - A_{UB}^2)$: once the induction circuit is on, the unigram strength decreases.

14.5 CONSOLIDATION. (Without a sink.) Remove the sink from the circuit, so that layer 2 attends only to $s \in \{1, \dots, t\}$. (a) Show that as $\beta_1, \beta_2 \to \infty$ the output tends to the empirical distribution $\nu_{x_t j}/\nu_{x_t}$ when $\nu_{x_t} > 0$, and to the in-context unigram frequencies when $\nu_{x_t} = 0$. (b) Show that the expected cross-entropy at every position $t \geq 1$ is $+\infty$ for all β_1, β_2, because the output gives probability zero to every token absent from $x_{1:t}$. (c) On the event that $\nu_{x_t} > 0$, that x_{t+1} occurs in $x_{1:t}$, and that $\nu_{x_t x_{t+1}} = 0$, show that for fixed large β_1 the loss $-\log \hat{y}_t(x_{t+1})$ grows linearly in β_2.

14.6 VARIATION. (Second-order chains.) Let x_{s+1} depend on (x_{s-1}, x_s) through a random $K^2 \times K$ matrix with $\mathrm{Dir}(\kappa \mathbf{1}_K)$ rows. (a) Derive the Bayes predictor. (b) Build a two-layer circuit with two heads in layer 1, writing x_{s-1} and x_{s-2} into two subspaces, and a layer-2 head whose logit is $\beta_2(\langle e_{x_t}, h_s \rangle + \langle e_{x_{t-1}}, h'_s \rangle - 1)$, and prove the analogue of Theorem 14.6. (c) What does the analogue of the caricature predict about the order in which unigram, bigram and trigram statistics are learned?

14.7 VARIATION. (Unbalanced initialization.) Start the caricature from $a(0) = \alpha$, $b(0) = 0$. (a) Show that $a^2 - b^2$ is conserved and that $\dot{v} = \sqrt{\alpha^4 + 4v^2}\, R$. (b) Show that v first grows linearly and then exponentially at rate $4G$, and that the escape time is still $\log(1/\alpha)/(2G) + O(1)$ as $\alpha \to 0$. Compare with the unbalanced deep linear networks of Chapter 2.

14.8 VARIATION. ★ (No in-context structure.) Suppose all rows of P are equal to one random vector $\pi \sim \mathrm{Dir}(\kappa \mathbf{1}_K)$, so that tokens are independent given π. (a) Show that in the long-context limit $G = 0$. (b) Assume $G \leq 0$ and $A_{UU}A_{BB} > A_{UB}^2$. Using (14.10) and (14.11), show that $v(t) \leq \alpha^2 e^{2A_{UB}c^\star/A_{UU}}$ for all t, and that the flow converges to $(c^\star, 0, 0)$: the induction circuit never forms. *(Hint: at a time when v reaches its running maximum V, bound $c^\star - c \leq c^\star e^{-2A_{UU}t} + A_{UB}V/A_{UU}$*

and use $A_{UB}^2 < A_{UU}A_{BB}$ *to show that* $\frac{\mathrm{d}}{\mathrm{d}t}\log v \le 4A_{UB}c^\star e^{-2A_{UU}t}$ *there.)* Relate this to the role of burstiness in Chan et al. (2022).

14.9 VARIATION. ★ (The induction gain for nearly uniform rows.) Write $P_{i\cdot} = \frac{1}{K}\mathbf{1}_K + \Delta_i$. (a) Show that $\pi = \frac{1}{K}\mathbf{1}_K + \frac{1}{K}\sum_i \Delta_i + O(\|\Delta\|^2)$. (b) Using Lemma 14.8, show that $G = \frac{K-1}{K^2}\mathbb{E}\|\Delta_1\|^2 + O(\mathbb{E}\|\Delta\|^3)$, and with $\mathbb{E}\|\Delta_1\|^2 = (K-1)/(K(K\kappa+1))$ conclude $G \sim (K-1)^2/(K^3(K\kappa+1))$ as $\kappa \to \infty$. (c) Compare with the Monte Carlo values 0.021 and 0.0115 for $K = 3$, $\kappa = 2, 4$. *(Hint: the terms of first order in* Δ *cancel because* $\langle \Delta_i, \mathbf{1}_K\rangle = 0$*.)*

14.10 SIMULATION. (Composition causes the delay.) In the script of Figure 14.3, at initialization 10^{-2}, (a) freeze layer 1 as a previous-token head (relative-position logit 6 at offset 1 and 0 elsewhere, value matrix I_K). The reader should observe that the loss crosses the midpoint between the unigram and Bayes levels within 100 steps instead of 1200: with one factor fixed, the induction path loses its saddle. (b) Remove the sink, flooring the attention output at 10^{-6} before the logarithm. The reader should observe that the loss fluctuates far above the unigram level and never approaches the Bayes level, because tokens absent from the context, and unseen transitions, receive near-zero probability (Exercise 14.5).

14.11 SIMULATION. (Plateau length and the prior.) Run the script of Figure 14.2 with $\kappa \in \{\frac{1}{2}, 1, 2\}$ and measure the slope of the escape time against $\log_{10}(1/\alpha)$. The reader should find slopes close to $\log(100)/(4G(\kappa))$ with G computed from the pool moments at $T = 24$, increasing with κ roughly as $1/G \propto K\kappa + 1$ for large κ.

14.12 SIMULATION. (The Bayes level is reached.) Train the transformer of Figure 14.3 for 10^4 steps and record the learned sink logit ℓ_0 and the layer-2 score of a matching position. Theorem 14.6 predicts that their difference approaches $\log(K\kappa) = \log 3$ when the circuit is sharp. Report how close the trained network comes, and why the readout γ, which is not part of the theorem, can absorb part of the discrepancy.

14.13 RESEARCH. (Dynamics of softmax induction heads.) Prove a version of Theorem 14.9 for the softmax network of Figure 14.3, from a small Gaussian initialization: identify the unigram plateau, show that its length is $\Theta(\log(1/\alpha))$, and compute the constant. Nichani et al. (2024) and Bietti et al. (2023) analyze related models with staged training or a few gradient steps; a theorem that predicts the plateau length of a trained softmax network is open.

14.14 RESEARCH. (Stages for higher-order chains.) For kth-order chains, do transformers learn the 1-gram, 2-gram, . . . , $(k+1)$-gram statistics in successive stages with their own plateaus, and are the plateau lengths predicted by a caricature with k composed strengths? Edelman et al. (2024) and Akyürek et al. (2024) study such sources experimentally; a quantitative theory of the stage lengths is open.

Notes and References

Induction heads were named by Elhage et al. (2021), in a study of attention-only transformers that introduced the residual-stream view, the W_{QK} and W_{OV} circuits and K-composition; Olsson et al. (2022) connected their formation to an abrupt improvement of in-context learning. Both appeared in the Transformer Circuits Thread, and the second also on arXiv. The data-dependence of in-context learning is from Chan et al. (2022) and Singh et al. (2023), and mechanistic studies of formation from Reddy (2024) and Singh et al. (2024).

The in-context Markov chain task is that of Edelman et al. (2024), who observed the uniform, unigram and bigram stages, studied the effect of the prior over chains, and extended the task to n-grams; Nichani et al. (2024) and Bietti et al. (2023) proved results on gradient dynamics for related tasks, and Makkuva et al. (2025) and Rajaraman et al. (2024) studied the loss landscape and the depth needed for Markov data. Theorem 14.3 is classical Dirichlet–multinomial conjugacy, and the add-$\frac{1}{2}$ estimator and its $\frac{K-1}{2}\log n$ redundancy go back to Krichevsky and Trofimov (1981). The construction of Theorem 14.6 follows the circuits of Elhage et al. (2021) and Edelman et al. (2024); the sink that carries the prior and the explicit error bound are, as far as we know, new here, as is the caricature with its escape theorem, which transfers the analysis of Chapter 2 to a composition of attention layers.

Sanford et al. (2024) (unrefereed) prove the separation between one and two layers by communication complexity. Xiao et al. (2024) introduced the term attention sink for the first tokens that absorb attention in language models. Function vectors and task vectors (Hendel et al. 2023; Todd et al. 2024) were found independently and posted to arXiv within a day of each other in October 2023.

TRANSLATING THE LITERATURE. Elhage et al. (2021) write $W_{QK}^h = W_Q^{h\top} W_K^h$ and $W_{OV}^h = W_O^h W_V^h$ for the circuits of head h; in our circuit layer 1 has $W_{QK} = \beta_1 \times$(the shift on one-hot positions) and W_{OV} copying the token into a fresh subspace, and layer 2 has $W_{QK} = \beta_2 \times$(the identity from the token subspace of the query to the layer-1 subspace of the key) and W_{OV} copying the token. Their pattern $[A][B]\ldots[A] \to [B]$ is our $x_{s-1} = x_t = A$, $x_s = B$. Where a source writes k for the number of states or α for the Dirichlet parameter, read our K and κ; the "statistical induction head" of Edelman et al. (2024) is our layer 2, and their unigram and bigram strategies are our unigram predictors and p_t^{B}. Olsson et al. (2022) measure context use by the loss difference between tokens 500 and 50, which for the Bayes predictor of our setting is about $\frac{K(K-1)}{2}(\frac{1}{50} - \frac{1}{500})$ by (14.14). Nichani et al. (2024) call our disentangled layout a disentangled transformer. Our α is the initialization scale of Chapter 2, and our G plays the role of its singular value s.

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Part VI

Training at Scale

The last two models concern large steps and large models. A product of two numbers shows why gradient descent can run at the edge of stability, and power-law spectra and Zipf's law show where power-law scaling can come from.

15 The Edge of Stability

Large step sizes and the catapult

THE MODEL — THE TWO-PARAMETER PRODUCT MODEL

IN THIS CHAPTER. Gradient descent on a quadratic diverges when the step size exceeds two over the largest curvature, yet networks trained by full-batch gradient descent spend most of their training at exactly that curvature, and large first steps make the loss spike and the curvature fall. The product of two numbers, $\mathcal{L}(u, v) = \frac{1}{2}(uv - y)^2$, is the smallest model whose minimizers differ in curvature. We prove that gradient descent almost never converges to a minimizer sharper than $2/\eta$; that it leaves a sharp region only through steps of large loss, because gradient flow conserves $u^2 - v^2$; and that an invariant ellipse separates the catapult from divergence exactly at sharpness $4/\eta$. A heuristic derivation, checked against the exact map, predicts where a small catapult lands. The model has no progressive sharpening, so it shows the catapult but not the sustained edge of stability.

15.1 The phenomenon: sharpening to the edge

Gradient descent on a quadratic with step size η converges if and only if every curvature is below $2/\eta$; above that threshold the stiffest mode flips sign at every step and grows (the prologue). The largest eigenvalue of the Hessian of the training loss,

$$S(\theta) = \lambda_{\max}\big(\nabla^2 \widehat{\mathcal{L}}(\theta)\big),$$

is called the *sharpness*, and the classical theory of optimization treats $\eta S < 2$ as a constraint on the user: if the step size is below 2 divided by the largest curvature met along the way, then the descent lemma guarantees that every step decreases the loss.[1]

For neural networks the constraint describes neither practice nor what training does. Cohen et al. (2021) measured the sharpness along full-batch gradient descent for fully connected, convolutional and transformer networks, with the square loss and the cross-entropy loss, and found the same two-stage pattern almost everywhere.

OBSERVATION 15.1 (Progressive sharpening and the edge of stability; Cohen et al. 2021). Train a neural network by full-batch gradient descent with a fixed step size η. (i) While $S(\theta_t) < 2/\eta$, the sharpness tends to increase. (ii) Once it reaches $2/\eta$, it stops increasing and hovers at or slightly above $2/\eta$ for the rest of training, while the training loss continues to decrease over long time scales, although not monotonically.

[1] Sharpness entered deep learning through generalization rather than optimization: Hochreiter and Schmidhuber (1997) argued that flat minima generalize better, and Keskar et al. (2017) found that large-batch training lands in sharper minima and generalizes worse. This chapter is about optimization only.

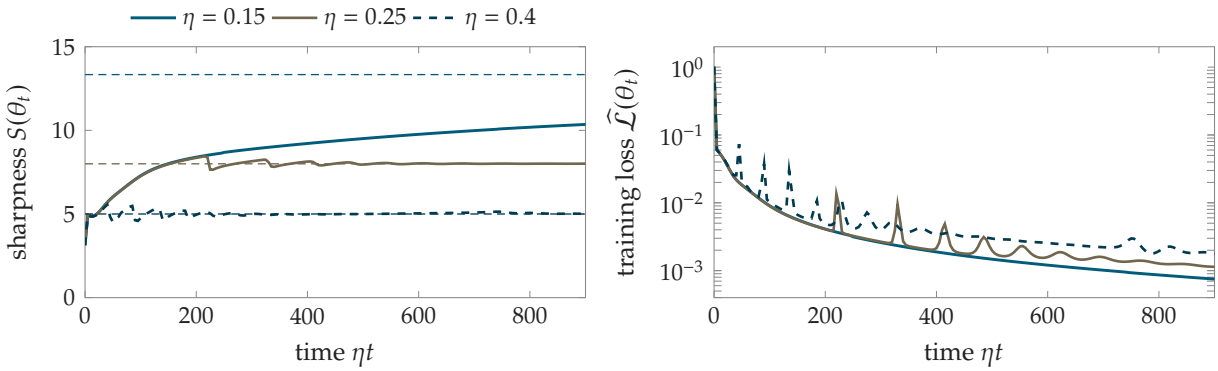


Figure 15.1. Progressive sharpening and the edge of stability in a small network: a tanh network with two hidden layers of width 64, trained by full-batch gradient descent on the square loss with 500 Gaussian inputs in $\mathbb{R}^{10}$ labelled by a random two-layer teacher (seeds fixed in `15-stability-mlp.py`). Left: the sharpness, by power iteration with Hessian–vector products every 5 units of time; dashed lines at $2/\eta$. The runs follow one curve until the sharpness reaches their own $2/\eta$; the run with $\eta = 0.15$ does not reach $2/\eta \approx 13.3$ by time 900. Right: the training loss keeps falling at the edge, with spikes.

The first stage is *progressive sharpening*, the second the *edge of stability*. The observation is robust for full-batch gradient descent, and its scope is known in three directions. Progressive sharpening is weaker for wider networks, and with the cross-entropy loss the sharpness typically drops at the end of training, as the margins grow (Cohen et al. 2021). With minibatches the sharpness stays lower, at a level set by the step size and the batch size (Jastrzębski et al. 2019, 2020). With adaptive optimizers it is a preconditioned sharpness that sits at an edge, according to an unrefereed preprint (Cohen et al. 2022). Figure 15.1 shows the observation in a network small enough to train in seconds.

A second observation concerns the first steps of training with a large step size.

Observation 15.2 (The catapult; Lewkowycz et al. 2020). Let λ_{init} be the curvature at initialization. If $\eta < 2/\lambda_{\text{init}}$, wide networks train much as under gradient flow. If $2/\lambda_{\text{init}} < \eta < \eta_{\max}$, the loss first grows for some steps, the curvature falls below $2/\eta$, and training then converges in a flatter region. Above $\eta_{\max}$ training diverges. In a two-layer linear model that Lewkowycz and coauthors solved, $\eta_{\max} = 4/\lambda_{\text{init}}$; for wide ReLU networks they measured $\eta_{\max} \approx 12/\lambda_{\text{init}}$.

The *catapult* is the name they gave to the loss spike that throws the parameters out of a sharp region.[2] The prologue's theorem says that gradient descent with $\eta S > 2$ must diverge, and it does not. Where can gradient descent end up, given its step size, and how does it get there from a region that is too sharp for it? The model of this chapter answers both questions exactly; what it cannot reproduce, a sharpness that stays at $2/\eta$ for thousands of steps, points to what it leaves out.

[2] The idea that the step size selects among minima by their sharpness is older. Wu et al. (2018) showed that gradient descent can be linearly stable only at minima with $\eta S \le 2$, and proposed this dynamical stability as the reason that gradient methods find flat minima. Jastrzębski et al. (2019) observed that the sharpness along the trajectory adapts to the step size early in training.

15.2 The model: a product of two numbers

Definition 15.3 (The product model). Fix a label $y > 0$ and a step size $\eta > 0$. The *product model* is the loss

$$\mathcal{L}(u, v) = \tfrac{1}{2}(uv - y)^2, \qquad \theta = (u, v) \in \mathbb{R}^2,$$

trained by gradient descent $\theta_{t+1} = G(\theta_t)$, where $G(u, v) = (u - \eta z v,\ v - \eta z u)$ and $z = uv - y$.

The model is a linear network $f(x) = vux$ with one hidden unit, input weight u and output weight v, trained on the single example $(x, y) = (1, y)$; the number z is its residual. The gradient is $\nabla\mathcal{L} = z\,(v, u)$, and the Hessian is

$$\nabla^2\mathcal{L}(u, v) = \begin{pmatrix} v^2 & uv + z \\ uv + z & u^2 \end{pmatrix}.$$

The minimizers form the hyperbola $\mathcal{M} = \{uv = y\}$, with one branch in the positive and one in the negative quadrant. At a minimizer the Hessian $\left(\begin{smallmatrix} v^2 & y \\ y & u^2 \end{smallmatrix}\right)$ has the eigenvalue 0 along the tangent $(u, -v)$ of $\mathcal{M}$ and the eigenvalue $u^2 + v^2$ along the normal (v, u), so the sharpness of a minimizer is $u^2 + v^2$.

Two quantities organize everything that follows: the *imbalance* $c = u^2 - v^2$ and the trace $\tau = u^2 + v^2 = \operatorname{tr}\nabla^2\mathcal{L}$. Because $(u^2 + v^2)^2 = (u^2 - v^2)^2 + 4(uv)^2$, they are related by

$$\tau^2 = c^2 + 4(z + y)^2. \tag{15.1}$$

On $\mathcal{M}$ we have $z = 0$, so the minimizer with imbalance c has sharpness

$$\Lambda(c) = \sqrt{c^2 + 4y^2}. \tag{15.2}$$

Each branch of $\mathcal{M}$ contains exactly one minimizer with each imbalance, the sharpness grows with $|c|$, and the flattest minimizers are the balanced ones, $u = v = \pm\sqrt{y}$, with $\Lambda(0) = 2y$. A quadratic analysis at a minimizer calls it stable for gradient descent when $\eta\Lambda(c) < 2$, that is, when $|c| < c_{\mathrm{e}}$, with the *edge imbalance*

$$c_{\mathrm{e}} = 2\sqrt{\eta^{-2} - y^2}.$$

Stable minimizers exist only when $\eta y < 1$. Two more facts about the scales: away from $\mathcal{M}$ the sharpness is not τ but $S(\theta) = \frac{1}{2}\left(\tau + \sqrt{c^2 + 4(uv + z)^2}\right)$ (Exercise 15.1); and the rescaling $(u, v) \mapsto (u, v)/\sqrt{y}$ turns the model with label y and step η into the model with label 1 and step ηy, so the dynamics depends on η and y only through the dimensionless step size ηy.

DICTIONARY

IN THE MODEL	IN A TRAINED NETWORK
the loss $\frac{1}{2}(uv - y)^2$	the square loss of a network on its training set
the hyperbola $\mathcal{M}$ of minimizers	the manifold of parameters that interpolate the data
the imbalance $c = u^2 - v^2$	the layer differences $W_{\ell+1}^\top W_{\ell+1} - W_\ell W_\ell^\top$ of a linear network (their traces for ReLU), conserved by gradient flow
$\Lambda(c)$, the sharpness of the minimizer with imbalance c	the sharpness of the minimizer that gradient flow would reach
the trace $\tau = u^2 + v^2$	the neural tangent kernel $\|\nabla_\theta f\|^2$, the Gauss–Newton part of the Hessian
the residual z, flipping sign at every step	the oscillation along the top Hessian eigenvector at the edge of stability
the dimensionless step size ηy	the step size in units of the scale of the outputs

WHAT WAS THROWN AWAY. Almost everything: the data (one example), the width and the depth (one hidden unit, two layers), the nonlinearity, and all directions of parameter space but two. What is kept is the smallest structure that the phenomenon needs. A quadratic has a constant Hessian, so its sharpness cannot respond to the dynamics; a one-parameter model has isolated minimizers, so there is nowhere flatter to go. Two parameters give a curve of minimizers along which the sharpness varies, and a Hessian that changes along the oscillation, because the third derivative of $\mathcal{L}$ is not zero; every overparameterized network has both.

WHAT ONE HAD TO NOTICE. Gradient flow on this loss conserves c, so it lands at the minimizer with the imbalance it started with, sharp or not. Gradient descent does not conserve c, and the only thing that changes it is the residual: each step multiplies c by $1 - \eta^2 z^2$. To invent the model one had to see that an oscillation of the residual, which a quadratic would treat as the prelude to divergence, is the mechanism that moves the parameters along the valley to a flatter place.

15.3 The solution: where gradient descent can land

We begin with the baseline, a quadratic loss, in the form we shall need. Let $\mathcal{L}(\theta) = \frac{1}{2}(\theta - \theta_\star)^\top H(\theta - \theta_\star)$, where H is symmetric positive semidefinite with eigenvalues $\lambda_1 \geq \cdots \geq \lambda_p \geq 0$ and orthonormal eigenvectors $e_1, \ldots, e_p$.

Proposition 15.4 (Linear stability). *Gradient descent with step size η on this quadratic converges from θ_0 if and only if $\langle \theta_0 - \theta_\star, e_k \rangle = 0$ for every k with $\eta\lambda_k \geq 2$. Consequently it converges from every initialization if $\eta\lambda_1 < 2$; if $\eta\lambda_1 \geq 2$, it converges only from a proper affine subspace of $\mathbb{R}^p$, which is a Lebesgue-null set.*

PROOF. By the normal-mode formula of the prologue, $\langle \theta_t - \theta_\star, e_k \rangle =$

$(1-\eta\lambda_k)^t\langle\theta_0-\theta_\star, e_k\rangle$. A geometric sequence with a nonzero first term converges if and only if its ratio lies in $(-1,1]$, which here means $0 \le \eta\lambda_k < 2$. So the iterates converge exactly when the components with $\eta\lambda_k \ge 2$ vanish, that is, when θ_0 lies in $\theta_\star + \operatorname{span}\{e_k : \eta\lambda_k < 2\}$, an affine subspace that is proper when $\eta\lambda_1 \ge 2$. ∎

At every minimizer of the product model the Hessian has the eigenvalue $\Lambda(c)$ in the direction normal to $\mathcal{M}$, so the quadratic approximation predicts that gradient descent is stable near the minimizer exactly when $\eta\Lambda(c) < 2$. To see what gradient descent actually does we need its exact form, which is simple in the right coordinates.

Lemma 15.5 (Gradient descent in product coordinates). *Let $\theta = (u, v)$ and $\theta' = G(\theta)$, and write $a = u + v$ and $b = u - v$, so that $c = ab$. Then*

(i) $a' = (1-\eta z)\,a$ *and* $b' = (1+\eta z)\,b$;

(ii) $c' = (1-\eta^2 z^2)\,c = \big(1 - 2\eta^2\mathcal{L}(\theta)\big)\,c$;

(iii) $z' = z\big(1 - \eta\tau + \eta^2 z(z+y)\big)$;

(iv) $\tau' = \tau + \eta z^2(\eta\tau - 4) - 4\eta y z$.

Proof. Adding and subtracting the two coordinates of $G(\theta)$ gives (i): $u'+v' = (u+v)-\eta z(v+u)$ and $u'-v' = (u-v)+\eta z(u-v)$. Multiplying the two identities of (i) gives (ii). For (iii) and (iv), expand $u'v' = (u-\eta z v)(v-\eta z u) = uv - \eta z\tau + \eta^2 z^2 uv$ and $u'^2 + v'^2 = \tau - 4\eta z uv + \eta^2 z^2\tau$, and substitute $uv = z + y$. ∎

By (ii), the imbalance changes only through the loss, and $|c|$ can grow only at steps with $\mathcal{L}(\theta) > 1/\eta^2$. By (iii), near $\mathcal{M}$ the residual is multiplied at each step by about $1 - \eta\Lambda(c)$: it decays when $\eta\Lambda(c) < 2$, and flips sign and grows when $\eta\Lambda(c) > 2$. By (iv), during a large oscillation the trace falls if $\eta\tau < 4$ and rises if $\eta\tau > 4$; for $y = 0$ this is the whole update, and the equation from which Lewkowycz et al. (2020) read off their threshold $4/\lambda_{\text{init}}$. For small steps the factor in (ii) tends to 1.

Proposition 15.6 (Gradient flow conserves the imbalance). *Along gradient flow $\dot\theta = -\nabla\mathcal{L}(\theta)$ the imbalance c is constant. If $u(0) + v(0) \ne 0$, then $\theta(t)$ converges to the minimizer with imbalance $c(0)$ on the branch of $\mathcal{M}$ on which $u + v$ has the sign of $u(0) + v(0)$, and the sharpness of the limit is $\Lambda(c(0))$. If $u(0) + v(0) = 0$, then $\theta(t)$ converges to the origin, a saddle point.*

Proof. The plan is to show that c is constant and z decays, and then to identify the limit. Differentiating, $\dot c = -2uzv + 2vzu = 0$, $\dot z = \dot u v + u\dot v = -z\tau$, $\dot a = -za$ and $\dot b = zb$. Suppose first that $c \ne 0$. Then $\tau \ge |c|$ by (15.1), so $|z(t)| \le |z(0)|e^{-|c|t}$, and $\tau(t) = (c^2 + 4(z(t)+y)^2)^{1/2}$ stays bounded. The speed $|\dot\theta| = |z|\sqrt{\tau}$ is therefore integrable, so $\theta(t)$ has a limit, at which $z = 0$ and $u^2 - v^2 = c$. The two minimizers with imbalance c differ in the sign of a, and $a(t) = a(0)\exp(-\int_0^t z)$ keeps its sign. Suppose next that $c = 0$ and $a(0) \ne 0$. Then $b = c/a = 0$ for all t, so $u = v = a/2$, $z = a^2/4 - y$, and the

scalar equation $\dot{a} = -(a^2/4 - y)a$ drives a to $\pm 2\sqrt{y}$, a balanced minimizer. Finally, if $a(0) = 0$, then a stays 0, so $z = -b^2/4 - y \le -y$ and $\dot{b} = zb$ gives $|b(t)| \le |b(0)|e^{-yt}$. ∎

So gradient flow ends wherever its imbalance points it, and when $\eta\Lambda(c(0)) > 2$ it ends at a minimizer where gradient descent is linearly unstable. To show that gradient descent almost never ends there, we need one fact from measure theory.

Lemma 15.7 (Preimages of null sets). *Let $P\colon \mathbb{R}^n \to \mathbb{R}^n$ be a polynomial map whose Jacobian determinant $J = \det DP$ is not the zero polynomial. Then $P^{-1}(N)$ is a Lebesgue-null set for every null set $N \subseteq \mathbb{R}^n$.*

Proof. The plan is to discard the zero set of J, which is null, and to cover the rest by countably many open sets on which P has a C^1 inverse, since such an inverse maps null sets to null sets. We use three facts.

(a) The zero set of a nonzero polynomial q on $\mathbb{R}^n$ is null. For $n = 1$ it is finite. For $n > 1$, write $q(x', x_n) = \sum_k q_k(x')x_n^k$ with some q_k not the zero polynomial. For x' outside the zero set of q_k, which is null by induction, $x_n \mapsto q(x', x_n)$ is a nonzero polynomial with finitely many zeros, and Fubini's theorem shows that the zero set of q is null.

(b) A C^1 map F from an open set $W \subseteq \mathbb{R}^n$ into $\mathbb{R}^n$ maps null sets to null sets. Write W as a countable union of closed cubes Q; on each, F is Lipschitz with constant $K_Q = \max_Q \|DF\|_{\mathrm{op}}$, by the mean value inequality. Let $N \subseteq W$ be null, let $\delta > 0$, and cover $N \cap Q$ by cubes C_i of total volume less than δ. Each $F(C_i \cap Q)$ has diameter at most $K_Q\sqrt{n}$ times the side of C_i, so it lies in a cube of volume $(2K_Q\sqrt{n})^n \operatorname{vol}(C_i)$. Hence $F(N \cap Q)$ is null, and so is $F(N)$, a countable union of such sets.

(c) Local inverses. Let $U = \{J \neq 0\}$. By the inverse function theorem each point of U has an open neighbourhood $W \subseteq U$ that P maps diffeomorphically onto an open set. Because $\mathbb{R}^n$ has a countable base, countably many such sets $W_1, W_2, \dots$ cover U; write ψ_i for the inverse of P on W_i.

Now $P^{-1}(N) \subseteq \{J = 0\} \cup \bigcup_i \psi_i\big(N \cap P(W_i)\big)$. The first set is null by (a), and each $\psi_i(N \cap P(W_i))$ is null by (b). ∎

Theorem 15.8 (Where gradient descent can land). *Let $y > 0$ and $\eta > 0$, and let $\theta_t = G^t(\theta_0)$ be gradient descent on the product model.*

(i) For every θ_0 and t, the imbalance is $c_t = c_0 \prod_{s<t}\big(1 - 2\eta^2\mathcal{L}(\theta_s)\big)$.

(ii) For every θ_0 outside a Lebesgue-null set, either the iterates do not converge, or they converge to a minimizer θ_∞ with $\eta S(\theta_\infty) \le 2$, that is, with $|c_\infty| \le c_{\mathrm{e}}$.

(iii) Suppose that $\eta y \le 1$ and $\eta\Lambda(c_0) > 2$. If θ_0 lies outside the null set of (ii) and the iterates converge, then

$$\prod_{t\ge 0}\left|1 - 2\eta^2\mathcal{L}(\theta_t)\right| = \frac{|c_\infty|}{|c_0|} \le \frac{c_{\mathrm{e}}}{|c_0|} < 1.$$

PROOF. Part (i) follows from Lemma 15.5(ii) by induction on t.

Part (ii) carries the weight. The plan is to show that convergence to a fixed point at which gradient descent is linearly unstable forces the iterates to land exactly on one of two curves after finitely many steps, and that the initializations whose orbits ever meet those curves form a null set. If $\theta_t \to \theta_\infty$, then $\theta_\infty = G(\theta_\infty)$ by continuity, so $z_\infty \cdot (v_\infty, u_\infty) = 0$: the limit is a minimizer or the origin.

A sharp minimizer. Let $\theta_\infty \in \mathcal{M}$ with $\eta S(\theta_\infty) > 2$. By Lemma 15.5(iii), $z_{t+1} = m_t z_t$ with $m_t = 1 - \eta\tau_t + \eta^2 z_t(z_t + y)$. As $t \to \infty$, $z_t \to 0$ and $\tau_t \to \tau(\theta_\infty) = S(\theta_\infty)$, so $m_t \to 1 - \eta S(\theta_\infty) < -1$, and there is a T with $|m_t| \geq 1$ for all $t \geq T$. If z_T were nonzero, then $|z_t| \geq |z_T| > 0$ for all $t \geq T$, contradicting $z_t \to 0$. Hence $z_T = 0$, that is, $\theta_T \in \mathcal{M}$.

The origin. Let $\theta_\infty = 0$. By Lemma 15.5(i), $a_{t+1} = (1 - \eta z_t)a_t$, and $1 - \eta z_t \to 1 + \eta y > 1$ because $z_t \to -y$. The landing argument for a sharp minimizer, applied to a_t in place of z_t, gives $a_T = 0$ for some T: the iterate θ_T lies on the line $\mathcal{D} = \{u + v = 0\}$.

The exceptional set is null. The curves $\mathcal{M}$ and $\mathcal{D}$ are null, being graphs of continuous functions. The map G is polynomial, and its Jacobian determinant $\det\big(I - \eta\nabla^2\mathcal{L}(u, v)\big)$ equals $1 - \eta^2 y^2 - \eta u^2$ at $v = 0$, so it is not the zero polynomial. By Lemma 15.7 and induction on T, each set $G^{-T}(\mathcal{M} \cup \mathcal{D})$ is null, and so is their union over $T \geq 0$, which contains every initialization whose iterates converge to the origin or to a minimizer with $\eta S > 2$. At the remaining limits $\eta S(\theta_\infty) \leq 2$, and since $S = \Lambda(c)$ on $\mathcal{M}$, this means $|c_\infty| \leq c_e$.

For part (iii), part (ii) says that the limit is a minimizer with $|c_\infty| \leq c_e$. By part (i) and the continuity of $c = u^2 - v^2$, the partial products $\prod_{s<t}|1 - 2\eta^2\mathcal{L}(\theta_s)| = |c_t|/|c_0|$ converge to $|c_\infty|/|c_0|$. Finally, $\eta\Lambda(c_0) > 2$ means $|c_0| > c_e$. ∎

If $\eta y > 1$, then every minimizer has $\eta S \geq 2\eta y > 2$, and part (ii) says that gradient descent from almost every initialization never converges. For general smooth losses the same conclusion follows from the stable manifold theorem when the Jacobian determinant of G vanishes nowhere (Lee et al. 2019, 2016); here it does vanish, at some points, and the landing argument together with the polynomial structure of G replaces the stable manifold theorem. For a class of matrix factorization problems that contains this model, Wang et al. (2022) proved convergence for step sizes well beyond $2/S(\theta_0)$, together with a balancing effect: the limit is more balanced than the initialization.

Parts (i) and (iii) say why the catapult is forced. Gradient flow from θ_0 would end at sharpness $\Lambda(c_0)$. If $\eta\Lambda(c_0) > 2$, gradient descent cannot end there, and it can move its destination only through the factors $1 - 2\eta^2\mathcal{L}(\theta_t)$, which differ from 1 only when the loss is not small: a run that starts near a sharp minimizer and converges must pass through a phase of large loss. While $0 < \mathcal{L}(\theta_t) < 1/\eta^2$, the sharpness $\Lambda(c_t)$ of the flow's destination can only decrease.[3] Below the edge, gradient descent lands almost where the flow lands.

[3] Kreisler et al. (2023) call $\Lambda(c_t)$ the sharpness of the gradient flow solution, and prove that it decreases monotonically along gradient descent for scalar linear networks of every depth, under conditions on the step size and the initialization.

Proposition 15.9 (Below the edge, gradient descent follows the flow). *Let $\eta y < 1$ and let θ_0 satisfy $\eta\Lambda(c_0) < 2$. Put $q = \max\{\eta\Lambda(c_0)-1,\ 1-2\eta y\}$, which lies in $[0,1)$, and suppose that $|z_0| \le (1-q)/(8\eta)$. Then $|z_t| \le \left(\frac{1+q}{2}\right)^t |z_0|$ for every t, $|c_t|$ is nonincreasing, the iterates converge to a minimizer, and*

$$|c_0| \ge |c_\infty| \ge |c_0|\Big(1 - \frac{2\eta^2 z_0^2}{1-q}\Big).$$

Proof. The plan is an induction showing that the residual contracts by the factor $\rho = (1+q)/2$ at every step, which keeps the iterates where the imbalance cannot grow. The number q is at least $|1-2\eta y| \ge 0$ because $\Lambda(c_0) \ge 2y$, and it is less than 1 because $\eta\Lambda(c_0) < 2$ and $\eta y > 0$.

Suppose that $|c_t| \le |c_0|$ and $|z_t| \le |z_0|$, as holds at $t = 0$. Compare the factor m_t of Lemma 15.5(iii) with $m_t^0 = 1 - \eta\Lambda(c_t)$. Since $2y \le \Lambda(c_t) \le \Lambda(c_0)$, we have $1 - \eta\Lambda(c_0) \le m_t^0 \le 1 - 2\eta y$, so $|m_t^0| \le q$. By (15.1), τ_t and $\Lambda(c_t)$ are the lengths of the vectors $(c_t, 2(z_t+y))$ and $(c_t, 2y)$, which differ by $(0, 2z_t)$, so $|\tau_t - \Lambda(c_t)| \le 2|z_t|$. Hence, using $\eta|z_t| \le \eta|z_0| \le \frac{1}{8}$ and $\eta y < 1$ in the last step,

$$\begin{aligned} |m_t - m_t^0| &\le \eta|\tau_t - \Lambda(c_t)| + \eta^2|z_t|\big(|z_t| + y\big) \\ &\le \eta|z_t|\big(2 + \eta|z_t| + \eta y\big) \le 4\eta|z_t| \le \frac{1-q}{2}. \end{aligned}$$

Therefore $|m_t| \le q + \frac{1-q}{2} = \rho$, and $|z_{t+1}| \le \rho|z_t| \le |z_0|$. Moreover $\eta^2 z_t^2 \le \frac{1}{64}$, so the factor $1 - \eta^2 z_t^2$ of Lemma 15.5(ii) lies in $[0,1]$ and $|c_{t+1}| \le |c_t| \le |c_0|$. This completes the induction.

Consequently $|z_t| \le \rho^t |z_0|$. The step length $|\theta_{t+1} - \theta_t| = \eta|z_t|\sqrt{\tau_t}$ is at most a constant times ρ^t, since $\tau_t \le (c_0^2 + 4(|z_0| + y)^2)^{1/2}$, so (θ_t) is a Cauchy sequence, and its limit has $z = 0$. Finally, the factors $1 - \eta^2 z_t^2$ lie in $[0,1]$, and the inequality $\prod_t (1 - x_t) \ge 1 - \sum_t x_t$ for numbers $x_t \in [0,1]$ gives

$$|c_\infty| \ge |c_0|\Big(1 - \sum_{t\ge 0} \eta^2 z_t^2\Big) \ge |c_0|\Big(1 - \frac{\eta^2 z_0^2}{1-\rho^2}\Big) \ge |c_0|\Big(1 - \frac{2\eta^2 z_0^2}{1-q}\Big),$$

because $1 - \rho^2 \ge 1 - \rho = (1-q)/2$. ∎

15.4 The solution: the catapult

Theorem 15.8 says where gradient descent can end, not whether it gets there. Start at a minimizer with $\eta\Lambda(c_0)$ above 2 and perturb it slightly. The residual flips sign and grows, and every step with nonzero loss shrinks $|c|$, as long as the loss stays below $1/\eta^2$. Does the shrinking win before the oscillation runs away? An exact answer locates the boundary between the catapult and divergence, and an approximate one predicts where a small catapult lands.

Theorem 15.10 (The catapult ellipse). *Let $0 < \eta y < 2$, and define*

$$\Psi(\theta) = \eta\,(u^2 + v^2) - \eta^2 y\,(uv - y) - 4 = \eta\tau - \eta^2 yz - 4.$$

Then for every θ,

$$\Psi\big(G(\theta)\big) = \big(1 + \eta^2 z(z+y)\big)\,\Psi(\theta), \tag{15.3}$$

and the factor $1 + \eta^2 z(z+y)$ is at least $1 - \frac{1}{4}\eta^2 y^2 > 0$. Consequently, with $\mathcal{E} = \{\Psi = 0\}$ and $\mathcal{K} = \{\Psi < 0\}$:

(i) the map G sends $\mathcal{E}$, $\mathcal{K}$ and the exterior $\{\Psi > 0\}$ into themselves;

(ii) $\mathcal{K}$ is the interior of an ellipse with axes along the diagonals $u = \pm v$, and every iterate started in $\mathcal{K}$ satisfies $\eta|z_t| < 2$, that is, $\mathcal{L}(\theta_t) < 2/\eta^2$;

(iii) a minimizer θ lies in $\mathcal{K}$ if and only if $\eta S(\theta) < 4$, and on $\mathcal{E}$ if and only if $\eta S(\theta) = 4$;

(iv) for every θ_0 with $\Psi(\theta_0) \geq 0$ outside the null set of Theorem 15.8, the iterates do not converge.

Proof. The plan is to verify the identity with Lemma 15.5 and then read off the consequences. Write $Q = 1 + \eta^2 z(z+y)$. By Lemma 15.5(iv), $\eta\tau' - 4 = (1+\eta^2 z^2)(\eta\tau - 4) - 4\eta^2 yz$, and by Lemma 15.5(iii), $\eta^2 yz' = \eta^2 yz(1-\eta\tau) + \eta^4 yz^2(z+y)$. Subtract the second identity from the first, and collect the terms that contain $\eta\tau - 4$ (among them $\eta^2 yz \cdot \eta\tau - 4\eta^2 yz$):

$$\Psi(\theta') = (1+\eta^2 z^2 + \eta^2 yz)(\eta\tau - 4) - \eta^2 yz\big(1 + \eta^2 z(z+y)\big) = Q\,(\eta\tau - 4 - \eta^2 yz).$$

The right side is $Q\,\Psi(\theta)$. Completing the square, $Q = (\eta z + \frac{1}{2}\eta y)^2 + 1 - \frac{1}{4}\eta^2 y^2$, which is positive because $\eta y < 2$.

(i) By induction, $\Psi(\theta_t) = \Psi(\theta_0)\prod_{s<t} Q(\theta_s)$, and every factor is positive.

(ii) In the coordinates $a = u + v$ and $b = u - v$ we have $u^2 + v^2 = \frac{1}{2}(a^2 + b^2)$ and $uv = \frac{1}{4}(a^2 - b^2)$, so the inequality $\Psi < 0$ reads

$$\frac{\eta a^2}{4(2+\eta y)} + \frac{\eta b^2}{4(2-\eta y)} < 1,$$

the interior of an ellipse. For the bound on z, use $\tau \geq 2|uv| = 2|z+y|$. If $z \geq 0$, then $\Psi < 0$ gives $2\eta(z+y) < 4 + \eta^2 yz$, that is, $\eta z\,(2 - \eta y) < 2\,(2 - \eta y)$, so $\eta z < 2$. If $z < -y$, then it gives $-2\eta(z+y) < 4 + \eta^2 yz$, that is, $-\eta z\,(2+\eta y) < 2\,(2+\eta y)$, so $\eta z > -2$. If $-y \leq z < 0$, then $\eta|z| \leq \eta y < 2$.

(iii) At a minimizer $z = 0$ and $\tau = S$, so $\Psi = \eta S - 4$.

(iv) If $\Psi(\theta_0) \geq 0$ and $\theta_t \to \theta_\infty$, then $\Psi(\theta_\infty) \geq 0$ by (i) and continuity. The limit is a fixed point of G, so it is the origin, where $\Psi = \eta^2 y^2 - 4 < 0$, or a minimizer, which by (iii) has $\eta S(\theta_\infty) \geq 4 > 2$. The first is impossible, and Theorem 15.8(ii) confines the second to the null set. ∎

The theorem leaves two halves open: whether gradient descent started in $\mathcal{K}$ converges, and whether it is unbounded when started outside.[4] With Theorem 15.8 the theorem gives the phase diagram for gradient descent started near a minimizer $\theta_\star$ with imbalance c_0, when $\eta y < 1$ (Figure 15.2). If $\eta\Lambda(c_0) < 2$, it converges close to $\theta_\star$ (Proposition 15.9). If $2 < \eta\Lambda(c_0) < 4$, then $\theta_\star$ and a neighbourhood of it lie in $\mathcal{K}$; the iterates stay in $\mathcal{K}$, with loss below $2/\eta^2$, and for almost every start they can converge only to a

[4] For $y = 0$ the identity (15.3) reduces to the update of the curvature from which Lewkowycz et al. (2020) derived their threshold, and $\mathcal{E}$ is the circle $\eta\tau = 4$. For $y > 0$, Balasubramanian (2026) reports the invariant ellipse for $0 < \eta y < 2$, and chaos on it, in an unrefereed preprint that reduces linear self-attention to a map of this kind; Liang and Montúfar (2026) derive the exact critical step size for convergence in a scalar–vector factorization.

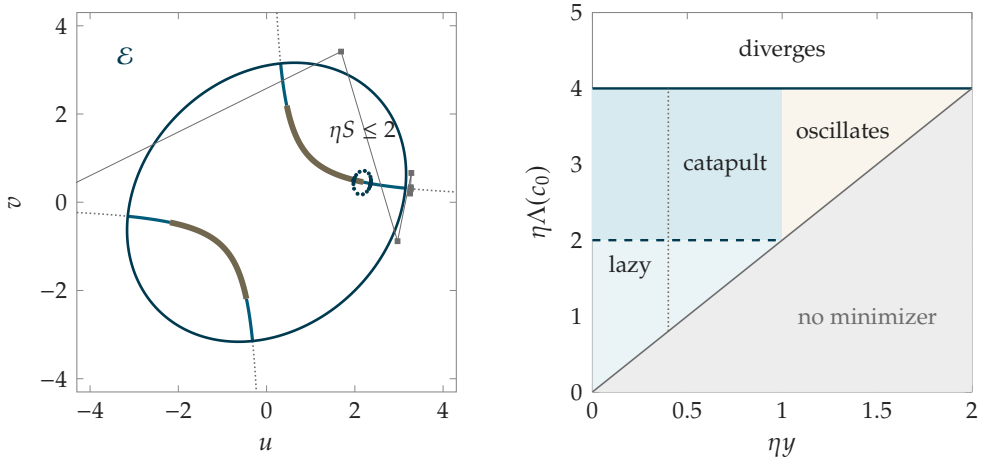


Figure 15.2. The geometry of the product model. Left: the parameter plane for $y = 1$ and $\eta = 0.4$. The hyperbola $uv = 1$ is thick where its minimizers are stable ($\eta S \le 2$), solid inside the catapult ellipse $\mathcal{E}$ ($2 < \eta S < 4$), and dotted outside. Circles: a catapult from next to the minimizer with $\eta S = 2.3$, which zigzags across the valley and lands at $\eta S = 1.67$. Squares: a run from next to the minimizer with $\eta S = 4.3$, which leaves the picture after seven steps. Right: the phase diagram for a start near a minimizer; the lines $\eta\Lambda = 2$ (dashed) and $\eta\Lambda = 4$ are exact boundaries, no minimizer is flatter than $2\eta y$, and in the band marked "oscillates" no minimizer is stable but the iterates stay bounded. The dotted line is $\eta y = 0.4$.

minimizer with $\eta S \le 2$. In all our runs they do converge: a catapult. If $\eta\Lambda(c_0) > 4$, they converge from almost no start near $\theta_\star$, and in our runs they diverge. The threshold $4/\lambda_{\text{init}}$ of Observation 15.2 is therefore exact in the product model as a threshold for convergence, and the same identity holds for the width-m model of Lewkowycz et al. (2020) with η^2/m in place of η^2 (Exercise 15.8).

The ellipse is also where the dynamics is most complicated. In the coordinates $a = 2\sqrt{(2+\eta y)/\eta}\,\sin\varphi$ and $b = 2\sqrt{(2-\eta y)/\eta}\,\cos\varphi$ on $\mathcal{E}$, gradient descent is the map $\varphi \mapsto \pi - 3\varphi$ of the circle (Exercise 15.3), an expanding map with a dense set of periodic orbits, whose only fixed points are the four minimizers with $\eta S = 4$, and whose topological entropy is $\log 3$; Liang and Montúfar (2026) prove the lower bound $\log 3$ for the topological entropy of gradient descent in their model. Inside $\mathcal{K}$, large oscillations behave chaotically for a while before they die out, and the landing point of a large catapult depends sensitively on where it started (Figure 15.3).

How far does a catapult throw? Near the edge the answer comes from a slow drift through a period-doubling instability, and it is derived, not proved.[5]

Heuristic derivation of the reflection law. Let $\eta y < 1$, and start near a minimizer just beyond the edge: $\varepsilon_0 = \eta\Lambda(c_0) - 2$ is small and positive, and z_0 is much smaller than ε_0. Follow the squared residual $A_t = z_t^2 = 2\mathcal{L}(\theta_t)$ and the excess $\varepsilon_t = \eta\Lambda(c_t) - 2$ of the sharpness of the flow's destination over the edge. Both stay small, A of order ε_0^2 and ε of order ε_0, and we keep

[5] The expansion below is the normal form of a flip (period-doubling) bifurcation, here with a parameter that drifts because of the oscillation itself (Kuznetsov 2004).

the leading terms.

The fast variable. Expand $\tau = (\Lambda(c)^2 + 8yz + 4z^2)^{1/2}$ in powers of z and substitute it in Lemma 15.5(iii), with $\eta\Lambda(c) = 2 + \varepsilon$ and with the coefficients of z^2 and z^3 evaluated at the edge:

$$z' = -(1+\varepsilon)\, z - \eta^2 y\, z^2 + \eta^4 y^2 z^3 + \cdots .$$

The residual flips sign at every step, and its size is multiplied by $1 + \varepsilon$. The quadratic term nearly cancels over two steps: composing $g(z) = -(1+\varepsilon)z + k_2 z^2 + k_3 z^3$ with itself gives $(1+\varepsilon)^2 z - 2(k_2^2 + k_3)z^3$ plus terms of order εz^2 and z^4. Here $k_2^2 + k_3 = 2\eta^4 y^2$, so over one step, on average,

$$A' = (1+2\varepsilon)A - \beta A^2 + \cdots , \qquad \beta = 4\eta^4 y^2.$$

The slow variable. By Lemma 15.5(ii), $c' = (1 - \eta^2 A)c$, and $d\varepsilon/dc = \eta c/\Lambda(c) = \eta^2 c/(2+\varepsilon)$. At the edge $c^2 = c_{\mathrm{e}}^2$, so

$$\varepsilon' = \varepsilon - \nu A + \cdots , \qquad \nu = \tfrac{1}{2}\eta^4 c_{\mathrm{e}}^2 = 2\eta^2(1 - \eta^2 y^2).$$

The flow. In one step A changes by a factor $1 + O(\varepsilon)$ and ε by $O(A)$, so both vary slowly, and we replace the recursions by differential equations in the step count:

$$\frac{dA}{dt} = 2\varepsilon A - \beta A^2, \qquad \frac{d\varepsilon}{dt} = -\nu A. \tag{15.4}$$

The oscillation grows while $\varepsilon > 0$ and decays once $\varepsilon < 0$, and its growth is what drives ε down. Relative to $2\varepsilon A$, the term βA^2 is of order $\beta\varepsilon_0/\nu$, a correction of relative order ε_0, and to leading order we drop it. Then $\frac{d}{dt}(\nu A + \varepsilon^2) = 2\nu\varepsilon A - 2\varepsilon\nu A = 0$, so $\nu A + \varepsilon^2$ is constant during the catapult. At the start $A \approx 0$ and $\varepsilon = \varepsilon_0$; at the end $A \to 0$ again, and ε, which only decreases, has become negative. Hence $\varepsilon_\infty = -\varepsilon_0$, which is the *reflection law*

$$\eta S(\theta_\infty) = 4 - \eta\Lambda(c_0). \tag{15.5}$$

The same first integral gives the height of the loss spike: A is largest when $\varepsilon = 0$, where $\nu A = \varepsilon_0^2$, so the largest loss is about $\varepsilon_0^2/(2\nu) = \varepsilon_0^2/\big(4\eta^2(1 - \eta^2 y^2)\big)$. ◇

Remark 15.11 (The next order). Each approximation in the derivation (dropping βA^2, replacing $(1+\varepsilon)^2$ by $1 + 2\varepsilon$, freezing $d\varepsilon/dc$ at the edge, and replacing the recursions by a flow) changes the dynamics by a relative amount of order ε, and so moves ε_∞ at order ε_0^2. Carrying all four to the next order (Exercise 15.5) gives

$$\varepsilon_\infty = -\varepsilon_0 + \Big(\frac{2\eta^2 y^2}{1 - \eta^2 y^2} - \frac{2}{3}\Big)\varepsilon_0^2 + O(\varepsilon_0^3). \tag{15.6}$$

With $s = \beta/\nu = 2\eta^2 y^2/(1 - \eta^2 y^2)$, and with $\log A$ and $\log c$ as the variables, the four contributions are $\frac{2}{3}s$ from the saturation, $\frac{1}{3}$ from the growth factor, $\frac{1}{3}(1+s)$ from the curvature of ε as a function of $\log c$, and $-\frac{4}{3}$ from the discreteness: the map updates A and c simultaneously, as the explicit Euler method does, and the first integral drifts. The exact map confirms (15.6) to five digits for every ηy from 0.05 to 0.8 (Figure 15.3, right). Keeping the saturation alone turns (15.4) into a system solvable with the Lambert W function, and gives the coefficient $\frac{2}{3}s$, which has the wrong sign whenever $\eta y < \frac{1}{2}$.

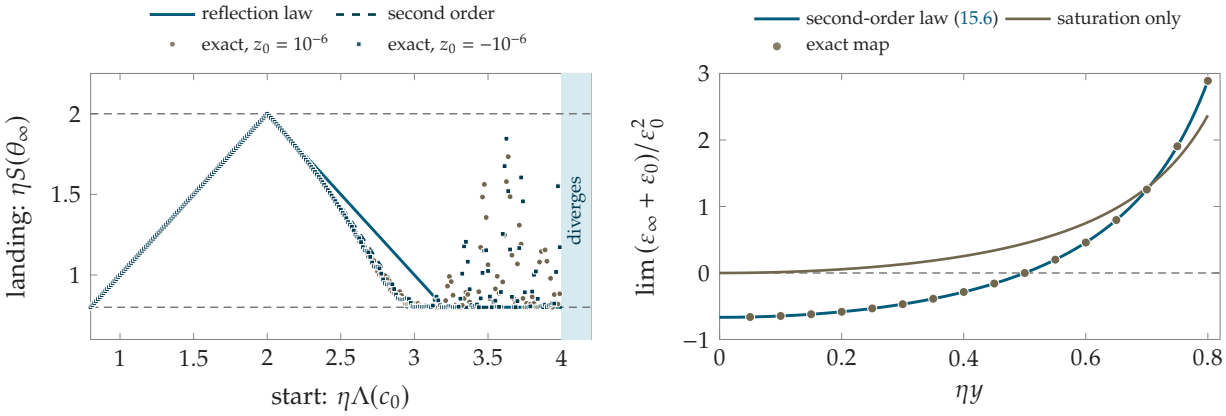


Figure 15.3. Where a catapult lands, for $y = 1$ and $\eta = 0.4$. Left: each point is one run of the exact map, started on the level set of c_0 at residual z_0 (grid spacing 0.0125) and iterated to convergence. Curves: the identity below the edge (Proposition 15.9), and the reflection law (15.5) and second-order law (15.6) above it, cut at the flattest minimizer $2\eta y$ (dashed). No run converges beyond $\eta\Lambda(c_0) = 4$. Right: the second-order coefficient, extrapolated from exact runs with $\varepsilon_0 = 0.01, 0.005, 0.0025$ and $z_0 = 10^{-9}$ (points), against (15.6) and the saturation term alone. Script: `15-stability-catapult.py`.

15.5 Reading the solution

THE CATAPULT CURVE. Figure 15.3 plots where gradient descent lands against where it starts, for $\eta y = 0.4$. Below the edge the landing sharpness equals the starting one, as Proposition 15.9 requires. Just above the edge the landing point is the mirror image of the start across $\eta S = 2$: a start at $\eta S = 2.08$ lands at 1.918, where the reflection law (15.5) predicts 1.92 and the second-order law (15.6) 1.9182. The reflection has a physical reading: the oscillation is an instability that feeds on the excess sharpness and destroys it, and since it dies out exactly as it grew, it overshoots the edge by as much as it started above it. The landing lies slightly below the mirror image, by the second-order term, and beyond $\eta\Lambda(c_0) \approx 2.4$ the gap widens as the expansion in powers of z loses accuracy. From about 2.9 on, most runs land near the flattest minimizer, $\eta S = 2\eta y = 0.8$, with a scatter that depends on the initial perturbation at the level of 10^{-6}: the oscillation grows large enough to behave chaotically before it dies out. At $\eta\Lambda(c_0) = 4$ the scatter ends in divergence, exactly where Theorem 15.10 puts it.

SIZES AND TIMES. The loss never exceeds $2/\eta^2$ during a catapult that starts inside $\mathcal{E}$ (Theorem 15.10). Near the edge the spike has height about $\varepsilon_0^2/(4\eta^2(1-\eta^2 y^2))$: for $\eta y = 0.4$ and $\varepsilon_0 = 0.08$ the prediction is 0.0119, and the exact map gives 0.0125. The spike comes after about $\log(A_{\text{peak}}/A_0)/(2\varepsilon_0)$ steps, the time an oscillation growing by the factor $(1+\varepsilon_0)^2$ per step needs to rise from its initial size A_0 to ε_0^2/ν: logarithmic in the initial perturbation and inversely proportional to the distance from the edge, the signature of an escape from an unstable fixed point that recurs throughout this book (Chapters 2 and 7). Figure 15.4 shows one catapult in time.

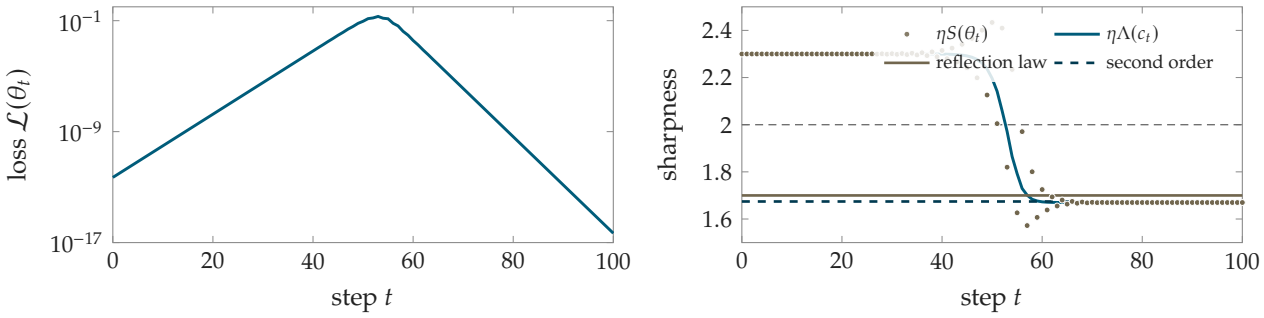


Figure 15.4. One catapult, for $y = 1$ and $\eta = 0.4$, started on the level set of the minimizer with $\eta S = 2.3$ at residual $z_0 = 10^{-6}$. Left: the loss grows geometrically for about fifty steps, peaks near 0.2 (the heuristic predicts 0.17), and decays at the rate set by the new minimizer. Right: the sharpness $\eta S(\theta_t)$ (points) and the sharpness $\eta\Lambda(c_t)$ of the flow's destination (curve), which decreases monotonically from 2.3 to 1.670; the lines mark the predictions 1.7 of (15.5) and 1.674 of (15.6). Script: `15-stability-trajectory.py`.

FROM THE CATAPULT TO THE EDGE OF STABILITY. The model has a catapult but no edge of stability: after one oscillation it lands below $2/\eta$ and stays there. The reason is that its gradient flow conserves c, so nothing pushes the sharpness of the flow's destination back up. In a network that is trained at the edge, progressive sharpening does exactly that. Add it to (15.4) as a constant rate $\gamma > 0$:

$$\frac{dA}{dt} = 2\varepsilon A, \qquad \frac{d\varepsilon}{dt} = \gamma - \nu A. \tag{15.7}$$

This is the Lotka–Volterra system. The function $\varepsilon^2 + \nu A - \gamma \log A$ is constant along its solutions, which are closed curves around the point $(A, \varepsilon) = (\gamma/\nu, 0)$. Over one period A and ε return to their initial values, so the time averages of $d \log A/dt = 2\varepsilon$ and of $d\varepsilon/dt = \gamma - \nu A$ vanish:

$$\langle \varepsilon \rangle = 0, \qquad \langle A \rangle = \gamma/\nu.$$

The sharpness of the flow's destination oscillates around $2/\eta$, its time average is exactly $2/\eta$, and the oscillation carries exactly the energy needed to cancel the sharpening. With the saturation term $-\beta A^2$ restored, the closed curves become, for small β, spirals into the fixed point $A = \gamma/\nu$, $\varepsilon = \beta\gamma/(2\nu)$, so the sharpness settles slightly above $2/\eta$, as in Observation 15.1. Up to scaling, (15.7) is the system that Damian et al. (2023) derived for a general loss near the edge, with A the squared displacement along the top eigenvector of the Hessian, ε the excess sharpness, γ proportional to the rate $-\langle \nabla\mathcal{L}, \nabla S \rangle$ of progressive sharpening, and ν proportional to $\|\nabla S\|^2$; they called the mechanism *self-stabilization*. With $\gamma = 0$ it is (15.4) without saturation, whose first integral gave the reflection law.

WHEN NO MINIMIZER IS STABLE. For $1 < \eta y < 2$ every minimizer is sharper than $2/\eta$, and gradient descent started in $\mathcal{K}$ stays bounded (Theorem 15.10) but almost never converges (Theorem 15.8). Numerically, its imbalance

tends to zero, and it settles on the balanced line $u = v$, on an orbit of period two around the balanced minimizer for ηy up to about 1.236. The orbit then period-doubles, and for most ηy above 1.3 the motion is chaotic; at $\eta y = 2$ the map on the balanced line is again a tripling map. In this band the loss never goes to zero: training is stuck at the edge, without the decrease of the loss that Observation 15.1 reports. Kalra et al. (2025) describe a period-doubling route to chaos at the edge in the width-m version of this model.

15.6 What the model gets wrong

WHAT WAS THROWN AWAY. One example, one hidden unit and two layers pin the top eigenvector of the Hessian to the normal of a single curve; in a network it rotates, and several eigenvalues can reach $2/\eta$ at once. The square loss has minima whose curvature does not change once they are reached, whereas along a minimizing sequence of the cross-entropy loss the curvature tends to zero. Gradient descent is full-batch with a fixed step: no noise, momentum, preconditioning, schedule or warmup. Above all, gradient flow conserves the imbalance, so the model has no progressive sharpening: it reproduces the catapult and the selection of minimizers flatter than $2/\eta$, but not the long phase at the edge. And the exact identities of Lemma 15.5 and Theorem 15.10 come from the bilinear structure; nothing guarantees them for deeper or nonlinear models (Exercise 15.12).

WHERE IT BREAKS. (1) The model does not explain why the sharpness rises in networks, although Kalra et al. (2025) report sharpness reduction, progressive sharpening and the edge of stability in its width-m version; the partial explanations use other models, second-order regression (Agarwala et al. 2023) and the growth of the output layer (Li et al. 2022). (2) With the cross-entropy loss the sharpness eventually falls (Cohen et al. 2021), and for logistic regression on separable data gradient descent converges with every step size, after a phase in which the loss oscillates (Wu et al. 2024, 2023); there the edge has no fixed location (Chapter 4). (3) With minibatches the full-batch sharpness stays lower (Jastrzębski et al. 2020), and what sits at the edge is a sharpness adapted to the minibatch (Lee and Jang 2023), the batch sharpness of an unrefereed preprint (Andreyev and Beneventano 2024). (4) With Adam, an unrefereed preprint reports that a preconditioned sharpness equilibrates near $38/\eta$ for $\beta_1 = 0.9$, the stability threshold of the momentum that Adam uses (Exercise 15.9), rather than at $2/\eta$ (Cohen et al. 2022). (5) The divergence threshold $4/\lambda_{\text{init}}$ belongs to bilinear models: for wide ReLU networks Lewkowycz et al. (2020) measured about $12/\lambda_{\text{init}}$. (6) Theorem 15.8 concerns limits. It says nothing about the thousands of steps that a network spends at the edge while its loss keeps falling, which is where the interesting optimization happens.

FINE STRUCTURE. Damian et al. (2023) expand the loss to third order around a point at the edge: the oscillation along the top eigenvector produces, through the third derivative, a gradient of the sharpness that pushes the iterates back below $2/\eta$, and gradient descent tracks gradient descent

projected onto $\{S \le 2/\eta\}$; (15.7) is the two-variable shadow of their analysis. Cohen et al. (2025) derive a *central flow*, a differential equation for the time-averaged trajectory at the edge that predicts the loss and the sharpness, also with several unstable directions. Minimalist models give complete proofs of edge-of-stability behaviour in low dimension: a two-parameter loss whose minimizers form a curve (Zhu et al. 2023), the learning of threshold neurons (Ahn et al. 2023), normalized gradient descent (Arora et al. 2022), scalar networks (Kreisler et al. 2023), and a bifurcation analysis of the trajectory (Song and Yun 2023). Large steps balance the layers of factorization models (Wang et al. 2022); in diagonal linear networks, the subject of Chapter 3, they help stochastic gradient descent find sparse solutions but can hinder gradient descent (Even et al. 2023). For scalar–vector factorization, Liang and Montúfar (2026) show that near the critical step size the minimizer reached depends sensitively on the initialization, and that with regularization the boundary of the convergence region becomes fractal. Open: a solvable model in which progressive sharpening is derived and the long phase at the edge is solved (Exercise 15.13), the convergence region of deeper products (Exercise 15.12), and a theory of the stochastic and adaptive edges.

15.7 Model card

MODEL CARD	
PHENOMENON	Full-batch gradient descent raises the sharpness to $2/\eta$ and then trains at that edge; large initial steps cause a loss spike and a drop in curvature, the catapult.
MODEL	$\mathcal{L}(u,v) = \frac{1}{2}(uv-y)^2$ trained by gradient descent with step size η; imbalance $c = u^2 - v^2$; minimizer sharpness $\Lambda(c) = (c^2 + 4y^2)^{1/2}$.
SOLUTION	Exact: $c' = (1 - 2\eta^2\mathcal{L})c$; convergence, for almost every start, only to minimizers with $\eta S \le 2$; an invariant ellipse $\Psi' = (1 + \eta^2 z(z+y))\Psi$. Derived: the reflection law.
PREDICTS	Phase boundaries at $\eta\Lambda = 2$ and $\eta\Lambda = 4$ (exact); landing at $\eta S \approx 4 - \eta\Lambda(c_0)$, with the second-order correction (15.6); spike height $\varepsilon_0^2/(4\eta^2(1-\eta^2 y^2))$; loss below $2/\eta^2$.
GETS WRONG	No progressive sharpening, hence no sustained edge of stability; square loss, full batch, plain gradient descent; the threshold 4 is special to bilinear models.
SUCCESSORS	Self-stabilization; central flows; minimalist models with proofs; stochastic and adaptive edges; large-step logistic regression.

The model is the prologue's quadratic with one more parameter, and the extra parameter turns the threshold $2/\lambda_1$ of the prologue from a constraint on the step size into a rule about which minimizers can be reached. The product uv is a diagonal linear network with one coordinate (Chapter 3), and its conservation law is the scalar case of the balancedness of deep linear networks (Chapter 2). The logarithmic delay before the catapult is an escape from an unstable fixed point, as in Chapters 2, 7 and 8; in the committee

machine of Chapter 8 the step size also enters through terms of order η^2. With the logistic loss, large steps lead to the maximum-margin solutions of Chapter 4 after an unstable phase.

Exercises

15.1 CONSOLIDATION. (The Hessian off the valley.) (a) Show that the eigenvalues of $\nabla^2\mathcal{L}(u,v)$ are $\frac{1}{2}\big(\tau \pm \sqrt{c^2+4(uv+z)^2}\big)$, so that $S(\theta) \geq \tau/2$, with $S = \tau$ on $\mathcal{M}$. (b) On the balanced line $u = v$, show that $S = \max\{3u^2 - y,\ y - u^2\}$, so that along gradient flow from a small balanced start the sharpness falls from y to $y/2$ and then rises to $2y$: a caricature of progressive sharpening.

15.2 CONSOLIDATION. (Edges in the valley.) Let $\eta = 0.4$ and $y = 1$. (a) Show that the minimizers on the positive branch of $\mathcal{M}$ with $2 < \eta S < 4$ are the points $(u, 1/u)$ with $u \in (0.3178, 0.4569) \cup (2.1889, 3.1463)$. (b) Show that $\mathcal{E}$ meets the line $u = v$ where $\eta u^2 = 2 + \eta y$, and the line $u = -v$ where $\eta u^2 = 2 - \eta y$. (c) For $\eta y < 2$, show that $\{\eta\tau < 4 - 2\eta y\} \subseteq \mathcal{K} \subseteq \{\eta\tau < 4 + 2\eta y\}$.

15.3 CONSOLIDATION. (The tripling map on the ellipse.) Let $\eta y < 2$, and parameterize $\mathcal{E}$ by $a = 2\sqrt{(2+\eta y)/\eta}\,\sin\varphi$ and $b = 2\sqrt{(2-\eta y)/\eta}\,\cos\varphi$. (a) Show that $\eta z = -2\cos 2\varphi$ on $\mathcal{E}$. (b) Using $\sin 3\varphi = \sin\varphi\,(1 + 2\cos 2\varphi)$ and $\cos 3\varphi = -\cos\varphi\,(1 - 2\cos 2\varphi)$, show that G acts on $\mathcal{E}$ as $\varphi \mapsto \pi - 3\varphi$. (c) Deduce that G has exactly four fixed points on $\mathcal{E}$, the minimizers with $\eta S = 4$, and exactly $3^k - (-1)^k$ points of $\mathcal{E}$ fixed by G^k for each $k \geq 1$. (d) Show that the two orbits of period two are $\{\theta, -\theta\}$ for the points θ where $\mathcal{E}$ meets the diagonals, with $\eta z = 2$ on $u = v$ and $\eta z = -2$ on $u = -v$.

15.4 CONSOLIDATION. (Volterra's averages.) (a) Show that $\varepsilon^2 + \nu A - \gamma \log A$ is constant along the solutions of (15.7), and that every solution with $A(0) > 0$ is periodic. (b) Show that small oscillations around $(\gamma/\nu, 0)$ have period $2\pi/\sqrt{2\gamma}$. (c) With the term $-\beta A^2$ added to the first equation, show that the fixed point moves to $(\gamma/\nu, \beta\gamma/(2\nu))$, that it is asymptotically stable, and that it is a focus if and only if $\beta^2\gamma < 8\nu^2$.

15.5 CONSOLIDATION. ★★ (The second-order reflection law.) Let $\ell = \log A$ and $w = \log c$, and model one step of gradient descent near the edge by

$$\ell' = \ell + F(w) - \beta e^{\ell}, \qquad w' = w - \eta^2 e^{\ell}, \qquad F(w) = 2\log\big(1 + \varepsilon(w)\big),$$

with $\varepsilon(w) = \eta(e^{2w} + 4y^2)^{1/2} - 2$. (a) Without the term βe^{ℓ}, show that $H = \eta^2 e^{\ell} + \int^{w} F$ is constant along the flow $\dot{\ell} = F(w)$, $\dot{w} = -\eta^2 e^{\ell}$, and that one step of the map increases it by $\frac{1}{2}g^2F'(w) + \frac{1}{2}gF(w)^2$ up to higher-order terms, with $g = \eta^2 e^{\ell}$. (b) Sum these increments over a catapult, converting sums into integrals with $dw = -g$ and using the leading-order solution $\nu A = \varepsilon_0^2 - \varepsilon^2$; add the effect of βe^{ℓ}, and derive (15.6). (c) Check with the exact map that $(\varepsilon_\infty + \varepsilon_0)/\varepsilon_0^2 \to -2/7$ for $\eta y = 0.4$.

(Hint: at the edge $d\varepsilon/dw = 2(1-\eta^2 y^2)$ and $d^2\varepsilon/dw^2 = 2(1-\eta^4 y^4)$; the increments of (a) contribute $-\frac{4}{3}$, and the expansion $F = 2\varepsilon - \varepsilon^2$ contributes $\frac{1}{3}$.)

15.6 VARIATION. (No label.) Let $y = 0$, so that $\mathcal{L} = \frac{1}{2}u^2v^2$ and the minimizers are the two axes, with sharpness τ. (a) Show that $\tau' = \tau + \eta z^2(\eta\tau - 4)$ exactly, and that $\mathcal{E}$ is the circle $\eta\tau = 4$. (b) If $\eta\tau_0 < 4$, show that $\mathcal{L}(\theta_t) \to 0$. (c) If $\eta\tau_0 > 4$, show that $\tau_t \to \infty$ for almost every θ_0. *(Hint: in (c), if τ_t stayed bounded it would converge, and $\sum_t \eta z_t^2(\eta\tau_t - 4) < \infty$ would force $z_t \to 0$; now argue as in Theorem 15.8.)*

15.7 VARIATION. (Three factors.) Let $\mathcal{L}(u,v,w) = \frac{1}{2}(uvw - y)^2$ and $z = uvw - y$. (a) Show that gradient flow conserves $u^2 - v^2$ and $v^2 - w^2$, and that gradient descent multiplies $u^2 - v^2$ by $1 - \eta^2 z^2 w^2$. (b) Show that the sharpness of a minimizer is $y^2(u^{-2} + v^{-2} + w^{-2})$, and that the flattest minimizers are balanced, with sharpness $3y^{4/3}$. (c) Adapt the proof of Theorem 15.8(ii) to show that the initializations from which gradient descent converges to a minimizer with $\eta S > 2$ form a null set. Deduce that if $\eta > 2/(3y^{4/3})$, then gradient descent from almost every initialization converges to no minimizer at all.

15.8 VARIATION. (Every width.) In the model of Lewkowycz et al. (2020), $u, v \in \mathbb{R}^m$, $f = \langle u, v\rangle/\sqrt{m}$, $\mathcal{L} = \frac{1}{2}(f - y)^2$, and the curvature is $\lambda = (\|u\|^2 + \|v\|^2)/m$. (a) With $z = f - y$, show that gradient descent satisfies $z' = z\big(1 - \eta\lambda + \frac{\eta^2}{m}z(z + y)\big)$ and $\lambda' = \lambda + \frac{\eta}{m}z^2(\eta\lambda - 4) - \frac{4\eta}{m}yz$. (b) Show that $\Psi_m = \eta\lambda - \frac{\eta^2 y}{m}z - 4$ satisfies $\Psi_m' = \big(1 + \frac{\eta^2}{m}z(z + y)\big)\Psi_m$. (c) Deduce that if $\eta y < 2\sqrt{m}$ and $\Psi_m < 0$ at initialization, then the iterates stay bounded, so that the threshold $4/\lambda_{\text{init}}$ is exact up to a correction of order $1/m$.

15.9 VARIATION. (Momentum and the adaptive edge.) Momentum in the form used by Adam updates $m_t = \beta_1 m_{t-1} + (1 - \beta_1)\nabla\mathcal{L}(\theta_t)$ and $\theta_{t+1} = \theta_t - \eta m_t$. (a) On a quadratic with curvature λ_1 in some direction, show that the iteration is stable in that direction if and only if $\eta\lambda_1 < 2(1 + \beta_1)/(1 - \beta_1)$. (b) For $\beta_1 = 0.9$, compare with the value $38/\eta$ at which Cohen et al. (2022) observed the preconditioned sharpness of Adam to equilibrate.

15.10 SIMULATION. (Other step sizes.) Run `15-stability-catapult.py` with $\eta y = 0.2$ and $\eta y = 0.8$. Check that near the edge the landing follows (15.6), with coefficients -0.583 and 2.889, and that no run converges beyond $\eta\Lambda(c_0) = 4$. For $\eta y = 0.8$, explain why the reflection law reaches the flattest minimizer already at $\eta\Lambda(c_0) = 2.4$.

15.11 SIMULATION. (A network at the edge.) Using `15-stability-mlp.py`: (a) train with $\eta \in \{0.3, 0.5\}$ and check that the sharpness levels off within a few percent of $2/\eta$; (b) check that the time ηt at which the sharpness first reaches $2/\eta$ lies on the sharpness curve of the run with the smallest step, so that before the edge gradient descent follows

gradient flow; (c) replace the full gradient by a minibatch gradient with 50 examples at $\eta = 0.25$, and observe that the full-batch sharpness levels off well below $2/\eta = 8$ (near 6.6 in our runs).

15.12 RESEARCH. (Deeper products.) For $\mathcal{L}(u, v, w) = \frac{1}{2}(uvw - y)^2$ the catapult survives (Exercise 15.7), but not the sharp boundary at $\eta S = 4$: in our runs from minimizers with $u = v > 1$ and $y = 1$, perturbed at relative size 10^{-4}, all runs converged for starts with ηS up to about 4.8 ($\eta = 0.2$) or 5.0 ($\eta = 0.05$), and the fraction that converged then fell gradually, to about 0.4 at $\eta S = 6$. Is there an invariant hypersurface that separates convergence from divergence, or is the boundary of the convergence region fractal, as Liang and Montúfar (2026) found for factorization with regularization? (From the other branch, $u = v < 1$, our runs at $\eta = 0.2$ all diverged already at $\eta S = 4.2$.) What is the largest ηS from which every nearby start converges? As far as we know, both questions are open.

15.13 RESEARCH. (Sharpening from first principles.) The product model must be given progressive sharpening by hand, as the rate γ of (15.7). Find a solvable model in which the rate is derived, and which then trains at the edge of stability with a decreasing loss. Agarwala et al. (2023) derive progressive sharpening in quadratic regression models; what is open is a model in which both the rate and the long phase at the edge are solved, and a derivation of the rate for a network.

Notes and References

Wu et al. (2018) proposed the selection of flat minima by linear stability, and Jastrzębski et al. (2019, 2020) observed that the sharpness reached early in training is set by the step size and the batch size. Lewkowycz et al. (2020) solved a wide two-layer linear model on one example, found its lazy, catapult and divergent phases, and named the catapult. Cohen et al. (2021) named progressive sharpening and the edge of stability and documented both across architectures and losses.

Gradient flow conserves the imbalance in every layered linear or homogeneous network (Du et al. 2018). For the product model and its relatives, Wang et al. (2022) proved convergence and balancing for large steps, Kalra et al. (2025) mapped the regimes of a linear model on one example, and Liang and Montúfar (2026) found the exact critical step size of a scalar–vector factorization, chaos near it, and a fractal boundary with regularization. Our proof of Theorem 15.8 uses the polynomial structure of G in place of the stable manifold theorem (Lee et al. 2019, 2016). We know no reference for the reflection law (15.5) or its second-order form (15.6); both are derived, not proved, and the reduction behind them is the case without sharpening of the analysis of Damian et al. (2023). The flip normal form is in Chapter 4 of Kuznetsov (2004).

TRANSLATING THE LITERATURE. Lewkowycz et al. (2020) write n for the width (our m); their curvature λ is the top eigenvalue of the neural tangent

kernel, our τ when $n = 1$, and λ_0 is its value at initialization. Damian et al. (2023) write x_t for the displacement along the top eigenvector (our $z_t/\sqrt{\tau}$), y_t for the excess sharpness (our ε_t/η), and α and β for the coefficients that correspond, up to constant factors, to our γ and ν. Kreisler et al. (2023) call our $\Lambda(c_t)$ the sharpness of the gradient flow solution (GFS sharpness).

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16 Scaling Laws

Power laws from spectra and from Zipf's law

THE MODEL — POWER-LAW LINEAR REGRESSION AND QUANTA

IN THIS CHAPTER. The test loss of language models falls as a power law in the number of parameters, the number of training tokens, and the compute, over many orders of magnitude, with small exponents that differ between parameters and data. Two solvable models produce such laws. In linear regression on Gaussian features whose target energy decays as $j^{-(1+b)}$ we compute the learning curve of least squares exactly at every model size N and sample size D, and read off a parameter exponent b and a data exponent $b/(1+b)$; ridge regression and the deterministic equivalent show that the spectrum of the features only converts a regularization or a training time into a number of fitted features. A memorizing learner on items with Zipf frequencies $p_k \propto k^{-(1+b)}$ gives the same two exponents, with an exact constant, and an elementary theorem allocates compute between parameters and data. The models get the mechanism of power laws but not the exponents: those are inputs, and fitted language-model data exponents exceed the models' prediction $\alpha_N/(1+\alpha_N)$, in some fits even the parameter exponent.

16.1 The phenomenon: smooth power laws over many decades

How much better does a model get when it is made ten times larger, or trained on ten times more data? For most of the history of machine learning the honest answer was "it depends". Hestness et al. (2017) measured it systematically in four domains, machine translation, language modeling, image processing, and speech recognition, and found that the generalization error falls as a power law in the size of the training set, that improvements of the architecture shift the curve without visibly changing its exponent, and that the model size needed to fit the data grows sublinearly with it.

Kaplan et al. (2020) made the observation quantitative for transformer language models. Measured in nats per token, the test loss of a model with N non-embedding parameters trained to convergence on a large corpus followed $\mathcal{L}(N) = (N_c/N)^{\alpha_N}$ with $\alpha_N \approx 0.076$ and $N_c \approx 8.8 \times 10^{13}$, and the loss of a large model trained with early stopping on D tokens followed $\mathcal{L}(D) = (D_c/D)^{\alpha_D}$ with $\alpha_D \approx 0.095$ and $D_c \approx 5.4 \times 10^{13}$. Some of the trends spanned more than seven orders of magnitude, and architectural details such as depth and width at fixed N mattered little.[1] From these fits they

[1] Kaplan et al. estimate the training compute as $C \approx 6ND$ floating-point operations: about $2N$ per token for the forward pass and twice that for the backward pass. We use $C = 6ND$ throughout.

concluded that a growing compute budget should be spent mostly on larger models, with the optimal number of parameters growing like $C^{0.73}$.

Hoffmann et al. (2022) revisited the allocation with more than 400 models and three estimation procedures. The third fitted the parametric form

$$\mathcal{L}(N, D) = \mathcal{L}_\infty + \frac{A}{N^{\alpha_N}} + \frac{B}{D^{\alpha_D}} \tag{16.1}$$

(they write E, α, β for our $\mathcal{L}_\infty$, α_N, α_D), and reported $\alpha_N \approx 0.34$ and $\alpha_D \approx 0.28$. All three procedures said that parameters and tokens should grow at about the same rate, and a 70-billion-parameter model trained on more tokens with the compute of a 280-billion-parameter one outperformed it.[2] The two studies disagree about the allocation, and later work traced much of the disagreement to how parameters were counted and to the small scale of the earlier fits (Pearce and Song 2024), and to the cost of the last layer, the warmup, and the tuning of the optimizer (Porian et al. 2024).

[2] The 70-billion-parameter model is Chinchilla, which is why (16.1) with fitted constants is often called the Chinchilla law; the larger model is Gopher.

OBSERVATION 16.1 (Hestness et al. 2017; Kaplan et al. 2020; Hoffmann et al. 2022). For language models trained on large corpora, the test loss, minus an irreducible part, decreases as a power law in the number of parameters when data are plentiful, and in the number of training tokens when the model is large, over several orders of magnitude, with exponents well below 1. The loss of compute-optimal models is a power law in the compute.

How robust is this? The power-law form is robust; the numbers are not. Besiroglu et al. (2024) reconstructed the data behind the third procedure of Hoffmann et al. (2022) from its plots and refitted it: the published estimates fit the reconstructed data poorly, were inconsistent with the other two procedures, and came with confidence intervals far too narrow for a few hundred runs, while their own fit had a larger data exponent and was compatible with the first two procedures. Curves of other quantities bend, and Caballero et al. (2023) fit them with smoothly broken power laws. Repeating data breaks the law in D slowly: up to about four epochs of repeated tokens are almost as good as fresh ones, and then the value of more compute decays (Muennighoff et al. 2023). Choosing the data instead of sampling it can beat the law (Sorscher et al. 2022). And while the loss is smooth, some downstream abilities appear abruptly with scale (Wei et al. 2022), although part of the abruptness comes from the choice of metric (Schaeffer et al. 2023).

Three questions follow. Why a power law at all, rather than an exponential or a logarithm? What sets the exponents, and why do parameters and data have different ones? And how should compute be split between them? The models of this chapter answer the first and third exactly and the second in a form that can be checked.

16.2 The models: power-law features and Zipf-distributed quanta

A power law in N needs a hierarchy: a model of size N must capture the N most important pieces of the task and miss the rest, and the missing

importance must decay as a power. Two hierarchies are natural. In regression the pieces are directions of feature space, and their importance is the variance they explain; in language the pieces are items of knowledge, and their importance is how often they are used. In this chapter N and D follow the scaling-law literature: N counts parameters, the p of Chapter 1, and D counts training examples, the n of that chapter.

Assumption 16.2 (Power-law features). The input is a sequence of independent features $x_j \sim \mathcal{N}(0, \lambda_j)$, $j = 1, \dots, P$, with $P \in \{1, 2, \dots\} \cup \{\infty\}$ and variances $\lambda_1 \geq \lambda_2 \geq \cdots > 0$. The label is $y = \sum_j w_{\star,j} x_j + \varepsilon$, with $\sum_j \lambda_j w_{\star,j}^2 < \infty$ and noise ε of mean 0 and variance σ^2, independent of x. In the *power-law case* $P = \infty$ and, for exponents $a > 1$ and $b > 0$,

$$\lambda_j = j^{-a}, \qquad \lambda_j w_{\star,j}^2 = j^{-(1+b)}. \tag{16.2}$$

The *model of size* N predicts with the first N features, $f(x) = \sum_{j \leq N} w_j x_j$, and is fitted by least squares on D independent examples (x_i, y_i); its data matrix $X_N \in \mathbb{R}^{D \times N}$ has rows $(x_{i1}, \dots, x_{iN})$, and for $N \leq D$ the estimator is $\hat{w}_N = (X_N^\top X_N)^{-1} X_N^\top y$. Because the features are independent, the population loss of a weight vector w supported on the first N coordinates is

$$\mathcal{L}(w) = \mathbb{E}\big(y - f(x)\big)^2 = \sigma^2 + T(N) + \sum_{j \leq N} \lambda_j (w_j - w_{\star,j})^2, \qquad T(N) = \sum_{j > N} \lambda_j w_{\star,j}^2. \tag{16.3}$$

The *approximation error* $T(N)$ is what the best model of size N still misses; the *excess loss* is $\mathcal{L} - \sigma^2$. In the power-law case $T(N)$ is a tail of $\zeta(1 + b)$, and b is called the *source exponent*: it measures how fast the importance of the features decays, and it does not involve a.

The second model replaces regression by recall.

Assumption 16.3 (Zipf-distributed quanta). A task consists of items $k = 1, 2, \dots$, the *quanta*. Each example, at training or test time, requires one item, drawn independently with probability

$$p_k = \frac{k^{-(1+b)}}{\zeta(1 + b)}, \qquad b > 0. \tag{16.4}$$

A learner answers correctly exactly when it has learned the item required. A learner of *capacity* N can hold N items; a *memorizing learner* learns every item that occurs in its D training examples.

Frequencies of words follow *Zipf's law*, $p_k \propto k^{-s}$ with s close to 1 (Zipf 1949); Hutter (2021) built the memorizing learner to show that power-law learning curves with every exponent arise from Zipf-distributed data, and Michaud et al. (2023) proposed that a language model's skills are such quanta, learned in order of decreasing frequency.[3]

[3] The exponent is written $1 + b$ so that $b > 0$ is the exponent of the tail $\sum_{k>N} p_k$, as b is the exponent of $T(N)$ in (16.2); Hutter (2021) writes α for our b.

DICTIONARY

IN THE MODEL	IN A TRAINED NETWORK
feature j, or quantum k	a direction of the representation, or a skill, ranked by importance
model size N	number of parameters
sample size D	number of training tokens, each seen once
variances $\lambda_j = j^{-a}$	spectrum of the data in the network's representation, e.g. of a tangent kernel
target energy $j^{-(1+b)}$, frequency p_k	share of the loss that each direction or skill carries
approximation error $T(N)$	loss of the best model of size N on infinite data
noise variance σ^2	irreducible loss $\mathcal{L}_\infty$, the entropy of the text
ridge parameter λ, training time t	weight decay, training time: how many directions are fitted
memorizing an item seen once	learning a skill from few examples
$C = 6ND$	training compute in floating-point operations

WHAT WAS THROWN AWAY, AND WHY FIRST. The features are fixed and independent, so the model learns no representation and its exponents are properties of the data in a given basis; but the question "why a power law" is already nontrivial for fixed features, and independence makes the learning curve exact. The model of size N uses the top N features, which assumes that a network spends its parameters on the most important directions first, the staged learning of Chapter 2. The noise is white, and it plays the role of the irreducible loss. Every example is fresh, as in one-epoch training of language models. The quanta are independent and are learned all at once; nothing is composed of anything.

WHAT ONE HAD TO NOTICE. Two things. A power law in the loss needs a power-law hierarchy of importance, and real data supply them: kernel spectra of image data decay as powers (Spigler et al. 2020),[4] and word frequencies follow Zipf's law. And a finite sample makes the hierarchy visible only down to a threshold: a direction or item worth less than about $1/D$ of the loss cannot be learned from D examples. The data exponent comes from where this threshold falls.

[4] For kernel regression, Spigler et al. (2020) measured learning-curve exponents of about 0.4 on MNIST and 0.1 on CIFAR-10, and estimated 0.36 and 0.07 from kernel principal components of the data.

16.3 The solution: exact learning curves of power-law regression

We need the size of power-law tails, then the learning curve at every N and D, and then its asymptotics, which are the scaling laws.

Lemma 16.4 (Power-law tails). *For $b > 0$ and every integer $N \geq 1$,*

$$\frac{(N+1)^{-b}}{b} \leq \sum_{j>N} j^{-(1+b)} \leq \frac{N^{-b}}{b}.$$

In particular, in the power-law case $T(N) = \frac{N^{-b}}{b}\,(1 + O(1/N))$.

PROOF. The function $t \mapsto t^{-(1+b)}$ decreases, so $\int_j^{j+1} t^{-(1+b)}\,\mathrm{d}t \leq j^{-(1+b)} \leq \int_{j-1}^{j} t^{-(1+b)}\,\mathrm{d}t$ for $j \geq 2$. Summing over $j > N$ gives $\int_{N+1}^{\infty} t^{-(1+b)}\,\mathrm{d}t \leq \sum_{j>N} j^{-(1+b)} \leq \int_N^{\infty} t^{-(1+b)}\,\mathrm{d}t$, and the integrals equal $(N+1)^{-b}/b$ and N^{-b}/b. Finally $(1 + 1/N)^{-b} = 1 + O(1/N)$. ∎

The main theorem says that, for least squares on the top N features, the variances λ_j do not matter at all: the learning curve depends on the target only through the tail $T(N)$.

Theorem 16.5 (Exact learning curve). *Under Assumption 16.2, let $1 \leq N \leq \min(P, D-2)$. The least-squares estimator on the first N features from D examples has test loss*

$$\mathbb{E}\mathcal{L}(\hat{w}_N) = \big(\sigma^2 + T(N)\big)\,\frac{D-1}{D-N-1}. \tag{16.5}$$

If $\sigma^2 + T(N) > 0$ and $N \in \{D-1, D\}$, then $\mathbb{E}\mathcal{L}(\hat{w}_N) = +\infty$.

PROOF. The plan is to treat the omitted features as noise, as in the weak-features model of Chapter 1, and to whiten the retained ones.

The omitted features are noise. For example i write $y_i = \sum_{j\leq N} w_{\star,j} x_{ij} + \xi_i$ with $\xi_i = \sum_{j>N} w_{\star,j} x_{ij} + \varepsilon_i$; the series converges in L^2 because $\sum_j \lambda_j w_{\star,j}^2 < \infty$. The features of example i are independent and ε_i is independent of them, so ξ_i is independent of $(x_{i1}, \dots, x_{iN})$, has mean 0, and has variance $T(N) + \sigma^2$; across examples the ξ_i are independent. Hence the vector $\xi = (\xi_1, \dots, \xi_D)$ is independent of X_N and satisfies $\mathbb{E}\xi\xi^\top = (\sigma^2 + T(N))I_D$.

Whitening. Let $\Lambda = \mathrm{diag}(\lambda_1, \dots, \lambda_N)$ and $Z = X_N \Lambda^{-1/2}$, a $D \times N$ matrix with independent $\mathcal{N}(0,1)$ entries, of rank N almost surely (Theorem A.7). The estimator is $\hat{w}_N = w_{\star,\leq N} + (X_N^\top X_N)^{-1} X_N^\top \xi$, and

$$\Lambda^{1/2}(\hat{w}_N - w_{\star,\leq N}) = \Lambda^{1/2}\big(\Lambda^{1/2} Z^\top Z \Lambda^{1/2}\big)^{-1} \Lambda^{1/2} Z^\top \xi = (Z^\top Z)^{-1} Z^\top \xi.$$

By (16.3) the last sum in $\mathcal{L}(\hat{w}_N)$ is the squared norm of this vector.

Averaging. Conditionally on Z, the squared norm has mean $(\sigma^2 + T(N))\,\mathrm{tr}\big((Z^\top Z)^{-1} Z^\top Z (Z^\top Z)^{-1}\big) = (\sigma^2 + T(N))\,\mathrm{tr}\big((Z^\top Z)^{-1}\big)$. By the inverse Wishart mean (Theorem A.7, or Lemma 1.5), $\mathbb{E}\,\mathrm{tr}\big((Z^\top Z)^{-1}\big) = N/(D-N-1)$ for $N \leq D-2$, and it is $+\infty$ for $N \in \{D-1, D\}$. Adding $\sigma^2 + T(N)$ from (16.3) gives $(\sigma^2 + T(N))\big(1 + \frac{N}{D-N-1}\big)$, which is (16.5). ∎

The degenerate size $N = 0$ (the zero predictor) has loss $\sigma^2 + T(0)$, which (16.5) also gives. The formula says that a model of size N pays its approximation error, and pays it again, together with the noise, inflated by the

variance factor $N/(D-N-1)$: the unlearned part of the target is noise for the learner. This self-consistent noise returns in the deterministic equivalent below. In the power-law case Theorem 16.5 turns into scaling laws.

Corollary 16.6 (Two scaling laws and the optimal size). *Under Assumption 16.2, suppose that $T(N) \sim cN^{-b}$ as $N \to \infty$, for some $c > 0$ and $b > 0$ (in the power-law case $c = 1/b$), and that $\sigma > 0$. Let $\mathcal{E}(N,D) = \mathbb{E}\mathcal{L}(\hat{w}_N) - \sigma^2$ and $\mathcal{E}^\star(D) = \min_{0\le N\le D-2}\mathcal{E}(N,D)$.*

(i) Parameter-limited. *For fixed N, $\mathcal{E}(N,D) \to T(N) \sim cN^{-b}$ as $D \to \infty$.*

(ii) Data-limited. *As $D \to \infty$,*

$$\mathcal{E}^\star(D) \sim (1+b)\, c^{\frac{1}{1+b}} \left(\frac{\sigma^2}{bD}\right)^{\frac{b}{1+b}}, \tag{16.6}$$

and every minimizer N_D satisfies $N_D \sim (bcD/\sigma^2)^{1/(1+b)}$.

Proof. Part (i) follows from (16.5), since $(D-1)/(D-N-1) \to 1$. For part (ii), the plan is to sandwich $\mathcal{E}(N,D)$ between multiples of $h_D(N) = cN^{-b} + \sigma^2 N/D$, whose minimum we compute first.

The comparison function. Let $M_D = (bcD/\sigma^2)^{1/(1+b)}$, the critical point of h_D, and $\epsilon_D = cM_D^{-b}$. Since $\sigma^2 M_D/D = \sigma^2 M_D^{1+b}/(DM_D^b) = b\,\epsilon_D$, substituting $N = sM_D$ gives $h_D(sM_D) = \epsilon_D\, g(s)$ with $g(s) = s^{-b} + bs$. The function g is strictly convex on $(0,\infty)$, with $g'(s) = b(1-s^{-1-b})$, so its minimum is $g(1) = 1+b$, and for every $\eta \in (0,1)$ the number $g_\eta = \min\{g(1-\eta), g(1+\eta)\}$ exceeds $1+b$ and bounds g from below off $(1-\eta, 1+\eta)$. The right side of (16.6) is $(1+b)\epsilon_D$, by the definition of M_D.

Upper bound. Take $N = \lceil M_D \rceil$. Since $M_D \to \infty$ and $M_D/D \to 0$, we have $T(N) = cN^{-b}(1+o(1)) = \epsilon_D(1+o(1))$ and $(\sigma^2 + T(N))\frac{N}{D-N-1} = \sigma^2\frac{M_D}{D}(1+o(1)) = b\epsilon_D(1+o(1))$. Hence $\mathcal{E}^\star(D) \le \mathcal{E}(N,D) = (1+b)\epsilon_D(1+o(1))$.

Lower bound. Since $\frac{N}{D-N-1} \ge \frac{N}{D}$ and $\sigma^2 + T(N) \ge \sigma^2$, every $N \le D-2$ has $\mathcal{E}(N,D) \ge T(N) + \sigma^2 N/D$. Fix $\delta \in (0,1)$ and choose N_0 with $T(N) \ge (1-\delta)cN^{-b} > 0$ for all $N \ge N_0$. For $N \ge N_0$, $\mathcal{E}(N,D) \ge (1-\delta)h_D(N) = (1-\delta)\epsilon_D g(N/M_D) \ge (1-\delta)(1+b)\epsilon_D$. For $N < N_0$, $\mathcal{E}(N,D) \ge T(N_0)$, because T is nonincreasing, and $T(N_0) > 0$ exceeds $(1+b)\epsilon_D$ once D is large, because $\epsilon_D \to 0$. Hence $\liminf \mathcal{E}^\star(D)/\epsilon_D \ge (1-\delta)(1+b)$ for every δ, which with the upper bound proves (16.6).

The minimizer. Fix $\eta \in (0,1)$, and choose $\delta > 0$ so small that $(1-\delta)g_\eta > (1+\delta)(1+b)$. For large D, the upper bound gives $\mathcal{E}^\star(D) < (1+\delta)(1+b)\epsilon_D$, while every $N \ge N_0$ with $|N/M_D - 1| \ge \eta$ has $\mathcal{E}(N,D) \ge (1-\delta)g_\eta\epsilon_D$, and every $N < N_0$ has $\mathcal{E}(N,D) \ge T(N_0)$; neither can be a minimizer. So $|N_D/M_D - 1| < \eta$ for large D. ∎

The proof shows where the data exponent comes from. At the optimum the approximation error cN^{-b} and the variance $\sigma^2 N/D$ are of the same order, so the model keeps exactly the features whose share of the loss, about $N^{-(1+b)}$ per feature, exceeds the estimation cost σ^2/D per feature; this gives $N_D \asymp D^{1/(1+b)}$ and $\mathcal{E}^\star \asymp N_D^{-b} = D^{-b/(1+b)}$. Without noise the variance has

nothing to amplify except $T(N)$ itself, the best model has a size proportional to D, and the data exponent rises to b (Exercise 16.2).[5]

[5] This is the distinction of Bahri et al. (2024) between variance-limited and resolution-limited scaling, here in one formula: the factor $(D-1)/(D-N-1)$ is the variance, and $T(N)$ the resolution.

16.4 The solution: the spectrum, ridge, and Zipf's law

Least squares on the top N features never sees the variances λ_j. A learner that uses all features and controls its complexity by a ridge parameter λ, or by its training time, does: then the spectrum decides how many features are fitted.

THE SEQUENCE MODEL. Strip the regression to its normal modes. For each j, observe $z_j = \theta_j + \frac{\sigma}{\sqrt{D}} g_j$, with $\theta_j = \sqrt{\lambda_j}\, w_{\star,j}$ the target in units of prediction and independent $g_j \sim \mathcal{N}(0,1)$: the per-feature estimation noise of Theorem 16.5 is σ^2/D when $N \ll D$. Ridge regression multiplies the jth coordinate by $\lambda_j/(\lambda_j+\lambda)$, as in Chapter 1; the estimator $\hat{\theta}_j = \frac{\lambda_j}{\lambda_j+\lambda} z_j$ has excess loss

$$\mathcal{E}_{\text{seq}}(\lambda) = \mathbb{E}\sum_j (\hat{\theta}_j - \theta_j)^2 = \underbrace{\sum_j \lambda_j w_{\star,j}^2 \Big(\frac{\lambda}{\lambda_j+\lambda}\Big)^2}_{\text{bias}} + \underbrace{\frac{\sigma^2}{D}\sum_j \Big(\frac{\lambda_j}{\lambda_j+\lambda}\Big)^2}_{\text{variance}}. \quad (16.7)$$

Proposition 16.7 (Ridge on a power-law spectrum). *In the power-law case (16.2), let $\frac{1}{2} < a$ and $0 < b < 2a$, and let $J = \lambda^{-1/a}$. There are constants $0 < c_1 < c_2$, depending only on a and b, such that for $0 < \lambda \le 1$*

$$c_1 J^{-b} \le \text{bias} \le c_2 J^{-b}, \qquad c_1 J \le \sum_j \Big(\frac{\lambda_j}{\lambda_j+\lambda}\Big)^2 \le c_2 J.$$

Consequently, if $\sigma^2 \le D$, then $\min_{\lambda>0} \mathcal{E}_{\text{seq}}(\lambda) \asymp (\sigma^2/D)^{b/(1+b)}$, attained up to constants at $J = (D/\sigma^2)^{1/(1+b)}$.

PROOF. The plan is to split the sums at $j = J$, where $\lambda_j = \lambda$, and compare each part with a power.

Two regimes of the filter. Write $r_j = \lambda j^a = (j/J)^a$, so that $\frac{\lambda}{\lambda_j+\lambda} = \frac{r_j}{1+r_j}$ and $\frac{\lambda_j}{\lambda_j+\lambda} = \frac{1}{1+r_j}$. For $j \le J$ we have $r_j \le 1$, so $\frac{r_j}{1+r_j} \in [\frac{r_j}{2}, r_j]$ and $\frac{1}{1+r_j} \in [\frac{1}{2}, 1]$; for $j > J$ we have $r_j > 1$, so $\frac{r_j}{1+r_j} \in [\frac{1}{2}, 1]$ and $\frac{1}{1+r_j} \in [\frac{1}{2r_j}, \frac{1}{r_j}]$. Since $\lambda \le 1$, $J \ge 1$.

The bias. By the first regime, the bias lies between $\frac{1}{4}$ and 1 times

$$\sum_{j\le J} j^{-(1+b)}\Big(\frac{j}{J}\Big)^{2a} + \sum_{j>J} j^{-(1+b)} = J^{-2a}\sum_{j\le J} j^{2a-1-b} + \sum_{j>J} j^{-(1+b)}.$$

The exponent $2a-1-b$ exceeds -1 because $b < 2a$, so $\sum_{j\le J} j^{2a-1-b}$ lies between two constant multiples of J^{2a-b} for $J \ge 1$ (compare with $\int_0^{J+1} t^{2a-1-b}\,\mathrm{d}t$ and with the single term $j = 1$ when $J < 2$); and $\sum_{j>J} j^{-(1+b)}$ lies between two constant multiples of J^{-b} by Lemma 16.4 applied at $\lfloor J \rfloor$. Both parts are of order J^{-b}.

The variance sum. By the two regimes it lies between $\frac{1}{4}$ and 1 times $\#\{j \le J\} + \sum_{j>J}(J/j)^{2a}$. The first term is $\lfloor J \rfloor \in [J/2, J]$, and the second is at most $J^{2a}\big(J^{-2a} + \int_J^\infty t^{-2a}\,\mathrm{d}t\big) = 1 + \frac{J}{2a-1}$, finite because $2a > 1$.

The optimum. With $\epsilon = \sigma^2/D \le 1$, the bounds give $\mathcal{E}_{\text{seq}} \asymp J^{-b} + \epsilon J$ on $\lambda \le 1$. At $J = \epsilon^{-1/(1+b)} \ge 1$ both terms equal $\epsilon^{b/(1+b)}$. Conversely, for every $J \ge 1$, either $J \le \epsilon^{-1/(1+b)}$ and $J^{-b} \ge \epsilon^{b/(1+b)}$, or $J > \epsilon^{-1/(1+b)}$ and $\epsilon J \ge \epsilon^{b/(1+b)}$. For $\lambda > 1$ every factor $\frac{\lambda}{\lambda_j+\lambda}$ is at least $\frac{1}{2}$, so the bias is at least $\frac{1}{4}\zeta(1+b)$, which is at least a constant times $\epsilon^{b/(1+b)}$. ■

So ridge on all features behaves like least squares on the top $J = \lambda^{-1/a}$ features: the spectrum exponent a only converts the regularization into an effective model size, and the exponents are those of Corollary 16.6. The same holds for training time. Gradient flow on half the population loss from zero fits the jth coordinate up to the factor $1 - e^{-\lambda_j t}$, so it has learned the features with $\lambda_j t \gtrsim 1$, an effective size $J = t^{1/a}$, and its excess loss decays like $t^{-b/a}$ (Exercise 16.6). Spectral bias, the order in which kernel gradient descent learns eigenfunctions (Chapter 6), becomes a law in time with exponent b/a. If $b > 2a$, the bias of ridge saturates at order λ^2 and the exponent drops to $2a/(1+2a)$ (Exercise 16.7): the target is smoother than ridge can exploit.

FROM THE SEQUENCE MODEL TO GAUSSIAN DESIGNS. With a finite Gaussian sample the spectrum of $\frac{1}{D}X^\top X$ is distorted, as the Marchenko–Pastur law of Chapter 1 showed for $\lambda_j \equiv 1$. The distortion is captured by a renormalized ridge parameter.

THE DETERMINISTIC EQUIVALENT. Under Assumption 16.2 with $\sum_j \lambda_j < \infty$, let $\hat{w}_\lambda = (X^\top X + D\lambda I)^{-1} X^\top y$ use all features. Let $\kappa > \lambda$ solve $\kappa = \lambda + \frac{\kappa}{D}\sum_j \frac{\lambda_j}{\lambda_j+\kappa}$ (for $\lambda = 0$ and $D < P$: $\sum_j \frac{\lambda_j}{\lambda_j+\kappa} = D$), and let $\gamma = \frac{1}{D}\sum_j \big(\frac{\lambda_j}{\lambda_j+\kappa}\big)^2 < 1$. The excess loss R of $\hat{w}_\lambda$ is then close to the solution of

$$R = \sum_j \lambda_j w_{\star,j}^2 \Big(\frac{\kappa}{\lambda_j+\kappa}\Big)^2 + \gamma\,(\sigma^2 + R), \qquad \text{that is,} \qquad R = \frac{\text{bias}(\kappa) + \gamma\sigma^2}{1-\gamma}. \tag{16.8}$$

In words: the Gaussian design behaves like the sequence model (16.7) with ridge κ instead of λ and with noise $\sigma^2 + R$ instead of σ^2, the unlearned signal acting as noise exactly as $T(N)$ does in (16.5). For $\lambda_j \equiv 1$ on P features, $\kappa/(1+\kappa)$ is the function v of Theorem 1.11, and (16.8) reproduces that theorem (Exercise 16.4). Cheng and Montanari (2024) prove nonasymptotic bounds of the form $(1 \pm \Delta)$ for the bias and the variance under conditions on effective ranks that hold for power laws; Atanasov et al. (2024) derive (16.8) as a renormalization, and Bordelon et al. (2020) obtained it for kernel regression with methods of statistical physics. ◇

For the power law, the equation for κ reads $\kappa - \lambda \asymp \kappa\,\kappa^{-1/a}/D$, so $\kappa \asymp \max(\lambda, D^{-a})$, and γ stays bounded away from 1; inserting Proposition 16.7 gives an excess loss of order $D^{-b/(1+b)}$ for optimally tuned ridge with noise, and of order $\kappa_0^{b/a} \asymp D^{-b}$ for the noiseless minimum-norm interpolator, the

two exponents of Corollary 16.6 and Exercise 16.2. Cui et al. (2021) describe the crossover between them as D grows, and Caponnetto and De Vito (2007) proved the noisy rate, with matching lower bounds, for kernel ridge regression under source and capacity conditions.

ZIPF'S LAW. The second model gives the same exponents with no regression at all.

Lemma 16.8 (Integral of a unimodal sequence). *Let $g\colon \mathbb{R} \to [0,\infty)$ be integrable, nondecreasing on $(-\infty, m]$ and nonincreasing on $[m,\infty)$. Then $\left|\sum_{k\in\mathbb{Z}} g(k) - \int g\right| \le 2\sup g$.*

PROOF. Let $k_0 = \lfloor m \rfloor$. For $k \le k_0 - 1$ the interval $[k, k+1]$ lies in $(-\infty, m]$, where g increases, so $g(k) \le \int_k^{k+1} g$; for $k \ge k_0 + 2$ the interval $[k-1, k]$ lies in $[m,\infty)$, so $g(k) \le \int_{k-1}^{k} g$. These intervals do not overlap (k_0, k_0+1), and the two remaining terms are at most $\sup g$ each, so $\sum_k g(k) \le \int g + 2\sup g$. In the other direction, $g(k) \ge \int_{k-1}^{k} g$ for $k \le k_0$ and $g(k) \ge \int_k^{k+1} g$ for $k \ge k_0 + 1$; these intervals cover $\mathbb{R}$ except (k_0, k_0+1), whose integral is at most $\sup g$. ∎

Theorem 16.9 (Zipf learning curves). *Under Assumption 16.3, the memorizing learner trained on D examples errs with probability*

$$M(D) = \sum_{k\ge 1} p_k (1-p_k)^D \sim C_b\, D^{-\frac{b}{1+b}}, \qquad C_b = \frac{\Gamma\left(\frac{b}{1+b}\right)}{(1+b)\,\zeta(1+b)^{\frac{1}{1+b}}}, \tag{16.9}$$

as $D \to \infty$. The learner of capacity N that holds the items $1,\dots,N$ errs with probability $\sum_{k>N} p_k \sim N^{-b}/\big(b\,\zeta(1+b)\big)$.

PROOF. The plan is to replace $(1-p)^D$ by e^{-Dp}, up to a factor that is harmless where the sum lives, and to evaluate the resulting sum as an integral.

The exact formula. The test item is independent of the training items, and the learner errs exactly when the test item is k and none of the D training items is k; this has probability $p_k(1-p_k)^D$. Summing over k gives $M(D)$.

The exponential sum. Let $s = 1+b$, $Z = \zeta(s)$, and $S(c) = \sum_k p_k e^{-cp_k}$ for $c > 0$. Then $S(c) = \sum_k g(k)$ with $g(x) = \frac{x^{-s}}{Z} e^{-cx^{-s}/Z}$ for $x > 0$ and $g(x) = 0$ for $x \le 0$. In the variable $u = cx^{-s}/Z$, which decreases from ∞ to 0 as x increases, $g = \frac{u}{c}e^{-u}$, which increases for $u > 1$ and decreases for $u < 1$; so g is unimodal in x, with $\sup g = \frac{1}{ec}$. Changing variables, with $x = (c/(Zu))^{1/s}$ and $\mathrm{d}x = -\frac{1}{s}(c/Z)^{1/s} u^{-1/s-1}\,\mathrm{d}u$,

$$\int_0^\infty g(x)\,\mathrm{d}x = \frac{1}{s}\left(\frac{c}{Z}\right)^{1/s}\frac{1}{c}\int_0^\infty u^{-1/s} e^{-u}\,\mathrm{d}u = \frac{\Gamma(1-\frac{1}{s})}{s\,Z^{1/s}}\,c^{\frac{1}{s}-1} = C_b\, c^{-\frac{b}{1+b}},$$

since $1 - \frac{1}{s} = \frac{b}{1+b}$. By Lemma 16.8, $|S(c) - C_b c^{-b/(1+b)}| \le \frac{2}{ec}$, so $S(c) \sim C_b c^{-b/(1+b)}$ as $c \to \infty$.

Comparing $(1-p)^D$ with e^{-Dp}. For $0 \le p < 1$, $\log(1-p) \le -p$ and $-\log(1-p) = \sum_{i\ge1} p^i/i \le p/(1-p)$, so

$$e^{-Dp/(1-p)} \le (1-p)^D \le e^{-Dp}.$$

The upper bound gives $M(D) \le S(D)$. For the lower bound fix $\delta \in (0,1)$ and let K be the number of items with $p_k > \delta$. Every $k > K$ has $e^{-Dp_k/(1-p_k)} \ge e^{-Dp_k/(1-\delta)}$, and the items $k \le K$ contribute at most $e^{-Dp_K/(1-\delta)} \le e^{-D\delta}$ to $S(D/(1-\delta))$ (if $K \ge 1$); hence $M(D) \ge S\big(\frac{D}{1-\delta}\big) - e^{-D\delta}$. With the asymptotics of S, $\liminf_D D^{b/(1+b)} M(D) \ge C_b(1-\delta)^{b/(1+b)}$, and letting $\delta \to 0$ proves (16.9).

Capacity. The learner holding the items $1,\dots,N$ errs exactly on the items $k > N$, with probability $\sum_{k>N} p_k = Z^{-1}\sum_{k>N} k^{-(1+b)}$; apply Lemma 16.4. ∎

The constants are explicit: $C_{1/2} \approx 0.942$, $C_1 = \sqrt{6}/(2\sqrt{\pi}) \approx 0.691$, and $C_2 \approx 0.425$. Once more the parameter exponent is b and the data exponent is $b/(1+b)$, and for the same reason: item k is learned once it has been seen, which happens when $Dp_k \gtrsim 1$, so the effective capacity after D examples is $N \asymp D^{1/(1+b)}$, and the error is the tail beyond it, $N^{-b} = D^{-b/(1+b)}$.[6]

[6] Cabannes et al. (2024) derive scaling laws of this kind for associative memories built from outer products of embeddings, where capacity and data limit the recall of Zipf-distributed associations together.

16.5 The solution: spending compute

If the loss has the additive form (16.1) and compute costs $C = \kappa ND$ (with $\kappa = 6$ for transformers), how should a budget C be split?

Theorem 16.10 (Compute-optimal allocation). *Let $A, B, \alpha_N, \alpha_D, \kappa > 0$, and for $C > 0$ minimize $\mathcal{L}(N,D) = \mathcal{L}_\infty + AN^{-\alpha_N} + BD^{-\alpha_D}$ over $N, D > 0$ subject to $\kappa ND = C$. Let $\nu_N = \frac{\alpha_D}{\alpha_N+\alpha_D}$, $\nu_D = \frac{\alpha_N}{\alpha_N+\alpha_D}$, and $G = \big(\frac{\alpha_N A}{\alpha_D B}\big)^{1/(\alpha_N+\alpha_D)}$. The minimizer is unique and equals*

$$N^\star = G\Big(\frac{C}{\kappa}\Big)^{\nu_N}, \qquad D^\star = G^{-1}\Big(\frac{C}{\kappa}\Big)^{\nu_D}. \tag{16.10}$$

At the minimum the two reducible terms are in the ratio $AN^{\star-\alpha_N} : BD^{\star-\alpha_D} = \alpha_D : \alpha_N$, and

$$\mathcal{L}(N^\star, D^\star) - \mathcal{L}_\infty = \frac{\alpha_N+\alpha_D}{\alpha_D}\, A\, G^{-\alpha_N}\Big(\frac{C}{\kappa}\Big)^{-\frac{\alpha_N\alpha_D}{\alpha_N+\alpha_D}}. \tag{16.11}$$

Proof. The plan is to eliminate D and minimize a strictly convex function of $u = \log N$. On the constraint, $D = C/(\kappa N)$, and the reducible loss is

$$\varphi(u) = Ae^{-\alpha_N u} + B\Big(\frac{\kappa}{C}\Big)^{\alpha_D} e^{\alpha_D u}.$$

Each term is strictly convex in u, and $\varphi(u) \to \infty$ as $u \to \pm\infty$, so φ has a unique minimizer, the unique zero of $\varphi'(u) = -\alpha_N AN^{-\alpha_N} + \alpha_D BD^{-\alpha_D}$ (with $N = e^u$ and $D = C/(\kappa N)$). The condition $\alpha_N AN^{-\alpha_N} = \alpha_D BD^{-\alpha_D}$ gives the ratio of the two terms. Substituting $D = C/(\kappa N)$ turns it into $N^{\alpha_N+\alpha_D} = \frac{\alpha_N A}{\alpha_D B}(C/\kappa)^{\alpha_D}$, which is the first formula of (16.10), and $D^\star = C/(\kappa N^\star)$ is the second, because $1 - \nu_N = \nu_D$. Finally the loss is $AN^{\star-\alpha_N}(1 + \frac{\alpha_N}{\alpha_D})$, and $N^{\star-\alpha_N} = G^{-\alpha_N}(C/\kappa)^{-\alpha_N\alpha_D/(\alpha_N+\alpha_D)}$. ∎

Four readings. The allocation exponents ν_N and ν_D depend only on the ratio of the exponents: the faster-decaying term gets the smaller share of the budget's growth. Parameters and data grow at the same rate exactly when $\alpha_N = \alpha_D$; otherwise the number of tokens per parameter, $D^\star/N^\star = G^{-2}(C/\kappa)^{(\alpha_N-\alpha_D)/(\alpha_N+\alpha_D)}$, drifts with the budget. The loss exponent in compute, $\frac{\alpha_N\alpha_D}{\alpha_N+\alpha_D}$, is half the harmonic mean of the two exponents, so it is smaller than both. And the constants A and B enter only G: the exponents of the allocation do not depend on them, but the number of tokens per parameter does, which is why it is so sensitive to the fit. With the rounded exponents $\alpha_N = 0.34$ and $\alpha_D = 0.28$ of Hoffmann et al. (2022), $\nu_N = 0.28/0.62 \approx 0.45$, $\nu_D \approx 0.55$, the tokens per parameter grow like $C^{0.10}$, and the compute-optimal loss falls like $C^{-0.154}$. A data exponent equal to the parameter exponent, closer to the refit of Besiroglu et al. (2024), gives $\nu_N = \nu_D = \frac{1}{2}$.

16.6 Reading the solution

Figure 16.1(a) compares Theorem 16.5 with simulations. Each fixed-size model follows a curve that falls like $1/D$ and then flattens at its approximation error $T(N)$: it is parameter-limited. The lower envelope of the curves, the best size at each D, is the data-limited law (16.6), with slope $-\frac{b}{1+b} = -\frac{1}{2}$ here; at $D = 2048$ the best size is $N = 96$, against $M_D = 101$ from Corollary 16.6. Below the envelope no model of the family can go, and above it each model wastes either parameters or data. The smooth law is thus the envelope of a family of curves that each saturate, which is the shape of the compute-efficient frontier in Kaplan et al. (2020).

Figure 16.1(b) shows the two exponents for ridge on all features: $D^{-b/(1+b)}$, here $D^{-1/2}$, with noise and optimal ridge, and D^{-b}, here D^{-1}, for the noiseless interpolator, as predicted by Proposition 16.7 and the deterministic equivalent. With noise, the local slope is steeper than $\frac{1}{2}$ at moderate D, because the unlearned signal adds to the noise in (16.8) and shrinks as D grows; for smaller b this transient lasts longer (Exercise 16.12).

WHAT SETS THE EXPONENTS. In both models the exponents are b for parameters and $b/(1+b)$ for data, and b is a property of the data: how the importance of directions or items decays. Neither the model class nor the spectrum exponent a appears, except through saturation when $b > 2a$ and through the time exponent b/a. This is the solvable version of the observation of Hestness et al. (2017) that better architectures shift learning curves without changing their exponents.[7] Three consequences can be checked. First, the data exponent is smaller than the parameter exponent, and determined by it: $\alpha_D = \alpha_N/(1+\alpha_N)$. Second, data that are less heavy-tailed (larger b) scale faster. Third, small exponents mean b close to 0, a nearly flat hierarchy: the frequencies of words, $p_k \propto k^{-s}$ with $s \approx 1$, correspond to $b = s - 1 \approx 0$, and a model limited by rare words would scale very slowly.

[7] A different route to the same conclusion: Sharma and Kaplan (2022) argue that a network regressing on a data manifold of intrinsic dimension d has a parameter exponent of about $4/d$.

EMERGENCE ON A SMOOTH CURVE. Figure 16.2(b) follows single quanta. Item k is known after D examples with probability $1-(1-p_k)^D \approx 1-e^{-Dp_k}$,

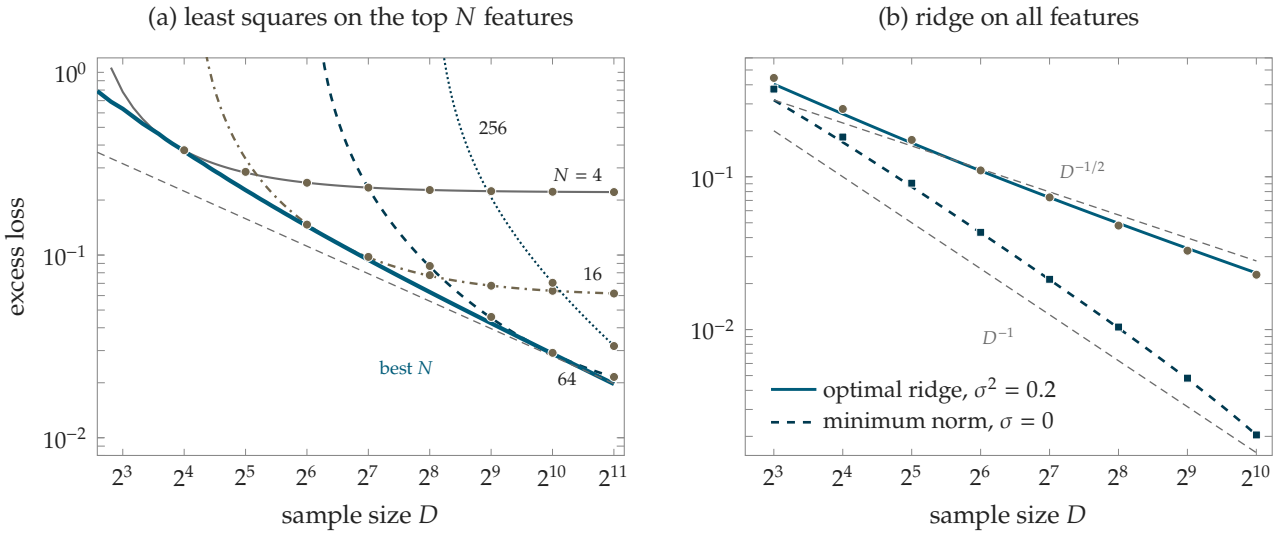


Figure 16.1. Learning curves on power-law features with $a = \frac{3}{2}$ and $b = 1$. (a) Least squares on the top $N \in \{4, 16, 64, 256\}$ of $P = 1024$ features, $\sigma^2 = 0.2$. Thin curves: Theorem 16.5; thick curve: its minimum over N; dashed: the asymptote (16.6) for $P = \infty$. Points: averages over 150 training sets (seed 1617), scored by their exact test loss; the largest deviation is 1.4 standard errors. Points with $D < 2N + 4$, where the simulated average has a large variance, are omitted. (b) Ridge on all $P = 4000$ features: minimum-norm interpolation without noise (squares), and ridge with the λ that minimizes the deterministic equivalent, $\sigma^2 = 0.2$ (circles); curves: the deterministic equivalent (16.8); 60 training sets per point. The equivalent is asymptotic: at $D \le 16$ the simulations lie up to 18% above it, and from $D = 64$ on within 4%.

which rises from 0.1 to 0.9 as Dp_k grows from 0.105 to 2.30, a factor of about 22 in data, at $D \approx 1/p_k$. Every skill turns on over the same window in $\log D$, at a position set by its frequency, while their sum, the loss, is a power law without a feature. A downstream task that needs a few specific rare quanta shows a jump; the loss does not. This is the quanta picture of emergence of Michaud et al. (2023).[8]

[8] Wei et al. (2022) call an ability emergent if it is absent in smaller models and present in larger ones, so that it cannot be predicted by extrapolating the smaller models.

ALLOCATION IN THE SOLVABLE MODELS. The form (16.1) is additive; Theorem 16.5 is not, because the variance $\sigma^2 N/D$ couples N and D. Minimizing $cN^{-b} + \sigma^2 N/D$ under $6ND = C$ gives $N^\star \propto C^{1/(2+b)}$ and $D^\star \propto C^{(1+b)/(2+b)}$: with $b = 0.34$, parameters grow like $C^{0.43}$ (Exercise 16.11). For a model trained by gradient flow on fresh data, whose cost is proportional to N times the training time, Exercise 16.6 gives $N^\star \propto C^{1/(1+a)}$ instead, and the spectrum exponent enters the allocation.

16.7 What the model gets wrong

WHAT WAS THROWN AWAY. *Feature learning.* The features, their spectrum, and the target's decay in their basis are fixed in advance, so the exponent b is an input; a network chooses its representation, and with it the basis in which importance is measured. *Optimization.* The estimators are closed-form; compute enters only through the accounting $C = 6ND$, not through the dynamics of stochastic gradient descent, its noise, its step size, or its

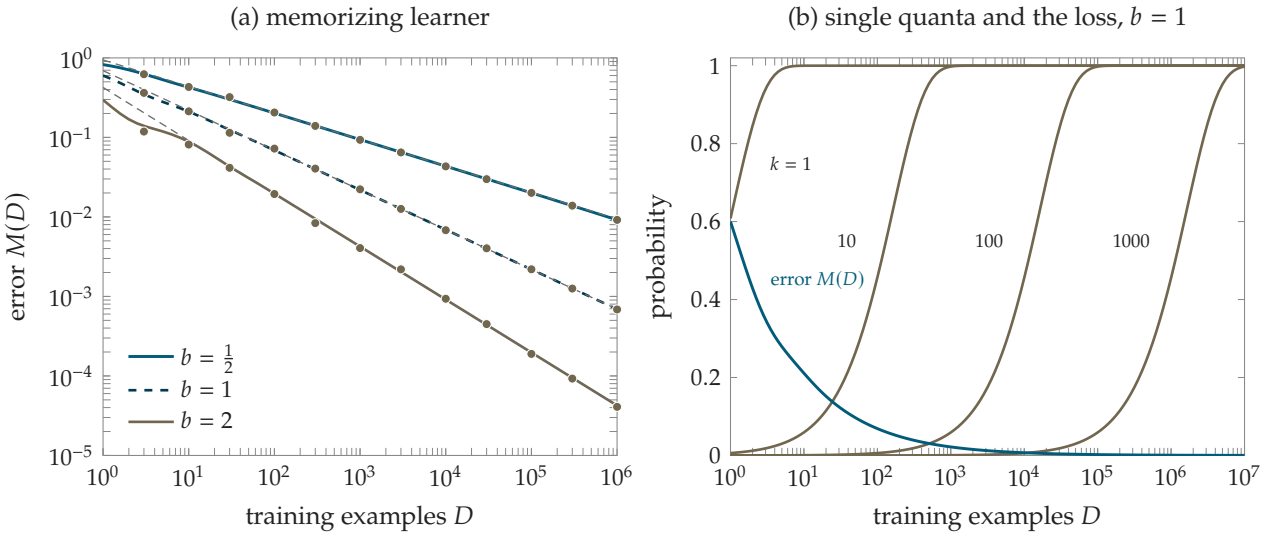


Figure 16.2. The quanta model. (a) Error of the memorizing learner for $b \in \{\frac{1}{2}, 1, 2\}$, items truncated at $K = 10^7$. Curves: the exact sum (16.9); dashed: the asymptote $C_b D^{-b/(1+b)}$ of Theorem 16.9 for the untruncated law; points: 20 simulated training sets per point (seed 1618), each scored by the exact mass of its unseen items, within 2.5 standard errors of the curves. (b) For $b = 1$: the probability that item $k \in \{1, 10, 100, 1000\}$ has been seen (sigmoids, each spanning a factor of about 22 in D from 0.1 to 0.9), and the error $M(D)$, which is smooth.

schedule. *Architecture.* Only the number of parameters matters, and they are spent on the top features in order. *The irreducible loss.* White noise of variance σ^2 stands for the entropy of the data, which in language is structured. *Repetition.* Each example is fresh and used once. *Composition.* Quanta are independent and are learned from one example; real skills share parts and need many examples (Exercise 16.8).

WHERE IT BREAKS. *The exponents are not derived.* The models say how a decay of importance becomes a decay of loss, not what the decay is. For kernels it can be measured: Spigler et al. (2020) estimated learning-curve exponents from the kernel spectra of MNIST and CIFAR-10 that came close to the measured ones. For trained networks there is no such recipe, and feature learning changes the exponents: in the solvable model of Bordelon et al. (2025), feature learning nearly doubles the exponent in time and compute for targets outside the reproducing-kernel space of the initial tangent kernel, and leaves it unchanged for easier targets.

The ratio of the exponents. Both models predict $\alpha_D = \alpha_N/(1 + \alpha_N) < \alpha_N$. With $\alpha_N = 0.076$, this gives 0.071, against the $\alpha_D \approx 0.095$ of Kaplan et al. (2020); with $\alpha_N = 0.34$ it gives 0.25, against the 0.28 of Hoffmann et al. (2022), and the refit of Besiroglu et al. (2024) has a data exponent that is larger, not smaller. The comparison is not clean, since the exponents of Kaplan et al. (2020) were measured in different regimes (convergence in N, early stopping in D), but the prediction fails in the direction that matters: data are worth more, relative to parameters, than the hierarchy alone allows. A learner that improves on many items from each example, rather than on one, would do this; memorization cannot.

The fits. The exponents of (16.1) are fitted, and the fits are fragile: the published estimates of Hoffmann et al. (2022) did not survive refitting (Besiroglu et al. 2024), and the allocations of Kaplan et al. (2020) and Hoffmann et al. (2022) disagree because of parameter counting, scale, warmup, and tuning (Pearce and Song 2024; Porian et al. 2024). Theorem 16.10 is exact for the additive form, but the solvable models are not additive, and the tokens per parameter depend on constants that the models do not predict.

Emergence. The quanta model makes each skill turn on over a factor of about 22 in data. Abilities that appear abruptly (Wei et al. 2022) may be sharper than this, or may look sharp because of the metric (Schaeffer et al. 2023); the model cannot tell which, since it has no notion of a task built from several quanta.

The data distribution. The law in D is a law of random sampling. If the training set contains each item once instead of sampling with repetition, the memorizing learner's error after D examples is $\sum_{k>D} p_k \asymp D^{-b}$: curation raises the data exponent from $b/(1+b)$ to b (Exercise 16.10). Sorscher et al. (2022) showed in theory and experiment that pruning data can beat power-law scaling, and Muennighoff et al. (2023) that repeated tokens are worth less than fresh ones after a few epochs; neither effect exists in a model of fresh, independent samples. And real curves bend: a spectrum that is a power law only over a range gives a broken power law (Caballero et al. 2023), with plateaus where the range ends (Maloney et al. 2022).

FINE STRUCTURE. Maloney et al. (2022) solved random-feature regression on data with a power-law latent spectrum: nonlinear random features extend the power law, the test loss scales in both N and D, and it plateaus when either exceeds the size of the latent spectrum. Bahri et al. (2024) separate variance-limited scaling, with exponent 1, from resolution-limited scaling with data-dependent exponents, in both N and D. Dynamics: Bordelon et al. (2024) solved a random-feature model trained by gradient descent, in which time and model size have different exponents and the gap between training and test loss builds up through reuse of data; Paquette et al. (2024) found four phases and three subphases of compute-optimal scaling for one-pass stochastic gradient descent in the plane of data and target complexity, with boundaries set by model capacity, optimizer noise, and the embedding of the features; and Lin et al. (2024) proved that for sketched linear regression trained by one-pass stochastic gradient descent the reducible loss is $\Theta(N^{-(a-1)} + D^{-(a-1)/a})$ for a Gaussian prior with $\lambda_j \asymp j^{-a}$, our $b = a - 1$, with no variance term growing with N: implicit regularization removes it. Sharma and Kaplan (2022) derive exponents of about $4/d$ from the intrinsic dimension d of the data manifold, and Cabannes et al. (2024) treat associative memories. Open: a model in which the exponent b of the learned representation is an output, computed from the data and the architecture, and checked against language models.

16.8 Model card

MODEL CARD	
PHENOMENON	Test loss falls as a power law in parameters, data, and compute, with small exponents; compute-optimal training scales parameters and data together.
MODEL	Least squares and ridge on independent Gaussian features with target energy $j^{-(1+b)}$; a memorizing learner on quanta with Zipf frequencies $p_k \propto k^{-(1+b)}$; an additive loss with $C = 6ND$.
SOLUTION	Exact learning curve $(\sigma^2 + T(N))\frac{D-1}{D-N-1}$ (Theorem 16.5); exponents b and $b/(1+b)$ (Corollary 16.6 and Proposition 16.7); Zipf curve with constant C_b (Theorem 16.9); allocation (Theorem 16.10).
PREDICTS	Optimal size $N \propto D^{1/(1+b)}$; data exponent $b/(1+b)$ with noise, b without; time exponent b/a; allocation exponents $\frac{\alpha_D}{\alpha_N+\alpha_D}$ and $\frac{\alpha_N}{\alpha_N+\alpha_D}$; skills emerging over a factor 22 in data.
GETS WRONG	Exponents are inputs; the predicted $\alpha_D = \alpha_N/(1+\alpha_N)$ is below fitted language-model data exponents; no feature learning, optimizer, repetition, or curation.
SUCCESSORS	Random-feature models; dynamical models and the 4+3 phases; one-pass stochastic gradient descent; feature learning that changes exponents.

This chapter closes the thread of spectra that began with the Marchenko–Pastur law of Chapter 1: Theorem 16.5 is the weak-features corollary of that chapter with an anisotropic design, and the deterministic equivalent is its ridge theorem with a renormalized penalty. The spectral bias of kernel gradient descent from Chapter 6, which learns eigenfunctions in order of their eigenvalues, becomes here a scaling law in time with exponent b/a, and the staged learning of Chapter 2, one singular mode after another, is the mechanism that the model of size N assumes.

The quanta connect smooth loss curves with the sudden transitions of earlier chapters: the plateaus of the committee machine (Chapter 8), hidden progress on parities (Chapter 9), grokking (Chapter 10), and the formation of induction heads (Chapter 14) are each one quantum, abrupt on its own scale, and a power law is what many such transitions look like when added up with Zipf weights. The capacity to hold more items than dimensions is the subject of Chapter 11.

Exercises

16.1 CONSOLIDATION. (Tails and constants.) (a) Show that $\sum_{k>N} p_k \sim N^{-b}/(b\zeta(1+b))$ and that $T(N) = N^{-b}/b + O(N^{-b-1})$ in the power-law case. (b) Show that $C_1 = \sqrt{6}/(2\sqrt{\pi})$, and compute $C_{1/2}$ and C_2 to three digits. (c) Show that $\sum_{k\geq 1} p_k e^{-cp_k}$ is within $2/(ec)$ of $C_b c^{-b/(1+b)}$ for every $c > 0$, and deduce that $D^{b/(1+b)}M(D) - C_b = O(D^{-1/(1+b)})$.

16.2 CONSOLIDATION. (Noiseless learning curves.) Under Assumption 16.2

with $\sigma = 0$ and $T(N) \sim cN^{-b}$, show that $\min_{N \le D-2} \mathbb{E}\mathcal{L}(\hat{w}_N) \sim c\,\frac{(1+b)^{1+b}}{b^b} D^{-b}$, attained at $N \sim \frac{b}{1+b} D$. Check numerically that for $b = 1$ and $T(N) = \sum_{j>N} j^{-2}$ the product $D \cdot \min_N \mathbb{E}\mathcal{L}(\hat{w}_N)$ is 4.000 at $D = 1000$. Why does the optimal model grow linearly with D here?

16.3 CONSOLIDATION. (Chinchilla arithmetic.) (a) Derive (16.10) with a Lagrange multiplier instead of the substitution $u = \log N$. (b) For $\alpha_N = 0.34$ and $\alpha_D = 0.28$, compute ν_N, ν_D, the compute exponent of the loss, and the exponent of $D^\star/N^\star$. (c) Show that at the optimum the parameter term carries the fraction ν_N of the reducible loss, and that a mistaken allocation $N = \rho N^\star$, $D = D^\star/\rho$ raises the reducible loss by the factor $\nu_N \rho^{-\alpha_N} + \nu_D \rho^{\alpha_D}$. Evaluate it for $\rho = 4$.

16.4 CONSOLIDATION. (The isotropic case.) Take $\lambda_j = 1$ for $j \le P$ and $P/D \to \gamma$. Show that the equation for κ in the deterministic equivalent becomes $\kappa = \lambda + \gamma\kappa/(1+\kappa)$, and that with $v = \kappa/(1+\kappa)$ it is the equation $\gamma v^2 + (1 - \gamma + \lambda)v - \lambda = 0$ of Chapter 1. Check that (16.8), in which γ becomes $\gamma(1-v)^2$, gives the bias $r^2(v - \lambda v')$ and the variance $\sigma^2 \gamma v'$ of Theorem 1.11, with $r^2 = \sum_j \lambda_j w_{\star,j}^2$. *(Hint: differentiate the quadratic equation in λ, and show that $v' = (1-v)^2/(1-\gamma(1-v)^2)$.)*

16.5 CONSOLIDATION. (Exact constants for ridge.) In Proposition 16.7, show that as $\lambda \to 0$ the bias is asymptotic to $\lambda^{b/a} \frac{1}{a} B\big(\frac{2a-b}{a}, \frac{b}{a}\big)$ and the variance sum to $\lambda^{-1/a} \frac{1}{a} B\big(\frac{1}{a}, 2 - \frac{1}{a}\big)$, with B the Beta function, and compute the optimal λ and the constant in $\min_\lambda \mathcal{E}_{\text{seq}} \sim c'(\sigma^2/D)^{b/(1+b)}$. *(Hint: $\int_0^\infty \frac{u^{s-1}}{(1+u^a)^2}\,\mathrm{d}u = \frac{1}{a} B\big(\frac{s}{a}, 2 - \frac{s}{a}\big)$; use Lemma 16.8-type comparisons on the two monotone parts of each summand.)*

16.6 VARIATION. (Training time.) Gradient flow on $\frac{1}{2}\mathcal{L}$ for the model of size N, from $w = 0$, gives $w_j(t) = (1 - e^{-\lambda_j t}) w_{\star,j}$, so the excess loss is $\sum_{j \le N} j^{-(1+b)} e^{-2\lambda_j t} + T(N)$ in the power-law case. (a) Show that for $t \ge 1$ and $N \to \infty$ the first sum is of order $t^{-b/a}$, and that the excess loss is of order $t^{-b/a} + N^{-b}$ when $N \ge t^{1/a}$ and of order N^{-b} when $N \le t^{1/a}$. (b) If one unit of compute buys one unit of time for one feature, $C = Nt$, show that the compute-optimal size is $N \asymp C^{1/(1+a)}$ and the loss is of order $C^{-b/(1+a)}$. Compare with Bordelon et al. (2024) and Paquette et al. (2024), where the model is also trained with finite data.

16.7 VARIATION. (Saturation.) In Proposition 16.7, let $b > 2a$. Show that the bias is of order λ^2, that the best excess loss is of order $(\sigma^2/D)^{2a/(1+2a)}$, and that least squares on the top N features, with N chosen well, does better. What happens at $b = 2a$?

16.8 VARIATION. (Several sightings.) Suppose an item is learned only after it has been seen m times. Show that the error is $\sum_k p_k \Pr\big(\mathrm{Bin}(D, p_k) < m\big) \sim C_{b,m} D^{-b/(1+b)}$ with

$$C_{b,m} = \frac{1}{(1+b)\,\zeta(1+b)^{1/(1+b)}} \sum_{i=0}^{m-1} \frac{\Gamma\big(i + \frac{b}{1+b}\big)}{i!}.$$

For $b = 1$, check numerically that $C_{1,2} \approx 1.036$ and $C_{1,3} \approx 1.296$. The exponent does not change; what does?

16.9 VARIATION. (Capacity and data together.) A learner of capacity N trained on D examples holds the items it has seen among $1, \dots, N$. (a) Show that its error is $\sum_{k>N} p_k + \sum_{k \le N} p_k (1 - p_k)^D$. (b) Show that this is of order $N^{-b} + D^{-b/(1+b)}$ uniformly in $N, D \ge 1$, and that the crossover lies at $N \asymp D^{1/(1+b)}$. (c) With $C = 6ND$, find the compute-optimal allocation and compare it with Theorem 16.10 for $\alpha_N = b$ and $\alpha_D = b/(1+b)$.

16.10 VARIATION. (Curated data.) (a) In Assumption 16.3, a curator shows the learner the items $1, \dots, D$ once each. Show that the error is $\sum_{k>D} p_k \sim D^{-b}/(b\zeta(1+b))$. (b) Now the curator removes the repetitions from a random sample of size D', keeping the D distinct items it contains. Express the error in terms of D', and show that the exponent in D' is still $b/(1+b)$ while the exponent in the number D of kept examples is b. Relate this to Sorscher et al. (2022).

16.11 VARIATION. (Compute for least squares.) Minimize $cN^{-b} + \sigma^2 N/D$ under $6ND = C$, and show that $N^\star \propto C^{1/(2+b)}$, $D^\star \propto C^{(1+b)/(2+b)}$, and that the loss falls like $C^{-b/(2+b)}$. Why does the variance term push the allocation toward data?

16.12 SIMULATION. (Ridge exponents.) With $a = \frac{3}{2}$, $\sigma^2 = 0.2$, and $P = 10^6$ features, compute the deterministic equivalent of optimally tuned ridge at $D = 10^3$ and $D = 10^4$ for $b \in \{\frac{1}{2}, 1, 4\}$, and the slope $-\log_{10}$ of their ratio. Observe slopes of about 0.39, 0.52, and 0.74, against $\frac{1}{3}$, $\frac{1}{2}$, and the saturated $\frac{3}{4}$. Explain why the slope for $b = \frac{1}{2}$ is too steep, using the effective noise $\sigma^2 + R$ in (16.8), and check that the sequence model (16.7), whose noise does not grow with R, gives about 0.34. Confirm one point by simulating ridge at $D = 10^3$ with 4000 features.

16.13 SIMULATION. (Fitting a law.) For $b = 1$, compute the local slope $-\,\mathrm{d}\log M/\,\mathrm{d}\log D$ of the exact Zipf curve at $D = 3$, 10, 100, and 10^4; observe about 0.48, 0.44, 0.50, and 0.50. Then draw D samples for $D = 10, 20, \dots, 320$, record the error of the memorizing learner, and fit $\log M$ against $\log D$ by least squares on 20 independent repetitions. How far is the fitted exponent from $\frac{1}{2}$, and how wide is the spread? Relate the result to the replication of Besiroglu et al. (2024).

16.14 RESEARCH. (Predicting an exponent.) Spigler et al. (2020) predicted kernel learning-curve exponents from kernel spectra of image data. Open: for a network that learns its features, define and measure the analogue of b in its learned representation, and predict its parameter and data exponents; a solvable model in which feature learning changes b during training, as in Bordelon et al. (2025), and which yields $\alpha_D > \alpha_N$, would be a first step.

Notes and References

Learning curves are older than scaling laws. In the statistical mechanics of learning, Amari et al. (1992) classified the asymptotic learning curves of parametric learning machines into four types, among them D^{-1} for a deterministic machine, a noiseless teacher, and a unique correct parameter, and $D^{-1/2}$ for a deterministic machine and a noisy teacher; these exponents do not depend on the data, because the model contains the target. The exponents of this chapter, which depend on the data, appear when the target is not representable by a finite model. In statistics, Caponnetto and De Vito (2007) proved the minimax rates of kernel ridge regression under a source condition, the decay of the target in the eigenbasis, and a capacity condition, the decay of the eigenvalues; Proposition 16.7 is the sequence-model version of their rate. Bordelon et al. (2020) and Spigler et al. (2020) brought these rates to wide networks and real data, and Cui et al. (2021) mapped the crossover from the noiseless rate D^{-b} to the noisy one.

Hestness et al. (2017) established power laws in data across domains for deep networks, Kaplan et al. (2020) in parameters, data, and compute for language models, and Hoffmann et al. (2022) the compute-optimal allocation that is used today, after correcting the earlier one. Theorem 16.10 is their allocation formula for the parametric fit, which they write as $N_{\text{opt}} \propto C^a$ and $D_{\text{opt}} \propto C^b$ with $a = \beta/(\alpha+\beta)$ and $b = \alpha/(\alpha+\beta)$. Besiroglu et al. (2024) is an unrefereed preprint, and our account of it rests on its abstract and its reported refit ($\alpha_N \approx 0.35$, $\alpha_D \approx 0.37$).

Solvable models of the exponents came in several waves: memorization of Zipf-distributed data (Hutter 2021) and quanta (Michaud et al. 2023); random features (Bahri et al. 2024; Maloney et al. 2022); dynamics (Bordelon et al. 2024; Lin et al. 2024; Paquette et al. 2024); and feature learning (Bordelon et al. 2025). The exact finite-sample curve of Theorem 16.5 is an elementary consequence of the inverse Wishart mean; we did not find it stated in this form in the scaling literature.

TRANSLATING THE LITERATURE. Hoffmann et al. (2022) write E, α, and β for our $\mathcal{L}_\infty$, α_N, and α_D, and a, b for our allocation exponents ν_N, ν_D, which are not our spectral exponents. Kaplan et al. (2020) use our α_N and α_D, but without an irreducible term, and count non-embedding parameters. Lin et al. (2024) write M for our N and N for our D; their spectral exponent is our a, and their Gaussian prior on the target makes our $b = a - 1$, so their rate $M^{-(a-1)} + N^{-(a-1)/a}$ has our two exponents b and $b/(1+b)$. Hutter (2021) writes $\theta_i \propto i^{-(1+\alpha)}$ and n for our $p_k \propto k^{-(1+b)}$ and D. In the kernel literature (Caponnetto and De Vito 2007; Cui et al. 2021) the capacity and source conditions are written with other exponents; with $\lambda_j \asymp j^{-a}$ and $\lambda_j w_{\star,j}^2 \asymp j^{-(1+b)}$, the noisy rate is always $D^{-b/(1+b)}$ below saturation, and translating a paper's exponents into a and b is the quickest way to compare results.

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Epilogue

Fine structure: what the solvable models get wrong together

Sixteen models, each solved in its own chapter, each with its own list of failures. Read one at a time, they look like sixteen separate tricks. Read side by side, they share a small number of mechanisms, and they also share their failures. A failure that one model has is a fact about that model. A failure that every model has is a fact about the method, and it is the fine structure that the next generation of models has to explain. This epilogue collects both, lists phenomena that still have no hydrogen atom, and ends with a recipe for building one.

Mechanisms that recur

The table below lists the mechanisms that carry more than one chapter, the chapters in which they do the main work, and what each one computes. None of them is new to this book; most are older than deep learning. What is striking is how few there are.

MECHANISM	CHAPTERS	WHAT IT COMPUTES
Escape from a saddle	2, 3, 7, 8, 9, 10, 13, 14	The length of a plateau: the logarithm of the initial distance from the saddle, divided by the unstable rate
Conservation laws from rescaling symmetries	2, 3, 13, 15	Which minimizer training reaches, how sharp it is, and which initializations make the dynamics closed
Reduction to order parameters	1, 5, 7, 8, 13	A closed recursion or ordinary differential equation in a few overlaps, from rotational invariance of Gaussian data
Harmonic analysis	5, 6, 7, 9, 10	What the gradient sees at each degree: Hermite coefficients, spherical harmonics, characters of $\mathbb{Z}_p$ and of the cube
Margins as attractors	4, 10, 12	The direction that an exponentially tailed loss selects after the training error reaches zero
Spectra of random and structured matrices	1, 6, 13, 16	Risk as a functional of a spectrum: Marchenko–Pastur, kernel eigenvalues, power-law tails
A limit that closes the equations	1, 5, 6, 7, 8, 16	Width, dimension or sample size taken to infinity in a fixed ratio or a fixed order

The first row deserves a comment, because it is the most reliable prediction in the book. Near a saddle the dynamics is linear, and an unstable mode with rate λ grows like $e^{\lambda t}$. Starting at distance ε, it needs a time $\lambda^{-1}\log(1/\varepsilon)$ to reach order one. This is the normal-mode theorem of the prologue with the sign of one eigenvalue reversed. It gives the half-learning time $\log(s/a_0)/(2s)$ of a two-layer linear network (Chapter 2), the time $\log(1/\alpha)/(2|w_{\star,i}|)$ at which a diagonal network learns coordinate i (Chapter 3), the $\log d$ plateau of the committee machine (Chapter 8), the delay $\lambda^{-1}\log(\alpha/\alpha_c)$ of grokking under weight decay (Chapter 10), and the escape times of linear attention from small initialization (Chapters 13 and 14). The single-index model and the

sparse parity (Chapters 7 and 9) are the same story at a degenerate saddle, where the unstable direction grows like a power rather than an exponential, and the logarithm becomes a power of the dimension. Whenever training shows a plateau followed by a drop, this is the first model to try, and the two numbers to measure are the initial distance and the unstable rate.

The second row is less visible but as useful. A rescaling symmetry of the parameterization, such as $(W_{\ell+1}, W_\ell) \mapsto (W_{\ell+1}A^{-1}, AW_\ell)$ for a linear network or $(u, v) \mapsto (cu, v/c)$ for a product, gives a conserved quantity under gradient flow: the balancedness of Chapter 2, the products $u_i v_i = \alpha^2$ of Chapter 3. Under gradient descent with a finite step the same quantity is no longer conserved but evolves by an exact law, and in Chapter 15 that law is what forces the catapult. In every case the conserved or slowly varying quantity selects among minimizers that the loss cannot tell apart.

The third and fourth rows are the reason the models are solvable at all. Gaussian data are rotationally invariant, so a high-dimensional state collapses to a few overlaps; uniform data on a group are invariant under translation, so the right basis is the group's characters. The fifth row is the mechanism of implicit bias in classification: once the data are separated, an exponentially tailed loss is dominated by its smallest margins, and the trajectory follows the maximum-margin problem. Three chapters of very different appearance, on logistic regression, modular addition and the last layer of a classifier, are three solutions of that problem.

What the models get wrong together

Each chapter ends with what its model gets wrong. Putting the lists side by side, six failures recur.

The data are isotropic, independent, or uniform. Most of the sixteen models assume Gaussian inputs with identity covariance, uniform inputs on a group or a cube, or independent features. This is what makes the order parameters close. Real data have structure at every scale, and in the models that structure enters, if at all, only through a spectrum (Chapters 1 and 16) or through a target that is given rather than learned. Where a chapter tests what happens without isotropy, the prediction changes: anisotropic inputs change the information exponent (Chapter 7), and structured inputs change or remove the plateau of the committee machine (Chapter 8).

One mechanism at a time. Every model isolates one phenomenon and removes everything that could compete with it. In a trained network the mechanisms run together. The delay of grokking in Chapter 10 had to be modelled by a different network from the one that explains the Fourier circuit; the induction head of Chapter 14 is one circuit among many that form during the same phase change; implicit bias, edge-of-stability dynamics and feature learning act on the same weights at the same time. No model in this book solves an interaction between two of its own mechanisms.

Limits in a convenient order. Solvability comes from a limit, and the limit is usually taken in the order that makes the equations close: width to infinity at fixed depth (Chapters 5 and 6), dimension to infinity before time

(Chapters 7 and 8), time to infinity for a margin (Chapter 4). The interesting behavior often lives where two limits compete. Signal propagation fails when depth is comparable to width; the committee machine's plateau length is set by fluctuations of order $d^{-1/2}$ that its own ordinary differential equation omits; the margin converges at a rate in $1/\log t$ that no training run reaches.

Gradient flow, or plain gradient descent. Almost every theorem is proved for gradient flow, online stochastic gradient descent with vanishing steps, or full-batch gradient descent with a small step. Only Chapter 15 takes a large step seriously, and no chapter treats an adaptive optimizer. Yet the chapters themselves record that the optimizer changes the answer: step size and label noise change the implicit bias of diagonal networks (Chapter 3), steepest descent and Adam change the margin that classification selects (Chapter 4), and reusing samples beats the information exponent (Chapter 7).

Special initializations. Several solutions require an initialization chosen to make the dynamics solvable: aligned and balanced for deep linear networks, symmetric for the committee machine and the sparse parity, small for diagonal networks and attention. The chapters show that random initializations keep the scaling and change the constants, but the exact solutions are exact only on a set of measure zero.

Minimal architectures. The models are linear, diagonal, two-layer, or one layer of attention. Residual connections, normalization layers and softmax attention appear only in the sections on what the models get wrong. The transformer simulation of Chapter 14 did not reach the bigram level with a standard linear readout and needed a modified one.

The book's own simulations are a fair test of these failures, because every chapter compares its model with a finite simulation, and several of them disagreed with the model in an instructive way. A small ReLU network trained in Chapter 12 collapsed only partially. The networks of Chapter 10 are far from the maximum-margin solutions that explain their Fourier features. The fitted exponents of language models violate the relation between data and parameter exponents that both models of Chapter 16 predict. And the transformers that match least squares on regression prompts cannot be performing the single gradient step that Chapter 13 proves optimal for one layer.

The pattern across these comparisons is consistent. What the models predict well are scalings and orders of events: a time that grows like $\log(1/\alpha)$ or a power of d, a threshold at $\eta\Lambda = 2$, the order in which modes are learned, the geometry of an optimum. What they predict badly are constants, and which mechanism is active in a given network. This is the same division as in physics, where the hydrogen atom gets the structure of the spectrum right and the fine structure wrong, and where the corrections are small. In deep learning the corrections are not always small.

Missing hydrogen atoms

Several phenomena that matter in practice have no model in this book, either because none exists or because the existing ones do not yet pass the four tests of the prologue.

Adaptive optimizers. Almost every large network is trained with Adam (Kingma and Ba 2015) or a relative of it. Sign descent and steepest descent in other norms have partial theories (Chapter 4), but there is no solvable model that predicts when Adam trains a network that gradient descent cannot, or the step-size schedules used in practice.

Progressive sharpening. The model of Chapter 15 explains why gradient descent at a large step does not diverge, but not why the sharpness rises to $2/\eta$ in the first place, which Cohen et al. (2021) observed in many networks. A model that produces sustained sharpening from its own dynamics is missing.

Normalization layers. Batch normalization (Ioffe and Szegedy 2015) and layer normalization (Ba et al. 2016) change the geometry of the loss, make it invariant to the scale of the weights, and interact with weight decay and the step size. They appear in this book only as things the models leave out.

Data mixtures and curricula. Real training sets are mixtures of sources with different difficulty and redundancy, and the order in which examples arrive can matter (Bengio et al. 2009). The models here draw every example from one fixed distribution.

Learning from feedback. Reinforcement learning from human preferences (Christiano et al. 2017; Ouyang et al. 2022) trains a pretrained model on a reward that is itself learned. Its dynamics, including how far the model moves from its pretrained state, have no solvable model of the kind collected here.

Chain of thought. Prompting a model to write intermediate steps improves its answers on reasoning tasks (Wei et al. 2022). The sparse parity of Chapter 9 suggests why intermediate steps can help a learner, since they break a high-degree target into low-degree pieces, but a model that predicts the size of the effect in a trained transformer does not exist yet.

Long-context retrieval. Language models use information at the start and end of a long input better than information in the middle (Liu et al. 2024). The models of Chapters 13 and 14 treat every position in the prompt alike.

A recipe for a new hydrogen atom

The chapters suggest a procedure, which a reader can apply to each of the phenomena above.

(i) *Choose a phenomenon with a number.* An exponent, a time scale, a threshold or a geometry, observed robustly in several networks. Without a number the model cannot fail, and a model that cannot fail cannot teach.

(ii) *Find the symmetry.* Look for the invariance that reduces the state to a few variables: rotations of Gaussian inputs, translations on a group,

a rescaling of the parameters. If no symmetry is visible, build it into the data.

(iii) *Choose the limit and the initialization.* Take the limit (dimension, width, time, initialization scale) and choose the starting point that makes the reduced equations close. Write down what this choice throws away; it will be the first section of "What the model gets wrong".

(iv) *Solve the smallest instance completely.* Find the saddle, the conserved quantity or the margin problem. Prove the main theorem, and label as heuristic whatever you cannot prove.

(v) *Predict a number and simulate with no fitted constant.* Compare the prediction with a finite simulation of the model, and then with a network that the model does not describe. The disagreement with the second is the result.

(vi) *Break one assumption on purpose.* Change the data, the optimizer, the architecture or the limit, one at a time, and solve again. This is what the variation exercises of every chapter ask for, and it is how one model becomes the next.

(vii) *Write the model card.* If the six lines cannot be filled honestly, in particular the line on what the model gets wrong, the model is not ready.

The hydrogen atom became the foundation of atomic physics not because it was right, but because every way in which it was wrong pointed to something real: relativity, the electron's spin, the quantized field. The models in this book are wrong in ways that are known and specific. The hope is that their failures, studied as carefully as their solutions, point to what deep learning theory needs next.

Notes and References

The table of mechanisms condenses material from the chapters, where each mechanism is proved and referenced. The observation that the time to escape a saddle scales like $\log(1/\varepsilon)/\lambda$ is classical in the theory of dynamical systems, and its role in the chapters is discussed in the notes to Chapters 2, 3 and 8. Adam is due to Kingma and Ba (2015); batch and layer normalization to Ioffe and Szegedy (2015) and Ba et al. (2016); curriculum learning was named by Bengio et al. (2009); learning from human preferences is due to Christiano et al. (2017) and was applied to language models by Ouyang et al. (2022); chain-of-thought prompting is due to Wei et al. (2022); the position dependence of long-context retrieval was measured by Liu et al. (2024); progressive sharpening was named by Cohen et al. (2021).

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A Toolkit

A

The mathematics the models share

IN THIS CHAPTER. Five pieces of mathematics recur across the models of this book: Gaussian integration by parts and Hermite polynomials, which turn expectations over Gaussian data into power series in an overlap; spherical harmonics, which do the same on the sphere; the Marchenko–Pastur law for the spectrum of a sample covariance matrix; martingale inequalities and the method of ordinary differential equations, which reduce a stochastic recursion in high dimension to a deterministic flow; and Fourier analysis on finite groups. This appendix states what the chapters use and proves what can be proved in a page.

The appendix is a reference, not a chapter to be read in order. Hermite polynomials (Section A.1) are used in Chapters 5 to 8; spherical harmonics (Section A.2) in Chapter 6; the Marchenko–Pastur law (Section A.3) in Chapters 1 and 16; martingales and the ODE method (Section A.4) in Chapters 7 and 8; Fourier analysis (Section A.5) in Chapters 9 and 10; and concentration inequalities (Section A.6) throughout.

A.1 Gaussian integration by parts and Hermite polynomials

Throughout this section z is a standard Gaussian random variable, with density $\varphi(z) = (2\pi)^{-1/2} e^{-z^2/2}$, and $L^2(\gamma)$ is the space of functions $f : \mathbb{R} \to \mathbb{R}$ with $\mathbb{E}[f(z)^2] < \infty$, with inner product $\langle f, g \rangle = \mathbb{E}[f(z)g(z)]$.

Theorem A.1 (Gaussian integration by parts). *Let $f : \mathbb{R} \to \mathbb{R}$ be absolutely continuous with $\mathbb{E}|f'(z)| < \infty$. Then $\mathbb{E}|z\,f(z)| < \infty$ and*

$$\mathbb{E}[z\,f(z)] = \mathbb{E}[f'(z)].$$

More generally, let $x \sim \mathcal{N}(0, \Sigma)$ in $\mathbb{R}^d$ and let $f : \mathbb{R}^d \to \mathbb{R}$ be continuously differentiable with $\mathbb{E}|f(x)|$ and $\mathbb{E}\|\nabla f(x)\|$ finite. Then $\mathbb{E}[x\,f(x)] = \Sigma\,\mathbb{E}[\nabla f(x)]$.

PROOF. The plan is to write f' as an integral against the Gaussian tail and exchange the order of integration. The density satisfies $\varphi'(u) = -u\varphi(u)$, so $\varphi(t) = \int_t^\infty u\varphi(u)\,du$ for $t \geq 0$ and $\varphi(t) = \int_{-\infty}^t (-u)\varphi(u)\,du$ for $t \leq 0$. On the positive half-line we substitute the first formula and exchange the order of integration, which Fubini's theorem allows because the same computation

with $|f'|$ in place of f' gives the finite value $\int_0^\infty |f'|\varphi$:

$$\int_0^\infty f'(t)\varphi(t)\,\mathrm{d}t = \int_0^\infty f'(t)\int_t^\infty u\varphi(u)\,\mathrm{d}u\,\mathrm{d}t = \int_0^\infty u\varphi(u)\big(f(u)-f(0)\big)\,\mathrm{d}u.$$

The negative half-line gives in the same way $\int_{-\infty}^0 f'\varphi = \int_{-\infty}^0 u\varphi(u)(f(u)-f(0))\,\mathrm{d}u$. Adding the two, $\mathbb{E}[f'(z)] = \mathbb{E}[z(f(z)-f(0))] = \mathbb{E}[zf(z)]$, since $\mathbb{E}z = 0$; the integrability of $zf(z)$ follows from the bound $\mathbb{E}|z||f(z)-f(0)| \le \mathbb{E}|f'(z)|$, which the same computation with absolute values provides.

For the vector form, write $x = \Sigma^{1/2}g$ with $g \sim \mathcal{N}(0, I_d)$ and $F(g) = f(\Sigma^{1/2}g)$, so that $\nabla F(g) = \Sigma^{1/2}\nabla f(x)$. Conditionally on the other coordinates, the scalar statement applied to the ith coordinate of g gives $\mathbb{E}[g_i F(g)] = \mathbb{E}[\partial_i F(g)]$, by Fubini's theorem and the integrability assumptions. Hence $\mathbb{E}[xf(x)] = \Sigma^{1/2}\mathbb{E}[gF(g)] = \Sigma^{1/2}\mathbb{E}[\nabla F(g)] = \Sigma\,\mathbb{E}[\nabla f(x)]$. ∎

Hermite polynomials are the orthogonal polynomials of the Gaussian measure. The quickest definition is through their generating function: for real s and z,

$$G(s,z) := e^{sz-s^2/2} = \sum_{k\ge 0} \mathrm{He}_k(z)\,\frac{s^k}{k!}. \tag{A.1}$$

Expanding $e^{sz}e^{-s^2/2}$ as a product of two power series shows that He_k is a monic polynomial of degree k: $\mathrm{He}_0 = 1$, $\mathrm{He}_1(z) = z$, $\mathrm{He}_2(z) = z^2 - 1$, $\mathrm{He}_3(z) = z^3 - 3z$, $\mathrm{He}_4(z) = z^4 - 6z^2 + 3$. The normalized Hermite polynomials are $h_k = \mathrm{He}_k/\sqrt{k!}$, and the *Hermite coefficients* of $f \in L^2(\gamma)$ are $\hat f_k = \mathbb{E}[f(z)h_k(z)]$.

Lemma A.2 (Hermite polynomials). *(i)* $\mathrm{He}_k' = k\,\mathrm{He}_{k-1}$ *and* $\mathrm{He}_{k+1}(z) = z\,\mathrm{He}_k(z) - k\,\mathrm{He}_{k-1}(z)$*; equivalently* $h_k' = \sqrt{k}\,h_{k-1}$ *and* $zh_k = \sqrt{k+1}\,h_{k+1} + \sqrt{k}\,h_{k-1}$.

(ii) $\mathbb{E}[\mathrm{He}_j(z)\,\mathrm{He}_k(z)] = k!\,\delta_{jk}$*, so the* h_k *are orthonormal in* $L^2(\gamma)$*; they form an orthonormal basis.*

(iii) If $f, f', \dots, f^{(k-1)}$ *are absolutely continuous and* $\mathbb{E}\big[|f^{(i)}(z)|(1+|z|)^{k-i}\big] < \infty$ *for* $0 \le i \le k$*, then* $\mathbb{E}[f(z)\,\mathrm{He}_k(z)] = \mathbb{E}[f^{(k)}(z)]$*, so* $\hat f_k = \mathbb{E}[f^{(k)}(z)]/\sqrt{k!}$.

Proof. For (i), differentiate the generating function: $\partial_z G = sG$ and $\partial_s G = (z-s)G$; comparing the coefficients of $s^k/k!$ gives $\mathrm{He}_k' = k\,\mathrm{He}_{k-1}$ and $\mathrm{He}_{k+1} = z\,\mathrm{He}_k - k\,\mathrm{He}_{k-1}$. Dividing by the factorials gives the normalized forms. For (ii), compute with the moment generating function of z:

$$\mathbb{E}[G(s,z)G(t,z)] = e^{-(s^2+t^2)/2}\,\mathbb{E}[e^{(s+t)z}] = e^{st} = \sum_{k\ge 0}\frac{(st)^k}{k!}.$$

The generating function also shows that $\mathrm{He}_k(z) = \mathbb{E}[(z+\mathrm{i}y)^k]$ for a standard Gaussian y independent of z, so $\sum_k |\mathrm{He}_k(z)||s|^k/k! \le e^{|s||z|}\mathbb{E}[e^{|s||y|}]$; since

$\mathbb{E}[e^{(|s|+|t|)|z|}]$ is finite, expectation and summation can be exchanged; comparing the coefficients of $s^j t^k$ gives $\mathbb{E}[\mathrm{He}_j\,\mathrm{He}_k]/(j!\,k!) = \delta_{jk}/k!$. That the h_k are complete, that is, that polynomials are dense in $L^2(\gamma)$, is classical (Bogachev 1998). For (iii), the recurrence and $\mathrm{He}'_{k-1} = (k-1)\,\mathrm{He}_{k-2}$ give $\mathrm{He}_k = z\,\mathrm{He}_{k-1} - \mathrm{He}'_{k-1}$, and Theorem A.1 applied to the absolutely continuous function $f\,\mathrm{He}_{k-1}$, whose derivative $f'\,\mathrm{He}_{k-1} + (k-1) f\,\mathrm{He}_{k-2}$ is integrable by the hypothesis, gives $\mathbb{E}[z f\,\mathrm{He}_{k-1}] = \mathbb{E}[f'\,\mathrm{He}_{k-1}] + \mathbb{E}[f\,\mathrm{He}'_{k-1}]$. Subtracting $\mathbb{E}[f\,\mathrm{He}'_{k-1}]$ from both sides,

$$\mathbb{E}[f\,\mathrm{He}_k] = \mathbb{E}[z f\,\mathrm{He}_{k-1}] - \mathbb{E}[f\,\mathrm{He}'_{k-1}] = \mathbb{E}[f'\,\mathrm{He}_{k-1}],$$

and induction on k, applied to f', whose derivatives satisfy the hypothesis with $k-1$ in place of k, finishes the proof. ∎

For example, the rectified linear unit $\mathrm{ReLU}(z) = \max(z, 0)$ has $\hat{\mathrm{ReLU}}_0 = \mathbb{E}[\mathrm{ReLU}(z)] = 1/\sqrt{2\pi}$, $\hat{\mathrm{ReLU}}_1 = \mathbb{E}[\mathbf{1}\{z > 0\}] = \frac{1}{2}$ by part (iii), and $\hat{\mathrm{ReLU}}_2 = \mathbb{E}[(z^3 - z)\mathbf{1}\{z > 0\}]/\sqrt{2} = \varphi(0)/\sqrt{2} = 1/(2\sqrt{\pi})$ by direct computation.

The basic fact that makes Gaussian data tractable is that Hermite polynomials of correlated Gaussians are orthogonal across degrees.

Theorem A.3 (Hermite polynomials of correlated Gaussians). *Let (u, v) be jointly Gaussian with $\mathbb{E}u = \mathbb{E}v = 0$, $\mathbb{E}u^2 = \mathbb{E}v^2 = 1$ and $\mathbb{E}uv = \rho$. Then $\mathbb{E}[\mathrm{He}_j(u)\,\mathrm{He}_k(v)] = \delta_{jk}\,k!\,\rho^k$, that is, $\mathbb{E}[h_j(u)h_k(v)] = \delta_{jk}\rho^k$. In particular, for unit vectors $a, b \in \mathbb{R}^d$ and $x \sim \mathcal{N}(0, I_d)$,*

$$\mathbb{E}\big[h_j(\langle a, x\rangle)\,h_k(\langle b, x\rangle)\big] = \delta_{jk}\,\langle a, b\rangle^k.$$

Proof. The moment generating function of $su + tv$ is $e^{(s^2+t^2+2st\rho)/2}$, so $\mathbb{E}[G(s,u)G(t,v)] = e^{st\rho} = \sum_k (st\rho)^k/k!$. Comparing coefficients of $s^j t^k$, as in the proof of Lemma A.2, gives the first claim. The pair $(\langle a, x\rangle, \langle b, x\rangle)$ is jointly Gaussian with unit variances and covariance $\langle a, b\rangle$. ∎

Corollary A.4 (Dual activations). *Let $f, g \in L^2(\gamma)$ and let (u, v) be as in Theorem A.3. Then*

$$\mathbb{E}[f(u)g(v)] = \sum_{k\ge 0} \hat{f}_k\,\hat{g}_k\,\rho^k,$$

and the series converges absolutely for $|\rho| \le 1$. In particular $\rho \mapsto \mathbb{E}[f(u)f(v)] = \sum_k \hat{f}_k^2 \rho^k$ is a power series with nonnegative coefficients.

Proof. The bilinear form $B(f, g) = \mathbb{E}[f(u)g(v)]$ satisfies $|B(f, g)| \le \|f\|\|g\|$ by the Cauchy–Schwarz inequality, because u and v are standard Gaussians; so B is continuous on $L^2(\gamma) \times L^2(\gamma)$. For the truncations $f_K = \sum_{k\le K} \hat{f}_k h_k$ and g_K, Theorem A.3 gives $B(f_K, g_K) = \sum_{k\le K} \hat{f}_k \hat{g}_k \rho^k$. As $K \to \infty$, $f_K \to f$ and $g_K \to g$ in $L^2(\gamma)$ by Lemma A.2(ii), so the left side tends to $B(f, g)$, while the right side converges absolutely because $\sum_k |\hat{f}_k \hat{g}_k| \le \|f\|\|g\|$. ∎

Lemma A.5 (Price's theorem). *Let f and g be absolutely continuous with $f, f', g, g' \in L^2(\gamma)$, and let (u, v) be as in Theorem A.3. Then $\rho \mapsto \mathbb{E}[f(u)g(v)]$ is differentiable on $(-1, 1)$, with*

$$\frac{\partial}{\partial \rho}\mathbb{E}[f(u)g(v)] = \mathbb{E}[f'(u)g'(v)].$$

Proof. The Hermite coefficients of f' are $\widehat{(f')}_{k-1} = \sqrt{k}\,\hat{f}_k$: the induction step in the proof of Lemma A.2(iii) gives $\mathbb{E}[f' \operatorname{He}_{k-1}] = \mathbb{E}[f \operatorname{He}_k]$, and the integrability it needs holds because f and f' are in $L^2(\gamma)$ and polynomials are in $L^2(\gamma)$; dividing by $\sqrt{(k-1)!}$ gives the claim. Corollary A.4 applied to f' and g' gives $\mathbb{E}[f'(u)g'(v)] = \sum_{k\geq 1} k\hat{f}_k\hat{g}_k\rho^{k-1}$, which is the termwise derivative of the power series $\sum_k \hat{f}_k\hat{g}_k\rho^k$. A power series with radius of convergence at least 1 may be differentiated termwise on $(-1, 1)$. ∎

A.2 Spherical harmonics

Let σ be the uniform probability measure on the unit sphere $\mathbb{S}^{d-1} \subset \mathbb{R}^d$, $d \geq 3$. A *spherical harmonic* of degree k is the restriction to the sphere of a homogeneous polynomial of degree k on $\mathbb{R}^d$ that is harmonic (its Laplacian vanishes). They form a finite-dimensional space $\mathcal{H}_k$, and the following facts are classical (Stein and Weiss 1971, Chapter IV); see also Atkinson and Han (2012) and Dai and Xu (2013).

Theorem A.6 (Spherical harmonics). *(i) $L^2(\mathbb{S}^{d-1}, \sigma)$ is the orthogonal direct sum of the spaces $\mathcal{H}_k$, $k \geq 0$, and*

$$N_{d,k} := \dim \mathcal{H}_k = \binom{d+k-1}{k} - \binom{d+k-3}{k-2},$$

with the second binomial coefficient read as 0 for $k < 2$. For fixed k, $N_{d,k} = \frac{d^k}{k!}\big(1 + O(1/d)\big)$ as $d \to \infty$.

(ii) (Addition theorem.) *Let $Y_1, \dots, Y_{N_{d,k}}$ be an orthonormal basis of $\mathcal{H}_k$ in $L^2(\sigma)$. The function $Z_k(x, y) = \sum_j Y_j(x)Y_j(y)$ does not depend on the choice of basis, is the kernel of the orthogonal projection onto $\mathcal{H}_k$, and depends only on $\langle x, y\rangle$: $Z_k(x, y) = N_{d,k}\,P_{d,k}(\langle x, y\rangle)$, where $P_{d,k}$ is the Gegenbauer polynomial of degree k for dimension d, normalized by $P_{d,k}(1) = 1$. In particular $Z_k(x, x) = N_{d,k}$ for every x.*

(iii) (Funk–Hecke formula.) *For every bounded measurable $F\colon [-1, 1] \to \mathbb{R}$ and every $Y \in \mathcal{H}_k$,*

$$\int F(\langle x, y\rangle)\, Y(y)\, \mathrm{d}\sigma(y) = \mu_k(F)\, Y(x),$$

with $\mu_k(F) = \frac{|\mathbb{S}^{d-2}|}{|\mathbb{S}^{d-1}|}\int_{-1}^{1} F(t)P_{d,k}(t)(1-t^2)^{(d-3)/2}\,\mathrm{d}t$. Consequently every kernel of the form $K(x, y) = F(\langle x, y\rangle)$ on the sphere has the spherical harmonics as eigenfunctions, with the eigenvalue $\mu_k(F)$ shared by all of $\mathcal{H}_k$.

Proof of the parts used in this book. We take the decomposition and the dimension formula in (i) from the references. The asymptotics follow from the formula: $\binom{d+k-1}{k} = \frac{d^k}{k!}(1 + O(1/d))$, while $\binom{d+k-3}{k-2} = O(d^{k-2})$.

For (ii), the function $Z_k(\cdot, y)$ satisfies $\langle Y, Z_k(\cdot, y)\rangle = Y(y)$ for every $Y \in \mathcal{H}_k$, which characterizes it as the reproducing kernel of the finite-dimensional space $\mathcal{H}_k$; so it does not depend on the basis. If U is an orthogonal matrix, then $Y \mapsto Y \circ U$ maps $\mathcal{H}_k$ onto itself and preserves $L^2(\sigma)$ inner products, because harmonicity, homogeneity and σ are rotation invariant; so $\{Y_j \circ U\}$ is another orthonormal basis, and $Z_k(Ux, Uy) = Z_k(x, y)$. Since every pair of points with the same inner product is mapped to every other such pair by some rotation, $Z_k(x, y)$ depends only on $\langle x, y\rangle$. In particular $Z_k(x, x)$ is a constant c, and integrating $\sum_j Y_j(x)^2$ against σ gives $c = \sum_j \|Y_j\|^2 = N_{d,k}$. The identification of $Z_k(x, y)/N_{d,k}$ with the Gegenbauer polynomial and part (iii) are in Stein and Weiss (1971, Chapter IV). ∎

For $d = 3$, the polynomial $P_{3,k}$ is the Legendre polynomial of degree k, and $N_{3,k} = 2k + 1$.

A.3 Wishart matrices and the Marchenko–Pastur law

Let $X \in \mathbb{R}^{n\times p}$ have independent $\mathcal{N}(0, 1)$ entries, with rows $x_1, \dots, x_n \in \mathbb{R}^p$, and let $\hat{\Sigma} = X^\top X/n$ be the sample covariance matrix; its population counterpart is I_p. The first result is exact at every size.

Theorem A.7 (Mean of the inverse Wishart matrix). *If $n \geq p + 2$, then $\mathbb{E}[(X^\top X)^{-1}] = I_p/(n - p - 1)$. If $n \in \{p, p + 1\}$, then $X^\top X$ is invertible almost surely and $\mathbb{E}\,\mathrm{tr}[(X^\top X)^{-1}] = +\infty$.*

Proof. The plan is to use rotational invariance to reduce to one diagonal entry, and then to compute that entry with a Schur complement. When $n \geq p$, the columns of X are linearly independent almost surely, so $X^\top X$ is invertible. For every orthogonal $U \in \mathbb{R}^{p\times p}$ the matrix XU has the same law as X, so $M = \mathbb{E}[(X^\top X)^{-1}]$, if finite (it is, once the diagonal entries have finite mean, because the off-diagonal entries of a positive definite matrix are bounded by the geometric mean of the corresponding diagonal entries), satisfies $U^\top MU = M$ for all U, and is therefore a multiple cI_p of the identity; also all diagonal entries of $(X^\top X)^{-1}$ have the same law. Let X_1 be the first column of X and P the orthogonal projection onto the span of the other $p - 1$ columns. By the formula for the inverse of a block matrix,

$$\left[(X^\top X)^{-1}\right]_{11} = \frac{1}{X_1^\top X_1 - X_1^\top P X_1} = \frac{1}{\|(I - P)X_1\|^2}.$$

Conditionally on the other columns, $I - P$ is a projection of rank $n - p + 1$ independent of $X_1 \sim \mathcal{N}(0, I_n)$, so $\|(I - P)X_1\|^2$ has the χ^2 distribution with $k = n - p + 1$ degrees of freedom. For $k \geq 3$, $\mathbb{E}[1/\chi^2_k] = \frac{\Gamma(k/2-1)}{2\Gamma(k/2)} = \frac{1}{k-2}$, and for $k \in \{1, 2\}$ the expectation is infinite, because the χ^2_k density does not vanish at 0. With $k - 2 = n - p - 1$ this gives $c = 1/(n - p - 1)$ when $n \geq p + 2$, and an infinite expected trace when $n - p \in \{0, 1\}$. ∎

In the proportional regime, $p/n \to \gamma \in (0,\infty)$, the eigenvalues of $\hat\Sigma$ do not concentrate at 1 but spread over an interval. We describe their distribution through the *Stieltjes transform*

$$s_p(z) = \frac{1}{p}\operatorname{tr}\left(\hat\Sigma - zI_p\right)^{-1} = \frac{1}{p}\sum_{j=1}^{p}\frac{1}{\lambda_j(\hat\Sigma) - z},$$

which for real $z = -\lambda < 0$ is a positive, decreasing, convex function of λ. Its derivative gives the other trace that appears in ridge regression: $\frac{d}{d\lambda}s_p(-\lambda) = -\frac{1}{p}\operatorname{tr}(\hat\Sigma + \lambda I)^{-2}$.

Theorem A.8 (Marchenko–Pastur law). *Let $n, p \to \infty$ with $p/n \to \gamma \in (0,\infty)$. For every $\lambda > 0$, almost surely,*

$$\frac{1}{p}\operatorname{tr}\left(\hat\Sigma + \lambda I_p\right)^{-1} \to m(-\lambda), \qquad \frac{1}{p}\operatorname{tr}\left(\hat\Sigma + \lambda I_p\right)^{-2} \to m'(-\lambda),$$

where $m(-\lambda)$ is the unique positive solution m of

$$\gamma z m^2 - (1 - \gamma - z)m + 1 = 0 \qquad \text{at } z = -\lambda, \tag{A.2}$$

namely $m(-\lambda) = \left(\gamma - 1 - \lambda + \sqrt{(\gamma - 1 - \lambda)^2 + 4\gamma\lambda}\right)/(2\gamma\lambda)$, and m' denotes the derivative of $m(z)$ in z. The function m is the Stieltjes transform of the Marchenko–Pastur law μ_γ, which has density $\sqrt{(b-x)(x-a)}/(2\pi\gamma x)$ on $[a,b] = [(1-\sqrt{\gamma})^2, (1+\sqrt{\gamma})^2]$ and, when $\gamma > 1$, an atom of mass $1 - 1/\gamma$ at 0. Consequently, almost surely, the empirical spectral distribution $\mu_p = \frac{1}{p}\sum_j \delta_{\lambda_j(\hat\Sigma)}$ converges weakly to μ_γ.

Proof. The plan is to derive an approximate quadratic equation for $\mathbb{E}s_p$ from a resolvent identity, using that each row of X is independent of the matrix built from the other rows, and then to show that s_p concentrates around its mean. Fix $\lambda > 0$, write $z = -\lambda$, $R = (\hat\Sigma + \lambda I)^{-1}$, $s_p = \frac{1}{p}\operatorname{tr} R$ and $\gamma_n = p/n$. Every resolvent below is positive definite with operator norm at most $1/\lambda$.

A resolvent identity. Since $\hat\Sigma R = I + zR$, taking normalized traces gives $1 + zs_p = \frac{1}{p}\operatorname{tr}(\hat\Sigma R) = \frac{1}{pn}\sum_i x_i^\top R x_i$. Let $R_{(i)} = (\hat\Sigma - \frac{1}{n}x_i x_i^\top + \lambda I)^{-1}$ be the resolvent without the ith row and $q_i = \frac{1}{n}x_i^\top R_{(i)} x_i \geq 0$. The Sherman–Morrison formula gives $Rx_i = R_{(i)}x_i/(1+q_i)$, hence $x_i^\top R x_i = n q_i/(1+q_i)$, and

$$1 + zs_p = \frac{1}{p}\sum_{i=1}^{n}\frac{q_i}{1+q_i} = \frac{1}{\gamma_n}\Big(1 - \frac{1}{n}\sum_{i=1}^{n}\frac{1}{1+q_i}\Big). \tag{A.3}$$

Removing one row changes little. Again by Sherman–Morrison, $R_{(i)} - R = \frac{1}{n}R_{(i)}x_i x_i^\top R_{(i)}/(1+q_i)$, a positive semidefinite matrix of rank one with trace $\frac{1}{n}x_i^\top R_{(i)}^2 x_i/(1+q_i) \leq \frac{1}{\lambda}\cdot\frac{q_i}{1+q_i} \leq \frac{1}{\lambda}$. Hence $0 \leq \operatorname{tr} R_{(i)} - \operatorname{tr} R \leq 1/\lambda$. As a consequence, replacing the row x_i by an independent copy changes s_p by at most $1/(p\lambda)$.

Concentration. By the rank-one bound, s_p is a function of n independent rows with bounded differences $1/(p\lambda)$, so McDiarmid's inequality (Theorem A.19) gives $\Pr(|s_p - \mathbb{E}s_p| \geq \varepsilon) \leq 2\exp(-2\varepsilon^2 p^2\lambda^2/n)$. Since $p^2/n \to \infty$ at a linear rate, these probabilities are summable, and the Borel–Cantelli lemma gives $s_p - \mathbb{E}s_p \to 0$ almost surely. The Efron–Stein inequality (Boucheron et al. 2013, Theorem 3.1) gives in the same way $\mathrm{Var}(s_p) \leq n/(2p^2\lambda^2)$.

The quadratic form concentrates. The row $x_i \sim \mathcal{N}(0, I_p)$ is independent of $R_{(i)}$, and for a fixed symmetric matrix A we have $\mathbb{E}[x^\top Ax] = \operatorname{tr} A$ and $\mathrm{Var}(x^\top Ax) = 2\operatorname{tr}(A^2)$ when $x \sim \mathcal{N}(0, I_p)$. Hence, with $s_{(i)} = \frac{1}{p}\operatorname{tr} R_{(i)}$, conditionally on $R_{(i)}$ the quadratic form q_i has mean $\gamma_n s_{(i)}$ and variance $\frac{2}{n^2}\operatorname{tr} R_{(i)}^2 \leq \frac{2p}{n^2\lambda^2}$. We split $q_i - \gamma_n\mathbb{E}s_p$ into $q_i - \gamma_n s_{(i)}$, $\gamma_n(s_{(i)} - s_p)$ and $\gamma_n(s_p - \mathbb{E}s_p)$, and bound the three in mean by the conditional variance, the rank-one bound, and the variance bound:

$$\mathbb{E}|q_i - \gamma_n\mathbb{E}s_p| \leq \frac{\sqrt{2\gamma_n/n}}{\lambda} + \frac{\gamma_n}{p\lambda} + \gamma_n\sqrt{\mathrm{Var}(s_p)} =: \epsilon_n,$$

and $\epsilon_n \to 0$.

The limiting equation. Take expectations in (A.3) and write $\bar{s} = \mathbb{E}s_p \in [0, 1/\lambda]$. Because $q_i \geq 0$ and $\gamma_n\bar{s} \geq 0$, the function $a \mapsto 1/(1+a)$ is 1-Lipschitz on the relevant range, so $\mathbb{E}[1/(1+q_i)] = 1/(1+\gamma_n\bar{s}) + O(\epsilon_n)$, and

$$1 + z\bar{s} = \frac{1}{\gamma_n}\Big(1 - \frac{1}{1+\gamma_n\bar{s}}\Big) + O(\epsilon_n) = \frac{\bar{s}}{1+\gamma_n\bar{s}} + O(\epsilon_n).$$

Multiplying by $1+\gamma_n\bar{s} \leq 1+\gamma_n/\lambda$ gives $(1+z\bar{s})(1+\gamma_n\bar{s}) - \bar{s} \to 0$. Along every subsequence on which $\bar{s}$ converges, its limit $s^* \geq 0$ satisfies $(1+zs^*)(1+\gamma s^*) = s^*$, which expands to (A.2). At $z = -\lambda$ the quadratic $\gamma z m^2 - (1-\gamma-z)m + 1$ has roots whose product is $1/(\gamma z) < 0$, so exactly one root is positive; and 0 is not a root. Hence every subsequential limit of $\bar{s}$ equals the positive root $m(-\lambda)$, so $\bar{s} \to m(-\lambda)$, and with the concentration step, $s_p \to m(-\lambda)$ almost surely.

Derivatives. The functions $\lambda \mapsto \frac{1}{p}\operatorname{tr}(\hat{\Sigma} + \lambda I)^{-1}$ are convex on $(0, \infty)$. Almost surely they converge on a countable dense set of λ simultaneously, hence, being monotone with a continuous limit, at every $\lambda > 0$. Pointwise convergence of convex functions on an open interval implies convergence of their derivatives at every point where the limit is differentiable (Rockafellar 1970, Theorem 24.5), and $\lambda \mapsto m(-\lambda)$ is differentiable. The derivative of $\frac{1}{p}\operatorname{tr}(\hat{\Sigma} + \lambda I)^{-1}$ in λ is $-\frac{1}{p}\operatorname{tr}(\hat{\Sigma} + \lambda I)^{-2}$, and the derivative of $m(-\lambda)$ in λ is $-m'(-\lambda)$, which gives the second limit.

The identification of m with the Stieltjes transform of μ_γ is a computation with the explicit density, which we omit; see Marchenko and Pastur (1967) and Bai and Silverstein (2010, Chapter 3). A probability measure on $[0, \infty)$ is determined by its Stieltjes transform on the negative half-line, because the transform extends analytically to $\mathbb{C} \setminus [0, \infty)$, where the Stieltjes inversion formula recovers the measure.

Weak convergence. The mean of μ_p is $\frac{1}{p}\operatorname{tr}\hat{\Sigma} = \frac{1}{np}\sum_{i,j} X_{ij}^2$, which has mean 1 and variance $2/(np)$; these variances are summable, so Chebyshev's inequality and the Borel–Cantelli lemma show that the mean of μ_p tends

to 1 almost surely. Fix an outcome in the probability-one event on which this holds and on which $\int \frac{\mu_p(\mathrm{d}t)}{t+\lambda} \to m(-\lambda)$ for every rational $\lambda > 0$. The measures μ_p are then tight, because their means are bounded, so every subsequence has a further subsequence that converges weakly to some probability measure ν on $[0,\infty)$. Since $t \mapsto 1/(t+\lambda)$ is bounded and continuous on $[0,\infty)$, the Stieltjes transform of ν equals m at every rational point of the negative half-line, hence at every point by continuity, and so $\nu = \mu_\gamma$. Every subsequence of (μ_p) therefore has a further subsequence converging to μ_γ, which is weak convergence of the whole sequence. ∎

Remark A.9. The edges of the Marchenko–Pastur law explain the double descent peak of Chapter 1: the smallest nonzero eigenvalue of $\hat{\Sigma}$ tends to $(1-\sqrt{\gamma})^2$, which vanishes at $\gamma = 1$, so the minimum-norm least squares solution, which inverts $\hat{\Sigma}$ on its range, becomes unstable exactly at the interpolation threshold. (The convergence of the extreme eigenvalues to the edges of the support needs an argument beyond the Stieltjes transform; see Bai and Silverstein (2010, Chapter 5).)

A.4 Martingales, Grönwall, and the ODE method

Several chapters analyze a stochastic recursion in which each step moves the state by an amount of order $1/d$, with random fluctuations of the same order, and claim that over d steps the state follows a deterministic ordinary differential equation. This section proves such a claim in the generality the chapters need. We write $\mathcal{F}_t$ for the σ-algebra generated by the first t steps.

Lemma A.10 (Doob's maximal inequality). *Let $(M_t)_{t=0}^T$ be a martingale with values in $\mathbb{R}^D$ and $\mathbb{E}\|M_T\|^2 < \infty$. Then for every $a > 0$,*

$$\Pr\Big(\max_{t\le T}\|M_t\| \ge a\Big) \le \frac{\mathbb{E}\|M_T\|^2}{a^2}, \qquad \mathbb{E}\Big[\max_{t\le T}\|M_t\|^2\Big] \le 4\,\mathbb{E}\|M_T\|^2.$$

If the increments satisfy $\mathbb{E}[\|M_{t+1}-M_t\|^2 \mid \mathcal{F}_t] \le v_t$, then $\mathbb{E}\|M_T\|^2 \le \mathbb{E}\|M_0\|^2 + \sum_{t<T}\mathbb{E}v_t$.

Proof. The process $\|M_t\|$ is a nonnegative submartingale, by Jensen's inequality for conditional expectations, and the two inequalities are Doob's maximal and L^2 inequalities for nonnegative submartingales (Williams 1991). For the last claim, increments of a square-integrable martingale are orthogonal: $\mathbb{E}\langle M_{t+1}-M_t, M_s - M_{s-1}\rangle = 0$ for $s \le t$, by conditioning on $\mathcal{F}_t$; so $\mathbb{E}\|M_T - M_0\|^2 = \sum_t \mathbb{E}\|M_{t+1}-M_t\|^2$, and the same conditioning shows $\mathbb{E}\langle M_T - M_0, M_0\rangle = 0$. ∎

Theorem A.11 (Freedman's inequality). *Let (M_t) be a real martingale with $M_0 = 0$ whose increments satisfy $M_{t+1} - M_t \le c$, and let $V_t = \sum_{s<t}\mathbb{E}[(M_{s+1}-M_s)^2 \mid \mathcal{F}_s]$. Then for all $a, v > 0$,*

$$\Pr\big(M_t \ge a \text{ and } V_t \le v \text{ for some } t\big) \le \exp\Big(-\frac{a^2}{2(v+ca)}\Big).$$

In particular (Azuma–Hoeffding), if $|M_{t+1} - M_t| \le c_t$ almost surely, then $\Pr(|M_T| \ge a) \le 2\exp\big(-a^2/(2\sum_{t<T} c_t^2)\big)$.

Freedman's inequality is proved by Freedman (1975). The Azuma–Hoeffding inequality follows from the exponential method; see Boucheron et al. (2013).

Theorem A.12 (Grönwall's inequality). *(i) If $u\colon [0,T] \to [0,\infty)$ is continuous and $u(t) \le a + \int_0^t b\,u(s)\,\mathrm{d}s$ for constants $a, b \ge 0$, then $u(t) \le a e^{bt}$.*

(ii) If $u_0, u_1, \dots$ are nonnegative and $u_t \le a + b\sum_{s<t} u_s$ for all t, then $u_t \le a(1+b)^t \le a e^{bt}$.

Proof. For (i), let $U(t) = a + \int_0^t bu$; then $U' = bu \le bU$, so $(e^{-bt}U)' \le 0$ and $u \le U(t) \le e^{bt}U(0) = ae^{bt}$. For (ii), let $S_t = a + b\sum_{s<t} u_s$; then $u_t \le S_t$ and $S_{t+1} = S_t + bu_t \le (1+b)S_t$, so $u_t \le S_t \le (1+b)^t S_0 = (1+b)^t a$. ∎

Theorem A.13 (The ODE method). *Let $U \subseteq \mathbb{R}^D$ be open and let $F\colon U \to \mathbb{R}^D$ be L-Lipschitz and bounded by B on U. Let $(Z_t)_{t\ge 0}$ be adapted random vectors with*

$$Z_{t+1} = Z_t + \frac{1}{d}\big(F(Z_t) + \xi_{t+1} + \epsilon_{t+1}\big),$$

where, on the event $\{Z_t \in U\}$, the noise satisfies $\mathbb{E}[\xi_{t+1} \mid \mathcal{F}_t] = 0$ and $\mathbb{E}[\|\xi_{t+1}\|^2 \mid \mathcal{F}_t] \le \varsigma^2$, and the error satisfies $\|\epsilon_{t+1}\| \le \epsilon$. Let $z\colon [0,T] \to U$ solve $\dot z = F(z)$, and suppose that every point within distance ρ of the path $z([0,T])$ lies in U. For $a > 0$ put

$$\Delta = \big(\|Z_0 - z(0)\| + a + T\epsilon + TLB/(2d)\big)e^{LT}.$$

If $\Delta < \rho$, then

$$\Pr\Big(\max_{0\le t\le Td}\|Z_t - z(t/d)\| > \Delta\Big) \le \frac{\varsigma^2 T}{a^2 d}.$$

Proof. The plan is to compare the recursion with the Euler scheme for the ODE, to control the accumulated noise with Doob's inequality, and to close the argument with Grönwall's inequality, using a stopping time to stay inside U. Write $D_t = Z_t - z(t/d)$. Integrating the ODE over one step gives $z(\frac{t+1}{d}) = z(\frac{t}{d}) + \frac{1}{d}F(z(\frac{t}{d})) + r_t$ with the Euler error

$$\|r_t\| \le \int_{t/d}^{(t+1)/d} \big\|F(z(u)) - F\big(z(\tfrac{t}{d})\big)\big\|\,\mathrm{d}u \le \frac{LB}{2d^2},$$

because $\|z(u) - z(t/d)\| \le B(u - t/d)$. Let τ be the first t with $\|D_t\| > \rho$; for $t < \tau$ the point Z_t lies within ρ of the path, hence in U. Let $M_t = \sum_{s<t\wedge\tau} \xi_{s+1}$, a martingale because τ is a stopping time. Subtracting the two recursions

and summing, for $t \le \tau \wedge \lfloor Td \rfloor$,

$$D_t = D_0 + \frac{1}{d}\sum_{s<t}\big(F(Z_s) - F(z(\tfrac{s}{d}))\big) + \frac{1}{d}M_t + \frac{1}{d}\sum_{s<t}\epsilon_{s+1} - \sum_{s<t} r_s.$$

Taking norms and using the Lipschitz bound,

$$\|D_t\| \le \Big(\|D_0\| + \frac{1}{d}\max_{u\le Td}\|M_u\| + T\epsilon + \frac{TLB}{2d}\Big) + \frac{L}{d}\sum_{s<t}\|D_s\|.$$

On the event $E = \{\max_{u\le Td}\|M_u\| \le ad\}$, the discrete Grönwall inequality (Theorem A.12(ii)) gives $\|D_t\| \le \Delta$ for every $t \le \tau \wedge \lfloor Td \rfloor$. Since $\Delta < \rho$, this rules out $\tau \le \lfloor Td \rfloor$, so on E the bound holds for all $t \le Td$. Finally, the increments of M have conditional second moment at most ς^2, so by Lemma A.10, $\Pr(E^c) \le \mathbb{E}\|M_{\lfloor Td\rfloor}\|^2/(ad)^2 \le \varsigma^2 Td/(a^2d^2) = \varsigma^2 T/(a^2 d)$. ∎

In the applications, ϵ and $1/d$ tend to zero as the dimension grows and a is chosen as $d^{-1/4}$, say; then, with L, B, ς and T fixed, the recursion stays within $O(\|Z_0 - z(0)\| + d^{-1/4} + \epsilon)$ of the ODE on the time interval $[0, T]$ with probability $1 - O(d^{-1/2})$. Choosing $a = d^{-1/2}\log d$ instead shows that the fluctuations are at most of order $d^{-1/2}$ up to a logarithmic factor.

A.5 Fourier analysis on finite groups

The cyclic group. Let $p \ge 2$ be an integer, $\mathbb{Z}_p = \{0, 1, \dots, p-1\}$ with addition modulo p, and $e_p(x) = e^{2\pi i x/p}$. For $f : \mathbb{Z}_p \to \mathbb{C}$ define the Fourier transform $\hat f(\omega) = \frac{1}{p}\sum_{a\in\mathbb{Z}_p} f(a)\, e_p(-\omega a)$ for $\omega \in \mathbb{Z}_p$, and the convolution $(f * g)(c) = \frac{1}{p}\sum_a f(a) g(c-a)$.

Theorem A.14 (Fourier analysis on $\mathbb{Z}_p$). *For all $f, g, u, u', v : \mathbb{Z}_p \to \mathbb{C}$:*

(i) (inversion) $f(a) = \sum_\omega \hat f(\omega) e_p(\omega a)$;

(ii) (Plancherel) $\frac{1}{p}\sum_a f(a)\overline{g(a)} = \sum_\omega \hat f(\omega)\overline{\hat g(\omega)}$;

(iii) (convolution) $\widehat{f * g} = \hat f\, \hat g$;

(iv) (triple correlation) $\frac{1}{p^2}\sum_{a,b} u(a)u'(b)v(a+b) = \sum_\omega \hat u(-\omega)\,\hat u'(-\omega)\,\hat v(\omega)$.

If f is real, then $\hat f(-\omega) = \overline{\hat f(\omega)}$, and $f(a) = \hat f(0) + \sum_\omega |\hat f(\omega)| \cos(2\pi\omega a/p + \arg \hat f(\omega))$, where for odd p the sum runs over $\omega = 1, \dots, (p-1)/2$ with each term doubled.

Proof. Everything follows from the orthogonality relation

$$\frac{1}{p}\sum_{a\in\mathbb{Z}_p} e_p(\omega a) = \mathbf{1}\{\omega = 0\},$$

which holds because for $\omega \ne 0$ the sum is a geometric series of pth roots of unity whose ratio is not 1. For (i), substitute the definition of $\hat f$ into the right side and sum over ω first. For (ii), substitute (i) for

f and g and use orthogonality. For (iii), $\widehat{f * g}(\omega) = \frac{1}{p^2}\sum_{a,c} f(a)g(c-a)e_p(-\omega a)e_p(-\omega(c-a)) = \hat{f}(\omega)\hat{g}(\omega)$, substituting $b = c-a$. For (iv), expand v by (i): $\sum_{a,b} u(a)u'(b)v(a+b) = \sum_\omega \hat{v}(\omega)\sum_a u(a)e_p(\omega a)\sum_b u'(b)e_p(\omega b) = p^2\sum_\omega \hat{v}(\omega)\hat{u}(-\omega)\hat{u}'(-\omega)$. The statements for real f follow from conjugating the definition, and from pairing the terms ω and $-\omega$ in (i). ∎

Some chapters use the unnormalized transform $p\hat{f}$ instead (for example Chapter 10); the identities above then acquire the corresponding powers of p and are otherwise unchanged.

THE BOOLEAN CUBE. For $S \subseteq [d] = \{1,\dots,d\}$ the *parity* or character $\chi_S(x) = \prod_{i\in S} x_i$ is a function on $\{\pm1\}^d$, with $\chi_\emptyset = 1$. Let x be uniform on $\{\pm1\}^d$ and $\langle f, g\rangle = \mathbb{E}[f(x)g(x)]$.

Theorem A.15 (Fourier–Walsh expansion). *The 2^d characters form an orthonormal basis of the functions on $\{\pm1\}^d$. Every f equals $\sum_S \hat{f}(S)\chi_S$ with $\hat{f}(S) = \langle f, \chi_S\rangle$, and $\mathbb{E}[f(x)^2] = \sum_S \hat{f}(S)^2$.*

PROOF. Since $x_i^2 = 1$, $\chi_S\chi_T = \chi_{S\triangle T}$, and by independence of the coordinates $\mathbb{E}[\chi_{S\triangle T}(x)] = \prod_{i\in S\triangle T}\mathbb{E}[x_i]$, which is 1 if $S = T$ and 0 otherwise. So the characters are orthonormal, and 2^d orthonormal vectors in a 2^d-dimensional space form a basis. ∎

The *degree* of f is the largest $|S|$ with $\hat{f}(S) \neq 0$. Low-degree functions cannot have heavy tails, a fact known as hypercontractivity. We prove the fourth-moment version, which suffices for most purposes, and state the general one.

Theorem A.16 (Bonami's lemma and hypercontractivity). *If $f\colon \{\pm1\}^d \to \mathbb{R}$ has degree at most k, then $\mathbb{E}[f^4] \le 9^k\big(\mathbb{E}[f^2]\big)^2$. More generally, for every $q \ge 2$, $\big(\mathbb{E}|f|^q\big)^{1/q} \le (q-1)^{k/2}\big(\mathbb{E}f^2\big)^{1/2}$, and consequently, for $t \ge (2e)^{k/2}$,*

$$\Pr\big(|f(x)| \ge t\,(\mathbb{E}f^2)^{1/2}\big) \le \exp\Big(-\frac{k}{2e}\,t^{2/k}\Big).$$

The same inequalities hold for polynomials of degree at most k in a standard Gaussian vector.

PROOF OF THE FOURTH-MOMENT BOUND. The plan is induction on the number d of variables. For $d = 0$, f is constant and the bound holds with equality. For $d \ge 1$, write $f(x) = x_d g(x') + h(x')$, where $x' = (x_1,\dots,x_{d-1})$, g has degree at most $k-1$ and h degree at most k. Expanding $(x_d g + h)^4$ and averaging over x_d removes the odd powers of x_d, and $\mathbb{E}[f^2] = \mathbb{E}[g^2]+\mathbb{E}[h^2]$ in the same way. Hence $\mathbb{E}f^4 = \mathbb{E}g^4 + 6\,\mathbb{E}[g^2h^2] + \mathbb{E}h^4$. We bound the outer terms by the induction hypothesis and the middle one by the Cauchy–Schwarz inequality and the induction hypothesis again:

$$\mathbb{E}f^4 \le 9^{k-1}(\mathbb{E}g^2)^2 + 6\cdot 3^{k-1}3^k\,\mathbb{E}g^2\,\mathbb{E}h^2 + 9^k(\mathbb{E}h^2)^2.$$

Since $6\cdot 3^{2k-1} = 2\cdot 9^k$ and $9^{k-1} \le 9^k$, the right side is at most $9^k(\mathbb{E}g^2+\mathbb{E}h^2)^2 = 9^k(\mathbb{E}f^2)^2$. ∎

The general moment bound is the Bonami–Beckner hypercontractive inequality, and the tail bound follows from it by Markov's inequality with an optimized q; see O'Donnell (2014, Chapters 9 and 11), where the Gaussian case is also treated.

A.6 Concentration inequalities

We collect the standard inequalities used in the chapters. We prove Hoeffding's inequality and quote the others without proof; proofs are in Boucheron et al. (2013) and Vershynin (2018).

Theorem A.17 (Hoeffding's inequality). *If $X_1, \dots, X_n$ are independent with $X_i \in [a_i, b_i]$ almost surely, then for every $t > 0$,*

$$\Pr\Big(\Big|\sum_i (X_i - \mathbb{E}X_i)\Big| \ge t\Big) \le 2\exp\Big(-\frac{2t^2}{\sum_i (b_i-a_i)^2}\Big).$$

Proof. The plan is to bound the moment generating function of each summand and then to apply Markov's inequality to an exponential. We may assume $\mathbb{E}X_i = 0$, replacing X_i by $X_i - \mathbb{E}X_i$ and $[a_i, b_i]$ by an interval of the same length. Let $\psi_i(s) = \log \mathbb{E}e^{sX_i}$. Then $\psi_i(0) = 0$, $\psi_i'(0) = \mathbb{E}X_i = 0$, and $\psi_i''(s)$ is the variance of X_i under the probability measure with density $e^{sX_i}/\mathbb{E}e^{sX_i}$ relative to the law of X_i. A random variable with values in $[a_i, b_i]$ has variance at most $(b_i - a_i)^2/4$, because its variance is at most its mean squared distance to the midpoint $(a_i + b_i)/2$, which is at most $((b_i - a_i)/2)^2$. Taylor's theorem therefore gives $\psi_i(s) \le s^2(b_i - a_i)^2/8$, which is Hoeffding's lemma. Write $S = \sum_i X_i$ and $V = \sum_i (b_i - a_i)^2$. For $s > 0$, Markov's inequality and independence give

$$\Pr(S \ge t) \le e^{-st}\,\mathbb{E}e^{sS} = e^{-st}\prod_i \mathbb{E}e^{sX_i} \le \exp\Big(-st + \frac{s^2 V}{8}\Big),$$

and the choice $s = 4t/V$ makes the right side $\exp(-2t^2/V)$. The same bound for $-S$ and a union bound give the factor 2. ∎

Theorem A.18 (Bernstein's inequality). *If $X_1, \dots, X_n$ are independent with $\mathbb{E}X_i = 0$, $|X_i| \le c$ almost surely and $\sum_i \mathbb{E}X_i^2 \le v$, then for every $t > 0$,*

$$\Pr\Big(\Big|\sum_i X_i\Big| \ge t\Big) \le 2\exp\Big(-\frac{t^2}{2(v + ct/3)}\Big).$$

Theorem A.19 (McDiarmid's bounded differences inequality). *Let $X_1, \dots, X_n$ be independent and let F satisfy $|F(x) - F(x')| \le c_i$ whenever x and x' differ only in the ith coordinate. Then* $\Pr(|F(X) - \mathbb{E}F(X)| \ge t) \le 2\exp\big(-2t^2/\sum_i c_i^2\big)$.

Theorem A.20 (Gaussian concentration). *If $x \sim \mathcal{N}(0, I_d)$ and $f : \mathbb{R}^d \to \mathbb{R}$ is L-Lipschitz, then $\Pr(|f(x) - \mathbb{E} f(x)| \geq t) \leq 2e^{-t^2/(2L^2)}$ for every $t > 0$.*

Gaussian concentration is due to Tsirelson et al. (1976); it shows, for example, that $\|x\|$ concentrates around $\sqrt{d}$ with fluctuations of order one, since the norm is 1-Lipschitz, and that $\|x\|^2/d = 1 + O(d^{-1/2})$ with high probability.

Exercises

A.1 CONSOLIDATION. Use Lemma A.2(iii) to compute the Hermite coefficients of $\phi(z) = |z|$ and of $\phi(z) = \operatorname{sgn}(z)$, and check that $\sum_k \hat{\phi}_k^2 = \mathbb{E}[\phi(z)^2]$ numerically for the first twenty coefficients.

A.2 CONSOLIDATION. Let u and v be standard Gaussians with correlation ρ. Show that

$$\mathbb{E}[\mathrm{ReLU}(u)\,\mathrm{ReLU}(v)] = \tfrac{1}{2\pi}\big(\sqrt{1-\rho^2} + (\pi - \arccos\rho)\rho\big).$$

(Hint: By Lemma A.5 the ρ-derivative of the left side is $\mathbb{E}[\mathbf{1}\{u > 0\}\mathbf{1}\{v > 0\}] = \tfrac{1}{4} + \tfrac{\arcsin\rho}{2\pi}$; integrate from $\rho = 0$.)

A.3 CONSOLIDATION. Prove that $N_{d,1} = d$, $N_{d,2} = d(d+1)/2 - 1$, and identify $\mathcal{H}_1$ and $\mathcal{H}_2$ explicitly.

A.4 CONSOLIDATION. ⋆ Show from the explicit formula for $m(-\lambda)$ that $\lambda\, m(-\lambda) \to 1 - 1/\gamma$ as $\lambda \downarrow 0$ when $\gamma > 1$, and $\lambda\, m(-\lambda) \to 0$ when $\gamma < 1$; interpret the limit through the atom of μ_γ at 0. *(Hint: $\lambda m(-\lambda) = \int \frac{\lambda}{x+\lambda}\,\mathrm{d}\mu_\gamma(x)$.)*

A.5 CONSOLIDATION. Check Theorem A.7 by simulation for $n = 30$ and $p \in \{5, 15, 25, 27\}$, and explain why the empirical average of $\operatorname{tr}(X^\top X)^{-1}$ is unstable at $p = 29$.

A.6 VARIATION. Prove the variance formula $\operatorname{Var}(x^\top A x) = 2\operatorname{tr}(A^2)$ used in the proof of Theorem A.8, and generalize it to $x \sim \mathcal{N}(0, \Sigma)$.

A.7 CONSOLIDATION. ⋆ Show that the error bound of Theorem A.13 is sharp in its dependence on d: for $D = 1$, $F = 0$ and ± 1 noise, the maximum of $|Z_t - Z_0|$ over $t \leq d$ is of order $d^{-1/2}$.

Notes and References

Gaussian integration by parts is often called Stein's lemma, after its role in Stein's method for normal approximation (Stein 1981); Hermite polynomials and their generating function are classical, and O'Donnell (2014) develops them alongside Fourier analysis on the cube in a way that matches this book's use. The expansion of $\mathbb{E}[\phi(u)\phi(v)]$ in powers of the correlation, which Daniely et al. (2016) called the dual activation, is a form of Mehler's formula. Price's theorem is from Price (1958).

The Marchenko–Pastur law appeared in Marchenko and Pastur (1967); the resolvent argument above follows the standard Stieltjes-transform proof (Bai and Silverstein 2010), restricted to the negative real axis, where every estimate is real and elementary. The ODE method for stochastic approximation goes back to Ljung (1977), and Kushner and Yin (2003) give a modern account; its high-dimensional form, in which the step size is $1/d$ and the state is a finite set of overlaps, is the method of Saad and Solla (1995) made rigorous, for example by Wang et al. (2017) and Goldt et al. (2019).

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Index

accelerated time, 64
addition theorem, 368
aligned initialization, 39
approximation error, 341
attention sink, 298

balanced initialization, 38
basis pursuit, 56
Bayes predictor, 300
benign overfitting, 25
Bernstein's inequality, 376
Bonami's lemma, 375
bounded differences inequality, 376
Bregman divergence, 60

catapult, 319
central flow, 332
chaotic phase, 101
clock, 214
coherence, 244
compute-optimal allocation, 348
correlation loss, 146
correlation map, 102
correlational statistical query, 156
critical line, 108
cubature rule, 215

deep linear network, 34
depth scale, 108
descent phase, 154
deterministic equivalent, 346
diagonal linear network, 57
directional convergence, 83
Doob's maximal inequality, 372
double descent, 12
  epoch-wise, 12
  sample-wise, 12
dual activation, 367
dual certificate, 67
dynamical isometry, 112

edge of chaos, 101
edge of stability, 319
effective model complexity, 13
effective overlap, 198
emergence, 190, 349
end-to-end matrix, 34
excess loss, 341
exponential loss, 78

feature dimensionality, 242
fine structure, 1
finite-width fluctuations, 111
Fourier analysis
  on $\mathbb{Z}_p$, 374
  on the Boolean cube, 375
Fourier gap, 193
Fourier neuron, 214
free energy, 80
Freedman's inequality, 372
Funk–Hecke formula, 368

Gaussian concentration, 377
Gaussian elimination, 188
Gaussian integration by parts, 365
Gaussian process limit, 104
generative exponent, 158
Grönwall's inequality, 373
gradient descent
  large step size, 91
gradient flow, 123
gradient independence assumption, 111
grokking, 211

hard-margin SVM, 79
Hermite coefficient, 366
Hermite polynomials, 366
  of correlated Gaussians, 367
hidden progress, 190
Hoeffding's inequality, 376
homogeneous network, 86, 213
hyperbolic entropy, 59
hypercontractivity, 375

imbalance, 320
implicit bias, 55, 95
importance of a feature, 235
importance weighting, 90
in-context learning, 277

in-context learning score, 296
incremental learning, 34, 56
induction gain, 304
induction head, 296
information exponent, 147
interference, 235
interpolation threshold, 12

K-composition, 296
Karush–Kuhn–Tucker point, 81
kernel
    of a random network, 102
kernel lower bound, 132
kernel regime, 56

law of equi-separation, 255
layer-peeled model, 256
lazy training, 55, 125
leap complexity, 159
length map, 102
linear self-attention, 279
linearized network, 124
loss staircase, 33

majority function, 193
Marchenko–Pastur law, 19, 370
margin, 78
    average, 218
    normalized, 79
    of a network, 218
    smoothed, 86
Markov chain
    in-context, 297
max-margin direction, 79
max-margin network, 218
mean-field limit, 134
memorizing learner, 341
minimum-norm interpolator, 14
mirror descent, 58
model size, 341
monomial neuron, 190

neural collapse, 255
neural tangent kernel, 122
NTK parameterization, 124
null risk, 14

obtuse configuration, 240
ODE method, 373
online SGD, 146
order parameters, 168
    of signal propagation, 102
order–chaos transition, 107
ordered phase, 101
overlap, 146

parity function, 375
plateau, 197, 304
polysemantic neuron, 233
power-law spectrum, 341
pre-activation, 102
pretraining loss, 279
previous-token head, 296
Price's theorem, 368
product model, 320
progressive differentiation, 33
progressive sharpening, 319
proportional limit, 18

quanta, 341

reflection law, 328
ReLU network
    He initialization, 109
rich regime, 56
ridge regression, 14
risk, 14

Saad–Solla equations, 172
saddle-to-saddle dynamics, 44, 65
scaling law, 340
search phase, 147
self-stabilization, 330
sequence model, 345
sharpness, 318
silent alignment, 44
simplex equiangular tight frame, 257
singular mode, 35
soft committee machine, 168
source exponent, 341
sparse autoencoder, 234
sparse parity, 188
sparsity, 234
specialization, 167
spectral bias, 123, 131
spherical harmonics, 368
spinodal, 240

statistical query, 193
    lower bound, 195
Stein's lemma, 365
Stieltjes transform, 18, 370
superposition, 233
support vector, 79
symmetric phase, 167

tangent kernel, 124
teacher–student setup, 167
terminal phase of training, 254
test error, 14

unconstrained features model, 256

vanishing and exploding gradients, 100

Wishart matrix, 369

Zipf's law, 341

How This Book Was Made

Colophon

Roles

The Hydrogen Atoms of Deep Learning was written by Claude Opus 5.5, an AI model made by Anthropic, in October 2026. Xiaoyu Li conceived the series *The Missing Textbooks* and proposed this volume: its subject, its gap (solvable models of deep learning scattered across papers in incompatible notation, and rarely accompanied by an account of where they fail), its format (a phenomenon, a minimal solvable model, a theorem with a complete proof, and a section on what the model gets wrong), a first list of models, and its position relative to existing books. Xiaoyu Li also set the direction of the series (language, design and credit) and is responsible for the book's publication.

Process

A written plan fixed the thesis, the sixteen models, the global notation, and a specification for each chapter, including the editor's plan for each main theorem. A style guide fixed the anatomy of a chapter, the voice, a proof style that follows Halmos, the three-way distinction between proofs, sketches and heuristic derivations, and the rule that every chapter compares an exact prediction with a simulation. Separate instances of the same model then drafted the chapters, at most five at a time, each re-deriving its results, checking them numerically, verifying its references with web search, and writing the scripts behind its figures. Five chapters were begun by one set of instances and completed by another; two of these received a separate pass to finish and check the draft before refereeing. A coordinating instance wrote the prologue, preface, notation, epilogue and this colophon, designed the volume's additions to the series typography, and integrated the whole.

Each chapter, the prologue and the appendix then received one independent referee pass from a separate instance, which re-derived the proofs, re-ran the figure scripts, checked constants and exercises numerically, and checked references and historical claims by web search. The referees corrected what they found directly. They found few errors in the main theorems, and several gaps in proofs, which they closed or marked; most of their corrections were to numbers quoted from simulations, to the statements and answers of exercises, and to claims about what other papers prove, many of which were softened or removed because they could not be confirmed. The epilogue was written last and was not refereed.

Limits

No human has yet checked every proof. Several results are presented as heuristic derivations, marked with ◇, because no proof is known to the author or because a proof would not fit; they are checked against simulations but not proved. Web search was available while the book was written, but the full texts of most papers could not be opened, so references were checked against search results, abstracts and secondary accounts. Some historical details, and some translations of the notation of source papers, rest on these or on the author's memory, and the chapters say so where it matters. One referee pass per chapter finds many errors but not all of them. Results from 2024 to 2026 are cited from preprints where no refereed version could be found, and are marked as such where a claim rests on them. Corrections are welcome and will be recorded in the repository of *The Missing Textbooks*.

Typeset with pdfLaTeX in `missingbook`, the series class built on KOMA-Script, with the volume style `hydrogen`; Palatino (newpx) for text and mathematics, biblatex and biber for references, Ti*k*Z and pgfplots for figures, whose data were computed with Python, NumPy, SciPy and PyTorch. The lines on the title page are the Balmer series of hydrogen, drawn at the wavelengths given by Rydberg's formula; the wave at the foot of each opening page rises from chapter to chapter.